



सत्यमेव जयते
भारत सरकार
Government of India
विद्युत मंत्रालय
Ministry of Power
पूर्वी क्षेत्रीय विद्युत समिति



Eastern Regional Power Committee

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सं /NO. ERPC/EE/OPERATION/2026/2042

DATE:30.01.2026

सेवा में /To
संलग्न सूची के अनुसार /As per list enclosed.

विषय : 21.01.2026 (बुधवार) को ईआरपीसी सचिवालय, कोलकाता में भौतिक रूप से आयोजित 235वीं ओसीसी बैठक के कार्यवृत्त – के संबंध में

Sub: Minutes of 235th OCC Meeting held on 21.01.2026 (Wednesday) physically at ERPC Secretariat, Kolkata – reg

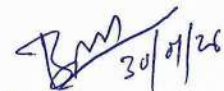
महोदय/महोदया,
Sir(s)/Madam,

कृपया अपनी जानकारी और आवश्यक कार्रवाई के लिए **21 जनवरी 2025 (बुधवार)** को आयोजित **235वीं ओसीसी बैठक** के **कार्यवृत्त** संलग्न देखें। यह ईआरपीसी वेबसाइट (www.erpc.gov.in) पर भी उपलब्ध है।
Please find enclosed **Minutes of 235th OCC Meeting** held on **21.01.2026 (Wednesday)** physically at **ERPC Secretariat, Kolkata at 10:30 hrs.** This is for your **kind information** and **necessary action**. The same is also available at ERPC website (www.erpc.gov.in).

टिप्पणियाँ, यदि कोई हों, कृपया यथाशीघ्र इस कार्यालय को अग्रेषित करें।
Observations, if any, may please be forwarded to this office at the earliest.

इसे सदस्य सचिव के अनुमोदन से जारी किया जाता है।
This issues with the approval of Member Secretary.

भवदीय /Yours faithfully


for (R.K.Meena)
SE(Operation)
एसई (ऑपरेशन)

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ERPC:: Kolkata

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4. मुख्य अभियंता, टीवीएनएल, डोरंडा, रांची - 834102 (फैक्स नंबर 06544-225414)
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9. महाप्रबंधक, टीटीपीएस, तालचेर, (फैक्स नंबर 06760-243212)
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20. कार्यकारी निदेशक, ईआरएलडीसी, पोसोको, कोलकाता, (फैक्स नंबर 033-2423-5809)
21. महाप्रबंधक, एफएसटीपीपी, एनटीपीसी, फरक्का, (फैक्स नंबर 03512- 224214/226085/226124)
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32. एजीएम (ऑपरेशंस), मैथन पावर लिमिटेड।
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34. मुख्य विद्युत अभियंता, पूर्वी रेलवे, कोलकाता-700 001 (फैक्स नं.: 033-22300446)
35. मुख्य विद्युत अभियंता, दक्षिण पूर्व रेलवे, कोलकाता-43 (फैक्स: 033-24391566)।
36. उप निदेशक, पूर्वी आरपीएसएच, 14, गोल्फ क्लब रोड, टॉलीगंज, कोलकाता-700033
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38. एसोसिएट वाइस प्रेसिडेंट, जीएमआर केईएल, भुवनेश्वर-751007। (फैक्स नंबर: 0674-2572794)
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सीसी:

मुख्य अभियंता, ओपीएम, सीईए	मुख्य अभियंता, एनपीसी, सीईए	सहायक सचिव,ईआरपीसी
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ईआरपीसी:: कोलकाता



Eastern Regional Power Committee

MINUTES OF 235th OCC MEETING

Venue: ERPC Secretariat, Kolkata

Date: 21.01.2026

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EASTERN REGIONAL POWER COMMITTEE

MINUTES OF 235TH OCC MEETING HELD ON 21.01.2026 (WEDNESDAY) AT 10:30 HRS

Member Secretary, ERPC welcomed all OCC members and other participants who were present physically as well as those who had joined virtually to the **235th OCC Meeting**.

There after he requested all the participants to give brief introduction.

□ He outlined the performance of ER grid during **December 2025** highlighting the following points:

❖ In **December-2025**, Energy consumption of ER was **13646 MU** which is **7.4 % more** as **December -2024**.

❖ In **December -2025**, Peak demand met of ER was **24054 MW** which is **4.3% more** than **December -2024**.

❖ Thermal **PLF** of **ER** during **December -2025** was **71%** w.r.t All India PLF of **66%**.

❖ Generating stations whose PLF was more than **90%** during **December -2025**:

Utility	Generating station	PLF (%)
WBPDC	SANTALDIH TPS	98
IPP	KAMALANGA TPS (GMR)	92
	HALDIA TPP (HEL)	94

❖ During **December -2025**, **77.5%** of the time, the ER grid frequency was within IEGC Band (49.90Hz-50.05Hz).

Coal stock position (As on 17.01.2026) was as follows:

SL.	Name of States/Power Stns.	% of Actual Stock vis-à-vis Normative Stock
1.	Jharkhand (TVNL)	228%
2.	Odisha (IB TPS)	88%
3.	West Bengal (WBPDC)	82% (Min. Santaldih TPS-38 %,
4.	D.P.L. TPS	Max. Sagardighi TPS –120%)
5.	DVC	35%
6.	NTPC	73%(Min.Meija TPS-71%,

□ He also highlighted the following:

✓ **CERC** has issued guidelines on **virtual PPA** (Power Purchase Agreement) dated **24.12.2025**. This could be evolved as another instrument to enable the Designated Consumers to meet their RCO (Renewable Consumption Obligation) targets.

✓ As per the decision taken in the meeting dated 30.12.2025 chaired by **Chairperson CEA**, the interim LILO arrangement may be continued upto **30th June 2026**. However, JSWEUL shall endeavour to complete the work of DTL by May, 2026 on best efforts basis.

To review the progress of DTL, another meeting would be convened in **February 2026**.

Transmission line (220 kV & above) commissioned during December -2025:

- **400KV-GAYA-NORTH KARANPURA** (circuit-1 & 2) commissioned by NKTLL on 25.12.2025 and 29.12.2025 respectively.
- **400KV-ALIPURDUAR (PG)-PUNASANGCHUNU** circuit-1 commissioned by Powergrid ER-II on 16.12.2025.

All thermal generating units in ER were advised to carry out necessary maintenance activities by availing shutdown during the lean period so as to avoid forced outages in the upcoming summer.

At the outset, ED, ERLDC welcomed all participants to the 235th OCC Meeting and underscored the following:

- ✓ *For FY 2026-27, All India demand is expected to be around 250-285 GW (in peak solar hours) and 250-255 GW (in non-solar hours).A significant shortage of 6-10 GW is expected in non-solar hours of the said period.*
- ✓ *Eastern region is likely to experience a maximum demand surge of the order of 6% in June 2026 followed by 4% in April 2026 while the expected peak demand of ER may lie in the range of 32GW to 35 GW during the period of April-June 2026.*

❖ *List of participants is attached at **Annexure-A**.*

1. PART-A : CONFIRMATIONOFMINUTES

1.1. Confirmation of Minutes of 234th OCC Meeting held physically at ERPC Secretariat on 23rd December 2025

The Minutes of 234th Operation Coordination Sub-Committee meeting held on 23.12.2025 was circulated vide letter dated 08.01.2026.

Members may confirm the minutes of 234th OCC meeting.

Deliberations in the meeting

Since no comments have been received so far, the Minutes of 234th OCC meeting were confirmed.

2. PART-B:ITEMSFORDISCUSSION

2.1 Update on follow up agenda: ERPC

a) Reconductoring of 400kV Talcher-Meramundali and 400kV Farakka-Kahalgaon D/C

□ Reconductoring of 400kV Talcher-Meramundali D/C (Important for Odisha system)

Reconductoring with HTLS conductors under ERES-43 is being executed by POWERGRID Odisha under the RTM route. Initially, it was decided to complete the 400 kV Talcher–Meramundali Circuit-I (shorter line of 52km). However, at the request of the POWERGRID Odisha project, shutdown of the LILO portion of Circuit-II was availed from 21.11.2025 onwards due to non-availability of the required materials for Circuit-I.

Deliberation in 234th OCC:

Powergrid Odisha submitted:

- ✓ Presently 4 teams have been deployed and from 28th Dec 2025, additional 4 teams will join after which S/D of circuit-1 will be availed.
- ✓ ROW issues has been resolved now.
- ✓ There are a number of crossings with LT lines which is hindering progress as per planned schedule.
- ✓ After completion of the LILO portion, reconductoring in circuit-1 will start as all pending materials have been received.
- ✓ Only 15 ckm out of 35 ckm had been completed in the LILO portion due to various ROW issues and delays in obtaining permissions for railway crossings

SLDC Odisha submitted:

- ✓ Reconductoring of the LILO portion may not be required as of now.
- ✓ Every possible cooperation will be extended i.r.o availing S/D of distribution feeders by Powergrid for reconductoring in line crossings.

SLDC Odisha also urged that at least one ckt (ckt#1) of 400KV Talcher-Meramundali D/C must be restored by 15th Feb 2026.

ERLDC deliberated that simultaneous shutdown of the LILO portion of Circuit-2 and Circuit-1 is not feasible, as it would necessitate curtailment at Talcher STPP due to higher power flows in the event of tripping of the HVDC Talcher–Kolarbipole.

234th OCC Decision

- PowergridOdisha was advised to expedite reconductoring works and make circuit-I of 400KV Talcher-Meramundali D/C along with bay equipment ready for service by 15th Feb 2026.
- PowergridOdisha was advised to share the detailed work schedule i.r.o reconductoring, clearly delineating the details of constraints at each location.
- PowergridOdisha was advised to coordinate with NTPC and OPTCL i.r.o bay upgradation at Talcher end in due course.
- POWERGRID was advised to return the LILO portion of Circuit-II and commence work on Circuit-I from the first week of January 2026 so that the valuable lean-demand period could be effectively utilized to complete reconductoring of at least one circuit before Summer-2026. Accordingly, shutdown of Circuit-I was availed on 06.01.2026, with a target completion by 15th February-2026
- SLDC Odisha was advised to grant shutdown of all transmission line crossings the Powergrid lines on priority on D-2 basis.
- OCC advised ERLDC to convene weekly monitoring meeting to assess the progress.

In line with the decisions of the 234th OCC, ERLDC is conducting weekly monitoring meetings, and so far, two meetings have been held on 08.01.2026 and 15.01.2026 respectively.

As on 15.01.2026, only 5 km out of 52 km has been completed.

As these lines serve as important drawal points for the Odisha system and are crucial links for evacuation of NTPC Talcher generation, both lines are required to be in service before Summer-2026.

Further, additional SPS at Talcher-Kaniha as proposed by ERLDC for facilitating 400 kV TSTPS-Meramundali line reconductoring work may also be approved.

POWERGRID/NTPC/OPTCL may update. Members may discuss.

Deliberations in the meeting

ERLDC informed that weekly monitoring meetings were being conducted. However, progress as on 15 January 2026 was reported to be only about 5 km out of 52 km, which was considered extremely slow.

Powergrid (Odisha) submitted :

- *About **9.3 km** had been completed and **5.3 km** was under progress.*
- *Five gangs are currently deployed, with plans to deploy two to three additional gangs within a few days.*
- *Works in Circuit-II and LILO portion shall be taken up only after charging of Circuit-I.*
- *The target for completion of Circuit-I (52 km) by **15th February 2026** was reiterated, subject to deployment of additional manpower.*

OCC Decision

- ✓ *OCC expressed serious concern over the slow pace of work and emphasised that this line is critical for reliability of Odisha system, particularly during Summer.*
- ✓ *Power Grid was advised to deploy additional gangs to expedite progress in reconductoring so as to ensure completion of Circuit-I (52 km) by 15th February 2026. The timeline must be strictly adhered to.*
- ✓ *It was also reiterated that Section-wise schedules may be shared by Powergrid on weekly basis and any change in execution strategy may be communicated immediately. Constraints at each location should be clearly delineated.*

- ✓ *OPTCL was advised to coordinate with Powergrid Odisha for facilitating shutdowns at power line crossings.*
- ✓ *OCC consented to implement SPS at Talcher-Kaniha as proposed by ERLDC for facilitating 400 kV TSTPS-Meramandali line reconductoring work.*
- ✓ *It was advised that ERLDC may convene weekly monitoring meeting to assess the progress and share the information with ERPC.*

□ **Reconductoring of 400KV Farakka-Kahalgaon D/C (Important for WB System)**

HTLS reconductoring of 400 kV Farakka–Kahalgaon D/C under ERES-43 is being executed by POWERGRID ER-I/ER-II. Shutdown was availed from 02.12.2025 after initial delays due to material non-availabilities with an agreed completion date of February 2026.

While discussing the progress of the lines during last OCC, it was emerged that for Bay upgradation works both buses shutdown required at NTPC FSTPP and KHSTPP end. The required planning to be done to complete the shutdown before Summer 2026.

Deliberation in 234th OCC:

Powergrid ER-I submitted:

- ✓ Shutdown has been availed on 2nd Dec 2025 and reconductoring work has commenced in 400KV Farakka-Kahalgaon D/C.
- ✓ Out of 94 ckt km of 400 kV Farakka-Kahalgaon D/C, reconductoring in 9 km has already been completed and work is under progress in 6 km.
- ✓ As materials have been received now, 2 additional gangs will be deployed to expedite progress in work.

234thOCC decision:

- Power Grid was advised to share the detailed work schedule i.r.o reconductoring, clearly delineating the details of constraints at each location.
- NTPC was advised to finalize the plan for Jackbus replacement at Farakka and accordingly communicate to Powergrid. ERLDC was requested to hold a separate meeting for further deliberation on the same.
- It was advised ERLDC to convene weekly monitoring meeting to assess the progress.

Accordingly, ERLDC is conducting weekly monitoring meetings for reconductoring progress and so far, two meetings have been held on 08.01.2026 and 15.01.2026 respectively.

As of 15.01.2026, only 24 km out of 95 km has been completed.

As 400 kV Farakka–Kahalgaon D/C serve as important drawal points for the WB system, both lines are required to be in service before Summer-2026.

Power Grid/ NTPC may update. Members may discuss.

Deliberations in the meeting

Powergrid updated:

- ✓ *Reconductoring for around 24 km is completed and about 10 km is under execution for one circuit.*

- ✓ 5 gangs to deployed soon.

NTPC submitted that circuit breakers with PIR have already been procured for Farakka station and opined that at Kahalgaon station also similar circuit breaker may be installed by Powergrid.

On the Issue of PIR Requirement for Circuit Breakers, Powergrid apprised:

- As per CEA guidelines(2022), PIR is not mandatory for transmission lines shorter than 200 km.
- Insisting on PIR would lead to a 36-month procurement timeline, negating near-term benefits of reconductoring.
- Technical experience indicates that PIRs on short lines can sometimes create operational complications rather than benefits.

OCC Decision

- ✓ OCC took serious note of slow pace of reconductoring and emphasized that this line is critical for ISTS and West Bengal system reliability, particularly during Summer.
- ✓ OCC emphasized that in order to restore the line with HTLS by March 2026, Powergrid shall take all necessary measures to complete reconductoring as per previously agreed timelines.
- ✓ OCC also opined that shutdown of any circuit beyond March 2026 will not be permitted.
- ✓ OCC suggested that NTPC may abide by CEA Technical Standards for Construction of Electrical Plants & Lines Regulations, 2022 for procurement of circuit breakers. This will avoid further delay in completion of reconductoring works.
- ✓ It was advised that ERLDC may convene weekly monitoring meeting to assess the progress and share the information with ERPC.

b) Bus split operationalization at NTPC Kahalgaon

As decided in **219th OCC** Meeting, a committee comprising of members from ERPC and ERLDC visited NTPC Kahalgaon on 17-10-2024 to assess the status of Bus splitting at 400kV level and way forward for operationalization of 400 KV Bus sectionalizer.

Following works need to be done to complete the installation of ICT 3 & 4:

1. Determination of underground cable conduit path for 400/132 kV ICT-3, 4 and 5 allocated for stage 2 supply.
2. Excavating the existing cable and relaying from Stage-1 132kV to New Stage-2 132kV switchyard, where ICT 3 & 4 will be connected.
3. Laying of additional 22.8 ckt. km control cable for STs.
4. Jumpering of ICTs in 132kV & 400kV level.
5. Bay equipment testing.

As per 54th TCC:

As the issue is long pending, TCC sought a firm timeline from NTPC for its completion.

- NTPC informed that the delay was caused by the intermittent rain and failure of a 132 kV power cable.
- He updated that to expedite the process, NTPC approached Bihar for providing the cable joining kit. Bihar agreed to supply the required kit within a week and accordingly, the work is expected to be completed by August 2025.
- TCC advised NTPC to strictly adhere to the timeline and implement the bus splitting scheme at

Kahalgaon within Aug-2025.

As per 229th OCC:

NTPC informed that 132 kV cable joining kit has been received from BSPTCL and as per the present work progress, KSTPP ICT 3 charging will be done by **15.08.2025**.

The expected bus splitting operationalization was expected to be completed by **Oct, 2025**.

As per 55th TCC:

NTPC informed the forum that, as per the current progress of works, idle charging of ICT-3 and ICT-4 is expected to be completed by **December 2025**, and bus splitting at NTPC Kahalgaon is tentatively scheduled for completion by **April 2026**.

55th TCC Decision

TCC took serious note of the inordinate delay in implementation of the bus splitting operational scheme at NTPC Kahalgaon, which is critical for reduction of fault level at Kahalgaon.

TCC advised NTPC to:

- Share weekly progress reports with ERPC and ERLDC.
- Expedite the bus splitting works to ensure completion strictly as per the submitted timeline.

NTPC may update. Members may discuss.

Deliberations in the meeting

NTPC updated on the long-pending bus split scheme at Kahalgaon, aimed at reducing fault levels.

It was informed that:

- *Delays had occurred due to failure of station transformers.*
- *Shutdowns of 400 kV bus (Bus-3 & 4) are now planned on **30th & 31st January 2026 respectively** for testing and commissioning activities.*
- *Charging of new ICTs is planned by **mid-February 2026**, followed by phase-wise shifting of station transformers to new 132 kV system.*
- *Completion of the extension package is targeted by **April 2026**.*

OCC Decision

- ✓ *In view of continued operation of 400 kV bus in synchronized mode at NTPC Kahalgaon at high fault level, OCC expressed serious concern over the lack of desired progress and advised NTPC to strictly adhere to the submitted timeline.*
- ✓ *Further, NTPC was advised to submit fortnightly progress report.*

c) Update on Reconductoring of ISTS lines under Eastern Region Expansion Scheme-44

In 230th OCC meeting, PowerGrid informed that the reconductoring of Chuka Transmission System involves significant financial implications. As such, the Letter of Award (LOA) will be issued after the approval of the POWERGRID Board, the meeting for which is tentatively scheduled for the end of August 2025. Additionally, PowerGrid apprised that once the LOA is awarded, the reconductoring work of the Malda–Gazole section (18 km) will commence. The work is planned to be completed by the end of the upcoming winter, as the line is heavily loaded during the summer season.

As per 55th TCC:

PGCIL apprised the forum that as per the earlier commitments in 230th OCC meeting reconductoring of 220KV Malda–Gazole section (18 km) will be commenced on priority. However, LOA is not awarded yet.

WBSLDC submitted the following,

- The shutdown of the said line is only can be allowed during winter period since no progress has been made by PGCIL in this winter so the reconductoring work is now delayed by another one year which will have a serious impact on catering the increased load of WB leading to overloading of these lines.
- As a mitigating measure, Bus split operation at Gazole is required for limiting the line loading.

55th TCC Decision

- ✓ TCC critically noted the inordinate delay caused in carrying out the HTLS reconductoring of 220KV Malda–Gazole D/C which is very much essential to aid in operational flexibility WB.
- ✓ TCC also advised Powergrid to explore all the possibilities for diverting HTLS conductors from other projects on loan basis so that Reconductoring can be completed at the earliest and after LOA has been awarded the HTLS conductor may be returned to the concerned utility.
- ✓ TCC also opined that PowerGrid may take the help of WB for procurement of the required HTLS conductor through a separate tender.

As per 55th ERPC:

Powergrid stated that **LoA** will be placed by **Jan'26**.

ERPC advised **Powergrid** to complete the reconductoring work **before onset of coming summer**.

Powergrid may update. Members may discuss.

Deliberations in the meeting

Powergrid submitted:

Bid evaluation currently under corporate review and LOA is yet to be awarded,

WB SLDC submitted :

- ✓ *Despite initiation in late-2023, the reconductoring package has not been finalized for past two years. Concerns were raised on two years being lost due to procedural delays, thereby increasing system vulnerability.*
- ✓ *In some cases, bus splitting is being forced to mitigate transmission constraints.*
- ✓ *The condition has even worsened after prolonged forced outage of major hydro stations in Sikkim i.e Teesta-II & Teesta-V since 2022.*

OCC Decision

- *OCC took a serious note on such inordinate delay in commencement of reconductoring works despite receipt of approvals from all concerned forums well in advance. There should be a mechanism for structured planning so that critical works are not held up by prolonged paperwork.*
- *OCC also advised ERLDC to communicate to higher authorities of Powergrid to seek firm timelines on completion of reconductoring of Chuka Transmission System.*

d) Intrastate Transmission Network Assessment & Mitigation – DVC:

□ Restoration of Koderma ICT

400/220KV, 315 MVA Koderma ICT-2 has been under outage since 02.06.2025 due to burnout. DVC has transported the 315 MVA regional spare ICT kept at Muzaffarpur as a replacement. At present, Koderma S/S is N-1 non-compliant due to the availability of only one ICT, which is also critically loaded especially during Solar hours. Further, the existing Koderma ICT-1 (in service) is experiencing DGA violations. Restoration of ICT-2 has therefore become critical to cater to the summer load requirements of the DVC system.

□ Implementation of SPS scheme for N-1 compliance of ICTs at Bokaro

Currently, Bokaro 400/220KV, 2x 315MVA ICTs are experiencing N-1 non-compliant. One SPS was proposed to safeguard the cascade tripping inside DVC system. A joint study was conducted on 22nd October 2025 where SPS proposal and logic was discussed. Matters have been deliberated in all OCC meetings since 231st OCC meeting. **This SPS needs to be implemented before Summer 2026 on priority basis.**

As per 234th OCC:

DVC informed:

- ✓ The SPS will be implemented in two phases sequentially.
- ✓ In the first phase, SPS will be operational in two lines of Ramgarh (within next 20 days)
- ✓ The Second phase of SPS can be implemented only after Koderma ICT is put in service (expected by 15th Feb 2025).

234th OCC decision:

OCC advised DVC to implement SPS as per submitted timeline and expedite in putting the Koderma ICT to service.

DVC may update. Members may discuss.

Deliberations in the meeting

DVC informed:

- ✓ *ICT-I will be put to service by end of January, 2026. Delay due to multiple rectification works followed by inspection*
- ✓ *The 2nd ICT will be charged by 15th Feb 2026 as foundation works have been completed and new ICT will be commissioned in 1.5 years.*
- ✓ *The SPS logic has been finalized and will be operational once the ICTs come to service.*

OCC Decision

It was advised that DVC may implement SPS as per submitted timeline and expedite putting the Koderma ICTs to service.

e) Intrastate Transmission Network Assessment & Mitigation – West Bengal

Reference:

Modification of Existing SPS Scheme at Subhasgram (PG) with Undervoltage Logic

ERLDC in **231st OCC** Meeting dated **22.09.2025** proposed a modification to the existing SPS at Subhasgram (PG) to incorporate undervoltage logic with time delay, aimed at preventing voltage collapse in the Kolkata area by shedding some non-critical loads. **A state-level meeting was initially held by SLDC, West Bengal followed by another meeting on 08.12.2025 attended by ERLDC, POWERGRID, WBSLDC, WBSIEDCL, and CESC.** In these meetings, it was agreed that 200 MW of load would be configured under this modified scheme in the initial stage.

As per 234th OCC:

Powergrid ER-II updated that the undervoltage logic of SPS has been finalized that shall be shared by next week and the same shall be implemented at Subhasgram (PG) by the last week of January 2026, just after commissioning of the 6th ICT.

234th OCC Decision

OCC advised Powergrid along with all concerned stakeholders to strictly adhere to the timeline for UVLS implementation in intra-state network of WB, as submitted in the 55th TCC meeting dated 16.12.2025.

WB and Powergrid may share the update. Members may discuss.

Deliberations in the meeting

Power Grid explained the SPS logic(Annex B.2.1.e) for Subhasgram(PG), incorporating modifications as suggested by ERLDC.

ERLDC submitted that the signal should be communicated to their end on operation of the SPS at Subhasgram(PG) to aid better grid monitoring.

OCC Decision

- *OCC granted in-principle approval for implementing the SPS logic, subject to technical finalization and coordinated shutdown planning.*
- *It was also suggested that SPS signal needs be transmitted to ERLDC, CESC and Powergrid are required to ensure this jointly.*

f) Intrastate Transmission Network Assessment & Mitigation-Odisha

Reference:

Implementation of the Under Voltage Load Shedding (UVLS) scheme in the Odisha system has been under review since the 231st, 232nd, 233rd, and 234th OCC Meetings held on 22.09.2025, 24.10.2025, 22.11.2025, and 23.12.2025 respectively. The matter was also discussed in the recently concluded 55th TCC/ERPC meeting held on 16.12.2025 and 17.12.2025 at Kalimpong, West Bengal. The continued delay in implementation is posing increasing risks not only to the Odisha system but also to the Eastern Region as a whole during the forthcoming Summer-2026 period.

As per 234th OCC:

SLDC Odisha submitted:

The UVLS scheme will be made operational by March 2026.

234th OCC Decision

OCC directed SLDC Odisha to install the UVLS before summer 26 i.e. March 26.

SLDC Odisha may update. Members may discuss.

Deliberations in the meeting

OPTCL updated that an internal meeting was scheduled on 21.01.2026 with DISCOM to finalize UVLS scheme.

OCC Decision

- *OPTCL was advised to convey outcome of the meeting to ERPC/ ERLDC within a week.*
- *In view of safeguarding reliability of Odisha intra-state network as well as ER grid as a whole, OCC urged Odisha to expedite UVLS implementation. A concrete plan and timeline may be submitted in the next OCC.*

g) Construction of Nawada–Durgapur–Jeerat (New) 765kV corridor for improved reliability in the Eastern Region and improve reliability of power supply to Kolkata:

- ✓ To improve the reliability of power supply to Kolkata and adjacent areas, the ERES-47: Nawada – Durgapur – Jeerat (New) 765 kV corridor was planned following joint studies by the CTU. This scheme, aimed at strengthening the 765 kV interconnection, was approved during the 47th CMETS-ER (29-09-2025), with the final scope of work finalized in the 48th CMETS-ER (30-10-2025).
- ✓ The TCC has since advised the CTU to conduct a comprehensive joint study with WBSETCL and ERLDC to evaluate the implementation's pros and cons. Furthermore, this project is to be integrated with the Bikaner Phase-VI RE power evacuation scheme (extending to the Eastern Region).
- ✓ A detailed report was scheduled for submission to the ERPC by 31st December 2025.

As per 234th OCC:

WBSLDC submitted:

The proposed 765 kV corridor will definitely aid in ensuring reliable power supply to Kolkata if it is extended upto 765 kV New Jeerat(PG). Along with the proposed 765 kV corridor, associated 400 kV lines are also needed to support the system in event of crisis by catering to all critical load points.

234thOCC Decision

As directed in the 55th ERPC meeting, it was decided that a special meeting shall be convened involving ERPC, CTU, ERLDC, WBSETCL and WB SLDC wherein joint study report by CTU shall be discussed for stakeholder views/opinion. CTU was also requested to complete the study at the earliest.

WBSETCL may update. Member may discuss.

Deliberations in the meeting

Reiterating its request for finalization of the originally proposed scheme, WBSETCL submitted:

- ✓ *Studies indicate that the power flow direction is predominantly from 400 kV to 765 kV, and not in the reverse.*
- ✓ *The existing ICTs are lightly loaded (200–300 MW) in solar hours and thus, there is no immediate justification for additional 765/400 kV transformation. Any need, if at all, would arise only for very short durations*
- ✓ *Provisioning of switching arrangements appears sufficient as an interim measure.*

OCC Decision

It was advised that CTU/WBSETCL may incorporate the suggestions in the joint study report.

h) Establishment of proposed TCF-II 220/132/33 KV SS by WBSETCL

The proposal for the TCF-II 220/132 kV intra-state substation with ISTS connectivity (through D/C LILO of Siliguri PG–Kishanganj PG 220 kV HTLS line) proposed to meet the rising demand in the Ghoshpukur–TCF and adjacent areas. The scheme has undergone detailed discussion at CMETS-ER since May 2025, considering multiple operational scenarios. Studies indicate that, under N-1 contingency, the Binaguri PG–Siliguri PG line would load to 426 MW, which is above within 90% of the HTLS line's thermal limit (450 MW) assuming a 1500 MW APD–Agra HVDC link setting.

The same matter was again discussed in West Bengal Resource Adequacy Studies, and it was proposed to set up a 400/220/132kV substation. The same matter was discussed in **55th TCC & ERPC meeting**. The matter was referred to West Bengal Resource Adequacy Studies held on **22nd December 2025**.

As per 234th OCC:

WBSLDC stressed the urgent need of **TCF-II 220/132/33 KV S/S** as there is no other source(adjacent generating station/line) to meet the load of upcoming 132 kV S/S at Ghoshpukur. This S/S is most likely to cater to the load of Data Centres to be setup in line with extant norms of Govt of India.

234thOCC Decision

OCC opined that the issue will be taken up in the special meeting to be convened by CEA on 24.12.2025. Based on the deliberations, further course of action will be framed in the next OCC.

WBSETCL may update. Member may discuss.

Deliberations in the meeting

WBSETCL submitted:

- ✓ *In the CEA meeting on resource adequacy dated 24.12.2025, 220 kV substation has been proposed that may be upgraded to 400 kV as per future requirement. This was not agreed by CEA and establishment of 400 kV system was insisted in the present timeframe.*
- ✓ *Upgrading to 400 kV will be convenient in absence of ROW issues owing to close proximity of 400 kV line (within 500m-1 km).*

WBSLDC submitted:

- ✓ *Many critical transmission proposals are being repeatedly deferred by linking them to long-term (2034–35) resource adequacy studies, resulting in unacceptable delays.*
- ✓ *For meeting a load of 110 MW, opting for 400 kV supply is not possible as per WBERC guidelines.*
- ✓ *Without timely establishment of Ghospukur S/S, data centre cannot be setup on priority in line with Govt of India policy. Moreover, there is no other source(adjacent generating station/line) to meet the load of upcoming 132 kV S/S at Ghoshpukur.*
- ✓ *With load requirement of only ~110 MW, there is adequate N-1 compliance with Clear provision of land and corridor for future 400 kV upgradation*

- ✓ All relevant data have already been shared with CEA and nothing is pending.

OCC Decision

OCC acknowledged the requirement of 132 kV S/S at Ghoshpukur for meeting the immediate power requirement of the upcoming data center and of adjoining areas.

It was advised that matter may be taken up with CEA by WBSETCL.

i) Restoration of 2nd ICT at Tenughat and upgrading 400 KV PVUNL- Tenughat line end termination at Tenughat:

Reference:

- ✓ Jharkhand is meeting a maximum demand of about 2,100 MW during Winter (December 2025 and January 2026). However, on several days, shortages in the range of 200-300 MW have been reported in Jharkhand despite commissioning of the Patratu 800 MW unit.
- ✓ These shortages are primarily attributed to frequent tripping of Tenughat units and thereby load restrictions in the Dumka/Gobindpur area due to inadequate transmission capacity. The constraints are mainly due to the availability of only a single ICT at Tenughat and loading restrictions on the 400 kV PVUNL–Tenughat line (earlier charged as 220KV Tenughat-Patratu) arising from non-availability of terminal equipment at Tenughat for power evacuation at 400 kV level. Additionally with Integration of 2nd Unit of PVUNL this loading will increase further and will aggravate the situation.

The following issues need to be addressed before Summer-2026:

1. Restoration of the second 400/220 kV, 315 MVA ICT at Tenughat.
2. Upgradation of the 400 kV PVUNL–Tenughat line-end terminations at the Tenughat end.

As per 234th OCC:

TVNL apprised:

- The 250 MVA ICT is very old (procured around 1996) and the same has not been charged till now.
- Electrical clearance of the ICT is pending as test reports are still awaited. Once electrical clearance is received, the ICT will be charged tentatively by 15th Jan 2026.

ERLDC informed:

One of the two ICTs procured at the same time, i.e 1996 was charged two years ago.

234thOCC Decision

- ✓ OCC expressed serious concern on the healthiness of the ICT which has not been loaded since procurement that may have exhausted its useful life.
- ✓ OCC also opined that this 250 MVA ICT is utmost essential to ensure reliable power evacuation from upcoming PVUNL unit-II.
- ✓ TVNL was urged to put the ICT to service but only after prudently assessing its healthiness and useful life to avoid recurrent outage followed by replacement in future. The same set of procedure for charging may be followed as for the other ICT charged two years back.

JUSNL may update. Members may discuss.

Deliberations in the meeting

TVNL submitted;

Third party testing of the 2nd ICT has been completed followed by inspection on 19.01.2026.

So, the ICT would be charged after rectification of the defective NIPS.

OCC Decision

- OCC advised TVNL to put the ICT to service by 15th February 2026.
- JUSNL and TVNL were advised to intimate the enhanced load carrying capacity of associated 220 kV lines so that the additional transformation capacity can add value to grid reliability.

j) Methodology for Cost calculation with respect to the already commissioned 7th ICT (500 MVA,400/220 KV) at Subhashgram(PG) Substation

- As per MOM of the Meeting held at Vidyut Bhavan on 25.08.2023 under the Chairmanship of the Secretary, Dept. of Power, Govt. of West Bengal, it was decided that incremental charges associated with this will be borne by CESC. But...*"as Powergrid cannot claim the cost related to this interim arrangement directly from CESC and can claim only from WBSEDCL as DIC Discom, WBSEDCL shall pay all cost / incremental charges to PGCIL after realising the same from CESC through bilateral arrangement between CESC & WBSEDCL"*...
- It was also decided in the said Meeting that realisation of incremental cost from CESC shall commence after the subject ICT is commissioned as interim measure ICT at Subhashgram till commissioning of the 6th 500 MVA 400/220 KV ICT thereafter.
- Now the said 7th ICT has already been commissioned on 21.06.2024 and as per the latest communication received from PGCIL, the 6th ICT is scheduled to be commissioned thereafter by the second half of January, 2026.
- So, guidance on the methodology for calculation of the incremental cost was sought by CESC.

Deliberation in 234th OCC:

ERLDC summarized the deliberation in the special meeting convened by CEA on 22.12.2025 wherein the continuation of 7th ICT on permanent basis at Subhasgram(PG) after 2030 came up, though final decision to be taken by Member (Power System), CEA.

It was apprised that the exact load relief at Subhasgram(PG) can be ascertained in study only after getting the information on which lines will be shifted from Subhasgram(PG) to New Laxmikantapur .

WBSLDC submitted:

CEA's prediction on non-requirement of the 7th ICT at Subhasgram(PG) after 2030 takes into account the commissioning of New Laxmikantapur S/S by 2029 that is expected to provide around 30% load relief to Subhasgram(PG) . But, if timely commissioning of New Laxmikantapur S/S doesn't take place, there shall be extreme network congestion in absence of 7th ICT and hindering reliable power supply to Kolkata. Also this load relief to Subhasgram(PG) is computed considering all generators in service and in absence of any of the generators, data on power drawl pattern of CESC from all the drawl points is needed for predicting the exact quantum of load relief.

CESC affirmed the need of 7th ICT at Subhasgram(PG) to meet the rising demand of Kolkata city. As per their study, rise in their load beyond 2400 MW will entail increased power drawl from Subhasgram (PG).

It was further intimated that CESC has already paid Rs 12 crores to WBSETCL (comprising installation and PGCIL consultancy charges) and guidance was sought on modalities of incremental cost & VAR charges sharing with reference to the discussion held in the meeting dated 25.08.2023 under chairmanship of Secretary(Power), Govt of WB.

234th OCC Decision

- ✓ As the planned transformation capacity is 500X2 MVA at the proposed New Laxmikantapur S/S, maximum 400 MW of load relief will be feasible from Subhasgram(PG) even after commissioning of New Laxmikantapur S/S(considering N-1 contingency).Also considering steady load growth, OCC assessed the permanent need of 7th ICT at Subhasgram(PG) that needs to be conveyed to CEA with all accessory data.
- ✓ OCC opined that after the final decision is taken i.r.o. the 7th ICT at Subhasgram(PG) by CEA , the modalities of cost sharing between the two DISCOMs i.e WBSEDCL and CESC shall be finalized.

Members may discuss.

Deliberations in the meeting

WBSEDCL affirmed of paying the installation charges to PGCIL i.r.o 7th ICT at Subhasgram(PG).Rest of the cost sharing is to be done as per MOU between CESC & WBSEDCL.

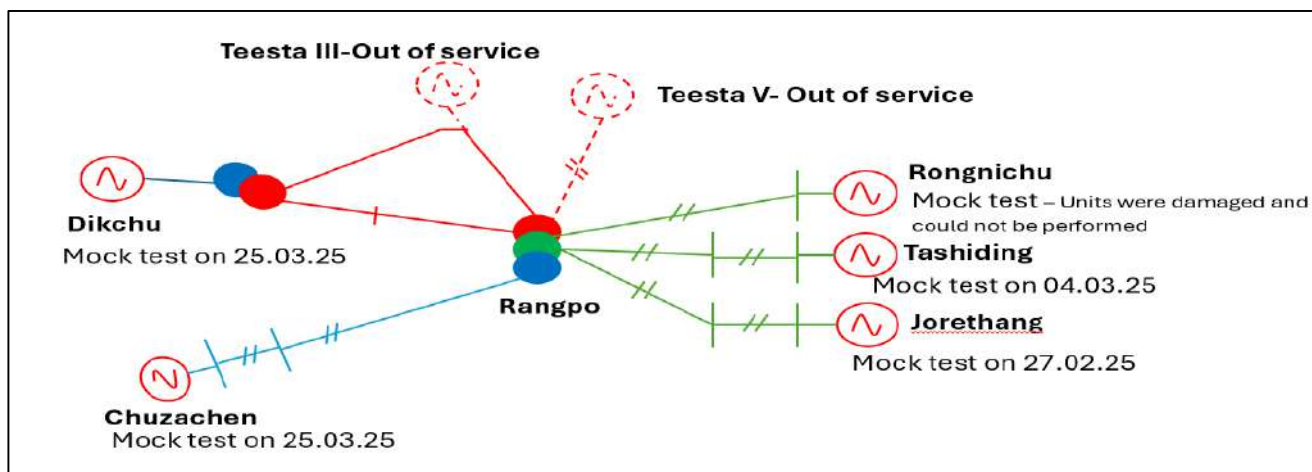
OCC Decision

- *OCC suggested that after the final decision is taken i.r.o. temporary/permanent status of the 7th ICT at Subhasgram(PG) by CEA , the modalities of incremental cost sharing between the two DISCOMs i.e WBSEDCL and CESC shall be finalized.*
- *It was also suggested that VAR charges to be computed as per norms irrespective of temporary/permanent status of the ICT. Sharing between the DISCOMs, i.e CESC & WBSEDCL will be as per the MOU signed.*

k) Black start of ER Units in Sikkim Complex

A black start capable resource and its readiness are essential for grid resilience. As mandated under Clause 34(3) of the IEGC 2023, all users must conduct mock black start exercises annually, coordinated by the SLDC for state-level stations and by the RLDC for ISGS and IPP stations. In the Eastern Region, all black start capable ISGS and IPP units are located in Sikkim and are connected to the 400/220/132 kV Rangpo substation.

In eastern region all black start capable ISGS and IPP units are located in Sikkim and connected to 400/220/132 kV Rangpo substation as shown below:



During 2024–25, mock black start exercises were conducted at Jorethang, Tashiding, Chuzachen, and Dikchu HEPs. Only Tashiding could synchronize with Rangpo, and that too after an hour's delay, while the other HEPs could not complete synchronization.

In view of the recent difficulties in synchronizing hydro units during restoration trials, it is proposed that the implementation of auto-synchronization at Rangpo(PG) be prioritized. Without this facility, the manual synchronization process remains too restrictive, preventing hydro stations in the Sikkim cluster from successfully completing the mock black start exercises mandated by IEGC 2023. This poses a significant risk to the reliable restoration of the Sikkim grid and its integration with the rest of the regional system.

Deliberation in 234th OCC:

ERLDC apprised:

- ✓ Attempt was made for mock black start of Tashiding HEP on 12th December 2025 but synchronization was not possible due to large magnitude of angular difference (160 degrees) between the incoming machine and Rangpo(PG) bus. The BCU witnessed technical disruption and leading to failure of synchronization.
- ✓ Substantial delay is caused due to manual command in BCU, making frequency band too stringent for feasible synchronisation. Increasing frequency range in manual synchronization poses risk of machine falling out of step with the bus.
- ✓ On the other hand, implementation of auto-synchronization facilities at Rangpo(PG) will aid in smooth and faster synchronization with a wider range of frequency. Auto-synchronization facility is highly essential for successfully carrying out actual black start.
- ✓ This is leading to failure of mock black start exercise(as mandated in IEGC 2023) of all surrounding hydro generating stations in Sikkim connected to 400 kV Rangpo bus, which acts as the only synchronization point with rest of the grid.
- ✓ Last year mock black start of Tashiding HEP was successful but with long delay and generation loss.
- ✓ The issue was put in NCT meeting but was not agreed for the proposed modifications facilitating Auto-synchronization.

Powergrid stated:

- Implementation of Auto-synchronization facility is not feasible as per prevailing technical specifications and guidelines of CEA transmission planning criteria. With modification in technical specifications, procurement of dedicated synchronization panel can be done accordingly.
- Parameters and settings at the generator end should also be checked for proper synchronization.

234th OCC Decision

Powergrid was advised to explore feasibility of implementing auto-synchronization facility through BCU upgradation or other means at Rangpo in consultation with concerned OEM as well as considering the prevailing procedure being

practised in other regions of the country for mock black start with auto-synchronization. **A detailed assessment report should be shared in the next OCC for further deliberation.**

An update was taken by ERLDC from different regions, regarding availability of the Auto-Synchronisation relay, which is as follows:

- **NER (North Eastern Region):** On 19th December 2025, a successful mock black start for Lower Subansiri HEP was conducted. Synchronization was successfully achieved at the BNC (PG) substation using the auto-synchronization facility available within the BCU. It is also gathered that other SAS-based substations in the NER are also equipped with this BCU-based auto-sync capability.
- **NR (Northern Region):** On 8th December 2024, synchronization for the NathpaJhakri and Rampur Hydro Stations was successfully carried out at the 400/220 kV Nalagarh (PG) substation. This was performed through the bus coupler utilising its integrated BCU-based auto-synchronisation facility.

In view of this, it is suggested either to expedite the implementation or upgradation of the BCU-based **auto-synchronization facility** at the Rangpo (PG) substation. This would align the substation with standard SAS-based architectures found at other PG stations.

Or

Alternatively, the **full auto-synchronization facilities** already available at the hydro generating stations could be deployed. By integrating these existing generator capabilities with the Rangpo (PG) bay controls, the system can achieve a wider frequency and phase-angle lock range, ensuring a smooth and reliable connection to the grid.

Powergrid and ERLDC may update. Members may discuss.

Deliberations in the meeting

ERLDC submitted :

- ✓ *Black start preparedness is critical under Clause 34.3 of IEGC-2023. The same is pending in ER due to repeated failure of mock black start through Rangpo(PG) station.*
- ✓ *Black start with auto-sync capability as carried out in NR and NER was put forth as reference.*
- ✓ *Rangpo(PG) being the main pooling station connected to surrounding hydro stations, having adequate synchronisation facility without manual intervention is utmost essential.*

Powergrid apprised :

- ✓ *Faulty BCU has been set right and after firmware upgradation, now synchronization has been checked to occur within a frequency band of 0.02 Hz.*
- ✓ *Next Mock black start activity will be conducted on 31.01.2026.*
- ✓ *Exact mechanism for synchronization as available in any generating station cannot be expected in a Grid substation.*

- ✓ *New BCU is under procurement whose LOA is expected in 2026 so that it can be availed for mock black start in 2027.*

OCC Decision

It was advised Power Grid to expedite procurement of new BCU (with higher storage and resolution) at Rangpo to allow sufficient frequency band for reliable synchronisation.

I) Restoration of 220KV FSTPP-LALMATIA-I

The line is out of service since long due to tower collapse. Presently 220 kV Farakka-Lalmatia line is charged(from loc no 241 to loc 84) at 132 kV voltage level for anti-theft purpose by tapping at loc. No. 100-101.

As per 55th TCC:

JUSNL updated that tower no 1 to 18 falls under West Bengal area out of these, foundation of tower 1 to 12 are pending due to local issues. They have taken up the matter with concerned District Administrations of West Bengal.

TCC advised WBSETCL to extend all support to JUSNL in this regard.

JUSNL may update. Members may discuss.

Deliberations in the meeting

JUSNL updated:

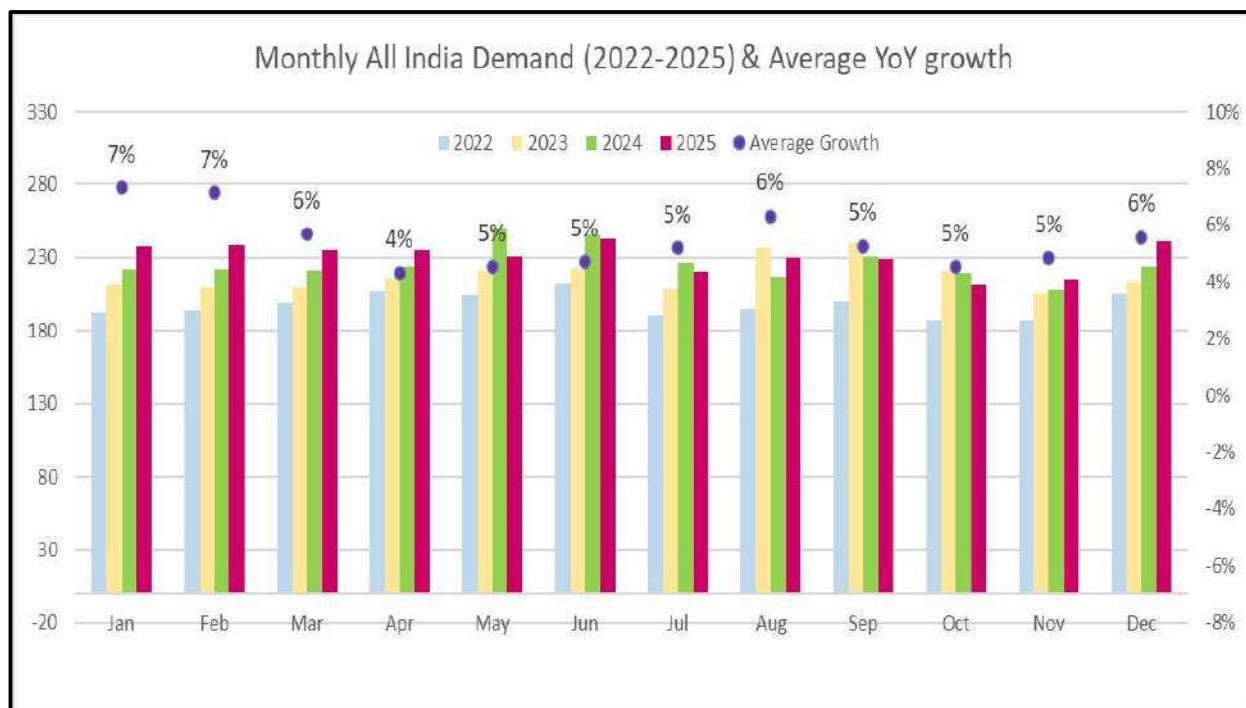
- ✓ *Presently the line is charged over location 248 to Location 33 from Lalmatia end*
- ✓ *Foundation is reaming in 8 locations.*
- ✓ *Tower erection is pending in 17 locations.*

OCC Decision

OCC advised JUSNL to expedite restoration of remaining towers and share the progress update regularly. The line may be brought to service before summer 2026.

2.2 `State-wise review and expeditious resolution of critical issues for ensuring reliable operation during Summer-2026:ERLDC

- All-India electricity
- demand has been witnessing a sharp and sustained growth over the past few years, with an annual growth rate in the range of 8–10%, driven by rapid urbanization, increased penetration of space-cooling loads, industrial recovery and changing consumption patterns. Notably, in December 2025, all India peak demand crossed 240 GW, registering an unprecedented 7.6% growth over the previous winter peak. As on 09.01.2026, all India demand crossed 245W mark, which is higher than summer peak 2025.



- In view of this emerging trend, Summer 2026 is expected to witness demand growth exceeding recent years trend, thereby placing exceptional stress on the transmission network and voltage support infrastructure. Anticipating these challenges, Eastern Region has proactively identified several critical pockets that are vulnerable under high-load and contingency conditions and require focused intervention to ensure secure, reliable, and resilient power supply. Timely implementation of identified corrective measures, particularly in load-dense and congestion-prone areas, is therefore essential to mitigate the risk of voltage instability, cascading outages, and large-scale supply disruptions during the forthcoming summer period.
- State-wise pending issues discussed in the OCC/TCC forums, which are still pending implementation are highlighted below:

West Bengal System:

1. Intrastate Transmission Network Assessment & Mitigation – West Bengal

Detailed in Agenda point B.1.(e).

2. Commissioning of 500MVA ICT VI at Subhasgram (PG):

On 10.01.2026, the 500 MVA ICT-6 at Subhasgram (PG) SS was successfully charged under no-load condition from the 400 kV side (with the 220 kV side EHV cables isolated), and no abnormality was observed. However, due to failure of the R-phase cable termination on the 220 kV side, the transformer could not be put into on-load operation and is likely to require additional time for normalization.

POWERGRID/SLDC WB/CESC may update. Members may discuss.

Deliberations in the meeting

Power Grid informed that faulty cable would be replaced shortly.

OCC Decision

It was advised that Power Grid may take appropriate action for charging the ICT #6 at the earliest.

3. Commissioning of Sagardighi Unit#5 (500MW)

Sagardighi Unit-5 (500MW) was first synchronized with the grid on 27.09.2025 at 21:01 hrs, after which multiple test synchronizations were carried out. As per records, the unit has been kept out of service since 20:16 hrs on 20.12.2025. Commissioning of the unit before Summer-2026 would significantly benefit the WB system, as it would reduce dependency on the 400 kV FSTPP–KHSTPP D/C line, which is currently under reconductoring.

SLDC WB/WBPDCL may update. Members may discuss.

Deliberations in the meeting

WBPDCL apprised:

Trial run of Sagardighi Unit#5 will be done in end of January 2026 and COD is expected by this financial year.

OCC Decision

WBPDCL was advised to adhere to the committed timeline.

Odisha System

1. Intrastate Transmission Network Assessment & Mitigation-Odisha

Detailed in Agenda point B.1.(f).

2. Committee Reports: Near Miss Event in Odisha System – 12th August 2025

- ✓ On 12th August 2025, at around 22:36 hrs, multiple tripping incidents occurred involving several 400 kV transmission lines in Odisha System. This resulted in a stressed system condition and a near-miss situation, wherein any further contingency could have led to a total system collapse in the **Meramundali, Mendhasal, New Duburi, Pandiabilli, and Baripada** areas, with a potential load loss of about **2,500 MW**.
- ✓ Pursuant to the discussions held in the 231st OCC meeting on 22.08.2025 at OPTCL Centre, Bhubaneswar, a committee comprising members from ERPC, ERLDC, SLDC, and OPTCL was constituted to frame short-term and long-term measures to prevent a blackout due to similar events in the future.
- ✓ The committee report is yet to be submitted.

SLDC Odisha may update. Members may discuss.

Deliberations in the meeting

OCC Decision

Regarding the near-miss event in the Odisha system, a committee has already been constituted. The committee convener will coordinate meetings and finalize findings to be put up in OCC forum.

3. Commissioning of New ACP-II Plant -Requirement of SPSs:

- ✓ Based on the decisions taken in the 233rd OCC meeting held on 21.11.2025 at the ERPC Secretariat, Kolkata, a special meeting to discuss SPS implementation at ACP-II/JSPL was convened on 26.11.2025 under the aegis of ERPC in hybrid mode, with participation from JSPL, ERLDC, OPTCL/SLDC Odisha, and the ERPC Secretariat.
- ✓ As per the deliberations and minutes of the meeting, Member Secretary, ERPC, advised OPTCL to complete the site visit to M/s JSPL within the committed timeline to assess the existing protection scheme and its adequacy in handling grid contingencies and internal network disturbances.
- ✓ As per the latest information, the site visit was conducted on 11.01.2026 by OPTCL, and a meeting is proposed to be held for scheme finalization. The SPS implementation has already been delayed and needs to be expedited.

SLDC Odisha/STU Odisha/ACPP-II JSPL may update. Members may discuss.

Deliberations in the meeting

ERLDC informed:

- ✓ *One SPS meant for injection has already been implemented by ACP-II.*
- ✓ *For the other SPS to restrict JSPL drawl at Meramandali end, power flow data of Talcher-Meramandali Line needs to be transmitted to JSPL end by OPTCL.*

OPTCL apprised:

RTU shall be installed at Meramandali to ensure data visibility at JSPL end but will take two months for the same.

OCC Decision

It was advised that JSPL and OPTCL may finalize the procurement bilaterally, so that the SPS gets implemented without further delay.

Jharkhand system:

1. Restoration of 2nd ICT at Tenughat and upgrading 400 KV PVUNL- Tenughat line end termination at Tenughat

Detailed in Agenda point B.1.(i).

2. Reliable power evacuation of PVUNL

- ✓ Currently, PVUNL is evacuating its 800 MW (Unit-1) via the 400 kV PVUNL–Patratu–New Ranchi D/C link and 400 kV PVUNL–Tenughat S/C (limited to 200 MW). Notably, the conductor type of 400 kV PVUNL–Patratu section is Quad Moose, while the upstream 400KV Patratu–New Ranchi section is of Twin Moose. In the event of an outage of any one circuit of the 400 kV Patratu–New Ranchi D/C, the evacuation of 800 MW becomes critical.
- ✓ As information gathered, PVUNL (3 x 800MW) is scheduled to synchronize its second unit in mid-February 2026, with a tentative commissioning date in March 2026. It will be difficult to evacuate 1600 MW through the existing corridor. It is noteworthy to mention that All India demand is expected to touch 267GW in the ensuing Summer 2026, where every single MW would be crucial as part of resource adequacy.

- ✓ 400 kV Patratu–Latehar-Chandwa D/C will serve as an important line of the PVUNL evacuation in the absence of planned evacuation system, in which 400 kV Chandwa–Latehar D/C is already operational. With the commissioning of the 400KV Patratu–Latehar link, PVUNL will be able to evacuate its entire generation, which would be a great help for PVUNL and Jharkhand System. Therefore, JUSNL needs to expedite the commissioning of the 400 kV Patratu–Latehar D/C before Summer 2026.

SLDC Jharkhand/JUSNL may update. Members may discuss.

Deliberations in the meeting

ERLDC informed :

On commissioning of 2nd unit of PVUNL, 1600 MW can't be evacuated through the Patratu-Ranchi twin moose lines. Both units then need to operate at reduced capacity. Hence, 400 kV Patratu–Latehar D/C needs to be commissioned on urgent basis for reliable power evacuation from PVUNL units.

OCC Decision

It was advised that ERLDC may initiate devising SPS logic in consultation with PVUNL, JUSNL and SLDC Jharkhand. This should be done in consensus with all stakeholders in the interest of Jharkhand system.

DVC system:

1. Restoration of Koderma ICT

Detailed in Agenda point B.1.(d).

2. SPS at Bokaro S/S

Detailed in Agenda point B.1.(d).

ISTS system

Reconductoring of 400kV Talcher-Meramundali and 400kV Farakka-Kahalgau D/C

Detailed in Agenda point B.1.(a).

2.3 Power Export/import through Meramundali JSPL D/C: OPTCL

M/s JSPL has requested approval for export of 1000 MW and import of 700 MW through the existing 400 kV D/C connectivity from 400/220/132/33 kV GSS, Meramundali under different scenarios. The said proposal has a significant impact on the **400 kV TSTPP- Meramundali** transmission line and, therefore, requires detailed scrutiny.

OCC is requested to deliberate to assess the system impact.

In this regard, detailed observation and study report of OPTCL attached at **Annex B.2.3**

OPTCL may update. Members may discuss.

Deliberations in the meeting

OPTCL submitted:

- *Export up to 850 MW is permitted as per joint study with ERLDC.*
- *Import limits will be studied after SPS implementation.*

OCC Decision

OCC noted.

2.4 Appropriate bus arrangement at Dalkhola (PG) Sub-station: WB SLDC

- ✓ To ensure compliance with the (N–1) contingency criterion for the **220 kV Malda (PG)–Gajol D/C** lines, the existing summer-season operating practice involves a split-bus arrangement at Gajol(WBSETCL) on both the 220 kV and 132 kV sides, and a split-bus arrangement on the 220 kV side at **Dalkhola (PG)**. Under this configuration, the load at **Gajol (WBSETCL)** is distributed as follows:

Bus reference at Gajol	Source	220 kV side Transformers connected	132 kV side feeder/Trf connected
220 KV Bus I	Malda (PG) ckt I, II	160 MVA * 2 nos,	Gangarampur, Balurghat, Harischandrapur D/C, Kusmundi, Habibpur, Malda 160 MVA * 2 nos, 50 MVA Trf 1
220 KV Bus II	Dalkhola (PG) I, II	160 MVA *1 no	Samsi D/C, 50 MVA Trf 2

Bus arrangement in Dalkhola(PG) as per prevailing summer practice:

Bus Reference	Lines connected
220 KV Bus I	220 KV Kishanganj D/C, Dalkhola(WBSETCL) D/C
220 KV Bus II	220 KV Purnea D/C, Gajol D/C

- ✓ In view of the anticipated load growth in and around the Malda and Dinajpur districts, the forecasted load at Gajol from Dalkhola (PG) is expected to be in the range of 150–155 MW during the ensuing summer season. Further, in the event of upgradation of the concerned ICT to 200 MVA prior to summer, the load may increase up to 170 MW
- ✓ In this context, it is requested that ERLDC may kindly undertake the necessary system studies to finalise the appropriate bus arrangement at Dalkhola (PG) Sub-station, along with any other related Power Grid sub-stations, so as to ensure secure and reliable grid operation during the upcoming summer period.

WB SLDC may update. Members may discuss.

Deliberations in the meeting

WB SLDC apprised:

- *During peak summer, loading of the ICTs may reach 160–170 MVA*
- *Power flow generally occurs from Malda → Gajol → Dalkhola → Purnea*
- *220 kV Malda–Gajol section becomes heavily loaded under loop operation whose reconductoring is still pending.*

OCC Decision

It was advised ERLDC to conduct a load flow study so that appropriate bus splitting / arrangement at Dalkhola (PG) may be finalized before summer.

2.5 Discrepancy i.r.o Reactive Energy billing: WB SLDC

- ✓ During the month of December 2025, the Reactive Energy billing data indicates a reversal in the direction of MVAR flow between two groups of ICTs at Subhasgram (PG) Sub-station. One group comprising ICT-I and ICT-II shows positive MVAR flow, while the other group comprising ICT-III, ICT-IV, ICT-V, and ICT-VII shows negative MVAR flow.
- ✓ Since all ICTs are operated in parallel, such divergence in the direction of reactive power flow appears anomalous and technically inconsistent. An Excel statement is enclosed as an Annexure to substantiate the observed discrepancies(**Annex B.2.5**)

WB SLDC may update. Members may discuss.

Deliberations in the meeting

WB SLDC briefed the observed discrepancy in energy billing data w.r.t Subhasgram (PG) and submitted:

- ✓ *In two ICTs, MVAR charges are negative while in other four ICTs, the charges are positive*
- ✓ *All transformers are operating in parallel, so this trend is unusual. This behaviour was not seen last year. Comparative data for the last few weeks shows a consistent anomaly.*
- ✓ *The discrepancy should be rectified to ensure accurate meter data and should be implemented in billing with retrospective effect.*

ERLDC apprised that the discrepancy is predominant in those ICTs where meters replaced from Genus make to Secure make.

Power Grid submitted:

- ✓ *Different OEM meters may be using different sign conventions. Some calculate “import – export” while Others calculate “export – import”.*
- ✓ *DLMS protocol allows OEM flexibility, but billing logic must be standardized*

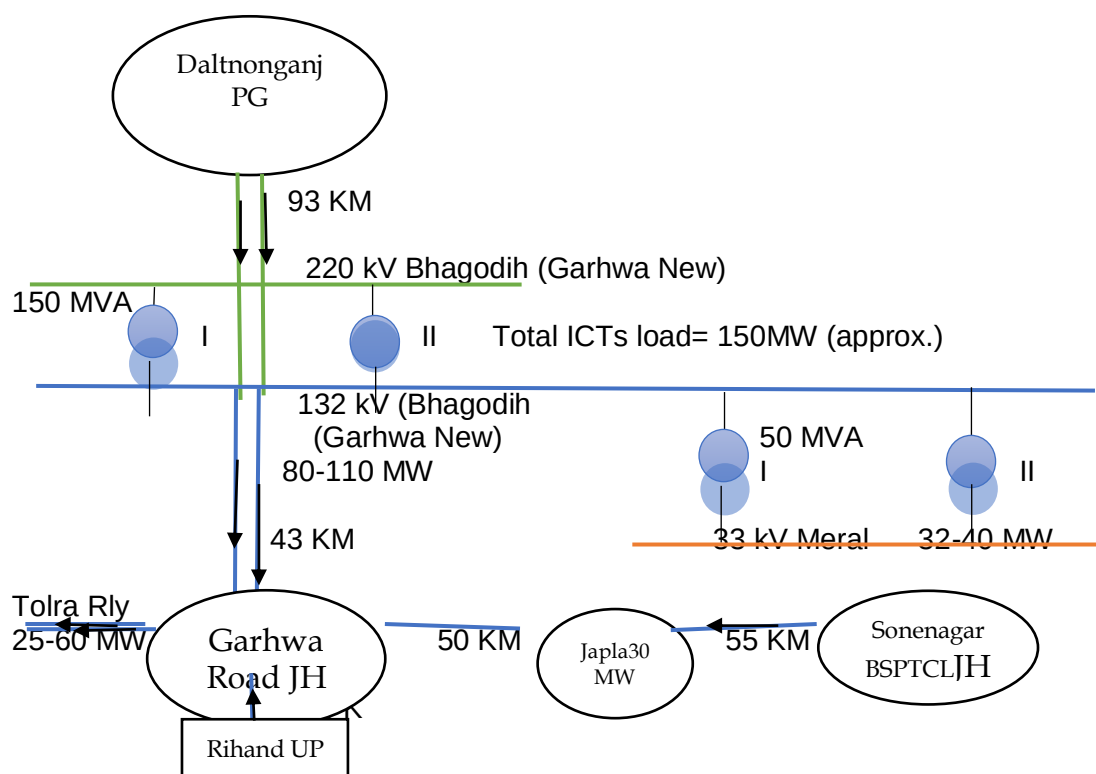
OCC Decision

- *It was advised Power Grid to address the issue in coordination with concerned OEMs of meters and ERLDC.*
- *Further, OCC opined that corrective logic to be applied retrospectively so no party is financially impacted.*

2.6 Persistent low voltage profile at 220/132kV Grid sub-station, Garhwa (Bhagodih): JUSNL

- ✓ Since last two months i.e from December 2025, 220/132 KV Garhwa New (Bhagodih) GSS is facing low voltage issue.
- ✓ The incoming voltage coming from 400/220kV Daltonganj PGCIL Grid Sub-Station lowers up to 204 kV or lower during morning and evening hours at 220/132kV Grid sub-station, Garhwa (Bhagodih). Due to which all connected consumers specially EC Railway faces difficulty in their routine operational activities of goods train.

- ✓ The issue may be resolved for quality power supply to all valuable consumers including EC Railway.



Report of incoming system voltage attached at [Annex B.2.6](#)
JUSNL may update. Members may discuss.

Deliberations in the meeting

JUSNL highlighted the low voltage issue being encountered in 220/132kV Garhwa station which is connected to 220 kV side of 400 kV Daltnonganj (PG). Low voltage to the order of 204 kV is recorded during solar peak hours.

ERLDC affirmed that voltage recorded at 400 kV Daltnonganj (PG) is mostly in the permissible range but downstream network with mostly radial load may be the reason of under voltage.

OCC Decision

- *SLDC Jharkhand was advised to communicate to ERLDC for appropriate remedial measures to mitigate low voltage issue.*
- *It was advised ERLDC to explore reactor switching at Sasaram (765 kV) to increase voltage slightly beyond 400 kV at Daltnonganj (PG) so as to boost voltage in the downstream network.*
- *OCC also recommended for adequate reactive compensation and other network strengthening measures after detailed system study.*

2.7 Status of spare Transformer/ICT in Eastern Region: ERPC

As per CEA guidelines for **availability of spares** and inventories for power transmission system (transmission lines & substation/switchyard) assets, adequate cold spare for ICTs has to be maintained at regional as well as state level.

□ Key guidelines for determining spare as per the guidelines are provided below:

- **Regional level spare:** For regional power utilities (POWERGRID & Transmission licensees), the spare at regional level would be required for these assets. These spares should be increased, optimized and limited to double the quantities mentioned for State Level based on transmission line assets in that region in order to avoid unnecessary storage of inventories. (**Annexure B.2.7**)
- **State level spare:** The spares at 'State level' can be maintained at a centralized location which could be conveniently accessed to meet the emergency requirement of various substations/switchyards spread across the State.
 - **Requirement of state level:** ICT and Shunt Reactor: One number single phase/three phase unit of each rating, as applicable
 - **Utility for State level spare:** If there are five or more substations/switchyards (of same voltage class) of a utility in a State, the 'State Level' spares shall be maintained by the utility.
 - **Replenishment of Consumed spare:** Replenishment of the consumed mandatory spares shall be made at the earliest but in any case, not later than six months from the date of its consumption depending on the criticality of equipment component/material.

With a significant rise in state demands and regional demand along with the number of ICTs, it would be desirable to have an adequate spare to improve reliability and resilience in case of any exigency. Recently, a substantial delay in restoration of damaged ICTs in eastern region has been observed. Thus, maintaining adequate regional and state level cold spare is important.

As per 55th TCC Meeting

- ✓ Powergrid representative informed that as per CEA guidelines adequate cold spare for ICTs has to be maintained at regional as well as state level. He added that the existing fleet of transformers of ER are considerably old and therefore sufficient spare management is required to ensure reliable power supply.
- Director(Op), OPTCL informed that they are assessing their spare requirements based on following factors:
 - n-1 criteria at each substation
 - Projected load growth
 - Standardization of transformers as per CEA guidelines
 - RLA of the existing transformersWBSETCL suggested that an SOP may be formulated for assessment of spare requirement of the region.

55thTCC Decision

- ✦ TCC advised Powergrid to reassess the spare requirement at ISTS level and submit it to ERPC secretariat for further deliberation in OCC meeting.
- ✦ TCC opined that a SOP for evaluation of spare requirement at state level may be formulated taking into consideration the factors adopted by OPTCL.

Consolidated details of spare ICTs in Eastern Regional pool maintained by Powergrid (ER-I,ER-II & Odisha Projects) attached at **Annex B.2.7**

Powergrid and other Transmission licensees may update. Members may discuss.

Deliberations in the meeting

Power Grid presented the spare requirement for Eastern region at ISTS level.

It was also apprised that optimal inventory requirement should not be based on operational studies.

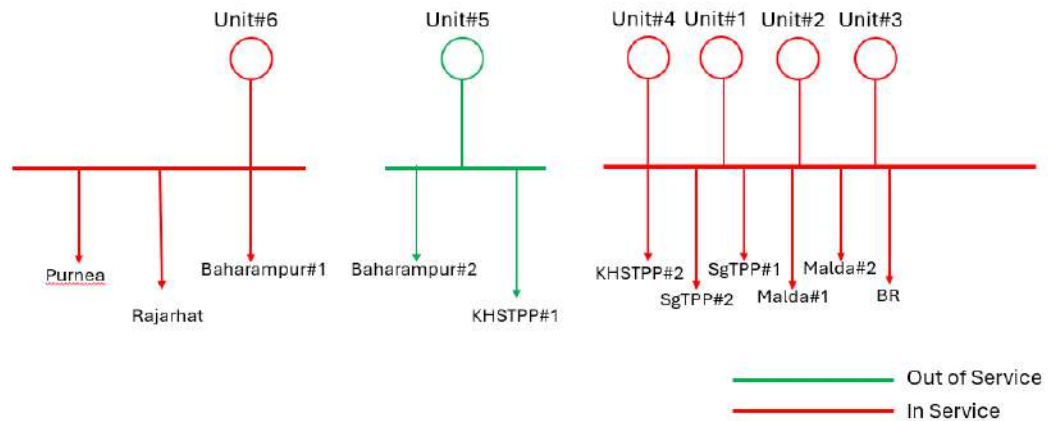
OCC Decision

- *It was advised Power Grid to justify the spare requirement at ISTS level as per CEA guidelines with due consideration of N-1 criteria, predicted rise in load as well as RLA of existing transformers.*
- *All ER states were advised to assess and furnish their spare requirement in line with CEA guidelines for availability of spares. Ensuring N-1 criteria and projected load growth should be considered in the assessment.*
- *The issue was referred to TCC for further deliberation.*

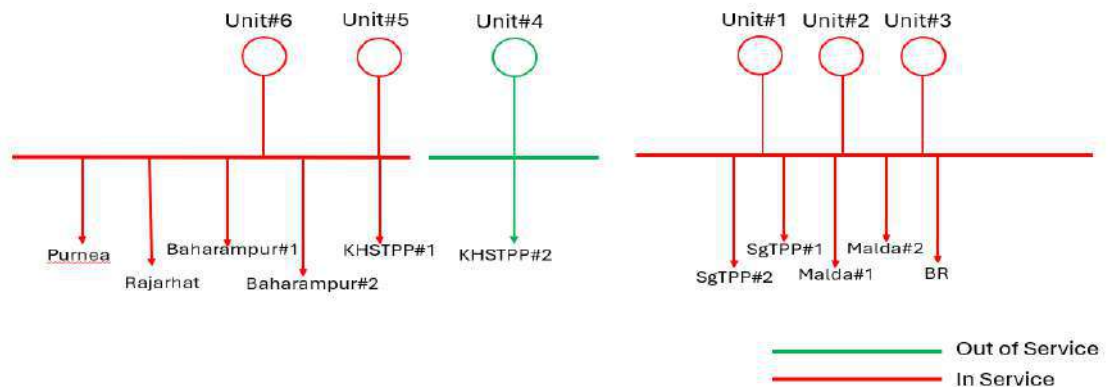
2.8 Bus Segregation at NTPC Farakka to facilitate Jack Bus reconductoring works:ERLDC

- As a part of reconductoring works of 400kV Farakka-Kahalgaon D/C, Bay upgradation is to be done at both NTPC FSTPP and KHSTPP SS for which jack bus along with other bay conductors also needs to be changed from Twin Moose to Twin HTLS. In the 234th OCC meeting, POWERGRID cited that the need of both 400kV buses shutdown at NTPC Farakka thereby shutdown of all the units of FSTPP to execute the reconductoring job.
- Subsequently, an online meeting was conveyed by NTPC on 26th Dec'2025 to discuss the issue and find optimal solutions to reduce the unit outages. In the meeting, it was decided to explore feasibility of Bus segregation to reduce the unit outages. NTPC vide mail dtd. 30th Dec 2025 has intimated detailed plan for facilitating bus shutdowns at Farakka in Feb'26 In which it was proposed that bus segregation may be done at 400kV Farakka bus by isolating only the portion required for reconductoring work. This would require shutdown of Unit#5(500MW) along with Baharampur#2 and ST#3 during reconductoring works for 400kV Farakka-Kahalgaon #1 for three days as per the shutdown plan. Similarly, shutdown of Unit#4(500MW) would be required for three days during reconductoring works in terminal bays at Farakka for 400kV Farakka-Kahalgaon #2.
- Power flow studies on the proposed Bus segregation scheme were conducted by ERLDC and no major issues observed. However, necessary protection coordination during the shutdown period shall be ensured by NTPC Farakka.
- The Proposed Scheme as under:

Bus arrangement at Farakka during SD for Reconductoring of Kahalgaon Ckt#1



Bus arrangement at Farakka during SD for Reconductoring of Kahalgaon Ckt#2



ERLDC may explain. Members may discuss.

Deliberations in the meeting

NTPC shared the plan for Jack bus replacement at Farakka end (**Annex B.2.8**).

Further concern was raised on tripping of Farakka-Kahalgaon line which may cause unwanted outage of Unit-04 at Farakka STPS.

OCC Decision

OCC took note of NTPC's concern regarding the possible generation outage. However, in view of the criticality of reconductoring of 400 kV Farakka-Kahalgaon D/C, opportunity based maintenance activities may be carried out by NTPC in event of any forced unit outage.

2.9 Techno-Commercial implications of Bus shutdown at Farakka: NTPC

Farakka vide mail dated **30th December 2025** has intimated a detailed plan for facilitating **bus shutdowns at Farakka in February 2026** for carrying out scheduled maintenance activities.

ERLDC has intimated that **power flow studies for the proposed segregated bus operation have been carried out** and **no major issues have been observed** from a system operation perspective. POWERGRID ER-II has also **concurred with the proposed shutdown schedule** and has intimated that **required materials will be made available by the first week of February 2026**, enabling execution of the planned maintenance works within the scheduled timeframe.

Need for Approval

While the shutdown plan has been technically assessed and agreed upon by all concerned agencies, **during the execution of maintenance works**, the following operational contingencies **cannot be completely ruled out**:

- Development of **hotspots** during maintenance activities
- **Unit tripping** or partial generation loss during segregated bus operation
- Associated impacts due to **forced outage**, if any

Such contingencies may lead to **commercial implications for NTPC**, including but not limited to:

- Loss of **Declared Capacity (DC)**.
- Increase in **oil consumption due to FO or Partial Outage**.
- **DSM charges** or loss of incentive.
- Other incidental commercial losses attributable to maintenance-related system constraints

Submission

In view of the above, it is submitted that **any commercial losses incurred by NTPC during the said maintenance period**, arising due to:

- Bus shutdown–related operational constraints
- Development of hotspots
- Unit tripping or forced outages attributable to the approved maintenance activities.

may be **covered / compensated**, so that **NTPC does not suffer financial disadvantage while undertaking system-critical maintenance works agreed upon by all stakeholders**.

The proposed maintenance is essential for system reliability and has been duly coordinated with ERLDC and POWERGRID. Granting approval for coverage of potential commercial losses will facilitate smooth execution of the shutdown activities without exposing NTPC to undue financial risk.

Approval of the **OCC** is solicited for:

Coverage of commercial losses to NTPC, including DC loss, oil consumption, DSM impact, and any other incidental losses, if incurred during the execution of the approved bus/line shutdown and maintenance activities at Farakka.

NTPC may explain. Members may discuss.

Deliberations in the meeting

NTPC submitted:

- ✓ *While the shutdown plan has been technically agreed upon by all concerned agencies, during the execution of maintenance works, several operational contingencies cannot be completely ruled out, such as, development of hotspots during maintenance activities and Unit tripping or partial generation loss during segregated bus operation.*
- ✓ *These contingencies may result in Loss of Declared Capacity (DC), Increase in oil consumption due to FO or Partial Outage or DSM charges or loss of incentive.*

OCC Decision

- OCC opined that reconductoring of 400 kV Farakka-Kahalgaon D/C is crucial for improved system reliability despite concerns raised by NTPC on forced complete/partial outage of Farakka units and consequent commercial implications.

2.10 Portal for entry of Schedule discrepancies and feedback Mechanism: NTPC

Schedule discrepancies such as ramping issues, change in schedule in same block, software issues or any other technical issues observed during real time is being informed to **ERLDC** control room through mail by Shift In charge and same is again verified and shared to **ERLDC** final Scheduling Group by Station through email on D+1 day, however it has been observed many a times issues remain unresolved and no feedback received for resolution of the discrepancy. Later on, during corroboration with REA, such discrepancies come into notice without resolution which leads to financial losses.

SRLDC & NRLDC have developed portal with login credentials to generators in their control for better tracking, resolution & feedback on real time schedule discrepancies

Portals snapshots are as below:

Forecast Monitoring & Schedules Correction System

GRID-INDIA

Username

Password

J D 4 9 8

Enter above text

Log In

New User

New User (RE Category)

- ✓ Portal may be developed by **ERLDC** for entry /posting of such issues of schedule discrepancies by the generator for better tracking, resolution & feedback In line with portal developed by SRLDC & NRLDC.
- ✓ Partial outage and Forced outage communication may also be included in the portal with mechanism to upload proof regarding the same.

NTPC and ERLDC may update. Members may discuss.

Deliberations in the meeting

OCC Decision

ERLDC was requested to incorporate modifications in the scheduling software to the best feasible extent for reducing the financial losses to NTPC. Further, NTPC was advised to approach ERLDC and resolve the issue bilaterally.

2.11 Reliable evacuation of power from PVUNL: NTPC

1. Standardized Shutdown Approval Mechanism for 400 kV GIS at PVUNL

A standardized and well-defined mechanism may kindly be finalized for availing shutdowns of 400 kV GIS–connected elements at PVUNL. The proposed framework should address the following categories:

- a) Preventive maintenance activities
- b) Short-notice critical works
- c) Emergencies.

The procedure should also define the workflow for shutdown application and approval, including access to the ERLDC online portal, coordination protocols, and communication requirements among PVUNL, SLDC, and RLDC.

2. Prior Intimation of Line Shutdowns and Power Evacuation Constraints

Any planned or emergent shutdown of transmission lines or power evacuation systems associated with PVUNL may kindly be communicated well in advance. Timely intimation is essential for

coordinated operational planning and to avoid adverse conditions, including the risk of blackout at PVUNL.

3. Reliability and Redundancy of Communication Links

Frequent failures of the existing communication link with PVUNL (Real time data CH-1/2, PMU, AGC CH-1/2, VOIP and AMR are communicated over these links), coupled with the absence of a redundant communication path, pose a significant operational risk. Immediate measures are requested to establish communication redundancy and enhance the overall reliability of real-time data exchange.

4. Integration of PVUNL Substation and Connected Network into UNMS

PVUNL substation, along with substations under the jurisdiction of SLDC connected to PVUNL, may kindly be fully integrated into the Unified Network Management System (UNMS). This will facilitate centralized monitoring, faster fault identification, and improved system visibility.

5. Third-Party Protection Audit of Katia Substation

A comprehensive third-party protection audit of Katia Substation is requested to evaluate the adequacy, coordination, and reliability of the existing protection schemes and to recommend corrective measures, wherever required.

6. Shutdown Requirement of Bay-417

A shutdown of Bay-417 is required to rectify an identified issue wherein the polarity of one core of the Current Transformer (CT) is reversed. Timely approval of the shutdown is requested to enable necessary corrective action and ensure protection integrity.

NTPC may explain. Members may discuss.

Deliberations in the meeting

OCC Decision

- OCC opined that standardized Outage procedure of Eastern region shall be followed by SLDC Jharkhand in processing any shutdown requests i.r.o PVUNL.
- The issue of redundancy of Communication Links as well as Integration into UNMS was referred to TeST meeting for deliberation.
- The requirement of Third-Party Protection Audit of Katia Substation was referred to PCC meeting for deliberation.

2.12 Shutdown proposal of Thermal generating units from the month of January to March 2026: ERPC

The approved shutdown schedule as per 234th OCC is as below:

Maintenance Schedule of Thermal Generating Units of ER during 2025-26									
System	Station	Unit	Capacity (MW)	LGBR Approved		No. of Days(as per OC C)	OCC Approved		Reason
				From	To		From	To	
	NABINAGAR STPP	2	660	5-Jan-26	18-Feb-26	—	Not required.		AOH

NTPC	BARAUN I TPS	9	250	01.12.2025	30.12.2025	45	01.01.2026	15.02.2026	AOH
	BARAUN I TPS	8	250	15.03.26	31.03.26	–	Not required.		AOH
DVC	MEJIA TPS	8	500	11-Jan-26	4-Feb-26	–	Not required.		BOH
	MEJIA TPS	6	250	03-12-2025	06-01-2026	34	28.12.2025	31.01.2026	AOH-R&M
	MEJIA TPS	7	500	29-08-2025	25-09-2025	34	06.02.2026	12.03.2026	AOH-R&M
	KODARM A TPP	2	500	11-Feb-26	17-Mar-26	–	Not required.		AOH
	KODARM A TPP	1	500	05-10-2025	01-11-2025	34	10.12.2025	13.01.2026	AOH
WBPDCCL	BANDEL TPS	5	210	3-Feb-26	9-Mar-26	7	05.02.2026	12.02.2026	Boiler License renewal
	SANTAL DIH	5	210	27.12.25	30.01.26	34	03.01.2026	06.02.2026	BTG OH+De-Nox
	SAGARD IGH	1	210	05.08.25	08.09.25	34	04.01.2026	07.02.2026	AOH
	KOLAGH AT	3	210	10.02.26	16.03.26	35	08.12.2025	11.01.2026	AOH
	KOLAGH AT	4	210	15-07-2025	08-08-2025	25	Not required now. To be availed in next FY.		ESP R&M

NTPC vide mail dated 15.01.2026 has requested:

- ✓ The agency engaged for OH works of Boiler of Unit-6 has now refused to take up the job. So we are in the process of engaging new agency for the work and propose to defer the OH. It is now proposed to take the AOH of **Unit-6** from 01.03.2026.
- ✓ Overhauling of **FSTPS Unit-6** may please be allowed from **01.03.2026** to **31.03.2026**.

All ER thermal generating units may peruse and update on any modification.

Members may discuss/update.

Deliberations in the meeting

- ✓ *NTPC requested to avail the Shutdown of Farakka STPS Unit#6(500 MW) from 01.03.2026 to 31.03.2026 for 30 days to carry out overhauling activities. The shutdown of this Unit was earlier approved from 15.01.2026 for a period of 30 days. Since, the agency engaged for OH works of Boiler of Unit-6 has recently refused to take up the job and engagement of new agency is in process, thus, deferment of the OH was proposed.*
- ✓ *West Bengal SLDC and WBSEDCL(West Bengal DISCOM) expressed strong dissent over the request of NTPC keeping in view projected high demand in Summer 2026 and reliable power availability during upcoming state board exams and ensuing State Assembly elections.*

OCC Decision

- *It was suggested that shutdown of NTPC Farakka Unit-06 may be allowed subject to ensuring equivalent quantum of power (as per share allocated) to West Bengal DISCOM during the said period i.e 01.03.2026 to 31.03.2026. For this purpose, ERPC may write a letter to CEA for allocating 170MW UA power to West Bengal from ER UA pool.*

- The shutdown schedule of other thermal generating units remains unchanged as detailed above.

2.13 Shutdown Program of Hydro power plants

Annual maintenance of RANGIT HEP

Deliberations in the meeting

OCC Decision

OCC approved the Shutdown of three units of Rangit HEP as per schedule detailed below:

Power Station	Unit	Capacity	Nature of outage (Partial/ Complete)	Proposed Shutdown Period		Reason for Shutdown
				From	To	
RANGIT	UNIT #1	20 MW	Complete	03.04.2026	26.04.2026	Annual Maintenance
	UNIT #2	20 MW	Complete	05.01.2026	28.01.2026	Annual Maintenance – already in progress
	UNIT #3	20 MW	Complete	31.01.2026	31.03.2026	Capital Maintenance

2.14 Resource Adequacy assessment using Probabilistic Resource Adequacy Suite (PRAS): ERLDC

- ERLDC has been carrying out Resource Adequacy assessment for the Eastern Region on Intra-Day, Day-Ahead, Week-Ahead, Month-Ahead, and Year-Ahead timeframes using a stacking-based deterministic approach based on ERLDC demand forecasts, planned generation outages as per LGBR, and historical trends.
- ERLDC has started using doing Probabilistic Resource Adequacy assessment for Month ahead basis for individual States using the Probabilistic Resource Adequacy Suite (PRAS). The assessment is carried out using Monte Carlo simulation with 10,000 scenarios, wherein generator forced outages are modelled probabilistically.
- The assessment considers ERLDC demand forecasts (based on historical SCADA data), forced outage rates derived from past 7-year outage data, planned outages as per LGBR, and hydro/storage generation based on past 3-year trends.
- Based on simulation results of PRAS, a consolidated report of month ahead resource adequacy is being generated and communicated to states via email from January-2026 onwards.

A summary table for **Feb 2026** Assessment is shown below.

State	Maximum Shortage MW	Maximum Shortage MW Day	Maximum MU shortage	Maximum MU shortage Day	Average LOLE	Maximum Surplus MW	Maximum Surplus MW Day
West Bengal	713	17-02-2026	1.71	17-02-2026	0.03	3,159	22-02-2026
Bihar	0.26	07-02-2026	0.00042	07-02-2026	0.00	5,686	18-02-2026
Jharkhand	4	18-02-2026	0.01	18-02-2026	0.00	1,384	23-02-2026
DVC	331	06-02-2026	3.54	07-02-2026	0.32	711	04-02-2026
Odisha	313	04-02-2026	1.39	28-02-2026	0.09	2,458	06-02-2026
Sikkim	4	18-02-2026	0.00	18-02-2026	0.00	113	02-02-2026

The detailed report is attached in **Annexure B.2.14**

ERLDC may explain. Members may discuss.

Deliberations in the meeting

ERLDC presented the simulation results of Probabilistic Resource Adequacy that shall be communicated to all ER states from January 2026 onwards.

OCC Decision

OCC noted.

2.15 Verification of Average Generation/Demand data for assessment of Frequency Response Obligation (FRO) for FY 2026-27: ERLDC

•IEGC Reg. 30(10) (f) mandates that the “Frequency Response Obligation (FRO) of each control area shall be assessed by NLDC and the same shall be informed to all control areas by 15th of March every year for the next financial year.” Accordingly, NLDC is in the process of calculating the FRO for each control area for the FY 2026-27. Annexure-2 of IEGC provides that the minimum FRO of each control area in MW/Hz shall be calculated as:

•
$$FRO = \frac{\text{Control Area average Demand} + \text{Control Area average Generation}}{\text{Sum of average demand of all control areas} + \text{Sum of average generation of all control areas}} \times \text{minimum all India Target Frequency Response Characteristic}$$

•Accordingly, ERLDC has computed the Average generation of all the ISGS/Central Sector IPPs and Average Generation and Demand data of State control areas of Eastern Region which has been communicated to the concerned utilities for verification vide mail dated 12th January,2026.

Following data has been used for calculating the average value of Demand and generation:

1. Daily power supply position demand MU is collected for 365 days, and then average MW demand is calculated for state control area
2. Daily power supply position state internal generation MU is collected for 365 days, and then average MW generation(A) is calculated for state control area.

Similarly for internal RE generation(B) of state is calculated. State internal average generation without RE is then calculated by A-B.

3. For all generating station under RLDC jurisdiction which are eligible for primary frequency response, average generation is calculated by 15min SEM data average over one year time period.

All the ISGS/IPP and states are requested to verify the data at the earliest for finalization of FRO calculation for the year 2026-27 at the earliest.

ERLDC may explain. Concerned constituents may update. Members may discuss.

Deliberations in the meeting

ERLDC submitted:

- ✓ *Average demand and generation data for each control area has been calculated as per procedure detailed in IEGC 2023 and circulated to all ER states as well as generating stations.*
- ✓ *Only West Bengal has verified the data so far.*

OCC Decision

- *All other states (except WB) along with generating stations were advised to verify generation/demand data and respond within the stipulated timeline.*
- *OCC also opined If no comments are received by the due date, the shared data may be treated as final for respective control areas.*

2.16 Type Test Report for spare HT equipment: MPL

- In last U#1 outage, GT R phase was replaced with spare GT.
- Apart from various test reports, RIO had asked for type test report (**not older than 5 years**) of the **spare GT** for taking the same in service.
- The GT was procured a decade back and it was not possible to furnish type test report of the same. so type test report of similar higher rating transformer from Mundra was furnished.
- The OCC forum is requested to discuss and finalize on necessity, difficulty, and alternatives of type test report for taking any spare HT equipment in Grid.

MPL may explain. Members may discuss.

Deliberations in the meeting

MPL submitted that they do not have the type test report for the GT in question and sought guidance from the forum.

OCC Decision

*OCC informed that new guidelines have been formulated by CEA regarding type test of similar equipment (**Annex B.2.16**) and advised MPL to take up the matter with CEA.*

2.17 Certification of Non-ISTS lines carrying ISTS power: ERPC

DVC&OPTCL have requested certification of their non-ISTS lines carrying ISTS power vide letter dated 03.12.2025 and 15.12.2025 respectively. (attached at **Annex B.2.17**)

Members may discuss.

Deliberations in the meeting

OCC Decision

It was advised that the availability of these non-ISTS lines will be certified at ERPC once their carrying of ISTS power is confirmed by CEA.

2.18 Data Collection for monitoring Pan-India Captive Generating Capacity: ERPC

- In the meeting taken by **Secretary (Power)**, Govt of India on **17.12.2025**, it was decided that the State Chief Electrical Inspectors (CEIs) / State Load DespatchCentres (SLDCs) shall act as the nodal agencies for collection of **Captive Generation&Open Access** data for their respective States.
- It was further decided that the **Regional Power Committees (RPCs)** shall act as the nodal coordinating agencies for consolidation and compilation of the data at the regional level on **monthly basis**.

Hence,all **SLDCs** are requested to send the data of the particular month by **10th** of the subsequent month as per the format shared via email.

Members may discuss.

Deliberations in the meeting

ERPC requested all SLDCs to send the CPP generation data at the earliest.

OCC Decision

*All SLDCs were advised to collate captive generation data of CPPs (as per format enclosed in **Annex B.2.18**) in their respective states and share it with ERPC secretariat on monthly basis regularly.*

ADDITIONAL AGENDA

2.19 Shutdown request of 400 kV DC Jeypore–Gajuwaka for ongoing works on NH-130 : NHAI

- ✓ NHAI has requested for shutdown of the **400 kV DC Jeypore–Gajuwaka** EHT transmission line (Tower 179–180) at Project Chainage Ch. 319+330 under NH-130CD, Package OD-6, for facilitating the shifting works essential for completion of the project.
- ✓ The proposed shutdown period is **15th February 2026 to 1st March 2026 (15 days)**, as detailed in the attached communication. (**Annex B.2.19**)
- ✓ This is for kind consideration and confirmation at the earliest to enable timely execution of the shifting work.

Powergrid Odisha may respond.Members may discuss.

Deliberations in the meeting

NHAI briefed the OCC regarding the Raipur–Visakhapatnam Economic Corridor (Bharatmala Project) and sought shutdown of an inter-regional 400 kV D/C transmission line for completion of road works at multiple tower crossing locations. Criticality of project completion by stringent timelines was highlighted.

ERLDC informed:

- The concerned transmission line is an inter-regional (ER–SR) element.
- Such requests must be processed through NLDC, with concurrence of both ERLDC and SRLDC.

ERLDC also submitted that the proposed shutdown window (mid-February to early March) may have limited impact on Eastern Region, though concerns were raised about simultaneous outages on SR side amid increased vulnerability with rising summer load.

OCC Decision

It was advised that during summer, ER–SR power flows may increase, particularly with higher solar generation in ER. At times, power may also flow from ER to SR, so coordination is critical. The subject line being inter-regional, OCC advised NHAI to urgently approach NLDC and SRPC for seeking necessary consent before coming Summer.

2.20 Decapitalization of 50 MVAR Bus Reactor -1 (3 X 16.67 MVAR) at Durgapur S/S on account of upgradation: POWERGRID ER-II

According to office Memorandum dated on 17.01.2023 (Ref ; C/CTU/AI/00/10th CCTP). new 420KV, 1X125 MVAR three phase bus Reactor along with associated bay equipment has been installed at Durgapur Substation.

- i) 125 MVAR CGL make (SI No – BHR11495-02) Bus Reactor -V has energized on 05.12.2025 and subsequently declared for commercial operation.
- ii) Existing 3x16.67 MVAR Bus Reactor -I (Make – CGL; SI No R Phase – 24893; Y Phase - 24894; B Phase – 24895) which was Commissioned on 05.03.1991 and already elapsed more than 34 Years of service. Further as per CTU OM, it is deliberated that, on successful charging of 125 MVAR Bus Reactor-V, subject 50 MVAR Bus Reactor-1 will be discontinued from service.
- iii) In this regard it is placed before forum for information that, subject Reactor will be decapped from service w.e.f 05.12.2025.

Members may note.

Deliberations in the meeting

OCC noted.

2.21 Installation of Three New Towers in the Multi-Circuit Section of 400 KV Gaya –Koderma and 400 KV Gaya-Maithon Lines (Tower Nos. 39-41) to Mitigate Frequent Conductor Snapping: POWERGRID ER-I

- The existing spans between towers 39-40 and 40-41 of the 400 KV Gaya –Koderma and 400 kv Gaya Maithon multi-circuit lines are unusually long (approximately 750 and 820 m). These stretches pass through hilly terrains and dense forest, resulting in high mechanical tension on the HTLS conductors.

- The line was constructed in 2013 under severe constraints, including significant security concerns due to high Maoist activity. Owing to these limitations, several optimal towers locations could not be utilized, compelling the adoption of exceptionally long spans.
- The 400 KV Gaya –Koderma line also connects to the Koderma DVC Power Plant and is operationally critical. Frequent conductor snapping continues to occur in these spans due to excessive mechanical stress, and repeated replacements of HTLS conductors is not a sustainable long-term solution.
- To resolve this issue permanently, the installation of three additional towers in these two spans has been proposed. This will reduce span lengths, lower conductor tension, and enhance the reliability of power evacuation from DVC Koderma.

Powergrid ER-I may explain. Members may discuss.

Deliberations in the meeting

OCC Decision

Powergrid was advised to put up the proposal in monthly shutdown meetings or as emergency shutdown.

3. PART-C:ITEMSFORUPDATE/FOLLOW-UP/INFORMATION

3.1. ERGridperformanceduringDecember2025

The average and maximum consumption of Eastern Region and Max/Min Demand (MW), Energy Export for the month December-2025 were as follows:

AVERAGE CONSUMPTION (MU)	MAXIMUM CONSUMPTION(MU)/ DATE	MAXIMUM DEMAND (MW)	MINIMUM DEMAND (MW)	SCHEDULE EXPORT	ACTUAL EXPORT
		DATE / TIME	DATE / TIME	(MU)	(MU)
464 MU	488 MU, 31.12.2025	24054 MW, 25.12.2025 at 17:56 Hrs.	14086 MW, 08.12.2025 at 03:14 Hrs.	5691	5804

ERLDC/ERPC may highlight the performance of the ER grid.

Deliberations in the meeting

The grid performance of ER for the month of December 2025 was highlighted.

3.2. Non-Submission of FRC data in stipulated time-frame: ERLDC

Adhering to IEGC clauses 30.8 and 30.10.(a) to 30.10.(q), generating stations within the Eastern region are required to submit essential data to ERLDC within two days of receiving a notification regarding a reportable frequency event. Additionally, according to clause 30.10.(n), all control areas within the eastern region must assess their frequency response characteristics and share the evaluation, along with high-resolution data, with the ERLDC. Therefore, timely submission of primary response data is crucial for compliance with the IEGC.

Hence all are again requested to follow the stipulated timeline and submit the data to ERLDC and also fill the google sheet below to include the email address where notifications of reportable events should be sent.

The latest data receipt status is given below: (as on **14.01.2026**):

STATIONS		11-05-2025 16:51 HRS	12-06-2025 13:34 HRS	16-06-2025 11:51 HRS	22-07-2025 19:46 HRS	29-07-2025 14:55 HRS	01-09-2025 14:57 HRS	24-09-2025 11:04 HRS	24-09-2025 11:32 HRS	15-10-2025 12:11 HRS	16-12-2025 06:34 HRS
FSTPP #STG 1 & 2	ISGS										
FSTPP # STG 3	ISGS										
KhSTPP #STG 1	ISGS										
KhSTPP #STG 2	ISGS										
TSTPP #STG 1	ISGS										
Barh stage-1	ISGS										
Barh stage-2	ISGS										
BRBCL	ISGS										
Darlipalli	ISGS										
North Karanpura	ISGS										
NPGC	ISGS										
TEESTA V	ISGS										
Dikchu											
IBEUL (JSW UTKAL)/INDBHARAT	IPP										
GMR	CPP										
MPL	CPP										
ADHUNIK	CPP										
JITPL	CPP										
TEESTA III	CPP										
Bihar	STATE										
Jharkhand	STATE										
DVC	STATE										
OPTCL	STATE										
WB	STATE										
Updated as on	14.01.2026										
	Received										
	Not Received										
	Plant Out										
	Data freeze at plant										

Hence all are again requested to follow the stipulated timeline and submit the data to ERLDC and also fill in the google sheet below to include the email address where notifications of reportable events should be sent.

https://docs.google.com/spreadsheets/d/1slvAOmQIEQVIMn0LnB78eKMa2sz2QYICZ-sPEpeV_jk/edit?usp=sharing

ERLDC may explain and all SLDCs may update. Members may discuss.

Deliberation in the meeting

OCC Decision

- All generators were advised to regularly share high resolution data against each reportable frequency event with ERLDC on time to facilitate accurate assessment of FRP for respective control areas.
- All generating utilities were also urged to update the google sheet (link mentioned above) with email address where notifications of reportable events will be shared.

3.3. Regarding Non-Submission of Forecasting Data from States: ERLDC

Clause 2 of Regulation 31 of IEGC 2023 has mandated all the SLDCs to timely submit the demand estimate data to the respective RLDC and RPC.

Current data submission status is given in the table below: Hence it is again requested to all the concerned for timely submission of demand estimation data to ERLDC. This collaboration is essential for effective planning and preparedness to meet the region's electricity demands efficiently and reliably.

Latest Forecast and Resource Adequacy Data receipt status at ERLDC is shown below:

4. PART-D:OPERATIONALPLANNING

4.1. Anticipated power supply position for February-2026

The abstract of peak demand (MW) vis-à-vis availability and energy requirement vis-à-vis availability (MU) for the month of February-2026 is prepared by ERPC Secretariat (Annexure D.1) on the basis of LGBR for 2025-26 and feedback of constituents, keeping in view that the units are available for generation and expected load growth etc.

Members may update.

Deliberations in the meeting

All states were requested to provide their anticipated power supply position for February 2026 within a week. Updated anticipated power supply position will be shared accordingly.

4.2. Major Thermal Generating Units/Transmission Element outages/shutdown in ER Grid (as on 13-01-2026)

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	NABINAGAR (NPGC)	BIHAR	NTPC	1	660	ANNUAL OVERHAULING	01-Dec-2025
2	NABINAGAR (BRBCL)	BIHAR	NTPC	1	250	GENERATOR STATOR EARTH FAULT	04-Dec-2025
3	KOLAGHAT	WEST BENGAL	WBPDC	3	210	CAPITAL OVERHAULING	11-Dec-2025
4	TSTPP	ODISHA	NTPC	1	500	ANNUAL OVERHAULING	15-Dec-2025
5	KODERMA	DVC	DVC	1	500	ANNUAL OVERHAULING	17-Dec-2025
6	NABINAGAR (BRBCL)	BIHAR	NTPC	2	250	GENERATOR END WINDING VIBRATION IS RUNNING HIGH	31-Dec-2025
7	SAGARDIGHI	WEST BENGAL	WBPDC	1	300	ANNUAL OVERHAULING	04-Jan-2026
8	BUDGE-BUDGE	WEST BENGAL	CESC	2	250	ANNUAL OVERHAULING	07-Jan-2026
9	MEJIATPS	DVC	DVC	2	210	STATOR EARTH FAULT	07-Jan-2026
10	NORTH KARANPURA	JHARKHAND	NTPC	1	660	TRIPPED ON BOILER TUBE LEAKAGE	11-Jan-2026
11	SANTALDI	WEST	WBPDC	5	250	ANNUAL	12-Jan-2026

	H TPS	BENGAL	L			OVERHAULING	
12	PVUNL	JHARKHA ND	PVUNL	1	800	TRIPPED ON BOILER TUBE LEAKAGE	12-Jan-2026
13	Sterlite	ODISHA	SEL	3	600	TRIPPED ON APH PROBLEM	12-Jan-2026
14	TENUGHA T	JHARKHA ND	TVNL	1	210	CLASS A PROTECTION OPERATED DUE TO EXCITATION TRANSFORMER FAILURE	12-Jan-2026
15	BOKARO- A'	DVC	DVC	1	500	TRIPPED ON BOILER TUBE LEAKAGE	12-Jan-2026
16	BARH	BIHAR	NTPC	1	660	TRIPPED ON BOILER TUBE LEAKAGE	13-Jan-2026
17	HEL HIRANMA YEE	WEST BENGAL	HEL	2	150	TRIPPED DUE TO ASH CONVEYING ISSUE	13-Jan-2026

All Generating stations are requested to update expected restoration time and reason outage to ERLDC/ERPC on weekly basis in case of any change at their end.

Major Generating stations Out on Reserve Shutdown due to low system demand:

SL No	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	SOUTHERN	WEST BENGAL	CESC	1	67.5	RESERVE SHUT DOWN	03-Oct-2025

Hydro Unit Outage Report: -

S. NO	STATION	STATE	AGENCY	UNIT NO	CAPACITY (MW)	REASON(S)	OUTAGE DATE
1	TEESTA STG III Hep	SIKKIM	TUL	1 to 6	200 X6	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in	04-Oct-2023
2	TEESTA STG III Hep	SIKKIM	TUL				
3	TEESTA STG III Hep	SIKKIM	TUL				

4	TEESTA STG III Hep	SIKKIM	TUL			Teesta River and damage of Teesta III Dam & downstream Powerhouses	
5	TEESTA STG III Hep	SIKKIM	TUL				
6	TEESTA STG III Hep	SIKKIM	TUL				
7	TEESTA HPS	SIKKIM	NHPC	1 to 3	170 X3	Sudden cloudburst at glacier fed LOHNAK Lake followed by huge inrush of water in Teesta River and damage of Teesta III Dam & downstream Powerhouses	04-Oct- 2023
8	TEESTA HPS	SIKKIM	NHPC				
9	TEESTA HPS	SIKKIM	NHPC				
10	CHIPLIMA HPS / HIRAKUD II	ODISHA	OHPC	1	24	CAPITAL OVERHAULING	15-Dec- 2023
11	BALIMELA HPS	ODISHA	OHPC	6	60	INITIALLY UNIT WAS OUT DUE TO SEVERE WATER LEAKAGE FROM TURBINE, LATER UNIT WAS TAKEN UNDER REPAIR AND MAINTENANCE WORK FROM 00:00 HRS OF 16.01.25	06-Jan- 2025
12	BALIMELA HPS	ODISHA	OHPC	5	60	REPAIR AND MAINTENANCE WORK	16-Jan- 2025
13	SUBARNREKHA HPS	JHARKHAND	JUUNL	1	65	DAMAGE IN CIVIL STRUCTURE NEAR PENSTOCK BLOCKING WATER FLOW.	20-Mar- 2025

14	SUBARNREKHA HPS	JHARKHAND	JUUNL	2	65	DAMAGE IN CIVIL STRUCTURE NEAR PENSTOCK BLOCKING WATER FLOW.	20-Mar-2025
15	BURLA HPS/HIRAKUD I	ODISHA	OHPC	7	37.5	ABNORMAL SOUND FROM SLIP RING AREA	18-Sep-2025
16	BALIMELA HPS	ODISHA	OHPC	4	60	ANNUAL MAINTENANCE	25-Oct-2025
17	INDRAVATI	ODISHA	OHPC	4	150	MIV REPLACEMENT WORK	12-Dec-2025
18	TASHIDING	SIKKIM	DANS	2	48.5	ANNUAL OVERHAULING	15-Dec-2025
19	RENGALI HPS	ODISHA	OHPC	2	50	ANNUAL MAINTENANCE	29-Dec-2025
20	CHUZACHEN	SIKKIM	GATI	1	55	ANNUAL MAINTENANCE	01-Jan-2026
21	RANGIT HPS	SIKKIM	NHPC	2	20	EARLIER OUT DUE MIV SEAL REPLACEMENT, TAKEN UNDER ANNUAL OVERHAULING FROM 05/01/2026	02-Jan-2026
22	TASHIDING	SIKKIM	DANS	1	48.5	WATER LEAKAGE FROM PENSTOCK (UPSTREAM SIDE)	06-Jan-2026

4.3. Long outage report of transmission Element (MORE THAN 01 WEEK) (As on 13.01.2026):

Transmission Element / ICT	Outage From	Reasons for Outage
220/132 KV 100 MVA ICT II AT LALMATIA	22-01-2019	220/132KV, 100MVA Transformer (NTPC side) is charged on 07.02.2024 from HV side on no load. Now, it is in idle charged condition
220KV-FSTPP-LALMATIA-I	21-04-2021	Two nos. of tower collapsed on 29.05.2024 near to Lalmatia GSS in the Loc. No. 246 &

		247.
220KV S/C Farakka-Lalmatia Transmission Line is in anti theft charging condition		
from Loc no 248 (Lalmatia end) to Loc no 33.Foundation ,erection and stringingin progress from loc 1 to 32		
132KV-BARHI-RAJGIR-1	25-03-2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earthwire between tension tower no. 218-237 in same line.The lines from Barhi(DVC) will be terminated at Barachatti(BH) and new line to be constructed from Barachatti to Rajgir(BH).
132KV-NALANDA-BARHI(DVC)-1	25-03-2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earthwire between tension tower no. 218-237 in same line.The lines from Barhi(DVC) will be terminated at Barachatti(BH) and new line to be constructed from Barachatti to Nalanda(BH).
400KV-RANGPO-TEESTA-V-1	04-10-2023	Tower near gantry of Teesta V HEP collapsed during GLOF event in Oct 2023 also leading to damage in powerhouse. Tower subsequently erected on 15.06.2024. Teesta V HEP GIS damaged due to hill sinking on 20.08.2024. Presently, GIS under restoration and generation expected by 31.03.2026.
400KV-RANGPO-TEESTA-V-2	04-10-2023	Tower near gantry of Teesta V HEP collapsed during GLOF event in Oct 2023 also leading to damage in powerhouse. Tower subsequently erected on 15.06.2024. Teesta V HEP GIS damaged due to hill sinking on 20.08.2024. Presently, GIS under restoration and generation expected by 31.03.2026.
400KV/220KV 315 MVA ICT 1 AT NORTH KARANPURA	12-09-2024	Tripped on Differential protection
400KV/220KV 315 MVA ICT 1 AT TSTPP	01-11-2024	Tripped on PRD protection. Current status:The failed transformer has reached the vendor, the repair scope has been finalized, the PR is created, and the PO is in the advanced stage of processing. The repair, transportation, installation, and commissioning are expected

		to take about six months, with the unit likely to be available by 30.06.2026. A spare 315 MVA ICT on loan from PGCIL is being explored, and their response is awaited.
132KV-PATRATU-PATRATU-1	16-11-2024	Taken out due to Rail-way diversion and height raising work between loc 11-12, the bottom conductor of Ckt#2 has been swapped with the middle phase conductor of Ckt#1 (which was under S/D since long).
Currently Ckt-1 is anti-theft charged from DVC end.		
-As a long-term measure, DVC has sought two nos of 132kV bays at newly constructed 400/220/132kV S/S of JUSNL at Patratu (JH) for termination of the lines.		
400KV/220KV 315 MVA ICT 2 AT MEJIA-B	20-01-2025	315 MVA ICT-2 at MTPS-B got damaged while charging from 220kV GIS bay. New procurement of ICT has been taken up & installation of the same may complete by end of Mar'28.
400KV/220KV 315 MVA ICT 2 AT LATEHAR	16-04-2025	400/220 ICT - II at Latehar idle charged from 400 kV Side at 01.01.2026 at 12:22 Hrs
400KV/220KV 315 MVA ICT 2 AT KODERMA	02-06-2025	315MVA ICT-2 at KTPS tripped & got damaged due to major fire inside the tank.
One spare ICT from PGCIL Mujaffarpur has been allocated to DVC from pool spare. The ICT has already reached at site & expected to get installed by Feb'26.		
220KV-RAJARHAT-NEW TOWN(AA-II)-1	10-07-2025	Emergency shutdown for BCU replacement work at Rajarhat. Charging attempted but tripped on SOTF. B_ph cable faulty,Cable order pending ,expected by Feb 2026
400KV-DIKCHU-RANGPO-2	05-08-2025	Damaged insulator replacement work. While charging the line bus bar protection operated at Dikchu. Issue in GIS chamber of Y ph Isolator between line cb and bus 2,Powder formation inside isolator chamber ,Revival Expected by december 25 as per availability of GE person.Presently negotiation in place for offer
400KV MAIN BUS - 2 AT DIKCHU	05-08-2025	Bus bar protection operated, Issue in GIS chamber of Y ph Isolator between rangpocket 2 line cb and bus 2,Powder formation inside isolator chamber ,Revival Expected by

		december 25 as per availability of GE person. Presently negotiation in place for offer
400KV-ARAMBAGH-NEW PPSP-2	04-01-2025	Damage in GIS chamber at New PPSP ss for Arambagck. Expected in January 2026
220KV-PATNA-KHAGAUL-1	24-09-2025	LBB relay operated during rectification of DC grounding defect by M/S KRR at GSS khagaul. Earlier w.e.f 02-08-2025 12:06 Hrs, Tower No. 63 has bent significantly on one side
220KV-DALTONGANJ-LATEHAR(JUSNL)-2	23-10-2025	To avoid overloading of 400/200 kV ICT-I at Latehar
400KV/220KV 315 MVA ICT 1 AT INDRAVATI HEP	25-10-2025	Due to oil leakage from Tan delta test tap of R phase 400 kV Bushing
220KV-BIDHANNAGAR-WARIA-1	29-10-2025	To control loading of 220 kV Waria-Mejia D/C (Anti-theft charged from Waria end.)
220KV-BIDHANNAGAR-WARIA-2	29-10-2025	Initially line was opened to control line loading. In between B-phase CT Blast at Bidhannagar end. Now Line is charged as anti-theft from Waria end to control loading of 220 kV Waria-Mejia D/C.
220KV-BALIMELA-UPPER SILERU-1	21-11-2025	Idle charged from U. Sileru end. Power drawl by Odisha halted due to non-concurrence by Andhra Pradesh.
400KV-FSTPP-KHSTPP-1	02-12-2025	Reconductoring works by HTLS Conductor.
220KV-KATAPALLI-BOLANGIR(PG)-1	20-12-2025	To restrict loading of 220kV Budhipadar-Lapanga ckt-1&2 due to hotspot observed in 220kV Budhipadar-Lapanga ckt-1.
132KV-MADHEPURA (BH)-SAHARSA-1	18-12-2025	To control the line loading. Line kept idle charged from Saharsa.
400KV-BINAGURI-TALA-4	25-12-2025	H/T on voltage regulation
HVDC 800KV ALIPURDUAR (PG) Pole 4	28-12-2025	For system requirement
HVDC 800KV ALIPURDUAR (PG) Pole 3	28-12-2025	For system requirement
400KV-ALIPURDUAR (PG)-PUNASANGCHUN-1	01-01-2026	H/T on voltage regulation
132KV-RAXAUL(NEW)-PARWANIPUR-2	01-01-2026	For system requirement
132KV-RIHAND-GARWAH-1	04-01-2026	For re-sagging work of conductor at different locations
132KV-NABINAGAR-NAGARUNTARI-1	04-01-2026	For re-sagging work of conductor at different locations
220/132 KV 100 MVA ICT II AT LALMATIA	22-01-2019	220/132KV, 100MVA Transformer (NTPC side) is charged on 07.02.2024 from HV side on no load. Now, it is in idle charged condition

220KV-FSTPP-LALMATIA-I	21-04-2021	Two nos. of tower collapsed on 29.05.2024 near to Lalmatia GSS in the Loc. No. 246 & 247.
220KV S/C Farakka-Lalmatia Transmission Line is in anti theft charging condition		
from Loc no 248 (Lalmatia end) to Loc no 33.Foundation ,erection and stringingin progress from loc 1 to 32		
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132KV-NALANDA-BARHI(DVC)-1	25-03-2023	Dismantling of tower no. 227, 228, and 229 crossing the premises of Mahabodhi Cultural centre along with Destraining of conductor of both circuits and Earthwire between tension tower no. 218-237 in same line.The lines from Barhi(DVC) will be terminated at Barachatti(BH) and new line to be constructed from Barachatti to Nalanda(BH).

Transmission licensees/Utilities are requested to update expected restoration date & work progress regarding restoration regularly to ERPC/ERLDC on monthly basis by 5th of each month so that status of restoration can be reviewed in OCC. Utilities are also requested to update outage of any elements within their substation premises like isolator/breaker to ERPC/ERLDC regularly. (Reported as per Clause 5.2(e) of IEGC).

Members may note.

Deliberations in the meeting

Members noted.

4.4. Commissioning of new units and transmission elements in Eastern Grid in the month of December-2025.

The details of new units/transmission elements commissioned in the month of December-2025 based on the inputs received from beneficiaries:

Generating Units							
Sl. No	Location / Pooling	Owner/Unit Name	Unit No/Source	Capacity added (MW)	Total/Installed Capacity	Date	Remarks

o	Station				(MW)		
NIL							
Interconnecting/Generator/Station Transformers							
Sl. No	Agency/ Owner	Sub-Station	ICT/GT/ST No.	Voltage Rating (kV)	Capacity (MVA)	Date	Remarks
1	SJVN	Buxar TPP	ST-02	400/11	110	02-12-2025	
New transmission Lines							
Sl. No	Agency/ Owner	Transmission Line Name		Line Length (km)	Conductor Type	Date	Remarks
1	NKTL	400KV-GAYA-NORTH KARANPURA-2		111.68	ACSR Moose	29-12-2025	
2	NKTL	400KV-GAYA-NORTH KARANPURA-1		111.68	ACSR Moose	25-12-2025	
3	PGCIL ER-II	400KV-ALIPURDUAR (PG)-PUNASANGCHUN-1		145.5	145.50 kms (Quad moose-64 Kms + Twin Moose - 81.50 Kms)	16-12-2025	
Anti-theft charging of new transmission Lines							
Sl. No	Agency/ Owner	Transmission Line Name		Line Length (km)	Conductor Type	Date	Remarks
NIL							
LILO/Re-arrangement/Re-conductoring of Existing Transmission Lines							
Sl. No	Agency/ Owner	Transmission Line Name {LILO at/ Re-arrangement of/Reconductoring}		Line Length (km) after LILO/ Re-arrangement	Conductor Type	Date	Remarks

1	BSPTCL	132KV-CHANDAULI-KARMANASA-2	17.7	Panther	07-12-2025	After the LILO of 132 kV Sahupuri-Karmnasa ckt-2 at Chandauli (UP)
Bus / Line Reactors						
Sl. No	Agency/Owner	Element Name	Voltage Level (kV)	Rating (MVar)	Date	Remarks
1	PGCIL ER-II	125MVAR 400KV B/R-5 AT DURGAPUR	DURGAPUR	125	01-12-2025	
HVDC / AC Filter bank / FACTS DEVICE associated System						
Sl. No	Agency/Owner	Element Name	Substation	Voltage Level (kV)	Date	Remarks
NIL						
Bus /Bay						
Sl. No	Agency/Owner	Element Name	Substation	Voltage Level (kV)	Date	Remarks
1	SJVN	400KV TIE BAY OF (ST-2 and ICT-2) AT BUXAR TPP	BUXAR TPP	400	02-12-2025	
2	SJVN	400KV MAIN BAY OF 110 MVA ST-02 AT BUXAR TPP	BUXAR TPP	400	02-12-2025	
3	PGCIL ER-II	400KV MAIN BAY OF 125MVAR B/R-5 AT DURGAPUR	DURGAPUR	400	01-12-2025	

Members may note.

Deliberations in the meeting

Members noted.

4.5. UFR operation during the month of December 2025

Frequency profile for the month as follows:

MONTH	MAX	MIN	% LESS IEGC BAND	% WITHIN IEGC BAND	% MORE IEGC BAND
	(DATE/TIME)	(DATE/TIME)			
December 2025	50.42 (on 13-Dec-25 at 06:03 Hrs.)	49.47 (on 06-Dec-25 at 17:45 Hrs.)	6.0	77.5	16.5

Hence, no report of operation of UFR has been received from any of the constituents.

Members may note.

Deliberations in the meeting

Members noted.

The meeting ended with a vote of thanks to the chair.

ANNEX-A
Participants in 235th OCC Meeting

Time :10:30 Hrs

Date:21.01.2026

Venue : ERPC Conference Hall, Kolkata

Sl No.	Name	Designation	Organisation
1	N.S.Mondal	Member Secretary	ERPC
2	S.Banerjee	Executive Director	ERLDC
3	R.K.Meena	S.E (O)	ERPC
4	P.K. De	S.E (C)	ERPC
5	S Chandra De	CGM	ERLDC
6	D.Biswas	GM	ERLDC
7	B.Achari	DGM	ERLDC
8	Chandan Mallick	CM(SO)	ERLDC Grid India
9	Alok Pratap Singh	CM(SO)	ERLDC Grid India
10	Manas Kr. Mallick	DGM (MO)	ERLDC Grid India
11	Rakesh Kr. Pradhan	CM (SO)	ERLDC Grid India
12	Jayathram R.F	DGM (EM)	NTPC Farakka
13	G.Baskar	AGM/EMD	NTPC Kahalgaon
14	M.Srinadh	GM	NTPC Kahalgaon
15	K.Sureshkumar	AGM (EHD)	PVUNL
16	Nishant Kumar	Sr. M	PVUNL
17	Shouvik Banerjee	CE,SLDC, WB	WBSETCL
18	R. Das	CE, CPD	WBSETCL
19	Sanjib Roy	ACE,SLDC,WB	WBSETCL
20	Utsab Aditya	SE,SLDC, WB	WBSETCL
21	Arnab Basu	DE,SLDC, WB	WBSETCL
22	Rajat Kr. Koley	SM (OS)	WBPDCL
23	Manoj Podder	AGM (SO)	WBPDCL
24	Debarshi De	DGM (SO)	CESC
25	Sumeet Narang	Sr. Manager	NTPC
26	Manas Guha	DGM (ALDC)	DPL
27	Palash Sen	SM (Comm)	DPL
28	Shiladitya Chatterjee	A.M (P.T)	APNRL
29	Tridib Mondal	DM (E & P)	IPCL
30	Arindam Kundu	A.M (O & M)	IPCL
31	Rahul Dev Banerjee	Group Head (EMD)	Maithon Power Limited
32	Sudipta Chowdhury	Sr. Manager (EMD)	JSW Energy Limited
33	Raju Kachhap	Sr. Manager	SLDC,Ranchi
34	Sailesh Prakash	DGM (OP) SLDC	SLDC,Ranchi
35	Umesh	GM ,SLDC	SLDC,Ranchi
36	Tuneshwar Kumar	Sr. Manager	STU,JUSNL
37	Rajendra Prasad	EEE	TVNL
38	Himanshu Kumar	EEE	SLDC,BSPTCL
39	Debi Prasad Kar	GMCE	SLDC, Odisha
40	Sanjaya Kr Mishra	Sr. GM	SLDC, Odisha
41	Sanjay Kr. Sharma	Sr. GM /HOD	DVC LDC

Participants in 235th OCC Meeting

Time :10:30 Hrs

Date:21.01.2026

Venue : ERPC Conference Hall, Kolkata

Sl No.	Name	Designation	Organisation
42	Santosh .Kr. Pande	Manager	DVC LDC
43	Sanjai Kr. Singh	Ch.Manager	PowerGrid ER-1
44	Mohsin Raza	DY.GM	PowerGrid ER-1
45	Partha Ghosh	DGM	ER-II
46	A.Chatterjee	AD-I	ERPC
47	B.S.Ray	DD	ERPC
48	I.K.Mehra	Director	ERPC
49	Manjunath .M	DD	ERPC
50	Mithun Gayen	Manager	PowerGrid ER-2
51	Anik Sen	Dy.Manager	PowerGrid ER-3
52	Debarati Basu	ED	PRDC
53	Sourav Bera	P.M	TCS, AMR
54	Jibanalal Mallick	S.B	WBSEDCL
55	Samim Mondal	A.M	ERLDC
56	Ankur Jain	C.M	ERLDC

Participants joined ONLINE				
Name	First Join	Last Leave	In-Meeting Duration	Email
ERPC Kolkata	1/21/26, 10:25:09 AM	1/21/26, 2:37:40 PM	4h 12m 30s	ERPC@KolkataMST.onmicrosoft.com
Raut Pravin (External)	1/21/26, 10:25:41 AM	1/21/26, 2:37:40 PM	3h 59m 37s	rautpv@tatapower.com
Satyajit Parida (External)	1/21/26, 10:25:41 AM	1/21/26, 1:32:20 PM	3h 6m 38s	SATYAJITPARIDA@NTPC.CO.IN
SLDC,ODISHA (Unverified)	1/21/26, 10:25:42 AM	1/21/26, 11:19:25 AM	53m 42s	
AVINASH SHUKLA (External)	1/21/26, 10:25:47 AM	1/21/26, 12:45:52 PM	2h 18m 17s	AVINASHSHUKLA@NTPC.CO.IN
Surya Pratap Rath, AGM OPTCL (Unverified)	1/21/26, 10:27:03 AM	1/21/26, 2:37:40 PM	4h 10m 36s	
Kumar Ashwini [Powerlinks] (External)	1/21/26, 10:29:21 AM	1/21/26, 11:40:15 AM	1h 10m 53s	ashwinikumar@tatapower.com
Ashok Singh HGIEL (Unverified)	1/21/26, 10:29:51 AM	1/21/26, 11:00:12 AM	30m 20s	
Sumeet (Unverified)	1/21/26, 10:31:03 AM	1/21/26, 2:37:40 PM	4h 6m 37s	
Somnath Chatterjee (External)	1/21/26, 10:31:19 AM	1/21/26, 2:26:39 PM	3h 55m 19s	schatterjee@tatapower.com
G Narasimha Naik {जी नरसिम्हा नेइक} (External)	1/21/26, 10:31:27 AM	1/21/26, 11:00:09 AM	28m 42s	gnarasimhanaik@powergrid.in
PD NHAI KORAPUT (Unverified)	1/21/26, 10:33:43 AM	1/21/26, 11:00:04 AM	26m 20s	
Chilakalapalli Mohana Rao {सी एच मोहन राव} (External)	1/21/26, 10:33:44 AM	1/21/26, 11:08:13 AM	34m 29s	mohan.rao@powergrid.in
Deepak Swaroop (Unverified)	1/21/26, 10:33:53 AM	1/21/26, 10:54:55 AM	21m 1s	
Satyadeep Rai	1/21/26, 10:34:09 AM	1/21/26, 12:42:23 PM	2h 4m 50s	
Bimal (Unverified)	1/21/26, 10:34:10 AM	1/21/26, 2:09:19 PM	3h 35m 8s	
Singh Parminder (External)	1/21/26, 10:34:19 AM	1/21/26, 11:47:48 AM	1h 13m 28s	parminder.singh@tatapower.com
SGM(CP), OPTCL (Unverified)	1/21/26, 10:34:31 AM	1/21/26, 1:12:20 PM	2h 37m 49s	
Rongnichu HEP (Unverified)	1/21/26, 10:35:21 AM	1/21/26, 2:37:40 PM	4h 2m 19s	
piyush jain	1/21/26, 10:35:53 AM	1/21/26, 11:00:16 AM	24m 22s	
UTTAM KUMAR (External)	1/21/26, 10:39:27 AM	1/21/26, 12:16:21 PM	1h 36m 54s	uttamkumar@dvc.gov.in
Ranjan Kumar Biswal {रंजन कुमार बिस्वाल} (External)	1/21/26, 10:40:16 AM	1/21/26, 2:37:40 PM	3h 57m 23s	ranjankumar@powergrid.in
Sangram Keshari Bhoi {संग्राम केशरी भोई} (External)	1/21/26, 10:40:36 AM	1/21/26, 1:58:16 PM	3h 17m 39s	sangramkbhoi@powergrid.in
Tanmaya Kumar Pattanayak {तन्मय कुमार पटनायक} (External)	1/21/26, 10:43:41 AM	1/21/26, 11:11:10 AM	27m 29s	tanmaya@powergrid.in
Sandip Lenka Nayagarh (Unverified)	1/21/26, 10:44:27 AM	1/21/26, 2:37:40 PM	3h 53m 13s	
Saswati Sarkar {सास्वती कुंडू} (External)	1/21/26, 10:46:10 AM	1/21/26, 11:11:34 AM	25m 24s	saswati.kundu@powergrid.in
Partha Ghosh {पार्थ घोष} (External)	1/21/26, 10:46:17 AM	1/21/26, 11:13:25 AM	27m 7s	partha.ghosh@powergrid.in
SHWETA YADAV (External)	1/21/26, 10:48:57 AM	1/21/26, 2:37:40 PM	3h 48m 43s	SHWETAYADAV@NTPC.CO.IN
Shabari Pramanick (External)	1/21/26, 10:49:05 AM	1/21/26, 2:37:40 PM	3h 48m 35s	shabari.pramanick@erldc.onmicrosoft.com
Saurabh Vijay Agarwal (External)	1/21/26, 10:50:16 AM	1/21/26, 1:44:47 PM	2h 54m 31s	saurabhvagarwal@erldc.onmicrosoft.com
Debadatta Padhy {देवदत्त पाढी} (External)	1/21/26, 10:53:54 AM	1/21/26, 2:37:40 PM	3h 43m 46s	debdatta.padhy@powergrid.in
SANJEEV RANJAN (External)	1/21/26, 10:54:57 AM	1/21/26, 12:16:23 PM	1h 21m 26s	SANJEEVRANJAN@NTPC.CO.IN
prahlada bhola (Unverified)	1/21/26, 10:56:10 AM	1/21/26, 1:11:33 PM	2h 15m 23s	
V Premkumar (External)	1/21/26, 10:57:13 AM	1/21/26, 2:37:40 PM	3h 40m 26s	VPREMKUMAR@NTPC.CO.IN
Ankit Jain (External)	1/21/26, 10:59:16 AM	1/21/26, 10:59:59 AM	42s	ankitjain@erldc.onmicrosoft.com
GRIDCO (Unverified)	1/21/26, 11:02:13 AM	1/21/26, 12:10:34 PM	1h 8m 21s	
Gitesh Patel (External)	1/21/26, 11:04:12 AM	1/21/26, 11:07:58 AM	3m 45s	giteshpatel@erldc.onmicrosoft.com
Laldhari Kumar (External)	1/21/26, 11:04:36 AM	1/21/26, 1:58:21 PM	2h 53m 45s	laldhari@erldc.onmicrosoft.com
Nishant Kumar Shankwar	1/21/26, 11:05:51 AM	1/21/26, 2:17:04 PM	3h 11m 12s	Nishant.Kumar@energy-sel.com
Dipankar Padhi (External)	1/21/26, 11:22:01 AM	1/21/26, 12:45:09 PM	1h 23m 7s	DPADHI@NTPC.CO.IN
Ponda tudu (Unverified)	1/21/26, 11:22:32 AM	1/21/26, 2:10:55 PM	2h 48m 22s	

Participants joined ONLINE				
Name	First Join	Last Leave	In-Meeting Duration	Email
PRABHAT KUMAR -AVP-SPTL (Unverified)	1/21/26, 11:26:00 AM	1/21/26, 2:37:40 PM	3h 11m 39s	
amit kumar (Unverified)	1/21/26, 11:27:38 AM	1/21/26, 11:56:58 AM	29m 20s	
shahi raza (Unverified)	1/21/26, 11:30:54 AM	1/21/26, 11:56:06 AM	25m 11s	
SLDC,ODISHA (Unverified)	1/21/26, 11:30:54 AM	1/21/26, 2:37:40 PM	3h 6m 45s	
R K (Unverified)	1/21/26, 11:38:00 AM	1/21/26, 11:44:11 AM	6m 11s	
R K (Unverified)	1/21/26, 11:42:19 AM	1/21/26, 11:44:42 AM	2m 23s	
shabari pramanick	1/21/26, 11:48:05 AM	1/21/26, 2:08:14 PM	1h 7m 50s	
Shanker choudhry	1/21/26, 11:50:59 AM	1/21/26, 2:08:57 PM	2h 17m 57s	
RK (Unverified)	1/21/26, 12:01:41 PM	1/21/26, 1:17:28 PM	1h 15m 47s	
Priya JUSNL	1/21/26, 12:18:09 PM	1/21/26, 12:35:36 PM	17m 26s	
Pranab Kumar Nayak (Unverified)	1/21/26, 12:26:02 PM	1/21/26, 12:30:20 PM	4m 17s	
ULDC, JUSNL (Unverified)	1/21/26, 12:28:01 PM	1/21/26, 2:37:40 PM	2h 9m 38s	
Pranab Kumar Nayak (Unverified)	1/21/26, 12:35:07 PM	1/21/26, 1:09:35 PM	34m 28s	
P P Jena	1/21/26, 12:56:14 PM	1/21/26, 2:37:40 PM	1h 41m 25s	
Kumar Sanjay (External)	1/21/26, 1:08:40 PM	1/21/26, 2:27:00 PM	1h 18m 20s	sanjayk@tatapower.com
ULDC (Unverified)	1/21/26, 1:42:53 PM	1/21/26, 2:37:40 PM	54m 47s	

Annex B.2.16



सत्यमेव जयते

Guidelines for Model Quality Assurance Plan (MQAP) for Major Equipment of Power Sector



**भारत सरकार
Government of India
विद्युत मंत्रालय
Ministry of Power
केन्द्रीय विद्युत प्राधिकरण
Central Electricity Authority**

January 2026

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1. Background

Quality Assurance aims to keep the quality of a product/ equipment above a specified level and ensures that the product not only meets all customer requirements but also meets the suppliers' objectives and targets. Further, following a good Quality Assurance Plan (QAP) reduces the chance of costly mistakes and mitigates risk. This is specially required for Power Sector for ensuring quality, reliable and economic power supply as it serves billions of people on a daily basis besides serving commercial and industrial sectors in their contribution to economic growth of country. To achieve this, manufacturers and utilities rely on QAP. However, it has often been observed that there is variance in QAPs being followed by different manufacturers, utilities, etc. suiting their requirements. It appears that there is a need for uniform guidelines for QAPs which shall be followed by all stakeholders throughout the country. Further, these Model QAPs (MQAPs) shall be efficient in terms of number and type of tests being performed during the entire cycle from raw material/ sub-component sourcing to production to their commissioning so as to reduce unnecessary testing, avoid redundancy at various stages and consequently save time and cost of the project.

2. Quality Assurance Plan (QAP): Introduction

Quality of an equipment can be improved by taking effective steps at the initial stage itself, which include 'use of high quality raw materials' and 'improved manufacturing processes'. The performance of an equipment largely depends on the excellence of design. However, all good designs may not yield good end product unless they are well supported by good materials, good and healthy machines and skilled workmen (operators)/ workmanship. The formulation and implementation of an efficient QAP comprises of a properly designed quality assurance program along with proper documentation at every stage, an efficient quality assurance test and finally implementation of quality assurance plans.

3. Broad Categories of Tests conducted on E&M Equipment in Power Sector

3.1 Type Tests:

These tests are conducted for validation of design i.e. for confirming the main and basic design expectation of the equipment. These tests are done mainly on a prototype unit and not on all manufactured units in a lot.

This is a test conducted on an equipment, which is representative of other equipment, to demonstrate that these comply with specified requirements not covered by routine tests. It needs to be noted here that an equipment is considered to be representative of others, if it is fully identical in rating and construction, but the Type Test may also be considered valid, if it is made on an equipment which has minor deviations of rating or other characteristics. These deviations should be subject to agreement between the manufacturer and the purchaser.

The Type Tests are conducted as per relevant applicable Standard.

3.2 Routine Tests:

These tests are conducted mainly for confirming the operational performance of the individual unit in a production lot.

These are at present conducted as per relevant applicable Standard.

3.3 Special Tests:

These are the tests other than a Type Test and/ or Routine Test as agreed by the manufacturer and the purchaser in the contract.

These tests are done as per customer requirement to obtain information useful to the user during operation or maintenance of the equipment.

These are conducted as per relevant applicable Standard (as specified under Contract or to meet certain requirement of Contract).

3.4 These above mentioned tests are further classified on the basis of the stage at which they are conducted, either at the works of the contractor or at site as given below:

- (i). The Type Tests/ Routine Tests/ Special Tests as the case may be, if conducted at the works of the manufacturer before Manufacture Dispatch Clearance Certificate (MDCC) of the equipment from the works to the site are called as Factory Acceptance Tests (FAT). The objectives of FAT are generally as follows:

- a) Factory Acceptance Test (FAT)

A FAT helps to verify that newly manufactured and packaged equipment meets its intended purpose. The FAT validates the operation of the equipment and makes sure the customers' purchase order specifications

and all other requirements have been met.

This is a test that runs on the equipment or components before it is delivered to its intended destination.

FATs are usually done at the factory works to make sure that certain requirements are met. Further, these tests are normally done with the customer, and also, in certain more demanding cases, with a third party inspection agency.

These are conducted as per relevant applicable Standard.

- (ii). The Routine tests/ Special tests as the case may be, if conducted at the site during erection and commissioning are called as Site Acceptance Tests (SAT). The objectives of SAT are generally as follows:

- a) Site Acceptance Test (SAT)

These tests are done to make certain that specified quality is met by the equipment at site and to offer customers quality testing and documents, and also further ensuring that nothing has changed or was damaged during shipment from supplier premises, storage and its installation.

SATs are done at the specific places where usually commissioning is done.

While FATs can use simulations to demonstrate as to how the equipment will function in its future operating environment, it may be difficult to mimic the actual operating conditions.

SATs are only conducted after the FAT has taken place and the item/ equipment has been delivered. This is the only way to truly determine whether a piece of new equipment will function as required.

These are conducted as per relevant applicable Standard.

4. Purpose of Uniform Guidelines for Model Quality Assurance Plan (MQAP):

There is an imperative need to have convergence in the preparation of Model Quality Assurance Plan (MQAP) as this will avoid variance among different stakeholders (manufacturers/ purchaser/ utilities, etc.), significantly save time & money and reduce burden on suppliers & testing facilities and cost to the end consumers by removing redundancies in testing procedures. Therefore, there is an urgent requirement to frame guidelines to make a uniform MQAP.

The major objective behind publication of these guidelines for MQAP of major E&M

equipment of Power Sector are broadly as given below:

- (i) **To prepare Model QAPs, which are efficient and simultaneously have overall coverage of tests.** Unnecessary and repetitive tests at different stages would be optimized and the tests for only important/ major equipment would be included in these MQAPs. The sample size for testing shall also be standardized in the MQAPs. This would help in saving cost, time and conflicts for not only the vendors/ manufacturers but also the purchaser/ consumer.
- (ii) **To moderate/ optimize Customer Hold Points (CHPs) for non-critical tests.** For this purpose, model QAPs are being framed and accordingly those tests are identified for which only the Test Certificates (TCs) shall be sufficient and acceptable upon its review, thereby reducing the overall number of CHPs in inspections.
- (iii) To bring the convergence in QAPs being prepared by various manufacturers or approved by utilities and **set uniform guidelines of MQAP for all major Electro-Mechanical equipment manufacturers/ suppliers and purchasers in the country.**
- (iv) **To avoid overloading of testing facilities.** This shall be achieved in conjunction with the guidelines for the validity period of Type Test(s) for major E&M equipment of Power Sector. Specifying the validity of Type Test(s) and removal of repetition of non-critical tests as per applicable Standards and optimizing CHPs in MQAPs would reduce the burden on the existing testing facilities leading to reduction in waiting time and henceforth optimal utilization of testing facilities which are available with limited number of laboratories.

5. Broad Guidelines for Model QAP of Power Sector

- (i) All the utilities/ purchaser/ developers/ suppliers in Power sector shall prepare and follow the Model QAPs (enclosed as **Annexure A-** for equipment of Substations, Switchyards and Transmission Lines, **Annexure B** – for major equipment in Hydro Power Plant, **Annexure C-** for major equipment in Thermal Power Plant & **Annexure D-** for major equipment used in Distribution System).
- (ii) MQAPs of those E&M equipment which have not been covered in the **Annexures** shall be subject to the terms of mutual agreement between manufacturer and purchaser.
- (iii) Test Certificates (TCs) of Bought-out items establishing the purchaser's

requirement upon review shall be acceptable to the purchaser.

- (iv) Any change in practice or acceptance norms (with reference to various tests / parameters in respective National / International standard) would be suitably incorporated by manufacturer from time to time and submit the same for approval of purchaser / utility.
- (v) These guidelines of MQAPs specify the maximum numbers of CHPs and highest limit of sampling size that could be included in the specification by the developer/ purchaser while conforming to the applicable Standard.
- (vi) In order to save time, both the supplier and the purchaser shall be free to identify lesser number of tests as CHPs as per specific needs of purchaser subject to the agreement between them.
- (vii) The utilities/ purchaser may request for additional tests which have not been covered in MQAPs of listed E&M equipment, at their own cost. However, the utilities should refrain from making it a regular practice. The utility must clearly specify as to which additional test(s) would be required during the awarding of the contract and quotation for such tests should be invited separately in the price bid. In such case, the utility shall provide extra time for repetition of such test(s). In case there is change in the governing regulations/ applicable standards, the supplier would conduct the required test at his own cost.
- (viii) The purchaser reserves the right to get the test conducted again at the supplier's works/ accredited laboratory at supplier's cost in case the Test Certificates are not establishing/ meeting the purchaser's requirement.
- (ix) The TCs at initial/ raw material/ in-process stage by customer shall be prepared and maintained by the supplier. Further, the same shall be considered as accepted only after its acceptance by the purchaser upon review at appropriate stage of CHP.
- (x) The quantum of checks shall be strictly kept as specifically quantified in the column of 'Quantum of Check' and shall be governed by the document referred under column of 'Acceptance Norms' in QAP. However, wherever the quantum of check has been mentioned as 'sample per lot' (except Distribution sector), the same shall be minimum 2 numbers or 5% of the lot whichever is higher except in the case of Type Tests.
- (xi) No material shall be dispatched from the manufacturer's works before the same is accepted, subsequent to pre-dispatch final inspection including verification of

records of all previous tests/ inspections by Purchaser's authorised Representative and duly authorised for dispatch. However, the purchaser may waive-off inspection depending upon the requirement. In case material is dispatched from manufacturer's works before the same is accepted by Purchaser's authorized representative, a separate procedure to resolve such cases may be made in the contract document.

- (xii) Manufacturer shall carry out all tests/ inspection required to establish that the items/ equipment conform to requirements of the specification and the relevant codes/ Standards specified in the specification, in addition to carrying out tests as per the approved quality plan.
- (xiii) Quality approval of the results of the tests and inspection will not, however, prejudice the right of the Purchaser to reject the equipment if it does not comply with the specification, when erected or does not give complete satisfaction in service and the above shall in no way limit the liabilities and responsibilities of the manufacturer in ensuring complete conformance of the materials/ equipment supplied to relevant specification, standard, data sheets, drawings (approved by the Purchaser), and minutes of various meetings with customer / Purchaser etc.
- (xiv) Any repair/ rectification procedures to be adopted to make the job acceptable shall be subject to the approval of the Purchaser/ authorised representative.
- (xv) All inspection, measuring and test equipment used by manufacturer shall be calibrated periodically as per NABL guidelines, depending on its use and criticality of the test/ measurement to be done. The manufacturer shall maintain all the relevant records of periodic calibration and instrument identification, and shall produce the same for inspection by purchaser. In case repair is carried out in the measuring and test equipment it should be compulsorily recalibrated. All calibrated measuring and test equipment must be properly sealed after calibration to stop any kind of manipulation with the equipment. Wherever mutually agreed between manufacturer & purchaser, the manufacturer shall re-calibrate the measuring and test equipment in the presence of the Inspector.
- (xvi) The latest CEA (Technical Standards for Construction of Electrical Plants and Electric Lines) Regulations shall also be complied with for ensuring the minimum technical requirements for construction of Generating Stations, Transmission & Distribution Lines and Substations.

ANNEXURE A

MODEL QUALITY ASSURANCE PLAN (MQAP) FOR EQUIPMENT OF SUBSTATION, SWITCHYARDS AND TRANSMISSION LINES

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11.	Model QAP of Surge Arrestor	A-210
12.	Model QAP of Transformers/ Reactors	A-227

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W -
Witness Actual testing, verify and accept

Bus Post Insulator

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of Check	Quantum of Check / Sampling with basis	Reference Document for Testing	Acceptance Norms	Format of Record	Category of Responsibility		
							Sub-Vendor	Manufacturer	Customer
1. RAW MATERIAL INSPECTION									
1.1	Alumina								
(a)	Visual Appearance	Physical	100 gm/bag from 5 % bags/ Consignment/ As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Residue on 63 µm	Physical					P	V	-
(c)	Chemical Analysis	Chemical					P	V	-
1.2	Feldspar								
(a)	Visual Appearance	Physical	100 gm/bag from 5 % bags/ Consignment/ As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Sieve Residue on 63 µm	Physical					P	V	-
(c)	Chemical Analysis	Chemical					P	V	-
1.3	Clays								
(a)	Visual Appearance	Physical					P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(b)	Appearance of Fired sample	Physical	Approx 30 kgs sample from 5% of bags / Consignment/ each type clay/ As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(c)	Sieve Residue on 38 microns	Physical		Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(d)	Moisture of Extrusion	Physical					P	V	-
(e)	Dry Shrinkage	Physical					P	V	-
(f)	Fired shrinkage	Physical					P	V	-
(g)	Dry bending strength	Physical					P	V	-
(h)	Warpage	Physical					P	V	-
(i)	Water absorption	Physical					P	V	-
(j)	Chemical Analysis	Chemical					P	V	-
2.	Glaze Materials								
2.1	Quartz Powder								
(a)	Visual Appearance	Physical	100 gm/bag 5% of bags/ consignment/ As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Residue on 63 µm	Physical					P	V	-
(c)	Chemical Analysis	Chemical					P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

2.2	Manganese Carbonate/ Iron Oxide/Chromium Oxide								
(a)	Visual Appearance of fired sample	Physical	100 gms / Individual Bags/ As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Sieve Residue (63 microns)	Physical					P	V	-
(c)	Glaze Color and Appearance of fired sample.	Physical					P	V	-
(d)	Chemical Analysis								
	1) Loss on ignition	Chemical					P	V	-
	2) Purity	Chemical					P	V	-
2.3	<u>Wollastonite</u>								
(a)	Visual Appearance	Physical	100 gm/bag, 5% of bags/ consignment/ As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Appearance of fired sample.	Physical					P	V	-
(c)	Sieve Residue (63 microns)	Physical					P	V	-
(d)	Chemical Analysis								
	1) Loss on ignition	Chemical					P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

	2) Silica & Calcia	Chemical					P	V	-
	3) Iron Oxide	Chemical					P	V	-
2.4	Dolomite								
(a)	Visual Appearance	Physical	100 gm/bag, 5% of bags from each consignment/ As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Appearance of fired sample.	Physical					P	V	-
(c)	Sieve Residue (63 microns)	Physical					P	V	-
(d)	Chemical Analysis								
	1) Loss on ignition	Chemical					P	V	-
	2) Calcia	Chemical					P	V	-
	3) Magnicia	Chemical					P	V	-
3.0	Auxiliary materials								
3.1	Cement								
(a)	Visual Appearance	Physical		IS 12269/ Plant Standard	IS 12269/ Plant Standard		P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(b)	Initial Setting time	Physical	500 gm/bag 5% of bags from each consignment/ As per Customer requirement/ Manufacturer's Standard				P	V	-
(c)	Final setting time	Physical					P	V	-
(d)	Soundness	Physical					P	V	-
(e)	Compressive Strength	Physical					P	V	-
(f)	Chemical Analysis						P	V	-
	1) Loss on Ignition	Chemical					P	V	-
(g)	Fitness Analysis	Physical							
3.2	Quartz (Assembly)								
a)	Appearance	Physical	100 gm/bag 5 bags/ lot/ As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
b)	Sieve Residue (Dry Method)	Physical					P	V	-
SECTION: 2. INCOMING INSPECTION									
1.0	Metal Parts								
1.1	MCI/ SGI Caps / B. Pins (As Cast)								

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(a)	Visual Inspection	Physical	100%	Technical Spec./ Approved Drawings	Technical Spec./ Approved Drawings		P	V	-
(b)	Dimensional Check	Physical	2/lot or As per Customer requirement/ manufacturer's Standard	IS: 1865 & Relevant Drg	Technical Spec./ Approved Drawings		P	V	-
(c)	M.P.I. Test	Physical	100%	IS 3703	No cracks		P	V	-
(d)	Tensile Load Test (Test Bar)	Physical	1 Test bar/ heat or As per Customer requirement/ manufacturer's Standard	IS: 1865	IS: 1865 Min.400N /mm ² (As per grade mentioned in drawings)		P	V	-
(e)	% Elongation (Test Bar)	Physical	1 Test bar/ heat or As per Customer requirement/ manufacturer's Standard	IS: 1865	Min.15% IS: 1865 (As per grade mentioned in drawings)		P	V	-
(f)	Hardness	Physical	1/heat or As per Customer requirement/ manufacturer's Standard	IS: 1865	Max.130-180 BHN (As per grade mentioned in drawings)		P	V	-
(g)	Chemical Analysis	Chemical	1/heat or As per Customer requirement/	IS: 1865	Phosphorous Max.0.12%		P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

			manufacturer's Standard		(As per grade mentioned in drawings)				
(h)	Microstructure (Only Caps)	Physical	1/heat or As per Customer requirement/ manufacturer's Standard	IS: 1865	IS: 1865 (As per grade mentioned in drawings)		P	V	-
(i)	Soundness Test	Physical	100 %		Ringing Sound		P	V	-
(j)	Proof Load Test	Mechanical	As per Customer requirement/ manufacturer's Standard	IS 1865	(As per grade mentioned in drawings)		P	V	-
(k)	Malleablising Test	Physical	As per Customer requirement/ Manufacturer's Standard	IS 14329	As per 14329/ (As per grade mentioned in drawings)		P	V	-
1.2	SGI Caps / B. Pins (Galvanised)								
(a)	Purity of Zinc	Chemical	As per Customer requirement/ Manufacturer's Standard	IS 209/ Technical Specification	IS 209/ Technical Specification		P	V	-
(b)	Visual Inspection	Physical	100%	IS: 2629/ 2633	Technical Spec./ Approved Drawings		P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(c)	Gauge Check	Physical	100%	Relevant Drg	Technical Spec./ Approved Drawings		P	V	-
(d)	Other Dimension	Physical	2%	Relevant Drg	Technical Spec./ Approved Drawings		P	V	-
(e)	Adhesion Test	Physical	1 sample/ Consignment or As per Customer's requirement/ Manufacturer Standard	IS: 2629	No Peeling		P	V	-
(f)	Uniformity of Zinc Coating.	Chemical	1% random or As per Customer's requirement/ Manufacturer Standard	IS: 2633	Min.4 dips or Technical Specifications/ Approved Drawings		P	V	-
(g)	Thickness of Zinc coating (Elcometer)	Physical	2% / lot	IEC 60168	Min.85 µm or Technical Specifications/ Approved Drawings		P	V	-
2.1	Corona Rings (Where Applicable)								
(a)	Visual Inspection	Physical	2 / lot or As per Customer's requirement/	As per Specifications/	No dent, scratch, breaking of welded joints.		P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

			Manufacturer Standard	Relevant Drawings					
(b)	Dimensions	Physical	IS 2500/ IL-3 AQL 0.4%	Relevant Drawing	Relevant Drawing		P	V	-
(c)	Dye Penetration Test on welded joints.	Chemical	IS 2500/ IL-3 AQL 0.4%	Technical Specifications/ IS 3658	No cracks		P	V	-
(d)	Chemical composition	Chemical	1/ lot or As per Customer's requirement/ Manufacturer Standard	Relevant Drawing/ IS 5082	Relevant Drawing		P	V	-
(e)	Mechanical load test	Physical	1/ lot or As per Customer's requirement/ Manufacturer Standard	Relevant Drawing	Relevant Drawing		P	V	-
2.2	Hardware Bolts-Nuts & Washers								
a)	Visual Inspection	Physical	As per Customer's requirement/ Manufacturer Standard	IS-6639/12427 and IS 1367 Part 3, IS 3063	Relevant Drawing/ Smooth, free from defect & no extra zinc.		P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

b)	Dimensions	Physical	As per IS: 1367 Part 17 and IS 6821	IS 6639/ 12427 and IS 1367 Part 3	Relevant Drawing		P	V	-
c)	Uniformity of Zinc Coating	Chemical	As per IS: 1367 Part 17	IS 2633	Relevant Drawing/ Withstood 4 dips of min each		P	V	-
d)	Thickness of Zinc coating by elcometer	Physical	IEC 60168	IEC 60168	Technical Specification/ Approved Drawings		P	V	-
e)	Permanent Set on spring washer	Mechanical	IS 1367- Part 17	`			P	V	-
f)	Twist Test on spring washer	Mechanical		IS 6639/ 12427 and IS 1367- Part 3			P	V	-
g)	Hardness of Bolt, Nut and Washer	Mechanical		IS 6639/ 12427 and IS 1367- Part 3, IS 3063			P	V	-
h)	Tensile Load Test (Bolt)	Mechanical					P	V	-
i)	Shear Test (Bolt) (If applicable)	Mechanical					P	V	-
j)	Proof Load Test on Bolt and Nut	Mechanical				P	V	-	
SECTION: 3. INPROCESS INSPECTION									
1.0	Body Preparation								
1.1	WEIGHTMENT								

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(a)	Weight	Physical		Manufacturer's Standard	Manufacturer's Standard			P	V
(b)	Moisture content	Physical	Weekly					P	V
1.2	BALL MILLS								
(a)	Litre Weight	Physical	Each Ball mill	Manufacturer's Standard	Manufacturer's Standard			P	V
(b)	Sieve Residue on 63 microns	Physical	Each Ball mill	Manufacturer's Standard	Manufacturer's Standard			P	V
1.3	Agitator Slurry								
(a)	Liter Weight	Physical		Manufacturer's Standard	Manufacturer's Standard			P	V
(b)	Sieve Residue on 63 microns	Physical	Each Shift					P	V
(c)	Free Iron particle	Physical	Each Shift					P	V
2.0	Glaze Preparation								
(a)	Liter Weight	Physical			Manufacturer's Standard			P	V
(b)	Sieve Residue on 38 micron	Physical	Each Batch					P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(c)	Colour & appearance of fired sample	Physical	Each cycle	Manufacturer's Standard	As per insulators fired in the same kiln			P	V
(d)	Weighment	Physical	Each Batch		Manufacturer's Standard			P	V
(e)	Glaze flow	Physical	As per Customer's requirement/ Manufacturers Standard					P	V
(f)	Density	Physical						P	V
3.0	Filter Press								
(a)	Moisture Content	Physical		Manufacturer's Standard	Manufacturer's Standard		P	V	-
4.0	<u>EXTRUSION</u>								
(a)	Pug Moisture content	Physical or As per Customer requirement/ manufacturer's standard	Each shift		Manufacturer's Standard			P	V
(b)	Free Iron particles	Physical	Weekly					P	V
(c)	De-airing check	Physical	Each shift					P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(d)	Vacuum	Physical	Each shift	Manufacturer's Standard				P	V
(e)	Physical properties	Physical	Weekly					P	V
(f)	Chemical Analysis	Chemical	Monthly					P	V
(g)	Penetrometer Reading/Stiffness	Physical	As per Customer requirement/ Manufacturer's Standard					P	V
5.0	Pug Drying								
(a)	Penetrometer Reading	Physical	Each Shift	Manufacturer's Standard	Manufacturer's Standard			P	V
(b)	Moisture Content	Physical	As per Customer requirement/ Manufacturer's Standard					P	V
6.0	Shaping								
(a)	PMR reading	Physical	As per Customer requirement/ Manufacturer's Standard	Manufacturer's Standard	Manufacturer's Standard			P	V
(b)	Dimensions	Physical	Each shift or As per Customer requirement/	Relevant Drawings	Manufacturer's Standard			P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

			Manufacturer’s Standard						
(c)	Visual check	Physical	100%	Manufacturer’s Standard	Free from defects			P	V
(d)	Covering after shaping	Physical	Random		Manufacturer’s Standard			P	V
7.0	Drying								
(a)	Moisture content	Physical		Manufacturer’s Standard	Manufacturer’s Standard			P	V
(b)	Temp. & Humidity	Physical	Every hour					P	V
(c)	Visual checks	Physical	Every item					P	V
8.0	Glazing								
(a)	Litre Weight	Physical		Manufacturer’s Standard	Manufacturer’s Standard			P	V
(b)	Density	Physical	As per Customer’s requirement/					P	V
(c)	Fluidity	Physical						P	V
(d)	Visual check before glazing	Visual						P	V
(e)	Glaze thickness	Physical						P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(f)	Visual check after glazing	Visual	Manufacturer's Standard					P	V
8.1	Gravel								
(a)	Sieve Analysis	Physical		Manufacturer's Standard	Manufacturer's Standard			P	V
8.2	Gravel Application								
(a)	Sanding Gap (Height)	Physical		Manufacturer's Standard/ relevant drawings	Manufacturer's Standard/ relevant drawings			P	V
9.0	Firing								
(a)	Temperature control	Physical	As per Customer's requirement/ Manufacturer's Standard	Manufacturer's Standard/ relevant drawings	Manufacturer's Standard/ relevant drawings			P	V
(b)	Firing schedule	Physical						P	V
(c)	Pressure Control	Physical	As per Customer's requirement/	Manufacturer's Standard	Manufacturer's Standard			P	V
(d)	Fuel oil control	Physical						P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

			Manufacturer's Standard						
10.0	Sorting (Fired Insulators)								
(a)	Visual checks	Physical							
	i) Free from cracks	Physical	100%	Manufacturer's Standard/ relevant drawings	No cracks as Applicable IS /IEC standard			P	V
	ii) Free from surface Defects.	Physical	100%		Manufacturer's Standard/ relevant drawings			P	V
	iii) Uniformity of Glaze Appearance	Physical	100%					P	V
(b)	Dimensional Check	Physical	2 Nos/ drg / Batch					P	V
(c)	Porosity	Physical	2 Nos / Kiln		Non -porous			P	V
(d)	Fired MOR (Glazed Specimen)	Physical	As per Customer's requirement/ Manufacturer's Standard	Manufacturer's Standard	Manufacturer's Standard			P	V
(e)	Bulk Density	Physical						P	V
11.0	Cutting & Grinding								
(a)	Dimensional check (Cutting length)	Physical	First piece set up approval of m/c	Manufacturer's Standard/	Manufacturer's Standard/			P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(b)	Ultrasonic Test	Physical	100%	relevant drawings	relevant drawings			P	V
12.0	<u>ASSEMBLY</u>								
(a)	Cement Cubes Test	Physical	Weekly	Manufacturer's Standard/ relevant drawings	Manufacturer's Standard/ relevant drawings			P	V
(b)	Jig verification	Physical	3 Month					P	V
(c)	Cement mortar preparation	Physical	Each shift					P	V
(d)	Curing temp. control	Physical	Each shift	Technical Specifications/ Manufacturer's Standard/ relevant drawings	Technical Specifications/ Manufacturer's Standard/ relevant drawings			P	V
SECTION: 4. TESTING AND FINAL INSPECTION									
1.0	Product Tests								
1.1	<u>ROUTINE TEST</u>								
(a)	Bending Test in four directions (50% min failing load)	Mechanical	100%	IEC-60168/ Technical Specifications/				P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

				IS-2544	Technical Specifications/ relevant drawings				
(b)	Visual Examination Checks for surface defects & glazing defects	Physical	100%	Technical Specifications/ relevant drawings				P	V
(c)	Dimensional Check	Physical	As per customer requirement/ Manufacturer's Standard	Technical Specifications/ Relevant Drawing				P	V
1.2	Acceptance Test								
(a)	Visual Check	Physical	IS/IEC- 60168, IEC-60273 & IS-2544/ 5350	Technical Specifications/ / relevant drawings	IS/IEC- 60168, IEC-60273 & IS-2544/ 5350 Technical Specifications/ / relevant drawings		-	P	W
(b)	Dimensional Check	Physical					-	P	W
(c)	Temp. Cycle Test	Physical					-	P	W
(d)	Mechanical Failing Load Test (50% in 3 direction & 100% in 4 th) Deflection under load	Physical					-	P	W
(e)	Porosity Test	Physical					-	P	W
(f)	Eccentricity & Parallelism	Physical					-	P	W
(g)	Deflection test at 20%, 50%, 70% min. bending failing	Mechanical							

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

	load on complete stack.								
(h)	Torsion Test	Mechanical					-	P	W
(i)	Galvanizing Test Mass of Zinc coating	Physical					-	P	W
(j)	(OR) Thickness of zinc Coating (Elcometer)	Chemical					-	P	W
(k)	Uniformity of Zn coating	Chemical	IEC- 60168, IEC-60273 & IS-2544/ 5350 or As per Customer requirement/ Manufacturer's Standard	Technical Specification/ IS 2633	Technical Specifications/ Approved Drawing		-	P	W
(l)	Adhesion test	Chemical		IS 2629	As per specification		-	P	W
2.0	Packing & Dispatch								
(a)	Visual Inspection	Visual	100%	Tech. spec.	Manufacturer's Drawing	Manufacturer's Drawing	-	P	-
(b)	Dimensions	Measurement	100%	Tech. spec.	Manufacturer's Drawing	Manufacturer's Drawing	-	P	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

Porcelain Long Rod Insulator

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of Check	Quantum of Check / Sampling with basis	Reference Document for Testing	Acceptance Norms	Format of Record	Category of Responsibility		
							Sub- Vendor	Manufacturer	Customer
1. RAW MATERIAL INSPECTION									
1.1	Alumina								
(a)	Visual Appearance	Physical	100 gm/bag from 5 % bags/ Consignment./ As per customer requirement/ Manufacturer's standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Residue on 63 µm	Physical					P	V	-
(c)	Chemical Analysis	Chemical					P	V	-
1.2	Feldspar								
(a)	Visual Appearance	Physical	100 gm/bag from 5 % bags/ Consignment. / As per customer requirement/ Manufacturer's standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Sieve Residue on 63 µm	Physical					P	V	-
(c)	Chemical Analysis	Chemical					P	V	-
1.3	Clays								
(a)	Visual Appearance	Physical	Approx 30 kgs sample from 5% of bags /				P	V	-
(b)	Appearance of Fired sample	Physical					P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(c)	Sieve Residue on 38 microns	Physical	Consignment/ each type clay/ As per customer requirement/ Manufacturer's standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(d)	Moisture of Extrusion	Physical					P	V	-
(e)	Dry Shrinkage	Physical					P	V	-
(f)	Fired shrinkage	Physical					P	V	-
(g)	Dry bending strength	Physical					P	V	-
(h)	Warpage	Physical					P	V	-
(i)	Water absorption	Physical					P	V	-
(j)	Chemical Analysis	Chemical					P	V	-
2.	Glaze Materials								
2.1	Quartz Powder								
(a)	Visual Appearance	Physical	100 gm/bag 5% of bags/ consignment/ As per customer requirement/ Manufacturer's standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Residue on 63 µm	Physical					P	V	-
(c)	Chemical Analysis	Chemical					P	V	-
2.2	Manganese Carbonate/ Iron Oxide/Chromium Oxide								
(a)	Visual Appearance of fired sample	Physical	100 gms / Individual Bags/ As per customer requirement/				P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(b)	Sieve Residue (63 microns)	Physical	Manufacturer's standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(c)	Glaze Color and Appearance of fired sample.	Physical					P	V	-
(d)	Chemical Analysis								
	1) Loss on ignition	Chemical					P	V	-
	2) Purity	Chemical					P	V	-
2.3	<u>Wollastonite</u>								
(a)	Visual Appearance	Physical	100 gm/bag, 5% of bags/ consignment/ As per customer requirement/ Manufacturer's standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Appearance of fired sample.	Physical					P	V	-
(c)	Sieve Residue (63 microns)	Physical					P	V	-
(d)	Chemical Analysis								
	1) Loss on ignition	Chemical					P	V	-
	2) Silica & Calcia	Chemical					P	V	-
	3) Iron Oxide	Chemical					P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

2.4	Dolomite								
(a)	Visual Appearance	Physical	100 gm/bag, 5% of bags from each consignment/ As per customer requirement/ Manufacturer's standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
(b)	Appearance of fired sample.	Physical					P	V	-
(c)	Sieve Residue (63 microns)	Physical					P	V	-
(d)	Chemical Analysis								
	4) Loss on ignition	Chemical					P	V	-
	5) Calcia	Chemical					P	V	-
	6) Magnesia	Chemical					P	V	-
3.0	Auxiliary materials								
3.1	Cement								
(a)	Visual Appearance	Physical	500 gm/bag 5% of bags from each consignment/ As per customer requirement/ Manufacturer's standard	IS 12269/Plant Standard	IS 12269/ Plant Standard		P	V	-
(b)	Initial Setting time	Physical					P	V	-
(c)	Final setting time	Physical					P	V	-
(d)	Soundness	Physical					P	V	-
(e)	Compressive Strength	Physical					P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(f)	Chemical Analysis						P	V	-
	1) Loss on Ignition	Chemical					P	V	-
(g)	Fineness Analysis	Physical	As per customer requirement/ Manufacturer's standard				P	V	-
3.2	Quartz (Assembly)								
a)	Appearance	Physical	100 gm/bag 5 bags/ lot	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		P	V	-
b)	Sieve Residue (Dry Method)	Physical	100 gm/bag 5 bags/ lot				P	V	-
SECTION: 2. INCOMING INSPECTION									
1.0	Metal Parts								
1.1	MCI/ SGI Caps / B. Pins (As Cast)								
(a)	Visual Inspection	Physical	100%	Technical Spec./ Approved Drawings	Technical Spec./ Approved Drawings		P	V	-
(b)	Dimensional Check	Physical	2/lot / As per customer	Approved Drg	Technical Spec./ Approved Drawings		P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

			requirement/ Manufacturer's standard						
(c)	M.P.I. Test	Physical	100%	IS: 3703	No cracks		P	V	-
(d)	Tensile Load Test (Test Bar)	Physical	1 Test bar/ heat/ As per customer requirement/ Manufacturer's standard	IS 1865	As per grade mentioned in approved drawings		P	V	-
(e)	% Elongation (Test Bar)	Physical	1 Test bar/ heat/ As per customer requirement/ Manufacturer's standard				P	V	-
(f)	Hardness	Physical	1/heat				P	V	-
(g)	Chemical Analysis	Chemical	1/heat		Phosphorous Max.0.12%		P	V	-
(h)	Microstructure (Only Caps)	Physical	1/heat		IS: 1865		P	V	-
(i)	Soundness Test	Physical	As per customer requirement/ Manufacturer's standard		Ringing Sound		P	V	-
(j)	Proof load test (Yield strength on test bar)	Mechanical		IS 1865	As per grade mentioned in approved drawings		P	V	-
(k)	Malleablising (For MCI)	Physical		IS 14329	IS 14329/ As per grade mentioned in approved drawings		P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

1.2	SGI Caps / B. Pins (Galvanised)								
	Purity of Zinc	Chemical	As per customer requirement/ Manufacturer's standard	IS 209/ Technical Specification	Technical Specification				
(a)	Visual Inspection	Physical	100%	IS: 2629/ 2633	Technical Spec./ Approved Drawings		P	V	-
(b)	Gauge Check	Physical	As per customer requirement/ Manufacturer's standard	Relevant Drg			P	V	-
(c)	Other Dimension	Physical	2%	Relevant Drg			P	V	-
(d)	Adhesion Test	Physical	1 sample/ Consignment/ As per customer requirement/ Manufacturer's standard	IS: 2629	No Peeling		P	V	-
(e)	Uniformity of Zinc Coating.	Chemical	1% random/ As per customer requirement/ Manufacturer's standard	IS: 2633	Min.4 dips/ Technical Spec./ Approved Drawings		P	V	-
(f)	Thickness of Zinc coating (Elcometer)	Physical	2% / lot	IEC 60168	Min.85 µm/ Technical Spec./ Approved Drawings		P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

2.1	Corona Rings (Where Applicable)								
(a)	Visual Inspection	Physical	2 / lot/ As per customer requirement/ Manufacturer's standard	As per Technical Specifications/ Approved drawings	No dent, scratch, breaking of welded joints.		P	V	-
(b)	Dimensions	Physical	IS 2500	Relevant Drawing	Relevant Drawing		P	V	-
(c)	Dye Penetration Test on welded joints.	Chemical	IS 2500	No cracks/ Technical Specifications/ IS 3658	No cracks		P	V	-
(d)	Chemical composition	Chemical	1/ lot/ As per customer requirement/ Manufacturer's standard	Relevant Drawing/ IS 5082	Relevant Drawing/ As per IS-5082/ As per Grade in approved Drawing		P	V	-
(e)	Mechanical load test	Physical	1/ lot/ As per customer requirement/ Manufacturer's standard	Relevant Drawing	Relevant Drawing		P	V	-
2.2	Hardware Bolts-Nuts & Washers								

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(a)	Visual Inspection	Physical	100 %	IS-6639/12427 and IS-1367 Part-3, IS-3063	Relevant Drawing/ Smooth, Free from defect & no extra zinc.		P	V	-
(b)	Dimensions	Physical	As per IS 1367- Part 17 and IS 6821	IS 6639 /12427 and IS-1367 Part-3	Relevant Drawing		P	V	-
(c)	Uniformity of Zinc Coating	Chemical	As per IS 1367 Part 17	IS 2633	Relevant Drawing/ Technical Spec./ Approved Drawings		P	V	-
(d)	Thickness of Zn Coating by Elcometer	Physical	As per IEC-60168	As per IEC-60168	Technical Spec./ Approved Drawings		P	V	-
(e)	Permanent Set on spring washer	Mechanical	As per IS: 1367- part-17	IS-6639/12427 and IS-1367 Part-3			P	V	-
(f)	Twist Test on spring washer					P	V	-	
(g)	Hardness of Bolt, nut and washer					P	V	-	
(h)	Tensile Load Test (Bolt)					P	V	-	
(i)	Shear Test (Bolt) (If applicable)					P	V	-	
(j)	Proof Load Test on Bolt and Nut					P	V	-	
SECTION: 3. INPROCESS INSPECTION									
1.0	Body Preparation								
1.1	WEIGHMENT								

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(a)	Weight	Physical		Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		-	P	V
(b)	Moisture content	Physical	Weekly				-	P	V
1.2	BALL MILLS								
(a)	Litre Weight	Physical	Each Ball mill	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		-	P	V
(b)	Sieve Residue on 63 microns	Physical	Each Ball mill				-	P	V
1.3	Agitator Slurry								
(a)	Liter Weight	Physical		Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		-	P	V
(b)	Sieve Residue on 63 microns	Physical	Each Shift				-	P	V
(c)	Free Iron particle	Physical	Each Shift				-	P	V
2.0	<u>Glaze Preparation</u>								
(a)	Liter Weight	Physical			Technical Specifications/ Manufacturer's Standard		-	P	V
(b)	Sieve Residue on 38 micron	Physical	Each Batch				-	P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(c)	Colour & appearance of fired sample	Physical	Each cycle	Technical Specifications/ Manufacturer's Standard	As per insulators fired in the same kiln		-	P	V
(d)	Weighment	Physical	Each Batch		Technical Specifications/ Manufacturer's Standard		-	P	V
(e)	Glaze flow	Physical	As per Customer requirement/ Manufacturer's Standard				-	P	V
(f)	Density	Physical	As per Customer requirement/ Manufacturer's Standard				-	P	V
3.0	Filter Press								
(a)	Moisture Content	Physical	As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		-	P	V
4.0	<u>EXTRUSION</u>								
(a)	Pug Moisture content	Physical	Each shift/ As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		-	P	V
(b)	Free Iron particles	Physical	Weekly				-	P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(c)	De-airing check	Physical	Each shift	Manufacturer's Standard			-	P	V
(d)	Vacuum	Physical	Each shift				-	P	V
(e)	Physical properties	Physical	Weekly				-	P	V
(f)	Chemical Analysis	Chemical	Monthly				-	P	V
(g)	Reading/Stiffness	Physical	As per Customer requirement/ Manufacturer's Standard				-	P	V
5.0	Pug Drying								
(a)	Penetrometer Reading	Physical	Each Shift	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		-	P	V
(b)	Moisture Content	Physical	As per Customer requirement/ Manufacturer's Standard				-	P	V
6.0	Shaping								
(a)	PMR reading	Physical	As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		-	P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(b)	Dimensions	Physical	Each shift/ As per Customer requirement/ Manufacturer's Standard	Relevant Drawings	Technical Specifications/ Manufacturer's Standard		-	P	V
(c)	Visual check	Physical	100%	Technical Specifications/ Manufacturer's Standard	Free from defects		-	P	V
(d)	Covering after shaping	Physical	Random	Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		-	P	V
7.0	Drying								
(a)	Moisture content	Physical		Technical Specifications/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard		-	P	V
(b)	Temp. & Humidity	Physical	Every hour				-	P	V
(c)	Visual checks	Physical	Every item			Free from visual defects	-	P	V
8.0	Glazing								
(a)	Litre Weight	Physical			Technical Specifications/		-	P	V
(b)	Density	Physical					-	P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

(c)	Fluidity	Physical	As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard	Manufacturer's Standard		-	P	V
(d)	Visual check before glazing	Visual					-	P	V
(e)	Glaze Thickness	Physical					-	P	V
(f)	Visual check after Glazing	Visual					-	P	V
9.0	Firing								
(a)	Temperature control	Physical	As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard/ relevant drawings	Technical Specifications/ Manufacturer's Standard/ relevant drawings		-	P	V
(b)	Firing schedule	Physical					-	P	V
(c)	Pressure control	Physical					-	P	V
(d)	Fuel Oil Control	Physical					-	P	V
10.0	Sorting (Fired Insulators)								
(a)	Visual checks	Physical							
	i) Free from cracks	Physical	100%	Technical Specifications/	No cracks as Applicable IS /IEC standard		-	P	V
	ii) Free from surface Defects.	Physical	100%		Technical Specifications/		-	P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

	iii) Uniformity of Glaze Appearance	Physical	100%	Manufacturer's Standard/ relevant drawings	Manufacturer's Standard/ relevant drawings		-	P	V
(b)	Dimensional Check	Physical	2 Nos/ drg / Batch	Manufacturer's Standard/ relevant drawings			-	P	V
(c)	Porosity	Physical	2 Nos / Kiln		Non -porous		-	P	V
(d)	Glazed Specimen	Physical	As per Customer requirement/ Manufacturer's Standard	Technical Specifications/ Manufacturer's Standard/	Technical Specifications/ Manufacturer's Standard/		-	P	V
(e)	Bulk Density	Physical					-	P	V
11.0	<u>Cutting & Grinding</u>								
(a)	Dimensional check (Cutting length)	Physical	First piece set up approval of m/c	Technical Specifications/ Manufacturer's Standard/ relevant drawings	Technical Specifications/ Manufacturer's Standard/ relevant drawings		-	P	V
(b)	Ultrasonic Test	Physical	100%				-	P	V
12.0	<u>ASSEMBLY</u>								
(a)	Cement Cubes Test	Physical	Weekly	Technical Specifications/ Manufacturer's Standard/ relevant drawings	Technical Specifications/ Manufacturer's Standard/ relevant drawings		-	P	V
(c)	Cement mortar preparation	Physical	Each shift				-	P	V
(d)	Curing temp. control	Physical	Each shift				-	P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

SECTION: 4. TESTING AND FINAL INSPECTION									
1.0	Product Tests								
1.1	<u>ROUTINE TEST</u>								
(a)	Mechanical Routine Test	Mechanical	100%	IEC 60383/ Technical Specification/ IS 731	IEC 60383/ Technical Specification/ IS 731		-	P	V
(b)	Visual Examination Checks for surface defects & glazing defects	Physical	100%	Technical Specifications/ relevant drawings	Technical Specifications/ relevant drawings		-	P	V
(c)	Dimensional Checks	Physical	As per Customer requirement/ Manufacturer's Standard				-	P	V
1.2	Acceptance Test								
(a)	Visual Check	Physical	IEC 60383	IEC 60383/ Technical Specifications/ Relevant drawings	IEC 60383/ Technical Specifications/ Relevant drawings		-	P	W

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	Dimensional Check	Physical	IEC 60383/IEC 60433/IS731/ 3 pcs for lot <300 nos or As per Customer requirement/	IEC 60383/ 60433/IS 731/ Technical Specifications/ relevant drawings	IEC 60383/ IEC 60433/ IS 731/ Technical Specifications/ relevant drawings				
(b)	Temp. Cycle Test	Physical					-	P	W
(c)	Mechanical Failing Load Test (50% in 3 direction & 100% in 4th) Deflection under load	Physical					-	P	W
(d)	Porosity Test	Physical					-	P	W
(e)	Eccentricity & Parallelism	Physical					-	P	W
(f)	Mechanical performance test	Mechanical					-	P	W
(g)	Warpage, Creepage, appearance check	Physical					-	P	W
(h)	Ultrasonic Test with angular probe	Physical	IEC 60383/IEC 60433/IS 731/1% of lot or As per Customer requirement/	No flaws	No flaws		-	P	W
(i)	Thermal Mechanical Performance test	Mechanical	5 samples/every 3260 nos. or As per Customer requirement/ Manufacturer's Standard	Technical Specifications	Technical Specifications		-	P	W

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(j)	Test on locking device for ball and socket coupling	Mechanical	IEC 60372/ 3 pcs for lot <300 nos or As per Customer requirement/ Manufacturer's Standard				-	P	W
(k)	Galvanizing Test Mass of zinc coating.	Physical		Technical Specifications/ relevant drawings	Technical Specifications/ relevant drawings		-	P	W
(l)	(OR) Thickness of zinc Coating (Elcometer)	Chemical			Min. 86 Microns		-	P	W
(m)	Uniformity of Zn coating	Chemical	IEC- 60168, IEC-60273 & IS-2544/ 5350	Customer specifications/ IS 2633	Withstood Min 6 dips., 1 min each		-	P	W
(n)	Adhesion test	Chemical	3 pcs/lot <300 nos. or As per Customer requirement/ Manufacturer's Standard	IS 2629	As per Customer specifications		-	P	W
1.2.1	Additional test on samples brought from site (if applicable as per Customer specification)								
(a)	Electro Mecha test	Electrical	3 samples/every 1500 nos. or As per Customer requirement/	Customer specifications	Customer specifications		-	P	W

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			Manufacturer's Standard						
2.0	Packing & Dispatch								
(a)	Visual Inspection	Visual	10%	Tech.spec.	Manufacturer's Drawing	Manufacturer's Drawing	-	P	-
(b)	Dimensions	Measurement	10%	Tech.spec.	Manufacturer's Drawing	Manufacturer's Drawing	-	P	-

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Disc Insulators

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S. No.	Components / Operations & Description of Test	Type of Check	Quantum of Check / Sampling with basis	Reference Document for Testing	Acceptance Norms	Format of Record	Category of Responsibility		
							Sub- Vendor	Manufacturer	Customer
1. RAW MATERIAL INSPECTION									
1.01	CALCINED ALUMINA POWDER								
(a)	Visual Inspection & Physical Properties	Visual/ Physical	5kgs/consignment/ 200 tones or part thereof/ As per Customer requirement/ Manufacturer’s Standard	IS9700 & Plant Standard	White or Off White powder, Retention on 300 B.S mesh-1% (Max.), Magnetic Iron- 0.01% (Max.).	Online/ TC	-	P	V
(b)	Chemical Analysis Al ₂ O ₃ Na ₂ O	Chemical		IS9700 & Plant Standard	95.0% Min. 0.10 % Max.		P/-	V/P	-/V
1.02	QUARTZ POWDER								
(a)	Visual Inspection	Visual	5kgs/consignment/ 200tonnes or part thereof/ As per Customer requirement/ Manufacturer’s Standard	Plant Std & IS 11465	White or Off White color, free from any foreign impurities./ As per plant standard	Online/ TC	-	P	V
(b)	Inspection of Fired samples (Button Test)	Visual			White or Off White color, free from any inclusion. / As per plant standard		-	P	V
(c)	Free magnetic Iron Content	Physical			0.01% Max. / As per plant standard		-	P	V
(d)	Grain Size	Physical			Retention on 300 BS Mesh - 4% Max. / As per plant standard		-	P	V
(e)	Chemical Analysis SiO ₂	Chemical			98.0% Min. / As per plant standard		P	V	-

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(f)	Sintering/ fired behaviour	Physical		Plant Standard	Should not stick to the container and no specks		P	V	-
1.03	FELDSPAR POWDER								
(a)	Visual Inspection & Physical Properties	Visual / Physical	5kgs/consignment/ 200 tonnes or part thereof/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	Off White powder with pinkish tinge. Retention on 300BS Mesh 5% Max.	Online/ TC	-	P	V
(b)	Inspection of Fired samples (Button Test)	Visual			Translucent Felspathic glass free from any foreign inclusions.		-	P	V
(c)	Chemical Analysis K ₂ O Na ₂ O K ₂ O+Na ₂ O	Chemical			A 10%Min. 4.0%Max. 4.0%Max 13%Min. 11%Min		-	P	V
(d)	Sintering/ fired behaviour	Physical			Translucent (glossy white) with low impurities		P	V	-
1.04	CHINA CLAY								
(a)	Grain Size	Physical	5kgs/consignment/ 200 tonnes or part thereof/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	Retention on 300 BS Mesh3%Max./ As per plant standard	Online/ TC	-	P	V
(b)	Fired colour	Visual			Whiter/Light Creamy Biscuit Color, free from black or brown specks		-	P	V

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(c)	Chemical Analysis L.O.I at 1000°C Al ₂ O ₃ Fe ₂ O ₃ TiO ₂	Chemical			15.0%Max. 30.0%Min. 2.0%Max. 2.0%Max. Or As per plant standard		-	P	V
1.05	BALL CLAY								
(a)	Grain Size	Physical	5kgs/consignment/ 200 tonnes or part thereof/ As per Customer requirement/ Manufacturer’s Standard	Plant Standard	Retention on 300 BS Mesh3%Max./ as per plant standard	Online/ TC	-	P	V
(b)	Free Moisture Content	Physical			Hymond Blue - 25% Max. Other Clays - 10% Max/ as per plant standard		-	P	V
(c)	Dry Shrinkage	Dimensional			8% Max. / as per plant standard		-	P	V
(d)	Total Shrinkage	Dimensional			18% Max. / as per plant standard		-	P	V
(e)	Green MOR	Physical			Min 35 Kg/cm2 / as per plant standard		-	P	V
(f)	Chemical Analysis L.O.I at 1000°C Al ₂ O ₃ Fe ₂ O ₃ TiO ₂	Chemical			One sample in every six month / As per Customer requirement/ Manufacturer’s Standard		13% Max. 23% Min. 2.5% Max. 3% Max. / as per plant standard	-	P
2. RAW MATERIAL (Glaze - Brown Color)									
2.01	Dolomite Powder								
(a)	Visual Inspection	Visual	500gms/consignm ent/20tonnes or		White in color, free from impurities	Online/ TC	-	P	V

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(b)	Inspection of Fired Samples	Visual	part thereof/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	Free from ferrogineous tint and foreign inclusions.		-	P	V
(c)	Grain Size	Physical			Retention on 200 BS Mesh 5% Max./ As per plant standard		-	P	V
(d)	Chemical Analysis Fe ₂ O ₃ CaO MgO	Chemical			0.2% Max. 28% Min. 18% Min./ As per plant standard		-	P	V
2.02	Talc Powder								
(a)	Visual Inspection	Visual	500gms/consignment/20tonnes or part thereof/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	White or Off white colour	Online/ TC	-	P	V
(b)	Fired Appearance	Visual			Free from ferrogeneous tint or foreign inclusion		-	P	V
(c)	Chemical Analysis L.O.I at 1000°C SiO ₂ Fe ₂ O ₃ CaO MgO	Chemical			6.0% Max. 58% Min. 1% Max. 5% Max. 30% Min./ As per plant standard		-	P	V
(d)	Grain Size	Physical			Retention on 200 BS Mesh 2%Max./ As per plant standard		-	P	V
2.03	Iron Oxide								
(a)	Visual Inspection	Visual	500gms/consignment/20tonnes or part thereof/ As per Customer requirement/	Plant standard	Red coloured powder	Online/ TC	-	P	V
(b)	Grain Size	Physical			Retention on 200 BS Mesh 2% Max.or 5%		-	P	V

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			Manufacturer's Standard		on 300 BS mesh/ As per plant standard				
(c)	Colored Intensity Fired	Visual			Should match std. Reference		-	P	V
(d)	Chemical Analysis Fe ₂ O ₃	Chemical			90% Min./ As per plant standard		-	P	V
2.04	Manganese Dioxide								
(a)	Visual Inspection	Visual			Black coloured powder.		-	P	V
(b)	Grain Size	Physical	500gms/consignment/20tonnes or part thereof/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	Retention on 200 BS Mesh 2% Max.or 5% on 300 BS mesh/ As per plant standard	Online/ TC	-	P	V
(c)	Color of Fired samples	Visual			Should match std. Reference		-	P	V
(d)	Chemical Analysis L.O.I at 1000°C MnO ₂ Acid Insolubles	Chemical			7% Max, 80% Min. 10% Max./ As per plant standard		-	P	V
2.05	Chromium Oxide								
(a)	Visual inspection		500gms/consignment/20tonnes or part thereof/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	Green Coloured powder	Online/ TC	-	P	V
(b)	Grain Size	Physical			Retention on 200 BS Mesh 2% Max.or 5% on 300 BS mesh/ As per plant standard		-	P	V
(c)	Fired Color	Visual			Should match std. Reference		-	P	V

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(d)	Chemical Analysis L.O.I at 1000°C Cr ₂ O ₃	Chemical			3% Max. 90% Min./ As per plant standard		-	P	V
2.06	Zircon Opacifier								
(a)	Visual Inspection	Visual	500gms/consignm ent/20tonnes or part thereof/ As per Customer requirement/	Plant standard	White or Off white coloured powder.	Online/ TC	-	P	V
(b)	Chemical Analysis ZrO ₂ SiO ₂	Chemical			25 to 40 % 50 to 70%/ As per plant standard		-	P	V
2.07	Sodium Carboxyl Methyl Cellulose (SCMC)								
(a)	Visual Inspection	Visual	500gms/consignm ent/20tonnes or part thereof/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	Buff /White /Off white coloured powder.	Online/ TC	-	P	V
(b)	Solubility Test	Visual			100% soluble in water and shall give clear solution.		P	V	-
(c)	Viscosity of 1% Solution.	Physical			330° to 360° overswing		P	V	-
(d)	Free Moisture Content	Physical			10% Max.		P	V	-
3. RAW MATERIAL (Assembly Material)									
3.01	Bituminous Paint								
(a)	Visual Inspection	Visual	Each consignment 1 litre at random from a lot of 20 drums or less/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	Black coloured liquid.	Online/ TC	-	P	V
(b)	Drying Time	Physical			8 Hours Max./ As per plant standard		-	P	V

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3.02	Rubberised cork sheet								
(a)	Specific Gravity	Physical	Avg. of 5 pcs / 5000 sheets or less/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	0.85gm/cm³ (min) ./ As per plant standard	Online/ TC	-	P	V
(b)	Thickness	Dimensional	5 pcs / 5000 sheets or less/ As per Customer requirement/ Manufacturer's Standard		1.0mm (+/- 10%)./ As per plant standard		-	P	V
3.03	Portland Cement (53 S)								
(a)	Visual Inspection	Visual	5kgs/consignment/ 50tonnes or part thereof/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	No lumpiness & Solidification	Online/ TC	-	P	V
(b)	Compressive Strength - (25mm cubes avg of 5 -10 cubes)	Mechanical			550kg/cm² Min.(2days water + 3days Air Curing) /As per plant standard		-	P	V
(c)	Optimum Water Content	Physical			26.0 to 31.0% /As per plant standard		-	P	V
(d)	Initial Setting Time	Physical			150 minutes Max /As per plant standard		-	P	V
(e)	Final Setting Time	Physical			400 minutes Max /As per plant standard		-	P	V
(f)	Auto Clave Expansion	Physical			0.2% Max. /As per plant standard		-	P	V
4. IN-PROCESS INSPECTION									
4.01	Glaze Making (Brown Glaze)								

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(a)	Glaze Flow	Physical	Each Batch as per Plant Standard	Plant Standard	40mm (Min.) /As per plant standard	Online	-	P	V
(b)	Physical Appearance	Visual			Should Match Std. Ref. Samples		-	P	V
(c)	Particle Size (< 10 micron)	Physical			65 to 85%/As per plant standard		-	P	V
4.02	Ceramic Slurry properties:								
(a)	Specific Gravity	Physical	Once in each Shift/ As per Customer requirement/ Manufacturer's Standard	Plant Standard		Online	-	P	V
	Silica Body				1.30 Min. /As per plant standard		-	P	V
	Alumina Body				1.35 Min. /As per plant standard		-	P	V
(b)	Free Iron Content	Physical	Twice a week/ As per Customer requirement/ Manufacturer's Standard	0.001% Max. /As per plant standard	-		P	V	
(c)	Particle Size	Physical	Once in a Day/ As per Customer requirement/ Manufacturer's Standard	(< 10 micron) /As per plant standard	-		P	V	
	Silica Body			65 to 75%/As per plant standard	-		P	V	
	Alumina Body			75 to 85%/As per plant standard	-		P	V	
4.03	Filter Pressing								
(a)	Moisture Content	Physical	Once in each Shift/ As per Customer requirement/ Manufacturer's Standard	Plant Standard			-	P	V
	Silica Body				15 to 25%/As per plant standard		-	P	V
	Alumina Body				15 to 25 %/As per plant standard		-	P	V
4.04	Pug Mill Blanks								
(a)	De-airing Check	Physical	Each Shift 2 to 3 nos. Slices/ As per	Plant Standard	No Lamination Crack		-	P	V

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(b)	Moisture Content	Physical	Customer requirement/ Manufacturer's Standard		16 to 24 %/As per plant standard		-	P	V
4.05	Prop. Of Unfired Porcelain								
(a)	Dry MOR	Physical	5 Nos. a day/ As per Customer requirement/ Manufacturer's Standard	Plant Standard	35kg/cm² Min. /As per plant standard		-	P	V
(b)	Dry Shrinkage	Dimensional	Twice a Week/ As per Customer requirement/ Manufacturer's Standard			7% Max/As per plant standard	-	P	V
	Silica Body				-		P	V	
	Alumina Body				-		P	V	
(c)	Total Shrinkage	Dimensional	Twice a Week/ As per Customer requirement/ Manufacturer's Standard			-	P	V	
	Silica Body				-	P	V		
	Alumina Body				-	P	V		
4.06	Shaping								
(a)	Making - Moisture Content (Wet Basis)	Physical	Each Shift	Plant Standard	Si & Al 16 to 24 % /As per plant standard	Online	-	P	V
							-	P	V
							-	P	V
(b)	Finishing Dimensional Check	Dimensional	Twice a day	As per Prod. Drg.	As per relevant Prod. Drg.		-	P	V
4.07	Drying of Green Ware								
(a)	Temp. Control	Physical	Once each Shift	Plant Standard	40°C to 120°C/As per plant standard		-	P	V
(b)	Moisture Content	Physical	Once each Shift		< 1.0%/As per plant standard		-	P	V

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4.08	Glazing (Brown Glaze)								
(a)	Density	Physical	At least Once in a Shift	Plant Standard	1.45 to 1.58 gm/c.c./As per plant standard		-	P	V
(b)	Fluidity (Degree Overswing)	Physical	At least Once in a Shift		330 to 350/As per plant standard		-	P	V
(c)	Visual Examination	Visual	100%		/As per plant standard		-	P	V
4.09	Grit / Sand								
(a)	Grain Size	Physical	Each Batch	Plant Standard	- 8 to 16 BS Mesh/As per plant standard		-	P	V
	Green Grit				-10 to 14 BS mesh/As per plant standard		-	P	V
	Fired Grit (Alumina body only)						-	P	V
4.10	Firing of Green Wares								
(a)	Temp. Control	Physical	Constant Recording	Plant Standard	As per Plant Std.		-	P	V
(b)	Pressure Control	Physical	Per Shift				-	P	V
(c)	Fuel Control	Physical	Per Shift				-	P	V
4.11	Sorting of Fired Insulators								
(a)	Visual Examination	Visual	100%	Plant Standard, IS:13305 & IEC: 60383.1	IS:13305 , IS: 731 & IEC: 60383.1		-	P	V
(b)	Porosity	Physical	1 No. / Car	Plant Standard, IS:731	No Dye Penetration		-	P	V

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(c)	Dimensional Check	Dimensional	2 item's/Day	Plant Standard, IS:731& IEC:60383.1	Should conform with the drawing		-	P	V
4.12	Tests on shells								
(a)	Hyd. Proof load Test	Mechanical	100 %	Tech.Spec. & Plant Standard	For 120kN-130 ± 10, 160kN - 150 ± 10 and 210kN - 200 ± 10/ As per Plant standard/ As per Technical Specification		-	P	V
4.13	Assembly								
(a)	Cement Mix Check	Physical	Daily	Plant Std.			-	P	V
(b)	Curing Time	Physical	Daily		At least 2 days water & 3 days air curing/ As per plant standard		-	P	V
(C)	Cube Test (Compressive Strength Test after - 6 days on 25.4 mm Cube)	Mechanical	Once in a day	IS : 4031, BS : 12, Plant Standard	500kg/cm2 Min./ As per Technical Specification		-	P	V
5.00 RAW MATERIAL (Metal Parts)									
5.01	Steel round for Ball pins								
(a)	Visual Inspection	Visual	100%	Plant Standard/ BS:970 (Gr. EN-8D)	Shall be straight, without rust & free from cracks, folds & similar surface defects.	CMTC	P	V	-

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(b)	Forgeability	Mechanical	2 Nos/ Lot	Plant Standard	No cracks on the surface		P	V	-
(c)	Non Metallic Inclusion rating	Metallurgical	2 Nos/ Lot	IS:4163	Max. 2 thick series/ As per Technical Specification		P	V	-
(d)	Chemical Analysis	Chemical	1 No/ Lot	BS:970	As per Gr. EN-8D/ As per Technical Specification		P	V	-
5.02	Forged Steel Ball Pins (Black condition)								
(a)	Visual inspection	Visual	100% by Manufacturer & 5% by insulator manufacturer	IS: 2486 & Plant standard	Shall be free from visual defects like surface defects, flakes & other visual defects that make the surface undulated.	CMTC	P	V	-
(b)	Heat Treatment	Physical	100% (100% Review of Documents by customer)	Time & Temperature graph	Should be output of recorder. Gradation as per plant standard/ As per Technical Specification	CMTC	P	V	-
(c)	M.P.I	Electro magnetic	100% by Manufacturer & 5% by insulator manufacturer	IS: 3703 & Plant Standard	No cracks		P	V	-
(d)	Dimension checking	Dimensional	IS:2486 & Plant Stabdard/ MP, Cl.19.8L	Approved Drawing	Should conform with the drawing	CMTC/	P	V	-
(e)	Gauge checking	Dimensional	100% by Manufacturer & 5% by insulator manufacturer	IS:2486	Should answer "GO and "NO GO" Gauges		P	V	-
(f)	Failing Load test	Mechanical	4 Nos./ 2000/ heat	Plant Standard	Should withstand at least 10% above Min.	CMTC	P	V	-

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					rated EMS/ As per Technical Specification				
(g)	Yield strength, Tensile strength & % Elongation on test bar	Physical	4 Nos./ 2000/ heat	IS:1608, BS:970, Tech.spec. & Plant Standard	Y.S - 28 kg/mm ² (min.)-Upto 120KN, 47 kg/mm ² (min.)-160 & 210KN T.S - 55 kg/mm ² . (min) upto 120KN, 71 kg/mm ² . (min)- 160-210KN %E - 16% min. upto 210KN/ As per Technical Specification		P	V	-
(h)	Hardness	Physical	4 Nos./ 2000/ heat	IS: 1500, BS 970, PG Spec. & Plant Standard	152 - 207 BHN - upto 120KN. 201 - 255 BHN - 160-210KN/ As per Technical Specification		P	V	-
(I)	Grain Size	Metallurgical	4 Nos./ 5000/ heat	IS:4748 & Plant Standard	Min. 6, (100X Magnification) / As per Technical Specification		P	V	-
(j)	Non Metallic Inclusion rating	Metallurgical	4 Nos/ 5000/ heat	IS:4163	Max. 2 thick series/ As per Technical Specification		P	V	-
(K)	Chemical Analysis	Chemical	1 No/ Lot	BS:970	Gr. EN-8D/ As per Technical Specification	CMTC/Third Party Lab.Test cert.	P	V	-
5.03	SGCI caps (Black & Galvanizing condition)								

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(a)	Visual inspection	Visual	100% by Manufacturer & 5% by Insulator manufacturer	Plant Standard	Shall be free from defects & shall be sound, fettled & free from blow holes, porosity, distortion & harmful defects.	CMTC	P	V	-
(b)	Gauge checking	Physical	100% by Manufacturer & 5% by Insulator manufacturer	Plant Standard, IS:2486 Part-II, IEC-120	Should answer "GO" and "NO GO" gauges		P	V	-
(c)	Dimension checking	Physical	IS:2486, Plant Standard	Approved Drawing	Should conform with the drawing		P	V	-
(d)	Failing Load test	Mechanical	4 Nos./ 2000/ annealing cycle	Plant Standard	Should withstand at least 10% above Min. rated EMS/ As per Tech. Specs.		P	V	-
(e)	Hardness Test	Physical	4 Nos./ 2000/ annealing cycle	IS: 1500 , IS:1865(Gr. 450/10)/ Plant Standard	SGCI- 130 BHN (Min.) / As per Tech. Specs.	CMTC	P	V	-
(f)	Microstructure	Metallurgical	2 Nos./ 5000 Nos	IS:1865/Tech. Spec.	SGCI- Spheroidal graphite nodules in ferrite matrix. / As per Tech. Specs.	CMTC	P	V	-
(g)	Tensile strength, %Elongation & 0.2% Proof strength test on test bar	Physical	2 Nos./ Lot of 5000 Nos./ Heat	IS: 1865 (Gr. 450/10)/ Plant standard.	For SGCI: T.S. -450 Mpa(min.), 0.2% PST- 310 Mpa(Min.) %E- 10%,(Min) / As per Tech. Specs.	CMTC	P	V	-
(h)	Chemical Analysis	Chemical	1 No/ 5000 Nos.	IS:228/ Tech. Spec./ Plant Standard	Phosphorous- 0.12% Max. / As per Tech. Specs.	CMTC	P	V	-

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(I)	Heat Treatment	Physical	100%	Time & temperature Graph	Should be output of recorder	CMTC	P	V	-
(j)	M.P.I	Electro magnetic	2nos/5000nos	IS 3703 & Tech. Spec.	IS 3703 & Tech. Spec.	CMTC/EM records	P	V	-
5.04	Hot Dip Galvanizing (Caps & Pins)								
(a)	Visual Inspection	Visual	100% by Manufacturer & 5% by insulator manufacturer	IS: 2629 & Plant Standard	Surface should be reasonably bright, smooth, free from rust, black spots, extra zinc.	CMTC/EM records	P	V	-
(b)	Gauge checking	Dimensional	100% by Manufacturer & 5% by insulator manufacturer	IS:2486, Plant Standard	Should answer "GO" and "NO GO" gauges	CMTC/EM records	P	V	-
(c)	Uniformity of Zn coating	Chemical	3Nos./5000	IS:2633& Plant Standard	6 dips Min./ As per Technical Specs.	CMTC/EM records	P	V	-
(d)	Mass of Zn coating	Chemical	3Nos./5000	IS:6745 & Plant Standard	610gm/m ² Min. / As per Technical Specs.		P	V	-
(e)	Adhesion test	Chemical	3Nos./5000	IS:2629 & Plant Standard	No Peel-Off of Zinc Coating/ As per Technical Specs.		P	V	-
(f)	Purity of Zinc (Ingot)	Chemical	1 No./ Lot	IS: 209 & Plant Standard 0	99.95% Min. / As per Technical Specs.	Third Party Lab.Report/ EM	P	V	-
(g)	For Ball Pin								
	i) Grain size	Metallurgical	4nos/5000nos/heat	IS 4748 & Plant Standard	Min 6,(100X magnification) / As per Technical Specs.	CMTC/EM records	P	V	-
	ii) Non Metallic Inclusion rating	Metallurgical	4nos/5000nos/heat	IS 4163	Max 2 thick series / As per Technical Specs.	CMTC/EM records	P	V	-

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	iii) Chemical analysis	Chemical	1no/lot/heat	BS:970 Carbon Manganese silicon Phosphorous Sulphur	Gr. EN-8D 0.40 - 0.45% 0.70 - 0.90% 0.05 - 0.35% 0.06% Max 0.05%Max / As per Technical Specs.	CMTC/EM records	P	V	-
(h)	For cap								
	I) Microstructure	Metallurgical	2 nos/5000nos	IS:1865/Tech. Spec./ Plant Standard	SGCI- Spheroidal graphite nodules in ferrite matrix - Min 85%/ As per Technical Specs.	CMTC/EM records	-	P	V
	ii) Chemical Analysis	Chemical	1 no/5000 nos	IS:1865/ Tech. Spec./ Plant Standard	IS:14329/ Tech. Spec./Plant Standard/ Phosphorous-0.12% max	CMTC/EM records	P	V	-
	iii) M.P.I	Electro magnetic	2 nos/5000nos	IS 3703 & Tech. Spec.	IS 3703 & tech. Spec.	CMTC/EM records	P	V	-
5.05	Zinc sleeve on Ball Pin								
(a)	Unfused area of surface	Visual	5 Nos./ 5000 Nos.	Plant Standard & Tech.spec.	Not more than 20% or as per Technical Specifications	CMTC/EM records	P	V	-
(b)	Load test	Mechanical	2 Nos./ 5000 Nos	Plant Standard	Sleeve should not be loose after applying 85% EMS or as per Technical Specifications	EM records	-	P	V
(c)	Blow holes on surface	Visual	5 Nos./ 5000 Nos.	Plant Standard	No blow holes on visual inspection	EM records	-	P	V
(d)	Deformation of sleeve (including thickness)	Visual	5 Nos./ 5000 Nos.	Plant Standard	No deformation on visual inspection	CMTC/EM records	P	V	-

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(e)	Foreign material	Visual	5 Nos./ 5000 Nos.	Plant Standard	No foreign materials visible on visual inspection	CMTC/EM records	P	V	-
(f)	Dimension checking	Dimensional	5 Nos./ 5000 Nos.	Approved Drawing	Should conform with the drawing	CMTC/EM records	P	V	-
(g)	Flash on head & Ball side	Visual	5 Nos./ 5000 Nos.	Plant Standard	No flash or as per Technical Specifications	CMTC/EM records	P	V	-
(h)	Zinc purity	Chemical	1 Nos./ 5000 Nos.	Plant Standard	99.8% Min. or as per Technical Specifications	Third Party Lab Report/EM records	P	V	-
5.06	Security clips (stainless steel /Phosphor Bronze)								
(a)	Visual	Visual	As per IEC:60372, Cl.12, Table 1	IEC:60372/ Plant Standard	surface cracking or incipient cracks, AQL=1.5%. For rough surface & blisters, AQL=6.5% or as per Technical Specifications	CMTC/EM records	P	V	-
(b)	Dimension checking	Dimensional	As per IEC:60383/ IS:2486	IEC:60372/ IS:2486/ Approved drawing/As per Tech. Spec.	Should conform with the drawing		P	V	-
(c)	Operation Test	Mechanical	As per IEC:60384/ IS:2486	IEC:60383/ IS:2486	50-500 N or as per Technical Specifications		-	P	V
(d)	Resistance to bending	Mechanical	As per IEC:60372, Cl.13/ IS:2486	IEC:60372/ IS:2486/ Plant Standard	No cracks at 75 degree bend. or as per		P	V	-

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					Technical Specifications				
(e)	Hardness	Physical	As per IEC:60372, Cl.13/ IS:2486	IEC:60372/ IS:2486/ Plant Standard	152 BHN Min. or as per Technical Specifications		P	V	-
(f)	Intergranular corrosion test (for stainless steel)	Metallurgical	1 No./ Lot	IS:10461 & ASTM: A 262A	No corrosion	Third Party Lab.Report/ EM Records	P	V	-
(g)	Chemical Analysis	Chemical	1 No./ Lot	AISI:304/IS78 14-Gr-I	Should conform standard		P	V	-
6.00	Final Inspection & Testing								
6.01	Routine Tests on Disc Insulators								
(a)	Visual Inspection	Visual	100 %	IS : 731	As per Spec.		-	P	V
(b)	Mechanical Routine test (At 65% of rated EMS)	Mechanical	100 %	IS : 731	As per Spec.	EM-records	-	P	V
(c)	Electrical Routine Test	Electrical	100 %	IS : 731	As per Spec.		-	P	V
(d)	High Frequency Test	Electrical	100 %	IEC : 60383-1, ANSI : 29.1 & Plant Standard	4 to 5 sec. or as per Technical Specifications		-	P	V
(e)	Power Frequency Test	Electrical	100 %	IEC : 60383-1, IS : 731 & Plant Standard	5 mins (Min.) or as per Technical Specifications		-	P	V
6.02	Acceptance Test								
(a)	Visual examination	Visual	IEC 60383 on lot Size of 10000 Nos. or less (E1 + E2)	Plant Standard, IS 731, IS :13305 & IEC: 60383-1	No Non permissible defect On glazed surface	EM-Format		P	W

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(b)	Verification of dimension	Dimensional	IEC 60383 on lot Size of 10000 Nos. or less(E2)	Approved drg. & IS 731, IEC 60383-I	As per Approved Drg.			P	W
(c)	Temperature cycle test	Thermal	IEC 60383 on lot Size of 10000 Nos. or less (E1 + E2 + Sample for Mech Test)	IEC : 60383-1, IS : 731	As per spec.			P	W
(d)	Mech. Performance test	Mechanical	20 Nos. Min on Lot size of 10000 Nos.	IEC 60575, Tech.Spec.	As per spec.			P	W
(e)	Test on locking devices for Ball & Socket coupling.	Mechanical	IEC 60383 on lot Size of 10000 Nos. - E2	IS : 2486 & IEC 60372/ 60383 & Tech.Spec.	As per spec.	EM-Format		P	W
(f)	Eccentricity Test (Axial & Radial runout)	Dimensional	IEC 60383 on lot Size of 10000 Nos. or less (E1 + E2)	IEC 60383-1 & Tech.approved Drg. & Spec.	As per Approved Drg.	EM-Format	P	V	W
(g)	Residual strength test (Preceded by Temperature cycle test)	Mechanical	25 Nos. Min on Lot size of 25000 Nos.	IEC 60797 & Tech.Specification.	As per spec.	EM-Format	P	V	W
(h)	Electromechanical strength Test.	Electrical - Mechanical	20 Nos. Min on Lot size of 10000 Nos.	Tech.spec., IS : 731 & IEC 60383	As per spec.	EM-Format	P	V	W
(i)	Puncture Test	Electrical	IEC : 60383/ Tech.Spec on Lot Size of 10000 Nos. or less - E2	Tech.spec., IS : 731 & IEC 60383	As per spec.		P	V	W
(j)	Porosity Test	Physical	IEC 60383 on lot Size of 10000 Nos. - E1	Tech.spec., IS : 731 & IEC 60383	As per spec.		P	V	W

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(k)	IR Test (using 5 kV Megger)	Electrical	25 Nos. on lot size of 10000 Nos.	Tech.spec. > 2000 Mega ohms	As per spec.		P	V	W
(l)	Galvanizing Test (Cap & Pin)		IEC 60383 on lot Size of 10000 Nos. - E2	IS : 2633/ 4759 / 6745		EM-Format	P	V	W
	i) Mass of zinc coating	Chemical		610 g/m ² min	As per spec.		P	V	W
	ii) Uniformity of zinc Coating	Chemical		6 dips min.	As per spec.		P	V	W
	iii) Adhesion test	Physical		IS : 731 & IS 2629	As per spec.		P	V	W
(m)	Chemical analysis of zinc Sleeve ball pin	Mechanical	1no/10000nos/LO T	99.8%min purity Tech.spec.	As per spec.	Third party Lab/ EM LAB	P	V	-
(n)	Impact Test	Mechanical	3 Nos. / Lot size of 10000 Nos.	Tech.spec. ANSI : C-29.2	As per spec.		P	V	W
(o)	Alumina Content (only if Power Arc Test is included in contract requirement)	Chemical	1 Sample / Lot	30% min, Tech.spec.	As per spec.		P	V	W
(p)	Bulk Density	Chemical	1 Sample / Lot	≥2.35g/cc for Si body and ≥2.5g/cc for Al body	As per spec.	EM-Format	P	V	W
(q)	Closed Porosity	Chemical	1 Sample / Lot	5.5 % max, by gravimetric method , Tech.spec.	As per spec.		P	V	W
(r)	Additional EMS Test (If applicable)	Mechanical	20 Nos. / 10000 Nos.	Tech. Spec./ IS:731/ IEC:60383	As per spec.		P	V	W

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6.03	Repeat Acceptance Test:-								
(a)	Steep wave front	Electrical	20 Nos. / 50000 Nos.	Tech.spec.	As per spec.	EM-Format		P	W
(b)	Thermal mechanical performance test	Thermal / Mechanical	20 Nos. / 50000 Nos.	Tech.spec.	As per spec.			P	W
7.00	Packing & Dispatch								
(a)	Visual Inspection	Visual	10%	Tech.spec.	EM - Drawing	EM - Drawing		P	-
(b)	Dimensions	Measurement	10%	Tech.spec.	EM - Drawing	EM - Drawing		P	-

EM : Equipment Manufacturer CM : Component Manufacturer CMTC :- Component manufacturer test certificate EMTR :- Equipment manufacturer test Report

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Wave trap

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S. No.	Components / Operations & Description of Test	Type of Check	Quantum of Check / Sampling with basis	Reference Document for Testing	Acceptance Norms	Format of Record	Category of Responsibility		
							Sub-Vendor	Manufacturer	Customer
A	Raw Material inspection								
1	Aluminium Conductor (Grade 63400)								
1.1	Tensile Strength [min] [max]	Mechanical	1 Sample / lot	Manufacturer Plant Specn./ As per specn / IS 733	Manufacturer Plant Specn./ As per specn / IS 733	Component Mfg's TC	P	V	-
1.2	Proof Stress [0.2%] [min] [max]	Mechanical	1 Sample / lot	Manufacturer Plant Specn./ As per specn / IS 733	Manufacturer Plant Specn./ As per specn / IS 733	Component Mfg's TC	P	V	-
1.3	Electrical Conductivity/ Electrical Resistivity [min] [max]	Electrical	1 Sample / lot	Manufacturer Plant Specn./ As per specn / IS 733/ IS 398	Conductivity: 48 % IACS to 56 % IACS Electrical Resistivity for annealed conductor (O) = at 20°C= 0.0280 ohm mm ² / m Max.	Component Mfg's TC	P	V	-

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1.4	Chemical Composition	Chemical	1 Sample / lot	Manufacturer Plant Specn./ As per specn / IS 733	Si : 0.3 - 0.7 % Fe : 0.6 % max Cu : 0.1 % max Mn : 0.3 % max Mg : 0.4 - 0.9 % Zn : 0.2 % max Ti : 0.2 % max Cr : 0.1 % max Balance - aluminium	Component Mfg's TC	P	V	-
1.5	Dimensional check	Physical	1 sample / lot	Plant Drg	Plant Drg	Eqpt. Mfg's TC	-	P	V
2	CROSS ARM [BOX SECTION]								
2.1	Chemical Composition (Grade 64430)	Chemical	1 Sample / lot	Manufacturer Plant Specn. As per specn / IS 617/ IS 1285	Si : 0.6 - 1.3 % Fe : 0.6 % max Cu : 0.1 % max Mn : 0.3 % max Mg : 0.4 - 1.0 % Zn : 01 % max Ti : 0.2 % max Cr : 0.25 % max Balance - Aluminium	Component Mfg's TC	P	V	-

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2.2	Tensile Strength	Mechanical	1 Sample / lot	Manufacturer Plant Specn./As per specn / IS 617/ IS 1285	Min. 31.5 Kg / mm ²	Component Mfg's TC	P	V	-
2.3	Proof Stress (0.2%)	Mechanical	1 Sample / lot	Manufacturer Plant Specn. /As per specn / IS 617/ IS 1285	Min. 27.5 Kg / mm ²	Component Mfg's TC	P	V	-
2.4	Elongation	Mechanical	1 Sample / lot	Manufacturer Plant Specn./As per specn / IS 617/ IS 1285	7% Min.	Component Mfg's TC	P	V	-
3	END RINGS								
3.1	Dimension	Physical	1 Sample / lot	Manufacturer Plant Drawing	Manufacturer Plant Drawing/ Compliance to Plant Drg	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
3.2	Visual Inspection	Visual	100%	Manufacturer Plant Specn.	Manufacturer Plant Specn.		P	V	-
3.3	Temperature Compatibility	Thermal	One Sample from each lot of 500 Nos or thereof	Manufacturer Plant Specn./ As per specn	Manufacturer Plant Specn./ As per specn No Deformation when left at 155		P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

					Deg C for 24 hrs				
3.4	Bending Strength	Mechanical	One Sample from each lot of 500 Nos or thereof	As per specn	Manufacturer Plant Specn./ As per specn		P	V	-
	[min]				40 Kg / mm ²				
4	PRE PREG FOIL								
4.1	Visual Check	Visual	100%	Manufacturer Plant Specn.	Manufacturer Plant Specn.	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
5	PERMA GLASS ROD (FRP ROD)								
5.1	Dimension	Physical	1 Sample / Lot	Manufacturer Plant Drawing	Manufacturer Plant Drawing Compliance to Plant Drg	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
5.2	Visual Inspection	Visual	100%	Manufacturer Plant Specn.	Manufacturer Plant Specn.	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
5.3	Tensile Strength	Mechanical	1 Sample / Lot	Manufacturer Plant Specn./ As per specn	Manufacturer Plant Specn./ As per specn	Component Mfg's TC	P	V	-
6	DISTANCE PIECE [FRP SPACER]								

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6.1	Dimension	Physical	5 samples / lot	Manufacturer Plant Drawing	Manufacturer Plant Drawing/ Compliance to Plant Drg	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
6.2	Visual Inspection	Visual	100%	Manufacturer Plant Specn.	Manufacturer Plant Specn.		P	V	-
7.0	STAINLESS STEEL TIE ROD								
7.1	Property of Non - Magnetism	Electrical	100%	Manufacturer Plant Specn. Should be Non Magnetic	Manufacturer Plant Specn.	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
7.2	Dimension	Physical	1 sample / lot	Manufacturer Plant Drawing	Manufacturer Plant Drawing/ Compliance to Plant Drg	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
7.3	Visual Inspection	Visual	100%	Manufacturer Plant Specn.	Manufacturer Plant Specn.	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
8	LIFTING EYE								
8.1	Capacity	Mechanical	One sample from each lot of 2000 Nos or part thereof	Manufacturer Plant Specn.	Manufacturer Plant Specn.	Eqpt. Mfg's TC / Component Mfg's TC /	P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

						Third Party Report			
8.2	Dimension	Physical	1 sample / lot	Manufacturer Plant Drawing	Manufacturer Plant Drawing/ Compliance to Plant Drg	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
8.3	Visual Inspection	Visual	100%	Manufacturer Plant Specn.	Manufacturer Plant Specn.	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
9	CORONA RING								
9.1	Dimension	Physical	1 sample / lot	Manufacturer Plant Drawing	Manufacturer Plant Drawing/ Compliance to Plant Drg	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
9.2	Visual Inspection	Visual	100%			Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
10	Surge Arrestor								
	Routine Test :								
10.1	Reference Voltage Test	Electrical	100%	IEC - 60099 - 4 / As per specn	IEC - 60099 - 4 / IS:15086-4/ As per specn	Component Mfg's TC	P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

10.2	Residual Voltage Test	Electrical	100%	As per specn	IEC - 60099 - 4 / IS:15086-4/ As per specn	Component Mfg's TC	P	V	-
10.3	Partial Discharge Test	Electrical	100%	IEC - 60099 - 4 / As per specn	IEC - 60099 - 4 / IS:15086-4/ As per specn	Component Mfg's TC	P	V	-
10.4	Measurement of Power	Electrical	100%	IEC - 60099 - 4 /	IEC - 60099 - 4 / IS:15086-4	Component Mfg's TC	P	V	-
10.5	Frequency Reference Voltage Test	Electrical	100%	As per specn	As per specn	Component Mfg's TC	P	V	-
10.6	Lightning Impulse Residual	Electrical	100%	IEC - 60099 - 4	IEC - 60099 - 4 / IS:15086-4	Component Mfg's TC	P	V	-
10.7	Voltage Test	Electrical	100%	As per specn	As per specn	Component Mfg's TC	P	V	-
10.8	Dimensional verification	Visual	1 sample	Manufacturer Drg	Manufacturer Drg	Component Mfg's TC	P	V	-
11	CAPACITORS								
11.1	Capacitance	Electrical	5 samples / lot	Manufacturer Plant Specn./ As per specn	Manufacturer Plant Specn. As per specn + / - 10 %	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
11.2	P. F. Voltage Test	Electrical	1 sample / lot		Manufacturer Plant Specn. /As per specn 1.5 kV for 5 Secs	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-

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11.3	Loss Angle (Tan Delta)	Electrical	1 sample / lot		Manufacturer Plant Specn./ As per specn	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
	[max]				0.002				
12	BIRD BARRIER								
12.1	Dimension	Physical	1 sample / lot	Manufacturer Plant Drawing	Manufacturer Plant Drawing/ Compliance to Plant Drg	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
12.2	Temperature Compatibility	Thermal	1 sample / lot	Manufacturer Plant Specn. /As per specn	Manufacturer Plant Specn./ As per specn No deformation at 155° C. for 24 hours	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
12.3	Visual Inspection	Visual	100%	Manufacturer Plant Specn.	Manufacturer Plant Specn.	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
13	TERMINAL CLAMP								
13.1	Major Dimensions	Physical	1 sample / lot	Plant Drawing / As per specn.	Plant Drawing / As per specn.	Component Mfg's TC / Eqpt. Mfg's	P	V	-

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						TC			
13.2	All routine test as per IS 5561	Mechanical	1 sample / lot	IS : 5561 / As per specn.	IS : 5561 / As per specn.	Component Mfg's TC	P	V	-
14	SUSPENSION HARDWARES								
14.1	Verification of Major Dimensions	Physical	1 set / lot	Approved Drg.	Approved Drg.	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
14.2	Visual Examination	Visual	100%	IS : 2486 Part 1 / As per specn	IS : 2486 Part 1 / As per specn	Component Mfg's TC / Eqpt. Mfg's TC	P	V	-
14.3	Routine Mechanical Test	Mechanical	1 No / Lot	IS : 2486 Part 1 / As per specn	IS : 2486 Part 1 / As per specn 40 % of min. failing load for 30 sec. 40 % of 120 KN =48 KN No damage should be noticed.	Component Mfg's TC	P	V	-
15	STRUCTURES								
a)	Hardwares - Bolts, Nuts, Flat & Spring	Physical	One sample / lot	Approved Drawing / As per	Approved Drawing / As	Component Mfr's TC /	P	V	-

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	Washers			specn.	per specn.	Eqpt. Mfg's TC			
b)	Dimensions	Electrical	One sample section / lot	Approved Drawing / As per specn	Approved Drawing / As per specn.	Component Mfr's TC / Eqpt. Mfg's TC	P	V	-
b)	Mechanical Properties	Mechanical	One sample section / lot /	Manufacturer Plant Specn.	Manufacturer Plant Specn./ IS 2062 Grade E250A	Component Mfr's TC / Third Party Report	P	V	-
	i) Tensile Strength[min]				410 N / sq. mm				
	ii) Yield Strength [min] for thickness less than 20mm				250 N / sq. mm				
	iii) Elongation		One sample / section / lot	Manufacturer Plant Specn.	Manufacturer Plant Specn.	Component Mfr's TC / Third Party Report			
	[min]				23%				
c)	Chemical Composition	Chemical	One sample	IS 2062 / Manufacturer Plant Specn.	maximum values: Carbon -0.23 % Sulphur-0.045 % Phosphorus- .045 % manganese-1.5% Silicon-0.4%	Component Mfr's TC /	P	V	-

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d)	Hot Dip Galvanising								
	i) Thickness	Physical	On Component	IS – 2633, 2629/ As per specn.	IS - 2633, 2629 /As per specn.	Component Mfr's TC / Eqpt. Mfg's TC	P	V	-
					5mm thick & above - 85 micron				
					2mm thick to below 5mm - 64 mic				
	ii) Adhesion Test [Pivot Hammer Test]	Mechanical	One sample / lot	IS - 2633, 2629 /As per specn.	IS - 2633 , 2629/ As per specn.	Component Mfr's TC	P	V	-
					Coating should not peel off				
	ii) Strip Test [min] [Wt. Of Zinc coating]	Chemical	One sample /section / lot	IS - 2633, 2629, 4759 As per specn./	IS - 2633, 2629, 4759 As per specn.	Component Mfr's TC	P	V	-
	a) 5mm thick & above				610 gm / m ²				
	b) 2mm thick to below 5mm				460 gm / m ²				
	c) 1.2mm thick to below 2mm				340 gm / m ²				
	iii) Prece Test [Uniformity of Zn coating]	Chemical	One sample / lot	IS - 2633 , 2629 As per specn	IS - 2633 , 2629 /As per specn.	Component Mfr's TC	P	V	-

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					No reddish deposit of copper after 6 dips in CuSo4 Solution				
16.0	HARDWARES [Stainless Steel - APL, KUN, LPS, SAI]								
16.1	Verification of Make	Visual	10%	Approved List	Approved List	Eqpt. Mfg's TC	P	V	-
B	<u>INPROCESS INSPECTION</u>								
1.0	WINDINGS								
1.1	Diameter	Physical	100%	APPROVED DRAWING	APPROVED DRAWING	Equip. Mfg's Log Book	-	P	V
1.2	Number Of Turns	Physical	100%	PLANT DRAWING	PLANT DRAWING	Equip. Mfg's Log Book	-	P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

2.0	Assembly Of Coil								
2.1	Dimensions	Physical	1 Sample / lot	PLANT STANDARD	PLANT STANDARD	MPR Card	-	P	V
3.0	CURING								
3.1	Gluing	Thermal	100%	PLANT STANDARD	PLANT STANDARD	MPR Card	-	P	V
4	WELDING								
4.1	Welding Procedure Specification			As per Manufacturer Internal Specification	As per Manufacturer Internal Specification		-	P	V
4.1	Welders Procedure Qualification						-	P	V
5.0	TUNING UNIT (Before Foaming)								
5.1	Adjustment for R. min (Blocking Impedance)	Electrical	100%	As per Specn.	As per Specn.	Equip. Mfg's Log Book	-	P	V
6.0	TUNING UNIT (After Foaming)								

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

6.1	Measurement of R. min (Blocking Impedence)	Electrical	100%	As per Specn.	As per Specn.	Equip. Mfg's Log Book	-	P	V
6.2	DP Test On load bearing members	Chemical	1 Sample / lot		No Dye Penetration / No chemical come out after foaming	Equip. Mfg's Log Book	-	P	V
C	<u>FINAL ACCEPTANCE & TESTING</u>								
1.0	Dimensional & Visual Check	Physical	1 Sample / lot	As per specn / Approved Drawings	As per specn / Approved Drawings	Eqpt. Mfg's TC	-	P	V
2.0	Measurement of Blocking Resistance & blocking Impedance	Electrical	100%	IEC - 60353 / IS 8792/ IS 8793/ As per specn	IEC - 60353 / IS 8792/ IS 8793/ As per specn	Eqpt. Mfg's TC	-	P	W
3.0	Measurement of rated inductance of the main coil	Electrical	100%			Eqpt. Mfg's TC	-	P	W
4.0	Measurement of Power Frequency Inductance of the main coil	Electrical	100%			Eqpt. Mfg's TC	-	P	W
5.0	Power Frequency voltage test on tuning device	Electrical	100%			Eqpt. Mfg's TC	-	P	W

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

6.0	Measurement of tapping loss and tapping loss based on blocking resistance (calculation method)	Electrical	100%	IEC - 60353 / IS 8792/ IS 8793	IEC - 60353 / IS 8792/ IS 8793	Eqpt. Mfg's TC	-	P	W
D	<u>PACKING & DESPATCH</u>								
i)	Verification of Completeness of Stage as appearing In MQP	Physical	100%	Manufacturer Packing Instruction	Manufacturer Packing Instruction	-	-	P	V
ii)	Check for Finish	Visual	100%			-	-	P	V
iii)	Check for Packing	Visual	100%			-	-	P	V

EM : Equipment Manufacturer

CM : Component Manufacturer

CMTC :- Component manufacturer test certificate

EMTR :- Equipment manufacturer test Report

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

Oil filled Current Transformer

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check /Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub- Vendor	Manufacturer	Customer
A. Section RAW MATERIAL INSPECTION									
1.A	Porcelain Insulators (If applicable)								
a	Routine Tests	Electrical & Mechanical	100%	IS : 5621/IEC 62155	IS : 5621/IEC 62155	CMTC	P	V	-
b	Dimensions and visual	Measurement	01 sample/lot	Relevant Drawing	Relevant Drawing	CMTC/ EMTR	P	V	-
c	Porosity Test	Physical	01 sample/lot	IS : 5621/IEC 62155	IS : 5621/IEC 62155	CMTC	P	V	-
d	Temp. Cycle Test	Physical	01 sample/lot	IS : 5621/IEC 62155	IS : 5621/IEC 62155	CMTC	P	V	-
1.B	Composite Insulator (If applicable)						P	V	-
	Dimension	physical	1 No/Lot	IEC 61462/ Tech. Specn./ Mfr Drg	IEC 61462/ Tech. Specn./ Mfr Drg	CMTC	P	V	-

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	Routine Mechanical test	Mechanical	100%	IEC 61462/ Tech. Specn.	IEC 61462/ Tech. Specn.	CMTC	P	V	-
	Silicon Content	Chemical	1 No/mfg batch			CMTC	P	V	-
	Ultrasonic test	Mechanical	100%			CMTC	P	V	-
							P	V	-
2	Transformer Oil						P	V	-
a	Kinematic Viscosity @ 40°C	Test	01 sample/Lot	Mfg.std./IEC 60296/IS 335	Mfg.std./IEC 60296/IS 335	CMTC	P	V	-
b	Pour Point	Test	01 sample/lot			CMTC	P	V	-
c	Water content	Test	01 sample/lot			CMTC	P	V	-
d	Electric Strength BDV (New unfiltered)	Test	01 sample/lot			CMTC/EMTR	P	V	-
e	Density @ 20degC	Test	01 sample/lot			CMTC	P	V	-
f	Dielectric Dissipation Factor (Tan Delta)@90°C	Test	01 sample/lot			CMTC/EMTR	P	V	-
g	Neutralization Value/Acidity	Test	01 sample/lot			CMTC	P	V	-
h	Interfacial Tension	Test	01 sample/lot			CMTC	P	V	-
i	Total Sulphur Content	Test	01 sample/lot			CMTC	P	V	-

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j	Corrosive Sulphur	Test	01 sample/lot			CMTC	P	V	-
k	Dibenzyl Disulphide	Test	01 sample/lot			CMTC	P	V	-
l	Antioxidant Additives,	Test	01 sample/lot			CMTC	P	V	-
m	Metal Passivator	Test	01 sample/lot			CMTC	P	V	-
n	Furfural Content	Test	01 sample/lot			CMTC	P	V	-
o	Oxidation Stability	Test	01 sample/lot			CMTC	P	V	-
p	Total acidity	Test	01 sample/lot			CMTC	P	V	-
3	Copper Wire								
a	Dimension	Measure	01 sample/lot	Mfg.std./IS 13730	Mfg.std./ IS 3730	CMTC	P	V	-
b	Mechanical Properties	Mechanical	01 sample/lot			CMTC	P	V	-
c	Thermal Properties	Thermal	01 sample/lot			CMTC	P	V	-
d	Chemical Properties	Chemical	01 sample/lot			CMTC	P	V	-
e	Electrical Properties	Electrical	01 sample/lot			CMTC	P	V	-
q	Total sludge	Test	01 sample/lot	Mfg.std./IEC 60296/IS 335	Mfg.std./IEC 60296/IS 335	CMTC	P	V	-
4	Primary Terminal								

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a	Visual	Visual	10 %/ lot	App. Drg./MFR PO.	Free from Damage	CMTC/EMTR	P	V	-
b	Dimension	Measure	5 % / lot	App. Drg./MFR PO.	As per Drg.	CMTC/EMTR	P	V	-
r	Flash Point	Test	01 sample/lot	Mfg.std./IEC 60296/IS 335	Mfg.std./IEC 60296/IS 335	CMTC	P	V	-
5	ALUMINIUM FOIL (if applicable)								
a	Thickness Tolerance	Physical	1 Sample / lot	Mfg. plant std.	Mfg. plant std.	CMTC	P	V	-
b	Purity (min)	Chemical	1 Sample / lot			CMTC	P	V	-
c	Visual	Visual	1 Sample / lot			CMTC	P	V	-

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6	Insulation Crepe Paper								
a	Thickness	Measure	01sample/size/lot	Material Spec./MFR PO	Material Spec./MFR PO	CMTC/ EMTR	P	V	-
b	Hillcount	Measure		Material Spec./IS 9335/IEC 60554	Material Spec./IS 9335/IEC 60554		P	V	-
c	Substance	Measure					P	V	-
d	Elongation at break MD	Mechanical					P	V	-
e	Ash Content	Mechanical					P	V	-
f	Moisture Content	Mechanical		IS 9335/IEC 60554	IS 9335/ IEC 60554		P	V	-
g	PH Value	Chemical					P	V	-
h	Conductivity	Electrical					P	V	-
i	Tensile Strength	Mechanical					P	V	-
7	Semi conducting Crepe Paper								
a	Visual	Visual		Mfg.std./Data Sheet	Free from Damage		P	V	-
b	Dimension Thickness	Measure					P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

c	Tensile Strength	Mechanical	01 sample/ lot		Mfg.std./Data Sheet	CMTC/ EMTR	P	V	-
d	Elongation	Mechanical					P	V	-
8	Core								
a	Visual / dimensions check	Measure	100%	As per Tech. Spec./ Drawing/ P.O	As per Tech. Spec./ Drawing/ P.O	CMTC/ EMTR	P	V	-
b	Test at specified flux density for losses.	Electrical	100%	Within limit as per specified grade of Core/ P.O.	Within limit as per specified grade of Core/ P.O.	CMTC/ EMTR	P	V	-
c	Magnetizing Force (Max)	Electrical	1 Sample / lot	As per Tech. Spec./Drawing/ P.O	As per Tech. Spec./Drawing/ P.O	CMTC/ EMTR	P	V	-
d	Type and Grade of material	Electrical				CMTC/ EMTR	P	V	-
e	Stacking Factor	Electrical				CMTC/ EMTR	P	V	-
9	MS Tanks								
a	Leakage Test	Mechanical	100%	Leakage check at 30 psi	No Leakage at 30 P.S.I	CMTC	P	V	-
b	Dimensions	Measure	01 sample/size/lot	App. Drg./MFR drg.	App. Drg./MFR drg.	EMTR	-	P	-

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10	Galvanising Checks								
a	Thickness of Zinc coating	Measure	1 Sample /Lot	IS 4759-1984/ App.Drg./ Tech Spec	Min Avg. thickness $\geq 5\text{mm}$ - 86 μm , < 5mm - 65 μm ,	CMTC	P	V	-
b	Mass of Zn coating on sample	Measure	1 Sample /Lot	IS 4759-1984/ App.Drg. /Tech Spec	Plant Std ,Min Avg. $\geq 5\text{mm}$ 610g/m ² < 5mm 460g/m ²	CMTC	P	V	-
c	Uniformity of Zinc coating	Measure	1 Sample /Lot	IS 2633/ App. Drg./ Tech Spec	IS 2633/PG App.Drg./Tech Spec	CMTC	P	V	-
d	Adhesion test	Mechanical	1 Sample /Lot	IS 2629/App. Drg./ Tech Spec	No peeling of Zinc	CMTC	P	V	-
11	Painting Of Tanks								
a	Visual /Shade	Visual	1 Samples / lot	App.Drg./Tech Spec/ MFR P.O	App.Drg. /Tech Spec /GTP /Mfg.std	CMTC	P	V	-
b	Thickness (DFT)	Measure	DO	App.Drg./ Tech Spec/ GTP/ Mfg.std.	App.Drg. /Tech Spec /GTP /Mfg.std.	CMTC	P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

12	Sealing Element Rubber 'O' Ring								
a	Dimensions	Measure	1 Sample /Lot	Mfg. Std./Drawing	Mfg. Std./Drawing	CMTC/ EMTR	P	V	-
b	Hardness	Mechanical				CMTC/ EMTR	P	V	-
c	Elongation at break	Mechanical				CMTC	P	V	-
d	Aging in transformer oil for 70 hrs.	Mechanical				CMTC	P	V	-
e	Compression Set	Mechanical				CMTC	P	V	-
13	Bellow								
a	Major Dimensions	Physical	1 Sample / lot	Mfg. Std./Drawing	Mfg. Std./Drawing	CMTC/ EMTR	P	V	-
b	Visual	Visual	100%	Mfg. Std./Drawing	Mfg. Std./Drawing	CMTC/ EMTR	P	V	-
14	TEFZEL CABLE/PTFE CABLE (if applicable)								

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

a	Di electric withstand voltage test	Physical	1 Sample / lot	Mfg. Std./Drawing	Mfg. Std./Drawing	CMTC/EMTR	P	V	-
15	Terminal Blocks								
a	Voltage Grade	Electrical	1 Sample / lot	Supplier Catalogue/ Tech. Specn.	Supplier Catalogue/ Tech. Specn.	CMTC	P	V	-
b	HV test	Electrical					P	V	-
c	IR b/w - adjacent terminal & b/w terminal & Channel	Electrical					P	V	-
B. Section INPROCESS INSPECTION - CT									
1	Primary Winding								
a	Visual check, Dimension & Cleanliness of Primary conductor	Physical	100%	Free from damage/PG App.Drg./Tech Spec	Free from damage/PG App.Drg./Tech Spec	EMTR	-	P	V
b	Overlap of Insulation		100%	As per Plant Standard	As per Plant Standard	EMTR	-	P	V
2	<u>Core Winding (Secondary) & Coil Insulation</u>								
a	Interturn Over Voltage test	Electrical					-	P	V

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b	Accuracy	Electrical	100%	IS 16227	IS 16227	EMTR	-	P	V
c	Polarity Marking & Verification of Core Sequence	Electrical					-	P	V
d	Secondary H.v Test	Electrical					-	P	V
e	Interturn Over Voltage test	Electrical					-	P	V
4	CT Assembly								
a	Fitment of Primary Terminal	Mechanical	100%	As per Approved Drawing	As per Approved Drawing	EMTR	-	P	V
b	Fitment of Bushing	Mechanical	100%	As per Approved Drawing	As per Approved Drawing	EMTR	-	P	V
c	Secondary H.V.test	Electrical	100%	IS 16227	IS 16227	EMTR	-	P	V
d	Verification of Accuracy Class	Electrical	100%			EMTR	-	P	V
e	Polarity Marking & Verification of Core Sequence	Electrical	100%			EMTR	-	P	V

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f	Leakage Test	Mechanical	100%	leakage testing is done on complete assembled unit at 30 psi.	No Leakage	EMTR	-	P	V
5	Vacuum Drying & Oil impregnation								
a	Temperature of Oven	Physical	100%	Mfr's Plant Standard	Mfr's Plant Standard	EMTR	-	P	V
b	Vacuum & vacuum Leakage	Electrical			Mfr's Plant Standard	EMTR	-	P	V
c	Measurement of Water in Vapor Condenser	Electrical			No water for 6 hours in vapour condensor.	EMTR	-	P	V
d	BDV of Insulating Oil	Electrical	One sample/lot	As per IEC 60296/IS 335	As per IEC 60296/IS 335	EMTR	-	P	V
e	Tan delta	Electrical				EMTR	-	P	V
f	Moisture/Water Content	Electrical				EMTR	-	P	V
C. Section FINAL INSPECTION & TESTING - CT									

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

1	Routine Test :- All Routine/Acceptance test as per Contract Technical Specification need to carried out without fail								
a	Verification of Terminal marking & Polarity	Electrical	100%	Tech. Specs / IS16227/ IEC61869	Tech. Specs / IS16227/ IEC61869	Log Sheet	-	P	V
b	High voltage power frequency withstand test on primary winding.	Electrical					-	P	V
c	High voltage power frequency withstand test on secondary winding.(All cores to be tested seperatly)3kV for 1 min	Electrical					-	P	V
d	Partial Discharge test	Electrical					-	P	V
e	Accuracy Test -Ratio and phase angle error test on metering & protection class cores.	Electrical					-	P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

f	Resistance ,Knee point voltage ,Exciting Current measurement on special protection (Class PS) cores.	Electrical	100%	Tech. Specs / IS16227/ IEC61869	Tech. Specs / IS16227/ IEC61869	Log Sheet	-	P	V
g	Turns ratio measurement (Ratio error at no load)for special purpose protection (Class PS)cores.	Electrical					-	P	V
h	Interturn over voltage test (all cores will be tested separately)	Electrical					-	P	V
i	Determination of composite error	Electrical					-	P	V
j	Measurement of capacitance & tan delta at 10kV & Um/root3 kV.	Electrical	100%		Min 7000 M Ohms./mfg std	Log Sheet	-	P	V
k	I.R. Measurement Between primary & secondary (Earth)	Electrical				Log Sheet	-	P	V

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

1	Oil level check	visual		Tech. Specs / Mfr's Drawing/ Standard	Tech. Specs / Mfr's Drawing/ Standard	Log Sheet	-	P	V
2	<u>Acceptance Test</u>								
a	Verification of Terminal marking & Polarity	Electrical	10 %	Tech Specs / IS16227/ IEC61869	Tech. Specs / IS16227/ IEC61869	Test Report	-	P	W
b	High voltage power frequency withstand test on primary winding.	Electrical					-	P	W
c	Transformer Oil from CT After HV Test for each rating								
i	BDV Test (Min.)	Electrical	01sample /lot	IEC 60422	Min.60 kV	Test Report	-	P/V	V
ii	Tan Delta at 90°C Max.	Electrical		IEC 60422	IEC 60422	Test Report	-	P/V	V
iii	Moisture Content	Electrical		IS 1866 /IEC 60422/ Tech. spec.	less than 10 ppm	Test Report	-	P/V	V
iv	Interfacial Tension(Min.)	Electrical		IEC 60422	Min. 35 mN/mtr	Test Report	-	P/V	V

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d	High voltage power frequency withstand test on secondary winding.(All cores to be tested seperatly)3kV for 1 min	Electrical	10 %	Tech. Specs / IS16227/ IEC61869	Tech. Specs / IS16227/ IEC61869	Test Report	-	P	W
e	Partial Discharge test	Electrical					-	P	W
f	Accuracy Test -Ratio and phase angle error test on metering & protection class cores.	Electrical					-	P	W
g	Resistance ,Knee point voltage ,Exciting Current measurement on special protection (Class PS) cores.	Electrical					-	P	W
h	Turns ratio measurement (Ratio error at no load)for special purpose protection (Class PS)cores.	Electrical					-	P	W

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

i	Interturn over voltage test (all cores will be tested separately)	Electrical	10 %	Tech. Specs / IS16227/ IEC61869	Tech. Specs / IS16227/ IEC61869	Test Report	-	P	W
j	Determination of composite error	Electrical					-	P	W
k	Measurement of capacitance & tan delta at 10 kV, 0.3,0.7,1.0 &1.1 Um/root3 kV.before & after HV Test	Electrical					-	P	W
l	Instrument Security Factor Test	Electrical					-	P	W
m	IR measurement between Primary & Secondary (Earth)	Electrical			Min 7000 M Ohms.		-	P	W
n	Painting Shade & Thickness Check	Measure	10 %	Tech. Specs./ 85 Micron Min.	Tech. Specs./85 Micron Min.	Test Report	-	P	W
o	Oil level check	visual	10 %	Tech. Specs / Mfr's Drawing/standard	Tech. Specs / Mfr's Drawing/standard	Log Sheet	-	P	W

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D. Section PACKING & DESPATCH - CT									
1	N2 Gas Filling	Mechanical	100%	N2 is filled at 10 psi.	Pressure is observed after 1 hour. To check leakage	EM RECORD	-	P	-
2	Checking for completeness of the equipment and accessories including spares as per order.	Visual		As per MFR Process Specification	As per MFR Process Specification	EM RECORD	-	P	-
3	Check for proper packing	Visual				EM RECORD	-	P	-
4	Ensure supply of operation manual, packing list and test reports along with copy of inspection reports, CIP and MICC	Visual				EM RECORD	-	P	-

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1.1 kV Grade PVC Power Cable

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub- Vendor	Manufacturer	Customer
A. RAW MATERIAL INSPECTION									
1.0	ALUMINIUM WIRE ROD								
1.1	Chemical Analysis	Chem	One Sample per heat per lot	IS: 4026 (Gr. II) and Tech. Specification Al-99.6% Min. Si-0.13% Max. Fe-0.30% Max. Cu-0.04% Max. (Ti+Va)-0.02% Max.	IS: 4026 (Gr. II) and Tech. Specification Al-99.6% Min. Si-0.13% Max. Fe-0.30% Max. Cu-0.04% Max. (Ti+Va)-0.02% Max.	Manuf. T.C./ Raw Material (RM) Test Report	-	P	V
1.2	Diameter of Aluminium Wire Rod	Meas.		IS: 5484 & Tech. Specification Min. 9.00 mm Nom. 9.50 mm Max. 10.00 mm	IS: 5484 & Tech. Specification Min. 9.00 mm Nom. 9.50 mm Max. 10.00 mm	Manuf. T.C./ RM Test Report	-	P	V
1.3	Breaking Load	Mech.		IS : 5484 & Tech. Specification	IS : 5484 & Tech.	Manuf.			

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

				Range I to IV Tensile strength- 65 to 150 N/mm ²	Specification Range I to IV Tensile strength- 65 to 150 N/mm ²	T.C./ RM Test Report	-	P	V
1.4	Resistivity / Conductivity	Elect.		IS : 5484 & Tech. Specn Max. resistivity 0.028264 Ohm x mm ² /meter at 20 deg C. Min. Conductivity 61% of IACS	IS : 5484 & Tech. Specn Max. resistivity 0.028264 Ohm x mm ² /meter at 20 deg C. Min. Conductivity 61% of IACS	Manuf. T.C./ RM Test Report	-	P	V
1.5	Cleanliness & Surface Smoothness (Visual Check)	Visual	100% on each Coil	IS : 5484 & Tech.Specification The rods shall be bright and clean. It shall be free from blisters, scale, fins, spills cracks and other defects.	IS : 5484 & Tech. Specification The rods shall be bright and clean. It shall be free from blisters, scale, fins, spills cracks and other defects.	Manuf. T.C./ RM Test Report	-	P	V
2.0 PVC COMPOUND FOR INSULATION (TYPE-A)									
2.1	Type / Specific Gravity	Meas.	One Sample per Lot of 5 MT or part	IS: 5831 & Approved data Sheets	IS: 5831 & Approved data Sheets	Manuf. T.C./ RM Test Report	-	P	V

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			thereof						
2.2	Visual Check for PVC granules i.e. burns etc	Visual	One sample per lot of 5 MT or part thereof	IS: 5831 & Approved Data sheets. Should be generally free from burn particles	IS: 5831 & Approved Data sheets. Should be generally free from burn particles	Manuf. T.C./ RM Test Report	-	P	V
2.3	Tensile Strength and elongation before ageing	Mech	One sample per lot of 5 MT or part thereof	IS: 5831 & Approved Data Sheets. Min. TS- 12.5 N/mm ² Elongation 150% (Min.)	IS: 5831 & Approved Data Sheets. Min. TS- 12.5 N/mm ² Elongation 150% (Min.)	Manuf. T.C./ RM Test Report	-	P	V
2.4	Tensile Strength and elongation after ageing	Mech	One sample per lot of 5 MT or part thereof	IS: 5831 & Approved Data Sheets. Min. TS- 12.5 N/mm ² Max. Variation±20% from the actual values before ageing Elongation 150% (Min.) Max.variation ± 20% from the actual values before ageing	IS: 5831 & Approved Data Sheets. Min. TS- 12.5 N/mm ² Max. Variation±20% from the actual values before ageing Elongation 150% (Min.) Max.variation ± 20% from the actual values before ageing	Manuf. T.C./ RM Test Report	-	P	V
2.5	Loss of Mass in air oven	Therm							

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			One sample per lot of 5 MT or part thereof	IS: 5831 & Technical Specification Max. 2 mg/cm ²	IS: 5831 & Technical Specification Max. 2 mg/cm ²	Manuf. T.C./ RM Test Report	-	P	V
2.6	Thermal Stability	Therm	One sample per lot of 5 MT or part thereof	IS: 5831 & Technical Specification. The indicating paper shall not change colour when the insulation is kept at a temperature of 200 deg. C for a period of Min. 80 Minutes	IS: 5831 & Technical Specification. The indicating paper shall not change colour when the insulation is kept at a temperature of 200 deg. C for a period of Min. 80 Minutes	Manuf. T.C./ RM Test Report	-	P	V
2.7	Volume Resistivity	Elect	One sample per lot of 5 MT or part thereof	IS: 5831 & Approved Data Sheets. Min. at 27 deg. C - 1x10 ¹³ Ohm cm. Min at 70 deg. C - 1x10 ¹⁰ Ohm.cm.	IS: 5831 & Approved Data Sheets. Min. at 27 deg. C - 1x10 ¹³ Ohm cm. Min at 70 deg. C - 1x10 ¹⁰ Ohm.cm.	Manuf. T.C./ RM Test Report	-	P	V
3.0	PVC COMPOUND FOR SHEATH (TYPE ST-1)								

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

3.1	Type / Specific Gravity	Meas.	One sample per lot of 5 MT or part thereof	IS: 5831 & Approved data sheet	IS: 5831 & Approved data sheet	Manuf. T.C./ RM Test Report	-	P	V
3.2	Visual Check for PVC granules i.e. burns etc.	Visual	One sample per lot of 5 MT or part thereof	IS: 5831 & Tech. Specification. Should be generally free from burn particles.	IS: 5831 & Tech. Specification. Should be generally free from burn particles.	Manuf. T.C./ RM Test Report	-	P	V
3.3	Tensile strength and elongation before ageing	Mech	One sample per lot of 5 MT or part thereof	IS: 5831 & Tech. Specification. Min. TS 12.5 N/mm ² Elongation 150% (Min.)	IS: 5831 & Tech. Specification. Min. TS 12.5 N/mm ² Elongation 150% (Min.)	Manuf. T.C./ RM Test Report	-	P	V
3.4	Tensile strength and elongation after ageing	Mech	One sample per lot of 5 MT or part thereof	IS: 5831 & Tech. Specification. Min. TS 12.5 N/mm ² (Max.) Variation $\pm 20\%$ from the actual value before ageing Elongation 150% (Min.) Max. variation $\pm 20\%$ from the	IS: 5831 Tech. Specification. Min. TS 12.5 N/mm ² (Max.) Variation $\pm 20\%$ from the actual value before ageing Elongation 150% (Min.) Max. variation $\pm 20\%$ from the	Manuf. T.C./ RM Test Report	-	P	V

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				actual values before ageing	actual values before ageing				
3.5	Loss of Mass in air oven	Therm	One sample per lot of 5 MT or part thereof	IS: 5831 & Tech. Specification Max. 2 mg/cm ²	IS: 5831 & Tech. Specification Max. 2 mg/cm ²	Manuf. T.C./ RM Test Report	-	P	V
3.6	Thermal Stability	Therm	One sample per lot of 5 MT or part thereof	IS: 5831 & Tech. Specification. The indicating paper Shall not change colour when the sample of sheath is kept at 200 deg. C for a period of min. 40 Minutes	IS: 5831 & Tech. Specification. The indicating paper Shall not change colour when the sample of sheath is kept at 200 deg. C for a period of min. 40 Minutes	Manuf. T.C./ RM Test Report	-	P	V
3.7	Oxygen Index Test on PVC for Outer Sheath Only.	Envir.	One sample per lot of 5 MT or part thereof	ASTM-D-2863 & Approved Data Sheets Min. 29%	ASTM-D-2863 & Approved Data Sheets Min. 29%	Manuf. T.C./ RM Test Report	-	P	V
3.8	Temperature Index Test on PVC for Outer Sheath Only	Envir.	One sample per lot of 5 MT or part thereof	ASTM-D-2863 & Approved Data Sheets Min. 250 Deg.C	ASTM-D-2863 & Approved Data Sheets Min. 250	Manuf. T.C./ RM Test Report	-	P	V

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					Deg.C				
4.0	ARMOUR WIRE / STRIP								
4.1	DIMENSION	Meas.	One Sample for every 10 coils	IS : 3975 & Approved Data Sheets	IS : 3975 & Approved Data Sheets	Manuf. T.C./ RM Test Report	-	P	V
4.2	Tensile Stength	Mech					-	P	V
4.3	Elongation	Mech					-	P	V
4.4	Torsion / Winding	Mech					-	P	V
4.5	Wrapping Test	Mech		IS : 3975 & Approved Data Sheet Wrap-8 Unwrap-6 & Wrap-6 on a mandrel as per above IS. The wire shall not crack or break	IS : 3975 & Approved Data Sheet Wrap-8 Unwrap-6 & Wrap-6 on a mandrel as per above IS. The wire shall not crack or break	Manuf. T.C./ RM Test Report	-	P	V
4.6	Uniformity of Zinc coating (Preece Test)	Chem.		IS : 4826, IS : 3975 & approved data sheets. At the end of specified number of dips as per IS, the specimen shall not	IS : 4826, IS : 3975 & approved data sheets. At the end of specified number of dips as per IS, the specimen shall	Manuf. T.C./ RM Test Report	-	P	V

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				show any red deposit of copper upon base metal	not show any red deposit of copper upon base metal				
4.7	Weight of Zinc Coating	Chem.	One Sample for every 10 coils	IS : 4826, IS : 3975 & Approved Data Sheets	IS : 4826, IS : 3975 & Approved Data Sheets	Manuf. T.C./ RM Test Report	-	P	V
4.8	Adhesion Test	Mech		IS: 4826, IS : 3975 & Approved Data Sheets. The Zinc Coating shall remain adherent to the steel wire.	IS: 4826, IS : 3975 & Approved Data Sheets. The Zinc Coating shall remain adherent to the steel wire.	Manuf. T.C./ RM Test Report	-	P	V
4.9	Resistivity	Elect.		IS : 3975 & Approved Data Sheets.	IS : 3975 & Approved Data Sheets.	Manuf. T.C./ RM Test Report	-	P	V
4.10	Check for Purity of Zinc	Chem	One Sample for every lot of 50 MT or part thereof	IS : 209 & Tech. Specification Min. purity of Zinc 99.95%	IS : 209 & Tech. Specification Min. purity of Zinc 99.95%	Manuf. T.C./ RM Test Report	-	P	V
B INPROCESS INSPECTION									
5.0	ALUMINIUM DRAWN WIRE (H2								

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	grade)								
5.1	Surface finish and winding (Visual check)	Visual	100% on each spool	IS: 8130 & Approved data sheet. The wire shall be smooth and free from scratches	IS: 8130 & Approved data sheet. The wire shall be smooth and free from scratches	In Process Log Sheet	-	P	V
5.2	Diameter of drawn Aluminium Wire	Meas.	One Sample from each spool	IS : 8130 & Approved data Sheet.	IS : 8130 & Approved data Sheet.	In Process Log Sheet	-	P	V
5.3	Tensile Strength	Mech					-	P	V
5.4	Wrapping Test	Mech		IS : 8130 & Approved data Sheets. Wrap-8 Unwrap-6 & Wrap-6 on its own dia. The wire shall not crack or break	IS : 8130 & Approved data Sheets. Wrap-8 Unwrap-6 & Wrap-6 on its own dia. The wire shall not crack or break	In Process Log Sheet	-	P	V
6.0	STRANDING								
6.1	Lay Direction & Compactness	Visual	Each length at the beginning	IS : 1554 / Approved data sheet.	IS : 1554 / Approved data sheet.	In Process Log Sheet	-	P	V
6.2	Smoothness / Surface scratches	Visual	100%	The surface shall be smooth and free from	The surface shall be smooth and free from	In Process Log Sheet	-	P	V

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	(Visual check)			Scratches	Scratches				
6.3	Resistance / Km at 20°C	Elect	One Sample from each spool	IS : 8130 & Approved data Sheet.	IS : 8130 & Approved data Sheet.	In Process Log Sheet	-	P	V
6.4	Dimensions	Meas.	One Sample from each spool	Approved data Sheet	Approved data Sheet	In Process Log Sheet	-	P	V
7.0	INSULATED CORE						-	P	V
7.1	Thickness of Insulation Average Minimum	Meas . Meas .	Each setting during running of machine	IS: 1554 (Part-1) Table 2 & Approved data sheet.	IS: 1554 (Part-1) Table 2 & Approved data sheet.	In Process Log Sheet	-	P	V
7.2	Type of Insulating Material	Visual	Each setting during running of machine	IS: 1554 (Part-1) Table 2 & Approved data sheet. Type-A	IS: 1554 (Part-1) Table 2 & Approved data sheet. Type-A	In Process Log Sheet	-	P	V
7.3	Surface Finish	Visual	Each setting during running of machine	IS: 1554 (Part-1) The Surface shall be smooth and free from any defects	IS: 1554 (Part-1) The Surface shall be smooth and free from any defects	In Process Log Sheet	-	P	V
7.4	Spark Test	Elect.	100%	Min. 6 KV (r.m.s.)	Min. 6 KV		-	P	V

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					(r.m.s.)	In Process Log Sheet			
7.5	Core Identification / Colour of Cores	Visual	100%	IS : 1554 (Part-1) & Approved data sheet.	IS : 1554 (Part-1) & Approved data sheet.	In Process Log Sheet	-	P	V
7.6	Eccentricity	Meas.	100%	Max. 10% / Approved Data Sheet	Max. 10% / Approved Data Sheet	In Process Log Sheet	-	P	V
8.0	LAID UP CORES								
8.1	Direction of Lay	Visual	Each setting during process	The outer most layer shall be right hand and other layers opposite to previous one.	The outer most layer shall be right hand and other layers opposite to previous one.	In Process Log Sheet	-	P	V
8.2	Lay Length	Meas.	Each setting during process	Approved data sheet	Approved data sheet	In Process Log Sheet	-	P	V
8.3	Sequence of Laying	Visual	Each setting during process	Approved data sheet	Approved data sheet	In Process Log Sheet	-	P	V
8.4	Tightness of Cores	Visual	Each setting	The cores shall be	The cores shall	In Process	-	P	V

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			during process	reasonable tight.	be reasonable tight.	Log Sheet			
8.5	Diameter	Meas.	Each setting during process	Approved Data Sheet	Approved Data Sheet	In Process Log Sheet	-	P	V
9.0	INNER SHEATHING BY EXTRUSION								
9.1	Thickness of Inner Sheath	Meas.	Each setting during process	IS : 1554 (Part-1) Table -4 & Approved Data Sheets.	IS : 1554 (Part-1) Table -4 & Approved Data Sheets.	In Process Log Sheet	-	P	V
9.2	Dia Over Inner Sheath	Meas.		Approved data sheet	Approved data sheet		-	P	V
9.3	Type of Sheathing Material	Visual		IS : 1554 (Part-1) & Approved data sheet	IS : 1554 (Part-1) & Approved data sheet		-	P	V
9.4	Surface Finish	Visual	100%	IS : 1554 (Part-1) The Surface shall be smooth & free from any defects.	IS : 1554 (Part-1) The Surface shall be smooth & free from any defects.		-	P	V

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10.0	ARMOURING								
10.1	No. of Armour Wires & Construction	Meas.	Starting of M/C and during running	Approved Data Sheet	Approved Data Sheet	In Process Log Sheet	-	P	V
10.2	Dimensions or Armour Wire / Strip	Meas.	Starting of M/C and during running	Approved Data Sheet	Approved Data Sheet	In Process Log Sheet	-	P	V
10.3	Surface Finish	Visual	during running	Should be reasonably smooth	Should be reasonably smooth		-	P	V
10.4	Lay Direction	Visual		Left Hand Lay	Left Hand Lay		-	P	V
10.5	Gap between two Wire / Strip	Meas.		Not more than one Wire dia/ width of Strip	Not more than one Wire dia/ width of Strip		-	P	V
10.6	Cross over / Over Riding	Visual		No cross - over	No cross - over		-	P	V
10.7	Armour Coverage / Face Bend	Meas.		Min. 90% coverage. No face bend	Min. 90% coverage. No face bend		-	P	V
10.8	Dia Over Armour	Meas.		Approved Data Sheet	Approved Data Sheet		-	P	V
11.0	FINISHED CABLES								

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	(Outer Sheathing)								
11.1	Thickness of Outer Sheath	Meas	Starting of M/C and during running	IS : 1554 (Part-1) Table -7 and Approved Data Sheet	IS : 1554 (Part-1) Table -7 and Approved Data Sheet	In Process Log Sheet	-	P	V
11.2	Type of Sheathing Material	Visual		IS : 1554 (Part-1) & Approved Data Sheet	IS : 1554 (Part-1) & Approved Data Sheet	In Process Log Sheet	-	P	V
11.3	Surface Finish	Visual		The surface shall be smooth and free from any defects	The surface shall be smooth and free from any defects		-	P	V
11.4	Overall Dia	Meas.		Approved Data Sheets.	Approved Data Sheets.		-	P	V
11.5	Porosity	Visual		No porosity	No porosity		-	P	V
11.6	Embossing	Visual		IS : 1554 (Part-1) and Tech. Specification	IS : 1554 (Part-1) and Tech. Specification		-	P	V
11.7	Progressive Sequential Marking	Visual		Tech. Specification	Tech. Specification		-	P	V
12.0	DRUM WINDING								

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12.1	Drum Condition	Visual	100%	Tech. Specification	Tech. Specification	In Process Log Sheet	-	P	V
12.2	Bituminized Water Proof Paper Covering	Visual		Tech. Specification	Tech. Specification		-	P	V
12.3	Cable Length in Meters	Meas.		Approved Data Sheets 500 / 1000 $\pm$ 5%	Approved Data Sheets 500 / 1000 $\pm$ 5%		-	P	V
12.4	Drum Marking	Visual		IS : 1554 (Part-1) and Tech. Specification	IS : 1554 (Part-1) and Tech. Specification		-	P	V
C FINAL INSPECTION & TESTING									
13.0	ROUTINE TESTING								
13.1	Conductor Resistance at 20°C	Elect.	100% Drums	IS : 8130 Table-2/ Approved Data Sheet	IS : 8130 Table-2/ Approved Data Sheet	Routine Test Certificate	-	P	W
13.2	High Voltage Test at Room Temperature	Elect.	100% Drums	IS : 1554 (Part-1) Shall withstand 3 KV (rms) for 5 Minute	IS : 1554 (Part-1) Shall withstand 3 KV (rms) for 5 Minute	Routine Test Certificate	-	P	W

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14.0	ACCEPTANCE TESTS								
14.1	FINISHED CABLE								
14.1.1	Visual Check for	Visual	All Drums	IS : 1554 (Part-1) / Approved Data Sheet	IS : 1554 (Part-1) / Approved Data Sheet		-	P	W
i)	Core Identification						-	P	W
ii)	Embossing						-	P	W
iii)	Printing Quality of sequential Marking Check						-	P	W
iv)	Surface Finish						-	P	W
v)	Top & Bottom Sequential Marking Check						-	P	W
vi)	Length Measurement	Measurement through rewinding 5% of drum from total lot offered.					-	P	W
14.1.3	Conductor Resistance at 20°C	Elect.	10% Drums Test value to tally with original routine test	IS : 8130 Table-2/ Approved Data Sheet	IS : 8130 Table-2/ Approved Data Sheet	Acceptance Test Certificate	-	P	W

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			value of drums.						
14.1.4	Insulation Resistance Test and Volume Resistivity	Elect.	Every 10 Drums or Part thereof per size	IS: 5831 / Approved Data sheets. Min. at 27°C 36.7 Mohm/Km	IS: 5831 / Approved Data sheets. Min. at 27°C 36.7 Mohm/Km	Acceptance Test Certificate	-	P	W
14.1.5	Flammability Test	Thermal	1 Sample per lot for each size	IS : 1554 (Part-1) / Tech. Specn.	IS : 1554 (Part-1) / Tech. Specn.	Acceptance Test Certificate	-	P	W
14.1.6	High Voltage test at room temperature		1 Sample per lot for each size	IS : 1554 (Part-1)	IS : 1554 (Part-1)	Acceptance Test Certificate			
14.2	ALUMINIUM CONDUCTOR								
14.2.1	Tensile Strength	Mech	1 sample for every 10 drums or part thereof per size	IS : 8130 / Tech. Specification Min. 100 N/mm ²	IS : 8130 / Tech. Specification Min. 100 N/mm ²	Acceptance Test Certificate	-	P	W
14.2.2	Wrapping Test	Mech	1 sample for every 10 drums or part thereof per size	IS : 8130 / Tech. Specification. Wrap-8, Unwrap-6 & Wrap-6 on its own dia. The wire shall not crack or break	IS : 8130 / Tech. Specification. Wrap-8, Unwrap-6 & Wrap-6 on its own dia. The wire shall not crack or break	Acceptance Test Certificate	-	P	W

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14.3	PVC INSULATION & SHEATH								
14.3.1	Thickness of Insulation of Core	Meas.	1 sample for every 10 drums or part thereof per size	IS : 1554 (Part-1) Table -2/ Approved Data Sheets.	IS : 1554 (Part-1) Table -2/ Approved Data Sheets.	Acceptance Test Certificate	-	P	W
i)	Average								
ii)	Minimum								
14.3.2	Thickness of Inner Sheath	Meas.	1 sample for every 10 drums or part thereof per size	IS : 1554 (Part-1) Table -4/ Approved Data Sheets.	IS : 1554 (Part-1) Table -4/ Approved Data Sheets.	Acceptance Test Certificate	-	P	W
14.3.3	Thickness of Outer Sheath	Meas.	1 sample for every 10 drums or part thereof per size	IS : 1554 (Part-1) Table -7/ Approved Data Sheets.	IS : 1554 (Part-1) Table -7/ Approved Data Sheets.	Acceptance Test Certificate	-	P	W
14.3.4	Tensile Strength & Elongation for Insulation	Mech	1 sample for every 10 drums or part thereof per size	IS : 5831 Table-1/ Approved Data Sheets. Min. TS 12.5 N/mm ² Elongation 150% (Min.)	IS : 5831 Table-1/ Approved Data Sheets. Min. TS 12.5 N/mm ² Elongation 150% (Min.)	Acceptance Test Certificate	-	P	W

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14.3.5	Tensile Strength & Elongation for Inner & Outer Sheath	Mech	1 sample for every 10 drums or part thereof per size	IS : 5831 Table-2/ Approved Data Sheets. Min. TS 12.5 N/mm ² Elongation 150% (Min.)	IS : 5831 Table-2/ Approved Data Sheets. Min. TS 12.5 N/mm ² Elongation 150% (Min.)	Acceptance Test Certificate	-	P	W
14.3.6	Cold Bend Test for Diameter upto 12.5 mm	Mech	1 sample of any size per order	IS : 1554 (Part-1) No signs of cracks or scales	IS : 1554 (Part-1) No signs of cracks or scales	Acceptance Test Certificate	-	P	W
14.3.7	Cold Impact Test for diameter more than 12.5 mm	Mech	1 sample of any size per order	IS : 1554 (Part-1) No signs of cracks or scales	IS : 1554 (Part-1) No signs of cracks or scales	Acceptance Test Certificate	-	P	W
14.3.8	Oxygen Index Test on Outer Sheath	Envir.	02 sample per lot per type	ASTM-D-2863 / Approved Data Sheet Min. 29%	ASTM-D-2863 / Approved Data Sheet Min. 29%	Acceptance Test Certificate	-	P	W
14.3.10	Flame Retardant Test On Single Cables	Thermal	1 sample per lot for each size	IS : 1554 Cl. 16.6 No visible damages on the test specimen within 300 mm from its upper end	IS : 1554 Cl. 16.6 No visible damages on the test specimen within 300 mm from its upper end	Acceptance Test Certificate	-	P	W

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14.4	ADDITIONAL TEST FOR FRLS CABLES								
14.4.1	Smoke Density Rating	Envir.	1 sample per lot for each size	ASTM-D-2843 / Approved Data Sheet Max. 60%	ASTM-D-2843 / Approved Data Sheet Max. 60%	Acceptance Test Certificate	-	P	W
14.4.2	Acid Gas Generation	Envir.	1 sample per lot for each size	IEC - 754-1/ Tech. Specn. Max. 20% by weight	IEC - 754-1/ Tech. Specn. Max. 20% by weight	Acceptance Test Certificate	-	P	W
14.4.3	Flammability	Thermal	1 sample per lot for each size	IEEE-383 Swedish Chimney SS-424-14-75 Class F3 & Approved Data Sheet	IEEE-383 Swedish Chimney SS-424-14-75 Class F3 & Approved Data Sheet	Acceptance Test Certificate	-	P	W
14.5	ARMOURING								
14.5.1	Resistivity for Armour Wire	Elect.	4 Sample per type per size of armour	IS : 3975 / Approved Data Sheets Max. 14.5× 10-6 Ohm.cm at 20 Deg. C	IS : 3975 / Approved Data Sheets Max. 14.5× 10-6 Ohm.cm at 20 Deg. C	Acceptance Test Certificate	-	P	W
14.5.2	Tensile Strength	Mech	4 Sample per	IS : 3975 /	IS : 3975 /	Acceptance	-	P	W

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			type per size of armour	Approved Data Sheet	Approved Data Sheet	Test Certificate			
14.5.3	Closeness	Visual	4 Sample per type per size of armour	IS : 3975 / Approved Data Sheet. Gap between two wires shall not be more than one wire dia/ width of strip	IS : 3975 / Approved Data Sheet.Gap between two wires shall not be more than one wire dia/ width of strip	Acceptance Test Certificate	-	P	W
14.5.4	Uniformity of Zinc Coating	Meas.	4 Sample per type per size of armour	IS: 3975 & IS : 4826/ Approved Data Sheet. At the end of specified number of dips as per IS, the specimen shall not show any red deposit of Copper upon base metal.	IS: 3975 & IS : 4826/ Approved Data Sheet. At the end of specified number of dips as per IS, the specimen shall not show any red deposit of Copper upon base metal.	Acceptance Test Certificate	-	P	W
14.5.5	Weight of Zinc Coating	Chem	4 Sample per type of offered lot	IS : 3975, IS: 4826 / Approved Data Sheets	IS : 3975, IS: 4826 / Approved Data Sheets	Acceptance Test Certificate	-	P	W
14.5.6	Torsion/Winding	Mech	4 sample per	IS:3975/approved	IS:3975/approve	Acceptance	-	P	V

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			type of offered lot	data sheet	d data sheet	Test Certificate			
14.6	DRUMS								
14.6.1	Visual Check for defects	Visual	100%	Tech. Specification	Tech. Specification	Acceptance Test Certificate	-	P	W
14.6.2	Dimensions	Meas.	1 sample per every 10 Drums or part thereof per size	Approved Drawings	Approved Drawings	Acceptance Test Certificate	-	P	W
14.6.3	Sealing of Drums	Visual	100%			Test Report	-	P	W
14.7	Additional Test Physical test for Insulation and outer Sheath								
14.7.1	Shrinkage Test	Meas.	1 sample per lot per size	Tech. Specn./As Per IS : 1554 (Part-1)	Tech. Specn./As Per IS : 1554 (Part-1)	Test Report	-	P	W
14.7.2	Hot Deformation Test	Meas	1 sample/Lot/S ize	Tech. spec/As per IS:1554(Part-1)	Tech. spec/As per IS:1554(Part-1)	Test Report	-	P	V

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14.7.3	Heat Shock Test	Meas	1 sample/Lot/Size	Tech. spec/As per IS:1554(Part-1)	Tech. spec/As per IS:1554(Part-1)	Test Report	-	P	V
14.7.4	Thermal Stability	Meas.	1 sample per lot per size	Tech. Specn./As Per IS : 1554 (Part-1)	Tech. Specn./As Per IS : 1554 (Part-1)	Test Report	-	P	W
14.7.5	High Voltage Test (Water Immersion Test)	Meas.	1 sample per lot per size	Tech. Specn./As Per IS : 1554 (Part-1)	Tech. Specn./As Per IS : 1554 (Part-1)	Test Report	-	P	W
14.7.6	Insulation resistance measurement with 500 V DC Calibrated IR tester								
a	Between Core to Core	Measurement	100% of the cable drum offered	During the measurement following parameters will be recorded Ambient Temperature at the time of measurement Length of the cable drum under measurement Actual Value of the IR measured in Mohm			-	P	W
b	Individual Core to armour								

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15.0	PACKING & DISPATCH	Visual	100%			Check List	-	P	V
15.1	Application of bituminized water proof paper								
15.2	Space between outer most layer & drum edge								
15.3	Proper Packing								
15.4	Sealing of ends								
15.5	Manufacturer's Name								
15.6	Name & Address of the Consignee								
15.7	Contract Number								
15.8	Drum Number								
15.9	Type, Size and length of the cable								
15.10	Gross Weight								
15.11	Tare Weight								
15.12	Nett Weight								
15.13	Arrow Marking for unwinding								
15.14	CIP/MICC Nos Marking								

EM : Equipment Manufacturer

CM : Component Manufacturer

CMTC :- Component manufacturer test certificate

EMTR :- Equipment manufacturer test Report

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SF6 Gas Circuit Breaker

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub- Vendor	Manufacturer	Customer
A. RAW MATERIAL INSPECTION									
1	Chamber & Support Insulators								
1.1	Visual Inspection	Visual	100%	IEC 62155	IEC 62155	CM's TC	P	V	-
1.2	• Dimensional Check after conducting routine tests • Paralism, height • Concentricity • Creepage, warpage	Mechanical	2 sample per lot of each type	As per manufacturer drawing/IEC 62155	As per manufacturer drawing/IEC 62155	CM's TC	P	V	-
1.3	Ultrasonic test	Electrical	100%	IEC 62155	IEC 62155	CM's TC	P	V	-
1.4	Hydraulic Pressure test (Chamber Insulator.)	Mechanical	100%				P	V	-
1.5	Bending load test in 4 direction at 70 % specified bending load for 10 sec	Mechanical	100%				P	V	-
1.6	Porosity test	Mechanical	One sample per firing cycle				P	V	-
1.7	Temperature cycle test	Mechanical	one sample per year				P	V	-
1.8	Burst pressure test	Mechanical	one sample per six months				P	V	-

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1.90	70 % load in all 3 direction for 10 sec & 100% load in 4th direction for 1 min	Mechanical	one sample per year	IEC 62155	IEC 62155	CM's TC	P	V	-
2	Insulating rod.								
2.1	Visual inspection for damage on surface and internal defects.	Visual	100%	Plant drawings	Certificate of conformance / TC	Certificate of conformance / TC	P	V	-
2.2	Pull Test	Mechanical	100%	Plant drawings	Certificate of conformance / TC	Certificate of conformance / TC	P	V	-
2.a	PIR Tube								
2.1a	Visual inspection for damage on surface and internal defects.	Visual	100%						
	Internal thread	Visual	100%						
3	Puffer Cylinder (Heat Cylinder)								
3.1	Visual check for damages, surface defects, surface finish	Visual	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard	-	P	V	-
3.2	Dimensional check	Mechanical	10%			CM's TC	P	V	-
3.3	Chemical composition	Chemical	1 sample/batch			CM's TC	P	V	-
3.4	Silver plating thickness	Chemical	5%			CM's TC	P	V	-
3.5	Adhesion of silver plating	Mechanical	100%			CM's TC	P	V	-
3.6	Tensile test & elongation	Mechanical	1 sample per batch			CM's TC	P	V	-

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4	Finger contacts								
4.1	Visual check for damage & surface finish	Visual	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard		P	V	-
4.2	Hardness	Mechanical	1 no per batch			CM's TC	P	V	-
4.3	Conductivity	Electrical	1 no per batch			CM's TC	P	V	-
4.4	Chemical composition	Chemical	1 no per batch			CM's TC	P	V	-
4.5	Dimensional check	Mechanical	10% of each lot	Drawing	Drawing	CM's TC	P	V	-
4.6	Silver plating thickness	Chemical	5% of each batch	Drawing	Drawing	CM's TC	P	V	-
4.7	Adhesion of silver plating	Mechanical	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard	CM's TC	P	V	-
5	Fixed Contact support (Al. Alloy casting)								
5.1	Visual aspects- blow holes, surface finish	Visual	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard	CM's TC	P	V	-
5.2	Chemical analysis	Chemical	1 sample per batch			CM's TC	P	V	-
5.3	Tensile test & elongation	Mechanical	1 sample per batch			CM's TC	P	V	-
5.4	Dimensional check	Mechanical	100%			CM's TC	P	V	-
6	Moving contact support (Al. Alloy casting)								
6.1	Visual aspects- blow holes, surface finish	Visual	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard	CM's TC	P	V	-
6.2	Chemical analysis	Chemical	1 sample per batch			CM's TC	P	V	-

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6.3	Tensile test & elongation	Mechanical	1 sample per batch			CM's TC	P	V	-
6.4	Dimensional check	Mechanical	100%			CM's TC	P	V	-
7	Arcing contacts								
7.1	Visual inspection	Visual	100%	Plant drawings	Certificate of conformance / TC	Certificate of conformance / TC	P	V	-
	Dimensional check		100%						
	Hardness (Cu, Tungsten)		1 sample per lot						
	Conductivity		1 sample per lot						
	Chemical test		1 sample per lot						
8	Multi lam contact								
8.1	Visual inspection & other tests	Visual	100%	Plant drawings	Certificate of conformance / TC	Certificate of conformance / TC	P	V	-
9	NOZZLE								
9.1	Visual check for damage and presence of foreign particles	Visual	100%	Plant drawings	Certificate of conformance / TC	Certificate of conformance / TC	P	V	-
	Dimensional check		2 sample per lot						
	Tensile Strength		1 sample per lot						
	Elongation		1 sample per lot						
	Hardness		1 sample per lot						
10	Mechanism Housing								
10.1	Visual aspects- blow	Visual	100%	Manufacturer's	Manufacturer's				

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	holes, surface finish			Plant Standard	Plant Standard				
10.2	Chemical analysis	Chemical	1 sample per batch			CM's TC	P	V	-
10.3	Tensile test & elongation	Mechanical	1 sample per batch			CM's TC	P	V	-
10.4	Dimensional check	Mechanical	2 sample per lot			CM's TC	P	V	-
11	Springs								
11.1	Visual checks for damages	Visual	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard				
11.2	Dimensional check	Mechanical	100%			CM's TC	P	V	-
11.3	Load characteristics (For tensile / compression springs)	Mechanical	1 sample per lot			CM's TC	P	V	-
11.4	Chemical check	Chemical	1 sample per lot			CM's TC	P	V	-
11.5	Magnetic particles test	Mechanical	100%			CM's TC	P	V	-
11.6	Hardness test (for spiral spring)		1 sample per lot						
12	Holding Latches								
12.1	Visual aspects- blow holes, surface finish	Visual	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard				
12.2	Chemical analysis	Chemical	1 sample per lot			CM's TC	P	V	-
12.3	M.P.I. Test	Mechanical	100%			CM's TC	P	V	-
12.4	Dimensional check	Mechanical	100%			CM's TC	P	V	-
12.6	Surface hardness & Case depth after heat treatment	Mechanical	1 sample per lot			CM's TC	P	V	-
12.7	Microstructure test		1 sample per lot						

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13	Trip & Close coils								
13.1	Coil resistance	Electrical	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard	Routine test certificate	P	V	-
13.2	HV test 2kv for 1 minute	Electrical	100%			Routine test certificate	P	V	-
13.3	Functional test	Electrical	100%			-	P	V	-
14	Auxiliary Switch								
14.1	Insulation resistance test	Electrical	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard	CM's TC	P	V	-
14.2	HV test 2kv for 1 minute	Electrical	100%			CM's TC	P	V	-
14.3	Contact resistance	Electrical	100%			CM's TC	P	V	-
14.4	Functional test	Electrical	100%				-	P	V
15	Motor								
15.1	Insulation resistance test	Electrical	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard	CM's TC	P	V	-
15.2	HV test 2kv for 1 minute	Electrical	100%			CM's TC	P	V	-
15.3	No load test	Electrical	100%			CM's TC	P	V	-
15.4	Functional test	Electrical	100%	-	-		-	P	V
16	Control Panel								
16.1	Visual check	Visual	100%	Drawing	Drawing	CM's TC	P	V	-
16.2	Dimensional check	Mechanical	15%	Drawing	Drawing	CM's TC	P	V	-
16.3	Coating thickness	Mechanical	1 sample per batch	Drawing	Coating thickness 80- 120 microns	CM's TC	-	P	V
16.4	Adhesion test	Mechanical	1 sample per batch	P.S.	P.S.	CM's TC	P	V	-
17	Density Monitor Switch								
17.1	Visual aspects	Visual	100%	-	-		-	P	V
17.2	Verification of different settings of alarm &	Electrical	100%	-	-	Routine Test Record	-	P	V

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	lockout								
17.3	HV Test at 2 KV for one minute	Electrical	100%	-	-	Routine Test Record	-	P	V
18	Gas pressure gauge								
18.1	Checks for visual defects.	Visual	100%	-	No damage	CM's TC	P	V	-
18.2	Accuracy ($\pm 1.5\%$ on full scale deflection)	Mechanical	100%	Manufacturer's Plant Standard	$\pm 1.5\%$	CM's TC	P	V	-
19	Gas stop valve								
19.1	Visual & Dimensional check	Visual	100%	Manufacturer's Plant Standard	Manufacturer's Plant Standard	CM's TC	P	V	-
19.2	Metal- Metail leak test	Mechanical	100%			CM's TC	P	V	-
19.3	Rubber- Metal leak test	Mechanical	100%			CM's TC	P	V	-
20	Sealing 'O' ring								
20.1	Visual inspection for damage on surface and internal defects.	Visual	100%	Plant Drawings	Certificate of conformance	Certificate of conformance	P	V	-
21	SF6 gas with cylinder								
21.1	Properties as per IEC 60376	Chemical	One no. per batch	IEC 60376	IEC 60376	CM's TC	P	V	-
21.2	Leakage test on cylinders	Mechanical	10%	-	No leakage with probe	-	P	V	-
22	Absorbent								
22.1	Visual inspection for damage on surface and internal defects.	Visual	100%	Plant Drawing	No damage	Certificate of conformance	P	V	-

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23	Base frame & structure								
23.1	Visual check	Visual	100%	Drawing	No damages		P	V	-
23.2	Dimensional check	Mechanical	100%	Drawing	Tolerances within limits		P	V	-
23.3	Coating thickness	Mechanical	1 sample per batch	Technical spec. min 85	Technical spec. min 85	CM's TC	P	V	-
				micron coating thickness.	micron coating thickness.		P	V	-
23.4	Uniformity of coating	Mechanical	1 sample per batch	IS2633	IS2633	CM's TC	P	V	-
23.5	Mass of coating	Mechanical	1 sample per batch	Tech spec.	Tech spec.	CM's TC	P	V	-
				610 gm/sq.m	610 gm/sq.m		P	V	-
23.6	Adhesion test	Mechanical	1 sample per batch	IS 2629	IS 2629	CM's TC	P	V	-
24	Grading Capacitor								
24.1	Visual check	Visual	100%	Drawing	No damages		P	V/P	-/V
24.2	Capacitance & Tan delta before & after 1 minute HV test	Electrical	100%	Measurement at 10kV, 250kV	C=±5% of rated value Tan delta less than 0.5%	CM's TC	P	V	-
24.3	Partial discharge before & after 1 minute HV test	Electrical	100%	Measurement at 349kV	less than or equal to 3 pC	CM's TC	P	V	-
24.4	One minute power frequency withstand test on the Capacitor unit	Electrical	100%	544kV	No breakdown of Insulation	CM's TC	P	V	-
24.5	Leakage test	Electrical	100%	Visual check	No leakage from any part	CM's TC	P	V	-
25	Resistor Disks								

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25.1	Visual inspection & other tests	Visual	100%	Plant drawings	Certificate of conformance	Certificate of conformance	P	V/P	-/V
26	PIR Operating rod								
26.1	Visual inspection & other tests	Visual	100%	Plant drawings	Certificate of conformance / TC	Certificate of conformance / TC	P	V/P	-/V
27	PSD (Phase synchronizing Device)								
27.1	Visual check	Visual	100%	-	No damages		P	V/P	-/V
27.2	Adjustment & test of I/O ports	Electrical	100%	Manufacturer internal document	Manufacturer internal document	CM's TC	P	V	-
27.3	Functional test	Electrical	100%	-do-	-do-	CM's TC	P	V	-
27.4	Compensation functions	Electrical	100%	-do-	-do-	CM's TC	P	V	-
27.5	Output signals	Electrical	100%	-do-	-do-	CM's TC	P	V	-
27.6	Voltage withstand test 1 kV AC	Electrical	100%	-do-	-do-	CM's TC	P	V	-
28	Contactors & Overload Relay								
28.1	HV Test at 1 KV for one minute	Electrical	100%	Manufacturer Internal Document	Manufacturer Internal Document		-	P	V
28.2	Functional test	Electrical	100%				-	P	V
28.3	Functional test of Overload relay	Electrical	100%			CM's TC	P	V	-
29	Plug in Connectors(For Interpole wiring) (for 800kV)								
29.1	Visual check	Visual	100%	Manufacturer	No damages	CM's TC	P	V	-

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				Internal Document					
29.2	HV Test (1kV for 1 minute) (pin to ground)	Electrical	100%	Manufacturer Internal Document	Manufacturer Internal Document	CM's TC	P	V	-
B. Section : IN PROCESS INSPECTION									
30	Interrupter Assembly								
30.1	Clutch plate movement by manual operation	Mechanical	100%		Free movement	NA	-	P	V
31	Drive Mechanism Assembly								
31.1	Verification of Tightening Torques	Mechanical	100%	Manufacturer internal document	Manufacturer internal document	Assembly Records	-	P	V
31.2	Stroke Length Measurement of Shock Absorber	Mechanical	100%			Assembly Records	-	P	V
31.3	1.2mm gap setting of Drive	Mechanical	100%			Assembly Records	-	P	V
31.4	Free rolling of bearings after sub assembly	Mechanical	100%			Assembly Records	-	P	V
31.5	In process quality check	Mechanical	100%			Assembly Records	-	P	V
32	Control Panel with wiring								
32.1	Functional Test	Mechanical	100%	As per IEC 62271-100	Routine test certificate	-	-	P	V
32.2	Continuity Test on Wiring	Electrical	100%	As per IEC 62271-100	Routine test certificate	-	-	P	V
32.3	HV Test on Wiring	Electrical	100%	As per IEC	Routine test	Routine test	-	P	V

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				62271-100	certificate	certificate			
33	Plug in Connectors(For Interpole wiring)(for 800kV)								
33.1	Visual check	Visual	100%	-	No damages		-	P	V
33.2	HV Test (2kV for 1 minute) (pin to ground)	Electrical	100%	As per IEC 62271-100	Routine test certificate	Routine test certificate	-	P	V
34	FINAL ASSEMBLY OF COMPLETE BREAKER								
34.1	Completeness of assembly as per check list	Mechanical	100%	Manufacturer internal document	Manufacturer internal document	Control card	-	P	V
34.2	Visual Check of switch lever safety plate bending	Mechanical	100%			PDI check	-	P	V
C. Section : FINAL TESTING									
35	Routine Tests								
35.1	Mechanical Operations at rated SF6 gas pressure at 20 deg.C *						-	P	W
a)	At 100% V & 110%V : 5-	Electrical	100%	IEC-62271-100 & Specification	IEC-62271-100 & Specification	Routine Test Record	-	P	W
b)	Closing,5-Opening								
c)	At 70% V 5-								
d)	Opening								
e)	At 85% V 5-Closing								
	At 100%V 5-CO & 10-C-O								
	At 50% voltage for								

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	opening.								
35.2	a) Measurement of speed & time (Curves to be maintained by operational analyser) At 100% V Opening Operation	-do-	100%			-do-	-	P	W
	b) At 100% V Closing Operation	-do-	100%			-do-	-	P	W
	c) At 110% V Opening Operation	-do-	100%			- do-	-	P	W
	d) At 110% V Closing Operation	-do-	100%			- do-	-	P	W
	e) At 70% V Opening Operation	-do-	100%			- do-	-	P	W
	f) At 85% V Closing Operation	-do-	100%			- do-	-	P	W
	g) At 100% V O-C-O Operation	-do-	100%			- do-	-	P	W
	h) At 100% V CO Operation	-do-	100%			- do-	-	P	W
35.3	MV Drop (Contact Resistance)	-do-	100%			- do-	-	P	W
35.4	Dynamic Contact Resistance Measurement 1 Close-Open operation	-do-	100%	As per Specification	As per Specification	-do-	-	P	W
35.5	Gas Leakage	-do-	100%	IEC 62271-100	IEC 62271-100	- do-	-	P	W
35.6	Dielectric test on the main circuit		100%			- do-	-	P	W
35.7	Tests on auxiliary and control circuits		100%			- do-	-	P	W
35.8	Measurement of the		100%			- do-	-	P	W

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	resistance of the main circuit								
35.9	Design and visual checks		100%			- do-	-	P	W
35.6	ELECTRICAL SEQUENCE CHECK								
a)	Antipumping / Pole discrepancy	Electrical	100%	IEC 62271-100	IEC 62271-100	Routine Test Record	-	P	W
b)	Other A.C.Circuit Check	-do-	100%			- do-	-	P	W
35.7	1 kV test on i) Control Circuit ii) Aux. Circuit	-do-	100%			- do-	-	P	W
35.8	Verification of settings of density monitor switch at Siemens	-do-	100%			- do-	-	P	W
	at Alarms & Lockout condition	-do-				- do-	-	P	W
35.9	HV Test (at Lock out pressure)	-do-				- do-	-	P	W
a)	Between contacts (Breaker in open condition)	-do-	100%			- do-	-	P	W
b)	Between live parts to ground (Breaker in closed condition)	-do-	100%			- do-	-	P	W
36	Grading Capacitor								
36.1	One minute power frequency withstand test on the GC along with CB (on each break)	Electrical	100%	IEC-62271-100	IEC-62271-100	Routine Test Record	-	P	W
37	Controlled Switching equipment (If								

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	applicable)								
37.1	Controlled switching equipment validation tests								
a)	Tests with PSD & CB, 1. Phase sequence check with PSD.2. Indication/display on PSD 3. Mech operations check for external tripping/closing thru PSD as per PSD requirements with inputs for validation testing with adaptive features as per annexure 1 attached. 4. Check for close / open operations- bypass operation of PSD. 5. Self-monitoring function check. 6. Potential free contact check.	Electrical	100%	As specification	per As specification	per Routine test record	-	P	W
38	Pre Insertion Resistor (If applicable)								
38.1	Measurement of Resistance	Electrical	100%	IEC-62271-100	IEC-62271-100	Routine Test Record	-	P	W
38.2	PIR Contact timings	Electrical	100%	- do-	- do-	- do-	-	P	W

Alternatively the mechanical tests can be conducted on Phantom arrangement.

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Control and Relay & SAS Panels

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S. No.	Components / Operations & Description of Test	Type of Check	Quantum of Check / Sampling with basis	Reference Document for Testing	Acceptance Norms	Format of Record	Category of Responsibility		
							Sub- Vendor	Manufacturer	Customer
A. RAW MATERIAL INSPECTION for Enclosures (Panel, Trolley, Kiosk)									
1	Sheet steel CRCA / HR								
1.1	Dimension conformity including thickness	Measurem ent	Minimum of 1 sheet of each size per lot.	IS/ISO 16162 (For CRCA) IS 1852(For HR), IS 513	IS/ISO 16162 (For CRCA) IS 1852(For HR), IS 513	CMTC	P	V	-
1.2	Surface finish (Dent, line marks, Waviness)	Visual	100%	IS 513 / IS 1079 IS 513 for CRCA & IS 1079 for HR	IS513 / IS 1079 IS 513 for CRCA & IS 1079 for HR	CMTC	P	V	-
1.3	Chemical composition (Grade CR2 & HR2 as per relevant IS)	Chemical	1 sample / lot	IS 513 for CRCA & IS 1079 for HR	IS 513 for CRCA & IS 1079 for HR	CMTC	P	V	-
1.4	Mechanical Test (Tensile Test, Hardness Test & Bend Test)	Mechanica l	1 sample / lot	IS 513 for CRCA & IS 1079 for HR	IS 513 for CRCA & IS 1079 for HR	CMTC	P	V	-
2	Fabrication								

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2.1	Dimensional conformity & appearance	Visual / Measurement	10% / Lot	As per Approved Drg.	As per Approved Drg.	CMTC	P	V	-
3	Surface preparation / Pretreatment.(7 Tank Process)								
3.1	Hot Degreasing, derusting/ / pickling, Hot phosphating, , rinsing with water after each process or equivalent	Visual / Measurement	100%	IS6005/IS 3618	IS6005/IS 3618	CMTC	P	V	-
3.2	Weight of Phosphate Coating	Measurement	once in month	IS6005/IS 3618	MIN. 4.3g/m2	CMTC	P	V	-
3.3	Visual check including nail scratch test	Visual	once in month			CMTC	P	V	-
4	Powder coating and baking.								
4.1	Shade	Visual	100%	As per Approved Drg.	As per Approved Drg.		P	V	-
4.2	Adhesion/ scratch hatch test.	Visual / Measurement	Test Coupon	ASTM-D-3359	ASTM-D-3359		P	V	-
4.3	Coating thickness/ Surface finish	Measurement	Random	As per Approved Drg.	As per Approved Drg.		P	V	-
5	Cubicle Components								

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5.1	EARTH BUS BAR Dimensional conformity Hardness & surface defects	Visual / Measurement	One sample per lot	IS 1897 / IS 613	IS 1897 / IS 613	CMTC	P	V	-
5.2	FASTENERS- (Bolts/Nuts/Washers) Dimensional conformity Surface finish	As per Sampling Plan	10% sample per lot	IS 1363, 1364, 1367, 2016	IS 1363, 1364, 1367, 2016		P	V	-
5.3	ASSEMBLED PANEL WITH ASSOCIATED COMPONENTS (Door Switch, Space Heater, Thermostats, 3 Pin Socket with Switch, Fluorescent/ Incandescent Lamp LED Lamp) Position of component assembly Type & Quantity of components assembled Quality of assembly	conformity tests	Sampling	Approved project specific Panel Construction drg, GA Drg	Approved project specific Panel Construction drg, GA Drg	CMTC	P	V	-
5.4	GASKET (Neoprene/EPDM) Physical properties, compressibility, Shore hardness, tensile and	Visual / Measurement	One sample per lot	TECH. Spec/ IS 11149	TECH. Spec/ IS 11149	CMTC	P	V	-

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	elongation test								
5.5	OTHER COMPONENTS (Bought out items) MCBs, ,Aux relays, Semaphore indicators, Indicating Lamp, Fuses, Selector Switches / Control Switches, terminal Blocks and Push buttons, Hooter, Bell & Buzzer	Functional / Conformity test	100%	Approved BOM, Panel Construction , GA drg ,	Project specific. approved BOM, Panel Construction , GA drg ,	CMTC	P	V	-
5.6	Electrical sundry items likes, PVC Channel,Test switches, short ckt tools, Extractor tools, Branch connectors (as applicable)	Functional / Visual ,Dimensional Check.	100%	Approved BOM, Panel Construction , GA drg ,	Project specific. approved BOM, Panel Construction , GA drg ,	CMTC	P	V	-
6	PVC/FRLS INSULATED WIRES (ISI Marked)								
6.1	Conductor Resistance, Strands/ Color, Elongation, Type of	Measurement - Dimensional	Sampling	Project Specific Panel Construction / IS 694 / IS	Project Specific Panel Construction / IS 694 / IS	CMTC	P	V	-

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	cable, Material, Shielding (if applicable), Overall Diameter	conformity Check/ Visual		5831	5831				
6.2	High voltage Spark test	Electrical	100%	Project Specific Panel Construction / IS 694 / IS 5831	Project Specific Panel Construction / IS 694 / IS 5831	CMTC	P	V	-
6.3	Check for thickness of insulation & Overall Dimension, Insulation, Tensile strength, Elongation at Break test, Shrinkage Test, Heat Shock Test, Hot Deformation Test, Loss of Mass Test, Thermal Aging Test in Air	Measurement / Electrical / Visual	1 coil/Type/ As per IS 694 / IS 5831			CMTC	P	V	-
6.4	Flammability (As applicable) & High Voltage test (water immersion test)	Electrical	Sampling			CMTC	P	V	-
7	MEASURING INSTRUMENTS (Analog/Digital), (Voltmeter, Ammeter, Wattmeter, VAR meter, Frequency Meter, Synchroscope, Strip Chart Recorders)								
7.1	Visual Inspection for,	Visual	100%			CMTC	P	V	-

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	Type, Range, Movement type, Angle of indication, Graduation Marking			Project specific approved BOM / IS 1248	Project specific approved BOM / IS 1248				
7.2	Operational & Calibration check- For accuracy test	Electrical/ Functional	100%			CMTC	P	V	-
7.3	H.V. Test: 2 kV for 1 min. in Panel assembly	Electrical/ Functional	100%			CMTC	P	V	-
7.4	Energy Meter	Visual Check, Measurement, Calibration & Electrical functional checks.	100%	As per 14697 & IEC 62053/52 As per MQP of component Manufacturer/ Tech. specification & GTP.	As per 14697 & IEC 62053/52 As per MQP of component Manufacturer/ Tech. specification & GTP.		P	V	W
8	TRANSDUCER Operational test, Visual, Dimensions check, HV Test 2 kV for 1min in panel assembly	Measurement / Electrical / Visual	100%	Project specific approved BOM / IS 1248	Project specific approved BOM / IS 1248	CMTC	P	V	-

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9	CURRENT/ VOLTAGE TRANSFORMER								
9.1	Name plate, CT/PT Ratio, Rated burden, accuracy Class, Termination, Mounting accessories	Visual check & Dimensional conformity	100%	Project specific approved BOM	Project specific approved BOM	CMTC	P	V	-
9.2	Functional check for Polarity check, Ratio check, Accuracy class, No load current, HV Test 2 kV for 1min	Functional / Electrical				CMTC	P	V	-
10	ANNUNCIATORS								
10.1	Type, No. of windows, Rows/ columns, Aux. Supply voltage, Initiation by NO/NC, Accessories, Text (Inscription)	Visual check & Dimensional conformity	100%	Project specific approved BOM / schematic drawing	Project specific approved BOM / schematic drawing	CMTC	P	V	-
10.2	Functional check for, Lamp/ Alarm test, Trip/ Non trip Segregation, Fault indication order, Any other sp. Features, HV Test 2 kV for 1min.	Functional / Electrical	100%			CMTC	P	V	-

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11	UPS / INVERTER/ BULK POWER SUPPLY UNITS - Type, Make, Rating, Operation Check, Load test, High Voltage Test 2 KV for 1Min	Visual check & Measure ment - Dimension al conformity / Electrical - Functional checks	100%	Project specific approved BOM/ GTP	Project specific approved BOM/GTP	CMTC	P	V	-
12	TIME SYNCHRONISATION EQUIPMENT Functional	Visual, Measure ment & Dimensi onal conform ity / Function al	100%	Project specific approved BOM/ GTP	Project specific approved BOM/ GTP	CMTC	P	V	-
12.1	HV Test 2kV for 1 min in panel assembly	Electrical	100%			CMTC	P	V	-
13	RECORDERS								
13.1	DISTURBANCE RECORDERS (Stand alone	Visual, Measureme nt,	100%	Project specific approved	Project specific approved	CMTC	P	V	-

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	functional) Functional Test HV Test 2 kV for 1min.	Dimensional conformity / Functional Electrical		BOM/ GTP	BOM/ GTP				
14	EVENT LOGGER Functional Test HV test 2KV for 1 min in panel assembly	Visual, Measurement, Dimensional conformity / Functional	100%	Project specific approved BOM/ GTP	Project specific approved BOM/ GTP	CMTC	P	V	-
15	COMPUTER HARDWARES & APPLICABLE SOFTWARES								
15.1	Printers, workstations computers, modems: Type, Make, Communication & configuration check, operating systems (as part of integrated test)	Visual, Measurement, Dimensional conformity / Functional	100% (As part of Integrated Test)	Project specific approved BOM/ GTP	Project specific approved BOM/ GTP	EM/C M- TC	P/-	V/P	-/W
15.2	Ethernet switch, Armoured communication cables						P	V	-

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15.3	Communication cables (FO cable / Patch chord), Extender and accessories.	visual / Functional	100%	Project specific approved BOM/ GTP	Project specific approved BOM/ GTP	EM/C M- TC	P/-	V/P	-/W
16	PROTECTION RELAYS(Convention al/Numerical)								
16.1	Verification of Hardware , Software version, Functional Calibration check /Accuracy Drop- out / Reset Test (As applicable) Verification of pickup voltage of BI and Ensuring Suitable BI cards as per scheme DC level. Indication Check High Voltage Test Communication Check (If applicable) Check for secured arrangement of Moving Parts for during transport	Electrical / Visual Check	100%	Tech. specification , GTP, BOM, and Approved drawings /, CM's Documents	Tech. specification , GTP, BOM, and Approved drawings /, CM's Documents	EM-TC	P	V	-

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17	PROTECTION RELAYS(Convention al/Numerical) including Bay controller								
17.1	Verification of Hardware , Software version, Functional/ Calibration check /Accuracy Drop-out / Reset Test (As applicable) Indication Check, High Voltage Test, Communication Check (If applicable), Check for secured arrangement of Moving Parts for during transport	Electrical / Visual Check	100%	Tech. specification , GTP, BOM, and Approved drawings /, CM's Documents	Tech. specification , GTP, BOM, and Approved drawings /, CM's Documents	EM TC	P	V	-
18	Software / OEM Software								
	Simulation Check Time synch check Alarm & Events General interrogation checks	Functional	100%	Tech. specification , GTP, BOM, and Approved drawings , software	Tech. specification , GTP, BOM, and Approved drawings , software	EM TC	-	P	V

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				manual, as a part of integrated test / software manual	manual, as a part of integrated test / software manual				
19	RELAY TEST KITS								
19.1	Visual check, functional	Electrical/ Visual	100%	Project specific approved Equipment list	Project specific approved Equipment list	CM /EM TC	-	P	V
20	RELAY TOOL KIT	Visual	100%	Tech. specification / Approved drawings/ Test instructions	Tech. specification / Approved drawings/ Test instructions	CM /EM TC	-	P	V
20.1	Time sync. Check	Functional	100%	Tech. specification , GTP, BOM, and Approved drawings,	Tech. specification , GTP, BOM, and Approved drawings, OEMs.	EM TC	-	P	V
B. IN PROCESS INSPECTION									

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

1	Auxiliary relays								
1.1	Functional and calibration check, High voltage test	Electrical/ Visual	100%	Tech. Specification Approved Drg/BOM, GTP	Tech. Specification Approved Drg/BOM, GTP	EM TC	-	P	V
2	Static relays: Current relays, voltage relays, trip ckt supervision relay, timer relays, flasher relays, fuse fail relays								
2.1	Functional and calibration check, Drop-out test flag indication check High Voltage test 2KV for 1 min check for secured arrangement of moving parts for transport	Electrical/ Visual	100%	Tech. Specification Approved Drg/BOM, GTP	Tech. Specification Approved Drg/BOM, GTP	EM TC	-	P	V
3	Protection Relays								
3.1	Functional test: Hardware test,	Functional	100%	Tech. specification	Tech. specification	RTR	-	P	V

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	Software loading & Measurement			, GTP, BOM, and Approved drawings	, GTP, BOM, and Approved drawings				
3.2	High Voltage Test 2 KV AC/DC for 1 min.	Functional	100%			RTR	-	P	V
4	PROCESS CHECKS PANEL WIRING & ASSEMBLY								
4.1	Availability of all Items and their position	visual	100%	Project specific approved Panel Construction / GA / BOM / scheme drg / Quality Check list	Project specific approved Panel Construction / GA / BOM / scheme drg / Quality Check list	EMTC / VTC	P	V	W
4.2	Verification of labelling of Equipment, terminal block, ferruling etc.	visual	100%			EMTC / VTC	P	V	W
4.3	Proper fastening of Items	visual	100%			EMTC / VTC	P	V	W
4.4	Terminal Block Mountings	visual	100%			EMTC / VTC	P	V	W
4.5	Mounting Locations	visual	100%			EMTC / VTC	P	V	W
4.6	Type of Lug /Crimping Quality	visual/Functional	Random			EMTC / VTC	P	V	W
4.7	Physical Damages	visual	100%			EMTC / VTC	P	V	W
4.8	Completion of wiring	visual	100%			EMTC / VTC	P	V	W
4.9	Earthing of all Equipment	visual	100%			EMTC / VTC	P	V	W
4.10	Verification of applicable Routine	verification	100%			EMTC / VTC	P	V	W

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	Test Certificates.								
C.	Preliminary Acceptance Testing								
C-1	Final Assembly								
1.1	Visual Check for Basic assembly and Mimic Diagram/ Dimensional Conformity.	Visual Dimensional - conformity tests	100%	Approved Panel Construction / GA/ BOM / scheme drg/ Quality Check list/ panel assembly check list	Approved Panel Construction / GA / BOM / scheme drg / Quality Check list/ panel assembly check list	EMTC / VTC	P	V	W
1.2	Visual Check for Mounting of all equipment (approved makes)/ BOM Check	Visual	100%			EMTC / VTC	P	V	W
1.3	Visual Inspection for Paint shade, Paint thickness, Surface finish, adhesion.	Visual Measurement of coating Thickness	Coating thickness tested Random			EMTC / VTC	P	V	W
C-2	WIRING								
2.1	Cross section of wires/ cable ferruling/ Crimping of lugs & wires	Visual & Physical Checks	Random	project specific approved Panel	project specific approved Panel	EMTC / VTC	P	V	W

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2.2	Continuity Check	Functional	100%	Construction / GA / BOM / scheme drg / Quality Check list	Constructio n/ GA / BOM / scheme drg / Quality Check list	EMTC / VTC	P	V	W
2.3	Check for earthing continuity between each apparatus and earth bus by continuity tester	Functional	100%			EMTC / VTC	P	V	W
C-3	Functional Test								
3.1	All Functional / scheme test -protection, -control, - metering. All spare IO/DO wiring up to TB (to be done)	Electrical/ Functional	100%			EM TC	-	P	V
3.2	Scheme Check (I/O check, DC/AC logic, DC1- DC2 Mixing, Interlock, Event logging, VT Selection, Special logic if any) -Protection (IED Function, A/R, VT fuse Fail) -Metering (Synchro Check, Measurement Accuracy) -Gateway communucation using	Electrical/ Functional	100%	Appd drg, BOM, GTP, Tech. spec/ Internal drawing/ Routine Test Reports	Appd drg, BOM, GTP, spec/ Internal drawing/ Routine Test Reports	EM TC	-	P	V

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	protocol Simulator - All trip commands (for TC1 & TC2 of CB) of main protection relays to be checked up to panel terminal block as per the PSL & Scheme. - Auto reclosure scheme checks to include : Block logic including 3- phase trip, priority closing in case of Main & Tie CB. verification of Trip and initiation from protection functions as per trip matrix, verification of Relay LED signal list. verification of list of functions available in relays								
C-4	Integrated FAT of Protection Panels with SAS (For								

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	Substation Automation System)								
4.1	Check all IEDs loaded with Settings received from Customer and the Approved PSL Logic. - Verification of DR signal standardisation as per approved DR list. - Verification of SAS Alarm /Events signals as per standard List.	Visual / Functiona l	100%	Appd drg, GTP, Internal Check list, Approved PSL Logic .	Appd drg, GTP, Internal Check list, Approved PSL Logic .	EM TC	-	P	V
4.2	Functionality of all IEDs checked with Settings received from Customer and the Approved PSL.	Visual / Functiona l	100%			EM TC	-	P	V
4.3	SAS Database is updated as per Customer approved SLD & SAS drawings	Visual / Functiona l	100%			EM TC	-	P	V
4.4	Checks based on Pre I-FAT Check	Visual / Functiona l	100%			EM TC	-	P	V

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C-5	FINAL ACCEPTANCE TESTS								
5.1	Visual Check for -Basic assembly and Mimic Diagram/ Dimensional Conformity/ finish and workmanship / door alignment / dust proofing	Visual Dimensional conformity tests	100%	Appd drg, BOM, GTP, Tech. spec/ Internal drawing/ RTR Approved PSL Logic	Appd drg, BOM, GTP, Tech. spec/ Internal drawing/ RTR Approved PSL Logic	EM - TC	-	P	W
5.2	Visual Check for -Mounting of all equipment (approved makes)/BOM Check/ Check for hardware and software version of IED , check for MLFB nos in IED	Visual conformity tests	100%	.	.	EM TC	-	P	W
5.3	IR Value Test (before & after HV test)	Electrical / Measurement	100%	Appd drg, BOM,GTP	Appd drg, BOM,GTP	EM TC	-	P	W
5.4	High Voltage Test 2KV test for 1min between all terminals and earth -HV Test.	Electrical / Measurement	100%	Tech. spec/ Internal drawing/ RTR	Tech. spec/ Internal drawing/ RTR	EM TC	-	P	W
5.5	Visual inspection for paint shade, paint thickness, surface finish, adhesion	Visual / Measurement	100%	,Settings provided by Customer & Approved	,Settings provided by Customer & Approved	EM TC	-	P	W

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5.6	Check for availability of Spare Digital Inputs & outputs as per specification	Visual	100%	PSL Logic	PSL Logic	EM TC	-	P	W
5.7	Check for availability of spare terminals in each panel	Visual	100%			EM TC	-	P	W
5.8	Check all IEDs loaded with Settings received from purchaser and the Approved PSL Logic.	Visual	100%			EM TC	-	P	W
5.9	Functionality of all IEDs checked with Settings received from purchaser and the Approved PSL Logic.	Visual	100%			EM TC	-	P	W
5.10	Sas Database is updated as per purchaser approved SLD and SAS drawings	Visual	100%			EM TC	-	P	W
5.11	Individual BCU mimics are configured as per approved SLD	Visual	100%			EM TC	-	P	W
5.12	Event/Alarm list is configured in full and reporting of same checked	Visual	100%			EM TC	-	P	W
5.13	Scheme , Protection, Metering and Gateway	Visual	100%			EM TC	-	P	W

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	communication checked as per approved scheme								
5.14	Operational check- For protection relays, meters etc.	Visual check & Measurement/ Electrical - Functional checks	100%	Appd drg, BOM, GTP, Tech. spec/ Internal drawing/ RTR	Appd drg, BOM, GTP, Tech. spec/ Internal drawing/ RTR	EM - TC	-	P	W
5.15	Current injection test on all instrument / meter	Visual check & Measurement/ Electrical - Functional	100%			EM - TC	-	P	W
5.16	Annunciation sequence annunciation inscription details checks	Visual check & Measurement/ Electrical - Functional	100%			EM - TC	-	P	W
5.17	Operational Test : -Scheme Check (I/O check, DC/AC logic, DC1-DC2 Mixing, Interlock, Event logging, VT Selection, Special logic if any) - Protection (IED Function, A/R, VT fuse Fail, trip circuit	Visual & Electrical	100%			EM - TC	-	P	W

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	supervision) -Metering (Synchro Check, Measurement Accuracy) -Gateway communication using protocol Simulator								
5.18	The complete integration check for Communication, relay configuration, parameterization, with interfacing of necessary software/ hardwares. Printers	Visual check & Measurem ent/ Electrical - Functional checks	100%	Appd drg, BOM,,GTP, Tech. spech/ Internal drawing/ RTR, As per Check List (Distance relays, Transformer - protn & all non numerical relays)	Appd drg, GTP, Tech. spec/ Internal drawing/ RTR.	EM - TC	-	P	W
C-6	Time Synchronization Equipment	Visual check & Measurem ent/ Calibration Electrical - Functional	100%	Appd drg, BOM,,GTP, Tech. spech/ Internal drawing/ RTR, As per Check List (Distance relays,	Appd drg, GTP, Tech. spec/ Internal drawing/ RTR	EM - TC	-	P	W
C-7	Disturbance Recorder (Stand alone funtional)	Visual check & Measureme	100%			EM - TC	-	P	W

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		nt & Dimensiona l conformity /funtional		Transformer - protn & all non numerical relays)					
C-8	Event Logger	Visual check & Measureme nt & Dimensiona l conformity /funtional	100%			EM- TC	-	P	W
C-9	Energy Meter	Visual check & Measurem ent & communic ation checks.	100%	As per IS 14697 & IEC 62053/6205 2/As per MQP of component Manufacture r/ Tech. Specificatio n and GTP	As per IS 14697 & IEC 62053/6205 2/As per MQP of component Manufacture r/ Tech. Specificatio n and GTP	EM/C M -	-	P	W
C-10	SYSTEM TESTING As per IEC 61850 Substation Automation System*								
10.1	Disturbance Recorder function check including communication through Ethernet	Electrical - Functional checks	100%	Appd Specificatio n.GTP & drg/	Appd Specificatio n.GTP & drg/	EM - TC	-	P	W

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	switch(Part of panel testing)								
10.2	Verification of Bay unit function using stand alone central unit for Bus Bar protection system (Part of panel testing)	Electrical - Functional checks	100%	Appd Specificatio n.GTP & drg & as per checklist : BB Protn (Decentralis ed)		EM - TC	-	P	W
10.3	BAY CONTROLLER	Electrical - Functional checks	100%	Appd Specificatio n.GTP & drg & as per checklist : Bay controller	Appd Specificatio n.GTP & drg/	EM - TC	-	P	W
C-11	PART OF SAS TESTING DURING FAT (all checks to be done as per approved FAT document by Customer)								
11.1	SAS with IEDs Mounted in panel along with Ethernet Switches ,accessories and software key and hardware key of SAS for FAT (items including software	Visual check,Va lidity of software, Electrical - Function al checks	100%	Approved SAS BOM & Tech. Specificatio n/ approved FATProcedu re	Approved SAS BOM & Tech. Specificatio n/ approved FATProced ure	EM - TC	-	P	W

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	used Project specific)								
11.2	Verification of SAS Accessories	Visual Check / Functional	100%	Approved SAS BOM & Tech. Specification n/ approved FAT Procedure	Approved SAS BOM & Tech. Specification n/ approved FAT Procedure	EM - TC	-	P	W
11.3	DS AGILE/ AVIEW server/ client/ gateway simulation check Time Synch Check Alarm and event Information availability for telecontrol centre Device control	Functional	100%	Approved SAS BOM & Tech. Specification n/ approved FAT Procedure	Approved SAS BOM & Tech. Specification n/ approved FAT Procedure	EM - TC	-	P	W
C-12	Hardware Integration Test								
12.1	The complete Integration check for Third party IEDs, Meters with interfacing of necessary softwares's/hardwares -IED/Meter healthiness -Communication -I/O Check	Visual & Electrical	100%	Appd drg / BOM,GTP	Appd drg / BOM,GTP	EM - TC	-	P	W

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	-LED Indication. - Operation of Electromechanical relays (if any) -Accuracy check -Relay configuration & setting								
12.2	Energy meter	Visual & Electrical	100%	As per IS 14697 & IEC 62053/62052/ As per MQP of component Manufacturer/ Tech. Specification and GTP	As per IS 14697 & IEC 62053/62052/ As per MQP of component Manufacturer/ Tech. Specification and GTP	EM/C M - T	-	P	W
12.3	Disturbance Recorder (Stand alone functional)	Visual & Electrical	100%	Appd. Drg., BOM, GTP, EM internal scheme drg.	Appd. Drg., BOM, GTP, EM internal scheme drg.		-	P	W
12.4	Disturbance Recorder function check including communication through Ethernet switch(Part of panel testing)	Visual & Electrical	100%				-	P	W
12.5	Verification of Bay unit function using stand alone central unit for Bus Bar protection system (Part of panel	Electrical Functional Checks	100%				-	P	W

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	testing)								
12.6	Bay control unit (BCU) Test	Electrical Functional Checks	100%				-	P	W
12.7	Verification of SAS BOQ						-	P	W
C-13	DS AGILE/AVIEW SOFTWARE								
13.1	Time sync. Check	Functional	100%	Tech specification , GTP, BOM, and Approved drawings	Tech specification , GTP, BOM, and Approved drawings	EM TC	-	P	W
13.2	Alarm & Events		100%	Tech specification , GTP, BOM, and Approved drawings	Tech specification , GTP, BOM, and Approved drawings	EM TC	-	P	W
C-14	SAS Functional Test								
14.1	Time synchronization Equipment Test a) Time synchronization Equipment Test	Visual & Electrical	100%			EM TC	-	P	W

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	b)System time Synchronization Check			Tech. specn/ GTP, Approved drgs., Approved FAT Procedure	Tech. specn/ GTP, Approved drgs., Approved FAT Procedure				
14.2	SAS Functional Test a)Digital Input/ Output b)Software Interlock Logic Display c)Synchronization Check d)Auto Reclosure Check	Visual & Electrical	100%			EM TC	-	P	W
14.3	HMI / Server Functional Test a)Event & Alarm management b)Trends & Reports c)OWS Memory & Disk utilization d)HMI Server Redundancy Check	Visual & Electrical	100%			EM TC	-	P	W
14.4	Substation SAS server / Gateway functional test a)Communication of server / Client with Bay Level devices b)Data retrieval of Master Digital c)Data retrieval of Analog Inputs d)SAS Server/ Gateway redundancy check.	Visual & Electrical	100%			EM TC	-	P	W

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14.5	EWS Test a)Setting and Configuring of IED's b)Retrieval of Disturbance file from IED's c)Automatic downloading of Disturbance.	Visual & Electrical	100%			EM TC	-	P	W
14.6	LAN Redundacy Test :As per Ring formation & Ethernet LAN redundancy check list	Visual & Electrical	100%			EM TC Checkli st	-	P	W
C- 15	Integrated System Test								
15.1	100 hours, System stability Test.	Functional	100%	Tech Spec.	Tech Spec.	EM TC	-	P	W

EM : Equipment Manufacturer

CM : Component Manufacturer

CMTC :- Component manufacturer test certificate

EMTR :- Equipment manufacturer test Report

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Capacitive Voltage Transformer

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check Sampling basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
A.	RAW MATERIAL INSPECTION								
1	Aluminium Foil								
a	Visual check for presence of foreign particles and surface finish	Visual	100%	Technical Data sheet/ Mfr's specification	Technical Data sheet/ Mfr's specification	CMTC/ EMTC	P	V/P	-/V
b	Verification of Dimensions Thickness & width	Measure	01 sample/lot			CMTC/ EMTC/ TPC	P	V/P	-/V
c	Tensile Strength	Test				CMTC/ EMTC	P	V/P	-/V
d	Elongation	Test				CMTC/ EMTC	P	V/P	-/V
e	Purity of Aluminium	Measure				CMTC/ EMTC	P	V/P	-/V
2	Porcelain Insulators								
a	Routine Tests	Test	100%	As per IS 5621/IEC 62155	As per IS 5621/IEC 62155	CMTC	P	V	-
b	Verification of Dimensions (Creepage)	Measure	01 sample/lot	As per Drg.	As per Drg.	CMTC	-	P	V

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c	Porosity Test	Test		As per IS 5621/IEC 62155	As per IS 5621/IEC 62155	CMTC	P	V	-
d	Temp. Cycle Test	Test				CMTC	P	V	-
d	Visual Inspection	Test				CMTC	P	V	-
2A	Composite Insulators (If Applicable)								
a	Dimension	physical	1 No/Lot	Relevant Drawing	Relevant Drawing	CMTC	P	V	-
b	Routine Mechanical test	Mechanical	1 No/Lot	IEC 61462/AS PER Specn.	IEC 61462/AS PER Specn.	CMTC	P	V	-
c	Silicon Content	Chemical	1 No/mfg batch			CMTC	P	V	-
3	<u>Transformer Oil for EMU</u> Unit :- All test as per IEC 60296/IS 335 to be carried out								
a	Appearance	Visual	01 sample/lot	IS 335/ IEC 60296	Oil shall be clear & transparent IS 335/ IEC 60296	CMTC	P	V	-
b	Density	Test				CMTC	P	V	-
c	Kinematic viscosity	Test				CMTC	P	V	-
d	Interfacial tension	Test				CMTC	P	V	-
e	Flash Point	Test				CMTC	P	V	-
f	Pour Point	Test				CMTC	P	V	-
g	Total Acidity	Test				CMTC	P	V	-

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h	Water content(ppm)	Test		IS 335/ IEC 60296	IS 335/ IEC 60296	CMTC	P	V	-
i	Corrosive Sulpher	Test				CMTC	P	V	-
j	Oxidation stability	Test				CMTC	P	V	-
k	Electrical Strength (B.D.V.)	Test				CMTC	P	V	-
l	Dielectric dissipation factor	Test				CMTC	P	V	-
4	<u>Synthetic Oil for Capacitor Unit</u>								
a	Appearance	Visual	01 sample/lot	Limpid with light bluish shimmers, no suspended matter/IEC 60867	Limpid with light bluish shimmers, no suspended matter/IEC 60867	CMTC	P	V	-
b	Density	Test	01 sample/year	Technical Data sheet/Mfr's specification//IEC 60867	Technical Data sheet/Mfr's specification//IEC 60867	CMTC	P	V	-
c	Kinematic viscosity	Test	01 sample/year			CMTC	P	V	-
d	Water content	Test	01 sample/lot			CMTC	P	V	-
e	Epoxide	Test				CMTC	P	V	-
g	Electrical Strength (B.D.V.)	Test				CMTC	P	V	-
h	Dielectric dissipation factor	Test				CMTC	P	V	-
5	<u>Insulation Kraft Paper</u>								
a	Elongation at break MD	Test	01 sample/lot	IS 9335 -2 / IEC60554	IS 9335 -2 /	CMTC	P	V	-

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b	Ash Content	Test			IEC60554	CMTC	P	V	-
c	Moisture Content	Test				CMTC	P	V	-
d	PH Value	Test				CMTC	P	V	-
e	Conductivity	Test				CMTC	P	V	-
f	Tensile Strength	Test				CMTC	P	V	-
g	Thickness	Measure		Tech.spec./Mfg.Std.	Tech.spec./Mfg.St d.	CMTC	P	V	-
h	Density	Test		IS 9335 -2 / IEC60554	IS 9335 -2 / IEC60554	CMTC	P	V	-
i	Electric strength in air	Test				CMTC	P	V	-
6	<u>Copper Wire</u>								
a	Dimension	Measure	01sample/size/ lot	IS 13730/Mfr's PO	IS 13730/Mfr's PO	CMTC/E MTC	P	V	-
b	Mechanical Properties	Test		IS 13730- 3	IS 13730- 3	CMTC/E MTC	P	V	-
c	Thermal Properties	Test		IS 13730- 6	IS 13730- 6	CMTC/E MTC	P	V	-
d	Chemical Properties	Test		IS 13730- 4	IS 13730- 4	CMTC/E MTC	P	V	-
e	Electrical Properties	Test		IS 13730- 5	IS 13730- 5	CMTC/E MTC	P	V	-
7	<u>Capacitor Tissue Paper</u>								
a	Thickness & width	Measure	01sample/ lot	Technical Data sheet/ Mfr's specification	Technical Data sheet/ Mfr's specification	CMTC/E MTC/TP C	P	V	-

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b	Density	Test				CMTC	P	V	-
c	Ash Content	Test				CMTC	P	V	-
d	pH Value	Test				CMTC	P	V	-
e	Conductivity of aqueous extract	Test				CMTC	P	V	-
f	Dielectric Strength	Test				CMTC	P	V	-
8	<u>Bi-Axially Oriented Plain Polypropylene Film</u>								
a	Thickness, Space Factor	Measure	01sample/size/lot	Technical Data sheet/ Mfr's specification	Technical Data sheet/ Mfr's specification	CMTC	P	V	-
b	Dielectric Strength	Test				CMTC	P	V	-
c	Shrinkage	Test				CMTC	P	V	-
d	Visual	Visual	100%			CMTC	P	V	-
9	Tanks								
a	Leakage Test	Mechanical	100%	As pe Mfr's Spec	As pe Mfr's Spec	CMTC	P	V	-
b	Dimensions	Measure	One sample/lot	As per approved drawings / Mfr's drawings	Obtain manufacturers test certificate	CMTC	P	V	-
c	Welding Procedure & Welder's Qualification, Type of Electrode, DP Test on lifting lug welds, Visual check of Paint		One sample/ lot	As per approved drawings / Mfr's drawings	Obtain manufacturers test certificate	CMTC	P	V	-

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	shade & DFT - Inside and Outside								
10	<u>Galvanizing</u>								
a	Thickness of Zinc coating	Measure	1 Samples / lot	As per Customer Specification/ IS 4759	As per Customer Specification/ IS 4759	CMTC	P	V	-
b	Mass of Zinc coating on sample	Measure		As per Customer Specification/ IS 4759	As per Customer Specification/ IS 4759	CMTC	P	V	-
c	Uniformity of Zinc coating	Chemical		IS 2633	IS 2633	CMTC	P	V	-
d	Adhesion test	Mechanical		IS 2629	No peel of Zinc	CMTC	P	V	-
11	<u>Rubber 'O' Ring</u>								
a	Dimensions	Measure	1 sample/ size/ lot	As per Drawing/Mfr's P.O	As per Drawing/Mfr's P.O	CMTC	P	V	-
b	Hardness	Test		Tech. Spec. / Mfg. Std.	Tech. Spec. / Mfg. Std.	CMTC	P	V	-
c	Elongation at break	Test				CMTC	P	V	-
d	Aging in transformer oil	Test				CMTC	P	V	-
e	Compression Set	Test				CMTC	P	V	-

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12	<u>Cork Gasket (If Applicable)</u>								
a	Dimensions	Measure	1 sample/ size/ lot	As per Drawing/Mfr's P.O	As per Drawing/Mfr's P.O	CMTC	P	V	-
b	Hardness	Test		Tech.spec./ IS 4253	Tech.spec./ IS 4253	CMTC	P	V	-
c	Tensile Strength	Test				CMTC	P	V	-
d	Compression Set	Test		Tech.spec./ IS 4253/IS 3400	Tech.spec./ IS 4253/ IS 3400	CMTC	P	V	-
e	Compressibility	Test		Tech.spec./ IS 4253	Tech.spec./ IS 4253	CMTC	P	V	-
f	Flexibility	Test				CMTC	P	V	-
g	Resistance to transformer oil at 100 °C for 70 hrs.	Test				CMTC	P	V	-
h	pH of water extract	Test				CMTC	P	V	-
13	<u>Cut Core (Transformer & Choke Lamination)</u>								
a	Dimensions	Measure	1 sample/ size/ lot	As per Drawing/Mfr's P.O	As per Drawing/Mfr's P.O	CMTC	P	V	-
b	Core Loss	Test		Technical sheet/Mfr's specification	Technical sheet/Mfr's specification	CMTC	P	V	-
c	Stacking Factor	Test				CMTC	P	V	-
14	<u>Damping Device(if applicable)</u>								

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a	Dimensions	Measure	1 sample/ size/ lot	As per Drawing/Mfr's P.O	As per Drawing/Mfr's P.O	CMTC	P	V	-
b	Resistance	Test	1 sample/ size/ lot	Technical Data sheet/Mfr's specification	Technical Data sheet/Mfr's specification	CMTC	P	V	-
15	<u>HRC Fuse with carrier(If applicable)</u>								
a	Visual Check	Visual	100%	Supplier Catalogue/ Mfr's Specification	Supplier Catalogue/ Mfr's Specification	CMTC	P	V	-
16	<u>Intermediate Voltage Bushing (MV Tap Bushing)</u>								
a	Dimensions	Measure	1 sample/ size/ lot	As per Drawing/ Mfr's P.O	As per Drawing/ Mfr's P.O	CMTC	P	V	-
b	Visual Check	Visual	100%	Free From Damage	Free From Damage	CMTC	P	V	-
17	<u>Stainless Steel Bellow (If applicable)</u>								
a	Dimensions	Measure	1 sample/ size/ lot	As per Drawing/Mfr's P.O	As per Drawing/ Mfr's P.O	CMTC	P	V	-
b	Helium Leak Test	Test		Technical Data sheet/Mfr's specification	Technical Data sheet/Mfr's specification	CMTC	P	V	-
c	Chemical Composition	Measure		Technical Data sheet/Mfr's	Technical Data sheet/Mfr's	CMTC	P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

				specification	specification				
18	<u>Fasteners (H.D.G / Stainless Steel)</u>								
a	Verification of Make & Visual	Visual	100%	Mfr's Specification	Mfr's Specification	CMTC	P	V	-
B. Section INPROCESS INSPECTION									
1	Test on Capacitor								
a	Alignment of foil pp film(if applicable) & condenser paper	Visual	one sample/ New coil or other operator/ M/C	Mfr's Drg./ Mfg. std.	Mfr's Drg./ Mfg. std.	EMTR/ LOG BOOK	-	P	V
b	Stack Height	Measure	100%	Mfr's Drg./ Mfg. std.	Mfr's Drg./ Mfg. std.	DO	-	P	V
c	High Voltage test on Dry element-(if applicable)	Test	DO	Mfr's Drg./ Mfg. std	Mfr's Drg./ Mfg. std	DO	-	P	V
2	Drying of Capacitors								
a	Temperature of Oven		Hourly Reading	Equipment plant standard	Mfr's	Equipment plant standard	Mfr's	EMTR/ LOG BOOK	- P V
b	Vacuum Leakage		6 Hourly	Equipment plant standard	Mfr's	Equipment plant standard	Mfr's	DO	- P V

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3	Assembly								
a	Capacitance measurement	Test	100%	As per Job Card/Drg.	As per Job Card/Drg.	EMTR/LOG BOOK	-	P	V
b	Stack & Insulator Assembly	Verification	100%	As per Approved Drawing	As per Approved Drawing	EMTR/LOG BOOK	-	P	V
4	Oil Filling								
a	Oil filling in Capacitor Unit		100%	App. GTP/Drg.	App. GTP/Drg.	LOG BOOK	-	P	V
5	Test on EMU at assembly stage								
a	Turn ratio measurement & Polarity Test	Test	100%	Mfr's Drg./ Mfg. std.	Mfr's Drg./ Mfg. std.	CMTC/E MTC	P	V/P	-/V
b	Resistance of HT winding- If applicable	Test	100%	As per Job Card/Drg.	As per Job Card/Drg.	CMTC/E MTC	P	V/P	-/V
c	Megger Test	Test	100%	As per Job Card/Drg.	As per Job Card/Drg.	DO	P	V/P	-/V
6	Vacuum Drying & Oil impregnation (EMU)								
a	Temperature of Oven		Hourly Reading	Equipment plant standard Mfr's	Equipment plant standard Mfr's	LOG BOOK	-	P	V
b	Vacuum Leakage		6 Hourly	Equipment plant standard Mfr's	Equipment plant standard Mfr's	LOG BOOK	-	P	V

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7	Treated Transformer Oil For EMU								
a	BDV	Test	1 Sample/ lot	IS 335 / IEC 60296/TS	IS 335 / IEC 60296/TS	LOG BOOK	-	P	V
b	Tan Delta at 90° C	Test	1 Sample/ lot			LOG BOOK	-	P	V
c	Moisture Content	Test	1 Sample/ lot			LOG BOOK	-	P	V
8	Synthetic Oil for Capacitor Divider after treatment								
a	BDV & Tan Delta	Test	1 Sample/ lot	IEC 60867/Mfr's Specification	IEC 60867/Mfr's Specification	LOG BOOK	-	P	V
9	Sealing Test on Divider with Bellow	Test	100%	IEC 60867/Mfr's Specification	IEC 60867/Mfr's Specification	LOG BOOK	-	P	V
10	Check for Connection on Between Bellow & Capacitor Stack	Test	100%	IEC 60867/Mfr's Specification	IEC 60867/Mfr's Specification	LOG BOOK	-	P	V
C. Section FINAL INSPECTION & TESTING :- All Routine/Acceptance test as per applicable contract Tech Spec need to be carried out without fail									
1	<u>Routine Test (Test on Complete CVT)</u>								
a	Visual & Dimension check	Measure	01sample/lot	Tech.spec./App.Drg.	Tech.spec./App.Drg.	EM Log Sheet	-	P	V
b	Verification of Terminal marking & Polarity	Electrical	100%	IS 16227 / IEC 61869-5	IS 16227 / IEC 61869-5	EM Log Sheet	-	P	V

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c	Determination of errors (ratio & phase angle) according to the requirements of appropriate accuracy class	Electrical				EM Log Sheet	-	P	V
d	Ferro resonance Test	Electrical				EM Log Sheet	-	P	V
	Test on EMU								
e	High voltage power frequency withstand test on primary winding.(HT neutral to earth) at 4KV	Electrical	100%	IS 16227 / IEC 61869-5	IS 16227 / IEC 61869-5	EM Log Sheet	-	P	V
f	High voltage power frequency withstand test on secondary winding at 3 KV.	Electrical				EM Log Sheet	-	P	V
g	Induced overvoltage test	Electrical				EM Log Sheet	-	P	V
	(Test on capacitor assembly)								
h	Measurement of Capacitance & Loss Angle before HV test	Electrical	100%	IS 16227 / IEC 61869-5	IS 16227 / IEC 61869-5	EM Log Sheet	-	P	V
i	Power frequency Withstand test for one minute	Electrical				EM Log Sheet	-	P	V
j	Measurement of Partial Discharge	Electrical				EM Log Sheet	-	P	V
k	Measurement of Capacitance & Loss Angle after HV test	Electrical				EM Log Sheet	-	P	V
l	Power frequency Withstand test on LV	Electrical				EM Log Sheet	-	P	V
m	Enclosure tightness test at	Physical			No leakage	EM Log	-	P	V

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	ambient temperature					Sheet			
2	<u>Acceptance Test (Test on Complete CVT)</u>								
a	Visual & Dimension check	Measure	01sample/lot	Tech.spec./App.Drg.	Tech.spec./App.Drg.	Test report	-	P	W
b	Verification of Terminal marking & Polarity	Electrical	10 %	IS 16227 / IEC 61869-5	IS 16227 / IEC 61869-5	Test report	-	P	W
c	Determination of errors (ratio & phase angle) according to the requirements of appropriate accuracy class	Electrical				Test report	-	P	W
d	Ferro resonance Test	Electrical				Test report	-	P	W
	(Test on EMU)								
e	High voltage power frequency withstand test on primary winding.(HT neutral to earth) at 4KV	Electrical	10 %	IS 16227 / IEC 61869-5	IS 16227 / IEC 61869-5	Test report	-	P	W
f	High voltage power frequency withstand test on secondary winding at 3 KV.	Electrical				Test report	-	P	W
g	Induced over voltage test	Electrical				Test report	-	P	W
	(Test on capacitor assembly)								
h	Measurement of Capacitance & Loss Angle before HV test	Electrical	10 %	IS 16227 / IEC 61869-5	IS 16227 / IEC 61869-5	Test report	-	P	W

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i	Power frequency Withstand test for one minute (Repeated test to be done at 80 % of test voltage level)	Electrical				Test report	-	P	W	
j	Measurement of Partial Discharge (Repeated test to be done at 80 % of test voltage as per standard)	Electrical				Test report	-	P	W	
k	Measurement of Capacitance & Loss Angle after HV test	Electrical				Test report	-	P	W	
l	Power frequency Withstand test on LV	Electrical				Test report	-	P	W	
m	Enclosure tightness test at ambient temperature	Physical	1 Sample/ lot			No Leakage	Test report	-	P	W
n	Transformer Oil from EMU									
i	BDV Test (Min.)	Electrical	01sample /lot	IEC 60422	Min.60 kV	Test Report	P	V	W	
ii	Tan Delta at 90 ⁰ C Max.	Electrical		IEC 60422	IEC 60422	Test Report	P	V	W	
iii	Moisture Content	Electrical		IS 1866 /IEC 60422/As per spec.	less than 10 ppm	Test Report	P	V	W	
iv	Interfacial Tension(Min.)	Chemical		IEC 60422	Min. 35 mN/mtr	Test Report	P	V	W	

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D. Section PACKING & DESPATCH									
1	Checking for completeness of the equipment and accessories including spares as per order.	Visual	100%	As per Mfr's Process Specification	As per Mfr's Process Specification		-	P	-
2	Check for proper packing	Visual					-	P	-
3	Ensure supply of operation manual,packing list and test reports along with copy of inspection reports,CIP and MICC	Visual					-	P	-

EM : Equipment Manufacturer

CM : Component Manufacturer

CMTC :- Component manufacturer test certificate

EMTR :- Equipment manufacturer test Report

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Isolators (Disconnecting Switches)

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
SECTION 1. RAW MATERIAL INSPECTION									
A	ELECTRICAL ITEMS								
1.1	CONTACTORS	Visual		IS/IEC 60947 (Part 4/Sec 4) /IS 6875/ Tech. Specn./Apprd. Drgs./ Mfg Plant Standard		CMTC			
a.	Visual		100%		Ok		P	V	-
b.	Verification –15%, 10% Voltage, verification of the limits of operation	Electrical	10%		should pick up without trouble		P	V	-
c.	Pick-up & drop out voltage test	Electrical	10%		IS/IEC 60947 (Part 4/Sec 4) /IS 6875/ Tech. Specn./Apprd. Drgs./ Mfg Plant Standard		P	V	-
d.	High Voltage Test	H V Tester	10%				P	V	-
e.	I.R. Value	Megger tester	10%				P	V	-
1.2	OVERLOAD RELAYS	Visual		IS/IEC 60947 /IS 6875/ Tech. Specn./Apprd. Drgs./ Mfg Plant Standard	IS/IEC 60947 (Part 4/Sec 4) /IS 6875/ Tech. Specn./Apprd. Drgs./ Mfg Plant Standard	CMTC			
a.	Visual		100%				-	P	V
b.	High Voltage Test	Electrical	10%				P	V	-
c.	I.R. Value	Electrical	10%				P	V	-
d.	Operation test during MOM assembly	Electrical	100 % on each type / Lot				P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
1.3	PUSH BUTTONS	Visual	IEC 60947	IS 4794/IEC 60947/ Tech. Specn./Apprd. Drgs.		CMTC			
a.	Visual (Make & Type)	Visual	100%		AS PER PO		-	P	V
b.	High Voltage Test	H V Tester	10%		2 KV , 1min		P	V	-
c.	I.R. Value	Megger tester	10%		> 500 V DC Megger		P	V	-
1.4	TERMINAL BLOCKS	Visual	IEC 60947	IEC 60947 / IS 3669/ Tech. Specn./Apprd. Drgs.		CMTC			
a.	Visual (Make & Type)	Visual	100%	"	AS PER PO		-	P	V
b.	High Voltage Test	H V Tester	10%	"	2 KV , 1min		P	V	-
c.	I.R. Value	IR tester	10%	IS4007	> 500 V DC IR tester		P	V	-
d.	Torque Test	Visual	10%	110% of Rated Torque	No damage to the screws		P	V	-
e.	Continuity Check	Electrical	5% on each type /lot	"	"		P	V	-
1.5	AUXILIARY SWITCH (SNAP type)	Visual	IS 6875	IS 6875/ Tech. Specn./Apprd. Drgs.		CMTC			
a.	Identification of Switch	Visual	100%		Sticker with		-	P	V

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
					Serial No.				
b.	Rotary Contact Alignment	Visual	10%		ok		-	P	V
c.	High Voltage Test	H V Tester	10%		2 KV , 1min		P	V	-
d.	NO & NC Configuration & Change over	Multimeter	10%		All the contacts shall be maintained at NO & NC		P	V	-
e.	Insulation resistance test	Megger tester	10%	min 2 Mohms	> 500 V DC Megger		P	V	-
1.6	HEATERS	Visual		Tech.	Make &	CMTC			
				Specn./ Apprd. Drgs.	Product Code No				
a.	Visual (Make & Type)	Visual	100%		AS PER schematic		-	P	V
b.	High Voltage Test	H V Tester	10%		2 KV , 1min		P	V	-
c.	I.R. Value	Megger tester	10%		> 500 V DC Megger		P	V	-
d.	Functional Check	Electrical	10 % of each type /lot				P	V	-
1.7	FUSE BASE & CARRIER	Visual	IS 13703	IS 13703/ Tech. Specn./Apprd. Drgs.	Make & Product Code No				

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
a.	Visual		100%		No damage		-	P	V
b.	Dimensional Check		10%		As per manufacturer		-	P	V
c.	Resistance Check		10%		As per manufacturer		P	V	-
d.	Power Dissipation Test		10%		Test to be withstood		P	V	-
e.	Insulation Properties		10%		IS 13703-1993		P	V	-
1.8	INTERLOCK COILS			Tech. Specn./Apprd. Drgs.	Make & Product Code No	CMTC			
a.	Visual (Make & Type)	Visual	100%		AS PER schematic		-	P	V
b.	High Voltage Test	H V Tester	100%of each type /lot		2 KV , 1min		P	V	-
c.	I.R. Value	IR tester	100%of each type /lot		> 500 V DC IR tester		P	V	-
d.	Operation Test at rated voltage , + 10 % and -15 % of rated voltage	DC/AC voltage	10% of each type / lot		Should pickup for 85% of Rated Voltage		P	V	-
e	Resistance check	Electrical	One Sample of each type / lot				P	V	-
f	Pick up and drop out Voltage test	Electrical	50 % of Each Type / lot				P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
1.9	CIRCUIT BREAKER (MCB)	Visual		Tech. Specn./Apprd. Drgs./ IEC 60898 - 1-2002 / IS 8828-1996	Make & Product Code No	CMTC			
a.	Visual (Make & Type)	Visual	100%		AS PER schematic		-	P	V
b.	High Voltage Test	H V Tester	10%		2 KV , 1min		P	V	-
c.	I.R. Value	Megger tester	10%		> 500 V DC Megger		P	V	-
d.	Functional check	Electrical	5 % on each type /lot				P	V	-
1.10	MOTORS (S1 Duty Class)	Visual	IS 325	IS 325 / Tech. Specn./Apprd. Drgs./ Apprd. Tech Data Sheets		CMTC			
a.	Insulation resistance	Megger tester	100% of each Type /lot		>500 V DC Megger		P	V	-
b.	High Voltage Test	H V Tester	100% of each Type /lot		2 kV, 1min		P	V	-
c.	No load test	wattmeter, Multimeter & tachometer	100% of each Type /lot		Measurement of Input power current and speed		P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
d.	Locked rotor readings	wattmeter, Multimeter & tachometer	1 sample/batch		Measurement of Input power current and speed		P	V	-
e.	Visual inspection/Rating	Visual	100%		no damage		-	P	V
f	Reduced Voltage running test	Electrical	1 sample/batch				P	V	-
1.11	GASKET (EPDM RUBBER)	Visual	Manufacturer Standard / ASTM D 1171 / IS 11149	Manufacturer Standard / ASTM D 1171 / IS 11149/ Tech. Specn. / Apprd. Drgs.		CMTC			
a.	UTS	UTS Machine	3samples/lot		70 KG/cm ² min (ref std ASTM D412)		P	V	-
b.	OZONE RESISTANCE(50PPHM /40degree C /20% STRAIN/ 72 HRS)	MFR Practice	3samples/lot		NO CRACK (ref std ASTM D1149)		P	V	-
c.	Compression set % (70 degree C X 22hrs)	MFR Practice	3samples/lot		50 max(ref std ASTM D396)		P	V	-
d.	Ageing	Mechanical	1 sample/lot	Manufacturer std			P	V	-
e	Dimension		1 sample/lot				P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
1.12	THERMOSTAT	Visual	IS 3017	IS 3017 / Tech. Specn. / Apprd. Drgs.	Make & Product Code No	CMTC			
a.	Visual		100%				-	P	V
b.	High Voltage Test	H V Tester	10%		IS 3017		P	V	-
c.	I.R. Value	IR tester	10%	Min 2 M ohms	> 500 V DC IR tester		P	V	-
d.	Marking		10%		As Per IS:3017:1985		-	P	V
e	Functional Check		10 % per lot				P	V	-
1.13	CASTLE LOCKS	Visual	Manufacturer's Catalogue	Manufacturer's Approved Catalogue / Specn	Make & Product Code No	CMTC			
a.	Visual(Make & Type)	Visual	100%		AS PER schematic		-	P	V
b.	Dimension check	Vernier	10%		Should be Within limits		-	P	V
c.	Operation of Lock & Key / Functional Check	Check	100%	As per manufacturer	As per manufacturer		-	P	V
1.14	TOGGLE / CONTROL SWITCHES	Visual	IS 13947 Part 5 & 3 / IS/IEC 60947	IS 13947 Part 5 & 3 / IS/IEC 60947/ Tech. Specn.	Make & Product Code No	CMTC			

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
				/Apprd. Drgs					
a.	Visual(Make & Type)	Visual	100%	IS 13947 Part 5 & 3	No damage		-	P	V
b.	High Voltage Test	H V Tester	10%		2 KV , 1min		P	V	-
c.	Switching Sequence		10%	Approved Schematic	Approved Schematic		P	V	-
d.	IR Check	IR tester	10%	Min 2 M ohms	> 500 V DC IR tester		P	V	-
1.15	LIMIT SWITCHES	Visual	IS 6875 / IS/IEC 60947	IS 6875 / IS/IEC 60947/ Tech. Specn. /Apprd. Drgs.		CMTC			
a.	Visual inspection	Visual	100%	Clause 8.3.3.4 of IS 13947-5-1,1993	no damage		-	P	V
b.	NC to Body (Dielectric Test)	H V Tester	10%	Clause 8.3.3.4	2.5 kV,1min		P	V	-
c.	NO to Body (Dielectric Test)	H V Tester	10%	Clause 8.3.3.4	2.5 kV,1min		P	V	-
d.	NC to NO (Dielectric Test)	H V Tester	10%	Clause 8.3.3.4	1.3 kV,1min		P	V	-
e.	IR Check	IR tester	10%	Min 2 M ohms	> 500 V DC IR		P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
					tester				
B	OTHER ITEMS								
1.16	STRUCTURAL STEEL (Channels, Sheets, Plates, Rods & Flats)	Mechanical	IS 2062 / IS 1079 / IS 5986/ IS 1977	IS 2062 / IS 1079 / IS 5986/ IS 1977/ Tech. Spec.		CMTC			
a.	Grade / Quality of Steel	Chemical analysis	1 sample/Lot		Fe410W A(E250A) ,410W B(E250 B), Fe 410W C(E250 C) as per IS 2062		P	V	-
b.	Dimensional check Galvanizing Thickness (As per IS) Mass of Zinc Coating (As per IS)	Visual	1 sample/Lot		IS1852		P	V	-
c.	Chemical Composition	Chemical analysis	1 sample/50 tonnes		IS 2062/ IS4923		P	V	-
d.	Tensile Strength (UTS & Yield Stress)	UTS Machine	1 sample/50 tonnes		250Mpa Min		P	V	-
e.	% Elongation	UTS Machine	1 sample/50 tonnes		23 % Min		P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
f.	Bend Test	Visual	1sample/50 tonnes		no crack		P	V	-
1.17	GI PIPES "B" CLASS	Mechanical	One Sample per Lot	IS 1239 / 1161 - 1998	As per	CMTC			
			of Each Size	IS 2633 / IS 4759	Manufacturer's Internal Document				
a.	Dimensional check	Vernier/micrometer	10% of each size /lot		Nominal Bore(25NB,32 NB,40 NB,50 NB etc) as per IS 1239		-	P	V
b.	Marking	Visual	100%		Light tubes- Yellow Medium- Blue Heavy-Red		-	P	V
c.	UTS	UTS Machine	1sample/lot		320Mpa Min		P	V	-
d.	Elongatoin , Bend Test /flattening test	UTS Machine	1sample/lot		Up to 25NB mm-12 % Min& 25 NB mm to 150 NBmm-20%		P	V	-
e.	Chemical Composition	Chemical analysis	1sample/lot		IS 1239/ CM/L- 2247549		P	V	-
f.	wt of Zn g/m 2 (up to	Chemical	1sample/lot		360		P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
	6mm thick)	analysis							
g.	Uniformity	Chemical analysis	1sample/lot		4x 1 min dip		P	V	-
1.18	ALUMINIUM TUBES (Seamless), FLATS, PROFILES	Mechanical/ Electrical	IS 2678 / IS 3965/ IS 5082 / IS 733	IS 2678 / IS 3965/ IS 5082 / IS 733 / Tech. S		CMTC			
a.	Grade of Aluminium Alloy	Chemical analysis	1sample/lot		63401Grade IS 5082		P	V	-
b.	Freedom from defects	Visual	100%		Free from cross cutting and harmful defects		P	V	-
c.	Dimensional check	vernier/micrometer	ONE SAMPLE		AS PER schematic		-	P	V
d.	UTS	UTS Machine	1sample/lot		200Mpa Min		P	V	-
e.	Elongatoin	UTS Machine	1sample/lot		10 % Min		P	V	-
f.	Bend Test for Flats	Visual	1sample/lot		No cracks upto 90 Deg		P	V	-
g.	Conductivity test	TECHNOFOUR	1sample/lot		min 55 % IACS		P	V/P	-
h.	Chemical Composition (in weight %)	Chemical analysis	1sample/lot		IS 5082-1998 IS 1608-2005		P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
1.19	COPPER RODS PROFILES / FLATS / PIPES	Mechanical/ Chemical/ Electrical	IS 613 / IS 191 / I S 2501 / IS 1897	IS 613 / IS 191 / TECH. Specn.		CMTC			
a.	Grade of copper	Chemical analysis	1sample/300tubes, 2samples/lot up to 100 flats/bars		ETP(ELECTROLYTIC TOUGH PITCH) IS191 (PART 5)		P	V	-
b.	Condition of Supply	Mechanical Properties Check	1sample/300tubes, 2samples/lot up to 100 flats/bars		As Drawn (HD) for Tubes IS 2501 & Half Hard (HB) for Flats/BarsIS 613		P	V	-
c.	Freedom from defects	Visual	100%		Free from blisters, slivers, scale, fins, spills, cracks		P	V	-
d.	Dimensional check	vernier/micro meter	ONE SAMPLE		AS PER schematic		P	V	-
e.	UTS (for tubes)	UTS Machine	ONE SAMPLE		250Mpa Min		P	V	-
f.	UTS (for flats)	UTS Machine	ONE SAMPLE		230Mpa Min		P	V	-
g.	Elongation(for flats)	UTS	ONE SAMPLE		15 % Min		P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
		Machine							
h.	Bend test -180 degree (for flats)	fixure	ONE SAMPLE		no crack		P	V	-
i.	Conductivity test	TECHNOF OUR	1sample/300tubes, 2sam ples/lot up to 100 flats/bars		min 97 % IACS		P	V/P	-
j.	Chemical Composition (in weight %)	Chemical analysis	1sample/lot		IS 191:(Pt V)& IS 2501-1995 IS 613-2000		P	V	-
1.20	SILVER PLATED COMPONENTS	Mechanic al	5%per lot	TECH. Specn. (Plating Thickness)	P.S 001	CMTC	P	V	-
a.	Purity of Silver	Chemical Analysis	One Sample/Month	Min. 99%		CMTC	P	V	-
b.	Adhesion Test	Oven Test	100%	Mfr's Specification	Free from Peel, Bubbles	CMTC	P	V/P	-/W
c.	Thickness	Thickness Meter	5%	Specn. /Apprd. Drgs.		CMTC	P	V/P	-/W
1.20 .1	TIN Plated Components						P	V/P	-/W
a.	Adhesion Test	Oven Test	100%	Mfr's Specification	Free from Peel, Bubbles	CMTC	P	V/P	-/W
b.	Thickness	Thickness Meter	5%	Specn. /Apprd. Drgs.		CMTC	P	V/P	-/W

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
1.21	MOM / BOM BOXES		IS 9921	IS 9921 /	IS 9921 /				
a.	Operation Test(Close & Open)	Electrical /	100%	do	do	CMTC	P	V	W
		Mechanica 1							
b.	IR Value Check	Electrical	100%	More than 2 Mega	More than 2 Mega	CMTC	P	V	W
				Ohms	Ohms		P	V	W
c.	High Voltage TEST	Electrical	100%	1.75 KV / 1min	1.75 KV / 1min	CMTC	P	V	W
d.	Dimensional Check	Mechanica 1	10%	IS 9921 / Specn. /Apprd. Drgs./Apprd. Data Sheet	IS 9921 / Specn. /Apprd. Drgs./Apprd. Data Sheet	CMTC	P	V	W
e.	Visual Check for gasketting	Visual	100%	do	FREE FROM DAMAGE		P	V	W
f.	Check for	Mechanica 1/	100%	do			P	V	W
	Manual Operation Interlock,	Electrical	100%		OK /NOT OK				
	O/L Relay Operation,	Electrical	100%		OK/NOT OK				
	Degree of Rotation	Mechanica 1/	100%		OK/NOT OK				
	Current Drawn by Motor,	Electrical	100%		NOT MORE THAN RATED CURRENT				
	Painting Thickness	Mechanica	5%		As per				

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
		I/			Drg/GTP/TS				
g.	Check for Pick-up & Dropout voltage on contactors.	Electrical	10%	pick up 85% of rated dropout 10to75% of rated D.C Voltage			P	V	W
h	Water jet test	Mechanical	1 NO/ LOT	12.5 lt/min -jet	No leakage		P	V	W
i.	Paper test for Gasket (4 direction)	Visual	10%	Specn./Apprd. Drgs.	Specn./Apprd. Drgs.		P	V	W
1.22	BASE CHANNEL Including Bearing Housing Assembly / Main Frame Structure Assembly			Approved Drawing & Data Sheet	Approved Drawing & Data Sheet				
a.	Visual Check	Visual	100%	do	do		-	P	V
b.	Dimensional Check	Mechanical	5%	do	do		-	P	V
c.	Check for Thickness of Zinc Coating	Mechanical	One Sample Per Lot	86 µ Min./ TECH. Specn.	86 µ Min./ TECH. Specn.		-	P	V
d.	Level / Straightness Check on base Channel by Spirit level	Mechanical	100%	Approved Drawing & data sheet	Approved Drawing & data sheet		-	P	V

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
e.	Check Free movement of rotating Shafts	Mechanical	100%	do	do		-	P	V
1.23	Spring (SS)			AISI standard	AISI standard				
				AISI 304 / AISI 301					
a.	Visual	Visual	10%			CMTC	-	P	V
b.	Dimensional	Dimensional	5 % of lot of each size				-	P	V
c.	Chemical	Chemical	1 sample per lot of each size				P	V	-
d.	Mechanical (tensile strength / Elongation	Mechanical	1 sample per lot of each size				P	V	-
1.23 .1	Spring (Spring Steel)			IS 4454 (Grade 2 & Grade 3)	IS 4454 (Grade 2 & Grade 3)				
a.	Visual	Visual	10%	do	do		-	P	V
						CMTC			
b.	Dimensional	Dimensional	5 % of lot of each size	do	do		P/-	V/P	-/W
c.	Chemical	Chemical	1 sample per lot of each	do	do		P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
			size						
d.	Mechanical (tensile strength / Elongation	Mechanical	1 sample per lot of each size	do	do		P	V	-
e.	Coating Thickness	Mechanical	1 sample per lot of each	Mfr's Drawing	min 50 μ		P	V	-
1.24	Bearing								
a.	Visual	Visual	10%	Product code	Product code		-	P	V
b.	Dimensional	Dimensional	5% samples per lot				-	P	V
1.25	Gear Box								
A	Gear Components								
	Raw Material Test Report	Chemical Analysis	One Sample / Batch	Component Manufacturer specification	Component Manufacturer specification	CMTC	P	V	-
	Mechanical (tensile strength / Elongation	Mechanical	One Sample / Batch	Component Manufacturer specification	Component Manufacturer specification	CMTC	P	V	-
	Surface Treatment report	Mechanical	One Sample / Batch	Component Manufacturer specification	Component Manufacturer specification	CMTC	P	V	-
B	Casting						P	V	-
	Raw Material Test Report	Chemical Analysis	One Sample / Batch	Component Manufacturer specification	Component Manufacturer specification	CMTC	P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
	Mechanical (tensile strength / Elongation	Mechanical	One Sample / Batch	Component Manufacturer specification	Component Manufacturer specification	CMTC	P	V	-
	Surface Treatment report	Mechanical	One Sample / Batch	Component Manufacturer specification	Component Manufacturer specification	CMTC	P	V	-
1.26	Aluminium Casting								
	Raw Material Test Report	Chemical Analysis	One Sample / Batch	Mfr's Specification	Mfr's Specification		P	V	-
	Mechanical (tensile strength / Elongation	Mechanical	One Sample / Batch	Mfr's Specification	Mfr's Specification		P	V	-
SECTION 2. INPROCESS INSPECTION									
2.1	MALE & FEMALE CONTACT ASSEMBLY		100%	IEC 62271-102 Technical Specn./Approved Drawings/ Approved Tech. Data Sheet	IEC 62271-102 Technical Specn./Approved Drawings/ Approved Tech. Data Sheet				
a.	Visual Check	Visual	100%	do	do		-	P	V
b.	Dimensional Check	Mechanical	5%	do	do		-	P	V
c.	Contact Resistance / MV Drop Test	Electrical	100%	1.2 Times Type Test Value Max.	1.2 Times Type Test Value Max.		-	P	V

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
2.2	EARTH SWITCH					EM Records			
	Visual & Dimensional Check	Visual/ Mechanical	5%	IEC 62271-102	IEC 62271-102/ Technical Specn./Approved Drawings		-	P	V
2.3	GEAR BOX ASSEMBLY			Approved Drawing & Data Sheet	Approved Drawing & Data Sheet	EM Records			
a.	Gear and Pinion movement	Visual/Manual	100%	Free Movement	Free Movement		P	V	-
b.	b.Motor run trial	Electrical	100%	5 min	No abnormality during 5 min		P	V	-
2.4	MOM SHELL MANUFACTURING								
a.	Fabrication of MOM shell	Visual	100%	Squareness shall be maintained	Free from Dents, Sharp edges		P	V	-
b.	Dimensional Check	Measurement	5%	As per approved GA	As per approved GA		P	V	-
2.5	PAINTING				PS:003				
a.	Pre Treatment of Shell	Chemical	100%	As per Powder coating Manufacturer	As per Powder coating Manufacturer		P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
b.	Baking of Shell after powder spray by Electrostatically	Measurement	100%	170-180 degrees for 15 minutes	170-180 degrees for 15 minutes		P	V	-
c.	Thickness measurement	Measurement	5%	As per GTP/Drg/TS	As per GTP/Drg/TS		P	V	-
2.6	GALVANISING								
a.	Pre Treatment of Structural Components	Chemical	100%	AS per IS 2629/2633/PG specifications	AS per IS 2629/2633/PG specifications		P	V	-
b.	Dipping of structural parts in molten Zinc	Measurement	Daily check	Zinc bath temperature shall be 445 to 465 centigrade	Zinc bath temperature shall be 445 to 465 centigrade		P	V	-
c.	Thickness measurement	Mechanical	One Sample Per Lot	86 µ Min./ Tech Specn.	86 µ Min./ Tech Specn.		P	V	-
2.7	SILVER PLATING								
a.	Pre-treatment of component	Chemical	100%	As per Manufacturer practice	As per Manufacturer practice	CMTC	P	V	-
b.	Silver strike	Chemical	Every Week	As per Manufacturer practice	As per Manufacturer practice		P	V	-
c.	Silver Plating	Chemical	Every Week	Verification of	As per		P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
				Silver & Free Cynide	Manufacturer practice				
d.	Thickness	Mechanical	5%per lot	Tech Specn.	Tech. Specn.		P	V	-
2.8	Lubrication								
a.	For Contacts	Visual		Mfr's Specification	Graphite Oil / Centoplex	CMTC	P	V	-
b.	For Gears	Visual		Mfr's Specification	Multipurpose Grease (MP3)/ Molykote long term 2 plus	CMTC	P	V	-
SECTION 3. FINAL INSPECTION **- All Routine/Acceptance test as per applicable contract Tech Spec need to be carried out without fail									
3.1	ACCEPTANCE TEST on Complete Isolator Assembly & All Accessories as per TECH. Spec./Apprd. Drgs. With * Insulators Forming Part of the Same Isolator with MOM (All acceptance Test as Per Contract to be Conducted)	Visual/ Mechanical/ Electrical	10 % of each type (kV) per lot (to nearest whole no) min. One	IEC 62271-102/ Tech. Specn. /Apprd. Drgs./Apprd. Tech. Data Sheet	IEC 62271-102/ Tech. Specn. /Apprd. Drgs./Apprd. Tech. Data Sheet		-	P	W
	* Mfr's Standard Insulators can be used for the acceptance test if they are equal (in height) to the approved drawings.								
a.	Mechanical Operation Test	Mechanical /	One Set of each	IEC 62271-102/ Tech. Specn.	IEC 62271-102/ Tech.	EM Test Reports	-	P	W

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
	-	Electrical	type (kV) per Lot / LOA / Project One Set of randomly Selected Sample with higher rating of Current and earthswitches per inspection lot	/Apprd. Drgs./Apprd. Tech. Data Sheet	Specn. /Apprd. Drgs./Apprd. Tech. Data Sheet no removal of silver coating in contact area(MS/ES)				
	i. Contact Resistance Measurment	Electrical		Tech. Specn. & Apprd. Data Sheet / Drgs	Tech. Specn. & Apprd. Data Sheet / Drgs	EM Test Reports	-	P	W
	ii. Check for Avg Power Drawn by Motor & Operating Time along with travel Characteristics	Electrical		do	±10 % variation after mechanical operations	EM Test Reports	-	P	W
	iii. Check for Manual (Mech.) Operation& Interlocks (elect & Mech).	Electrical/ Mechanical		do	do	EM Test Reports	-	P	W
	iv. Vertically Alignment Check & Clearance Check	Electrical/ Mechanical		do	do	EM Test Reports	-	P	W
	v. General Assembly	Electrical/		do	do	EM Test Reports	-	P	W

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
	Dimensional Check & Aux. Switch and Limit Switch Operation.	Mechanical							
	vi. Visual Check for Completeness	Visual		IS 9921 / IEC 62271-102/ Tech. Specn. /Apprd. Drgs./Apprd. Tech. Data Sheet	IS 9921 / IEC 62271-102/ Tech. Specn. /Apprd. Drgs./Apprd. Tech. Data Sheet	EM Test Reports	-	P	W
b.	Dimension measurement between Phase to Earth & between Conductive Parts.	Mechanical		do	do	EM Test Reports	-	P	W
c	CRM	Electrical/ Mech	5%	do	do		-	P	W
d	Di-Electric test on Aux Ckt (2 kV)	Electrical/ Mech	5%	do	do		-	P	W
e	Witness of Silver Plating	Mechanical	One sample / Inspection lot	Tech. Specn. & Apprd. Data Sheet/Drgs.	do	EM Test Reports	-	P	W
SECTION 4.PACKING & DISPATCH									
4.1	Verification of Completeness of all Items as per Contract		100%	Tech. Specn./Apprd. Drgs.	Tech. Specn./Apprd. Drgs.		-	P	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
4.2	Check for Soundness of Packing		100%	Tech. Specn./Apprd. Drgs.	Tech. Specn./Apprd. Drgs.		-	P	-
4.3	Verification of CIP/MDCC Reference on Packing / Crates		100%	do	do		-	P	-

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Surge Arrestor

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility			
							Sub-Vendor	Manufacturer	Customer	
SECTION 1. Raw Materials/ Bought out Components										
1	Porcelain Insulator									
a)	Visual Examination (surface defect)	Visual	100%	Manufacturer's Plant Standard/IS 5621/IEC 62155/IEC 99-4 As per spec /GTP	Manufacturer's Plant Standard/IS 5621/ IEC 62155	CMTC/EMTC	P	V/P	-/V	
b)	Verification of Dimensions including Creepage distance & Parallelity	Dimensional	2 Samples/Lot		Drawing/Manufacturer's Plant Standard/IS 5621/IEC 62155	CMTC/EMTC	P	V/P	-/V	
c)	Porosity Test	Physical	1 Sample/Lot		No Dye penetration should occur	CMTC/EMTC	P	V/P	-/V	
d)	Temperature Cycle Test	Physical	2 Samples/Lot		Cracks / Glaze defects should not occur	CMTC/EMTC	P	V/P	-/V	
e)	Routine Pressure Test	Mechanical	100%		Manufacturer's Plant Standard/IS 5621/IEC 62155/IEC 99-4 As per spec /GTP	Manufacturer's Plant Standard/IS 5621/IEC 62155/IEC 99-4 As per spec /GTP	CMTC/EMTC	P	V/P	-/V
f)	Ultrasonic Test	Mechanical	100%				CMTC/EMTC	P	V/P	-/V
g)	Burst Pressure Test	Mechanical	1 Sample/Lot				CMTC/EMTC	P	V/P	-/V
h)	Bending Load Test in 3 directions at 50% specified	Mechanical	1 Sample/Lot				CMTC/EMTC	P	V/P	-/V

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
	bending load and in 4th direction at 100% load								
2	Polymer Insulator (Silicon)								
a)	Visual	Visual	100%	Manufacturer's Plant Standard/ Purchase Drg/IEC 61462/IEC 62217/AS PER Spec	Manufacturer's Plant Standard/ Purchase Drg/IEC 61462/IEC 62217/AS PER Spec	CMTC/EMTC	P	V/P	-/V
b)	Overall Height	Mechanical	100%			CMTC/EMTC	P	V/P	-/V
c)	Inner Diameter	Mechanical	100%			CMTC/EMTC	P	V/P	-/V
d)	Parallesim of flanges	Mechanical	100%			CMTC/EMTC	P	V/P	-/V
e)	Creepage distance	Mechanical	100%			CMTC/EMTC	P	V/P	-/V
g)	Inner pressure	Mechanical	100%			CMTC/EMTC	P	V/P	-/V
h)	bending load	Mechanical	100%			CMTC/EMTC	P	V/P	-/V
i)	Gas Leakage test	Mechanical	100%			CMTC/EMTC	P	V/P	-/V
j)	NDT check to ensure the quality of jointing of the housing interface with core	Mechanical	1 Sample/Shift			CMTC/EMTC	P	V/P	-/V
	All routine test					CMTC/	P	V/P	-/V

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
k)	(Electrical & Mech) as per IEC 61462 as applicable	Electrical	100%			EMTC			
3	Aluminium Castings								
a)	Visual Examination	Visual	100%	Manufacturer's Plant Standard/ DRG/AS PER Spec/BS 1490	Manufacturer's Plant Standard/ DRG /AS PER Spec/ BS 1490	CMTC/ EMTC	P	V/P	-/V
b)	Verification of Dimensions	Dimensional	5 Samples/Lot			CMTC/ EMTC	P	V/P	-/V
c)	Tensile Test on Test Bar	Mechanical	1 Test piece/1000kgs castings			CMTC	P	V/P	-/V
d)	Elongation Test on Test Bar	Mechanical	1 Test piece/1000kgs castings			CMTC	P	V/P	-/V
e)	Chemical Analysis	Chemical	1 Test piece/2000kgs castings			Third party TC	P	V/P	-/V
f)	Hardness	Physical	1 Test piece/1000kgs castings			CMTC	P	V/P	-/V
4	Gaskets / 'O' rings								

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
a)	Verification of Dimensions	Dimensional	2 Samples/Lot	Manufacturer's Plant Standard/DRG/AS PER Spec/IS 11149/ IS 3400	Manufacturer's Plant Standard/DRG/AS PER Spec/IS 11149/ IS 3400	CMTC/EMTC	P	V/P	-/V
b)	Hardness Test (Shore A)	Physical	2 Samples/Lot			CMTC	P	V	-
c)	Elongation	Mechanical	2 Samples/Lot			CMTC	P	V	-
d)	Tensile Strength	Mechanical	2 Samples/Lot			CMTC	P	V	-
e)	Density	Mechanical	2 Samples/Lot			CMTC	P	V	-
f)	Comprssion Set Test	Physical	2 Samples/Lot			CMTC	P	V	-
g)	Accelerated Ageing Test	Physical	2 Samples/Lot			CMTC	P	V	-
h)	Ozone Resistance test	Physical	2 Samples/Lot			CMTC	P	V	-
5	Springs								
a)	Verification of Dimensions	Dimensional	2 Samples/Lot	Manufacturer's Plant Standard/DRG/IS 4454 (Part-4)	Manufacturer's Plant Standard/DRG	CMTC/EMTC	P	V/P	-/V
b)	Surface Condition	Visual	100%		Should be smooth and free from burrs	CMTC	P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
c)	Load deflection test	Mechanical	2 Samples/Lot		Manufacturer's Plant Standard/DRG/IS 4454 (Part-4)	CMTC	P	V	-
d)	Chemical Analysis	Chemical	2 Samples/Lot			CMTC	P	V	-
6	Pressure Relief Diaphragm								
a)	Visual Examination	Visual	100%	Manufacturer's Plant Standard/DRG/AS PER Spec	Manufacturer's Plant Standard/DRG/AS PER Spec	CMTC/EMTC	P	V/P	-/V
b)	Verification of Dimensions	Dimensional	1 Sample/Lot			CMTC	P	V	-
c)	Burst Pressure Test	Pneumatic	1 Sample/Lot			CMTC	P	V	-
7	Sulphur Cement (if applicable)								
a)	Compressive Strength after aging for 48hrs	Mechanical	2 Cubes of 50mm / lot of 2500kg	Manufacturer's Plant Standard/IS 4832	Manufacturer's Plant Standard/IS 4832	CMTC	P	V	-
b)	Tensile strength after aging for 48hrs	Mechanical	2 Samples/lot			CMTC	P	V	-
c)	Flextural strength after aging for 48hrs	Mechanical	2 Samples/lot			CMTC	P	V	-
d)	Bonding strength after aging for 48hrs	Mechanical	2 Samples/lot			CMTC	P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
8	Silicon Rubber								
a)	Specific gravity	Physical	1 /month	Manufacturer's Plant Standard/AS PER Spec	Manufacturer's Plant Standard/AS PER Spec	CMTC	P	V	-
b)	Hardness	Physical	1 /month			CMTC	P	V	-
c)	Tensile strength	Mechanical	1 /month			CMTC	P	V	-
d)	Elongation at break	Mechanical	1 /month			CMTC	P	V	-
e)	Tear resistance	Electrical	1 /half year			CMTC	P	V	-
f)	Dielectric strength	Mechanical	1 /month			CMTC	P	V	-
g)	Volume resistivity	Electrical	1 /month			CMTC	P	V	-
h)	Surface resistivity	Electrical	1 /month			CMTC	P	V	-
i)	Dielectric constant, 50Hz	Electrical	1 /half year			CMTC	P	V	-
j)	Dissipation factor (tanδ), 50Hz	Electrical	1 /half year			CMTC	P	V	-
k)	Tracking resistance	Electrical	1 /half year			CMTC	P	V	-
l)	Flammability (FV-0)	Physical	1 /half year			CMTC	P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
9	Sheet Metal Pressed Components								
a)	Visual Examination	Visual	100%	Manufacturer's Plant Standard/DRG	Manufacturer's Plant Standard/DRG	CMTC/EMTC	P	V/P	-/V
b)	Verification of Dimensions	Dimensional	2 Samples/Lot			CMTC/EMTC	P	V/P	-/V
10	Grading/ Corona Ring								
a)	Visual Examination	Visual	100%	IS 733/ Manufacturer's Plant Standard/DRG	IS 733/ Manufacturer's Plant Standard/DRG	CMTC/EMTC	P	V	-
b)	Verification of Dimensions	Dimensional	2 Samples/Lot			CMTC/EMTC	P	V	-
c)	Chemical Composition	Chemical	1 Sample/Lot			Third party test report	P	V	-
d)	Tensile Strength	Mechanical	1 Sample/Lot			CMTC	P	V	-
11	FRP Tie Rods / Tubes								

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
a)	Visual Examination	Visual	100%	IEC 62217/Manufacturer's Plant Standard/DRG/AS PER Spec.	IEC 62217/Manufacturer's Plant Standard/DRG/AS PER Spec.	CMTC/EMTC	P	V	-
b)	Verification of Dimensions	Dimensional	100%			CMTC/EMTC	P	V	-
c)	Voltage With stand Test	Electrical	1 Sample/Lot			CMTC/EMTC	P	V	-
d)	Leakage Current Test	Electrical	1 Sample/Lot			CMTC/EMTC	P	V	-
e)	Specific Gravity	Physical	1 Sample/Lot			CMTC/EMTC	P	V	-
f)	Water Absorption/Diffusion	Physical	1 Sample/Lot			CMTC/EMTC	P	V	-
g)	Dye penetration test	Physical	1 Sample/Lot			CMTC/EMTC	P	V	-
12	Insulating Base								
a)	Visual Examination	Visual	100%	Manufacturer's Plant Standard/DRG	Manufacturer's Plant Standard/DRG	CMTC/EMTC	P/-	V/P	-/W
b)	Verification of Dimensions	Dimensional	2 Sample/Lot			CMTC/EMTC	P/-	V/P	-/W
c)	Insulation With stand Test	Electrical	2 Sample/Lot			CMTC	P	V	-
d)	Bending moment	Mechanical	2 Sample/Lot			CMTC	P	V	-
e)	Tensile test	Mechanical	100%			CMTC	P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
13	Milli-ammeter								
a)	Visual Examination	Visual	100%	IS:1248/Manufacturer's Plant Standard	IS:1248/Manufacturer's Plant Standard	CMTC/EMTC	P/-	V/P	-/W
b)	Verification of Dimensions	Dimensional	2 Sample/Lot			CMTC/EMTC	P/-	V/P	-/W
c)	Operational Test	Electrical	100%			CMTC	P	V	-
d)	HV test	Electrical	2 Sample/Lot			CMTC	P	V	-
e)	Insulation resistance	Electrical	2 Sample/Lot			CMTC	P	V	-
14	Cyclometric Counter								
a)	Visual Examination	Visual	100%	Manufacturer's Plant	Manufacturer's Plant Standard/ DRG	CMTC/EMTC	P/-	V/P	-/W
b)	Operational Test	Electrical	100%	Standard/ DRG		CMTC/EMTC	P/-	V/P	-/W
15	Raw materials for Metal Oxide Blocks (ZnO								

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
	Blocks)								
a	Zinc Oxide	Chemical	50 gms / batch	Manufacturer's Plant Standard/DRG	Manufacturer's Plant Standard/DRG	CMTC/EMTC	P/-	V/P	-/W
b	Bismuth Oxide	Chemical	50 gms / batch			CMTC/EMTC	P/-	V/P	-/W
c	Antimony Tri Oxide	Chemical	50 gms / batch			CMTC/EMTC	P/-	V/P	-/W
d	Cobalt Oxide	Chemical	50 gms / batch			CMTC/EMTC	P/-	V/P	-/W
e	Chromium Oxide	Chemical	50 gms / batch			CMTC/EMTC	P/-	V/P	-/W
f	Manganese Dioxide	Chemical	50 gms / batch			CMTC/EMTC	P/-	V/P	-/W
g	Silicone Dioxide	Chemical	50 gms / batch			CMTC/EMTC	P/-	V/P	-/W
h	PVA	Chemical	50 gms / batch			CMTC/EMTC	P/-	V/P	-/W
16	Pressure Release Plate (S.S – 304)								
a)	Visual Inspection	Physical	1 Sample/Lot	Manufacturer's Plant Standard/DRG	Manufacturer's Plant Standard/DRG	CMTC	P	V	-
b)	Dimensions	Physical	1 Sample/Lot			CMTC	P	V	-
c)	Chemical Analysis	Chemical	1 Sample/Lot			CMTC	P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
17	S.S. Hardware								
a)	Visual Inspection	Physical	100%	Manufacturer's Plant Standard/DRG	Manufacturer's Plant Standard/DRG	CMTC	P	V	-
b)	Magnetic Check	Physical	100%			CMTC	P	V	-
c)	Chemical Analysis	Chemical	1 Sample/Lot			CMTC	P	V	-
18	Terminal Clamps								
a)	Visual Examination	Physical	1 Sample/Lot	Manufacturer's Plant Standard/DRG/Tech. Spec/IS 733	Manufacturer's Plant Standard/DRG/Tech. Spec/IS 733	CMTC	P	V	-
b)	Dimensions	Physical	1 Sample/Lot			CMTC	P	V	-
c)	Tesnile test	Mechanical	1 Sample/Lot			CMTC	P	V	-
d)	Chemical Analysis	Chemical	1 Sample/Lot			CMTC	P	V	-
Section II : In process Checks									
	Zinc Oxide Blocks Inspections								
a)	Visual Examination	Visual	100%	IEC 60099-4/Manufacturer's Plant	IEC 60099-4/Manufacturer's	EMTC	P/-	V/P	-/W
b)	Verification of Dimensions	Dimensional	5 Sample/Lot			EMTC	P/-	V/P	-/W

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
c)	Energy Handling Capability test	Electrical	100%	Standard/DRG/ Tech. Spec.	Plant Standard/DRG/ Tech. Spec.	EMTC	P/-	V/P	-/W
d)	RDV Test	Electrical	100%			EMTC	P/-	V/P	-/W
e)	Accelerated aging test for 72hrs	Electrical	5 Sample/Lot			EMTC	P/-	V/P	-/W
f)	Thermal Stability Test	Electrical	5 Sample/Lot			EMTC	P/-	V/P	-/W
g)	Wattloss or Resistive Current	Electrical	5 Sample/Lot			EMTC	P/-	V/P	-/W
	Sample should be from ongoing production irrespective of contract								
Section III: Surge Arresters Assembly									
1	Surge Arresters								
a)	Seal leak Test	Physical	100%	IEC 60099-4/ IS : 3070/AS PER Tech. Spec./ Apporved	Megger value after dip test should be >10GΩ for L.A. >=66kV,>1GΩ for LA<66kV	EMTC	P	V	-
b)	Visual Examination	Visual	100%		IEC 99-4	EMTC	P	V	-

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S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
c)	Verification of Dimensions	Dimensional	2 Samples/Lot	GTP/Mfr's Plant Std.	Approved Drawing	EMTC	P	V	-
d)	Reference Voltage	Electrical	100%		> Surge Arrester Rating.	EMTC	P	V	-
e)	Lightning Impulse Residual Discharge Voltage Test at 10kA	Electrical	100%		IEC 99 - 4 GTP	EMTC	P	V	-
f)	Partial Discharge Test at MCOV x 1.05 times KV	Electrical	100%		< 10 pC	EMTC	P	V	-
g)	Leakage Current at MCOV	Electrical	100%		IEC 99 - 4/ Approved GTP	EMTC	P	V	-
h)	Verticality Check on	Mechanical	1 Sample/Lot		Max.10 mm deviation	EMTC	P	V	-
2	Surge Monitor								
a)	Visual Examination	Visual	100%	IEC 60099-4// AS PER Spec./Apporved GTP/Mfr's Plant Std.	IEC 60099-4/ AS PER Spec./Apporved GTP/Mfr's Plant Std.	EMTC	P	V	-
b)	Verification of Dimensions	Dimensional	1 Sample/Lot			EMTC	P	V	-
c)	Meter Operation	Electrical	100%			EMTC	P	V	-
d)	Counter Operation	Electrical	100%			EMTC	P	V	-

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
e)	Water Dip Test	Physical	100%		No condensation of water	EMTC	P	V	-

Section IV : Acceptance Tests (All Routine and Acceptance test mentioned in applicable Technical Specification of subject Contract in question need to be carried out)

1	Surge Arresters								
a)	Reference Voltage test	Electrical	Cube root of the Qty ordered	IEC 99 - 4/IS : 3070/appvd GTP/As per Spec	> Surge Arrester Rating	EMTC	P	V	W
b)	Lightning Impulse Residual Discharge Voltage Test at 10kA	Electrical	Cube root of the Qty ordered	IEC 99 - 4/IS : 3070/appvd GTP/As per Spec	IEC 99 - 4 GTP As per Spec	EMTC	P	V	W
c)	Partial Discharge Test	Electrical	Cube root of the Qty ordered	IEC 99 – 4/IS : 3070/appvd GTP/As per Spec	< 10 pC	EMTC	P	V	W
d)	Thermal Stability Test on ZnO blocks	Electrical	1 Samples per Lot	IEC 99 - 4/ /appvd GTP/As per Spec	IEC 99 - 4	EMTC	P	V	W

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
e)	Accelerated Aging test for 72 Hrs. on ZnO blocks	Electrical	3 Samples per Lot	IEC 99 - 4/appvd GTP/As per Spec	IEC 99 - 4/appvd GTP/As per Spec	EMTC	P	V	W
f)	Energy Handling Test on ZnO blocks	Electrical	3 Samples per Lot	IEC 99 - 4/appvd GTP/As per Spec	IEC 99 - 4/appvd GTP/As per Spec	EMTC	P	V	W
g)	Watt loss / IR measurement test on ZnO blocks	Electrical	3 Samples per Lot	IEC 99 - 4/appvd GTP/As per Spec	IEC 99 - 4/appvd GTP/As per Spec	EMTC	P	V	W
h)	Seal leak Test	Physical	Cube root of the Qty ordered	IEC 99 - 4/appvd GTP/As per Spec	Megger value after dip test should be >10GΩ for L.A. >=66kV, >1GΩ for LA<66kV	EMTC	P	V	W
i)	Verticality Check on Assembled Surge Arrester	Mechanical	1 Sample/ Lot	IEC 99 - 4/appvd GTP/As per Spec	Max.10 mm deviation allowed in verticality	EMTC	P	V	-
j)	Galvanizing Test								
(i)	Uniformity of Zn Coating	Chemical	2 Sample/ Lot	IS 2633	IS 2633	EMTC	P	V	W

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

S. No.	Components / Operations & Description of Test	Type of check	Quantum of Check / Sampling with basis	Reference document for Testing	Acceptance Norms	Format of Record	Category of responsibility		
							Sub-Vendor	Manufacturer	Customer
(ii)	Weight of Zn Coating	Chemical	2 Sample/ Lot	IS 2633	IS 2633	EMTC	P	V	W
(iii)	Adhesion Test	Physical	2 Samples/Lot	IS 2633	IS 2633	EMTC	P	V	W
2	Surge Monitor								
a)	Visual Examination	Visual	Cube root of the Qty ordered	IEC 60099-4/ AS PER Spec./Apporved GTP/Mfr's Plant Std.	IEC 60099-4/ AS PER Spec./Apporved GTP/Mfr's Plant Std.	EMTC	P	V	W
b)	Verification of Dimensions	Dimensional				EMTC	P	V	W
c)	Water Dip Test	Physical				EMTC	P	V	W
d)	Counter Operation at 10kA 8/20μs	Electrical				EMTC	P	V	W
e)	MiliiAmmeter Operation	Electrical				EMTC	P	V	W
Section V: Packing & Dispatch									
1	Physical	visual	AS PER Spec./ Apporved GTP/Mfr's Plant Std.			EMTC	P	V	-
2	Dimension	Physical				EMTC	P	V	-

EM : Equipment Manufacturer CM : Component Manufacturer CMTC :- Component manufacturer test certificate EMTR :- Equipment manufacturer test Report

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

Transformer/ Reactors

For Model QAP of Transformer/ Reactor (66 kV and above), Annexure-E of CEA's Standard Specification and Technical Parameters for Transformers and Reactors (66 kV & above Voltage Class) shall be referred.

Note: Purchaser reserves the right to review the records of all tests specified in this document at the time of final acceptance.

* Category of Responsibility: P - Actual Test Performance V - Verify and Accept W - Witness Actual testing, verify and accept

ANNEXURE B

MODEL QAPs OF MAJOR EQUIPMENT IN HYDRO POWER PLANT

List of Major Equipment in Hydro Power Plant for Model QAP Preparation:

Sl. No.	Name of Equipment
1)	Hydro Turbine/ Pump Turbine & Governing System a) Runner b) Stay Rings c) Discharge Ring d) Spiral Casing e) Head Cover f) Bottom Ring g) Turbine Shaft h) Turbine Shaft seal i) Servo Motor j) Turbine Guide Bearing k) Labyrinth l) Injector (Nozzle Assembly) m) Guide Apparatus n) Digital Governing System
2)	Generator a) Rotor b) Stator c) Generator Shaft d) Generator Bracket e) Generator Guide and/ or Thrust Bearings
3)	Main Inlet Valve (Spherical / Butterfly Valve)
4)	Penstock Valve (Butterfly Valve)
5)	i. Static Excitation System for Synchronous machines ii. Dynamic Excitation System for Asynchronous machines
6)	Isolated Phase Bus Duct (IPBD) / Segregated Phase Bus Duct (SPBD)
7)	EOT Crane
8)	Lubricating and Insulating Oil Handling System
9)	i. Static Frequency Converter (SFC) system ii. Variable-Frequency Drive (VFD)
10)	Current Limiting Reactors
11)	Generator Circuit Breaker

12)	Phase Reversal Disconnecting Switch
13)	i. Dynamic Braking cubicle ii. Starting Switch
14)	EHV XLPE cable and termination bushings
15)	Shunt Reactor/ Generator Transformer/ Distribution Transformer **
16)	Circuit Breaker**
17)	Isolator**
18)	Lightning Arrester**
19)	Wave Trap**
20)	Instrument Transformers (CT, PT)**
21)	LV and MV Switchgear**
22)	GIS & Hybrid Switchgear**
23)	Cables and associated Joints**
24)	Battery and Battery Charger**
25)	Transformer/ Reactor Fittings and Accessories**

** MQAPs of Transmission Sector shall be referred for these equipment.

Note:- MQAPs of those E&M equipment which have not been covered in this section shall be subject to the terms of mutual agreement between manufacturer and purchaser.

Index of Model QAPs of major Equipment in Hydro Power Plant:

Sl. No.	Name of Equipment	Page Nos.
1)	Hydro Turbine/ Pump Turbine & Governing System a) Runner b) Stay Rings c) Discharge Ring d) Spiral Casing e) Head Cover f) Bottom Ring g) Turbine Shaft h) Turbine Shaft seal i) Servo Motor j) Turbine Guide Bearing k) Labyrinth l) Injector (Nozzle Assembly) m) Guide Apparatus n) Digital Governing System	B-7-B-57
2)	Generator a) Rotor b) Stator c) Generator Shaft d) Generator Bracket e) Generator Guide and/ or Thrust Bearings	B-58-93
3)	Main Inlet Valve (Spherical/ Butterfly Valve)	B-94-B-107
4)	Penstock Valve (Butterfly Valve)	B-94-B-107
5)	i. Static Excitation System for Synchronous machines ii. Dynamic Excitation System for Asynchronous machines	B-108-B-118 B-119-B-125
6)	Isolated Phase Bus Duct (IPBD) / Segregated Phase Bus Duct (SPBD)	B-126-B-139
7)	EOT Crane	B-140-B-147
8)	Lubricating and Insulating Oil Handling System	B-148-B-157
9)	i. Static Frequency Converter (SFC) system ii. Variable-Frequency Drive (VFD)	B-158-B-163 B-164-B-167
10)	Current Limiting Reactors	B-168-B-170

11)	Generator Circuit Breaker	B-171-B-174
12)	Phase Reversal Disconnecting Switch	B-175-B-178
13)	i. Dynamic Braking cubicle ii. Starting Switch	B-179-B-181 B-182-B-186
14)	EHV XLPE cable and termination bushings	B-187-B-200
15)	Shunt Reactor/ Generator Transformer/ Distribution Transformer **	
16)	Circuit Breaker**	
17)	Isolator**	
18)	Lightning Arrester**	
19)	Wave Trap**	
20)	Instrument Transformers (CT, PT)**	
21)	LV and MV Switchgear**	
22)	GIS & Hybrid Switchgear**	
23)	Cables and associated Joints**	
24)	Battery and Battery Charger**	
25)	Transformer/ Reactor Fittings and Accessories**	

** MQAPs of Transmission Sector shall be referred for these equipment.

Abbreviations used in Model QAPs:

S. No	Abbreviation	Full Form
1.	IS	Indian Standard
2.	IEC	International Electrotechnical Commission
3.	NABL	National Accreditation Board for Testing and Calibration Laboratories
4.	RR	Review of Records
5.	TC	Test Certificate
6.	TR	Test Report
7.	JIR	Joint Inspection Report
8.	CHP	Customer Hold Point
9.	IR	Internal Report
10.	P	Perform
11.	W	Witness
12.	NDT	Non-Destructive Testing
13.	RT	Radiographic Testing
14.	UT	Ultrasonic Testing
15.	MPI	Magnetic Particle Inspection
16.	DPT	Dye Penetrant Test
17.	HVOF	High Velocity Oxygen Fuel

1) HYDRO TURBINE/ PUMP TURBINE & GOVERNING SYSTEM

उपस्कर का नाम (NAME OF EQUIPMENT) : Stay Ring & Spiral Casing					विक्रेता (VENDOR) :				
					एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :				
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORDED FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material								
1.1	Casting/Plate for Parallel Ring, Stay Vane & Spiral Casing			Appd. drawings/ Tech. specs./IS/IEC/Equivalent					
a)	Chemical Composition	Chemical Test	Sample per Lot	-do-	TC	3/2	-	1	TC
b)	Mechanical Analysis-As Applicable (Tensile Test, Impact Test, Hardness Test)	Mechanical Test	Sample per Lot	-do-	TC	3/2	-	1	TC
c)	NDT of Plate Material	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
2	In Process Stage								
2.1	Post weld Heat treatment	Heat Chart	-do-	-do-	TC	3/2	-	1	TC
2.2	Welder Certification	Visual	100%	-do-	TC	3/2	-	1	TC
2.3	NDT of Weld Joints after Heat Treatment	UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
2.4	Dimensional Check -PCD, Dia & orientation of Holes w.r.t. parting plane, Concentricity	Visual / Measurement	-do-	-do-	IR	3/2	-	1	RR
2.5	Marking of X-Y axis	Visual	-do-	-do-	IR	3/2	-	1	RR
3	Final Inspection								

3.1	Layout check of individual segments with dummy stay ring	Visual / Measureme nt	Critical Dim.	-do-	IR	3/2	-	1	TC
3.2	Dimensional check of Stay Ring after machining.	-do-	100%	-do-	JIR	3/2	1	-	CHP
4	Surface Preparation & Painting	-do-	-do-	-do-	TC	3/2	-	1	TC
Note :	a. In ‘Inspection Agency’column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate ‘supplier’ & 3- will indicate ‘sub-supplier’.								
	b. In ‘Remarks’ column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : - Bottom Ring				विक्रेता (VENDOR) :					
				एनआईटी/क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	

1	Raw Material								
1.1	Plates / Carbon Steel			Appd. drawings/ Tech. specs./IS/IEC/Equivalent					
a)	Test Piece Marking	Visual	Sample per Lot		IR	3/2	-	1	TC
b)	Chemical Composition	Chemical Test	Sample per Lot		IR	3/2	-	1	TC
c)	Mechanical Analysis- As applicable (Tensile Test, Impact Test and Hardness Test)	Mechanical Test	- do-	-do-	IR	3/2	-	1	TC
d)	NDT of Plates / Casting	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
2	In Process Stage								
2.1	Welder Certification	Visual	100%	-do-	TC	3/2	-	1	TC
2.2	Post Weld Stress Relieving	Heat chart	-do-	-do-	IR	3/2	-	1	TC
2.3	NDT on weld joints after Heat Treatment	UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
2.4	Dimensions check and Flatness of liner plate face	Measurement	- do-	-do-	IR	3/2	-	1	TC
3	Final Inspection before HP-HVOF Coating								
3.1	Marking of X & Y axis	Visual / Measurement	100%	-do-	IR	3/2	-	1	TC
3.2	Dimensional check & Surface finish check	Measurement	Critical Dim.	-do-	IR	3/2	-	1	TC
3.3	Proof Assly. with Guide Apparatus Assembly	Measurement	100%	-do-	IR	3/2	-	1	TC
3.4	Attachment/Tagging of atleast five sample pieces with Job before start of coating (If HVOF coating is applicable)	Visual	Sample	-do-	IR	3/2	-	1	TC
4	HVOF Coating (If applicable)								
4.1	Chemical Composition of coating Powder	Test	Sample per Lot	-do-	TC	3/2	-	1	TC
4.2	Detachment of sample pieces after coating	Visual	Sample	-do-	IR	3/2	-	1	TC
4.3	Mechanical Properties- Micro Hardness, Porosity, Bond Strength on sample	Visual & Measurement	Sample per Lot	-do-	IR	3/2	-	1	TC
4.4	Coating Thickness	Visual & Measurement	Sample	-do-	IR	3/2	-	1	TC

Note :	a. In ‘Inspection Agency’column figure 1,2,or 3 to be filled. 1- will indicate‘Employer’, 2- will indicate ‘supplier’ & 3- will indicate ‘sub-supplier’.									
	b. In ‘Remarks’ column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.									
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)									
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.									
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .									
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.									
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.									
			Signature			Signature & Seal				
			EMPLOYER (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : -Discharge Ring				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material								
1.1	Plates / Casting -			Appd. drawings/ Tech. specs./IS/IEC/Equivalent					
a)	Chemical Composition	Chemical Test	Sample per Lot		TC	3/2	-	1	TC
b)	Mechanical Analysis- As applicable (Tensile Test, Impact Test and Hardness Test)	Mechanical Test	- do-	-do-	TC	3/2	-	1	TC

c)	NDT of Plates	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
2	In Process Stage								
2.1	Post Weld Stress Relieving	Heat chart	-do-	-do-	IR	3/2	-	1	TC
2.2	Welder Certification	Visual	100%	-do-	TC	3/2	-	1	TC
2.3	NDT on weld joints after Heat Treatment	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
2.4	Dimensional Check of Machined Surface	Measurement	100%	-do-	IR	3/2	-	1	RR
3	Final Inspection								
3.1	Marking of X & Y axis	Visual / Measurement	100%	-do-	IR	3/2	-	1	TC
3.2	Dimensional check	Measurement	Critical Dim.	-do-	IR	3/2	-	1	TC
3.3	Attachment/Tagging of atleast five sample pieces with Job before start of coating (If HVOF coating is applicable)	Visual	Sample	-do-	IR	3/2	-	1	TC
4	HVOF Coating (If applicable)								
4.1	Chemical Composition of coating Powder	Test	Sample per Lot	-do-	TC	3/2	-	1	TC
4.2	Detachment of sample pieces after coating	Visual	Sample	-Do-	IR	3/2	-	1	TC
4.3	Mechanical Properties- Micro Hardness, Porosity, Bond Strength on sample	Visual & Measurement	Sample per Lot	-do-	IR	3/2	-	1	TC
4.4	Coating Thickness	Visual & Measurement	Sample	-do-	IR	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								

	f. Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.				
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.				
			Signature		Signature & Seal
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : -Head Cover				विक्रेता (VENDOR) :					
				एनआईटी/क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material								
1.1	Plates / Carbon Steel			Appd. drawings/ Tech. specs./IS/IEC/Equivalent					
a)	Chemical Composition	Chemical Test	- do-		IR	3/2	-	1	TC
b)	Mechanical Analysis(Tensile Test, Impact Test,Hardness Test & Bend Test -Plate) As applicable	Mechanical Test	- do-		IR	3/2	-	1	TC
c)	NDT of Plates / Casting	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
2	In Process Stage								
2.1	Post Weld Stress Relieving	Heat chart	-do-	-do-	IR	3/2	-	1	TC
2.2	Welder Certification	Visual	100%	-do-	TC	3/2	-	1	TC
2.3	NDT on weld joints after Heat Treatment	UT, MPI, DPT	-do-	-do-	TC	3/2	-	1	TC

		(as applicable)							
2.4	Dimensions check and Flatness of liner plate face	Measurement	- do-	-do-	IR	3/2	-	1	TC
3	Final Inspection before HP-HVOF Coating								
3.1	NDT of Facing Plate after machining		100%	-do-	IR	3/2	-	1	TC
3.2	Marking of X & Y axis	Visual / Measurement	100%	-do-	IR	3/2	-	1	TC
3.3	Dimensional check and Surface finish check	Measurement	Critical Dim.	-do-	IR	3/2	-	1	TC
3.4	Proof Assly. with Guide Apparatus Assembly	Measurement	100%	-do-	IR	3/2	-	1	TC
3.5	Attachment/Tagging of atleast five sample pieces with Job before start of coating (If HVOF coating is applicable)	Visual	Sample	-do-	IR	3/2	-	1	TC
4	HVOF Coating (If Applicable)								
4.1	Chemical Composition of coating Powder	Test	Sample per Lot	-do-	TC	3/2	-	1	TC
4.2	Detachment of sample pieces after coating	Visual	Sample	-do-	IR	3/2	-	1	TC
4.3	Mechanical Properties- Micro Hardness, Porosity, Bond Strength on sample	Visual & Measurement	Sample per Lot	-do-	IR	3/2	-	1	TC
4.4	Coating Thickness	Visual & Measurement	Sample	-do-	IR	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :					ग्राहक (CLIENT) :				
उपस्कर का नाम (NAME OF EQUIPMENT) : -Guide Vanes					विक्रेता (VENDOR) :				
					एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :				
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material								
1.1	Casting-			Appd. drawings/ Tech. specs./IS/IEC/Equivalent					
a)	Keel Block Identification	Visual	Sample per Lot		IR	3/2	-	1	TC
b)	Chemical Composition	Chemical Test	Sample per Lot	-do-	IR	3/2	-	1	TC
c)	Mechanical Analysis (Tensile Test, Impact Test and Hardness Test)	Mechanical Test	- do-	-do-	IR	3/2	-	1	TC
d)	NDT of Casting	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
2	In Process Stage								
2.1	Heat Treatment	Heat Chart	-do-	-do-	IR	3/2	-	1	TC
2.2	Profile Checking	Measurement	100%	-do-	IR	3/2	-	1	RR
2.3	Machining								
a)	Dimensions & Surface Finish Check	Visual / Measurement	100%	-do-	IR	3/2	-	1	TC
3	Final Inspection before HP-HVOF Coating								
3.1	Dimensional Check & Surface finish	Measurement	Critical Dim.	-do-	IR	3/2	-	1	TC
3.2	Profile Checking with Template (Top, Middle & Bottom)	Measurement	100%	-do-	IR	3/2	-	1	TC

3.3	Coaxiality of Journal Diameters /Run out Check	-do-	20%	-do-	IR	3/2	-	1	TC
3.4	Proof Assly. With Guide Appratus Assly.	-do-	100%	-do-	IR	3/2	-	1	TC
3.5	Attachment/Tagging of atleast five sample pieces with one lot before start of coating (If HVOF coating is applicable)	Visual	Sample	-Do-	IR	3/2	-	1	TC
4	HP-HVOF Coating (If Applicable)								
4.1	Chemical Composition of coating Powder	Test	Sample per Lot	-do-	TC	3/2	-	1	TC
4.2	Detachment of sample pieces after coating	Visual	Sample	-Do-	IR	3/2	-	1	TC
4.3	Mechanical Properties- Micro Hardness, Porosity, Bond Strength on sample	Visual & Measurement	Sample per Lot	-do-	IR	3/2	-	1	TC
4.4	Coating Thickness	Visual & Measurement	Sample	-do-	IR	3/2	-	1	TC
Note:	a. In ‘Inspection Agency’column figure 1,2,or 3 to be filled. 1- will indicate‘Employer’, 2- will indicate ‘supplier’ & 3- will indicate ‘sub-supplier’.								
	b. In ‘Remarks’ column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :	ग्राहक (CLIENT) :
उपस्कर का नाम (NAME OF EQUIPMENT) : Servomotor	विक्रेता (VENDOR) :
	एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :

क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material								
1.1	Cast / Forged Steel, Steel Plate and Rod			Appd. drawings/ Tech. specs./IS/IEC/Equivalent					
a)	Chemical Composition	Chemical Test	Sample per Lot		TC	3/2	-	1	TC
b)	Mechanical Analysis-As Applicable (Tensile Test, Impact Test, Hardness Test etc)	Mechanical Test	- do-		TC	3/2	-	1	TC
c)	NDT of Material	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
2	In Process Stage								
2.1	Heat Treatment	Heat Chart	- do-	-do-	IR	3/2	-	1	TC
2.2	Welder Certification	Visual	100%	-do-	TC	3/2	-	1	TC
2.3	NDT of weld joints after Heat Treatment	UT, MPI, DPT (as applicable)	- do-		TC	3/2	-	1	TC
2.4	Dimensional check after machining	Visual/Measurement	- do-	-do-	IR	3/2	-	1	RR
3	Final Inspection								
3.1	Dimensional check of Final Assembly	Measurement	Critical Dim.	-do-	JIR	3/2	1	-	CHP
3.2	Hydraulic pressure test of Servomotor-On test Pressure (Design Pressure)	Test	100%	-do-	JIR	3/2	1	-	CHP
3.3	Leakage across Piston Check in fully open& fully close Position-On Working pressure	-do-	100%	-do-	JIR	3/2	1	-	CHP
3.4	Stroke Length measurement-On Working Pressure	Visual / Measurement	-do-	-do-	JIR	3/2	1	-	CHP
3.5	Functional Test of servo-motor (Stroking 10 cycles)-on Working Pressure	-do-	-do-	-do-	JIR	3/2	1	-	CHP
4	Painting	-do-	-do-	-do-	TC	3/2	-	1	TC
Note	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								

:					
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.				
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)				
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.				
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .				
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.				
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.				
			Signature		Signature & Seal
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : - Francis Runner				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Runner (Runner casting, Plate/blade,Crown, Band)			Appd. drawings/ Tech. specs./IS/IEC/Equivalent					
1.1	Keel Block Identification	Visual	Sample per Lot		JIR	3/2	1	-	CHP
1.2	Chemical Composition	Chemical Test	- do-		JIR	3/2	1	-	CHP
1.3	Mechanical Analysis (Tensile Test, Impact Test, Hardness Test etc)	Mechanical Test	- do-	-do-	JIR	3/2	1	-	CHP
1.4	Microstructure of material	Metallurgical Test	- do-	-do-	JIR	3/2	1	-	CHP

2	In Process Stage Inspection							-	
2.1	Welder Certification	Visual	100%	-do-	TC	3/2	-	1	TC
2.2	Heat Treatment & S/R of Complete Runner	H-T Chart	-do-	-do-	TC	3/2	-	1	TC
2.3	NDT of Casting/blades of Runner after heat treatment & S/R	RT, UT, MPI, DPT (as applicable)	-do-	-do-	JIR	3/2	1	-	CHP
2.4	Dimensional and profile check before machining	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
3	Final Inspection before HVOF Coating.								
3.1	Dimension Check and Surface Finish	-do-	Critical Dim.	-do-	JIR	3/2	1	-	CHP
3.2	Concentricity & Run Out Check of labyrinth with Runner	-do-	100%	-do-	IR	3/2	-	1	TC
3.3	Profile checking with templates.	-do-	-do-	-do-	JIR	3/2	1	-	CHP
3.4	Inlet & Outlet Pitch	Measurement with Template	-do-	-do-	JIR	3/2	1	-	CHP
3.5	Water paths (Pr.and Suction Side) Profile Check		-do-	-do-	JIR	3/2	1	-	CHP
3.6	NDT for Surface / Sub-surface Defects after machining	UT, MPI, DPT (as applicable)	-do-	-do-	JIR	3/2	1	-	CHP
3.7	Attachment/Tagging of atleast five sample pieces with Job before start of coating (If HVOF is applicable)	Visual	Sample	-Do-	IR	3/2	-	1	TC
4	Static & Dynamic Balancing of Runner	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
5	Final Inspection after HVOF Coating (If Applicable)								
5.1	Chemical Composition of coating Powder	Measurement	Lot/ sample	-Do-	TC	3/2	-	1	TC
5.2	Detachment of sample pieces after coating	Visual	Sample	-Do-	IR	3/2	-	1	TC
5.3	Mechanical Properties- Micro Hardness, Porosity, Bond Strength on sample	Visual & Measurement	Sample per Lot	-do-	IR	3/2	-	1	TC
5.4	Coating thickness	Measurement	Sample	-do-	IR	3/2	-	1	TC
5.5	Dimension checking after coating	Measurement	Critical Dim.	-do-	IR	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								

	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.				
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .				
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.				
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.				
			Signature		Signature & Seal
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : - Labyrinth				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material								
1.1	Test Piece Marking	Visual	Sample per Lot	Appd. Drawings/Tech. specs./IS/IEC/Equivalent	IR	3/2	-	1	TC
1.2	Chemical Composition	Chemical Test	Sample per Lot		IR	3/2	-	1	TC
1.3	Mechanical Analysis-As applicable (Tensile test, Impact test, Hardness etc.)	Mech. Test	Sample per Lot	-do-	IR	3/2	-	1	TC
2	In Process Stage								
2.1	Heat Treatment	Heat chart	-do-	-do-	IR	3/2	-	1	TC

2.2	NDT of forged material after heat treatment	RT, UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
2.3	Dimension checking of forged Labyrinth after machining	Measurement	100%	-do-	IR	3/2	-	1	RR
3	Final Inspection before HVOF coating								
3.1	Dimensional Check	Measurement	Critical Dim.	-do-	IR	3/2	-	1	TC
3.2	NDT of forged/ casting/ fabrication material after machining.	UT, MPI, DPT (as applicable)	100%	-do-	IR	3/2	-	1	TC
3.3	Attachment/Tagging of atleast five sample pieces with Job before start of coating (If HVOF coating is applicable)	Visual	Sample	-Do-	IR	3/2	-	1	TC
4	HP-HVOF Coating (If applicable)								
4.1	Chemical Composition of coating Powder	Measurement	Lot/ sample	-Do-	TC	3/2	-	1	TC
4.2	Detachment of sample pieces after coating	Visual	Sample	-Do-	IR	3/2	-	1	TC
4.3	Mechanical Properties- Micro Hardness, Porosity, Bond Strength on sample	Visual & Measurement	Sample per Lot	-do-	IR	3/2	-	1	TC
4.4	Coating thickness	Measurement	Sample	-do-	IR	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :					ग्राहक (CLIENT) :				
उपस्कर का नाम (NAME OF EQUIPMENT) :-Turbine Shaft					विक्रेता (VENDOR) :				
					एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :				
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material								
1.1	Forged Shaft-Forged Carbon Steel			Appd. Drawings/Tech. specs./IS/IEC/Equivalent					
a)	Test Piece Marking	Visual	Sample per Lot		JIR	3/2	1	-	CHP
b)	Chemical Composition	Chemical Test	Sample per Lot	-do-	JIR	3/2	1	-	CHP
c)	Mechanical Analysis-As applicable (Tensile Test, Impact Test, Hardness Test & Bend Test)	Mechanical Test	- do-	-do-	JIR	3/2	1	-	CHP
d)	NDT of Shaft	UT, MPI, DPT (as applicable)	100%	- do-	IR	3/2	-	1	RR
2	In Process Stage								
2.1	Heat treatment	Heat Chart	- do-	- do-	IR	3/2	-	1	TC
2.2	Post Heat Treatment- NDT of Shaft (Radial & Axial Direction)	RT, UT, MPI, DPT (as applicable)	- do-	- do-	IR	3/2	-	1	RR
2.3	Dimensional examination after rough machining	Measurement	- do-	- do-	IR	3/2	-	1	RR
3	Final Inspection								

3.1	Boroscopic Examination of Shaft - Centre Bore Dia	Visual / Measurement	- do-	- do-	JIR	3/2	1	-	CHP
3.2	NDT for Surface / Sub-surface Defects	UT, MPI, DPT (as applicable)	- do-	- do-	JIR	3/2	1	-	CHP
3.3	Run out check of Turbine shaft, Flanges	Measurement	- do-	- do-	JIR	3/2	1	-	CHP
3.4	Dimension check for Shaft	Visual / Measurement	Critical Dim.	- do-	JIR	3/2	1	-	CHP
3.5	Concavity & Convexity of Flanges	Measurement	100%	- do-	JIR	3/2	1	-	CHP
3.6	Surface Finish Check of Guide Bearing Journals & Flanges	Visual / Measurement	Sample	- do-	JIR	3/2	1	-	CHP
3.7	Dimensional check for coupling holes (Both side)	Visual / Measurement	100%	- do-	JIR	3/2	1	-	CHP
4	Painting	Visual / Measurement	- do-	- do-	TC	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :	ग्राहक (CLIENT) :
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उपस्कर का नाम (NAME OF EQUIPMENT) : -Turbine Guide Bearing Assembly				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material			Appd. Drawings/Tech. specs./IS/IEC/Equivalent					
1.1	Plate for Housing, Pivot Bolt & Locking Nut								
a)	Chemical Composition	Chemical Test	Sample per Lot	-do-	TC	3/2	-	1	TC
b)	Mechanical Analysis-As applicable-(Tensile Test and Hardness Test)	Mechanical Test	- do-	-do-	TC	3/2	-	1	TC
c)	NDT of Plates	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
1.2	Bearing Pad Segment / Shell/SS Sleeve (As applicable)								
a)	Chemical Composition	Chemical Test	Sample per Lot	-do-	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical Test	-do-	-do-	TC	3/2	-	1	TC
c)	NDT of Casting / Plates	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
1.3	Babbit Material/Abrasion Resistent Elastomeric Material (As applicable)	Chemical Test	100%	-do-	TC	3/2	-	1	TC
1.4	Heat Exchanger/Oil Cooler (If applicable)								
a)	Make, Model & Capacity	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Chemical Composition & Mechanical Analysis for Tube	Chemical Test	Sample per Lot	-do-	TC	3/2	-	1	TC

c)	Hydrostatic Pr. test	Test	100%	-do-	TC	3/2	-	1	TC
2	In Process Stage								
2.1	Heat Treatment for Casted / Welded components	Temp. Regulation	100%	-do-	IR	3/2	-	1	TC
2.2	Welder Certification	Visual	100%	-do-	TC	3/2	-	1	TC
2.3	NDT of weld joints after Heat Treatment	UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
2.4	Bonding of Babbit material/Abrasion Resistant Elastomeric Material on Segment / Shell (Bearing Pad)		-do-	-do-	IR	3/2	-	1	TC
2.5	NDT of Machined Surface		-do-	-do-	TC	3/2	-	1	TC
3	Final Inspection								
3.1	Dimensional Check for Assembly of TGB, Thickness of Babbit material/Abrasion Resistant Elastomeric Material	Measuremen t	Critical Dim.	-do-	IR	3/2	-	1	TC
3.2	Surface defects of Babbit material/Abrasion Resistant Elastomeric Material	DPT	100%	-do-	IR	3/2	-	1	TC
3.3	Leakage Test of TGB Housing (Oil/Water Tank)	Gravity Test	100%	-do-	IR	3/2	-	1	TC
4	Painting	Visual / Measuremen t	-do-	-do-	TC	3/2	-	1	TC
Note :	a. In ‘Inspection Agency’column figure 1,2,or 3 to be filled. 1- will indicate‘Employer’, 2- will indicate ‘supplier’ & 3- will indicate ‘sub-supplier’.								
	b. In ‘Remarks’ column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :					ग्राहक (CLIENT) :				
उपस्कर का नाम (NAME OF EQUIPMENT) : -Shaft Seal Assembly					विक्रेता (VENDOR) :				
					एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :				
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material			Appd. Drawings/Tech. specs./IS/IEC/Equivalent					
1.1	Plates								
a)	Chemical Composition	Chemical Test	Sample per Lot	-do-	TC	3/2	-	1	TC
b)	Mechanical Analysis-As applicable (Tensile Test and Hardness Test)	Mechanical Test	- do-	-do-	TC	3/2	-	1	TC
1.2	NDT of Plates	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
2	In Process Stage								
2.1	Welder Certification	Visual	100%	-do-	TC	3/2	-	1	TC
2.2	NDT and Visual examination on weld joints	UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
2.3	Dimensional check of components after Machining	Measurement	-do-	-do-	IR	3/2	-	1	TC
3	Final Inspection								
3.1	Dimensional check of Assembly	Measurement	Critical Dim.	-do-	JIR	3/2	1	-	CHP
3.2	Co-axiality and Surface Finish of Revolving Sleeve	-do-	100%	-do-	JIR	3/2	1	-	CHP
3.3	Attachment/Tagging of atleast five sample pieces with one lot before start of coating (If HVOF coating is applicable)	Visual	Sample	-Do-	IR	3/2	-	1	TC

4	HVOF Coating (If applicable)								
4.1	Chemical Composition of coating Powder	Measurement	Lot/ sample	-Do-	TC	3/2	-	1	TC
4.2	Detachment of sample pieces after coating	Visual	Sample	-Do-	IR	3/2	-	1	TC
4.3	Mechanical Property- Bond Strength, Porosity, Hardness of coating	Visual & Measurement	Sample per Lot	-do-	IR	3/2	-	1	TC
4.4	Coating thickness	Measurement	Sample	-do-	IR	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT) :			
उपस्कर का नाम (NAME OF EQUIPMENT) : Pelton Turbine-Injector (Nozzle Assy. with Deflector)				विक्रेता (VENDOR) :			
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :			
क्र.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की	जाँच की	संदर्भ दस्तावेज़ /	रिकॉर्ड	निरीक्षण एजेंसी (INSP. AGENCY)	टिप्पणी REMARK

सं. SR. NO.		प्रकृति NATURE OF CHECKS	मात्रा QUANTUM OF CHECKS	स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	प्रारूप RECORD FORMA T	प्रदर्शन Perfor m	गवाह Witnes s	सत्या पन Verify	S
1	Raw Material								
1.1	Nozzle Body, Main Injector, Needle, Seat Ring, Deflector & Fasteners (above M32) etc			Appd. Drawings/Tech. specs./IS/IEC/Equivalent					
a)	Chemical Composition	Chemical Test	Sample per Lot		TC	3/2	-	1	TC
b)	Mechanical Analysis-As Applicable (Tensile Test, Impact Test , Hardness Test)	Mechanical Test	- do-		TC	3/2	-	1	TC
c)	NDT of Cast/Forged steel	UT, MPI, DPT (as applicable)	100%		IR	3/2	-	1	RR
1.2	Deflector- Bearing								
a)	Chemical Composition	Chemical Test	Sample per Lot	-do-	TC	3/2	-	1	TC
b)	Mechanical Analysis-As Applicable (Tensile Test, Impact Test , Hardness Test)	Mechanical Test	- do-	-do-	TC	3/2	-	1	TC
1.3	Nozzle Spring	Mechanical Test	- do-	-do-	TC	3/2	-	1	TC
a)	Mechanical Analysis-As Applicable (Tensile Test, Impact Test , Hardness Test)	Mechanical Test	- do-	-do-	TC	3/2	-	1	TC
2	In-process Inspection								
2.1	Welding as applicable								
a)	Fillet welds NDT	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
b)	Full strength welds NDT	UT	100%	-do-	TC	3/2	-	1	TC
2.2	HVOF Tungsten Carbide coating (Needle Tips & Seat Ring)- thickness	Measuremen t	100%	Appd. Drawings/Tech. specs./IS/IEC/Equivalent	TC	3/2	-	1	TC

2.3	Machining Dimension	Measurement	100%		TC	3/2	-	1	TC
3	Final Inspection								
3.1	Injector (Nozzle Assembly with Deflector)								
a)	Dimension check	Measurement	-do-	-do-	IR	3/2	-	1	TC
b)	Surface Finish check	Visual / Measurement	-do-	-do-	IR	3/2	-	1	TC
c)	Hydraulic Pressure Test on Nozzle, Servomotor Cylinder, Nozzle Body & Tip	Measurement	- do -	-do-	JIR	3/2	1	-	CHP
d)	Operational Test								
i)	Alignment and free movement of needle with servometer	Visual	- do -	-do-	JIR	3/2	1	-	CHP
ii)	Needle full opening and closing with servomotor	-do-	- do -	-do-	JIR	3/2	1	-	CHP
iii)	Nozzle Servomotor-Stroking & Stroke	Measurement	- do -	-do-	JIR	3/2	1	-	CHP
iv)	Free Movement of Deflector (with overhead crane)	Visual	- do -	-do-	JIR	3/2	1	-	CHP
3.2	Painting	Visual / Measurement	100%	-do-	TC	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								

			Signature			Signature & Seal
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)	

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : Pelton Runner				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material-Runner			Appd. Drawings/Tech. specs./IS/IEC/Equivalent					
1.1	Keel Block Identification	Visual	Sample per Lot		JIR	3/2	1	-	CHP
1.2	Chemical Composition	Chemical Test	- do-		JIR	3/2	1	-	CHP
1.3	Mechanical Analysis (Tensile Test, Impact Test and Hardness Test)	Mechanical Test	- do-	-do-	JIR	3/2	1	-	CHP
1.4	Microstructure of material	Metallurgical Test	- do-	-do-	JIR	3/2	1	-	CHP
1.5	Welder Certification	Visual	100%	-do-	IR	3/2	-	1	TC
1.6	Heat Treatment of Complete Runner	H-T Chart	-do-	-do-	IR	3/2	-	1	TC
1.7	NDT of Runner after heat treatment	UT, MPI, DPT (as applicable)	-do-	-do-	IR	3/2	-	1	TC
2	In Process Stage Inspection								
2.1	Dimension Checking after rough machining	Measurement	-do-	-do-	IR	3/2	-	1	TC

2.2	Bucket Pitch, Thickness & profile check	Measurement with template	-do-	-do-	IR	3/2	-	1	TC
3	Final Inspection								
3.1	Dimension Check and Surface Finish	Measurement	Critical Dim.	-do-	JIR	3/2	1	-	CHP
3.2	Bucket Pitch, Thickness & profile check	Measurement with template	-do-	-do-	JIR	3/2	1	-	CHP
3.3	NDT of finished Surface / Sub-surface Defects	UT, MPI, DPT (as applicable)	100%	-do-	JIR	3/2	1	-	CHP
3.4	Static Balancing of Runner	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
4	Painting	Visual/Measurement	100%	-do-	TC	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer. ', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			Employer (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : Guide Apparatus Assembly				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR · N O.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Shop Assembly of Turbine Guide Apparatus								
1.1	Matching & Doweling of Head cover, Bottom Ring and Discharge Ring	Visual / Measurement	100%	Appd. drawings/ Tech. specs./IS/IEC/Equivalent	JIR	3/2	1	-	CHP
1.2	Match Marking of X-Y axis	- do-	-do-	-do-	JIR	3/2	1	-	CHP
1.3	Co-axiality and Level check	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
1.4	Dimension check for Height between Top Cover and Bottom Ring liners	- do-	-do-	-do-	JIR	3/2	1	-	CHP
1.5	Top & Bottom clearances of Guide Vanes	- do-	-do-	-do-	JIR	3/2	1	-	CHP
1.6	Free movement of Guide Vanes (opening & closing) and Top & Bottom clearances	Visual / Measurement	-do-	-do-	JIR	3/2	1	-	CHP
1.7	Clearance between two consecutive guide vanes in closed position.	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
1.8	Gap between Consecutive Guide Vanes at 25%, 50%, 75% and 100% (opening & closing).	Visual / Measurement	-do-	-do-	JIR	3/2	1	-	CHP
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission & CHP - Customer Hold Point.								

	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			EMPLOYER (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)			

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : Digital Governing System				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र.सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS / ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Digital Electronic Controller								
a)	Type, Make and Rating	Visual	100%	Tech. Spec./ Apprd. Drawing / IS/IEC/Equivalent	TC	3/2	-	1	TC

b)	Range of adjustment	Test	100%	-do-	TC	3/2	-	1	TC
c)	Functional Test for Governor Characteristics / Parameter	Test	100%	-do-	TC	3/2	-	1	TC
1.1	Digital Micro Processor								
a)	Make,Type & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Performance	Test	100%	-do-	TC	3/2	-	1	TC
1.2	Governor Local control cubicle (Relay, Control and Indication etc)								
a)	Make,Type & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Functional Test	Test	100%	-do-	TC	3/2	-	1	TC
1.3	Power Supply Unit								
a)	Make,Type & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Routine Test	Test	100%	-do-	TC	3/2	-	1	TC
1.4	PLC Unit for control logic, Digital I/O								
a)	Make,Type & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Routine Test	Test	100%	-do-	TC	3/2	-	1	TC
1.5	Terminal Card, Signal Converter, Transducers, sensors, speed switches, communication facilities with control system, control and indication system, shielded cable, surge suppression devices, Actuator Lock & other accessories (As applicable)								
a)	Make,Type & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Routine Test	Test	100%	-do-	TC	3/2	-	1	TC
1.6	Mobile Work Station								
a)	Make,Type & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
2.	Digital Electronic Cubicle								
a)	Lay out and BOM Check	Visual	100%	-do-	JIR	3/2	1	-	CHP
b)	Dimension , Thickness of Sheet & Door lock arrangement and Paint Shed	Measurement	100%	-do-	JIR	3/2	1	-	CHP
c)	Scheme & Functional Checking	Measurement	100%	-do-	JIR	3/2	1	-	CHP

3.	Electro Hydraulic Actuator(Valve Table Assembly), Pilot/ Servomotor Valve, Relay Valve, Solenoid Valves, DC Switches, Limit Switches, Manifold Blocks, Emergency Shut Down Valve (As applicable)			Tech. Spec./ Apprd. Drawing / IS/IEC/Equivalent					
a)	Make, Type & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Routine Test	Test	100%	-do-	TC	3/2	-	1	TC
c)	Calibration / Setting of Electro Hydraulic Actuator	Test	100%	-do-	JIR	3/2	1	-	CHP
4	Valves and Pressure Switches, Mainfold Block etc.(As applicable)								
a)	Pressure Test	Test	100%	-do-	TC	3/2	-	1	TC
5	Dual oil filtration unit with changeover facility along with clogging indication								
a)	Mesh Size	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Functional Test	Test	100%	-do-	TC	3/2	-	1	TC
6	Pipes and Fittings								
a)	Pressure Test	Test	100%	-do-	TC	3/2	-	1	TC
7	Speed Signal Generator / Potential Transformer(Transducer)								
a)	Routine Test	Test	100%	-do-	TC	3/2	-	1	TC
8	Complete Assembly of Governor With Control and Electro- Hydraulic Actuator								
a)	Verification of Items as per Bill of Material & Scheme Check	Visual & Test	1 unit at random	-do-	JIR	3/2	1	-	CHP
b)	Check of Protection Devices	Test	1 unit at random	-do-	JIR	3/2	1	-	CHP
c)	Functional Check	Test	1 unit at random	-do-	JIR	3/2	1	-	CHP
9	Centrifugal Type Over Speed Limiter (Mechanical Overspeed Devices)								
a)	Performance Test	Test	100%	-do-	TC	3/2	-	1	TC
10	Control and Power Cable for interconnection of								

	Governor with SCADA								
a)	Make, Type & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Routine Test	Test	100%	-do-	TC	3/2	-	1	TC
Note:	a. In ‘Inspection Agency’ column figure 1,2,or 3 to be filled. 1- will indicate ‘Employer’, 2- will indicate ‘supplier’ & 3- will indicate ‘sub-supplier’.								
	b. In ‘Remarks’ column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, CHP - Customer Hold Point & JIR-Joint Inspection Report.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d.Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be replaced by "2" under the column " Perform" except testing of material.								
	f. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	g.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
			Signature			Signature & Seal			
			Employer (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : Digital Governing System - Oil Pressure Unit				विक्रेता (VENDOR) :					
				एनआईटी/क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS / ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Sump Tank			Appd. Drgs./ Tech. specs./IS/IEC/Equivalent					
1.1	Material for Tank				TC	3/2	-	1	TC

a)	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
b)	Mechanical Strength	Mechanical	- do-	-do-	TC	3/2	-	1	TC
1.2	Welder Certification	Visual	100%	-do-	TC	3/2	-	1	TC
1.3	NDT of weld joints	DPT	100%	-do-	TC	3/2	-	1	TC
1.4	Gravity Leakage test for tank	Test	- do-	-do-	TC	3/2	-	1	TC
1.5	Dimensional Check	Measurement	- do-	-do-	TC	3/2	-	1	TC
2	Pressure Vessel/Oil Pressure Accumulator								
2.1	Material for Tank	NDT	100%	-do-	TC	3/2	-	1	TC
a)	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
b)	Mechanical Strength	Mechanical	- do-	-do-	TC	3/2	-	1	TC
2.2	Welder Certification	Visual	100%	-do-	TC	3/2	-	1	TC
2.3	NDT of weld joints	UT/DPT	100%	-do-	TC	3/2	-	1	TC
2.4	Post Weld stress relieving	Temp. Regulator	- do-	-do-	TC	3/2	-	1	TC
2.5	Hydrostatic pressure test for pressure vessel	Test	- do-	-do-	JIR	3/2	1	-	CHP
2.6	Dimensional Check	Measurement	- do-	-do-	JIR	3/2	1	-	CHP
3	Pipes & Fittings								
3.1	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
3.2	Mechanical Strength	Mechanical	- do-	-do-	TC	3/2	-	1	TC
3.3	Hydrostatic pressure test	Test	100%	-do-	TC	3/2	-	1	TC
3.4	Dimensional Check	Measurement	- do-	-do-	IR	3/2	-	1	RR
4	Pressure gauge, Pressure Switch, Float Switch, Temp gauge,moisture indicator, temperature gauges, temperature switches, level transducer, level gauges, level switch etc. (As applicable)								
4.1	Make, Model & Type	Visual	100%	-do-	TC	3/2	-	1	TC
4.2	Calibration Test	Test	- do-	-do-	TC	3/2	-	1	TC
4.3	Functional Test	Test	- do-	-do-	TC	3/2	-	1	TC
5	Distributing valve, Proportional valve, control valve, emergency shutdown valve, Unloader Valve, Check Valve, Pilot valve, Safety relief								

	valve, Air blow off valve, By Pass Valve, Solonoid Valve and Isolating Valve, Nitrogen Charging kit etc.(As applicable)								
5.1	Make, Model & Type	Visual	- do-	-do-	TC	3/2	-	1	TC
5.2	Functional Check	Test	- do-	-do-	TC	3/2	-	1	TC
5.3	Hydrostatic pressure test	Test	- do-	-do-	TC	3/2	-	1	TC
6	Oil Pump			Appd. Drgs./ Tech. specs./IS/IEC/Equivalent					
6.1	Make, Model, Type & Capacity	Visual	- do-	-do-	TC	3/2	-	1	TC
6.2	Routine Test	Test	- do-	-do-	TC	3/2	-	1	TC
7	Electric Motor								
7.1	Make, Type & rating	Visual	- do-	-do-	TC	3/2	-	1	TC
7.2	Routine Test	Test	- do-	-do-	TC	3/2	-	1	TC
8	Strainer								
8.1	Mesh Size	Measurement	- do-	-do-	TC	3/2	-	1	TC
8.2	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
9	Dual oil filtration unit with changeover facility along with clogging indication								
9.1	Mesh Size	Visual	100%	-do-	TC	3/2	-	1	TC
9.2	Functional Test	Test	100%	-do-	TC	3/2	-	1	TC
10	Oil								
10.1	Make & type	Visual	100%	-do-	TC	3/2	-	1	TC
11	Nitrogen Cylinder								
11.1	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
11.2	Mechanical Strength	Mechanical	- do-	-do-	TC	3/2	-	1	TC
11.3	Hydrostatic pressure test & Leakage Test for Nitrogen Cylinder	Test	100%	-do-	TC	3/2	-	1	TC
12	Complete assembly of Oil Pressure Unit with Test/ Actual Control Panel								
12.1	Dimensional Check	Measurement	100%	-do-	JIR	3/2	1	-	CHP
12.2	Capacity	Measurement	- do-	-do-	JIR	3/2	1	-	CHP

12.3	Functional Test/ Operational Test of Pumps, Motors, Electromagnetic Valves, Pressure Switch, Temp switch & Accessories (As applicable)	Test	- do-	-do-	JIR	3/2	1	-	CHP
13	Control panel / Starter pnel for Oil Sump Tank								
13.1	Verification of bill of material	Visual	- do-	-do-	JIR	3/2	1	-	CHP
13.2	Functional test	Test	- do-	-do-	JIR	3/2	1	-	CHP
13.3	IR & HV Test	Test	- do-	-do-	JIR	3/2	1	-	CHP
14	Painting	Visual/ Measurement	- do-	-do-	TC	3/2	-	1	TC
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
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	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
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	f. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	g.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	h. Site/Field test shall be conducted in presence of Project's representative only.								
			Signature			Signature & Seal			
			Employer (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT) :						
उपस्कर का नाम (NAME OF EQUIPMENT) : Digital Governing System - Oil Leakage Unit (If required)				विक्रेता (VENDOR) :						
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :						
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS / ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS	
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify		
1	Oil Leakage Unit			Appd. Drgs./ Tech. specs./IS/IEC/Equivalent						
1.1	Pump									
a)	Make, Model, Type & Capacity	Visual	100%	-do-	TC	3/2	-	1	TC	
b)	Routine Test	Test	- do-	-do-	TC	3/2	-	1	TC	
1.2	Electric Motor									
a)	Make, Type & rating	Visual	- do-	-do-	TC	3/2	-	1	TC	
b)	Routine Test	Test	- do-	-do-	TC	3/2	-	1	TC	
1.3	Float, Level Switch									
a)	Make, Type & rating	Visual	- do-	-do-	TC	3/2	-	1	TC	
1.4	Oil Tank									
a)	Leakage test	Test	- do-	-do-	TC	3/2	-	1	TC	
b)	Dimensional Check	Measurement	- do-	-do-	TC	3/2	-	1	TC	
1.5	Functional test on complete oil leakage unit	Test	- do-	-do-	TC	3/2	-	1	TC	
1.6	Dimensional & BOM Check	Visual/ Measurement	- do-	-do-	TC	3/2	-	1	TC	
2	Painting	Visual/ Measurement	- do-	-do-	TC	3/2	-	1	TC	
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.									
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, CHP - Customer Hold Point & JIR-Joint Inspection Report.									

	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d.Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be replaced by "2" under the column " Perform" except testing of material.								
	f. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	g.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
			Signature			Signature & Seal			
			Employer (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN):									
परियोजना (PROJECT):				ग्राहक (CLIENT):					
उपस्कर का नाम (NAME OF EQUIPMENT): Digital Governing System & Accessories				विक्रेता (VENDOR):					
				एनआईटी/क्रय आदेश संदर्भ (NIT/P.O. REFERENCE):					
क्र. सं. SR. NO.	Activity & Operation of Component	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Digital Governing System & Accessories								

1.1	Leakage tests on all pressure oil and Nitrogen system piping,	Test	100%	Tech. Spec./ Apprd. Drawing /IS/IEC/Equivalent	TC/TR	3/2	1	-	TC/TR
1.2	Hydrostatic tests on pressure vessel, all pressure oil piping and all Nitrogen pressure piping systems at a pressure 50 % greater than the maximum design pressure after installation. (As per Standard)	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.3	Functional check of Measuring Instruments, Switches & Controller provided in the Governing System, Oil Pressure System & Nitrogen System	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.4	Leakage & Pressure Test on OPU Tank	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.5	Functional Tests of Level Switch, Pr. Switch & Pr. Relief Valve etc of OPU Tank	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.6	IR test on Electrical Panels	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.7	Test to verify logic control scheme from local / remote including start/stop, load control, locking & other features	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.8	Transducers/Detecting device of Guide Vane Servomotor & Main Distributor Valve (MDV) shall be installed, positioned and tested by the Contractor.	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.9	Check on the range of action of the set point generator	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.10	The sensitivity of the set point generator's in regard to positioning/speed to pulse orders from governor cabinet and to external orders shall be checked and comply with the agreed values. The allowable discrepancy in the values shall be guaranteed by the manufacturer in advance in load, no-load and exceptional conditions, viz black start, load rejection etc.	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
¶1.11	In case of load rejection, restoration of no-load operation after load rejection has to be tested, with the governor ensuring the return to no-load operation with maximum efficiency in complying with over speeding and maximum pressure guarantees and time taken to return to stabilized speed.	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR

1.12	On load mechanical tripping	Functional	100%	Tech. Spec./ Apprd. Drawing /IS/IEC/Equivalent	TC/TR	3/2	1	-	TC/TR
1.13	On load electrical tripping	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.14	Check on opening adjustment and the value of permanent speed droop,	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.15	Determination of the frequency insensitivity threshold	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.16	Adjustment of Main Distribution Valve (MDV) opening & checking of opening & closing times,	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.17	Checking the response to a set point load step in no-load operation	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.18	Recording of the speed build-up and loading	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.19	Functional check of Mobile Work Station (LAPTOP) for Programming/Diagnostic of the System at the time of commissioning.	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.20	Testing for Interfacing with Complete Governor System with Plant SCADA System & testing of FGMO, RGMO and Auto control mode of operation.	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.21	Checking the speed sensing sensor and feedback mechanism for proper operation and calibration.	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Representative of Engineer-In -Charge , Employer', 2- will indicate 'Contractor' & 3- will indicate 'sub-Contractor'.								
	b. In 'Remarks' column following abbreviations shall be used - TC-Test Certificate, TR- Test Report								
	c. At the time of Inspection the Contractor shall arrange the requisite calibrated measuring instruments/Tools .								
	d. The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	e. Any additional activity/test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.								
	f. The contractor shall furnish the details of test methods to the Engineer-In charge, Employer for approval in the time line as specified in contract agreement.								
	g. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.								
	Signature					Signature & Seal			
	Engineer-in-charge				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN) :									
परियोजना (PROJECT) :					ग्राहक (CLIENT) :				
उपस्कर का नाम (NAME OF EQUIPMENT) :Field Test on each Turbine shall include					विक्रेता (VENDOR) :				
					एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :				
क्र. सं. SR. NO.	Activity & Operation of Component	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORDED FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witnesses	सत्यापन Verify	
1	Final Assembly Inspection of Turbine & its associated Component			Appd. drawings/ Tech. specs./IS/IEC/Equivalent					
1.1	Check horizontality of draft tube flange, flange radius & elevation, knee liner elevation and the distance between machine center line & knee liner end, after erection and concreting of knee liner	Measurement /Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.2	Check match spiral members with top & bottom contour of stay ring & center line marked on spiral member with reference to center line of stay ring.	Measurement /Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.3	Check leveling & Centring of Spiral casing assembly before and after welding of Spiral case.	Measurement /Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.4	Checking of Spiral case final welding	RT,UT,MPI,DPT (as applicable)	100%	- do-	TC/TR	3/2	1	-	TC/TR
1.5	Hydrostatic pressure test of Spiral Casing	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.6	Check height & level of mating faces of Stay Ring with Head Cover & Bottom Ring before & after stay ring flange machining .	Measurement /Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR

1.7	Checking of Weld condition/bolts tightness of Stay Ring & Bottom Ring assembly and level checking of bottom ring after assembly.	Measurement /Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.8	Concentricity of guide vane bores in Bottom ring & Head Cover	Measurement /Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.9	Measure distance between Head Cover and Bottom Ring at 8 points.	- do-	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.10	Check Verticality & Free movement of Guide Vanes	- do-	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.11	Measure Guide Vane Bedding Gap, Top & Bottom Clearance.	Measurement /Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.12	Check Guide Vane Servomotor stroke length and Guide Vane opening & closing time	Measurement /Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.13	Check the dimensions between the Main shafts and guide bearing housing	- do-	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.14	Static components of the unit are centered along the rotation axis. Check Concentricity of Guide Bearing Housing, Head Cover Labyrinth, Bottom Ring Labyrinth, Discharge Ring/Draft Tube Upper Cone w.r.t Unit Vertical Axis.	- do-	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.15	Level Check, Relative heights of Bottom Ring, Head Cover, Guide Bearing housing top and Operating Ring.	Measurement	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.16	Checking of Radial Clearance between operating Ring and wearing Pads and check angular position of Operating Ring.	Measurement /Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.17	Checking of bolts tightness of Stay Ring & Bottom Ring assembly and level checking of bottom ring after assembly.	- do-	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.18	Check the fitment & alignment of Turbine shaft with Runner	- do-	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.19	Check tightness of Coupling Bolts between Runner & Turbine Shaft	- do-	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.20	Check tightness of Coupling Bolts between Turbine Shaft & Generator Shaft	- do-	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.21	Alignment, leveling and Centring of Complete Shaft line (Turbine & Generator Shaft)	- do-	100%	-do-	TC/TR	3/2	1	-	TC/TR
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Representative of Engineer-In -Charge , Employer', 2- will indicate 'Contractor' & 3- will indicate 'sub-Contractor'.								
	b. In 'Remarks' column following abbreviations shall be used - TC-Test Certificate, TR-Test Report								
	c. At the time of Inspection the Contractor shall arrange the requisite calibrated measuring instruments/Tools								
	d. The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								

	e. Any additional activity/test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.				
	f. The contractor shall furnish the details of test methods to the Engineer-In charge, Employer for approval in the time line as specified in contract agreement.				
	g. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.				
	Signature				Signature & Seal
	Engineer-in-charge				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN):									
परियोजना (PROJECT):					ग्राहक (CLIENT):				
उपस्कर का नाम (NAME OF EQUIPMENT): Commissioning Test of Unit					विक्रेता (VENDOR):				
					एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE):				
क्र. सं. SR. NO.	Activity & Operation of Component	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Commissioning Test (All applicable tests listed in IEC Guide for Commissioning Operation and Maintenance of Hydraulic Turbines shall be carried out in accordance with specified procedures. Pre-start tests and checks shall be carried out before filling of waterways as per approved procedure. After filling waterways, necessary checks shall be made before initial run.)			Appd. drawings/ Tech. specs./IS/IEC/Equivalent					
1.1	Run Test (Initial run test)								
a)	Check for Abnormal noise/sound all around the unit during slight opening and closing of Wicket gates manually	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR

1.2	Bearing run test								
a)	Check oil temperatures, Shaft seal flow, Shaft vibration at 25%, 50%, 75%, 100% & 110% of rated speed.	Measurement/Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
b)	Verify Proper operation of stopping sequence	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
c)	Record units slow down speed versus time until speed is close to zero.	Measurement/Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
d)	Function test of Brake	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
e)	Over speed test to be performed if temperatures, Vibrations and balance are correct	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.3	Heating run test								
a)	Heat run test shall be undertaken at full load if load rejection at 75% of rated power has been successfully performed.	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
b)	Bearing temperatures, shaft seal temperatures and shaft seal flow at every 5 minutes shall be checked and recorded.	Measurement/Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
c)	Check temperature rise versus load position	Measurement/Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
d)	Check shaft vibrations.	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.4	Over Speed Test							-	
a)	Function of mechanical over speed detector at desired tripping value.	Visual/Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.5	Load rejection test								
a)	Load rejection tests at 75, 100 and 110 percent load of the turbine output to verify speed rise and pressure rise,	Test	100%	Appd. drawings/ Tech. specs./IS/IEC/Equivalent	TC/TR	3/2	1	-	TC/TR
b)	Unit shut down –Check the entire unit stopping sequence at full output: normal stop, emergency stop due to mechanical fault, emergency stop due to electrical fault.	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.6	Gate opening versus turbine Output								

a)	Draw the gate opening vs output chart. Estimate the turbine flow and calculate the head losses and the net heads for all testing points. Outline the gate opening value corresponding to the contractual maximum output. Set the output limitation and the Gate opening limitation stored inside the governor.	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.7	Performance testing								
a)	If nothing unusual has been observed in load run and load rejection tests, the test service period of 72 hours shall follow. During this test service period, the unit must operate continuously at rated condition without any interruption except of those beyond the control of the Contractor. However, such interrupted period shall not be counted for in the test service period.	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.8	Field Efficiency Test (All field acceptance tests in accordance with provision in IEC 60041-1991/Latest standard shall be conducted at site before acceptance of turbine by the Engineer-in-charge or his representative.)								
b)	Output and Efficiency tests shall be conducted at different heads and guide vanes openings on any one turbine, chosen by the Engineer-in-charge or his representative, to determine guaranteed efficiency parameters .Efficiency shall be measured by ultrasonic or other acceptable method prescribed in IEC. The Contractor shall furnish details of test methods, agency which will conduct the test, provisions to be made for field testing, calibration of instruments for purposes of test and all other relevant details for approval.	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR

1.9	Testing for Interfacing of operation for Turbine and its component with Plant SCADA System .	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
Note:	a. In ‘Inspection Agency’ column figure 1,2,or 3 to be filled. 1- will indicate ‘Representative of Engineer-In -Charge , Employer’, 2- will indicate ‘Contractor’ & 3- will indicate ‘sub-Contractor’.								
	b. In ‘Remarks’ column following abbreviations shall be used - TC-Test Certificate, TR-Test Report								
	c. At the time of Inspection the Contractor shall arrange the requisite calibrated measuring instruments/Tools								
	d. The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	e. Any additional activity/test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.								
	f. The contractor shall furnish the details of test methods to the Engineer-In charge, Employer for approval in the time line as specified in contract agreement.								
	g. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.								
	Signature					Signature & Seal			
	Engineer-in-charge				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

2) GENERATOR

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Stator (Frame)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन VERIFY	
A	Raw Material (Carbon Steel Plate)								
i)	Chemical Composition	Chem.	Sample per Lot	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Mechanical Properties	Mech.	-do-	-do-	TC	3/2	-	1	TC
iii)	NDT on Steel Plate (>20 mm thick plates)	UT	-do-	-do-	TC	3/2	-	1	TC
B	In process Inspection								
i)	Check of Full Strength Welding	UT	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	NDT of welds	RT, UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
C	Final Inspection								
i)	Sole Plate Mounting Face - Level Checking & Check of Flatness	Measurement	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	Sole Plate- Dimensional & Flatness	-do-	-do-	-do-	JIR	3/2	1	-	CHP
iii)	Dimensional Check of Stator Frame	-do-	-do-	-do-	JIR	3/2	1	-	CHP
iv)	Surface finish of Machined surface	Visual	-do-	-do-	JIR	3/2	1	-	CHP

v)	Flatness Check of Seating Surface of Air Cooler	-do-	-do-	-do-	TC	3/2	-	1	TC
vi)	Painting	-do-	-do-	-do-	JIR	3/2	1	-	CHP
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column "Perform" .								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)					(VENDORS Q.C. DEPT. OR REPRESENTATIVE)			

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपष्कर का नाम (NAME OF EQUIPMENT) : Generator Stator (Stator Punchings)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	

A	Raw Material (CRNGO)								
i)	Material Properties								
	Grade of Material, Nominal Thickness, Specific Loss, Magnetic Polarization, Anisotropy of loss, Stacking factor, Conventional Density	Visual/ Measurement/ Electrical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
B	In Process								
i)	Check of Burr Level in Punching	Measurement	Sample	-do-	TC	3/2	-	1	TC
ii)	Stacking factor	Measurement	-do-	-do-	TC	3/2	-	1	TC
iii)	Check for Quality of Lamination Varnishing, Thickness & IR value	Measurement/ Electrical	-do-	-do-	JIR	3/2	1	-	CHP
iv)	Dimensional check of Punchings	Measurement	-do-	-do-	TC	3/2	-	1	TC
v)	Specific Total Loss	Electrical	-do-	-do-	TC	3/2	-	1	TC
vi)	Magnetic Polarization	-do-	-do-	-do-	TC	3/2	-	1	TC
vii)	Trial Assembly of Punching (Core dia , height etc)	Measurement	-do-	-do-	TC	3/2	-	1	TC for core assembly
	# If stacking done at site then above " CHP* " mentioned test shall be performed at site otherwise at manufacturer's works								
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			

	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)
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परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Stator (Unwound Stator)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ॉर्मेट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन VERIFY	
A	Studs & End Plate								
i)	Material Properties.	Chem/Mech.	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
B	Core Assembly								
i)	Dimensional checking (check of core for waviness, core dia, Packet size and height after assembly)	Measurement	Sample plan	-do-	JIR	3/2	1	-	CHP
ii)	Core Loss Test and Hot Spot Detection	Electrical	5% (TC for 100%)	-do-	JIR	3/2	1	-	CHP
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								

	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .				
	Signature				Signature & Seal
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Stator (Stator Winding Bars)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमेट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
1	Raw Material								
A.	Copper Wire Bars/ Ingots/ CC Wire Rods/ Extruded Sections Strands (Material Grade) (As applicable)								
i)	Chemical Composition	Chemical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Resistivity Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
B.	Fiber Glass Covered Polyesterimide/ Epoxy Varnish Bonded Rectangular Copper Conductor								
i)	Dimensional check of Bare & Covered Conductor & Insulation Thickness	Measurement	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Flexibility/ Bending , Tensile & Elongation	Mechanical	-do-	-do-	TC	3/2	-	1	TC

	Test								
iii)	Breakdown Voltage Test at Room & elevated temp.	Electrical	-do-	-do-	TC	3/2	-	1	TC
iv)	Cure Test	Thermal	-do-	-do-	TC	3/2	-	1	TC
v)	Electrical Resistivity Measurement	Electrical	-do-	-do-	TC	3/2	-	1	TC
C.	Mica Tape								
i)	Thickness	Measurement	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Tensile Strength	Mechanical	-do-	-do-	TC	3/2	-	1	TC
iii)	Mica content	Chemical	-do-	-do-	TC	3/2	-	1	TC
iv)	Bond/ Binder/ Resin Content	-do-	-do-	-do-	TC	3/2	-	1	TC
v)	Backing/ Glass Fabric	-do-	-do-	-do-	TC	3/2	-	1	TC
vi)	Shelf life (Tape life)	Chemical/Thermal	-do-	-do-	TC	3/2	-	1	TC
2	In Process Inspection								
A.	After Hot Press								
i)	Transposition angle verification	Visual	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Inter Strand Test	Electrical	100%	-do-	TC	3/2	-	1	TC
3)	Final Inspection								
i)	Tan delta measurement	Electrical	5% (TC for 100%)	-do-	JIR	3/2	1	-	CHP
ii)	Impulse Voltage Withstand Test (Type Test)	-do-	1 random bar for complete order	-do-	JIR	3/2	1	-	CHP
iii)	HV Test on Straight & Overhang portion	-do-	5% (TC for 100%)	-do-	JIR	3/2	1	-	CHP
iv)	IR after HV Test	-do-	5% (TC for 100%)	-do-	JIR	3/2	1	-	CHP
v)	Dimensional & Model Checking	Measurement/ Visual	1 full cradle after every 200 bars (5%)	-do-	JIR	3/2	1	-	CHP
vi)	Inter Turn Test (Applicable for coil design only)	Electrical	100%	-do-	TC	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								

	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)					(VENDORS Q.C. DEPT. OR REPRESENTATIVE)			

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Stator (Wound Stator)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमेट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन VERIFY	
1)	HV Test after placing the Bottom Bars in Stator Slot, Test Bottom Bar Only	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
2)	HV Test after placing the Top Bars in Stator Slot, Test Bottom Bar and Top Bar together	-do-	-do-	-do-	TC	3/2	-	1	TC
3)	RTD Laying, continuity test, HV & IR test and Resistance measurement	Visual/Elect.	-do-	-do-	TC	3/2	-	1	TC
4)	Insulation Resistance (IR) and Polarisation Index (PI) of Stator Winding of individual	Electrical	-do-	-do-	TC	3/2	-	1	TC

	sector								
5)	HV test on Segments	Electrical	-do-	-do-	JIR	3/2	1	-	CHP*
	# If winding of stator done at manufacturer's works then " CHP* " mark mentioned tests shall be performed at manufacturer works otherwise same shall be carried out at site.								
Note :	a. In ‘Inspection Agency’ column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate ‘supplier’ & 3- will indicate ‘sub-supplier’.								
	b. In ‘Remarks’ column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Rotor (Rim assembly)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFOR	गवाह WITNES	सत्यापन	

						M	S	VERIFY	
1	Rim Punching								
A	Raw Material (High Tensile Magnetic Steel Sheet)								
i)	Chemical Composition	Chemical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC
B	Bolts / Studs & Nuts								
i)	Material Properties	Chem/Mech	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
C)	Final Inspection								
i)	Visual inspection	Visual	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Thicknes of Punching & Burr level Measurement	Measurement	-do-	-do-	TC	3/2	-	1	TC
iii)	Trial Assembly of Rotor Rim Punching without spider (inner & outer diameter verification, slot width & depth measurement)	Visual/ Measurement	-do-	-do-	TC	3/2	-	1	TC
2	Rim Stud & Nuts								
A	Raw Material								
i)	Test Piece Marking	Visual	Sample per Lot	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Chemical Composition	Chemical Test	-do-	-do-	TC	3/2	-	1	TC
iii)	Mechanical Testing (Tensile , Impact , Hardness, Bend Test)- As applicable	Mechanical Test	-do-	-do-	TC	3/2	-	1	TC
3	Braking ring & Rotor Fan Assembly								
A	Raw Material								
i)	Chemical & Mechanical Properties	Chem/Mech	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
B	Final Inspection								

i)	Dimensions of Rotor Fan : i) Radial ii) Axial	Measurement	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR / TC	3/2 / 3/2	1 / -	- / 1	i) Radial - CHP ii) Axial - TC
ii)	UT of brake ring (After machining and before hole punching)	UT	-do-	-do-	JIR	3/2	1	-	CHP
iii)	NDT of Welds	RT, UT, MPI, DPT (as applicable)	-do-	-do-	JIR	3/2	1	-	CHP
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT):							
उपष्कर का नाम (NAME OF EQUIPMENT) : Generator Rotor (Pole Assembly)				विक्रेता (VENDOR):							
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :							
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks		
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन			

								VERIFY	
	Pole Punching, End Plate, Pole Coils & Damper Bars								
1	Pole punching								
A	Raw Material (High Tensile Magnetic Steel sheet & Copper)								
i)	Chemical Composition	Chemical	Sample per Lot	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC
iii)	Electrical Properties	Electrical	-do-	-do-	TC	3/2	-	1	TC
B	Final Inspection								
i)	Visual Finish & Dimensional Check	Visual/ Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
2	Material properties of End plate , damper winding bars, Nuts & Bolts	Chem/Mech/Elect (as applicable)	Sample	-do-	TC	3/2	-	1	TC
3	Pole coils								
A	Raw material (Copper)								
i)	Material Properties	Chem/Mech/Elect	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
B	In Process Inspection								
i)	Inter Turn Test / Millivolt Drop Test	Electrical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
C	Final Inspection (wound pole assembly)								
i)	Visual Finish , Weight & Dimensional Check	Visual/ Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	HV & IR Test	Electrical	-do-	-do-	JIR	3/2	1	-	CHP
iii)	DC Resistance Measurement	Electrical	-do-	-do-	JIR	3/2	1	-	CHP
iv)	Impedance Measurement	Electrical	-do-	-do-	JIR	3/2	1	-	CHP
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								

	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.				
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column "Perform".				
	Signature				Signature & Seal
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Rotor (Spider)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन VERIFY	
A	Raw Material (Steel Plate)								
i)	Chemical Composition	Chem	Sample per Lot	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Mechanical Properties	Mech	-do-	-do-	TC	3/2	-	1	TC
iii)	NDT of Steel Plates (>20 mm thick plates)	UT	-do-	-do-	TC	3/2	-	1	TC
B	In-process Inspection								
i)	NDT of welds	RT, UT, MPI, DPT	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC

		(as applicable)							
ii)	Post weld Stress Relieving-Heat Treatment (if applicable)	Heat Chart	-do-	-do-	TC	3/2	-	1	TC
iii)	Machining Dimensional check	Measurement	-do-	-do-	TC	3/2	-	1	TC
C	Final Inspection								
i)	Machining Dimensional check	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
ii)	Surface Finish of machined surface	Visual	-do-	-do-	JIR	3/2	1	-	CHP
iii)	Flange Flatness & Gap Checking	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
iv)	Painting	Visual	-do-	-do-	JIR	3/2	1	-	CHP
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column "Perform".								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)					(VENDORS Q.C. DEPT. OR REPRESENTATIVE)			

परियोजना (PROJECT) :				ग्राहक (CLIENT):				
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Shaft (Top & bottom)				विक्रेता (VENDOR):				
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference)				

				:					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
A	Raw Material								
	Forged Shaft-Forged Carbon Steel								
i)	Test Piece Marking	Visual	Sample per Lot	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	Chemical Composition	Chemical Test	- do-	-do-	JIR	3/2	1	-	CHP
iii)	Mechanical Analysis-As applicable (Tensile Test, Impact Test, Hardness Test & Bend Test)	Mechanical Test	- do-	-do-	JIR	3/2	1	-	CHP
iv)	NDT of Shaft	RT, UT, MPI, DPT (as applicable)	100%	- do-	TC	3/2	-	1	TC
B	Inprocess inspection								
i)	Heat Treatment	Heat Chart	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	NDT After Heat Treatment	RT, UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
C	Final Inspection								
i)	Boroscopic Examination of Shaft - Centre Bore Dia	Visual / Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	NDT for Surface / Sub-surface Defects	RT, UT, MPI, DPT (as applicable)	- do-	- do-	JIR	3/2	1	-	CHP
iii)	Runout Check of Shaft and Shaft with Thrust Collar Assembly (if applicable)	-do-	-do-	-do-	JIR	3/2	1	-	CHP
iv)	Dimension check of Shaft	Visual / Measurement	Critical Dim.	- do-	JIR	3/2	1	-	CHP
v)	Concavity & Convexity of Flanges	Measurement	100%	- do-	JIR	3/2	1	-	CHP

vi)	Surface Finish Check of Guide Bearing Journals & Flanges	Visual / Measurement	Sample	- do-	JIR	3/2	1	-	CHP
vii)	Interchangeability check & Dimensional check for coupling holes & bolts	Visual / Measurement	100%	- do-	JIR	3/2	1	-	CHP
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपष्कर का नाम (NAME OF EQUIPMENT) : Generator Bracket (Upper & Lower)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन VERIFY	
A)	Raw Material								
i)	Chemical Composition	Chemical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC

B)	Inprocess Inspection								
i)	Dimensional Check	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	NDT of Welds	RT, UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
C)	Final Inspection								
i)	Surface finish & Dimensional Check	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	Leakage Test	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
iii)	Flatness & Level checking of Sole Plate face	-do-	-do-	-do-	JIR	3/2	1	-	CHP
iv)	Trial assembly of bottom bracket arm with central part & upper bracket arm with central part	-do-	-do-	-do-	JIR	3/2	1	-	CHP
vi)	Painting	Visual	-do-	-do-	JIR	3/2	1	-	CHP
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column "Perform".								
	Signature				Signature & Seal				
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Bearing Pads (Thrust, Upper & Lower Guide Bearing Pads)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
A	Raw Material (Supports & Pads)								
i)	Chemical Composition	Chemical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC
iii)	NDT of Pad base /Cast Part	RT, UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
iv)	Babbitt Material - Chemical	Chemical	-do-	-do-	TC	3/2	-	1	TC
B	Final Inspection								
i)	UT & DP test of Bearing pads	UT & DPT	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	Dimensional & Surface finish check	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								

	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .				
	Signature				Signature & Seal
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Accessories & Instruments (Brake & Jacking)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन VERIFY	
	Brake and Jacking System (Brake-Jack Assembly, Brake-Jack Control Panel, Brake Liner and Seamless Pipe)								
A	Raw Material (Brake Liner and Seamless Pipe)								
i)	Material Chemical Properties	Chemical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC
B	Final Inspection								
i)	Functional Test of individual brake jacks	Functional/ Hydraulic	-do-	-do-	JIR	3/2	1	-	CHP
ii)	Dimensions & UT of Brake Liner	UT	-do-	-do-	JIR	3/2	1	-	CHP

C)	Brake & Jack Control Panel								
i)	BOM Verification	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	Scheme Checking	Visual	-do-	-do-	JIR	3/2	1	-	CHP
iii)	IR & HV Test	Electrical	-do-	-do-	JIR	3/2	1	-	CHP
iv)	Functional Check of Brake-Jack Control Panel	Functional/ Hydraulic	-do-	-do-	JIR	3/2	1	-	CHP
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)					(VENDORS Q.C. DEPT. OR REPRESENTATIVE)			

परियोजना (PROJECT) :				ग्राहक (CLIENT):				
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Accessories & Instruments (Shaft Coupling Bolts)				विक्रेता (VENDOR):				
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :				
क्र.	मद/घटक एवं विशेषता	जाँच की	जाँच की मात्रा QUANTUM OF	संदर्भ दस्तावेज़ / स्वीकृति	रेकॉर्ड	निरीक्षण एजेंसी Inspecting Agency		टिप्पणी Remarks

सं. S.N.	ITEM/ COMPONENT & CHARACTERISTICS	प्रकृति NATURE OF CHECKS	CHECKS	मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	फ़ारमैट RECORD FORMAT	प्रदर्शन PERFORM	गवाह WITNES S	सत्याप न VERIFY	
1)	Coupling Bolts								
A	Raw Material- Forged Coupling Bolts								
i)	Chemical Composition	Chemical Test	-do-	-do-	JIR	3/2	1	-	CHP
ii)	Mechanical Testing (Tensile , Impact , Hardness, Bend Test)- As applicable	Mechanical Test	-do-	-do-	JIR	3/2	1	-	CHP
iii)	NDT of Coupling Bolts before machining	RT,UT,MPI,DP T (As applicable)	100%	-do-	TC	3/2	-	1	TC
B	In Process Stage								
i)	Dimensional examination after machining	Measurement	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
C	Final Inspection								
i)	Dimensional check	Visual / Measurement	Critical Dim.	-do-	JIR	3/2	1	-	CHP
ii)	Painting	Visual / Measurement	100%	-do-	TC	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			

	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)
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परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Accessories & Instruments (Disc, Collar, Springs etc)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन VERIFY	
1	Disc, Collar & Spring								
A)	Raw Material								
i)	Chemical Composition	Chemical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC
B)	Inprocess Inspection								
i)	Heat Treatment procedure - Thrust Collar	Heat Chart	-do-	-do-	TC	3/2	-	1	TC
ii)	NDT of Thrust Collar after Heat Treatment (radial & axial)	RT, UT, MPI, DPT (as applicable)	-do-	-do-	JIR	3/2	1	-	CHP
iii)	Surface Finish of Thrust Collar Runner Disc	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
C)	Final Inspection								
i)	UT of Runner Disc	UT	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	Flatness & Parallelity of surfaces	Measurement	100%	-do-	JIR	3/2	1	-	CHP

iii)	Match marking of Thrust Collar, Shaft & Disc	Visual	-do-	-do-	JIR	3/2	1	-	CHP
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)					(VENDORS Q.C. DEPT. OR REPRESENTATIVE)			

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Accessories & Instruments (Slip Ring)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
A)	Raw Material (Steel)								
i)	Chemical Composition	Chemical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC

iii)	Carbon Brush								
iv)	Material Grade, Size, Current Density, Hardness, Appaarent Density, Resistivity, etc.	Chem/Mech./Elect.	-do-	-do-	TC	3/2	-	1	TC
B)	Inprocess Inspection								
i)	UT of Slip Ring	UT	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
C)	Final Inspection								
i)	Dimensional check of assembly after machining	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	HV & IR testing of slip ring assy.	Electrical	-do-	-do-	JIR	3/2	1	-	CHP
iii)	Run out measurement of assembly	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)					(VENDORS Q.C. DEPT. OR REPRESENTATIVE)			

परियोजना (PROJECT) :	ग्राहक (CLIENT):			
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Accessories & Instruments (Air & Oil Coolers)	विक्रेता (VENDOR):			

				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन VERIFY	
A)	Raw Material (Cooling Tubes)								
i)	Chemical Composition	Chemical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC
iii)	Dimensional Check	Measurement	-do-	-do-	TC	3/2	-	1	TC
B)	Final Inspection								
i)	Dimensional check of complete assembly of coolers	Measurement	Sample for each type	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	Hydraulic test of assembled coolers	Pressure	Sample for each type	-do-	JIR	3/2	1	-	CHP
iii)	Painting	Visual	Sample for each type	-do-	JIR	3/2	1	-	CHP
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			

	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)
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परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपस्कर का नाम (NAME OF EQUIPMENT) : Generator Accessories & Instruments (Other Devices)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
1	Instruments								
i)	Test of accuracy and calibration of RTD and DTT (Thermometer)	-do-	-do-	-do-	TC	3/2	-	1	TC
2	Other items/devices (As applicable)								
i)	Brake Dust Collection Device	Performance test	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Carbon Dust Collection Device	-do-	-do-	-do-	TC	3/2	-	1	TC
iii)	High Pressure Automatic lubrication system	Hydraulic test	-do-	-do-	TC	3/2	-	1	TC
iv)	Anti Condensation Heaters	Performance	-do-	-do-	TC	3/2	-	1	TC
v)	Vibration Detector with alarm and tripping contact	-do-	-do-	-do-	TC	3/2	-	1	TC
vi)	Partial Discharge Analyzer with permanent coupler	-do-	-do-	-do-	TC	3/2	-	1	TC
vii)	Moisture Detection System	-do-	-do-	-do-	TC	3/2	-	1	TC
viii)	Air Gap Measuring device	-do-	-do-	-do-	TC	3/2	-	1	TC
ix)	Speed Measuring Device	-do-	-do-	-do-	TC	3/2	-	1	TC

x)	Fire protection system	-do-	-do-	-do-	TC	3/2	-	1	TC
Note :	a. In ‘Inspection Agency’ column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate ‘supplier’ & 3- will indicate ‘sub-supplier’.								
	b. In ‘Remarks’ column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN):									
परियोजना (PROJECT):				ग्राहक (CLIENT):					
उपष्कर का नाम (NAME OF EQUIPMENT): Generator				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ(NIT / P.O. Reference):					
क्र.सं. S.N.	घटक की गतिविधि और संचालन ACTIVITY & OPERATION OF COMPONENT	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
A)	Field test on each Generator shall include:								
a)	Visual inspection and dimensional check.	Visual/ Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR

b)	Check of stator bore for exact roundness at the upper and lower edges of the Iron Core.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
c)	Checking of the uniformity of the air gap.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
d)	Rotor roundness measurement.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
e)	Measurement of deflection of thrust bearing on loading.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
f)	Checking of the alignment of the ready assembled turbine and generator shaft.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
g)	Checking of bearing clearances, Dimensions, Insulation resistance & blue matching of pads with shaft journal	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
h)	Checking of the bearing oil cooling installation for complete and correct assembly.	Visual	-do-	-do-	TC/TR	3/2	1	-	TC/TR
i)	Checking of the generator cooling system for complete and correct assembly.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
j)	Hydrostatic pressure and tightness tests of all equipment containing/carrying water, oil and compressed air.	Pressure	-do-	-do-	TC/TR	3/2	1	-	TC/TR
k)	Checking of jacks in braking and lifting operation respectively	Visual	-do-	-do-	TC/TR	3/2	1	-	TC/TR
l)	Measurement of the DC resistance of the field winding.	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
m)	Measurement of the stator winding DC resistance per phase.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
n)	Insulation test and H.V. test of the field winding.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
o)	Insulation test (1 min. and 10 min. value) & H. V. test of the stator winding and determination of	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR

	polarization factor.								
p)	Corona visibility test of stator for determination of corona initial and ending voltages and inspection of the corona at 120% of the rated voltage.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
q)	Operational tests on all generator auxiliary equipment, including calibration of related electric indicating, control and metering instruments, and checks for correctness of wiring and piping.	Electrical/ Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
r)	Checking of alarm and protection devices.	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
s)	Checking the functioning of Generator fire fighting devices i.e. valve, sensors, switches, relays, panels etc.	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
t)	Magnetization test of the ready assembled Iron Core to verify the absence of hot spots (local overheating, if core built at site).	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
u)	High voltage withstand test on individual windings in accordance with IEC 60034-1.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
v)	High voltage withstand test of the ready assembled stator winding in accordance with the IEC 60034--1.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
B)	At least the following tests shall be performed on each generator:								
a)	Test of shaft movement, bearing run and shaft eccentricity.	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Checking of start-up and shut-down sequence including mechanical braking.	Visual/ Mechanical	-do-	-do-	TC/TR	3/2	1	-	TC/TR

c)	Short -circuit and Open circuit test	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
d)	Measurement of shaft voltages and bearing currents.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
e)	Over speed test, noise and vibration measurement including balancing of the unit.	Mechanical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
f)	Synchronization with the grid system.	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
g)	Load rejection tests at different load steps, including at 110 % of rated unit load.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
h)	Rated output test of the unit including temperature run.	Electrical/ thermal	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
i)	Measurement of partial discharge of stator insulation.	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
j)	Testing of on-line monitoring systems and verification of various sensors and Instruments.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
C)	The following tests shall be performed on one of the generator selected by Engineer in Charge								
a)	Guaranteed Characteristics of Generator.	Electrical	On one Unit	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Stator and Rotor Winding Resistance Measurement.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
c)	Water Flow Measurement (If applicable).	Measurement	-do-	-do-	TC/TR	3/2	1	-	TC/TR
d)	No Load Saturation Curve.	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
e)	Wave form Deviation.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
f)	Shaft Voltage Measurement.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
g)	Temperature Rise test.	Thermal	-do-	-do-	TC/TR	3/2	1	-	TC/TR
h)	Measurement of the Total harmonic Distortion (THD) and evaluation of waveform irregularities	Electrical	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR

i)	Plotting of oscillograms for the examination of the no- load voltage sine wave.	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
j)	Measurement of the individual losses and determination of the efficiency	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
D)	Testing for interfacing of operation for Generator and its components with Plant Scada System.	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
Note:	a. In ‘Inspection Agency’column figure 1,2,or 3 to be filled. 1- will indicate Representative of Engineer-in-charge, 2- will indicate ‘Contractor’ & 3- will indicate ‘Sub-contractor’.								
	b. In ‘Remarks’ column following abbreviations shall be used - TC/TR - Test Certificate/ Test Report								
	c.The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	d. Any additional acitivity/ test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.								
	e. The contractor shall furnish the details of test method to the Engineer-in-Charge for approval in the time line as specified in contract agreement								
	f. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.								
	Signature					Signature & Seal			
	Engineer-in-charge				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

3) MAIN INLET VALVE (SPHERICAL/ BUTTERFLY VALVE) &
4) PENSTOCK VALVE (BUTTERFLY VALVE)

परियोजना (PROJECT) :					ग्राहक (CLIENT) :				
उपस्कर का नाम (NAME OF EQUIPMENT) : Spherical Valve/Butterfly Valve for MIV & Penstock Valve					विक्रेता (VENDOR) :				
					एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :				
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material								
1.1	Spherical Valve Body, Disc & Trunnion, Up stream & Downstream Connection pipe			Appd. Drg. / Tech. Spec. / IS/IEC/Equivalent					
a)	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC
c)	NDT of Cast/ Weldable Plate/ Forged steel	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
d)	Heat Treatment	Temp. Regulation	100%	-do-	IR	3/2	-	1	TC
1.2	Bush, Lever, Counter Weight								
a)	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC
c)	NDT of Cast/ Weldable plate/ Forged steel	RT, UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
1.3	Seal Seat, Seal Retainer & Fastener								
a)	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC

1.4	Seal								
a)	Mechanical Analysis & Chemical Composition	Test	Sample	-do-	TC	3/2	-	1	TC
b)	Hardness	Test	-do-	-do-	TC	3/2	-	1	TC
c)	Dimensional Check	Measurement	100%	-do-	IR	3/2	-	1	TC
1.5	By-Pass Valve								
a)	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC
c)	NDT of Cast/ Forged steel	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
d)	Heat Treatment	Temp. Regulation	100%	-do-	IR	3/2	-	1	TC
1.6	Grease, Water & Oil Pipe Line								
a)	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	-do-	-do-	TC	3/2	-	1	TC
c)	Hydro-Static pressure Test	Test	-do-	-do-	TC	3/2	-	1	TC
1.7	Guard Valve, Isolating Gate Valve, Drain Valve, Air Admission & Release Valve etc								
a)	Hydro-Static pressure Test	Test	-do-	-do-	TC	3/2	-	1	TC
2	In-process Inspection								
a)	NDT of weld joints before Stress Relieving	MPI,DPT,UT (As applicable)	-do-	-do-	IR	3/2	-	1	TC
b)	Valve Body & Disc - Stress Relieving after welding	Temp. Regulation	-do-	-do-	IR	3/2	-	1	TC
c)	NDT of all weld joints after stress relieving	MPI,DPT,UT (As applicable)	-do-	Appd. Drg. / Tech. Spec. / IS/IEC/Equivalent	TC	3/2	-	1	TC
d)	Dim. Check & Surface Finish of Machined Surface for Body, Disc, Journal, Seal & Seal Seat	Measurement	100%	-do-	IR	3/2	-	1	TC
3	Final Inspection								
3.1	Spherical Valve Assembly								
a)	Dimensional Check of complete assembly	Measurement	Critical dim.	-do-	JIR	3/2	1	-	CHP
b)	Operational Test - Free movement of Disc	Visual & Measurement	100%	-do-	JIR	3/2	1	-	CHP

c)	Hydro-Static Test on B.F. Valve Body & Disc with Upstream Pipes	-do-	-do-	-do-	JIR	3/2	1	-	CHP
d)	Leakage Test on Valve Seals	-do-	-do-	-do-	JIR	3/2	1	-	CHP
3.2	BY Pass valve Assembly								
a)	Hydro Static Pressure Test	Pressure Test	100%	-do-	IR	3/2	-	1	TC
b)	Leakage Test	Test	- do-	-do-	IR	3/2	-	1	TC
c)	Dimensional check	Measurement	- do-	-do-	IR	3/2	-	1	TC
d)	Operational Test	Test	- do-	-do-	IR	3/2	-	1	TC
3.3	Air Release Valve & Drain Valve etc.								
a)	Hydro Static Pressure Test	Pressure Test	100%	-do-	IR	3/2	-	1	TC
b)	Leakage Test	Test	- do-	-do-	IR	3/2	-	1	TC
c)	Dimensional check	Measurement	- do-	-do-	IR	3/2	-	1	TC
d)	Operational Test	Test	- do-	-do-	IR	3/2	-	1	TC
4	Control panel								
4.1	Verification of bill of material	Visual	- do-	-do-	JIR	3/2	-	1	TC
4.2	Functional test	Test	- do-	-do-	JIR	3/2	-	1	TC
4.3	IR & HV Test	Test	- do-	-do-	JIR	3/2	-	1	TC
5	Painting	Visual/ Measurement	- do-	-do-	TC	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f. Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			

			Employer (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)
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परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : - Oil Pressure Unit for MIV & Penstock Valve				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Sump Tank			Appd. Drg. / Tech. Spec. / IS/IEC/Equivalent					
1.1	Material for Tank								
a)	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
b)	Mechanical Strength	Mechanical	- do-	-do-	TC	3/2	-	1	TC
1.2	NDT of weld joints	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
1.3	Leakage test check for tank	Test	- do-	-do-	TC	3/2	-	1	TC
1.4	Dimensional Check	Measurement	- do-	-do-	TC	3/2	-	1	TC
2	Pressure Vessel/Oil Pressure Accumulator								
2.1	Material for Tank								
a)	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
b)	Mechanical Strength	Mechanical	- do-	-do-	TC	3/2	-	1	TC
2.2	NDT of weld joints	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
2.3	Post Weld stress relieving	Temp. Regulator	- do-	-do-	TC	3/2	-	1	TC
2.4	Hydrostatic pressure test for pressure vessel	Test	- do-	-do-	TC	3/2	-	1	TC

2.5	Dimensional Check	Measurement	- do-	-do-	TC	3/2	-	1	TC
3	Pipes & Fittings								
3.1	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
3.2	Mechanical Strength	Mechanical	- do-	-do-	TC	3/2	-	1	TC
3.3	Hydrostatic pressure test	Test	100%	-do-	TC	3/2	-	1	TC
4	Pressure gauge, Pressure Switch, Pressure Transducer, Moisture Indicator, Float Switch, Temp gauge, Temp switch, Level Transducer etc.								
4.1	Make, Model & Type	Visual	100%	-do-	TC	3/2	-	1	TC
4.2	Calibration Test	Test	- do-	-do-	TC	3/2	-	1	TC
4.3	Functional Test (wherever applicable)	Test	- do-	-do-	TC	3/2	-	1	TC
5	Unloader Valve, Check Valve, Pilot valve, Safety valve, Air blow off valve, By Pass Valve, Solenoid Valve and Isolating Valve etc.								
5.1	Make, Model & Type	Visual	- do-	-do-	TC	3/2	-	1	TC
5.2	Functional Check (wherever applicable)	Test	- do-	-do-	TC	3/2	-	1	TC
5.3	Hydrostatic pressure test (wherever applicable)	Test	- do-	-do-	TC	3/2	-	1	TC
6	Oil Pump								
6.1	Make, Model, Type & Capacity	Visual	- do-	-do-	TC	3/2	-	1	TC
6.2	Routine Test	Test	- do-	-do-	TC	3/2	-	1	TC
7	Electric Motor								
7.1	Make, Type & rating	Visual	- do-	-do-	TC	3/2	-	1	TC
7.2	Routine Test	Test	- do-	-do-	TC	3/2	-	1	TC
8	Strainer			Appd. Drg. / Tech. Spec. / IS/IEC/Equivalent					
8.1	Mesh Size	Measurement	- do-		TC	3/2	-	1	TC
8.2	Chemical Composition	Chemical	Sample	-do-	TC	3/2	-	1	TC
9	Oil								
9.1	Make & type	Visual	100%	-do-	TC	3/2	-	1	TC
10	Nitrogen Cylinder/Bottle								
10.1	Chemical Composition & Mech. Properties	Chem/Mech Test	Sample	-do-	TC	3/2	-	1	TC

10.2	NDT of weld joints	UT,DPT, MPI (As applicable)	100%	-do-	TC	3/2	-	1	TC
10.3	Hydrostatic pressure test & Leakage Test for Nitrogen Cylinder	Test	- do-	-do-	TC	3/2	-	1	TC
11	Heat Exchanger/Oil Cooler (If applicable)								
11.1	Make, Model & Capacity	Visual	100%	-do-	TC	3/2	-	1	TC
11.2	Chemical Composition for Tube	Chemical Test	Sample per Lot	-do-	TC	3/2	-	1	TC
11.3	Hydrostatic Test	Pressure Test	100%	-do-	TC	3/2	-	1	TC
12	Complete assembly of Oil Pressure Unit with Test/ Actual Control Panel								
12.1	Dimensional Check	Measurement	100%	-do-	JIR	3/2	1	-	CHP
12.2	Capacity	Measurement	- do-	-do-	JIR	3/2	1	-	CHP
12.3	Functional Test/ Operational Test of Pumps, Motors, Electromagnetic Valves, Pressure Switch, Temp switch & Accessories	Test	- do-	-do-	JIR	3/2	1	-	CHP
12.4	Pressure Tank relief valve setting	Visual	- do-	-do-	JIR	3/2	1	-	CHP at site
13	Control panel								
13.1	Verification of bill of material	Visual	- do-	-do-	JIR	3/2	1	-	CHP
13.2	Functional test	Test	- do-	-do-	JIR	3/2	1	-	CHP
13.3	IR & HV Test	Test	- do-	-do-	JIR	3/2	1	-	CHP
14	Painting	Visual/ Measurement	- do-	-do-	TC	3/2	-	1	TC
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f. Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								

	h. CHP*-Test shall be conducted at site in presence of Power Station representative.								
			Signature			Signature & Seal			
			Employer (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : Servomotor for MIV & Penstock Valve				विक्रेता (VENDOR) :					
				एनआईटी/क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material								
1.1	Piston & Cylinder			Appd. Drg. / Tech. Spec. / IS/IEC/Equivalent					
a)	Chemical Composition	Chemical	Sample per Lot		TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	- do-	-do-	TC	3/2	-	1	TC
c)	NDT of Cast / Weldable Steel	UT, MPI, DPT (as applicable)	100%	-do-	TC	3/2	-	1	TC
2	In Process Stage Inspection								
2.1	Heat treatment of Piston & cylinder	Temp. Regulation	-do-	-do-	IR	3/2	-	1	TC
2.2	NDT of weld joints after Heat Treatment	RT, UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC

2.3	Dimensional exam after machining	Measurement	-do-	-do-	IR	3/2	-	1	TC
3	Final Inspection								
3.1	Dimensional check of Final Assembly	Measurement	Critical Dim.	-do-	JIR	3/2	1	-	CHP
3.2	Hydraulic pressure test of Servomotor-On test Pressure (Design Pressure)	Test	100%	-do-	JIR	3/2	1	-	CHP
3.3	Leakage across Piston Check in fully open & fully close -On Working pressure	-do-	-do-	-do-	JIR	3/2	1	-	CHP
3.4	Stroke Length measurement-On Working Pressure	Visual / Measurement	-do-	-do-	JIR	3/2	1	-	CHP
3.5	Functional Test of servo-motor (Stroking 10 cycles)-on Working Pressure	-do-	-do-	-do-	JIR	3/2	1	-	CHP
4	Painting		-do-	-do-	TC	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	f. Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature				Signature & Seal		
			Employer (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)			

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : -Oil Leakage Unit for MIV & Penstock Valve (if required)				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORDED FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Oil Leakage Unit			Appd. Drg. / Tech. Spec. / IS/IEC/Equivalent					
1.1	Pump								
a)	Make, Model, Type & Capacity	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Routine Test	Test	- do-	-do-	TC	3/2	-	1	TC
1.2	Electric Motor								
a)	Make, Type & rating	Visual	- do-	-do-	TC	3/2	-	1	TC
b)	Routine Test	Test	- do-	-do-	TC	3/2	-	1	TC
1.3	Float Switch								
a)	Make, Type & rating	Visual	- do-	-do-	TC	3/2	-	1	TC
1.4	Oil Tank								
a)	Leakage test	Test	- do-	-do-	TC	3/2	-	1	TC
b)	Dimensional Check	Measurement	- do-	-do-	TC	3/2	-	1	TC
1.5	Functional test on complete oil leakage unit	Test	- do-	-do-	TC	3/2	-	1	TC
1.6	Dimensional & BOM Check	Visual/	- do-	-do-	TC	3/2	-	1	TC
2	Painting	Measurement	- do-	-do-	TC	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR-Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								

	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f. Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			Employer (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)			

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN) :									
परियोजना (PROJECT) :					ग्राहक (CLIENT) :				
उपस्कर का नाम (NAME OF EQUIPMENT) : Butterfly Valve/Spherical Valve for Main Inlet Valve & Penstock Valve					विक्रेता (VENDOR) :				
					एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :				
क्र. सं. SR. NO.	Activity & Operation of Component	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTITY OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witnesses	सत्यापन Verify	
1	Dry Test			Appd. Drg. / Tech. Spec. / IS/IEC/Equivalent					
1.1	Setting of MIV opening & Closing Time	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.2	Check tightness of Service Seal & Maintenance Seal	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR

1.3	Check Plug alignment with servomotor in full open & close position.	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.4	Record operating pressure of servomotor during closing & opening	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.5	Check clearance between the plug & seals in closed position of MIV	Visual/Measurement	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.6	Check seal ring mechanical locks & measure all pads strokes	Visual/Measurement	100%	-do-	TC/TR	3/2	1	-	TC/TR
2	Wet Test				TC/TR				TC/TR
2.1	Functional check of all detection devices (Limit switches, Pr. Switches etc)	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.2	Check bypass valve proper operation under water flow and operating time	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.3	Check bypass valve water leakage	Visual	100%		TC/TR	3/2	1	-	TC/TR
2.4	Verify sealing under mechanical lock application	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.5	Measurement of servomotor stroke	Visual/Measurement	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.6	Pressure test of all field installed piping for oil and air at 1.5 times the maximum design pressure (As per standard)	Test	100%		TC/TR	3/2	1	-	TC/TR
2.7	Leakage test of all field piping at requisite pressure	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.8	Opening and closing of valve and by-pass valve at rated pressure/head,	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.9	Tests to determine leakage from valve seal under specified head conditions	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.10	Tests to determine closing and opening time of butterfly valve/Spherical valve and by-pass valve at conditions defined,	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.11	Closure of the valve by use of counter weights for maximum turbine discharge conditions,	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.12	Pressure testing of service and maintenance seal and check Service and Maintenance seal stroke	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.13	Functional check of Control Circuit.	Functional Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.14	Testing for Interfacing of operation for MIV system with Plant SCADA System .	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Representative of Engineer-In -Charge , Employer 2- will indicate 'Contractor' & 3- will indicate 'sub-Contractor'.								

	b. In 'Remarks' column following abbreviations shall be used - TC-Test Certificate, TR-Test Report								
	c. At the time of Inspection the Contractor shall arrange the requisite calibrated measuring instruments/Tools .								
	d. The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	e. Any additional activity/test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.								
	f. The contractor shall furnish the details of test methods to the Engineer-In charge,Employer for approval in the time line as specified in contract agreement.								
	g. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.								
	Signature					Signature & Seal			
	Engineer-in-charge				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

5) (i) STATIC EXCITATION SYSTEM FOR SYNCHRONOUS MACHINES

परियोजना (PROJECT) :					ग्राहक (CLIENT):				
उपकरण का नाम (NAME OF EQUIPMENT) : Static Excitation System					विक्रेता (VENDOR):				
					एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :				
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
A)	Bought out items								
1	Thyrister, Surge Supression Device, HMI/MMI etc. (As applicable)								
a)	Make and Rating	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Functional Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
2	Power Supply								
a)	Make and Rating	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Performance Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
3	DC field circuit breaker								
a)	Make and Rating	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Performance Test	Electrical	-do-	-do-	TC	3/2	-	1	TC

4	Discharge Resistors								
a)	Make and Rating	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Performance Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
5	Digital/ Analog Metering Equipment								
a)	Dimensional check, Make & Rating	Measurement / Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Class and Accuracy	Visual	-do-	-do-	TC	3/2	-	1	TC
c)	Calibration	Electrical	-do-	-do-	TC	3/2	-	1	TC
d)	Routine Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
6	Excitation Cable & other Interconnection cables (If any)								
a)	Make and Rating	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Routine Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
7	Crowbar Circuit								
a)	Make and Rating	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Performance Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
8	Clipper Circuit (If applicable)								
a)	Make and Rating	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Performance Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
9	Earth fault detector (If applicable)								
a)	Make and Rating	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Routine Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
10	Rectifier cooling fan								
a)	Make and Rating	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Routine Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
B)	Monitoring & Protection features check								

a)	Thyristor Failure Detection	-do-	-do-	-do-	TC/ IR	3/2	-	1	TC/ IR
b)	DC Short Circuit	-do-	-do-	-do-	TC/ IR	3/2	-	1	TC/ IR
c)	Check of AC system overvoltage(if applicable) and overcurrent protection	-do-	-do-	-do-	TC/ IR	3/2	-	1	TC/ IR
d)	Thyristor cooling fan failure	-do-	-do-	-do-	TC/ IR	3/2	-	1	TC/ IR
e)	Transient overvoltage monitoring due to lightning or switching surges	-do-	-do-	-do-	TC/ IR	3/2	-	1	TC/ IR
f)	Thyristor and field Overvoltage protection	-do-	-do-	-do-	TC/ IR	3/2	-	1	TC/ IR
C)	Final Inspection of Excitation System								
1	Dimensional check of panels, Sheet thickness	Visual/ Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1		CHP
2	Painting								
a)	Paint Shade, Coating Thickness & Finish	Visual/ Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
3	BOM check	Visual	100%	-do-	JIR	3/2	1	-	CHP
4	IR Test of Excitation System	Electrical	100%	-do-	TC	3/2	-	1	TC
5	IP class	Mech.Test	100%	-do-	TC	3/2	-	1	TC
6	HV Test on Excitation System								
a)	Main Circuit	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Auxilliary circuits	-do-	-do-	-do-	TC	3/2	-	1	TC
7	Scheme & Interlocking checking	Electrical	100%	-do-	JIR	3/2	1	-	CHP
8	Functional Check of Protection and Trip Logics & Indications								
a)	Failure of DVRs, Loss of regulator, Thyristor firing circuit failure, Field ground detector and PT fuse failure detection,V/f protection for alarm & tripping etc. (As applicable)	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
9	Functional Check								
a)	Discharge Circuit	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP

b)	Field Flashing Circuit Contactor	-do-	100%	-do-	JIR	3/2	1	-	CHP
c)	Functional Tests in all relays and other devices of comand and control	-do-	-do-	-do-	JIR	3/2	1	-	CHP
10	Closed Loop simulation tests of the excitation system	Test	100%	-do-	JIR	3/2	1	-	CHP
11	Functional Test								
a)	Check of Trigger Pulse Equipment (Firing Module & Power Supply Fault)	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
b)	Light load test	-do-	-do-	-do-	JIR	3/2	1	-	CHP
c)	Rectifier Leakage Test	-do-	-do-	-do-	TC	3/2	-	1	TC
d)	Current sharing tests of parallel branches (if applicable)	-do-	-do-	-do-	JIR	3/2	1	-	CHP
e)	PIV Test on Thyristor	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
f)	Check of redundant Thyristor Stack/ Manual Thyristor Stack (As applicable).	-do-	-do-	-do-	JIR	3/2	1	-	CHP
12	Tests on DVRs :								
a)	Stator and Rotor Current Limiter Test	-do-	-do-	-do-	JIR	3/2	1	-	CHP
b)	Load Angle Limiter Test	-do-	-do-	-do-	JIR	3/2	1	-	CHP
c)	Over fluxing Limiter	-do-	-do-	-do-	JIR	3/2	1	-	CHP
d)	Check of Sequence/ Auto & Manual Operation	Electrical	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
13	Temperature rise test (Heat Run)	-do-	On 01 unit	-do-	JIR	3/2	1	-	CHP
14	Burn in Test of assembly	-do-	-do-	-do-	TC	3/2	-	1	TC (Wherever module level burn-in is checked, this may be exempted.)
15	Provision of necessary contacts and/or ports for integration with existing plant SCADA system (As	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP

	applicable)								
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	f. The type test may not be mandatory if similar equipment has been type tested and test certificate(s) for relevant tests are accepted by the Employer								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN) :									
परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Excitation System				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					

क्र. सं. S.N.	घटक की गतिविधि और संचालन ACTIVITY & OPERATION OF COMPONENT	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
a)	Accuracy Test on DVRs	-do-	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
A	Test of excitation and regulation with generator at standstill								
a)	Excitation equipment visual inspection	Visual/	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Connections check	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
c)	Insulation tests for cables	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
d)	Check of the AC & DC supplies	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
e)	Check of the transducers and protections circuit. Tripping of Generator breaker/ Generator transformer breaker and field breaker by simulated operation of unit lockout relay.	Functional	-do-	-do-	TC/TR	3/2	1	-	TC/TR
f)	Operational Test on Field Flashing Circuit	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
g)	Test of the self – excitation point of the converter (for changeover from field flashing to thyristor bridge supply)	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
h)	Test of the rectifier bridges connected to a separate supply.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
i)	i) Functional checking of commands to Excitation equipment, operation and status for Local commands ii) Functional checking of commands to Excitation equipment, operation and status for control room commands	Functional	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard/ Scheme	TC/TR	3/2	1	-	TC/TR

	iii) Excitation equipment status & alarm indications at control room and local. iv) Checking alarms, trip conditions and Sequence of Events.								
B	Test of excitation and regulation with generator at No Load / Reduced load (As applicable)								
a)	Automatic Generator Voltage regulation (Upper & Lower Limits, Step Response)	Functional	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard / Commissioning procedure	TC/TR	3/2	1	-	TC/TR
b)	Manual Field Current regulation	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
c)	Regulator changeover	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
d)	Limiters Test (Over Excitation Limit, Under -Excitation Limit)	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
e)	Generator Voltage Build up Test under Automatic and Manual regulator operating modes	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
f)	V/Hz limiter test	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
C	Test of excitation and regulation with generator at Load								
a)	Automatic Generator Voltage regulation	Functional	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard / Commissioning procedure	TC/TR	3/2	1	-	TC/TR
b)	Manual Field Current regulation	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
c)	Regulator changeover	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
d)	Limiters Test (Underexcitation , Overexcitation)	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
e)	Field Suppression Test (Field Suppression by inverter operation and Discharge element) with Generator at Generator on – Load under field forcing operation	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR (subject to permission from purchaser/ Grid authorities)
f)	PT failure Detection Verification	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR (subject to permission from purchaser/ Grid authorities)

g)	Positive & Negative Ceiling Curve	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR (subject to permission from purchaser/ Grid authorities)
h)	Load rejection tests	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR (subject to permission from purchaser/ Grid authorities)
i)	Transient response to a set point step in not – Synchronized operation & Isolated Network operation	Electrical	100%	-do-	TC/TR	3/2	1	-	TC/TR
j)	Transient response to a set point step on load in reactive Power supply	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
k)	Transient response to a set point step on load in reactive Power absorption	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
l)	Closed – loop frequency analysis of excitation control system	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR (subject to permission from purchaser/ Grid authorities)
m)	Positive & Negative Ceiling Voltage and Voltage Response Time, Voltage Time Response, Nominal Response tests of the Excitation system	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR (subject to permission from purchaser/ Grid authorities)
n)	Power System Stabilizer (PSS) check (for Unit size > 100 MW)	Functional	-do-	-do-	TC/TR	3/2	1	-	TC/TR
Note:	a. In ‘Inspection Agency’ column figure 1,2,or 3 to be filled. 1- will indicate Representative of Engineer-in-charge, 2- will indicate ‘Contractor’ & 3- will indicate ‘Sub-contractor’.								
	b. In ‘Remarks’ column following abbreviations shall be used - TC/TR - Test Certificate/ Test Report								
	c. The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	d. Any additional activity/ test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.								
	e. The contractor shall furnish the details of test method to the Engineer-in-Charge for approval in the time line as specified in contract agreement								
	f. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.								
	Signature					Signature & Seal			

	Engineer-in-charge				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)
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5) (ii) DYNAMIC EXCITATION SYSTEM FOR ASYNCHRONOUS MACHINES

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपष्कर का नाम (NAME OF EQUIPMENT) : AC Excitation Variable Speed				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
Sl. No.	Item / Components & Characteristics	Nature of Checks	Quantum of Checks	Reference Document / Acceptance Norms	Record Format	Inspection Agency			Remarks
						Sub supplier	Supplier	Employer	
Inspections and tests on AC excitation boards									
1.1	Work Shop tests								
1.1.1	Visual checks								
1.1.1.1	Visual inspection of sub assesmblies / modules / components before integration in converter	Mechanical	Sample - 30%	Acc. To drawings	TC	P	RR	RR	
1.1.1.2	Check of parts acc. Part lists	Mechanical	Sample - 10%	Acc. To drawings	TC	P	RR	RR	
1.1.1.3	Check of wiring (color, diameter, connection quality, labeling)	Mechanical	Sample	Acc. To drawings	TC	P	RR	RR	
1.1.1.4	Check of cubicle dimensions	Mechanical	Sample	Acc. To drawings	TC	P	RR	RR	
1.1.1.5	Check of cubicle finishing and labels	Mechanical	Sample	Acc. To drawings	TC	P	RR	RR	
1.1.2	Functional tests								
1.1.2.1	Verificaion of BOM	Mechanical	Complete converter	Acc. To drawings	JIR	P	W	CHP	
1.1.2.2	Insulation test of control circuits	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.1.2.3	Insulation test of power circuits	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.1.2.4	Voltage test of power circuits	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	

1.1.2.5	Test of selected binary signals	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.1.2.6	Monitoring of serial interfaces	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.1.2.7	Fiber optics - assignment and check of firing signals	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.1.2.8	Light load function test (W.reduced voltage)	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.1.2.9	Function test of auxiliary devices	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.1.2.10	Function test of protective devices	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.1.2.11	Check of properties of control equipment	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.2	Site tests								
1.2.1	Tests after erection								
1.2.1.1	Tests of external aux. power supplies	Electrical	100%	Acc. Manufacturers commissioning instruction	JIR	P	W	RR	
1.2.1.2	Tests of external binary input signals	Electrical	100%		JIR	P	W	RR	
1.2.1.3	Tests of internal binary input signals	Electrical	100%		JIR	P	W	RR	
1.2.1.4	Tests of external analog input signals	Electrical	100%		JIR	P	W	RR	
1.2.1.5	Test of binary output signals	Electrical	100%		JIR	P	W	RR	
1.2.1.6	Test of analog output signals	Electrical	100%		JIR	P	W	RR	
1.2.1.7	Test of serial interfaces	Electrical	100%		JIR	P	W	RR	
1.2.1.8	Check of protection trips	Electrical	100%		JIR	P	W	RR	
1.2.1.9	Check of insulation resistance of power circuit	Electrical	100%		JIR	P	W	RR	
1.2.1.10	Check of insulation resistance of power cables	Electrical	100%		JIR	P	W	RR	
1.2.1.11	Adjustment of line side current and voltage measurement	Electrical	100%		JIR	P	W	RR	
1.2.1.12	Adjustment of unit side current and voltage measurement	Electrical	100%		JIR	P	W	RR	
1.2.1.1	Operation of converter with reduced voltage	Electrical	100%		JIR	P	W	RR	

3									
1.2.1.1 4	Check of combined function of AC excitation system, control system and protection	Electrical	100%		JIR	P	W	RR	
1.2.2	Tests after first run of unit								
1.2.2.1	First start up of unit with AC excitaiton	Electrical	1 time	Acc. Manufacturers commissioning instruction	JIR	P	W	RR	
1.2.2.2	Startup of motor unitl rated speed	Electrical	2 times		JIR	P	W	RR	
1.2.2.3	Sinusoidal wave form distortion factor measurement	Electrical	All three phases		JIR	P	W	RR	
1.2.2.4	First electrical braking of unit	Electrical	1 time		JIR	P	W	RR	
Inspections and tests on Cooling unit									
2.1	Work Shop tests								
2.1.1	Visual checks								
2.1.1.1	Visual inspection of sub assesmblies / modules / components before integration in converter	Mechanical	Sample - 30%	Acc. To drawings	TC	P	RR	RR	
2.1.1.2	Check of parts acc. Part lists	Mechanical	Sample - 10%	Acc. To drawings	TC	P	RR	RR	
2.1.1.3	Check of wiring (color, diameter, connection quality, labeling)	Mechanical	Sample	Acc. To drawings	TC	P	RR	RR	
2.1.2	Functional tests								
2.1.2.1	Test rotation direction pumps	Mechanical	Complete converter	Acc. To factory standard	JIR	P	W	RR	
2.1.2.2	Functional test on sensors	Mechanical	Complete converter	Acc. To factory standard	JIR	P	W	RR	
2.1.2.3	Pressure test	Mechanical	Complete converter	Acc. To factory standard	JIR	P	W	RR	
Inspections and tests on AC excitation Transformers									
3.1	Work Shop tests								
3.1.1	Visual checks								
3.1.1.1	Goods receipt	Mechanical	100%	Acc. To factory standard	TC	P	RR	RR	
3.1.1.2	Quantity	Mechanical	100%	Acc. To factory standard	TC	P	RR	RR	

3.1.1.3	Visual appearance	Mechanical	100%	Acc. To drawings	TC	P	RR	RR	
3.1.1.4	Damages	Mechanical	100%	Acc. To factory standard	TC	P	RR	RR	
3.1.1.5	Check of Transformer finishing and labels	Mechanical	100%	Acc. To drawings	TC	P	RR	RR	
3.1.2	Routine Tests								
3.1.2.1	Lightning impulse test	Electrical	1 No. per lot	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.2	Separate-source AC withstand voltage test	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.3	Measurement of voltage ratio & check phase displacement	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.4	Measurement of winding resistance	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.5	Induced AC withstand voltage test	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.6	Measurement of no-load loss and current	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.7	Measurement of short-circuit impedance & load loss	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.8	Measurement of partial discharge	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.9	Measurement of A-weighted sound level by sound pressure method at no load	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.10	Temperature rise test	Electrical	1 No. per lot	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.11	Paint adhesion / thickness	Electrical	100%	Acc. To drawings	JIR	P	W	CHP	
3.1.2.12	Dimensional inspection	Electrical	100%	Acc. To drawings	JIR	P	W	CHP	
3.1.2.13	Functional test of auxiliary components	Electrical	100%	Acc. To manufacturer standard	JIR	P	W	CHP	
3.1.2.14	Measurement of insulation resistance to earth	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
3.1.2.15	Determination of capacitances: windings to earth and between windings	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	

3.2	Site tests								
3.2.1	Tests before energizing								
3.2.1.1	Visual inspection (painting, oil tightness, etc.)	Mechanical	100%	Acc. To manufacturer standard	JIR	P	W	RR	
3.2.1.2	Check of grounding connection between transformer parts and to ground	Mechanical	100%	Acc. To manufacturer standard	JIR	P	W	RR	
3.2.1.3	Chech of tightness of all primary and secondary connections	Mechanical	100%	Acc. To manufacturer standard	JIR	P	W	RR	
3.2.1.4	Measurement of winding resistance	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	RR	
3.2.1.5	Measurement of voltage ratio & check phase displacement	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	RR	
3.2.1.6	Measurement of insulation resistance to earth	Electrical	100%	'IS/ IEC/ equivalent or above'	JIR	P	W	RR	
3.2.1.7	Funtional test of auxiliary components	Electrical	100%	Acc. To manufacturer standard	JIR	P	W	RR	
3.2.1.8	Check of all hardwired and bus signals	Electrical	100%	Acc. To drawings	JIR	P	W	RR	
Inspections and tests on Crowbar									
4.1	Work Shop tests								
4.1.1	Visual checks								
4.1.1.1	Goods receipt	Mechanical	100%	Acc. To factory standard	TC	P	RR	RR	
4.1.1.2	Quantity	Mechanical	100%	Acc. To factory standard	TC	P	RR	RR	
4.1.1.3	Visual appearance	Mechanical	100%	Acc. To drawings	TC	P	RR	RR	
4.1.1.4	Damages	Mechanical	100%	Acc. To factory standard	TC	P	RR	RR	
4.1.2	Functional Tests								
4.1.2.1	Identification of main components	Mechanical	100%	Acc. To drawings	JIR	P	W	CHP	
4.1.2.2	Visual inspection	Mechanical	100%	Acc. To drawings	JIR	P	W	CHP	
4.1.2.3	Insulation test	Electrical	100%	Acc. To factory standard	JIR	P	W	CHP	
4.1.2.4	Test of auxiliary supply	Electrical	100%	Acc. To factory standard	JIR	P	W	CHP	
4.1.2.5	Break over diode test	Electrical	100%	Acc. To factory standard	JIR	P	W	CHP	
4.1.2.6	Crowbar trigger test	Electrical	100%	Acc. To factory standard	JIR	P	W	CHP	

Note:								
a. Abbreviations used - RR-Review of Records, TC - Test Certificate Submission, JIR - Joint Inspection Report & CHP - Customer Hold Point.								
b. Test Certificates shall be submitted at the time of final inspection. Valid calibration Certificate of Instruments used for Testing shall be provided during the Test.								
Signature								Signature & Seal
EMPLOYER (QA&I DEPT.)								(VENDORS Q.C. DEPT. OR REPRESENTATIVE)

6) ISOLATED PHASE BUS DUCT (IPBD) / SEGREGATED PHASE BUS DUCT (SPBD)

परियोजना (PROJECT) :					ग्राहक (CLIENT):				
उपकरण का नाम (NAME OF EQUIPMENT) : Bus Duct					विक्रेता (VENDOR):				
					एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :				
क्रम स. (SR. NO.)	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज / स्वीकृति मानदंड REFERENCE ACCEPTANCE DOCUMENTS NORMS	रिकॉर्ड फॉर्मेट RECORD FORMA T	निरीक्षण एजेंसी INSP. AGENCY			टिपणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्याप न Verify	
A	BUS DUCT								
1	Raw Material & Bought-out Items								
1.1	Aluminium for Enclosure								
a)	Dimension	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	Sample	-do-	TC	3/2	-	1	TC
c)	Chemical Composition (incl. Al Percent.)	Chemical	-do-	-do-	TC	3/2	-	1	TC
d)	Conductivity	Electrical	-do-	-do-	TC	3/2	-	1	TC
1.2	Aluminium for Conductor								
a)	Dimension	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	Sample	-do-	TC	3/2	-	1	TC
c)	Chemical Composition (incl. Al Percent.)	Chemical	-do-	-do-	TC	3/2	-	1	TC
d)	Conductivity	Electrical	-do-	-do-	TC	3/2	-	1	TC
1.3	Mounting Structure (Steel, Angle Sections and Plates)								
a)	Physical Properties (Ultimate Strength,Yield Stress,% Elongation & Bend Test	Mechanical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Chemical Composition	Chemical	-do-	-do-	TC	3/2	-	1	TC
1.4	Epoxy Insulator/Epoxy Bushing								

a)	Dimensional	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Creepage distance	-do-	Sample	-do-	TC	3/2	-	1	TC
c)	Dry PF Voltage Withstand & PD Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
d)	Test for Mechanical Strength	Mechanical	-do-	-do-	TC	3/2	-	1	TC
e)	Temperature Cycle Test	Test	-do-	-do-	TC	3/2	-	1	TC
f)	Impulse voltage withstand test(For Bushing)	Electrical	-do-	-do-	TC	3/2	-	1	TC
1.5	Expansion Compensator								
a)	Dimension	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	Sample	-do-	TC	3/2	-	1	TC
c)	Chemical Composition (incl. Al Percent.)	Chemical	-do-	-do-	TC	3/2	-	1	TC
d)	Conductivity	Electrical	-do-	-do-	TC	3/2	-	1	TC
1.6	Metal Bellows								
a)	Chemical Composition, Thickness, Finish	Test	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
1.7	Ethylene - Propylene Bellows								
a)	Dimensional	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	Sample	-do-	TC	3/2	-	1	TC
c)	Insulation Resistance	Electrical	-do-	-do-	TC	3/2	-	1	TC
2	IN PROCESS INSPECTION								
2.1	Bus Duct enclosure								
a)	Dimensional check	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	NDT (DPT) on welding	Test	Sample	-do-	TC	3/2	-	1	TC
2.2	Busbar mounting								
a)	Size of busbar	Visual	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Straightness of busbar	-do-	-do-	-do-	TC	3/2	-	1	TC
2.3	Al./Cu. Flexible								

a)	Alignment, uniformity freedom from defects	-do-	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Dimensions	-do-	-do-	-do-	TC	3/2	-	1	TC
2.4	Short Links for testing purpose								
a)	Chemical Composition, Mechanical ,Thickness, Finish	Test	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Conductivity	Electrical	-do-	-do-	TC	3/2	-	1	TC
2.5	Air Pressurisation System/Hot air blowing system								
a)	BOM & Functional Check	-do-	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
3	FINAL INSPECTION OF BUS DUCT								
3.1	Type Test								
a)	Temperature rise Test	Electrical	One Unit	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Short Time Current Test	-do-	-do-	-do-	TC	3/2	-	1	TC
c)	Impulse Voltage Test	-do-	-do-	-do-	TC	3/2	-	1	TC
d)	Degree of Protection (Air & Water Tightness Test)	Test	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
3.2	Routine Test								
a)	Check for assembly, dimension, tightness of busbar joints, busbar support, clearance & freedom from surface defects & painting	Visual	5% or 02 nos. whichever is higher	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
b)	One Minute Power Frequency Voltage Withstand Test	Electrical	-do-	-do-	JIR	3/2	1	-	CHP
c)	IR Test before and after HV Test	-do-	-do-	-do-	JIR	3/2	1	-	CHP
d)	Mili Volt Drop Test	-do-	Sample	-do-	JIR	3/2	1	-	CHP
e)	Fuctional & BOM verification of Air Pressuration System	Visual & Fuctional	-do-	-do-	TC	3/2	-	1	TC
B	LAVT & NG CUBICLE								

1	Raw Material								
1.1	Steel Sheet & angle								
a)	Dimension	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Bend Test	Visual	Sample	-do-	TC	3/2	-	1	TC
1.2	Copper Strip & Flats								
a)	Dimension	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Mechanical Test	Mechanical	Sample	-do-	TC	3/2	-	1	TC
c)	Conductivity	Electrical	-do-	-do-	TC	3/2	-	1	TC
2	Bought Out Items								
2.1	Epoxy Insulator								
a)	Dimensional	Measurement	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Creepage distance	-do-	-do-	-do-	TC	3/2	-	1	TC
c)	Dry PF Voltage Withstand & PD Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
d)	Test for Mechanical Strength	Mechanical	-do-	-do-	TC	3/2	-	1	TC
e)	Temperature Cycle Test	Test	-do-	-do-	TC	3/2	-	1	TC
2.2	Epoxy seal off bushing								
a)	Dimension	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Dry Power Frequency Withstand Test	Electrical	Sample	-do-	TC	3/2	-	1	TC
c)	Partial Discharge	-do-	-do-	-do-	TC	3/2	-	1	TC
d)	Creepage	-do-	-do-	-do-	TC	3/2	-	1	TC
2.3	Current Transformer								
a)	Routine Test								
i)	Make, Type, Rating & Visual Inspection	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
ii)	Terminal Marking & Polarity Verification	Electrical	-do-	-do-	TC	3/2	-	1	TC
iii)	Power Frequency Test on Primary & Secondary Winding separately.	-do-	-do-	-do-	TC	3/2	-	1	TC
iv)	Determination of Error according to the requirements of the Appropriate Accuracy Class	-do-	-do-	-do-	TC	3/2	-	1	TC

v)	Over Voltage Inter Turn Test	-do-	-do-	-do-	TC	3/2	-	1	TC
b)	Type Test (Impulse & Temp. Rise Test)	Electrical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
2.4	Disconnecting Switch (As applicable)								
a)	Type & No.of Poles	Visual	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Dimensional check	-do-	-do-	-do-	TC	3/2	-	1	TC
c)	Electrical & Mechanical Operation	Operation	100%	-do-	TC	3/2	-	1	TC
d)	Aux.Contact Function	Electrical	-do-	-do-	TC	3/2	-	1	TC
e)	Contact Resistance	-do-	-do-	-do-	TC	3/2	-	1	TC
f)	Power Frequency Votage Test	-do-	-do-	-do-	TC	3/2	-	1	TC
i)	Pole & Body	-do-	-do-	-do-	TC	3/2	-	1	TC
ii)	Across The Isolating Distance	Measurement	-do-	-do-	TC	3/2	-	1	TC
3	Neutral Grounding Cubicle								
3.1	Neutral Grounding Resistor								
	Routine Test								
a)	Visual & Dimensional Check	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Resistance at amb.temp	Electrical	-do-	-do-	TC	3/2	-	1	TC
c)	Insulation Resistance	-do-	-do-	-do-	TC	3/2	-	1	TC
d)	H.V withstand Test	-do-	-do-	-do-	TC	3/2	-	1	TC
	Type Test								
a)	Temperature Rise Test (Part Resistance test)	Electrical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
3.2	Neutral Grounding Transformer								
a)	Type Test (Impulse & Temp. Rise Test)	Electrical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Routine Tests								
i)	Measurement of Winding Resistance	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
ii)	Measurement of Voltage Ratio	-do-	-do-	-do-	JIR	3/2	1	-	CHP

iii)	Measurement of Short Circuit Impedance & Load Loss	-do-	-do-	-do-	JIR	3/2	1	-	CHP
iv)	Measurement of No Load Losses and current	-do-	-do-	-do-	JIR	3/2	1	-	CHP
v)	Measurement of Magnetizing Current (before and after Dielectric Test)	-do-	-do-	-do-	JIR	3/2	1	-	CHP
vi)	Measurement of Insulation Resistance and Polarization Index	-do-	-do-	-do-	JIR	3/2	1	-	CHP
vii)	Dielectric Test:								
	i) Separate source AC withstand voltage test	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
	ii) Induced AC voltage withstand test (IVW)	-do-	-do-	-do-	JIR	3/2	1	-	CHP
	iii) Partial discharge measurement	-do-	-do-	-do-	JIR	3/2	1	-	CHP
viii)	Complete Transformer - Overall Physical Verification & Dimensions	-do-	-do-	-do-	JIR	3/2	1	-	CHP
ix)	Paint shade, thickness & Adhesion Test	-do-	-do-	-do-	TC	3/2	-	1	TC
4)	LAVT Cubicle								
4.1	Voltage Transformer								
	Routine Tests								
a)	Make, Type, Rating & Visual Inspection	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Terminal Marking & Polarity Verification	Electrical	-do-	-do-	TC	3/2	-	1	TC
c)	Power Frequency Test on Primary & Secondary Winding separately	-do-	-do-	-do-	TC	3/2	-	1	TC
d)	Determination of Error according to the requirement of the Appropriate Accuracy Class	-do-	-do-	-do-	TC	3/2	-	1	TC
e)	Over Voltage Inter Turn Test	-do-	-do-	-do-	TC	3/2	-	1	TC
	Type Test								
a)	Impulse Test	Electrical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Heat Run Test	-do-	-do-	-do-	TC	3/2	-	1	TC
4.2	Lightning Arrestor								
a)	Visual & Dimension	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC

b)	Routine test (Measurement of reference Voltage, PD & Seal Leak Test etc.)	Electrical	-do-	-do-	TC	3/2	-	1	TC
c)	Type Test (Lighting Impulse Residual Voltage, Switching Surge Operating Duty Test)	-do-	Sample	-do-	TC	3/2	-	1	TC
4.3	Surge Capacitor								
a)	Visual	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Routine Test	Electrical	-do-	-do-	TC	3/2	-	1	TC
i)	Output Test (Capacitance & Tan Delta)	-do-	-do-	-do-	TC	3/2	-	1	TC
ii)	Voltage Test Between Terminals	-do-	-do-	-do-	TC	3/2	-	1	TC
iii)	Efficiency test-discharge upto min voltage	-do-	-do-	-do-	TC	3/2	-	1	TC
iv)	Sealing Test	-do-	-do-	-do-	TC	3/2	-	1	TC
c)	Type Test (Impulse Test, PD Test)	-do-	Sample	-do-	TC	3/2	-	1	TC
4.4	Earthing Switch								
a)	Type Test	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Routine Test	-do-	-do-	-do-	TC	3/2	-	1	TC
4.5	H.T. Fuse								
a)	Continuity Test	Electrical	100%	-do-	TC	3/2	-	1	TC
4.6	Mounting Structure								
a)	Dimensional check	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Uniformity, mass and adhesion of Zn coating	-do-	-do-	-do-	TC	3/2	-	1	TC
5	FINAL INSPECTION OF CUBICLE								
a)	Check of BOM, dimension, freedom from surface defect, layout, joints & clearance	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
b)	Scheme checking	Electrical	-do-	-do-	JIR	3/2	1	-	CHP
c)	HV & IR Test on main and auxiliary circuit, Continuity check	-do-	-do-	-do-	JIR	3/2	1	-	CHP
d)	Heater & Thermostat			-do-					CHP
e)	Type & Rating	Visual	100%	-do-	JIR	3/2	1	-	CHP
f)	Functional Test	Test	-do-	-do-	TC	3/2	-	1	TC
g)	Painting (Uniformity, Adhesion & Shade)	Visual	Sample	-do-	JIR	3/2	1	-	CHP

h)	Degree of Protection	-do-	-do-	-do-	TC	3/2	-	1	TC
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column "Perform" .								
	f. The type test may not be mandatory if similar equipment has been type tested and test certificate(s) for relevant tests are accepted by the Employer								
	Signature						Signature & Seal		
	EMPLOYER (QA&I DEPT.)						(VENDORS Q.C. DEPT. OR REPRESENTATIVE)		

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN) :									
परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Bus Duct				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	घटक की गतिविधि और संचालन ACTIVITY & OPERATION OF COMPONENT	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी INSP. AGENCY			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन VERIFICATION	
A	Bus Bar								
a)	Check assembly, dimension, tightness of busbar joints, busbar support, clearance & freedom from surface defects & painting	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR

b)	One Minute Power Frequency Voltage Withstand Test	Electrical	-do-	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
c)	Insulation Resistance Test for Enclosure Circuit	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
B	Current Transformer								
a)	Make, Type, Rating & Visual Inspection	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Polarity Verification	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
c)	Winding Resistance Test on Primary & Secondary Winding separately.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
d)	Insulation resistance Test	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
e)	Ratio & Knee Voltage Test	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
C	Potential Transformer								
a)	Make, Type, Rating & Visual Inspection	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Polarity Verification	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
c)	Winding Resistance Test on Primary & Secondary Winding separately.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
d)	Insulation resistance Test	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
e)	Ratio Test	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
D	Neutral Grounding Register								
a)	Make, Type, Rating & Visual Inspection	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Insulation resistance Test	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
c)	Resistance Test	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
E	Neutral Grounding Transformer								
a)	Make, Type, Rating & Visual Inspection	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR

b)	Winding Resistance Test on Primary & Secondary Winding separately.	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
c)	Terminal Marking & Polarity Verification	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
F	Lightning Arrestor								
a)	Dimension Check	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Insulation resistance Test	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
G	Capacitor								
a)	Visual Check	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Capacitance Measurement Test	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
H	Surge Protection Capacitor								
a)	Visual Check	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Capacitance Measurement Test	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
I	Surge Monitor								
a)	Visual Check	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate Representative of Engineer-in-charge, 2- will indicate 'Contractor' & 3- will indicate 'Sub-contractor'.								
	b. In 'Remarks' column following abbreviations shall be used - TC/TR - Test Certificate/ Test Report								
	c. The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	d. Any additional activity/ test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.								
	e. The contractor shall furnish the details of test method to the Engineer-in-Charge for approval in the time line as specified in contract agreement								
	f. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.								

	Signature					Signature & Seal
	Engineer-in-charge				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)	

7) EOT CRANE

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : EOT Crane				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS / ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORDED FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Performance	गवाह Witnesses	सत्यापन Verify	
1	Raw Material								
1.1	Structural Steel (Box Girder, End Trucks, Trolley, Rails, Beam, Cabin etc.)	Chemical Composition Mech. Properties Ultrasonic Test (UT for above 19 mm)	Sample/Batch 100%	Appd. drawings/ Tech. specs./IS/IEC/Equivalent	TC	3/2	-	1	TC
1.2	Cast/Forged Steel (Wheels, Shafts, Pins,Gears, Pinions, Brake Drum, Couplings, Hook, Buffer Stopper, etc.)	Chemical Composition Mech. Properties Ultrasonic Test (UT for above 50 mm)	Sample/Batch 100%	-do-	TC	3/2	-	1	TC
1.3	Bought Out Items								
a)	Bushes, Bearings, Pulleys, Fasteners etc.	Material Properties	Sample per Lot	-do-	TC	3/2	-	1	TC
b)	Drive Motor, Gear Box, Plumber Block, EHT Brakes, EM Beakes, Wire Rope, Electrical Cables, Remote Radio Control, Load cell system etc	Make, Type & Rating, Routine Test	100%	-do-	TC	3/2	-	1	TC

c)	Detection devices (Anti collision limit switch, Over load detection device, Load control device, Detection of Brake Wear, Over speed detection device, Circumferential cable staggering on drum, Unbalancing detection device etc) - As applicable	Make, Type & Rating, Routine Test	100%	-do-	TC	3/2	-	1	TC
2	In Process Stage Inspection								
2.1	Welder certification	Visual	100%	-do-	TC	3/2	-	1	TC
2.2	NDT-Welding Fillet Welds	UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
2.3	NDT-Butt Welds (Box Girder, Rope Drums and other full strength welds)		100%	-do-	TC	3/2	-	1	TC
2.4	Post Weld Heat Treatment (Fabricated Rope Drums, Gear Box Housing, Forged Crane hooks)	Heat Treatment Chart	-do-	-do-	TC	3/2	-	1	TC
2.5	Machining								
a)	Surface Finish (Rope drum shaft, Rope Drum grooves, Brake Drum)	Visual/Measurement	-do-	-do-	TC	3/2	-	1	TC
b)	Dimensions	Measurement	-do-	-do-	IR	3/2	-	1	RR
2.6	Sub-Assembly Stage- Drive Mechanism, Hoist / Trolley arrangements, Brakes								
a)	Alignment/Layout	Visual/Measurement	100%	-do-	IR	3/2	-	1	RR
b)	Fitment and Clearance	Visual/Measurement	-do-	-do-	IR	3/2	-	1	RR
c)	Dimensional check	Visual/Measurement	-do-	-do-	IR	3/2	-	1	RR
d)	Running Performance (Drive Motors, Gear Boxes)-Noise level, Vibration, Temp. rise Leakage Test - Gear Boxes	Visual/Measurement	-do-	-do-	TC	3/2	-	1	TC
e)	Crane Hook-Proof Load Test	Load Test	-do-	-do-	TC	3/2	-	1	TC
f)	NDT-after Proof Load Test Butt Welds (Box Girder, Rope Drums and other full strength welds)	UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
3	Final inspection/Tests on assembled Crane at manufacturer's works								
3.1	Operator's cabin- Dim. Check & its accessories check	Visual/Measurement	-do-	-do-	JIR	3/2	1	-	CHP
3.2	BOM Checking & Verification of TR of drawing	Visual	-do-	-do-	JIR	3/2	1	-	CHP
3.3	Examination of Welding	Visual	-do-	-do-	TC	3/2	-	1	TC

3.4	Welding- NDT	UT, MPI, DPT (as applicable)	-do-	-do-	TC	3/2	-	1	TC
3.5	Dimensional check on assembled crane - bridge structure (Box girders, end trucks) hoisting trolley frame, travelling mechanism	Measurement	100%	Appd. drawings/ Tech. specs./IS/IEC/Equivalent	JIR	3/2	1	-	CHP
3.6	Crane Girder Camber (If applicable)	-do-	-do-	-do-	TC	3/2	-	1	TC
3.7	Deflection test at full load/ Rated load & Over load	-do-	-do-	-do-	JIR	3/2	1	-	CHP
3.8	Hook Approach	-do-	-do-	-do-	JIR	3/2	1	-	CHP
3.9	Operational/functional test								
a)	Hoisting and LT & CT Speeds - without load	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
b)	Hoisting and LT & CT Speeds at Rated load	-do-	-do-	-do-	JIR	3/2	1	-	CHP
c)	Hoisting and LT & CT travel test at 125% of rated load	-do-	-do-	-do-	JIR	3/2	1	-	CHP
d)	Function of brakes for trolley and hoistings	Visual/Measurement	-do-	-do-	JIR	3/2	1	-	CHP
e)	Amperage at no load and full load for all motors	-do-	-do-	-do-	JIR	3/2	1	-	CHP
f)	Creepling speed/ Jog movement with rated load(VVVF Drive)	-do-	-do-	-do-	JIR	3/2	1	-	CHP
4	Control Panel (At Control Panel's Supplier premises)								
4.1	Resistors, Capacitors, Limit Switches, Master Controllers, Thyristors, VVVF Modules, Transformers, Circuit Breakers, Switches etc.	Electrical Test	-do-	-do-	TC	3/2	-	1	TC
4.2	BOM, Verification of scheme, make and rating of components	Visual	-do-	-do-	JIR	3/2	1	-	CHP
4.3	Cubicle Dimensions	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
4.4	Functional check including HV and IR tests	Visual/Test	-do-	-do-	JIR	3/2	1	-	CHP
4.5	Sequence of operation and interlock tests	Visual/Test	-do-	-do-	JIR	3/2	1	-	CHP
4.6	Protection Test (Single phase protection, over current protection, thermal over load, under voltage protection etc as per approved scheme.)	Visual/Test	-do-	-do-	JIR	3/2	1	-	CHP
4.7	Radio control tests of speed control of bridge travel, trolley travel & hoist motion, control of master contactor, indication and alarm (if applicable)	-do-	-do-	-do-	JIR	3/2	1	-	CHP
4.8	Pendant control test (if applicable)	-do-	-do-	-do-	JIR	3/2	1	-	CHP

5	Sand blasting & Painting	Visual / Measurement	-do-	-do-	TC	3/2	-	1	TC
Note :	a. In ‘Inspection Agency’ column figure 1,2,or 3 to be filled. 1- will indicate ‘EMPLOYER ’, 2- will indicate ‘supplier’ & 3- will indicate ‘sub-supplier’.								
	b. In ‘Remarks’ column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission , JIR- Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f. Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP. NDT shall be conducted from NABL accredited lab/Govt. approved lab/Level 2 or level 3 certified person.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	h. Site test shall be conducted in presence of Project's Representative only.								
			Signature			Signature & Seal			
			EMPLOYER (QA&I DEPT.)		(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN) :									
परियोजना (PROJECT) :					ग्राहक (CLIENT) :				
उपस्कर का नाम (NAME OF EQUIPMENT) : EOT CRANE					विक्रेता (VENDOR) :				
					एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :				
क्र. सं. SR. NO.	Activity & Operation of Component	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTITY OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Performance	गवाह Witness	सत्यापन Verify	

1	Installation and Pre Commissioning Test								
1.1	Operational and functional tests of components, sub-assemblies, auxiliaries i.e. lubrication etc	Functional	100%	Appd. Drg. / Tech. Spec. / IS/IEC/Equivalent	TC/TR	3/2	1	-	TC/TR
1.2	Control logic field tests to prove control logic scheme for operation and interlocks of system components and controls, including annunciation and shutdown scheme shall be done in so far as practical	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.3	Calibration test - All instruments, sensors, switches, relays etc. shall be calibrated and adjusted after installation, wherever practical	Visual/Measurement	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.4	Operational tests for verification of correct operation of protective relays and safety devices shall be conducted	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.5	Operational tests for drives for verification of correct operation of each and every component of the drives	Visual/Measurement	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.6	Deflection Test - The deflection test shall be carried out with test load of 100 % of rated load at rest and with the crab in central position. The measurement shall not be taken on the first application of load. Datum line for measuring the deflection should be obtained by placing the crab on the extreme end of the crane span with smaller hook approach.	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.7	Load tests at 100% and 125% of rated load on each crane	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.8	Verification of crane clearance and lifts	Visual	100%		TC/TR	3/2	1	-	TC/TR
1.9	Incremental movement of hoists/travels	Visual	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.10	Tandem operation	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.11	Check Speed : Hoisting speed, Long travel, Cross travel	Visual/Measurement	100%	-do-	TC/TR	3/2	1	-	TC/TR
2	Commissioning Tests								
2.1	Insulation resistance test & Earthing Test of complete crane	Test	100%	Appd. Drg. / Tech. Spec. / IS/IEC/Equivalent	TC/TR	3/2	1	-	TC/TR

2.2	Run all the motions of crane in idle conditions and following needs to be observed: (a) Eccentric running of moving parts. (b) Alignment of shafts. (c) Fauling of moving components. (d) Any un-usual noise. (e) Presence of lubricant at desired surfaces. (f) Performance of motors, limit switches,brakes, controllers etc	Visual/Measuremen t	100%	-do-	TC/TR	3/2	1	-	TC/TR
2.3	Operational test - Operational test with/without rated load in all hoisting/ travelling motions to check for any misalignment, improper adjustments, overheating and other possible defects. This shall also include operational test for limit switches correctness of circuits, interlocks, operation of protective devices, brakes etc	Functional	100%	-do-	TC/TR	3/2	1	-	TC/TR
Note :	a. In ‘Inspection Agency’column figure 1,2,or 3 to be filled. 1- will indicate 'Representative of Engineer-In -Charge , Employer', 2- will indicate ‘Contractor’ & 3- will indicate ‘sub-Contractor’.								
	b. In ‘Remarks’ column following abbreviations shall be used -TC-Test Certificate, TR- Test Report								
	c. At the time of Inspection the Contractor shall arrange the requisite calibrated measuring instruments/Tools								
	d. The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	e. Any additional activity/test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.								
	f. The contractor shall furnish the details of test methods to the Engineer-In charge, Employer for approval in the time line as specified in contract agreement.								
	g. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.								
	Signature					Signature & Seal			
	Engineer-in-charge				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

8) LUBRICATING AND INSULATING OIL HANDLING SYSTEM

परियोजना (PROJECT) :				ग्राहक (CLIENT) :					
उपस्कर का नाम (NAME OF EQUIPMENT) : Lubricating Oil purifier System				विक्रेता (VENDOR) :					
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :					
क्र. सं. SR. NO.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	रिकॉर्ड प्रारूप RECORD FORMAT	निरीक्षण एजेंसी (INSP. AGENCY)			टिप्पणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Centrifuge			Appd. Drawings/ Tech. specs./ IS/IEC/Equivalent					
1.1	Centrifugal Oil Purifier-Bowl Body, Hood & Lock Ring								
a)	Chemical & Mech Properties	Chem & Mech	Sample per Lot	-do-	TC	3/2	-	1	TC
1.2	In Process Control								
a)	Bowl Assembly---Dynamic Balancing	Measurement	100%	-do-	TC	3/2	-	1	TC
b)	Bearing Assembly, Idler Arm Assembly, Idler Bearing Assembly, Drag Assembly	Visual / Measurement	100%	-do-	TC	3/2	-	1	TC
i)	Smoothness/Freeness of Rotation	Visual	- do-	-do-	TC	3/2	-	1	TC
2	Bearings- Make, Type & rating	Visual	100%	-do-	TC	3/2	-	1	TC
3	Valves, Pipes, Fitting & Hoses etc.								
3.1	Chemical & Mech Properties	Chem & Mech	Sample per Lot	-do-	TC	3/2	-	1	TC
3.2	Hydrostatic pressure test	Measurement	100%	-do-	TC	3/2	-	1	TC
3.3	Dimensional Check	-do-	100%	-do-	TC	3/2	-	1	TC
3.4	HV, IR test for Solenoid Valve	-do-	100%	-do-	TC	3/2	-	1	TC
4	Temperature Gauge, Flow Meter, Pressure Gauge, Level Switch, Solenoid Valve, Thermostat, Vacuum gauge etc								
4.1	Make, Type & rating	Visual	1/Lot	-do-	TC	3/2	-	1	TC

4.2	HV & IR test (As applicable)	Test	100%	-do-	TC	3/2	-	1	TC
4.3	Performance, Accuracy, Repetability & Calibration(As applicable)	Test	100%	-do-	TC	3/2	-	1	TC
5	Circulating Pump, Oil Transfer Pump etc (As applicable)								
5.1	Make, Type & Capacity	Visual	100%	-do-	TC	3/2	-	1	TC
5.2	Physical Analysis.& Chemical Composition	Test	Sample per Lot	-do-	TC	3/2	-	1	TC
5.3	Leak Tightness of casing	Test	100%	-do-	TC	3/2	-	1	TC
5.4	Performance Test								
a)	Q v/s Head, Q v/s Power, Q v/s Pump efficiency	Type Test	1/type	-do-	TC	3/2	-	1	TC
b)	NPSH Test	-do-	- do-	-do-	TC	3/2	-	1	TC
c)	Leakage Test	-do-	- do-	-do-	TC	3/2	-	1	TC
d)	Vibration & Noise Test	-do-	- do-	-do-	TC	3/2	-	1	TC
5.5	Painting inspection (uniformity, shade, adhesion & DFT)	Visual/ Measuremen t	100%	-do-	TC	3/2	-	1	TC
6	Vacuum Pump								
6.1	Make, Type & Capacity	Visual	100%	-do-	TC	3/2	-	1	TC
6.2	Physical Analysis & Chemical Composition	Test	Sample per Lot	-do-	TC	3/2	-	1	TC
6.3	Leak Tightness of Casing	Test	100%	-do-	TC	3/2	-	1	TC
6.4	Performance Test								
a)	Bearing temp. rise Test	Type Test	1/type	-do-	TC	3/2	-	1	TC
b)	Free air displacement	-do-	- do-	-do-	TC	3/2	-	1	TC
c)	Ultimate Vaccume pulled by the pump	-do-	1/type	-do-	TC	3/2	-	1	TC
6.5	Painting inspection (uniformity, shade, adhesion & DFT)	Visual/ Measuremen t	100%	-do-	TC	3/2	-	1	TC
7	Motors								
7.1	Make, Type & rating	Visual	100%	-do-	TC	3/2	-	1	TC
7.2	Type test	Test	Sample	-do-	TC	3/2	-	1	TC
7.3	Routine Test	Test	100%	-do-	TC	3/2	-	1	TC
						प्रदर्श न Perfor m	गवाह Witnes s	सत्याप न Verify	
8	Filters/Strainers			Appd. Drawings/ Tech.					

8.1	Make, Type & Capacity	Visual	100%	specs./ IS/IEC/Equivalent	TC	3/2	-	1	TC
8.2	Physical Analysis & Chemical Composition	Test	Sample per Lot	-do-	TC	3/2	-	1	TC
8.3	Mesh Size	Measurement	100%	-do-	TC	3/2	-	1	TC
8.4	Hydrostatic Test & Performance Test	Test	100%	-do-	TC	3/2	-	1	TC
8.5	Painting inspection (uniformity, shade, adhesion & DFT)	Visual/ Measurement	100%	-do-	TC	3/2	-	1	TC
9	Direct type Electric Oil Heater & Degasing Chamber								
9.1	Make, Type & Capacity	Visual	100%	-do-	TC	3/2	-	1	TC
9.2	Leak Tightness Test	Leak Test	100%	-do-	TC	3/2	-	1	TC
9.3	IR test for heater	Elect. Test	100%	-do-	TC	3/2	-	1	TC
10	Vacuum & Oil Tank								
10.1	Chemical & Mech Properties	Chem & Mech	Sample per Lot	-do-	TC	3/2	-	1	TC
10.2	Dimension/Capacity	Measurement	100%	-do-	JIR	3/2	1	-	CHP
10.3	Leakage Test	Test	100%	-do-	JIR	3/2	1	-	CHP
11	Final Inspection								
11.1	Purification Test of Lubricating Oil (single Pass)								
a)	Moisture Content (Less than 50 PPM)	Measurement	100%	-do-	JIR	3/2	1	-	CHP for Sample Collection Only
b)	Filtration- Less than 1 micron	Test	100%	-do-	JIR	3/2	1	-	
c)	Free Water-100% Reduction	Test	100%	-do-	JIR	3/2	1	-	
11.2	Lubricating Oil Purifier system including Control panel								
a)	Dimension Check of Assembly	Measurement	100%	-do-	JIR	3/2	1	-	CHP
b)	Verification of Bill of material	Visual	-do-	-do-	JIR	3/2	1	-	CHP
c)	Capacity (Suction & Discharge Pressure)	Measurement	-do-	-do-	JIR	3/2	1	-	CHP
d)	Power Consumption Test for Motors & Heaters	Elect. Test	-do-	-do-	JIR	3/2	1	-	CHP
e)	Sequential Operation & Interlock Check	Functional Test	-do-	-do-	JIR	3/2	1	-	CHP
f)	Vacuum test at 650mm of Hg	Leak Test	-do-	-do-	JIR	3/2	1	-	CHP
g)	Vibration & Noise Measurement	Measurement	-do-	-do-	JIR	3/2	1	-	CHP

		t							
h)	Tests of Control Panel								
i)	Dimension Check of control panel	Measuremen t	-do-	-do-	JIR	3/2	1	-	CHP
ii)	HV & IR Test	Elect. Test	-do-	-do-	JIR	3/2	1	-	CHP
iii)	Wiring, Interlock & Sequence Check	Elect. Test	-do-	-do-	JIR	3/2	1	-	CHP
iv)	Degree of Protection	Test	-do-	-do-	TC	3/2	-	1	TC
v)	Painting inspection (uniformity, shade, adhesion & DFT)	Visual/ Measuremen t	Sample	-do-	JIR	3/2	1	-	CHP
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'EMPLOYER', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR- Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature			Signature & Seal			
			EMPLOYER (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)			

परियोजना (PROJECT) :				ग्राहक (CLIENT) :			
उपस्कर का नाम (NAME OF EQUIPMENT) : Insulating Oil purifier System				विक्रेता (VENDOR) :			
				एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :			
क्र.	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की	जाँच की	संदर्भ दस्तावेज़ /	रिकॉर्ड	निरीक्षण एजेंसी (INSP. AGENCY)	टिप्पणी REMARK

सं. SR. NO.		प्रकृति NATURE OF CHECKS	मात्रा QUANTUM OF CHECKS	स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	प्रारूप RECORD FORMA T	प्रदर्श न Perform	गवाह Witness	सत्याप न Verify	S
1	Instruments (As applicable)			Appd. Drawings/ Tech. specs./IS/IEC/Equivalen t					
1.1	Dial Thermometer	Accuracy Test	100%		TC	3/2	-	1	TC
1.2	Regulation Thermostat	-do-	-do-	-do-	TC	3/2	-	1	TC
1.3	Safety Thermostat	Elect.Test	-do-	-do-	TC	3/2	-	1	TC
1.4	Optical Level Thermometer								
a)	Continuity test	Elect.Test	-do-	-do-	TC	3/2	-	1	TC
b)	Performance test	Test	-do-	-do-	TC	3/2	-	1	TC
1.5	Flowmeter	Test	-do-	-do-	TC	3/2	-	1	TC
1.6	Float Switches	Functional Test	-do-	-do-	TC	3/2	-	1	TC
1.7	Pressure guage	Accuracy Test	-do-	-do-	TC	3/2	-	1	TC
1.8	Vacuum guage	-do-	-do-	-do-	TC	3/2	-	1	TC
1.9	Compound guage	-do-	-do-	-do-	TC	3/2	-	1	TC
1.10	Coarse Filter	Visual	-do-	-do-	TC	3/2	-	1	TC
1.11	Heater	Functional Test	-do-	-do-	TC	3/2	-	1	TC
2	Pumps								
2.1	Inlet gear pump								
a)	Make & Rating	Visual	-do-	-do-	TC	3/2	-	1	TC
b)	Leak tightness Test	Test	-do-	-do-	TC	3/2	-	1	TC
c)	Flow rate	Measuremen t	-do-	-do-	TC	3/2	-	1	TC
d)	Performance test	Test	-do-	-do-	TC	3/2	-	1	TC
e)	Type Test	Test	Sample	-do-	TC	3/2	-	1	TC
2.2	Outlet centrifugal Pump								
a)	Make & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Leak tightness Test	Test	-do-	-do-	TC	3/2	-	1	TC
c)	Flow rate	Measuremen t	-do-	-do-	TC	3/2	-	1	TC

e)	Routine Test of motor & pump	Elect.Test	100%	-do-	TC	3/2	-	1	TC
f)	Type Test	Elect.Test	Sample	-do-	TC	3/2	-	1	TC
2.3	Vacuum pump								
a)	Make & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Ultimate vacuum	Mechanical	-do-	-do-	TC	3/2	-	1	TC
c)	RPM Check	-do-	-do-	-do-	TC	3/2	-	1	TC
d)	Temperature rise	-do-	-do-	-do-	TC	3/2	-	1	TC
e)	Free air displacement	-do-	-do-	-do-	TC	3/2	-	1	TC
f)	Type Test	Test	Sample	-do-	TC	3/2	-	1	TC
2.4	Oil Transfer Pump								
a)	Make & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
b)	Performance Test	Functional Test	-do-	-do-	TC	3/2	-	1	TC
c)	Type Test	Test	Sample	-do-	TC	3/2	-	1	TC
3	Valves								
3.1	Make & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
3.2	Leak tightness Test	Test	-do-	-do-	TC	3/2	-	1	TC
3.3	Type Test	Test	Sample	-do-	TC	3/2	-	1	TC
4	Motors			Appd. Drawings/ Tech. specs./IS/IEC/Equivalent					
4.1	Make & Rating	Visual	100%		TC	3/2	-	1	TC
4.2	Routine test	Elect.Test	100%	-do-	TC	3/2	-	1	TC
4.3	Type Test	Test	Sample	-do-	TC	3/2	-	1	TC
5	Vessels (Tank) (Heater Vessel, Filter Vessel, Degassing chamber)								
5.1	Visual Check	Visual	100%	-do-	TC	3/2	-	1	TC
5.2	Dimensional Check	Measurement	100%	-do-	TC	3/2	-	1	TC
5.3	Leak tightness Test	Test	100%	-do-	TC	3/2	-	1	TC
6	Pressure Relief Valve								
6.1	Make & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
6.2	Functional Test	Test	100%	-do-	TC	3/2	-	1	TC
6.3	Type Test	Test	Sample	-do-	TC	3/2	-	1	TC
7	Oil Tank								

7.1	Chemical & Mech Properties	Chem & Mech	Sample per Lot	-do-	TC	3/2	-	1	TC
7.2	Dimension/Capacity	Measurement	100%	-do-	JIR	3/2	1	-	CHP
7.3	Leakage Test	Test	100%	-do-	JIR	3/2	1	-	CHP
8	Air Dryers								
8.1	Make & Rating	Visual	100%	-do-	TC	3/2	-	1	TC
8.2	Performance Test	Functional	100%	-do-	JIR	3/2	1	-	CHP
9	Oil hoses								
9.1	Make & Size	Visual	100%	-do-	TC	3/2	-	1	TC
9.2	Before and after ageing Dimensional Check	Measurement	Sample	-do-	TC	3/2	-	1	TC
9.3	Leakage under air pressure	Pressure	Sample	-do-	TC	3/2	-	1	TC
9.4	Leakage under vacuum	-do-	Sample	-do-	TC	3/2	-	1	TC
10	Trolly								
10.1	Overall Finish	Visual	100%	-do-	TC	3/2	-	1	TC
10.2	Dimensional Check	Measurement	100%	-do-	TC	3/2	-	1	TC
11	Final Assembly			Appd. Drawings/ Tech. specs./ IS/IEC/Equivalent					
11.1	BOM & Rating Check	Visual	100%	-do-	JIR	3/2	1	-	CHP
11.2	Overall Dimensions and Finish	Measurement	100%	-do-	JIR	3/2	1	-	CHP
11.3	Leak tightness under vacuum at 1 Torr	-do-	100%	-do-	JIR	3/2	1	-	CHP
11.4	Pressure Test on Pipes with assembly	-do-	100%	-do-	JIR	3/2	1	-	CHP
11.5	Capacity Check	-do-	100%	-do-	JIR	3/2	1	-	CHP
11.6	Power consumption of motor and heater	Elect. Test	100%	-do-	JIR	3/2	1	-	CHP
11.7	Interlock Test and Annunciation check.	-do-	100%	-do-	JIR	3/2	1	-	CHP
11.8	IR and HV Test	-do-	100%	-do-	JIR	3/2	1	-	CHP
11.9	Continuity & Functional Check	-do-	100%	-do-	JIR	3/2	1	-	CHP
11.10	Vacuum level	Test	100%	-do-	JIR	3/2	1	-	CHP
12	Performance Test of Oil Parameters (Single-Pass)								

12.1	Dielectric strength- 70 kV across 2.5 mm gap	Elect. Test	Sample	-do-	TC	3/2	-	1	TC
12.2	Moisture content (Less than 5 PPM)	Measurement	Sample	-do-	JIR	3/2	1	-	CHP
12.3	Gas content (Less than 0.1% by volume)	-do-	Sample	-do-	TC	3/2	-	1	TC
12.4	Filtration (Less than 1 micron)	Mech. Test	Sample	-do-	JIR	3/2	1	-	CHP
12.5	Dielectric dissipation factor (Tan delta at 90°C): 0.002	Elect. Test	Sample	-do-	TC	3/2	-	1	TC
12.6	Neutralisation value	-do-	Sample	-do-	TC	3/2	-	1	TC
Note :	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'EMPLOYER', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate Submission, JIR- Joint Inspection Report & CHP - Customer Hold Point.								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. At the time of Inspection the supplier shall arrange the requisite calibrated measuring instruments i.e.micrometer, vernier caliper, pie tape, surface roughness tester, dial gauge etc.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	f.Material testing shall be conducted from Govt. Lab, NABL Lab or NABL accredited lab having valid Scope of accreditation of all specified tests mentioned in approved QAP.								
	g. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
			Signature				Signature & Seal		
			EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)		

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN) :							
परियोजना (PROJECT) :					ग्राहक (CLIENT) :		
उपस्कर का नाम (NAME OF EQUIPMENT) : Oil Handling System					विक्रेता (VENDOR) :		
					एनआईटी/ क्रय आदेश संदर्भ (NIT/P.O. REFERENCE) :		
क्र. सं. SR.	Activity & Operation of Component	जाँच की	जाँच की	संदर्भ दस्तावेज़ /	रिकॉर्ड	निरीक्षण एजेंसी (INSP. AGENCY)	टिप्प

NO.		प्रकृति NATURE OF CHECKS	मात्रा QUANTU M OF CHECKS	स्वीकृति मानदंड REFERENCE DOCUMENTS/ ACCEPTANCE NORMS	प्रारूप RECO RD FORM AT	प्रदर्श न Perfor m	गवा ह Witne ss	सत्या पन Verify	णी REMA RKS
1	Oil Handling System								
1.1	Leakage and vacuum test	Test	100%	Tech. Spec./ Apprd. Drawing /IS/IEC/Equivalent	TC/TR	3/2	1	-	TC/TR
1.2	Operating condition of all pumps	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.3	Test to demonstrate operational capabilities of the system as per the intended requirement	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.4	Functional check of Measuring Instruments, Switches & Controller provided in the Oil Handling System	Functional Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.5	IR Test on Control Panels	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
1.6	Test to verify logic control scheme from Local / Remote including start/stop,Interlocks, Saftey Devices & other features	Test	100%	-do-	TC/TR	3/2	1	-	TC/TR
Note:	a. In ‘Inspection Agency’column figure 1,2,or 3 to be filled. 1- will indicate 'Representative of Engineer-In -Charge , Employer', 2- will indicate ‘Contractor’ & 3- will indicate ‘sub-Contractor’.								
	b. In ‘Remarks’ column following abbreviations shall be used - TC-Test Certificate, TR- Test Report								
	c. At the time of Inspection the Contractor shall arrange the requisite calibrated measuring instruments/Tools .								
	d. The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	e. Any additional activity/test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.								
	f. The contractor shall furnish the details of test methods to the Engineer-In charge, Employer for approval in the time line as specified in contract agreement.								
	g. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.								
	Signature					Signature & Seal			
	Engineer-in-charge				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

9) (i) STATIC FREQUENCY CONVERTER (SFC) SYSTEM

					ग्राहक (CLIENT):				
परियोजना (PROJECT) :					विक्रेता (VENDOR):				
उपष्कर का नाम (NAME OF EQUIPMENT) : Static Frequency Converter					एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :				
Sl. No.	Item / Components & Characteristics	Nature of Checks	Quantum of Checks	Reference Document / Acceptance Norms	Record Format	Inspection Agency			Remarks
						Sub supplier	Supplier	Employer	
1.1	Incoming goods inspection								
	BOUGHT OUT MATERIALS								
	Thyrister/IGBTs , Heat sink, Surge Supression Device, HMI/MMI etc. (As applicable)								
1.1.1	Make and Rating	Visual	100%	Tech specifications/approved drawing	TC	P	RR	-	
1.1.2	Functional test check	Electrical	100%	Tech specifications/approved drawing	TC	P	RR	-	
	Power Supply								
1.1.3	Make and Rating	Visual	100%	Tech specifications/approved drawing	TC	P	RR	-	
1.1.4	Performance test	Electrical	100%	Tech specifications/approved drawing	TC	P	RR	-	
	Digital/Analog Metering Equipment								
1.1.5	Dimensional Check , Make and Rating	Measurement/Visual	100%	Tech specifications/approved drawing	TC	P	RR	-	
1.1.6	Class of Accuracy	Visual	100%	Tech specifications/approved drawing	TC	P	RR	-	
1.1.7	Calibration	Electrical	100%	Tech specifications/approved drawing	TC	P	RR	-	
	Cooling Fan								
1.1.8	Make and Rating	Visual	100%	Tech specifications/approved drawing	TC	P	RR	-	

1.1.9	Routine test	Electrical	100%	Tech specifications/approved drawing	TC	P	RR	-	
1.2	In process inspection								
1.2.1	Check of identity	Mechanical	Complete converter	Internal specification	TC	P	RR	-	
1.2.2	Condition test	Mechanical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.2.3	Labeling, marking	Mechanical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.2.4	Check of electrical components	Electrical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.2.5	mechanical construction	Mechanical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.2.6	Screw connections	Mechanical	Complete Converter	Acc. To factory standard	TC	P	RR	-	
1.2.7	Clearance + Creepage distances	Mechanical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.2.8	Cable inspection	Electrical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.3	Internal routine test								
1.3.1	Visual check								
1.3.1.1	Check degree of protection	Mechanical	Complete Converter	IEC 60529	TC	P	RR	-	
1.3.1.2	Check earthing of cubicles & components of cubicles	Electrical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.3.1.3	Check marking of components in acc. With relevant drawings	Electrical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.3.1.4	Check rating plate	Electrical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.3.1.5	Check terminals & wiring for correct marking and diameters	Electrical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.3.1.6	wiring test, point to check	Electrical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.3.2	Insulation test								
1.3.2.1	check clearance and creepage	Electrical	Complete Converter	Acc. To drawings	TC	P	RR	-	
1.3.2.2	AC high voltage test	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	TC	P	RR	-	
1.3.2.3	insulation resistance test	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	TC	P	RR	-	

1.3.2. 4	Impulse voltage test & Partial discharge test as per IEC 60146-1-1(Performed only if agreed upon prior to order)	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	TC	P	RR	-	
1.5	Routine test (function test) control cubicle								
1.5.1	Visual inspection (Dimensions, Bill of material etc)	Electrical	Complete Converter	Acc. To drawings	TC	P	RR	RR	
1.5.2	Checking of Sheet Thickness & paint thickness of cubicle	Mechanical	Complete Cubicle	Acc. To drawings	TC	P	RR	RR	
1.5.3	Insulation resistance test	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.5.4	HV test on Auxiliary & Control Circuit	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.5.1	Check of power supplies	Electrical	Complete Converter	Acc. To drawings	TC	P	RR	RR	
1.5.2	check of modules with self test routine	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.5.3	check of operator panel	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.5.4	check of synchronizing module	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.5.5	check of actual values	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.5.6	check of binary inputs/outputs	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.5.7	Check of interlocking system	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.6	Routine test (function test) Converters & DC link (Internal)								
1.6.1	Check fibre optics	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.6.2	Check the earthing switch and /or disconnector on line side	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.6.3	Check the earthing switch and /or disconnector on motor side	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.6.4	Check door switches	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.6.5	Check current transducer and voltage sensing	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.6.6	voltage distribution in the thyristor valve arms	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.6.7	Functional test of thyristor firing and monitoring	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	

1.6.8	Monitoring and protective functions as far as possible	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.6.9	Converter operation in DC-Link pulsation	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
1.7	Routine test FAT								
1.7.1	Visual inspection	Mechanical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.7.2	Insulation test	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.7.3	Check of auxiliary devices	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.7.4	Check of protective devices	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.7.5	Checking the properties of control equipment	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.7.6	Functional test	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.7.7	Light load test	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
Inspections and tests on Cooling unit									
2	Water cooled units								
2.1	Hydraulic test	Mechanical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
2.2	Pressure difference between converter in/output at rated flow	Mechanical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
2.3	Rated flow rate on module tower at rated pressure difference	Mechanical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
2.4	Rated flow rate on DC link reactor at rated pressure difference	Mechanical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
2.5	Measurement on the PT100	Electrical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
2.6	Static pressure check	Mechanical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
2.7	check leakage detectors	Mechanical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
3	Air cooled units								
3.1	Fan test	Mechanical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
3.2	Check of fans correct rotation direction	Mechanical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
3.3	Check of fans correct current input	Mechanical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	
3.4	Check of differential pressure monitor	Mechanical	Complete Converter	Acc. To factory test plan	TC	P	RR	RR	

Note:									
a. Abbreviations used - RR-Review of Records, TC - Test Certificate Submission, JIR - Joint Inspection Report & CHP - Customer Hold Point.									
b. Test Certificates shall be submitted at the time of final inspection. Valid calibration Certificate of Instruments used for Testing shall be provided during the Test.									
	Signature			Signature & Seal					
	EMPLOYER (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)					

9) (ii) VARIABLE-FREQUENCY DRIVE (VFD)

					ग्राहक (CLIENT):				
	परियोजना (PROJECT) :				विक्रेता (VENDOR):				
उपष्कर का नाम (NAME OF EQUIPMENT) : Variable Frequency Drives					एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :				
Sl. No.	Item / Components & Characteristics	Nature of Checks	Quantum of Checks	Reference Document / Acceptance Norms	Record Format	Inspection Agency			Remarks
						Sub supplier	Supplier	Employer	
1.1	Incoming goods inspection								
	BOUGHT OUT MATERIALS								
	Thyrister/IGBTs , Heat sink, Surge Supression Device, HMI/MMI etc. (As applicable)								
1.1.1	Make and Rating	Visual	100%	Tech specifications/approved drawing	TC	P	RR	-	
1.1.2	Functional test check	Electrical	100%	Tech specifications/approved drawing	TC	P	RR	-	
	Power Supply								
1.1.3	Make and Rating	Visual	100%	Tech specifications/approved drawing	TC	P	RR	-	
1.1.4	Performance test	Electrical	100%	Tech specifications/approved drawing	TC	P	RR	-	
	Digital/Analog Metering Equipment								
1.1.5	Dimensional Check , Make and Rating	Measurement/Visual	100%	Tech specifications/approved drawing	TC	P	RR	-	
1.1.6	Class of Accuracy	Visual	100%	Tech specifications/approved drawing	TC	P	RR	-	
1.1.7	Calibration	Electrical	100%	Tech specifications/approved drawing	TC	P	RR	-	
	Cooling Fan								
1.1.8	Make and Rating	Visual	100%	Tech specifications/approved drawing	TC	P	RR	-	

1.1.9	Routine test	Electrical	100%	Tech specifications/approved drawing	TC	P	RR	-	
1.1	Visual checks								
1.1.1	Rating plate	Mechanical	100%	Acc. To drawings	JIR	P	W	CHP	
1.1.2	Housing degree of protection	Mechanical	1 per lot	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.1.3	Electrical cabinet components	Mechanical	Sample	Acc. To drawings	JIR	P	W	CHP	
1.1.4	Cable installation	Mechanical	Sample	Acc. To drawings	JIR	P	W	CHP	
1.1.5	Cable markings	Mechanical	Sample	Acc. To drawings	JIR	P	W	CHP	
1.1.6	Setting values of circuit breakers	Mechanical	Sample	Acc. To drawings	JIR	P	W	CHP	
1.1.7	Dimensions	Mechanical	100%	Acc. To drawings	JIR	P	W	CHP	
1.2	Functional tests								
1.2.1	Auxiliary voltages	Electrical	Complete converter	Acc. To factory standard	JIR	P	W	CHP	
1.2.2	Interrupted cable to the operator panel	Electrical	Complete Converter		JIR	P	W	CHP	
1.2.3	Temperature monitoring	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.2.4	Door limit switches	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.2.5	Door interlocking system	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.2.6	Ground fault monitoring	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.2.7	DC link precharging	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.2.8	Sensing the actual DC link voltage	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.2.9	Under voltage monitoring in the DC link	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.2.10	Output current measurement.	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.2.11	Overvoltage monitoring in the DC link	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
1.2.12	Failure of a power semi conductor module	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
1.2.13	Safety shutdown	Electrical	Complete Converter	'IS/ IEC/ equivalent or above'	JIR	P	W	CHP	
Inspections and tests on Cooling unit									
2.1	Air flow direction	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	
2.2	Air pressure monitoring	Electrical	Complete Converter	Acc. To factory standard	JIR	P	W	CHP	

Note:									
a. Abbreviations used - RR-Review of Records, TC - Test Certificate Submission, JIR - Joint Inspection Report & CHP - Customer Hold Point.									
b. Test Certificates shall be submitted at the time of final inspection. Valid calibration Certificate of Instruments used for Testing shall be provided during the Test.									
	Signature			Signature & Seal					
	EMPLOYER (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)					

10) CURRENT LIMITING REACTORS

					ग्राहक (CLIENT):				
	परियोजना (PROJECT) :				विक्रेता (VENDOR):				
उपकरण का नाम (NAME OF EQUIPMENT) : Current Limiting Reactor					एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :				
Sl. No.	Item / Components & Characteristics	Nature of Checks	Quantum of Checks	Reference Document / Acceptance Norms	Record Format	Inspection Agency			Remarks
						Perform	Witness	Verify	
A	Raw material								
1	Steel for End frame & Enclpsure sheet								
a)	Chemical composition	Chemical	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
b)	Mechanical test	Mechanical	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
c)	Thickness	Measurement	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
d)	Fittings and Accessories	Visual	Sample	Appd. Spec./drawing	TC	2/3	-	1	TC
2	Conductor								
a)	Dimension	Measurement	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
b)	Tensile strength	Mechanical	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
c)	Hardness	Mechanical	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
d)	Conductivity	Electrical	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
e)	Elongation	Mechanical	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
f)	Chemical composition	Chemical	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
3	Insulation								
a)	Physical properties	Physical	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
b)	Electrical Test	Electrical	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
c)	Insulating oil Testing (oil immersed current limiting reactor only)	Electrical	Sample	Appd. Spec./ 'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
B	Inprocess inspection								
1	Coil Assembly								

a)	No. of turns	Visual	100%	Tech Spec./Appd. Drg/'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
b)	Brazing	Mechanical	100%	Tech Spec./Appd. Drg/'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
c)	Inter turn insulation	Electrical	100%	Tech Spec./Appd. Drg/'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
d)	Continuity	Electrical	100%	Tech Spec./Appd. Drg/'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
e)	End insulation	Electrical	100%	Tech Spec./Appd. Drg/'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
f)	Coil clamping pressure	Measurement	100%	Tech Spec./Appd. Drg/'IS/ IEC/ equivalent or above'	TC	2/3	-	1	TC
C	Routine Tests								
i)	Measurement of winding resistance	Electrical	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
ii)	Measurement of reactance	-do-	100%	-do-	JIR	3/2	1	-	CHP
iii)	Measurement of loss	-do-	100%	-do-	JIR	3/2	1	-	CHP
iv)	Measurement of Insulation resistance	-do-	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above' (To be recorded only for reference)	JIR	3/2	1	-	CHP
v)	Lightning impulse test	-do-	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
vi)	Dimensional checks	Visual	100%	Approved General assembly drawings	JIR	3/2	1	-	CHP
vii)	Separate source ac withstand voltage test (For Liquid immersed reactors).	Electrical	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
viii)	Winding overvoltage test	Electrical	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
ix)	Capacitance and dissipation factor (tan delta) of the winding insulation to earth (For Liquid immersed reactors)	Electrical	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
Note:									
a. In 'Inspection Agency' column figure 1, 2 or 3 to be filled. 1- will indicate 'Owner', 2- will indicate 'Supplier' & 3- will indicate 'Sub-supplier'.									
b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, TC - Test Certificate Submission, JIR - Joint Inspection Report & CHP - Customer Hold Point.									

c. Test Certificates shall be submitted at the time of final inspection. Valid calibration Certificate of Instruments used for Testing shall be provided during the Test.									
	Signature			Signature & Seal					
	EMPLOYER (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)					

11) GENERATOR CIRCUIT BREAKER

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम (NAME OF EQUIPMENT) : Generator Circuit Breaker (GIS)				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
A)	CIRCUIT BREAKER								
	Routine Tests								
1	Pressure Test on Enclosure	Test	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
2	Gas Leakage Test	-do-	-do-	-do-	JIR	2/3	1	-	CHP
3	Auxiliary & Control Circuit								
a)	Wiring Check.	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	2/3	1	-	CHP
b)	HV Test (2kV for One Minute)	Electrical	-do-	-do-	JIR	2/3	1	-	CHP
c)	Insulation Resistance Measurement	-do-	-do-	-do-	JIR	2/3	1	-	CHP
4	Mechanical Operating Test								
a)	5 Open & 5 Close Operations at Minimum Supply Voltage and Minimum Pressure	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	2/3	1	-	CHP
b)	5 Open & 5 Close Operations at Maximum Supply Voltage and Maximum Pressure	-do-	-do-	-do-	JIR	2/3	1	-	CHP
c)	5 Close-Open Operating Cycles at Rated Supply Voltage and Rated Pressure	-do-	-do-	-do-	JIR	2/3	1	-	CHP
d)	Measurement of Opening/Closing Time	Measurement	-do-	-do-	JIR	2/3	1	-	CHP
5	Electrical Tests								
a)	Power Frequency Voltage Test of the Main Circuit.	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	2/3	1	-	CHP

b)	Measurement of Contact Resistance of Main Circuit	Measurement	-do-	-do-	JIR	2/3	1	-	CHP
c)	Measurement of Resistance of Circuit Breaker Closing and Trip Coils	-do-	-do-	-do-	JIR	2/3	1	-	CHP
d)	Partial Discharge Measurement	-do-	-do-	-do-	JIR	2/3	1	-	CHP
e)	Measurement of Power Consumption of Motor Operated Mechanism at Rated Supply Voltage	-do-	-do-	-do-	JIR	2/3	1	-	CHP
f)	Operational & Interlocks Check.	Functional	-do-	-do-	JIR	2/3	1	-	CHP
g)	Operational Check of Pressure Density Monitoring Switches	-do-	-do-	-do-	JIR	2/3	1	-	CHP
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f. The type test may not be mandatory if similar equipment has been type tested and test certificate(s) for relevant tests are accepted by the Employer								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN):									
परियोजना (PROJECT):					ग्राहक (CLIENT):				
उपकरण का नाम (NAME OF EQUIPMENT): Generator Circuit Breaker(GIS)					विक्रेता (VENDOR):				
एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference):									
क्र.	घटक की गतिविधि और संचालन ACTIVITY & OPERATION OF COMPONENT	जाँच की	जाँच की	संदर्भ दस्तावेज़ /	रेकॉर्ड	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks

सं. S.N.		प्रकृति NATURE OF CHECKS	मात्रा QUANTUM OF CHECKS	स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	फारमैट RECORD FORMAT	प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
A)	General Check								
a)	Equipment should be free from dirt/dust/rust/foreign material etc.	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	2/3	1	-	TC/TR
b)	All nut and bolts should be tightened correctly as per specified torque	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
c)	Leveling and alignment of structure and base frame should be checked	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
d)	Earthing and wiring connection should be completed	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
e)	All trenches should be cleaned and covered	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
B)	Commissioning check								
a)	SF6 Gas Quality test and checking of pressure monitoring system	Test	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	2/3	1	-	TC/TR
b)	Gas Leakage Test	Test	-do-	-do-	TC/TR	2/3	1	-	TC/TR
c)	Measurement of operating time	Test	-do-	-do-	TC/TR	2/3	1	-	TC/TR
d)	Contact Resistance Measurement	Test	-do-	-do-	TC/TR	2/3	1	-	TC/TR
e)	Insulation resistance measurement test of motor	Electrical	-do-	-do-	TC/TR	2/3	1	-	TC/TR
f)	Measurement of coil resistance	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
g)	Breaker operation counter reading	Visual	-do-	-do-	TC/TR	2/3	1	-	TC/TR
h)	Auxiliary & Control Circuit								
i)	Wiring Check.	Electrical	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	2/3	1	-	TC/TR
ii)	HV Test (2kV for One Minute)	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
iii)	Insulation resistance Measurement	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate Representative of Engineer-in-charge, 2- will indicate 'Contractor' & 3- will indicate 'Sub-contractor'.								
	b. In 'Remarks' column following abbreviations shall be used - TC/TR - Test Certificate/ Test Report								
	c. The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column " Perform" .								
	d. Any additional activity/ test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any								

	additional financial implication.					
	e. The contractor shall furnish the details of test method to the Engineer-in-Charge for approval in the time line as specified in contract agreement					
	f. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.					
	Signature					Signature & Seal
	Engineer-in-charge				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)	

12) PHASE REVERSAL DISCONNECTING SWITCH

					ग्राहक (CLIENT):				
	परियोजना (PROJECT) :				विक्रेता (VENDOR):				
उपष्कर का नाम (NAME OF EQUIPMENT) :Phase reversal Device					एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :				
Sl. No.	Item / Components & Characteristics	Nature of Checks	Quantum of Checks	Reference Document / Acceptance Norms	Record Format	Inspection Agency			Remarks
						Perform	Witness	Verify	
	RAW MATERIAL								
A	Aluminium casings								
1	Material test certificates	Mechanical	100%	Supplier report	TC	3	---	1/2	
2	Traceability markings	Mechanical	100%	End of Manufacturing report	TC	3	---	1/2	
	BOUGHT OUT MATERIAL								
B	Ceramic or composite insulator								
1	Bending test	Mechanical	100%		TC	3	---	1/2	
2	Visual examination / Dimensional check	Mechanical	100%		TC	3	---	1/2	
	Only for composite insulator				TC	3	---	1/2	
3	Tightness test	Mechanical	100%		TC	3	---	1/2	
4	Partial discharge measurement	Electrical test	100%		TC	3	---	1/2	
C	Control Cubicle								
i)	Visual and dimensional check	Visual	-do-	'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
ii)	Paint shade, thickness and adhesion Test	-do-	-do-	-do-	JIR	3/2	1	-	CHP

iii)	Dielectric tests on auxiliary circuits	Electrical	-do-	-do-	JIR	3/2	1	-	CHP
iv)	Insulation resistance test	-do-	-do-	-do-	JIR	3/2	1	-	CHP
v)	Wiring continuity Test	Electrical	-do-	-do-	JIR	3/2	1	-	CHP
vi)	Interlock and sequential operation	-do-	-do-	-do-	JIR	3/2	1	-	CHP
vii)	Check of interface with plant SCADA	-do-	-do-	-do-	JIR	3/2	1	-	CHP
vii)	Mechanical and electrical operation test	-do-	-do-	-do-	JIR	3/2	1	-	CHP
viii)	Checking of the number of auxiliary contact	Visual	-do-	-do-	JIR	3/2	1	-	CHP
ix)	Varification visual of protection against electrical shock	Visual	-do-	-do-	JIR	3/2	1	-	CHP
x)	Functioning verification of heaters	Electrical	-do-	-do-	JIR	3/2	1	-	CHP
xi)	Functioning verification of lighting	-do-	-do-	-do-	JIR	3/2	1	-	CHP
D	PRDS complete test								
	Mechanical operation test					3/2	1	-	CHP
1	- at maximum control voltage and springs fully charged	Mechanical	100%	'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
2	- at minimum control voltage and springs in lockout condition	Mechanical	100%	do	JIR	3/2	1	-	CHP
3	- at minimum control voltage and springs fully charged	Mechanical	100%	do	JIR	3/2	1	-	CHP
4	- at rated control voltage and springs fully charged	Mechanical	100%	do	JIR	3/2	1	-	CHP
5	- at rated control voltage and springs fully charged	Mechanical	100%	do	JIR	3/2	1	-	CHP
6	Drive operating sequence check	Mechanical	100%	do	JIR	3/2	1	-	CHP
E	Measurement of operating times at rated conditions								
1	Motor Operation tests								
a	At rated supply voltage	Electrical test	100%	IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP

b	measurement of the closing time and motor consumption	Electrical test	100%	'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
c	measurement of the opening time and motor consumption	Electrical test	100%	'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
2	Resistance measurement of the main circuit	Electrical test	100%	do	JIR	3/2	1	-	CHP
3	Power frequency withstand voltage test to earth and across open contacts	Electrical test	100%	do	JIR	3/2	1	-	CHP
F	Voltage Transformers								
1	Visual and dimensional check	Visual	-do-	Appd. Spec.	JIR	3/2	1	-	CHP
2	Verification of markings	Visual	-do-	Appd. Spec.	JIR	3/2	1	-	CHP
3	Dielectric tests on secondary circuits	Electrical	-do-	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
4	Partial discharge measurement	Electrical	-do-	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
5	Test for Accuracy	Electrical	-do-	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	3/2	1	-	CHP
Note:									
a. In 'Inspection Agency' column figure 1, 2 or 3 to be filled. 1- will indicate 'Owner', 2- will indicate 'Supplier' & 3- will indicate 'Sub-supplier'.									
b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, TC - Test Certificate Submission, JIR - Joint Inspection Report & CHP - Customer Hold Point.									
c. Test Certificates shall be submitted at the time of final inspection. Valid calibration Certificate of Instruments used for Testing shall be provided during the Test.									
	Signature			Signature & Seal					
	EMPLOYER (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)					

13) (i) DYNAMIC BRAKING CUBICLE

					ग्राहक (CLIENT):				
	परियोजना (PROJECT) :				विक्रेता (VENDOR):				
उपष्कर का नाम (NAME OF EQUIPMENT) :Dynamic Braking Equipment					एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :				
Sl. No.	Item / Components & Characteristics	Nature of Checks	Quantum of Checks	Reference Document / Acceptance Norms	Record Format	Inspection Agency			Remarks
						Perform	Witness	Verify	
	RAW MATERIAL								
A	Aluminium casings								
1	Material test certificates	Mechanical	100%	Supplier report	TC	3	---	1/2	
2	Traceability markings	Mechanical	100%	End of Manufacturing report	TC	3	---	1/2	
	BOUGHT OUT MATERIAL								
B	Ceramic or composite insulator								
1	Bending test	Mechanical	100%		TC	3	---	1/2	
2	Visual examination / Dimensional check	Mechanical	100%		TC	3	---	1/2	
	Only for composite insulator				TC	3	---	1/2	
3	Tightness test	Mechanical	100%		TC	3	---	1/2	
4	Partial discharge measurement	Electrical test	100%		TC	3	---	1/2	
1	Test for Assembled Panel.								
	Routine Test								
a)	BOM & Rating Check	Visual	100%	Tech. Spec./Appd.drg./'IS/ IEC/ equivalent or above'	JIR	3-2	1	-	CHP
b)	I.R Test (before & after HV Test)	-do-	-do-	Tech. Spec./Appd Drg/ 'IS/ IEC/ equivalent or above'	JIR	3-2	1	-	CHP
c)	H.V. Test	-do-	-do-	-do-	JIR	3-2	1	-	CHP

d)	Dimensional Check	Measurement	-do-	-do-	JIR	3-2	1	-	CHP
e)	Electric operation of the isolator motor								
f)	Functional/Operational Check for all items	Elect/Mech.	-do-	-do-	JIR	3-2	1	-	CHP
g)	Continuity Check	Electrical	-do-	-do-	JIR	3-2	1	-	CHP
h)	Scheme Checking & Interlocks Testing	-do-	-do-	-do-	JIR	3-2	1	-	CHP
i)	Checking of Earthing circuit.	Electrical	100%	Tech. Spec./Appd.drg./'IS/ IEC/ equivalent or above'	JIR	3-2	1	-	CHP
j)	Checking for clearances	Visual	100%	Tech. Spec./Appd.drg./'IS/ IEC/ equivalent or above'	JIR	3-2	1	-	CHP
k)	Busduct Interface dimensional checks	Visual	100%	Tech. Spec./Appd.drg./'IS/ IEC/ equivalent or above'	JIR	3-2	1	-	CHP
2	Verification of providing interfacing with SCADA System	Visual	100%	-do-	JIR	3/2	1	-	CHP
Note:									
a. In 'Inspection Agency' column figure 1, 2 or 3 to be filled. 1- will indicate 'Owner', 2- will indicate 'Supplier' & 3- will indicate 'Sub-supplier'.									
b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, TC - Test Certificate Submission, JIR - Joint Inspection Report & CHP - Customer Hold Point.									
c. Test Certificates shall be submitted at the time of final inspection. Valid calibration Certificate of Instruments used for Testing shall be provided during the Test.									
	Signature			Signature & Seal					
	EMPLOYER (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)					

13) (ii) STARTING SWITCH

-					ग्राहक (CLIENT):	-	-	-	
	परियोजना (PROJECT) :				विक्रेता (VENDOR):				
उपकरण का नाम (NAME OF EQUIPMENT) : Starting Switch					एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :				
Sl. No.	Item / Components & Characteristics	Nature of Checks	Quantum of Checks	Reference Document / Acceptance Norms	Record Format	Inspection Agency			Remarks
						Sub supplier	Supplier	Employer	
	RAW MATERIAL								
A	Aluminium casings (Pressurized with SF6 gas- If applicable)								
	Foundry parts under pressure								
1	Pressure & Tightness test ("HP" punch) by supplier / Traceability markings	Mechanical	100%	Supplier report	TC	P	RR	-	
	Corroded parts under pressure								
2	Traceability markings	Mechanical	100%	End of Manufacturing report	TC	P	RR	-	
	BOUGHT OUT MATERIAL								
B	Ceramic or composite insulator (pressurized with SF6 gas- If applicable)								
1	Pressure test by supplier ("P" punch)	Mechanical	100%	Supplier report	TC	P	RR	-	
2	Bending test ("F" punch)	Mechanical	100%		TC	P	RR	-	
3	Visual examination / Dimensional check	Mechanical	100%		TC	P	RR	-	
	Only for composite insulator								
4	Tightness test ("H" punch)	Mechanical	100%		TC	P	RR	-	
5	Partial discharge measurement ("D" punch)	Electrical test	100%		TC	P	RR	-	
C	Insulated Rod								
	By supplier								

1	General aspect / Dimensional control / Tesile test ("T" punch)	Mechanical	100%	Supplier report	TC	P	RR	-	
	In factory								
2	Power frequency voltage withstand test ("D" punch)	Electrical test	100%		TC	P	RR	-	
3	Partial discharge measurement ("D" punch)	Electrical test	100%		TC	P	RR	-	
D	Spring operating mechanism								
	By supplier			Supplier report					
1	Spring energy adjustments & Operational tests	Mechanical	100%		TC	P	RR	-	
2	Records of operating times (at rated, minimum and maximum voltage)	Mechanical							
	In factory								
3	Test report verification		100%		TC	P	RR	-	
4	Adjustments and conformity control versus electric diagram during routine tests	Mechanical	100%		TC	P	RR	-	
E	LV control cubicle								
	Auxiliary & Control Circuit								
1	HV Test	Electrical test	100%	Supplier report	TC	P	RR	-	
2	Insulation Resistance Measurement	Electrical test	100%		TC	P	RR	-	
3	Scheme Check (Interlock & operational)	Electrical test	100%		TC	P	RR	-	
	By supplier								
1	Conformity control versus electric diagram	Mechanical	100%		TC	P	RR	-	
	In factory								
2	Adjustments and conformity control versus electric diagram during routine tests	Mechanical	100%		TC	P	RR	-	
	Factory Acceptance tests								
A	Drive mechanism test								
1	Voltage test of aux. circuits	Electrical test	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	P	W	W	CHP
2	Performance test of pilot valves	Mechanical	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	P	W	W	CHP
3	Mechanical operation test	Mechanical	100%	do	JIR	P	W	W	CHP
4	Performance test of pump motor	Electrical test	100%	do	JIR	P	W	W	CHP

5	Adjustment control gate and check of overpressure relieve device	Mechanical	100%	do	JIR	P	W	W	CHP
6	Check of auxiliary switch & position indicators	Electrical test	100%	do	JIR	P	W	W	CHP
7	Leak test (spring travel without operation)	Mechanical	100%	do	JIR	P	W	W	CHP
8	Pressure test	Mechanical	100%	do	JIR	P	W	W	CHP
	Circuit breaker complete test								
1	Identification of drive mechanism	Mechanical	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	P	W	W	CHP
B	Mechanical operation test								
1	- at maximum control voltage and springs fully charged	Mechanical	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	P	W	W	CHP
2	- at minimum control voltage and springs in lockout condition	Mechanical	100%	do	JIR	P	W	W	CHP
3	- at minimum control voltage and springs fully charged	Mechanical	100%	do	JIR	P	W	W	CHP
4	- at rated control voltage and springs fully charged	Mechanical	100%	do	JIR	P	W	W	CHP
5	- at rated control voltage and springs fully charged	Mechanical	100%	do	JIR	P	W	W	CHP
6	Totally 50 operations	Mechanical	100%	do	JIR	P	W	W	CHP
7	Drive operating sequence check	Mechanical	100%	do	JIR	P	W	W	CHP
C	Measurement of operating times at rated conditions								
1	Contacts to close	Mechanical	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	P	W	W	CHP
2	Contacts to open - trip circuit 1	Mechanical	100%	do	JIR	P	W	W	CHP
3	Contacts to open - trip circuit 2	Mechanical	100%	do	JIR	P	W	W	CHP
4	Close-open time	Mechanical	100%	do	JIR	P	W	W	CHP
5	SF6-Gas tightness test loss rate	Mechanical	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	P	W	W	CHP
6	Resistance measurement of the main circuit	Electrical test	100%	do	JIR	P	W	W	CHP

7	Resistance measurements of operating coils and heater	Electrical test	100%	Appd. Spec./ 'IS/ IEC/ equivalent or above'	JIR	P	W	W	CHP
8	Power frequency withstand voltage test to earth and across open contacts	Electrical test	100%	Appd. Spec./'IS/ IEC/ equivalent or above'	JIR	P	W	W	CHP
Note:									
a. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, TC - Test Certificate Submission, JIR - Joint Inspection Report & CHP - Customer Hold Point.									
b. Test Certificates shall be submitted at the time of final inspection. Valid calibration Certificate of Instruments used for Testing shall be provided during the Test.									
	Signature			Signature & Seal					
	EMPLOYER (QA&I DEPT.)			(VENDORS Q.C. DEPT. OR REPRESENTATIVE)					

14) EHV XLPE CABLE AND TERMINATION BUSHINGS

परियोजना (PROJECT) :				ग्राहक (CLIENT):					
उपकरण का नाम नाम (NAME OF EQUIPMENT) : EHV XLPE Cable				विक्रेता (VENDOR):					
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					
क्र. सं. S.N.	मद/घटक एवं विशेषता ITEM/ COMPONENT & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
1	Raw Material								
1.1	Copper conductor								
a)	Diameter	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Surface Condition & Conductivity	Visual/ Electrical	Sample	-do-	TC	2/3	-	1	TC
c)	Tensile strength (T.S) and Elongation at break(E.B.)	Mechanical	-do-	-do-	TC	2/3	-	1	TC
d)	Chemical Composition	Chemical	-do-	-do-	TC	2/3	-	1	TC
1.2	Insulating Paper (If Applicable)								
a)	Thickness	Measurement	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Tensile Strength	Mechanical	-do-	-do-	TC	2/3	-	1	TC
c)	Elongation	-do-	-do-	-do-	TC	2/3	-	1	TC
1.3	Internal & External Semiconductive shield								
a)	Check of compound grade	Verification	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Volume Resistivity	Electrical	-do-	-do-	TC	2/3	-	1	TC
c)	Specific gravity/ Density test	Physical	-do-	-do-	TC	2/3	-	1	TC

1.4	XLPE Compound								
a)	Tensile strength & Elongation	Mechanical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Permittivity	Electrical	-do-	-do-	TC	2/3	-	1	TC
c)	Dielectric loss angle (Tan delta)	Electrical	-do-	-do-	TC	2/3	-	1	TC
1.5	Aluminium Tape / Cu Tape								
a)	Dimension	Measurement	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	resistivity	Electrical	-do-	-do-	TC	2/3	-	1	TC
c)	Chemical purity	Chemical	-do-	-do-	TC	2/3	-	1	TC
d)	TS & EB, Annealing test (Cu only)	Mechanical	-do-	-do-	TC	2/3	-	1	TC
1.6	Sheathing compound								
a)	Check for compound Grade	Visual	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Hardness Shore (If applicable)	Mechanical	-do-	-do-	TC	2/3	-	1	TC
c)	Tensile strength & Elongation, Ageing test	Physical	-do-	-do-	TC	2/3	-	1	TC
d)	Specific gravity/ Density test	Physical	-do-	-do-	TC	2/3	-	1	TC
e)	Oxygen Index and Temp index	Physical	-do-	-do-	TC	2/3	-	1	TC
f)	Melt Index test	Physical	-do-	-do-	TC	2/3	-	1	TC
1.7	Aluminium Wires (if applicable)								
a)	Surface Condition	Visual	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Mechanical Strength	Mechanical	-do-	-do-	TC	2/3	-	1	TC
c)	Elongation	Mechanical	-do-	-do-	TC	2/3	-	1	TC
d)	Diameter	Measurement	-do-	-do-	TC	2/3	-	1	TC
1.8	Semi Conductive/ Non semiconducting tape								
a)	Dimension	Measurement	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Swellable height and speed	Measurement	-do-	-do-	TC	2/3	-	1	TC
c)	Tensile Strength & Elongation	Mechanical	-do-	-do-	TC	2/3	-	1	TC
1.9	Online PD Monitoring System (If applicable)	Test	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
1.10	Cable Support system (Fixure, hardware etc.)	Physical	1 sample per lot/size	-do-	TC	2/3	-	1	TC

1.11	MVW Spray & LHS cable (if applicable)	Test	-do-	-do-	TC	2/3	-	1	TC
2	Electrical Type Test on Complete Cable Systems								
a)	Bending Test	Physical	1 sample per lot/size	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Tan Delta measurement	Electrical	-do-	-do-	TC	2/3	-	1	TC
c)	Heating Cycle Voltage Test	Thermal	-do-	-do-	TC	2/3	-	1	TC
d)	Partial Discharge Tests	Electrical	-do-	-do-	TC	2/3	-	1	TC
e)	Switching Impulse Voltage Test ($U_m \geq 300kV$)	-do-	-do-	-do-	TC	2/3	-	1	TC
f)	Lightning Impulse Voltage Test followed power frequency voltage test	-do-	-do-	-do-	TC	2/3	-	1	TC
g)	Tests for outer protections for joints	Physical	-do-	-do-	TC	2/3	-	1	TC
3	Non-Electrical Type Test								
3.1	XLPE insulation and Sheath								
a)	Thickness eccentricity of Insulation and Outer sheath	Measurement	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Tensile & Elongation of XLPE and Outersheath (before & after ageing)	Mechanical	-do-	-do-	TC	2/3	-	1	TC
c)	Loss of mass test on PVC oversheaths of type ST2	Physical	-do-	-do-	TC	2/3	-	1	TC
d)	Pressure test at high temperature on oversheaths	Physical	-do-	-do-	TC	2/3	-	1	TC
e)	Ageing Test	Physical	-do-	-do-	TC	2/3	-	1	TC
f)	Heat shock test for PVC oversheaths (ST1 and ST2)	Thermal	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
g)	Hot set test for XLPE insulations	Thermal	-do-	-do-	TC	2/3	-	1	TC
h)									
	Measurement of carbon black content of black PE oversheaths (ST3 and ST7)	Physical	-do-	-do-	TC	2/3	-	1	TC
i)	Water penetration test	-do-	-do-	-do-	TC	2/3	-	1	TC
j)	Tests on components of cables with a longitudinally applied metal tape or foil, bonded to the oversheath	-do-	-do-	-do-	TC	2/3	-	1	TC
k)	Shrinkage test of Insulation & Outer Sheath	-do-	-do-	-do-	TC	2/3	-	1	TC
l)	Hot deformation of Outer sheath	Thermal	-do-	-do-	TC	2/3	-	1	TC
3.2	Armouring material								

a)	Dimensions	Measurement	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Tensile & Elongation	Mechanical	-do-	-do-	TC	2/3	-	1	TC
c)	Wrapping Test	Physical	-do-	-do-	TC	2/3	-	1	TC
d)	Resistivity Test	Physical	-do-	-do-	TC	2/3	-	1	TC
3.3	Metallic Screen								
a)	Test for concentric copper wire	Electrical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Test for concentric copper tape	Measurement	-do-	-do-	TC	2/3	-	1	TC
c)	Semi-conducting Screen resistivity test	-do-	-do-	-do-	TC	2/3	-	1	TC
4	Routine Tests								
a)	Conductor Resistance Test	Electrical	100% Drum	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	High Voltage Test	-do-	-do-	-do-	TC	2/3	-	1	TC
c)	Partial discharge test	-do-	-do-	-do-	TC	2/3	-	1	TC
5	Acceptance Test								
a)	Conductor Examination (no.of strands, lay direction, lay length, dimension)	Measurement	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	2/3	1	-	CHP
b)	Electrical Resistance of Conductor and of Mettalic Sheath/Screen	Electrical	-do-	-do-	JIR	2/3	1	-	CHP
c)	Measurement of Thickness of Insulation and Oversheath	Measurement	-do-	-do-	JIR	2/3	1	-	CHP
d)	Measurement of Thickness of Metallic Sheath	-do-	-do-	-do-	JIR	2/3	1	-	CHP
e)	Measurement of Diameter of core and overall diameter of cable	-do-	-do-	-do-	JIR	2/3	1	-	CHP
f)	Hot Set Test and permanent set test for XLPE Insulations	Thermal	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	2/3	1	-	CHP
g)	Measurement of Capacitance	-do-	-do-	-do-	JIR	2/3	1	-	CHP
h)	Specific gravity/ density test of Outer Sheath	Physical	-do-	-do-	JIR	2/3	1	-	CHP
i)	Lightning Impulse Voltage Test	Electrical	-do-	-do-	JIR	2/3	1	-	CHP
j)	Water Penetration Test			-do-					
k)	IR & HV Test	Electrical	-do-	-do-	JIR	2/3	1	-	CHP
l)	Partial discharge test	Electrical	-do-	-do-	JIR	2/3	1	-	CHP

m)	Test on components of cables with a longitudinally applied metal tape or foil, bonded to the oversheath		-do-	-do-	JIR	2/3	1	-	CHP
n)	Void and contamination test	Measurement	-do-	-do-	JIR	2/3	1	-	CHP
6	Additional test (As applicable)								
a)	HCL Gas Generation Test	Chem.	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	2/3	1	-	CHP
b)	Oxygen Index Test	Thermal	-do-	-do-	JIR	2/3	1	-	CHP
c)	Temperature Index Test	-do-	-do-	-do-	JIR	2/3	1	-	CHP
d)	Smoke Density Test	-do-	-do-	-do-	JIR	2/3	1	-	CHP
e)	Flammability Test	Flam.	-do-	-do-	JIR	2/3	1	-	CHP
f)	Ladder Test	-do-	-do-	-do-	JIR	2/3	1	-	CHP
g)	Anti Rodent & Termite Test	Chem.	-do-	-do-	JIR	2/3	1	-	CHP
h)	Cold Impact test	Physical	-do-	-do-	JIR	2/3	1	-	CHP
7	Check for Embossing & Colour Coding etc.	Visual	-do-	-do-	JIR	2/3	1	-	CHP
8	Packing, Marking & End Sealing								
a)	Marking on Cable Drum	Visual	Each Drum	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	2/3	1	-	CHP
b)	End Sealing	-do-	-do-	-do-	JIR	2/3	1	-	CHP
9	Oil Immersed Sealing End								
a)	Appearance & Dimensional Check.	Measurement/ Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Dielectric Tests.	Electrical	-do-	-do-	TC	2/3	-	1	TC
c)	Pressure Leak Test.	-do-	-do-	-do-	TC	2/3	-	1	TC
d)	Partial Discharge Test.	-do-	-do-	-do-	TC	2/3	-	1	TC
10	SF6 Sealing End								
a)	Appearance & Dimensional check.	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Dielectric Tests.	Electrical	-do-	-do-	TC	2/3	-	1	TC
c)	Pressure Leak Test.	-do-	-do-	-do-	TC	2/3	-	1	TC
d)	Partial Discharge Test.	-do-	-do-	-do-	TC	2/3	-	1	TC
11	Pothole Sealing End (if applicable)								
a)	Appearance & Dimensional check.	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC

b)	Dielectric Tests.	Electrical	-do-	-do-	TC	2/3	-	1	TC
c)	Pressure Leak Test.	-do-	-do-	-do-	TC	2/3	-	1	TC
d)	Partial Discharge Test.	-do-	-do-	-do-	TC	2/3	-	1	TC
12	Joints								
a)	Appearance & Dimensional Check.	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Dielectric Tests.	Electrical	-do-	-do-	TC	2/3	-	1	TC
c)	Pressure Leak Test.	-do-	-do-	-do-	TC	2/3	-	1	TC
d)	Partial Discharge Test.	-do-	-do-	-do-	TC	2/3	-	1	TC
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e. This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then "3/2" shall be considered as "2" under the column "Perform" .								
	f. The type test may not be mandatory if similar equipment has been type tested and test certificate(s) for relevant tests are accepted by the Employer								
	Signature					Signature & Seal			
	EMPLOYER (QA&I Deptt.)					(Vendors Q.C Dept or Representative)			

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN):									
परियोजना (PROJECT):					ग्राहक (CLIENT):				
उपकरण का नाम (NAME OF EQUIPMENT): EHV XLPE Cable					विक्रेता (VENDOR):				
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference):					

क्र. सं. S.N.	घटक की गतिविधि और संचालन ACTIVITY & OPERATION OF COMPONENT	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESSES	सत्यापन VERIFY	
A)	General Checks								
a)	Visual Inspection	Visual	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Check fire detection/protection circuit	Electrical	-do-	-do-	TC/TR	3/2	1	-	TC/TR
B)	XLPE Cable								
a)	Continuity Test	Electrical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	3/2	1	-	TC/TR
b)	Conductor Resistance checking.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
c)	DC Voltage test of the oversheath	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
d)	Dielectric Test	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
e)	Insulation resistance measurement.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
f)	Verification of phase order.	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
Note :	a. In ‘Inspection Agency’ column figure 1,2,or 3 to be filled. 1- will indicate Representative of Engineer-in-charge, 2- will indicate ‘Contractor’ & 3- will indicate ‘Sub-contractor’.								
	b. In ‘Remarks’ column following abbreviations shall be used - TC/TR - Test Certificate/Test Report								
	c. The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	d. Any additional acitivity/ test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.								
	e. The contractor shall furnish the details of test method to the Engineer-in-Charge for approval in the time line as specified in contract agreement								
	f. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.								
	Signature					Signature & Seal			
	Engineer-in-charge					(Vendors Q.C Dept or Representative)			

परियोजना (PROJECT) :	ग्राहक (CLIENT):			
उपकरण का नाम (NAME OF EQUIPMENT) : Bushing	विक्रेता (VENDOR):			

					एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :				
क्रम स. (SR. NO.)	मद/घटक एवं विशेषता ITEM /COMPONENTS & CHARACTERISTICS	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज / स्वीकृति मानदंड REFERENCE ACCEPTANCE DOCUMENTS NORMS	रिकॉर्ड फॉर्मेट RECORD FORMAT	निरीक्षण एजेंसी INSP. AGENCY			टिपणी REMARKS
						प्रदर्शन Perform	गवाह Witness	सत्यापन Verify	
1	Raw Material & Bought-out Items								
1.1	Conductor								
a)	Dimension	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Mechanical Properties	Mechanical	Sample	-do-	TC	3/2	-	1	TC
c)	Chemical Composition (incl. Al Percent.)	Chemical	-do-	-do-	TC	3/2	-	1	TC
d)	Conductivity	Electrical	-do-	-do-	TC	3/2	-	1	TC
1.2	Condenser Tissue Paper								
	Visual, Dimensions, PH Value, Density, Dielectric Strength, Mineral Ash Content, Chloride Content, & Dielectric Loss Angle.	Test	One Sample per lot	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
1.3	Porcelain Housing								
a)	Dimension	Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Physical properties	Visual	-do-	-do-	TC	2/3	-	1	TC
c)	Routine Test	-do-	100%	-do-	TC	2/3	-	1	TC
1.4	Terminal Assembly								
a)	Dimension / Fitment	Measurement	-do-	-do-	TC	2/3	-	1	TC
b)	Physical (Galvanising or Tinning)	Chemical	-do-	-do-	TC	2/3	-	1	TC
1.5	Mounting Structure (Steel, Plates etc.), if any								
a)	Physical Properties (Ultimate Strength, Yield Stress, % Elongation & Bend Test)	Mechanical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Chemical Composition	Chemical	-do-	-do-	TC	2/3	-	1	TC
2	IN PROCESS INSPECTION								

(a)	Visual & Dimensional	Visual & Measurement	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	2/3	-	1	TC
b)	Electrical Routine Test	Electrical	-do-	-do-	TC	2/3	-	1	TC
3	FINAL INSPECTION								
3.1	Type Test								
a)	long duration power-frequency voltage withstand test	Electrical	one unit	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC	3/2	-	1	TC
b)	Dry lightning impulse voltage withstand test	-do-	-do-	-do-	TC	3/2	-	1	TC
c)	Dry or wet switching impulse voltage withstand test	-do-	-do-	-do-	TC	3/2	-	1	TC
d)	Electromagnetic compatibility	-do-	-do-	-do-	TC	3/2	-	1	TC
e)	Verification of thermal short-time current withstand	-do-	-do-	-do-	TC	3/2	-	1	TC
f)	Temperature rise test	-do-	-do-	-do-	TC	3/2	-	1	TC
g)	Cantilever load withstand test	Physical	-do-	-do-	TC	3/2	-	1	TC
h)	Tightness test on liquid-filled, compound-filled and liquid-insulated bushings	-do-	-do-	-do-	TC	3/2	-	1	TC
i)	Internal pressure test on gas-filled, gas-insulated and gas-impregnated bushings	-do-	-do-	-do-	TC	3/2	-	1	TC
j)	External pressure test on partly or completely gas-immersed bushings	-do-	-do-	-do-	TC	3/2	-	1	TC
k)	Verification of dimensions	Measurement	-do-	-do-	TC	3/2	-	1	TC
3.2	Routine Test								
a)	Measurement of dielectric dissipation factor ($\tan \delta$) and capacitance at ambient temperature	Electrical	Sample	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
b)	Dry lightning impulse voltage withstand test	-do-	-do-	-do-	JIR	3/2	1	-	CHP
c)	Dry power-frequency voltage withstand test	-do-	-do-	-do-	JIR	3/2	1	-	CHP
d)	Measurement of partial discharge quantity	-do-	-do-	-do-	JIR	3/2	1	-	CHP
e)	Tests of tap insulation	-do-	-do-	-do-	JIR	3/2	1	-	CHP
f)	Internal pressure test of gas-filled, gas-insulated and gas-impregnated bushings	Measurement	-do-	-do-	JIR	3/2	1	-	CHP

g)	Tightness test on liquid-filled, compound-filled and liquid-insulated bushings	Physical	-do-	-do-	JIR	3/2	1	-	CHP
h)	Tightness test on gas-filled, gas-insulated and gas-impregnated bushings	-do-	-do-	-do-	JIR	3/2	1	-	CHP
i)	Tightness test at the flange or other fixing device	-do-	-do-	-do-	JIR	3/2	1	-	CHP
j)	Visual inspection and dimensional check	Visual & Measurement	-do-	-do-	JIR	3/2	1	-	CHP
3.3	Special tests								
a)	Seismic test	Test	one unit	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	JIR	3/2	1	-	CHP
Note:	a. In 'Inspection Agency' column figure 1,2,or 3 to be filled. 1- will indicate 'Employer', 2- will indicate 'supplier' & 3- will indicate 'sub-supplier'.								
	b. In 'Remarks' column following abbreviations shall be used - RR-Review of Records, T.C. - Test Certificate, JIR - Joint Inspection Report & CHP - Customer Hold Point,								
	c. The firm shall obtain acceptance/clearance of finished product for further activities based on Test Certificates & Internal Report (IR)								
	d. Any additional test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from QAP will be carried out by the firm without any additional financial implication.								
	e.This QAP does not absolve the Contractor/sub-contractor of his responsibility to supply the correct products, strictly in conformity to the specifications given in the purchase order/contract. The above Model QAP stipulates the bare minimum requirements which shall be complied by the contractor. The above QAP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	f. The type test may not be mandatory if similar equipment has been type tested and test certificate(s) for relevant tests are accepted by the Employer								
	Signature					Signature & Seal			
	EMPLOYER (QA&I DEPT.)				(VENDORS Q.C. DEPT. OR REPRESENTATIVE)				

कार्य स्थल गुणवत्ता योजना (FIELD QUALITY PLAN) :									
परियोजना (PROJECT) :					ग्राहक (CLIENT):				
उपकरण का नाम (NAME OF EQUIPMENT) : Bushing					विक्रेता (VENDOR):				
				एनआईटी/ क्रय आदेश संदर्भ (NIT / P.O. Reference) :					

क्र.सं. S.N.	घटक की गतिविधि और संचालन ACTIVITY & OPERATION OF COMPONENT	जाँच की प्रकृति NATURE OF CHECKS	जाँच की मात्रा QUANTUM OF CHECKS	संदर्भ दस्तावेज़ / स्वीकृति मानदंड REFERENCE DOCUMENT/ ACCEPTANCE NORMS	रेकॉर्ड फ़ारमैट RECORD FORMAT	निरीक्षण एजेंसी Inspecting Agency			टिप्पणी Remarks
						प्रदर्शन PERFORM	गवाह WITNESS	सत्यापन VERIFY	
A)	General Checks								
a)	Equipment should be free from dirt/dust/rust/foreign material etc.	Visual	100%	Tech spec./ Appd Drg./ IS/IEC/Equivalent Standard	TC/TR	2/3	1	-	TC/TR
b)	All nut and bolts should be tightened correctly as per specified torque	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
c)	Leveling and alignment of structure and base frame should be checked	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
d)	Earthing and wiring connection should be completed	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
e)	All trenches should be cleaned and covered	-do-	-do-	-do-	TC/TR	2/3	1	-	TC/TR
f)	Tan delta measurement	-do-	-do-	-do-	TC/TR	3/2	1	-	TC/TR
Note:	a. In 'Inspection Agency'column figure 1,2,or 3 to be filled. 1- will indicate Representative of Engineer-in-charge, 2- will indicate 'Contractor' & 3- will indicate 'Sub-contractor'.								
	b. In 'Remarks' column following abbreviations shall be used - TC/TR - Test Certificate/Test Report								
	c.The above FQP stipulates the bare minimum requirements which shall be complied by the contractor. The above FQP has been made presuming subletting of work shall be allowed in the tender. In case subletting of work is not allowed in the tender, then"3/2" shall be considered as "2" under the column " Perform" .								
	d. Any additional acitivity/ test required as per the requirement of equipment in accordance with Contract Agreement/Relevant standard apart from FQP will be carried out by the firm without any additional financial implication.								
	e. The contractor shall furnish the details of test method to the Engineer-in-Charge for approval in the time line as specified in contract agreement								
	f. All the Test in FQP shall be witnessed by Engineer-in-charge or his representative at site.								
	Signature					Signatue & Seal			
	Engineer-in-charge					(Vendors Q.C Dept or Representative)			

ANNEXURE C

MODEL QUALITY ASSURANCE PLAN (MQAP) FOR MAJOR EQUIPMENT IN THERMAL POWER PLANT

QUALITY ASSURANCE PLAN

1.1 QUALITY ASSURANCE PROGRAMME

All materials, components and equipment covered under the scope of supply shall be procured, manufactured, erected and tested at all stages as per the comprehensive Quality Assurance (QA) programme. The QA document indicating the specific inspection and testing requirements to be followed shall be subject to the approval of the purchaser. The contractor shall submit his quality assurance programme for review by purchaser and shall generally cover the following:

- i) Organizational structure for management and implementation of the proposed Quality Assurance programme.
- ii) Quality system manual.
- iii) Documentation control system.
- iv) Qualification and experience data of contractor's key personnel.
- v) The procedure for procurement of materials, parts, components and equipment.
- vi) Procedure for inspection at source, inspection of incoming raw-materials/ parts/ components & verification of materials purchased.
- vii) Procedure for selection of sub-vendors and sub- contractors.
- viii) System for shop manufacturing including process controls, fabrication and assembly controls, and procedure for product identification & traceability.
- ix) System for inspection and testing during manufacture, processing, fabrication, welding, assembly and other activities.
- x) The reference documents, plant standards, acceptance norms, test and inspection procedure etc.
- xi) Control of calibration and testing of measuring and testing equipment.
- xii) System for quality audits, documentation to indicate conformance or non-conformance of the product to the specification and testing and inspection requirements; review and control of non- conforming items and system for corrective actions.
- xiii) System for indication and appraisal of inspection status.
- xiv) System for authorizing release of manufactured product to the purchaser.
- xv) System for packaging, handling, transportation, storage and delivery.
- xvi) System for generation and maintenance of inspection and test records.

- xvii) All the accepted non-conformance reports (major/ minor) / deviation including complete technical details/ repair procedure.
- xviii) Inspection reports duly signed by the Utilities and contractor for the agreed Customer Hold Points.
- xix) Certificate of Conformance (CoC) wherever applicable.
- xx) Inspection release note (IRN).

TECHNICAL REQUIREMENTS FOR QUALITY PLAN

1.2 STEAM GENERATOR AND AUXILIARIES

Testing and inspection requirements of major equipment over and above the respective code/standard requirements are given hereunder:

1.2.1 Pressure Parts

Only those materials shall be used in the manufacture of pressure parts which can be identified against mill sheet or manufacturer test certificates. Material shall meet all the mandatory requirements (and supplementary checks, if asked for) of specified specification, Indian Boiler Regulations (IBR), and relevant code/standard. All nondestructive testing as detailed against relevant equipment shall meet the requirement of ASTM section 3 Vol. 3.03 or equivalent BIS standard.

1.2.1.1 Drum / Water Separator / Storage Tank

- i) Each plate shall be subjected to a 100 % normal ultrasonic at the mill to meet the minimum requirements of BS: 5996 grade LC3/EN 10160:1999/ASTM or equivalent standards. Elevated temperature tensile tests shall also be carried out on plate material for each heat.
- ii) After cutting to size and removal of cut outs, the plates shall be subjected to magnetic particle examination (MPE) along the edges of the plate and on areas adjacent to the cutouts.
- iii) All forged connections shall be examined by 100% ultrasonic testing (UT) before machining.
- iv) Fully machined connecting pieces of internal diameter 100mm and above, except for forgings, shall be subjected to magnetic particle examination / liquid penetrant examination.
- v) Mechanical tests shall be carried out on specimens prepared from the production control test plates of the longitudinal welds.
- vi) Mechanical tests shall be conducted on the specimens from manhole cutouts of dished ends.
- vii) All butt welds shall be subjected to 100% UT/RT/PAUT+TOFD after stress relieving.
- viii) On completion of welding, the entire drum / Separator / Storage Tank shall be subjected to stress relieving in the furnace.
- ix) All butt welds shall be subjected to 100% UT/RT/PAUT+TOFD test and magnetic particle examination before stress relieving.
- x) All full penetration nozzle welds and Lifting lug welds shall be subjected to

ultrasonic examination before stress relief.

- xi) After stress relieving (SR) all welds, internal and external shall be examined by MPI methods depending on size and accessibility and all butt welds shall be subjected to 100% radiography.
- xii) Complete drum / Separator/ storage tank/ Mixing Sphere etc. shall be subjected to hydraulic pressure test and all compensating pads shall be pneumatically tested.
- xiii) All connecting tubes & pipes shall be subjected to UT prior to fabrication as per BS 3602 or equivalent with longitudinal calibration notch of depth 5% of wall thickness (0.3 mm min. and 1.5 mm max.)
- xiv) PT shall be carried out in inaccessible areas where MPI cannot be performed.
- xv) Forged components of water separator shall be tested as per applicable forging ASME code and IBR.

1.2.1.2 Headers

- i) Raw material for headers shall be subjected to UT prior to fabrication. Transverse test on headers (greater than 8") from one end (25% of lot). Flattening test on pipe from one end (25% of lot).
- ii) Butt welds in Carbon steel and low alloy steel shall be subjected to RT/PAUT+TOFD before stress relieving except for P91/92 and X20 grade material. P91/92 and X20 grade shall be subjected to MPE + RT /PAUT+TOFD after stress relieving.
- iii) Full penetration nozzle welds of carbon and low alloy steels shall be subjected to UT before PWHT . However, UT shall be carried out Gr 91/92 on full penetration welds after PWHT. All attachment welds shall be subjected to MPI/DPT.
- iv) All nozzles, branches, stubs and load bearing attachment shall be examined by MPE/ PT techniques after the toes of the weld have been ground smooth and stress relieved.
- v) Non-load bearing welds shall be examined by MPE/PT techniques after the toes of the welds have been ground smooth and stress relieved.
- vi) Completed closed end headers shall be subjected to hydraulic pressure tests and all compensating pads shall be pneumatically tested as per IBR.
- vii) All weld joints in alloy steel headers of P 91, X20 and X22 & other material of P15E group and above shall be checked for Hardness. 3% hardness check shall be carried out on welds of other alloy steel Headers.
- viii) Boroscopy examination shall be carried out for those header which will be interconnected with other headers by welding.
- ix) PT shall be carried out in inaccessible areas where MPI cannot be performed.

1.2.1.3 Tubes and tube elements

- i) Raw material for tubes for water wall, superheater, reheater, economiser, riser, supply and connecting tubes including nozzle/stubs, connections for drum, headers, pipe work etc., shall be subjected to 100% UT prior to fabrication as per IBR or ASME E 213 or EN10246:7 1996 or equivalent with the longitudinal calibration notch of depth 5% of wall thickness (0.3mm minimum and 1.5mm maximum).
- ii) All bent tubes, stubs shall be checked for ovality and thinning by ultrasonic method on first off lot and random checks on subsequent pieces. Critical bends, where PWHT is required after bending, shall be subjected to LPI/MPI. For FOT area reduction shall be calculated
- iii) All tubes, stubs, panels, coils shall be checked for clearance by steel ball test and for cleanliness by sponge passage.
- iv) Finished butt welds shall be subjected to RT or UT. Wherever the code/standard/process specifies random sampling, the same shall be minimum 20%. Finished field butt welds not covered under random sampling for RT/UT, shall be subjected to RT or PAUT.
Minimum 10 % of the fillet welds (including fins if any) shall be subjected to MPE or dye penetration test (DPT). However, fillet welds of material grades P15E and above or any other new material grade shall be subjected to 100% MPI / LPI.
- v) After erection tubes and fabricated panels, coils shall be subjected to hydraulic pressure test excluding loose tubes but including water wall panel, burner panels, reheaters, superheaters and economizers.
However, Hydro test shall be carried out at Shop on 100% of the coils, SOFA Panels, Burner panels. For other panels, hydro test at shop shall be 100% for tube thickness $\leq 5.5\text{mm}$; 20% for thickness $> 5.5\text{ mm}$ and $< 7\text{ mm}$; NIL if thickness $\geq 7\text{ mm}$.
- vi) 10% hardness survey on butt welds of P15E material group and above. 10% Hardness checks shall also be carried out on welds of T23 material grade. 3% Hardness checks shall also be carried out on welds of T22 material grade used in water wall panels.
- vii) In case of RT of tube welds with DWDI (elliptical view) number of exposure shall be as per relevant code / plant standard and will not be less than two exposures for each weld wherever there is no limitation in carrying out two RT shots.
- viii) In case of spiral water wall design, trial assembly of complete wall of each side including hopper shall be carried out.
- ix) Austenitic Stainless-Steel tubes & coils shall be Solution Heat Treated, Stored, preserved and handled so as to avoid Stress Corrosion Cracking due to Chloride contamination.

1.2.1.4 Integral piping

- i) All raw materials used shall have co-related mill test certificate meeting material specification. All pipe lengths shall be subjected to 100 % ultrasonic examination as per ASME E213 or BS 3602 or equivalent, with longitudinal calibration notch of

depth 5% of wall thickness (0.3mm minimum and 1.5mm maximum) or hydraulic tests and UT or RT on longitudinal welds at the tube mill.

- ii) All mother pipes shall be subjected to 100% UT prior to fabrication as per ASME E 213 or equivalent with longitudinal calibration notch of depth 5% of wall thickness (0.3mm minimum and 1.5mm maximum).
- iii) All cold bent pipes shall be checked for ovality and thinning by UT on first off lot and on random samples for subsequent pieces. Outer surface of hot/ induction bent pipes shall be subjected to MPE.
- iv) The edge preparation for shop and site welds in stainless steel and alloy steel shall be subjected to dye penetration check. Non-destructive examination of welds shall be carried out after post weld heat treatment, if any.
- v) All butt welds shall be subjected to UT or RT and MPE/PT. For weld on alloy steel piping, UT or RT shall be done after stress relieving. Wherever the code/ standard/ process specifies random sampling, the same shall be minimum 20 %.
- vi) All butt welds in alloy steel piping of P91, X20 and X 22 shall be checked for RT or UT and MPE after stress relieving (SR). UT shall be of digital recordable type.
- ix) All weld joints in alloy steel piping of P 91, X20 and X22 shall be checked for hardness. For preheating and post weld heat treatment (PWHT) induction heating shall be deployed. However, PWHT can be done in furnace also. 3% hardness check shall be carried out on welds of other alloy steel piping.
- x) All load-bearing attachment welds shall be subjected to MPE after stress relieving.
- xi) For HP piping, non-destructive examination of welds shall be carried out in accordance with the relevant design/manufacturing codes. However, apart from above, the following requirements shall be met. Further statutory requirement, wherever applicable shall also be complied with the following:
 - a) Temperature > 400⁰C and/ or pressure exceeding 71 bar
 - 100% RT or UT on butt welds and full penetration branch welds.
 - 100% MPE.
 - b) Temperature > 175⁰C up to 400⁰C and/or pressure exceeding 17 bar and up to 71 bar
 - 100% RT or UT on butt welds and full penetration branch welds for pipe diameter more than 100NB.
 - 10% RT or UT on butt welds and full penetration branch welds for pipe diameter up to 100NB.
 - 100% MPE.
 - c) All other pipes not covered above shall be subjected to 100% MPE or DPT in

case of underground pipes and 10% MPE or DPT in case of piping above the ground. Further, 10% of butt welds of underground piping shall be subjected to RT.

Wherever SR/PWHT is envisaged for alloy steel, above NDTs shall be before SR/PWHT except for Gr.91 /92 NDT shall be after PWHT.

1.2.1.5 Fittings:

- i) Raw material of all forged fitting shall be ultrasonically tested. All mother pipes used for formed fitting shall be ultrasonically tested as per ASME 213 or BS 3602 or equivalent with longitudinal calibration notch of depth 5% of wall thickness (0.3mm min. and 1.5mm max.) shall be adhered to.
- ii) Fittings shall be subjected to suitable NDT as per applicable standards. However following minimum NDE requirement shall also be applicable / met.
 - a) For fittings X20, P-91, P-92 and material group P15E & above
 - 100% MPI/ PT &
 - 10% hardness check &
 - For fittings of 200 NB & above 100% UT/RT
 - b) 100% UT/RT for fittings of 200 NB & above for boiler feed discharge, recirculation and spray piping of boiler feed system.
 - c) 100% UT/RT for fittings of all other piping of size OD 508mm & above.

1.2.1.6 Valves

- i) Pressure retaining parts of valves shall be subjected to NDT as per Table 1.
- ii) Hardened and stellitted valve disc and seat shall be subjected to DPT and hardness check.
- iii) All forgings rounds above diameter 40 mm shall be ultrasonically tested.
- iv) NDT requirements for pressure retaining components of valves shall be as per Table 1 below.
- v) Color matching of valve disc/plug and seat shall be carried out to ensure minimum 80% contact and no through passage. UT / RT of full penetration welds & MPE/ DPT of fillet joints (in case of cast valves).
- vi) Hydraulic pressure test and seat leak test shall be carried out as per ASME / ANSI 16.34 or IBR.
- vii) Air seat leak test shall be carried out as per applicable standards or codes.
- viii) Functional tests shall be carried out on each valve to check the following as per the approved valve data sheet

- a. Smooth operation.
 - b. Valve travel, closing and opening time.
 - c. Current drawn by actuators.
- ix) Springs for safety valves shall be tested with suitable NDT and for springstiffness rate.
- x) Safety relief valves shall be tested for performance. Lift set pressure, back pressure, reset pressure test in case of safety relief valves.

All critical valve components shall be tested for mechanical and chemical properties.

Table 1: NDT requirements for pressure retaining components of valves

Valve size NB in mm	ANSI Class up to 300	ANSI Class above 300 up to 600	ANSI Class above 600 below 900	ANSI Class 900 and above but below 4500
Less than 50	Visual	Visual	Visual	MPE / DPT
50 & above but below 100	Visual	Visual	MPE/DPT(for special class valves)	MPE/DPT and RT on 100% of valves area as per ANSI/ASME B16.34
100 & above but less than 300	Visual	MPE/DPT	MPE/DPT and RT on 10% of valves on change of section and weld ends	MPE/DPT and RT on 100% of valves area as per ANSI/ASME B16.34
300 and above	MPE/DPT	MPE/DPT	MPE/DPT and RT on change of sections and weld ends	MPE/DPT and RT on 100% of valves area as per ANSI/ASME B16.34

Note:

- i) For body and bonnet forgings, UT with MPE may be adopted in place of RT. For Austenitic steel MPI may be replaced with LPI.

1.2.1.7 Non pressure bearing attachments

Load bearing welds shall be subjected to UT and MPE after stress relieving. Non load bearing welds shall be subjected to MPE after stress relief. The toes of the welds adjoining the drum shall be ground smooth prior to stress relieving before carrying out this examination. LPI shall be carried out in inaccessible areas where MPI cannot be performed

1.2.1.8 Hydraulic test

- i) The drum and all components which are to be subjected to fluid pressure shall be tested to minimum of 150% of the design pressure. The duration of the pressure tests shall be sufficient, as approved by the purchaser, to show any leakage paths and to permit a thorough examination of the component whilst under pressure.
- ii) The temperature of the fluid used for the pressure test shall be such as to avoid any possibility of brittle fracture at a low temperature and the same shall be modified and submitted to the purchaser for approval, before commencing the test.
- iii) The fluid used shall be of a sufficient purity and where relevant, suitable inhibitors shall be used to avoid excessive corrosion and /or damage to temporary parts either during the test or prior to drying and cleaning.

1.2.1.9 Pneumatic test of compensating pads

All compensating pads shall be provided with two-threaded weep holes to test welds at 0.5 kg/cm² (g) with soap solution and no leakage shall be ensured.

1.2.2 Boiler water circulation pumps

- i) Raw material for casing, shaft and impeller shall be checked for high temperature physical properties, apart from mandatory and supplementary check of material specification.
- ii) All forging and castings shall be subjected to 100% UT or RT and MPE or DPT check.
- iii) Static and dynamic balancing of the rotary parts shall be carried out.
- iv) Hydraulic pressure test shall be conducted on pump casing at minimum 1.5 times the design pressure.
- v) Each pump shall be subjected to performance test at the manufacturer's works under as near actual site conditions as possible.
- vi) Following test shall be carried out on assembled units:
 - A) Type test:
 - a) NPSH test
 - b) Temperature rises test.
 - c) Under voltage test.
 - d) Quality assurance proof test.
 - e) Tests to establish unit functioning of pump at operating temperature and pressure.
 - f) Hot standstill and start up tests.

Note :- Type test if already done on the same model will not be repeated. Documents will be submitted for review and approval of the purchaser.

- B) Routine test:
 - a) Hydrostatic test of complete unit.
 - b) Over speed test.
 - c) Tests to determine unit characteristics.
 - d) Pump performance.
 - e) Unit run at rated voltage.
 - f) Starting current at rated voltage.
 - g) Cold start up test.
 - h) Endurance test of motor windings, joints and terminal seals
 - i) Noise level.
 - j) Inspection of dismantled unit.
 - k) High voltage test.
- vii) For heat exchanger for these pumps, butt welds on pressure parts shall be tested with RT or UT and all other welds shall be tested with MPE or DPT.
- viii) Hydraulic test shall be carried out both on tube side as well as shell side at minimum 1.5 times the design pressure.

1.2.3 Air Preheaters, Steam Coil Air Pre-Heater and Fuel Oil Heaters

1.2.3.1 Air Preheaters

- i) Forged shafts for air preheater like stub shaft, main rotor forging, housing hub shall be subjected to 100% UT at mill and magnetic particle examination after machining.
- ii) For non-modular design, trial assembly shall be carried out at shop prior to dispatch to site.
- iii) Critical welds of rotor post shall be subjected to radiographic examination.
- iv) Trial run of air preheater rotor drive assembly with gear box, pinion, motor at shop.

1.2.3.2 Steam coil air pre-heater and fuel oil heaters

Hydraulic pressure test shall be carried out on the heating coils. All pipes, valves, steam traps and mountings shall be subjected to hydraulic test as called for under IBR, BS or other approved codes.

1.2.4 Soot Blowers

- i) Butt weld between nozzle and lance tube shall be subjected to 20 % MPI/ LPI.
- ii) Hydraulic test shall be carried out on valve body.
- iii) Soot blower shall be subjected to operational checks as below:

- a) Smooth operation
- b) Long tube travel, closing and opening time.
- c) Current drawn.

1.2.5 ID, FD and PA fans

- i) Rotor components shall be subjected to ultrasonic test at mill and magnetic particle examination/dye penetration examination after rough machining.
- ii) Butt welds in rotor components shall be subjected to 100% UT and all welds shall be subjected to MPE or DPT after stress relieving.
- iii) All rotating components of fans shall be dynamically balanced to quality grade 2.5 of ISO 1940.
- iv) Test for natural frequency of all fan components, including fan blades shall be carried out for the fans.
- v) Full range performance test shall be carried out on one fan of each type and size as per BSEN ISO 5801 or equivalent standard.
- vi) Dimensional and profile checking of fan blades
- vii) Hydraulic coupling of ID fan shall be checked for string test i.e. operational check of one fan assembly using hydraulic coupling to check temperature rise, smooth operation, vibration and noise level. Dry run test shall preferably be carried out during string test.
- viii) Seal air Fan / Scanner air fans
 - (a) Rotor components i.e. shaft and hub shall be subjected to ultrasonic test at mill and magnetic particle examination after rough machining.
 - (b) 10% of Butt and fillet welds both in rotor and static components of the fan shall be subjected to MPI / DPT after stress relieving.
 - (c) Fan impeller shall be balanced dynamically to quality grade 2.5 of ISO 1940.
 - (d) All Fans performance test shall be carried out at the manufacturer's works under as near actual site conditions as possible.

1.2.6 Coal Mills, Pulverised Coal Piping and Burners

- i) Raw material for shaft, coupling, gears and pinions, top and bottom races and other rotating components shall be subjected to UT. MPE or DPT shall be carried out to check surface soundness.
- ii) Wear-resistant parts shall be subjected to UT or RT to check soundness after suitable heat treatment. Check for chemical composition and hardness shall be carried out. For ceramic materials check for various properties including hardness, density, wear rate and composition shall be carried out.

- iii) Butt welds in the tube/ separator /body casing of the mill shall be tested by UT / RT and MPI/LPI. All other welds in main tube/separators shall be tested by MPI/LPI for acceptance. The tube shall be statically balanced.
- iv) All gearboxes shall be run tested for adequate duration to check rise in oil temperature, noise level and vibration. Check for leak tightness of gear case also shall be performed.
- v) Fabricated pipe welds shall be examined by MPE/LPI.
- vi) Ceramic/ basalt lined piping and bends etc. shall be checked for proper layout.
- vii) Weldments on burner components shall be checked with suitable NDT. The burner assemblies shall be tested for operation at shop.
- viii) Trial assembly of at least one mill complete with all major components at shop.
- ix) All rotating components shall be dynamically balanced.

1.2.7 Coal Feeders

- i) All welds in the casing and pulley fabrication shall be checked with MPE/PT.
- ii) Type tests including degree of protection and routine tests shall be done as per relevant Indian Standards or equivalent International Standards.
- iii) All major items like plates for casings, head pulley, tail pulley, pulley shaft and major castings shall be procured with respective material test certificates.
- iv) Explosion proof test at 50 psi as per NFPA code shall be done as type test. Leak tightness test shall be done on individual feeder casing. Endurance test for load cell shall be carried out.
- v) Test for weighing accuracy, calibration and repeatability shall be carried out at various speeds by coal flow on one feeder.
- vi) Calibration check shall be carried out on all feeder cabinet and assemblies prior to dispatch.

1.2.8 Boiler structure, ducts, hoppers, dampers etc.

- i) Only those materials which have been identified against mill sheet or test certificates shall be used for construction. Structural steel and built up plate girders for main boiler shall be fully killed fine grained and normalised. All plates of tension and compression flanges and connection material and plates above 40mm thickness shall be 100% ultrasonically tested.
- ii) Visual inspection of all welds shall be performed in accordance with AWS D.1.1. Also the butt and fillet welds for built up plate girders shall be inspected 100% by magnetic particle examination.

- iii) RT or UT shall be performed on all butt welds of thickness 32mm and above. For thickness below 32mm and up to 25mm, 100% MPE shall be carried out along with 10% RT/UT and for thickness below 25mm 10% MPE or DPT shall be performed. Edge preparations for field welding shall be examined by MPE for plate thickness 32mm and above. Field welds ends in ceiling girder shall be subjected to Ultrasonic examination for 100mm depth from the edges. All fillet welds of built up plate girders shall be inspected 100% by MPI.
- iv) Ceiling girders, columns, ducts hoppers and tunnels shall be trial assembled and match marked prior to dispatch/erection. At least two consecutive girders along with cross member shall be assembled at a time. Production test coupons for Butt and Fillet welds of main columns and Ceiling girders shall be carried out.
- v) Drum sling rods
 - a) Sling rods forging shall be subjected to ultrasonic examination.
 - b) Welds shall be examined by UT before stress relief and MPE after stress relief.
 - c) Trial fitment of the rods with the drum shell shall be carried out to ensure proper contact.
 - d) Screw thread of the rods shall be suitably protected to avoid damage during handling and transport.
- x) Dampers
 - a) All the dampers shall be subjected to operational test/checks.
 - b) Gas tight dampers shall be subjected to shop leakage test to demonstrate the guaranteed tightness (minimum one damper of each type and size offered). In case such type test is already done, the reports of the same shall be submitted for review.
 - c) All dampers shall be checked for sealing dimensions to establish guaranteed tightness.

1.2.9 Electro- Static Precipitators (ESPs)

1.2.9.1 Discharge and collecting electrodes

- i) Work tests for discharge electrodes shall include the following (for the wire type electrodes):
 - a) Chemical and tensile tests.
 - b) Metallographic examination-longitudinal and transverse (250X)
 - c) Surface finish and surface purity from chloride ions.

- d) Spring back and surface finish after coiling (applicable to helical discharge electrodes)

For all other type of electrodes, testing to be carried out as per the applicable standards.

- ii) Work tests for collecting electrodes and rigid discharges electrode shall include the following:
 - a) Chemical and mechanical properties.
 - b) Check for profile and straightness.
 - c) Check for surface finish and dimensional accuracy.
 - d) Cupping test for deep drawn sheets as per IS 513.

1.2.9.2 ESP structure

- i) All plates above 40mm thickness shall be 100% ultrasonically tested.
- ii) Visual inspection of all welds shall be performed in accordance with AWS D1.1.
- iii) Also the butt and fillet welds for built up plates and columns shall be inspected by 100% MPI.
- iv) NDT requirements of structural steel welds shall be as under:-
 - a) 100% RT/UT on butt-welds of plate thickness ≥ 32 mm.
 - b) For plates of $25\text{mm} < \text{thickness} < 32\text{mm}$ - 10% RT/UT and 100% MPI.
 - c) For plates of thickness $< 25\text{mm}$ - 10% MPI/LPI.
 - d) All fillet welds of structural members shall be inspected 100% by MPI.
- v) Edge preparation for field welding shall be examined by MPI for thickness $\geq 32\text{mm}$.

1.3 STEAM TURBINE GENERATOR AND AUXILIARIES

Testing and inspection requirement of major equipment over and above the requirements of respective codes/standards are given hereunder:

1.3.1 Steam Turbine

1.3.1.1 High pressure (HP) and intermediate pressure (IP) enclosures

The following shall be applicable to high pressure cast steel enclosures (for example high pressure and intermediate pressure inner and outer cylinders, steam chests and liners, steam inlet pipes, nozzle boxes):

- i) Test pieces fully representative of the material and condition of the casting shall be provided to the purchaser to enable the determination of the properties of material to be used. Casting suppliers should have established practices to ensure requirements of creep and rupture for long exposure of the component/ equipment to high temperatures and pressures. In case of new or non-established vendors, creep data shall be made available by the contractor.
- ii) Tests shall be carried out on the casting material to establish its mechanical properties, chemical composition, and microstructure.
- iii) Each casting shall be subjected to magnetic particle examination on the entire inner and outer surfaces after heat treatment.
- iv) Each casting shall be subjected to visual examination and dimensional check before taking up the machining work on it. Dimensional check shall be carried out after machining also.
- v) Each casting shall be subjected to a 100% examination for internal flaws by UT or RT method after heat treatment and suitable preparation.
- vi) Cast enclosure shall be subjected to a hydraulic pressure test based on established practice of manufacturer.
- vii) Excavated area of all the defects shall be subjected to MPE to ensure excavation up to sound area. All the areas repaired/ upgraded by welding shall be examined by UT, RT (to confirm findings of UT wherever required) and MPE. Sketches/ reports of location of repair and reports of NDT carried out on repaired areas shall be submitted along with certificates. Hardness survey shall be carried out on the repaired areas.
- viii) Where stub pipes and transition pieces are welded to the main body of an enclosure the following shall be carried out:
 - a) RT and MPE or DPT of weld preparations.
 - b) MPE of finished welds after stress relieving.
 - c) RT or UT of finished welds.
 - d) 100% RT or UT and MPE of site weld preparations on the stub pipes and transition pieces before dispatch to site.
 - e) Hardness survey on the weld joint, heat affected zone (HAZ) and parent material.
- ix) Wall thickness measurement for critical and highly stressed zones of the casting shall be carried out by ultrasonic method.
- x) Colour matching of castings by putting two halves together and feeler gauge tightness check from both sides, i.e. inside and outside to ensure required contact

area and joint tightness shall be carried out.

- xi) Final dimensional check before dispatching should be carried out. Suitable arrangement should be taken care to avoid any distortion during transit/ transportation.

1.3.1.2 Low pressure (LP) enclosure (fabricated)

- i) Tests shall be carried out on the plate material to establish its mechanical properties and chemical composition.
- ii) Plates used for fabrication shall be subject to 100% UT.
- iii) Where welds are made by chipping and grinding back to the first side weld before completing the weld from second side, a magnetic particle or dye penetrant examination of the chipped area shall be carried out.
- iv) Fabricated enclosure shall be subjected to stress relieving as per manufacturer's standard practice.
- v) Dimensional check shall be carried out on the fabricated enclosure.
- vi) The following minimum requirements shall be met for NDT on the welds:
 - a) Butt welds & full penetration welds : 10% RT or UT and 10% MPE or DPT (100%)
 - b) Fillet welds : 10% MPE or DPT (100%)
 - c) Nozzle welds : 10% MPE or DPT (100%)
 - d) Lifting lug and other load bearing fillet welds : 100% MPE or DPT
 - e) Site weld edge preparations : 10% MPE or DPT
- vii) Fabricated enclosure shall be subject to hydraulic pressure tests. If it is not the manufacturer's practice, the justification for not carrying out hydraulic test shall be furnished for purchaser's approval.
- viii) Feeler gauge tightness check from inside and outside to ensure required joint tightness shall be carried out.
- ix) Suitable arrangement should be taken care to avoid any distortion during transit/ transportation.

1.3.1.3 Turbine rotors

1.3.1.3.1 Forgings

Rotor forgings (mono block and/ or discs), impulse wheel and nozzle box and coupling forgings:

- i) Fully representative tangential, radial and axial test pieces shall be provided at each end of the body, at each shaft end and from the trepanned core (when a core is

trepanned) to determine mechanical properties including impact, Brinell hardness etc. and tests for notch toughness (both transition temperature and room temperature impact values).

- ii) Forging supplier should have established practices to ensure requirements of creep and rupture for long exposure of the component to high temperatures and pressures.
- iii) Heat treatment shall be carried out in such a way so as to ensure minimum residual stress in the rotor. Residual stress measurement will be carried out.
- iv) Tests shall be carried out on the forging material to establish its mechanical properties, chemical composition and micro structure.
- v) Thermal stability tests shall be carried out on HP and IP rotor forgings to ensure the thermal stability of the rotors in service and at over speed.
- vi) Each forging shall be subjected to a 100% ultrasonic examination. Normal probes and angular probes with different probe angles shall be used for thorough examination to ensure complete soundness of the forging
- vii) Each rotor shall be subjected to a 100% MPE after final machining on journal areas and before gashing on other areas.
- viii) When a rotor forging is bored, a visual check and magnetic particle examination of the bore shall be carried out.
- ix) Following tests shall be carried out on the rotor welds:
 - Ultrasonic examination with normal and angular probes of the weld to ensure complete coverage and freedom from harmful defects.
 - Run out of rotor before and after welding.
 - MPE on finish welds.
 - Hardness survey on the welds.
 - Stress relieve annealing.
 - Test reports of filler material used.
 - Dimensional record of weld preparation.
- x) Dimensional examination of the rotor blade grooves and other important dimensions shall be carried out to ensure the conformance to drawing dimensions and log sheets/records shall be prepared for all important dimensions.

1.3.1.3.2 Complete rotors

- i) Axial & radial run-outs and surface finish checks shall be carried out before and after blading and after over speed tests. Run out examination shall be carried out at blade shrouds also.
- ii) Check shall be carried out for clearance between rotor groove and blade at the root.
- iii) Rotors shall be dynamically balanced at rated speed.
- iv) An over speed test shall be carried out during which the rotor shall withstand an over speed of 120% for five continuous minutes or 125% for two continuous minutes. If bidder's practice is different from that stated above, then the same shall be furnished for purchaser's approval. During over speed test, vibration measurement and analysis shall also be carried out.
- v) After blading and again after over speed testing, rotor stages with blades over 225 mm of active length are to be subject to standing vibration tests to determine natural frequencies in various vibration modes to ensure that the ranges are outside operating frequencies.
- vi) In case impulse stage and or blade discs are fitted on the rotor, fit up between such disc and rotor shall be checked up before and after over speed test.
- vii) Lock blade lift after the over speed test shall be checked and record for same shall be maintained.

1.3.1.4 Stator and rotor blades and shroud bands

- i) Fully representative test pieces shall be provided to enable mechanical properties of the material to be determined. In case of blades machined from bar stock, mechanical tests shall be carried out on the hardest and softest specimens of each heat treatment batch. Hardness test will be carried out on 100% basis.
- ii) Tests shall be carried out on the materials to establish its mechanical properties, chemical composition and micro structure. Creep requirement as per relevant standard.
- iii) Each bar stock for machining blades and forging shall be subjected to 100% ultrasonic examination.
- iv) When erosion shielded, the erosion shield and blade joint shall be radiographed. In case of flame & laser hardening MPI shall be done.
- v) Dye penetrant test shall be made on the erosion shield and blade joint in manufacture prior to fitting to the wheel and after over speed tests.
- vi) Magnetic particle examination or dye penetrant test (when MPE is not applicable) shall be carried out on finish machined blade profile, roots and shrouds.
- vii) All moving blades of over 225mm active length are to be moment weighed and assembled on shaft in a prescribed sequence to ensure optimum balancing of rotor.

- viii) Natural frequencies of the LP turbine blades shall be determined before mounting on rotors to ensure that the same are outside operating frequency range.
- ix) Shroud bands after punching and after riveting shall be subjected to 100% DPT to ensure freedom from harmful surface defects.
- x) In case of cast blades, following testing shall be done:
 - a) Chemical analysis and mechanical testing per heat/heat treatment batch.
 - b) MPE on rough machined and finish machined blade.
 - c) RT/UT on blades.
 - d) Before starting mass productions, following technological tests shall be carried out on the first lot of 10 to 15 blades:
 - 100% RT and 100% MPE on blades
 - 100% hardness testing.
 - Mechanical testing and metallurgical testing.
 - Weld repair shall not be permitted.

1.3.1.5 Diaphragms

- i) Welded and fabricated diaphragms
 - Concentricity checks shall be carried out on final machined diaphragms to ensure that there are no negative overlaps between guide and moving blades.
 - 10% UT and 100% MPE shall be carried out on finished, stress relieved and machined welds. For inaccessible areas, DPT shall be carried out in place of MPE.
- ii) Cast, forged and machined diaphragms
 - a) Tests shall be conducted to determine mechanical properties together with chemical analysis, metallographic/ metallurgical examination, and required heat treatment.
 - b) Concentricity, flatness, blade drop and area checks shall be carried out on finally machined diaphragms to ensure that there are no negative overlaps between guide and moving blades and port wall.
 - c) 100% ultrasonic examination shall be carried out on diaphragm materials. Blade junction areas with the side walls shall be checked by MPE or DPT.
- iii) Colour matching of all the diaphragms by putting two halves together and feeler gauge tightness check shall be carried out.

1.3.1.6 Exhaust hood

All castings shall be subjected to chemical and mechanical tests as per relevant material

standards. In case of fabricated construction of exhaust hood, butt welds shall be subjected to 10% RT and 100% DPT. Exhaust hood shall be pressure tested. No repair of welding shall be carried out on cast iron castings. Blue matching shall be carried out on exhaust hood parting planes.

1.3.1.7 Stop, control and bypass valves, actuators/ servo-motors and steam strainers

- i) Test pieces shall be provided to enable the determination of mechanical properties of valve bodies, bonnets, valve disc and seat, and valve spindle. Casting suppliers should have established practices to ensure requirements of creep and rupture for long exposure of the component/ equipment to high temperatures and pressures.
- ii) Tests shall be carried out on the materials to establish their mechanical properties, and chemical compositions.
- iii) Dye penetrant tests shall be carried out on stellite and nitrided areas of components and on stellite components in the finish ground or honed condition.
- iv) Hardness check shall be carried out to ensure required hardness.
- v) Valve body and bonnet castings or forgings shall be subjected to 100% RT or 100% UT. Body and bonnet shall also be subjected to 100% MPE on entire surface.
- vi) All pressure containing welds in body and bonnet shall be subjected to 100% RT or UT and 100% MPE.
- vii) Wall thickness of the body and bonnet after finish machining shall be measured by ultrasonic method and valve seat bore shall be checked for size and concentricity.
- viii) Dimensions of valve spindle shall be measured and valve lift shall be checked.
- ix) Bar stock of 40 mm and above size for valve stem shall be subjected to UT, and finish machined stem shall be subjected to MPE or DPT.
- x) Each valve body and bonnet shall be hydraulically tested at minimum 1.5 times the maximum working pressure after applying temperature corrections for minimum 30 minutes.
- xi) All the actuating cylinders and servomotors shall be performance tested.
- xii) Performance testing shall be carried out on valve operators and actuators to check functional requirements like trip closing and opening time, valve lift and hysteresis.
- xiii) Colour matching of the valve disc and seat to ensure the required contact area is to be carried out.

1.3.1.8 Cast and forged steel components (such as LP casing- in case of cast design, inlet, extraction and exhaust connections, shaft seal covers and rings, governor shaft, breach nut, threaded ring, angle ring, U-ring, servomotor parts such as body, piston, cover, yokes; turning gear casing and other items which are not specifically covered elsewhere)

- i) Results of tests conducted to determine mechanical properties, chemical analysis, metallurgical/ metallographic examination, and heat treatment procedures recommended and actually followed shall be recorded on certificates.
- ii) Each pressure containing enclosure shall be subjected to a hydraulic pressure test at 1.5 times the design pressure and after applying temperature corrections.
- iii) All castings and forgings shall be subject to suitable non-destructive examination by RT or UT and MPE or DPT methods to ensure freedom from harmful defects in line with clause 1.3.1.1.

1.3.1.9 Bolts and nuts for pressure retaining enclosures and rotor couplings

- i) Bar stock for bolts shall be subject to 100% UT (having more than 40mm dia.).
- ii) Finish machined bolts shall be subject to preferably 100% MPE to detect surface defects. However, DPT can also be done in place of MPE.
- iii) Coupling bolts and nuts shall be suitably identified after weight control checks.

1.3.1.10 Governing and protection system equipment (such as electro-hydraulic controller, hydraulic amplifier, hydraulic controller, electro hydraulic convertors, hydraulic convertors, hydraulic speed governor, trip devices etc.)

- i) All pressure retaining parts shall be subjected to hydraulic testing.
- ii) All the major castings and forgings shall be subjected to suitable NDT methods depending upon their application and criticality to ensure the freedom from harmful defects.
- iii) All the main assemblies and sub-assemblies shall be subjected to functional test.
- iv) All butt welds shall be subjected to minimum 10% RT or UT and all fillet and corner welds shall be subjected to MPE or DPT.
- v) All control equipment shall be subjected to rig testing, if it is not possible to test it on the steam turbine light run. The purpose of rig testing shall be as far as practical to prove that the functioning of the control equipment is in accordance with the approved design.
- vi) Nitrided and stellited components shall be subject to DPT and hardness check.

1.3.1.11 Inspection of completed turbine

- i) HP, IP and LP turbines

The steam turbine shall be assembled in the manufacturer's works to such an extent that a thorough inspection can be carried out. The purpose of this inspection shall be to ensure that the fit between mating components is correct and that all clearances are in accordance with the design requirement. The following minimum checks/

measurements shall be carried out on the assembled turbine:

- a) Check and measurement of clearance between shaft seal casing and shaft seal ring.
- b) Check and measurement of clearance in anti- rotational device in shaft seal casing joint.
- c) Check and measurement of axial and radial alignments of inner and outer casings.
- d) Check and measurement for radial and axial blade clearances in blading section.
- e) Check and measurement for axial and radial clearances in shaft seal.
- f) Check and measurement for tightness and elongation of horizontal joint inner casing bolts.
- g) Check and measurement of minimum axial clearances and minimum radial clearances in completely assembled steam turbine.
- h) Check for alignment of over speed governor.
- i) Check for axial distances for shroud bands for casings.
- j) No load running test on the steam turbine including functional tests for steam turbine control and emergency control equipment (as per manufacturer's practice).
- k) Pre- dispatch inspections including clearance check for transportation device.

ii) LP Turbines

For the LP turbine, the following minimum checks/ measurements shall be carried out on the assembled turbine at the manufacturer's works, as per manufacturer's standards:

- a) Check and measurement of axial and radial alignments of inner and outer casings.
- b) Check and measurement for radial and axial blade clearance in blading section.
- c) Check and measurement of minimum axial clearances and minimum radial clearances in completely assembled steam turbine.
- d) Check and measurement for alignment of stationary blade carriers.

- e) No load running test on the steam turbine including functional tests for steam turbine control and emergency control equipment.
- f) Pre- dispatch inspections including clearance check for transportation device.
- iii) No load running tests on complete steam turbine shall be carried out as per manufacturer's standard practice.

1.3.2 Integral Auxiliaries of Steam Turbine

1.3.2.1 Bearing pedestals, housings and bearings

- i) Leakage test shall be conducted on pedestals. For fabricated pedestals and housing, 10% weld shall be checked after stress relieving by magnetic particle test and minimum 10% of the butt welds shall be checked by RT or UT.
- ii) Bearing shell
 - a) The shell shall be subjected to suitable nondestructive examination like RT or UT and/or MPE as applicable.
 - b) Colour matching of the shell by putting two halves together and feeler gauge tightness checks from inside and outside to ensure required contact area and joint tightness shall be carried out.
 - c) The shell shall be subjected to hydraulic pressure test.
 - d) Chemical analysis of white metal shall be carried out. The effectiveness of the white metal adhesion shall be checked by UT or other approved method, and the exposed edges of the white metal shall be subject to DPT.
 - e) Hydraulic test shall be carried out for bearing oil inlet piping and jacking oil piping.
 - f) Alignment check shall be made for bearing in bearing pedestal.

1.3.2.2 Cross around pipes

- i) Weld edge preparation of shop and site welds shall be checked by 100% MPE/DPT.
- ii) All butt welds shall be subjected to 100% RT/UT.
- iii) 100% MPE shall be carried out on all welds.
- iv) Check for dimensions and visual inspection shall be carried out on finished pipes.

1.3.2.3 Lubricating oil, jacking oil and control oil systems

- i) Pumps

- a) Main oil pump shaft shall be subjected to ultrasonic examination. Butt welds shall be subjected to RT or UT.
 - b) Pump impeller shall be subjected to suitable NDT method like MPE or DPT for surface defect examination. Impeller of main oil pumps shall also be subjected to an over speed test at 120% of rated speed for 5 minutes (as per the guidelines followed for HP Rotor)
 - c) Pump casing shall be subjected to hydraulic pressure test at 2 times the working pressure or 1.5 times the pump shut off head whichever is higher.
 - d) Rotor assemblies shall be dynamically balanced.
 - e) All pumps shall be performance tested at the manufacturer's works. Test shall include check for vibration and noise levels also.
- ii) Coolers and lubricating oil tanks
- a) All raw materials used shall have co-related mill test certificate meeting mandatory and supplementary checks of material specification.
 - b) Material for tube plates shall be ultrasonically tested. Drilled tube plates shall be checked for ovality of holes, ligaments, surface finish etc.
 - c) Dished ends shall be subjected to 100% MPE and RT or UT on welded joints. Knuckle portion shall be checked by MPE for surface defects and thinning shall be checked by UT.
 - d) Butt welds and full penetration welds shall be checked by suitable RT or UT. Fillet welds shall be checked by MPE or DPT.
 - e) Tubes shall be tested as per the relevant codes or standards.
 - f) Before tubes expansion in the tube sheets, the mockup test for expansions shall be carried out, in case not done earlier. Torque setting of expander shall be based on mock up tests. Joints shall be checked for tube thinning.
 - g) Completed assemblies shall be pressure tested. The twin oil coolers shall be tested on both tube side and shell side. After hydro test, the coolers shall be suitably dried.
 - h) Atmospheric tanks shall be tested for leakage by water fill test for at least 12 hrs.
- iii) Oil purifiers
- a) All pressure parts shall be subjected to hydraulic pressure test.
 - b) Components/ parts of the equipment shall be subjected to suitable NDT

depending upon the criticality of the application to ensure freedom from surface and sub-surface defects.

- c) All rotating parts like bowl assembly etc. shall be subjected to static and dynamic balancing test.
- d) The complete purifier shall be tested at manufacturer's works for capacity, mechanical running, sequential operation and interlocks, moisture content, vapour tightness, vibration, noise level, quality improvements etc. Sample shall be drawn from inlet and outlet of purifier after works test and shall be tested for moisture content, chemical tests and particle size of impurities. In case, type test has already been carried out by the contractor for the offered model of the oil purifier, the test shall not be repeated and certificates of test carried out earlier shall be furnished for review of the purchaser.

1.3.3 Steam Condenser

1.3.3.1 Condenser

- i) Tests shall be carried out on the materials of plates, tubes and nozzles etc. to establish their mechanical properties, and chemical compositions.
- ii) Plates shall be subject to 100% ultrasonic testing.
- iii) Tubes shall be subject to dimensional check, 100% eddy current test and 100% hydraulic test. In place of hydraulic test, air under water or Helium leak detection test can also be carried out.
- iv) Forgings of the nozzles shall be subject to 100% UT and 100% MPE.
- v) All welds shall be visually examined. Radiographic examination or Ultrasonic testing of 10% of butt welds (including T-joints) shall be carried out. However, for vacuum containing welds, RT or UT on at least 10% of each butt weld shall be carried out. Surface defect examination by MPE or equivalent test method shall be carried out for minimum 10% weldments. Nozzle welds shall be subject to 100% MPE or DPT. These shall apply to site welds also.
- vi) All edge preparations shall be examined for surface defects. Edge preparation for welds to be carried out at site shall be checked by magnetic particle examination or DP method before dispatch.
- xi) In case of fabricated flanges, welds shall be checked by 100% RT or UT and 100% MPE to ensure freedom from internal and surface defects.
- xii) To ensure dimensional control of condenser, parts/ sub-assemblies shall be trial assembled at shop. Trial insertion of a few tubes through main tube plates and support plates shall be carried out to ensure alignment of tube plates and proper fitting and matching of parts and sub-assemblies.
- xiii) Dimensional check including tube hole diameter, ligament pitch etc. shall be carried out.

- xiv) Mock-up test for tube to tube sheet expansion shall be carried out. In case such a test is already conducted for similar tube/ tube sheet thickness and materials, record for the same shall be furnished for purchaser's review and approval.
- xv) Tube to tube sheet weld joints shall be subject to 100% DP test.
- xvi) Water fill test on shell side of condenser shall be carried out. No leakage shall be permitted.
- xvii) Water fill test of standpipe, Flash tank etc. shall be carried out. No leakage shall be permitted.
- xviii) Hydraulic test of tube side of condenser shall be carried out. No leakage shall be permitted.
- xix) All compensating pads (RF pads) shall be pneumatically tested.

1.3.3.2 Spring assembly

- i) Static load testing of the springs shall be carried out and spring characteristics shall be drawn and verified.
- ii) Surface defect test shall be carried out on all the springs after coiling and heat treatment.

1.3.3.3 Condenser air evacuation system

- i) Vacuum pumps
 - a) Tests shall be carried out on the materials of Head, Body, Rotor/Impeller, Shaft and Bearing Housing etc. to establish their mechanical properties, and chemical compositions.
 - b) Vacuum pump shafts shall be subject to ultrasonic test. After finish machining, shaft shall be subject to 100% MPE or DPT.
 - c) Pump casings and impellers shall be subject to MPE or DPT. Finished pump rotor shall be subject to dynamic balancing.
 - d) Pump casings shall be subjected to hydraulic test at 1.5 times the shut off pressure or twice the maximum operating pressure, whichever is higher.
 - e) The heat exchangers shall be tested on both tube side and shell side. After hydro test, the heat exchangers shall be suitably dried.
 - f) Each pump shall be tested at supplier's works at full speed and load conditions to demonstrate successful operation and performance in accordance with the design requirements. Visual cavitation test shall also be carried out to demonstrate that pump shall be operating under all operating condition including blank off condition without any cavitation.

- ii) The complete package shall be subjected to hydraulic pressure and leakage test and shop tested to check interlocks and functional requirements. The one complete unit shall also be subjected to demonstrate successful operation and performance testing, with saturated air conditions at condenser design vacuum point as well as vacuum pump design point with total minimum three points. The test shall be conducted with the respective motors to be supplied. The test shall include check for vibration and noise level also.

1.3.3.4 Air cooled condensers for main turbine and drive turbine

- i) Fabrication of condenser and accessories
 - a) All welds shall be visually examined. Radiographic examination of 10% of butt welds shall be carried out. However, for vacuum containing welds, R.T on at least 10% of each butt weld shall be carried out. Surface defect examination by magnetic particle inspection or equivalent test method shall be carried out for minimum 10% weldments. This shall apply to site welds also.
 - b) All edge preparations shall be examined for surface defects. Edge preparation for welds to be carried out at site shall be checked by magnetic particle inspection/Dye penetrant examination method before despatch.
 - c) In case of fabricated flanges, welds shall be checked by 100% radiographic/ultrasonic and 100% magnetic particle inspection methods to ensure freedom from internal and surface defects.
- ii) Fan
 - a) Ultrasonic Test shall be carried out on shaft material diameter ≥ 40 mm.
 - b) Fan hub and fan blade shall be tested as per relevant material code requirement and shall be tested for internal defects as per relevant code.
 - c) Moment weight test on blades, blade track variation, tip clearance shall be checked.
 - d) In case of fabrication of hub and blades by welding, the weld joint shall be tested by RT.
 - e) Assembly fit up and balancing shall be checked.
- iii) Gear box
 - a) Gear box shall be tested as per relevant standard for performance, noise, vibration etc.
 - b) Gear box shall be tested for no load run test for 4 hours.
- iv) Motor: The respective specification covered elsewhere is to be referred.

v) Piping

- a) All piping joint shall be tested for 10% RT/UT. IBR code regulations 1950 shall be ensured for IBR piping and relevant original IBR certificates shall be furnished.

vi) Tubes

- a) Condenser tubes/finned tubes shall be tested as per the requirements of relevant codes and standards.

1.3.4 Feed water heaters, drain coolers, gland steam condenser and deaerator

1.3.4.1 Heaters, drain coolers and gland steam condensers

- i) Tests shall be carried out on the materials of forgings, plates and tubes etc. to establish their mechanical properties, and chemical compositions.
- ii) Plates shall be subject to 100% ultrasonic testing.
- iii) Nozzle forgings shall be subject to suitable heat treatment and 100% ultrasonic testing and 100% MPE.
- iv) Tubes (U-tubes)
 - a) Tubes shall be subject to dimensional check including U-bending (bend radius & Mockup), Check for flatness, Min. thickness at minimum radius of bend for each thickness
 - b) 100% eddy current test shall be carried out for tube thickness < 3.6 mm and 100% UT for tube thickness ≥ 3.6 mm, 100% DPT on bend areas.
 - c) Tubes shall be subject to 100% hydraulic test.
 - d) Heat treatment after cold bending, U- bend thickness check, Hardness after bending check.
 - e) Check for Residual Chloride contamination on inside & outside of the tube.
 - f) Flattening and flaring tests, IGC (Practice-E), microstructure, residual stress measurement shall also be carried out for the tubes as per applicable codes.
 - g) Before tubes expansion in the tube sheets, the mockup test for expansions shall be carried out. Torque setting of expander shall be based on mock up tests. Joints shall be checked for tube thinning. Also, tube to tube sheet welding qualification tests (mock up) to check leak path.
- v) Tube sheets
 - a) Tube sheets shall be subject to visual examination and dimensional check.

- b) Impact test shall be carried out for the plate materials. Mechanical Properties of heat treated test coupon.
 - c) 100% UT and 100% DP test shall be carried out after overlay and machining. In case overlay is not applicable, only 100% UT shall be carried out.
 - d) For clad plates, bonding shall be checked by UT. Drilled tube plates shall be checked for ovality of holes, ligaments, surface finish etc.
 - e) Tube to tube sheet weld joints shall be subject to 100% DP test. These joints shall also be subject to air test.
- vi) Shell, dished ends and hemi heads
 - a) Impact test shall be carried out for the plate materials.
 - b) 100% UT shall be carried out on the finished and formed dished ends and hemi heads.
 - c) Dished ends shall be subject to dimensional check after dishing.
 - d) Knuckle portion shall be checked by 100% MPE or DPT for surface defects and check for thinning shall be carried out by UT.
 - e) Testing of production test coupons as applicable.
- vii) Welding
 - a) Root run of butt welds shall be examined by 100% DPT or MPE.
 - b) Butt welded and full penetration joints and nozzle welds shall be checked by 100% RT or UT and 100% MPE or DPT.
 - c) Fillet welds shall be checked by 100% MPE or DPT.
 - e) All weldments shall be given suitable heat treatment.
 - f) Dished ends shall be subjected to 100% MPI and RT/UT on welded joint.
- viii) Visual check and dimensional measurement shall be carried out on the completed equipment. All compensating pads (RF pads) shall be pneumatically tested.
- ix) Completed assembly shall be pressure tested with working-fluid using hydraulic and pneumatic method. The heat exchangers shall be tested on both tube side and shell side. After hydro test, the heat exchangers shall be suitably dried and nitrogen capped.

1.3.4.2 Deaerator

- i) Tests shall be carried out on the materials of forgings, plates and tubes etc. to establish their mechanical properties, and chemical compositions.

- ii) Plates shall be subject to 100% ultrasonic testing.
- iii) Forgings shall be subject to suitable heat treatment and 100% ultrasonic testing and 100% MPE.
- iv) Shell and dished ends
 - a) Impact test shall be carried out for the plate materials.
 - b) 100% UT shall be carried out on the finished/ formed plates.
 - c) Dished ends shall be subject to dimensional check after dishing.
 - d) Knuckle portion shall be checked by 100% MPE or DP test for surface defects and check for thinning shall be carried out by UT.
 - e) Testing of production test coupons as applicable.
- v) Welding
 - a) Root run of the butt welds shall be examined by 100% DPT or MPE.
 - b) Butt welded and full penetration joints and nozzle welds shall be checked by 100% RT or UT and 100% MPE or DPT.
 - Fillet welds shall be checked by 100% MPE or DPT.
 - Dished ends shall be subjected to 100% MPI and RT/UT on welded joint.
- vi) Visual check and dimensional measurement shall be carried out on the completed equipment.
- vii) Completed assembly shall be pressure tested with working-fluid using hydraulic method. After hydro test, same shall be suitably dried.
- viii) All weldments shall be given suitable heat treatment.
- ix) All compensating pads (RF pads) shall be pneumatically tested.

1.3.4.3 Valves on heaters and deaerator

- i) Tests shall be carried out on the materials of forgings, plates and tubes etc. to establish their mechanical properties, and chemical compositions.
- ii) 100% RT or UT shall be carried out on bodies, bonnets, nozzles and stem of valves of HP heaters.
- iii) 100% DPT or MPE shall be carried out on machined surfaces of valve body, bonnet, stem, disc and springs.
- iv) Valve body shall be subject to hydraulic testing.

- v) Valve seat shall be subject to hydro leak test as per ANSI 16.34.
- vi) Visual check and dimensional measurement shall be carried out on the completed valve assembly.

1.3.5 Boiler Feed Pumps

1.3.5.1 Main pump

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition. Further, heat treatment shall be given as required.
- ii) 100% UT shall be carried out on barrel casing, casing cover, suction & discharge branches, shaft, ring sections, and wearing rings.
- iii) 100% RT shall be carried out on suction branch, impellers and diffusers.
- iv) 100% MPE shall be carried out on barrel casing, casing cover, suction & discharge branches, gland housing, shaft, diffusers, balancing drum and spring disc.
- v) 100% DPT shall be carried out on welds and overlay preparation on barrel casing, and machined components after final machining.
- vi) Hydraulic test shall be carried out on barrel casing, discharge cover, and suction and discharge branches.
- vii) Individual impellers and completed rotor assembly shall be subject to dynamic balancing test. Rotor assembly shall be subject to run out test also.
- viii) Hardness test shall be carried out on wearing rings.
- ix) Visual check and dimensional measurement shall be carried out for all the components and completely assembled pump.
- x) Final tests
 - a) Performance testing shall be carried out as per Hydraulic Institute Standards (HIS) on each pump to determine its characteristic curve at design speed and to ensure compliance with design requirements. Tests shall be carried out with loop water at specified design temperature. Soften quality water shall be used for the performance testing.
 - b) Vibration on all pumps shall be measured in transverse, horizontal and vertical direction at all measuring points.
 - c) Noise Level on each pump shall be measured at a distance of 1.5 m above floor level in elevation and 1 m horizontally from the nearest surface of the equipment as per HIS. The measurement shall be taken at six points around the equipment for each flow condition.

d) Type tests

1. NPSH (R) test shall be carried out on one pump using cold water at pump flows of 25%, 50%, 80%, 100% and 125% of design flow at design speed. This shall be preferably done at 1% and 3% head break by suction throttling procedure. In case, NPSH (R) test has already been carried out by the contractor for the offered frame of BFP, the test shall not be repeated and certificates of test carried out earlier shall be furnished for review of the purchaser. The validity of the type test carried out earlier shall be limited to five (5) years.
2. Pressure pulsation and axial thrust measurement shall be carried out on one boiler feed pump at all measuring points. Pressure pulsation shall be measured at suction as well as at discharge in the operating range.

e) Strip down test

Complete strip down of one feed pump which undergoes NPSH test shall be done after completion of all the tests on it. The strip down shall check for the condition of thrust bearing and journal bearing, and problems such as internal rubbing damage, excessive wear on the components. For other feed pumps, strip down examination shall be restricted to inspection of bearings only. However, if excessive vibration, high noise, high bearing temperature etc. is observed during performance test of any feed pump, complete strip down shall be done for such pumps also.

Note: Tested pump parameters shall be within following tolerances.

At design head : + 10% of design capacity

At design capacity : + 5% of design head (for < 152.4 m)
+ 3% of design head (for ≥ 152.4 m)

The results of the performance test must show no minus tolerance with regard to flow and head. No minus tolerance on efficiency or positive tolerance on power input at motor terminals shall be allowed.

1.3.5.2 Booster pump

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition. Further, heat treatment shall be given as required.
- ii) 100% UT shall be carried out on shaft and wearing rings.
- iii) 100% MPE shall be carried out on casing, impeller and seal cooling jacket.
- iv) 100% DPT test shall be carried out on machined components after final machining.
- v) Hydraulic test shall be carried out on pump casing at 1.5 times the shut off pressure

or 2 times the working pressure whichever is higher.

- vi) Impeller and completed rotor assembly shall be subject to dynamic balancing test. Rotor assembly shall be subject to run out test also.
- vii) Visual check and dimensional measurement shall be carried out for all the components and completely assembled pump.
- viii) Final tests
 - a) Performance testing shall be carried out as per Hydraulic Institute Standards (HIS) on each pump to determine its characteristic curve at design speed and to ensure compliance with design requirements. Tests shall be carried out with loop water at specified design temperature. Soften quality water shall be used for the performance testing.
 - b) Vibration on all pumps shall be measured in transverse, horizontal and vertical direction at all measuring points.
 - c) Noise Level on each pump shall be measured at a distance of 1.5 m above floor level in elevation and 1 m horizontally from the nearest surface of the equipment as per HIS. The measurement shall be taken at six points around the equipment for each flow condition.
 - d) Type tests
 - 1. NPSH (R) test shall be carried out on one pump using cold water at pump flows of 25%, 50%, 80%, 100% and 125% of design flow at design speed. This shall be preferably done at 1% and 3% head break by suction throttling procedure. In case, NPSH (R) test has already been carried out by the contractor for the offered frame of booster pump, the test shall not be repeated and certificates of test carried out earlier shall be furnished for review of the purchaser. The validity of the type test carried out earlier shall be limited to five (5) years.
 - f) After performance testing, thrust bearing and journal bearing shall be subject to visual check.

Note: Tested pump parameters shall be within following tolerances.

At design head	:	+ 10% of design capacity
At design capacity	:	+ 5% of design head (for < 152.4 m) + 3% of design head (for ≥ 152.4 m)

The results of the performance test must show no minus tolerance with regard to flow and head. No minus tolerance on efficiency or positive tolerance on power input at motor terminals shall be allowed.

1.3.5.3 Gear box and hydraulic coupling

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition. Further, heat treatment shall be given to the internal components as required.
- ii) Internal components such as gears, pinions, wheels and shafts shall be examined by 100% UT and 100% DPT or 100% MPE.
- iii) Leak test shall be carried out for the casing.
- iv) Dynamic balancing test shall be carried out for the assembled rotating component.
- v) Full load speed and back to back locked rotor torque test shall be carried out on one gear box.
- vi) Visual check and dimensional measurement shall be carried out for the completely assembled equipment.

1.3.5.4 Strainers

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition.
- ii) In case of fabricated construction, the welds shall be examined for surface defects by 100% DPT.
- iii) The strainer body shall be subject to hydraulic test.
- iv) Pressure drop test shall be carried out for each type and size of the strainer assembly.

1.3.5.5 BFP drive turbine and associated equipment

The QA requirements for BFP drive turbine and associated auxiliaries shall be same as those applicable for main steam turbine and its auxiliaries described at clause 1.3.1 of this Section.

1.3.6 Condensate Extraction Pumps (CEPs)

1.3.6.1 Pump

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition. Further, heat treatment shall be carried out for materials of shaft and rotor.
- ii) 100% UT shall be carried out on pump shaft.
- iii) 10% RT shall be carried out for butt welds on fabricated components of thickness more than 10 mm.

- iv) 100% MPE shall be carried out on pump shaft.
- v) 100% DPT shall be carried out on welds on casing, suction bell, shaft, impeller and fabricated components.
- vi) Hydraulic test shall be carried out on casing and pressure containing fabricated parts at 1.5 times the pump shut off pressure or 2 times the working pressure whichever is higher.
- vii) Individual impellers and completed rotor assembly shall be subject to dynamic balancing test. Rotor assembly shall be subject to run out test also.
- viii) Visual check and dimensional measurement shall be carried out for all the components and completely assembled pump.
- ix) Final tests

- a) Performance testing shall be carried out as per Hydraulic Institute Standards (HIS) on each pump to determine its characteristic curve at design speed and to ensure compliance with design requirements. Tests shall be carried out using cold soften quality water.
- b) Vibration on all pumps shall be measured in transverse, horizontal and vertical direction at all measuring points.
- c) Noise Level on each pump shall be measured at a distance of 1.5 m above floor level in elevation and 1 m horizontally from the nearest surface of the equipment as per HIS. The measurement shall be taken at six points around the equipment for each flow condition.
- d) NPSH test

NPSH (R) test shall be carried out on one pump using cold water at pump flows of 25%, 50%, 80%, 100% and 125% of design flow at design speed. This shall be preferably done at 1 % and 3% head break. In case, NPSH (R) test has already been carried out by the contractor for the offered frame of CEP, the test shall not be repeated and certificates of test carried out earlier shall be furnished for review of the purchaser. The validity of the type test carried out earlier shall be limited to five (5) years.

- e) Strip down test

Complete strip down of one pump which undergoes NPSH test shall be done after completion of all the tests on it. The strip down shall check for the condition of bearings and problems such as internal rubbing, excessive wear. For other pumps, strip down shall be restricted to inspection of bearings only. However, if excessive vibration, high noise etc. is observed during performance test of any pump, complete strip down shall be done for such pumps also.

Note: Tested pump parameters shall be within following tolerances.

At design head	:	+ 10% of design capacity
At design capacity	:	+ 5% of design head (for < 152.4 m) + 3% of design head (for ≥ 152.4 m)

The results of performance test must show no minus tolerance with regard to flow and head. No minus tolerance on efficiency or positive tolerance on power input at motor terminals shall be allowed.

1.3.6.2 Strainers

The QA requirements for strainers at suction of the CEPs shall be same as those applicable for BFP strainers described at clause 1.3.5.4 of this Section.

1.3.7 Condensate Polishing Unit (CPU)

1.3.7.1 CPU service vessels

- i) Tests shall be carried out on the materials of the vessels, internals and rubber used for lining to establish their mechanical properties, and chemical compositions. Heat treatment as required shall be done as per ASME code.
- ii) Plates shall be subject to 100% ultrasonic testing.
- iii) Impact test shall be carried out for the plate materials.
- iv) 100% UT shall be carried out on the finished and formed dished ends and hemi heads. Knuckle portion shall be checked by 100% MPE or DP test for surface defects and check for thinning shall be carried out by UT.
- v) Welding
 - a) Root run of butt welds shall be examined by 100% DPT or MPE.
 - b) Butt welds full penetration joints and nozzle welds shall be checked by 100% RT and 100% MPE or DPT.
 - c) Fillet welds shall be checked by 100% MPE or DPT.
- vi) Rubber lining shall be subjected to following tests as per IS-4682 part-I or acceptable equivalent:
 - a) Tensile & Elongation, Specific Gravity, Ash content
 - b) Adhesion test
 - c) Measurement of thickness
 - d) Shore hardness test

- e) Visual examination and spark test at 5 kV/mm of thickness
- f) Bleeding resistance test with keeping the sample in 33% HCl, 48% NaOH and DM water for 72 hours.
- g) Ozone resistance test: (50 PPHM) / 40° C / 20 % Strain /70 Hrs
- h) Water Absorption Test, After ageing test & Compression Set test
- i) Spark test for rubber lining
- vii) Resin testing shall be carried out as per relevant IS.
- viii) Visual check and dimensional measurement shall be carried out on the completed equipment.
- ix) Internals of the vessel shall be subject to dimensional check and applicable tests as per relevant codes.
- x) The fabricated vessel shall be hydraulically tested at 1.5 times the working pressure before the rubber lining and at the working pressure after the rubber lining.

1.3.7.2 Acid/ alkali handling tanks

- i) Tests shall be carried out on the materials of the tanks to establish their mechanical properties, and chemical compositions
- ii) Plates shall be subject to 100% ultrasonic testing.
- iii) Welding
 - a) Root run of butt welds shall be examined by 100% DPT or MPE.
 - b) Butt welds, full penetration joints and nozzle welds shall be checked by 100% RT and 100% MPE or DPT.
 - c) Fillet welds shall be checked by 100% MPE or DPT.
- iv) Rubber lining shall be subject to relevant tests as described for CPU vessels.
- v) Visual check and dimensional measurement shall be carried out on the completed equipment.
- vi) The fabricated tank shall be subject to water fill test to check for the leakage.

1.3.7.3 Dosing pumps/ metering pumps

- i) Tests shall be carried out on the materials of the pumps to establish their properties, and chemical compositions

- ii) Pump casings shall be subject to hydraulic test at 1.5 times the shut off pressure or 2 times the working pressure whichever is higher.
- iii) 100% DPT or MPE shall be carried out for the screw set, shaft and machined surfaces of casing and impellers. UT on screw set, shaft (diameter greater than or equal to 40mm),
- iv) Safety relief valve setting to be checked
- v) Pumps shall be performance tested as per HIS, USA.

1.3.7.4 Horizontal centrifugal pumps

The QA requirements for horizontal centrifugal pumps shall be as per clause 1.5.1.1.17 of this Section.

1.3.7.5 Rotary blowers

- i) Tests shall be carried out on the materials of the rotary blowers to establish their properties, and chemical compositions.
- ii) 100% DPT or MPE shall be carried out for the rotor and machined surfaces of casing and impellers. UT on shaft (diameter greater than or equal to 40mm).
- iii) The shaft and impellers shall be dynamically balanced.
- iv) Assembly fit up check, and dimensional check shall be carried out for the completed blower assembly.
- v) The blower casing shall be subject to hydraulic test at 1.5 times the shut off pressure or 2 times the working pressure whichever is higher.
- vi) The blowers shall be performance tested as per relevant code/ standard for Free Air Delivery, Head, Power, Efficiency, Speed including Discharge Pressure and Temperature. The test shall include check for vibration and noise level also.

1.3.7.6 Valves

The QA requirements for high pressure valves such as for service vessels shall be as per clause 1.4.2 of this Section. The QA requirements for other valves shall be as per clause 1.4.6.2 of this Section.

1.3.8 Condenser On-Line Tube Cleaning System

1.3.8.1 Ball recirculation pump

- i) All rotating parts shall be dynamically balanced.

- ii) Pump casing shall be subjected to hydraulic test at 1.5 times the shut off head or twice the maximum working pressure whichever is higher.
- iii) Complete pump assembly shall be subjected to shop performance test at supplier's works.

1.3.8.2 Ball sorter/ fabricated body (housing)

- i) In the case of fabricated design, all butt welds shall be subject to 10% RT or UT. All welds shall also be subjected to 10% MPE to ensure freedom from surface and sub-surface defects.
- ii) Body shall be subject to hydraulic pressure test at 1.5 times the design pressure.
- iii) Performance test shall be carried out on ball sorter assembly.

1.3.8.3 Strainer

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition
- ii) Strainer mesh shall be checked for chemical composition and mesh size.
- iii) Strainer body shall be subject to hydraulic pressure test at 1.5 times the design pressure.
- iv) Strainer assembly shall be checked for its functional performance.

1.3.8.4 Piping and fittings

- i) Butt welds on piping shall be subject to 10% RT and 10% DPT. Butt welds on segmental flanges shall be checked by 100% RT and 100% DPT.
- ii) Fillet welds with load transfer shall be subject to 100% MPE or DPT and fillet welds without load transfer shall be subjected to 10% MPE or DPT.
- iii) Wrought and forged fittings shall be tested as per relevant codes or standards.

1.3.8.5 Coating/ lining

- i) Coating shall be checked for DFT and adhesion. Further, Contractor shall furnish his practice for testing of coating to ensure the uniformity and freedom from pinholes
- ii) Rubber lined items shall be hydraulically tested before rubber lining. All rubber lining shall be subjected to following tests as per IS-4682 part-I or acceptable equivalent:
 - a) Tensile & Elongation, Specific Gravity, Ash content
 - b) Adhesion test

- c) Measurement of thickness
- d) Shore hardness test
- e) Visual examination and spark test at 5 kV/mm of thickness
- f) Bleeding resistance test with keeping the sample in 33% HCl, 48% NaOH and DM water for 72 hours.
- g) Ozone resistance test: (50 PPHM) / 40° C / 20 % Strain /70 Hrs.
- h) Water Absorption Test, After ageing test & Compression Set test

1.3.8.6 Valves

The QA requirements for the valves shall be as per clause 1.4.6.2 of this Section.

1.3.9 Debris Filter

- i) Body, strainer mesh and other components shall be checked for chemical composition.
- ii) Strainer element shall be checked for mesh size.
- iii) Body shall be subject to hydraulic pressure test at 1.5 times the design pressure.
- iv) Filter assembly shall be checked for its functional requirements.
- v) Valves shall be tested as per relevant standards.

1.4 HIGH PRESSURE PIPING, VALVES, THERMAL INSULATION AND MISCELLANEOUS SYSTEMS/ EQUIPMENT

1.4.1 High Pressure Piping and Fittings

- i) All raw materials used shall have co-related mill test certificate meeting mandatory and supplementary checks of material specification.
- ii) The pipes and fittings shall be subject to visual examination, identification, correlation and stamping.
- iii) For alloy steel pipes (SA 335, P11, P22, & P91), 25% pipes per lot shall be subject to the following tests:
 - a) Product analysis
 - b) Transverse tension test on pipes from one end for pipe size 200 mm and above.
 - c) Flattening test on pipe from one end.

- iv) All pipe lengths shall be subjected to 100% UT.
- v) All mother pipes used for fittings shall be subject to a hydraulic test or an ultrasonic test. Raw material of all forged fittings shall be ultrasonically tested. Forged fittings shall be ultrasonically tested.
- vi) All alloy and carbon steel pipes shall be subject to 100% UT for pipe thickness ≥ 3.6 mm and 100% eddy current test for thickness < 3.6 mm as per material specification standard with acceptable notch depth of 5% of the selected wall thickness (1.5mm maximum)
- vii) Thickness of all pipe bends (cold/ hot formed) shall be checked by ultrasonic or other acceptable methods on sample basis for high pressure applications. Further, outer surface of bends shall be subject to 100% MPE or DPT. Pipe bend shall be checked for ovality and thinning by ultrasonic or other acceptable methods on first off lot & on random samples for subsequent pieces for high pressure applications. Outer surface of bends shall be subjected to magnetic particle examination/LPI.
- viii) Welded and cast fittings, if any, shall be subjected to suitable NDT as per applicable standards. However, as a minimum, 100% RT shall be carried out on all alloy steel fittings and on carbon steel fittings for use above 71 bar design conditions.
- ix) The edge preparation for stainless steel/ alloy steel shall be subject to a 100% DPT. For other piping, the edge preparation for the welds shall be checked by 100% MPE or DPT.
- x) Non-destructive examination of welds shall be carried out in accordance with the relevant design and/or manufacturing codes. However, as a minimum, the following requirements shall be met (except for oil piping):
 - a) For temperature $> 400^{\circ}\text{C}$ and/or pressure exceeding 71 bar, the butt welds and full penetration branch welds shall be subject to 100% RT or UT and 100% MPE.
 - b) For temperature $> 175^{\circ}\text{C}$ up to 400°C and/or pressure exceeding 17 bar and up to 71 bar, the butt welds & full penetration branch welds shall be subject to NDTs as below:

- pipe diameter > 100 NB	100% RT or UT and 100% MPE
- pipe diameter ≤ 100 NB	10% RT or UT and 100 % MPE

In addition to above, statutory requirement, wherever applicable, shall also be complied with.

- xi) Finished welding for alloy steel piping, BFP discharge piping, MS piping, HRH piping, CRH piping and associated piping shall be subject to the following NDTs:
- | | |
|---|----------------------------|
| a) Butt welds | 100% RT and 100% MT or DPT |
| b) Welds for nozzle, branch connections of size > 100NB | 100% RT and 100% MT or DPT |

- c) Welds for all other attachments 100% MT or DPT
 - d) Removal of weld defects 100% MT or DPT
- xii) Finished welding for Carbon Steel Class- I piping having design temperature > 218 deg. C or design pressure for steam > 17.6 kg/ cm² (g) or design pressure for feed water > 24.6 kg/cm² (g) shall be subject to the following NDTs:
 - a) Butt welds:
 - pipe size > 100 NB 100% RT and 100% MPE or DPT
 - pipe size ≤ 100 NB 10% RT and 10% MPE or DPT
 - b) Welds for nozzle, branch connections of size > 100NB and thickness > 19 mm 100% RT and 100% MPE or DPT
 - c) Welds for all other attachments 10% MPE or DPT
 - d) Removal of weld defects/weld repair 100% MPE or DPT
 - xiii) Finished welding for carbon steel class- I piping having design temperature ≤ 218 deg. C or design pressure for steam ≤ 17.6 kg/ cm² (g) or design pressure for feed water ≤ 24.6 kg/cm² (g) shall be subject to the following NDTs:
 - a) Butt welds:
 - pipe size > 100 NB 100% RT and 10% MPE or DPT
 - pipe size ≤ 100 NB 10% MPE or DPT
 - b) Welds for nozzle, branch attachments 10% MPE or DPT connections/
 - c) Removal of weld defects 100% MPE or DPT
 - xiv) Non-destructive examination of welds shall be carried out after post weld heat treatment, if any.
 - xv) Pre- heating, stress relieving and post weld heat treatment as applicable shall be carried out as per requirements of ASME B 31.1. For welds in P91, X20 & X22 materials, only induction type of heating shall be deployed for heat treatment.
 - xvi) Hardness survey of welds shall be carried out on alloy steel/ stainless steel piping. (100% hardness survey of welds on P91, X20 & X22 material grade piping).
 - xvii) All other pipes not covered above (except oil piping) shall be subjected 100% MPE or DPT in case of underground piping and 10% MPE or DPT in case of over ground piping. Further, 10% of butt welds of underground piping shall be subjected to RT.
 - xviii) Oil piping shall be subjected to following NDTs.
 - a) Butt welds of Oil piping shall be subjected to 10% RT/ UT and 10% DP Test. For Jacking oil lines 100% RT/UT and 100% DPT shall be carried out on butt

welds.

- b) Fillet welds with load transfer shall be subjected to 100% MPE or DPT and fillet welds without load transfer shall be subjected to 10% MPE or DPT.
- xix) Rubber lined pipes shall be hydraulically tested before rubber lining. All rubber lining shall be subject to following tests as per IS-4682 part-I or acceptable equivalent:
 - a) Tensile & Elongation, Specific Gravity, Ash content
 - b) Adhesion test
 - c) Measurement of thickness
 - d) Shore hardness test
 - e) Visual examination and spark test at 5 kV/mm of thickness
 - f) Bleeding resistance test with keeping the sample in 33% HCl, 48% NaOH and DM water for 72 hours.
 - g) Ozone resistance test: (50 PPHM) / 40° C / 20 % Strain /70 Hrs.
 - h) Water Absorption Test, After ageing test & Compression Set test

1.4.2 Power Cycle Valves

1.4.2.1 Valves other than extraction line valves and butterfly valves

- i) Bar stock/forging above 40mm diameter for valve trim shall be subjected to UT.
- ii) Hardened/ stellitted valve disc and seat are to be subjected to LPI and hardness check.
- iii) Colour matching of valve disc, plug and seat shall be carried out to ensure contact ensure min. 80% contact and no through passage.
- iv) Hydraulic pressure test and seat leak test shall be carried out as per ANSI 16.34.
- v) Air seat leak test shall be carried out as per applicable standards/ codes.
- vi) Pressure retaining parts of valves shall be subject to NDTs as below:

Valve size NB in mm	ANSI Class up to 300	ANSI Class above 300 up to 600	ANSI Class above 600 below 900	ANSI Class 900 & above & below 4500
Less than	Visual	Visual	Visual	MPI/DPT

50				
50 & above but below 100	Visual	Visual	MPI/DPT	MPI/DPT & RT (on 10% of valves on 100% area)
100 & above but less than 300	Visual	MPI/DPT	MPI/DPT & RT (on 10% of valves on change of section & weld ends)	MPI/DPT & RT (on 100% area)
300 and above	MPI/DPT	MPI/DPT	MPI/DPT & RT (on change of sections & weld ends)	MPI/DPT & RT (on 100% area)

- vii) Weld edge preparations shall be subject to MPE or DPT.
- viii) Functional testing shall be carried out on fully assembled valve to check the following:
 - a) Smooth operation
 - b) Valve travel, closing and opening time.
 - c) Current drawn by actuators.
 - d) Pressure resistance test of pneumatic actuator
 - e) CV test for control valve
- ix) Springs for safety valves shall be tested with suitable NDT and for spring rate.
- x) Safety and safety relief valves shall be tested for performance.

1.4.2.2 Extraction line valves

- i) Surface crack examination and hardness check shall be carried out on all hard faced surfaces and stellite surfaces, if any.
- ii) As a minimum requirement of castings for all valves on cold reheat and extraction lines shall be subjected to 100% MPE on all areas and RT on butt weld ends and change of section. For forgings minimum requirement shall be 100% UT and 100% MPE.

- iii) Bar stock for valves stem shall be subjected to UT. Finish machined valve stem shall be subjected to MPE or DPT.
- iv) Wall thickness measurement by ultrasonic for critical and highly stressed zones of the casting and forging shall be carried out.
- v) Colour matching of the valve disc and seat to ensure required contact area shall be carried out.
- vi) Hydraulic pressure tests shall be carried out on each valve to check body and bonnet strength. Seat leakage and back seat leakage test (wherever applicable) shall be carried out. Air seat leakage test shall also be carried out. Minimum test requirements of pressure shall be as per ANSI B 16.34.
- vii) Functional testing shall be carried out on each valve to check for freedom of movement, adherence to clearance, opening/ closing etc.

1.4.2.3 Butterfly Valves

- i) Tests shall be carried out on the materials of body, disc etc. for determination of chemical and mechanical properties.
- ii) In case of fabricated valves, the plates used for body, disc and flanges shall be subject to 100% UT.
- iii) Valve body, disc and shaft shall be checked for surface and sub-surface defects by 100% MPE.
- iv) For sea water application valves, austenitic stainless steel welds shall be subject to 100% IGC (inter-granular corrosion) test.
- v) All wetted SS 316 components shall be subject to Molybdenum check.
- vi) Stubs and driving shafts shall be tested for internal defects by ultrasonic method.
- vii) Dye penetration test shall be carried out on shafts, seat rings etc.
- viii) For fabricated components of the valves, all the longitudinal/ circumferential weld seams shall be subject to 100% RT. Further, all welds on magnetic material shall be subject to 100% MPE, and welds on non-magnetic material shall be subject to 100% DPT.
- ix) Test samples for rubber seal shall be subjected to tensile, elongation and hardness test for vulcanising and after ageing. Hydraulic stability test (Bleed Resistance) and ozone crack resistance tests also be carried out.
- x) Valve shall be subjected to hydraulic pressure test for body and air seat leakage tests as per AWWA-C504/ BS- 5155.
- xi) Proof of design tests for valves and actuator shall be carried out as per AWWA-C504/ BS- 5155. In case the test has already been carried out on previous supplies,

the contractor may submit the test certification of same for approval of purchaser.

- xii) Disc Strength test by Hydro (Forward and Backward) as per EN 12266-1
- xiii) After complete assembly each valve with actuator shall be subject to the performance test by opening and closing the valve from fully closed to fully open position and the reverse, under no flow for at least 25 cycles to check the following:
 - a) Smooth uninterrupted movement of valve.
 - b) Closing and opening time.
 - c) Current drawn by actuator.

3. Operation of tripping switch
and position indicator.

- xiv) After assembly, one valve of each size with respective actuator shall be shop operated over the full range of movement in both the directions, with the body subjected to the full hydrostatic pressure conditions, to demonstrate that the unit is in working order without any leakage through the joints and torque switches/clutches, limit switches are operating satisfactorily. During the test, hand wheel operation, opening and closing time and current drawn shall also be checked. The test shall be conducted for three consecutive cycles with valve shaft both in vertical and horizontal planes.

1.4.3 Metallic expansion bellows

- i) All raw materials used shall have co-related mill test certificate meeting mandatory checks of material specification.
- ii) Hydraulic pressure test shall be carried out on each pipe and expansion bellow.
- iii) Longitudinal butt weld on bellow shall be subject to MPE or DPT before forming and after forming.
- iv) All welds shall be subject to 100% MPE or DPT. Butt welds shall be subject to 100% RT.
- v) All the bellows subjected to vacuum service shall be vacuum tested.
- vi) The bellows shall be subjected to movement test to establish suitability to perform satisfactorily in site conditions. During this test, spring rate shall also be measured.
- vii) Life cycle test, meridional yield rupture test and squirm test to be carried out on a prototype/ expansion bellow as per Sec. D clause 3.2 of standards of Expansion Joint Manufacturer Association (EJMA).

1.4.4 Hangers and Supports

- i) All raw materials used shall have co-related mill test certificate meeting mandatory checks of material specification.
- ii) Completed springs shall be tested for sagging test and load versus deflection test. For diameter more than 25mm, MPE shall also be carried out.
- iii) Butt welds shall be tested for UT/RT and fillet welds shall be tested for MPI/DPT.
- iv) Dampers with viscous fluids shall be checked for viscosity of liquid used, damping resistance of the damper, stiffness of the damper etc.
- v) Turn-buckle, pipe clamps and hangers of thickness greater than 25mm shall be checked by MPE or DPT on bent portions.
- vi) Assembled hanger, one number of each type and size shall be checked for variation in deflection and travel versus load test.

1.4.5 Thermal insulation, refractory, lagging and cladding

- i) Thermal insulation
 - a) Pre-formed fibrous pipe insulation and LRB mattresses/ sections of rock wool/ mineral wool from approved manufacturing sources conforming to and tested as per relevant standards shall be used.
 - b) For resin bonded mineral wool insulation, testing shall be carried out as per IS: 8183.
 - c) For resin bonded rock wool insulation, testing shall be carried out as per IS: 9842.
 - d) For sprayed mineral wool, testing shall be carried out as per IS: 9724.
 - e) For ceramic fiber blankets and block insulation, testing shall be carried out as per IS: 15402.
 - f) Thermal conductivity shall be regularly carried out once in three months.
 - g) Thermal conductivity (K value) shall be measured in line with IS: 3346.
 - h) Wire mesh of diameter 0.71mm (minimum) shall only be used.
- ii) Castable refractory

Fire bricks or castable refractory from approved manufacturing sources conforming to and tested as per relevant standards shall be used. Castable refractory shall have proper identification, supplier name, customer name, batch no., date, material name and net weight in kg with proper instructions for handling.
- iii) Lagging and cladding

All insulation shall be protected by means of an outer covering of aluminum sheeting conforming to ASTM B-209-1060 temper H14 from reputed manufacturer.

1.4.6 Low Pressure Piping, Valves and Fittings etc.

1.4.6.1 Pipes, fittings and mitre bends

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition. Further, heat treatment shall be carried out for materials as required.
- ii) Dye penetration test of welds of pipes and fittings (including welds of rolled and welded pipes) shall be carried out.
- iii) All pipes and fittings shall be tested as per applicable codes/ standards at manufacturer's works.

1.4.6.2 Valves

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition. Further, heat treatment shall be carried out for materials as required.
- ii) Shaft/ spindle of size ≥ 40 mm diameter shall be subjected to ultrasonic test.
- iii) Machined surfaces of casing, disc and shaft shall be subjected to 100% MPI or DPT as applicable.
- iv) All valves shall be hydraulically tested for body, seat and back seat (wherever provided) at 1.5 times the maximum pressure to which respective valves can be subjected during plant operation. Check valves shall also be tested for leak tightness test at 25% of the specified seat test pressure. For rubber lined valves, hydraulic test shall be carried out before rubber lining.
- v) For butterfly valves, hydraulic test, seat and disc string test and proof of design test (if not carried out earlier) shall be carried out in accordance with latest edition of AWWA-C-504/ C-516 standard. Actuators shall also meet requirements of P.O.D. test of AWWA-C-504/AWWA-C-516.
- vi) Visual and dimensional check shall be carried out for all valves as per relevant code/ approved drawing.
- vii) Functional/ operational checks for and check for smooth opening and closing of the valves shall be carried out.
- viii) Gate, globe and swing check valves
 - a) Machined surfaces of castings and butt welds shall be subjected to MPE or DPT.

- b) Blue matching, wear travel for gates, valves, pneumatic seat leakage, reduced pressure test for check valves shall be done as per relevant standard.
- ix) Diaphragm valves
 - a) Seat leakage test for actuator operated valves, shall be done with by closing the valves with actuator.
 - b) Tests on rubber parts shall be conducted per batch of rubber mix for tensile, elongation, hardness, adhesion, spark test, bleed resistance test. In addition, type test for 50,000 cycles of each type of diaphragm shall also be conducted.
- x) Cast butterfly valves
 - a) Machined surfaces of casing, disc and shaft shall be subjected to MPE or DPT.
 - b) Actuator operated valves shall be checked for seat leakage by closing the valves with actuator. Seat leakage test shall be carried out in both directions.
- xi) Fabricated butterfly valves (refer 1.4.2.3)
 - a) UT as per ASTM A-435/IS 11630 & IS 4225 on plate material for body and disc shall be carried out for plate thickness 25mm and above.
 - b) Machined surfaces of casing, disc and shaft shall be subjected to MPE or DPT.
 - c) 100% RT and DPT as per ASTM, Section-VIII, Division-I, on butt joints of body and disc. 10% DPT on other welds shall be done. Post weld heat treatment as per ASME, Section-VIII, Division-I on butt joints of body and disc.
 - d) Actuator operated valves shall be checked for seat leakage by closing the valves with actuator. Seat leakage test shall be carried out in both directions.
- xii) Dual plate check valves
 - a) Machined surfaces of casing, disc and shaft shall be subjected to MPI or DPT. Dry Cycle Test on Dual Plate Check valve spring for one lakh Cycles shall be carried out as a type test. If Dry Cycle test carried out earlier for same material & diameter, Test report shall be reviewed.

1.4.6.3 Rubber lining of pipes and valves

- i) For rubber lining, the following tests shall be carried out as per IS-4682 part-I or acceptable equivalent standard:
 - a) Tensile & Elongation, Specific Gravity, Ash content
 - b) Adhesion test
 - c) Measurement of thickness

- d) Shore hardness test
- e) Visual examination and spark test at 5 kV/mm of thickness
- f) Bleeding resistance test with keeping the sample in 33% HCl, 48% NaOH and DM water for 72 hours.
- g) Ozone resistance test: (50 PPHM) / 40° C / 20 % Strain /70 Hrs
- h) Water Absorption Test, After ageing test & Compression Set test.
- ii) Dimensional check shall be carried out as per relevant code/ approved drawing.

1.4.6.4 Coating and wrapping of pipes

Spark test/ Holiday test, adhesion test and material test for primer and enameled and coal tar tapes, as applicable, shall be carried out as per AWWA-C-203-91/ IS 15337/ IS 10221 as applicable.

1.4.6.5 Rubber expansion joints

- i) Rubber compound test slab after vulcanising shall be tested for tensile strength, elongation and shore hardness. Tests on rubber compound shall also include hydro stability test as per ASTM D-3137, ozone resistance test as per ASTM D- 380 and kerosene dip test.
- ii) Fabric strength of synthetic fiber for reinforcement shall be checked, and test for rubber to fabric adhesion as per IS: 3400 or ASTM D- 413, rubber to metal adhesion as per IS 3100 or ASTM D-429 shall be carried out.
- iii) All expansion joints in assembled condition shall be subjected to vacuum test at 730 mm Hg under conditions to ensure its suitability to withstand deflection in each axial transverse and longitudinal direction. Duration of test shall be of minimum 10 minutes.
- iv) All bare bellows shall be subjected to hydraulic pressure test in normal condition at twice the design pressure for a duration of 30 minutes. Additionally, all bare bellows shall be subjected to deflection tests under pressure, pressure being raised from zero to the design value in regular steps and deflection measured at each step.
- v) All expansion joints in assembled condition along with control rod assembly shall be subjected to deflection test (axial compression & expansion test and lateral deflection) under design pressure.
- vi) Either during the hydraulic test or during the vacuum test, change in circumference at the top position of the arch shall not exceed 1.5% of measured circumference at normal position.

vii) Twenty-four (24) hours after the above tests, the permanent set (variation in dimensions with respect to its original dimension) shall be measured and recorded. The permanent set shall not be more than 0.5%.

vii) Proof of Design (Life cycle test and burst test) shall be carried out on bellows of each type, design and size.

1.4.7 Equipment Cooling Water (ECW) System

1.4.7.1 Primary side and secondary side pumps

i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition. Further, heat treatments shall be carried out for materials of shaft and rotor.

ii) 100% UT shall be carried out on pump shaft.

iii) 10% RT shall be carried out for butt welds on fabricated components of thickness more than 10 mm.

iv) 100% MPE shall be carried out on pump shaft.

v) 100% DPT shall be carried out on welds on casing, suction bell, shaft, impeller and fabricated components.

vi) Hydraulic test shall be carried out on casing and pressure containing fabricated parts at 1.5 times the pump shut off pressure or 2 times the working pressure whichever is higher.

vii) Individual impellers and completed rotor assembly shall be subject to dynamic balancing test. Rotor assembly shall be subject to run out test also.

viii) Visual check and dimensional measurement shall be carried out for all the components and completely assembled pump.

ix) Final tests

a) Performance testing shall be carried out as per Hydraulic Institute Standards (HIS) on each pump to determine its characteristic curve at design speed and to ensure compliance with design requirements.

b) Vibration on all pumps shall be measured in transverse, horizontal and vertical direction at all measuring points.

c) Noise level on each pump shall be measured at a distance of 1.5 m above floor level in elevation and 1 m horizontally from the nearest surface of the equipment as per HIS.

d) NPSH(R) test shall be carried out on one pump at pump flows of 25%, 50%, 80%, 100% and 125% of design flow at design speed. In case, NPSH (R) test

has already been carried out by the contractor for the offered frame of a pump, the test shall not be repeated and certificates of test carried out earlier shall be furnished for review of the purchaser.

- e) Complete strip down of one pump shall be done after completion of all the tests on it. The strip down shall check for the condition of bearings and problems such as internal rubbing, excessive wear.

1.4.7.2 Plate heat exchangers

- i) The material used for cover plates, heat exchange plates and tie rods shall be subject to chemical and mechanical tests on one per heat basis. For gasket, ageing test, shrinkage test and hardness to be carried out.
- ii) Each plate after pressing shall be subject to light box test or vacuum test or air chamber test as per manufacturer's practice.
- iii) UT shall be done for plates with thickness 25 mm or above.
- iv) DPT shall be conducted for 10% of the lot of heat exchanger plates. However, in case of any defects, entire lot shall be tested and only defect free plates shall be accepted.
- v) 100% DPT shall be conducted on all welds.
- vi) Each heat exchanger shall be subjected to hydraulic test.
- vii) Assembly fit up and dimensional checks shall be carried out for each heat exchanger.

1.4.7.3 Auto clean filters

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition.
- ii) In case of fabricated construction, the welds shall be examined for surface defects by 100% DPT.
- iii) The body shall be subject to hydraulic test at 1.5 times the design pressure.
- ii) Pressure drop, flow and particle size tests shall be carried out for the filter assembly.

1.4.7.4 Pipes, Valves and fittings

The QA requirements for piping, valves and fittings shall be as per clause 1.4.6 of this Section.

1.4.8 EOT Cranes and Hoists

1.4.8.1 Hooks

- i) All tests including proof load test as per relevant IS shall be carried out.

- ii) MPE or DPT shall be done after proof load test.

1.4.8.2 Steels castings

Steel castings shall be subjected to DPT on machined surface.

1.4.8.3 Girders, end carriage, crab, gear-box and rope drum

- i) The plates of thickness 25mm and above for girders, end carriage, crab, gear-box and rope drum shall be ultrasonically tested.
- ii) NDT requirements on weldments shall be as follows:
 - i) Butt welds in tension : 100% RT and 100% DPT
 - ii) Butt welds in compression : 10% RT and 100% DPT
 - iii) Butt weld in rope drum : 100% RT and 100% DPT
 - iv) Fillet welds : 10% DPT (random)

1.4.8.4 Forgings

- i) All forgings (wheel, gears, pinions, axles, hooks and hook trunnion) greater than or equal to 40mm diameter or thickness shall be subjected to ultrasonic testing.
- ii) DPT or MPE shall be done after hard facing and machining.

1.4.8.5 Wire rope

Wire rope shall be tested as per relevant standard including breaking force and no. of strands.

1.4.8.6 Reduction gear

Reduction gears shall be tested for reduction ratio, backlash and contact pattern. Gear box shall be subjected to no- load run test to check for oil leakage, temp. rise, noise and vibration.

1.4.8.7 Final testing

The cranes shall be completely assembled at shop for final testing. All tests for dimension, deflection, load, overload, hoisting motion, etc. as per IS-3177 shall be carried out at shop.

1.4.8.8 Electric hoists

All electric hoists shall be tested as per IS-3938 and chain pulley blocks shall be tested as per IS-3832.

1.4.9 Elevators

- i) Reduction gears shall be checked for reduction ratio, backlash and contact pattern. Run outs of wheel shafts and work shafts, tooth contact and running test shall also be carried out.
- ii) Breaking load test shall be carried out along with all other tests as per relevant standard for steel wire rope.
- iii) Buffer springs shall be subjected to load test as per relevant specifications.
- iv) All components prior to assembly shall be checked for dimensions.
- v) All rotating components shall be tested for dynamic balancing.
- vi) Car sling and car body in assembled condition shall be checked for position of all major components i.e. car sling, inside depth, width, height, positions of push box, indicator box lights, fans etc.
- vii) Vibration level shall be determined on work geared machine.
- viii) Mechanical balance test and determination of vibration level on lift and accessories shall be carried out.
- ix) In case the lift is provided with pressurized unit, the fan shall be dynamically balanced and complete unit shall be performance tested.
- x) Function test of Elevator assembly shall be carried out. All electrical equipment shall be of proven quality.

1.4.10 Air Conditioning System

1.4.10.1 Refrigerant compressor (reciprocating/ screw/ centrifugal)

- i) Hydraulic/ pneumatic test of castings for cylinder block, crank case and casings etc. shall be carried out. No leakage shall be permitted
- ii) DPT on connecting rod, piston, crankshaft, screw, impeller with shaft, vanes, crank case, cylinder and casing after machining shall be carried out.
- iii) All rotating parts of reciprocating/ screw and centrifugal compressor shall be statically and dynamically balanced to ISO 1940 Gr. 6.3/ IS 21940.
- iv) Leak tightness and vacuum check for chilling units and compressors inassembled condition shall be carried out. No leakage shall be permitted.
- v) Performance test of assembled compressor shall be done to check for following:
 - a) Capacity test for oil pump for reciprocating compressor.
 - b) No load air run (free run) test of all types of compressors and chilling units to check FAD (free air delivery), noise, vibration and temperature rise of bearing and body.

- c) Hydraulic/ leakage test for reciprocating compressor.
- d) Functional run test and capacity control (for part load performance) check shall be carried out

1.4.10.2 Condenser and evaporator

- i) DPT shall be carried out on finish welds.
- ii) 10% RT of butt weld joints on shell shall be carried out.
- iii) Dimensional check including tube hole diameter, ligament pitch etc. shall be carried out.
- iv) Mock-up test for tube to tube sheet expansion shall be carried out. In case such a test is already conducted for similar tube/ tube sheet thickness and materials, record for the same shall be furnished for purchaser's review and approval.
- v) Hydraulic/ pneumatic test of shell side and tube side of condenser and evaporator as applicable shall be carried out. No leakage shall be permitted.

1.4.10.3 Vapour absorption machine (VAM)

- i) All materials used for fabrication shall be of tested quality. Mill test certificates for chemical and mechanical properties shall be furnished by the manufacturer. In absence of correlated mill test certificates, check test shall be carried out.
- iii) Tubes for heat exchangers/ vessels and interconnecting pipes shall be tested as per the requirement of relevant code/standard.
- iv) All welding shall be performed as per approved Welding Procedure Specification and IBR qualified procedure and welders.
- v) Mock-up test for tube to tube sheet expansion shall be carried out. In case such a test is already conducted for similar tube/tube sheet thickness and materials, record for the same shall be furnished for purchaser's review.
- vi) RT on butt weld joints of heat exchangers/ vessels shall be carried out as per the requirement of design code/approved drawing.
- vii) DPT on Root run after back gouging and on finished welds shall be carried out.
- viii) Vessels/ heat exchangers like high temperature and low temperature generator/ condenser/ high temperature and low temperature heat exchanger/ evaporator/ absorber shall be subjected to hydraulic pressure test and leakage test under vacuum with nitrogen gas and helium gas, both tube side and shell side as applicable for individual component prior to assembly. Helium leak test shall also be conducted on

complete assembly under suitable cover to detect any leakage into the system.

- viii) The complete assembled unit shall be performance tested in shop for capacity (TR) and steam consumption at the rated conditions and part load conditions. Manufacturer shall furnish a detailed procedure along with calculation for conducting such test for approval and in case of any limitation same shall be clearly brought out in the bids. All the controls shall be tested for proper functioning during the above test.

1.4.10.4 Air handling unit (AHU)

- i) 20% DPT of welding on fan hub, blades, casing and impeller as applicable shall be carried out.
- ii) UT of fan shafts (diameter greater than or equal to 40mm) shall be carried out.
- iii) DPT of fan shafts after machining shall be carried out.
- iv) DPT of welding on shaft (in case of fabricated shaft) shall be carried out.
- v) Blower fan shall be statically and dynamically balanced to ISO 1940 Gr. 6.3/ IS 21940.
- vi) One fan of each type and size shall be performance tested as per AMCA / IS for air flow, static pressure, speed, efficiency, power consumption, noise and vibration.
- vii) One per type of assembled AHU (AHU casing and fan assembly) shall be subject to free run test. Noise, vibration and temperature rise of bearing shall be measured during run test.
- viii) All cooling coil shall be pneumatically tested and no leakage shall be permitted.

1.4.10.5 Centrifugal pump

The QA requirements for centrifugal pumps shall be as per clause 1.4.12.2 of this Section.

1.4.10.6 Cooling towers

- i) UT of fan shaft and drive shaft (diameter greater than or equal to 40mm) shall be carried out.
- ii) DPT of fan hub and shafts shall be carried out after machining.
- iii) Colors of fills shall be as per approved data sheet.
- iv) Fan assembly shall be statically balanced.
- v) Cooling towers being supplied to site in assembled condition shall be subjected to run test at shop to measure FAD, noise and vibration. For cooling towers being supplied in knocked-down condition, these tests shall be done at site.

1.4.10.7 Fans

- i) 20% DPT of welding on fan hub, blades, casing and impeller as applicable shall be carried out.
- ii) DPT of fan shafts shall be carried out after machining.
- iii) UT of fan shafts (diameter greater than or equal to 40mm) shall be carried out.
- iv) Rotating components of all fans shall be statically and dynamically balanced to ISO-1940 Gr. 6.3.
- v) All centrifugal fans shall be subjected to run test for 4 hour or till temperature stabilization is reached. Vibration, noise level, temperature rise and current drawn shall be measured during the run test.
- vi) One fan of each type and size will be performance tested as per corresponding IS code for airflow, static pressure, total pressure, speed, efficiency, power consumption, noise, vibration and temperature rise.

1.4.10.8 Low pressure air distribution system

- i) Functional test for fire damper along with solenoid shall be done.
- ii) Prototype tests for fire resistance rating as per UL-555 of each type and size of damper shall be carried out. In case prototype tests have already been done, the contractor shall submit the test report for verification and approval.
- iii) All ducting system shall be checked/tested for air leakages/tightness (smoke test) at site after completion.

1.4.10.9 Insulation

- i) Insulation material shall be tested for all mandatory tests only as per relevant code or standard.
- ii) Thermal conductivity tests (for thermal insulation only) shall be done once in six months for insulation material manufactured during six months period for the same density, outer diameter and thickness of material as applicable as per IS:3346 or equivalent standard.

1.4.10.10 Air filters

Pre and fine filters shall be tested for initial and final pressure drop versus flow and average synthetic dust weight arrestance as per the requirement of BS 6540 or ASHARE-52-76 or EN779.

1.4.10.11 Packaged, split and window air conditioners

- i) Compressor of packaged air conditioner (PAC) shall be tested as per relevant code/standard.
- ii) PAC shall be subjected to production routine test in accordance with IS: 8148 for the following.
 - a) General running test.

- b) Pressure or leakage test of refrigerant.
- c) Insulation resistance test.
- d) High voltage test.
- e) Performance test on one PAC of each type/size at ambient condition to check for following.
 - DBT and WBT of supply and return air.
 - Air flow
 - Current, voltage measurement and power consumption
 - Noise and vibration measurement
- ix) Manufacturer's standard test certificate or guarantee certificate shall be submitted for split and window air conditioners.

1.4.10.12 Pipes, valves and fittings

The QA requirements for piping, valves and fittings shall be as per clause 1.4.6 of this Section.

1.4.11 Ventilation System

The QA requirements for ventilation system equipment shall be same as those applicable similar equipment of air conditioning system equipment described at clause 1.4.10 of this Section.

1.4.12 Miscellaneous Items/ Equipment

1.4.12.1 Chemical dosing system

- i) Pumps of chemical dosing system shall be performance tested as per relevant codes.
- ii) In case of diaphragm type of pumps, the life cycle test shall be done on pumps.
- iii) Dosing skid shall be subjected to leakage test and functional test.

1.4.12.2 Centrifugal pumps

- i) Tests shall be carried out on the materials of the pumps to establish their properties, and chemical compositions.
- ii) 100% DPT or MPE shall be carried out for the rotor and machined surfaces of casing and impellers.
- iii) UT on pump shaft (diameter greater than or equal to 40mm), MPI or DPT on pump shaft and impeller after machining shall be carried out.
- iv) All rotating components of the pumps shall be statically and dynamically balanced to ISO-1940 Gr. 6.3 / IS 21940.

- v) Assembly fit up check, and dimensional check shall be carried out for the completed pump assembly.
- vi) Pump casings shall be subject to hydraulic test at 1.5 times the shut off pressure or 2 times the working pressure whichever is higher for a minimum duration of 30 minutes.
- vii) All pumps shall be tested at manufacturer's works for head, capacity, power and efficiency as per requirements of HIS, USA or equivalent standard.. Pump shall be given running test over the entire operating range covering from the shut-off head to the maximum flow. Acceptance shall be as per approved datasheet and HIS only.
- viii) Pumps shall be subjected to strip down examination visually to check for mechanical damages after testing at shop in case abnormal noise level and/or excessive vibration are observed during the shop test.

1.4.12.3 Blowers and exhausters

- i) Tests shall be carried out on the materials of the rotary blowers to establish their properties, and chemical compositions.
- ii) 100% DPT or MPE shall be carried out for the rotor and machined surfaces of casing and impellers.
- iii) Rotors shall be dynamically balanced as per ISO-1940 Gr. 6.3/IS 21490
- iv) Assembly fit up check, and dimensional check shall be carried out for the completed blower assembly.
- v) The blower casing shall be subject to hydraulic test at 1.5 times the shut off pressure or 2 times the working pressure whichever is higher.
- vi) Performance test including noise and vibration tests shall be carried out as per relevant standards and codes.

1.4.12.4 Filters and strainers

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition.
- ii) In case of fabricated construction, the welds shall be examined for surface defects by 100% DPT.
- iii) The body shall be subject to hydraulic test at 1.5 times the design pressure.
- iv) Pressure drop, flow and particle size tests shall be carried out for the filter assembly.

1.4.12.5 Tanks and vessels

- i) Tests shall be carried out on the materials on various components to establish their mechanical properties and chemical composition.
- ii) UT shall be carried out on plate material used for fabrication of tanks and vessels.
- iv) Hydraulic test for pressurised vessels and water fill test for atmospheric tanks shall be carried out as per relevant standards/ codes.
- iv) Butt welds and full penetration welds shall be checked by suitable RT or UT. Fillet welds shall be checked by MPE or DPT.

1.4.12.6 Lube oil system/ hydraulic power pack

Lube oil system/ hydraulic power pack shall be tested for performance.

1.5 FGD SYSTEM

1.5.1 Wet lime stone FGD System:

1.5.1.1 Mechanical Systems

1.5.1.1.1 Flue Gas System

I. Booster fans

- i) Rotor components shall be subjected to ultrasonic test at mill and magnetic particle examination/dye penetration examination after rough machining.
- ii) Butt welds in rotor components shall be subjected to 100% UT and all welds shall be subjected to MPE or DPT after stress relieving.
- iii) All rotating components of fans shall be dynamically balanced to quality grade 2.5 of ISO 1940.
- iv) Test for natural frequency of all fan components, including fan blades shall be carried out for the fans.
- v) Full range performance test shall be carried out on one fan as per BS 848, Part-1.
- vi) Hydraulic coupling of booster fan shall be checked for string test i.e., operational check of one fan assembly using hydraulic coupling to check temperature rise, smooth operation, vibration and noise level. Dry run test shall preferably be carried out during string test.

II. Thermal insulation

- i) Pre-formed fibrous pipe insulation and LRB mattresses/ sections of rock wool/ mineral wool from approved manufacturing sources conforming to and tested as per relevant standards shall be used.
- ii) For resin bonded mineral wool insulation, testing shall be carried out as per IS: 8183.
- iii) For resin bonded rock wool insulation, testing shall be carried out as per IS: 9842.

- iv) Type tests except thermal conductivity shall be regularly carried out once in three months.
- v) Type test for thermal conductivity shall be carried out by the manufacturer minimum once in six months. Thermal conductivity (K value) shall be measured in line with IS: 3346.

III. Dampers

- i) All the dampers shall be subjected to operational test/checks.
- ii) Gas tight dampers shall be subjected to shop leakage test to demonstrate the guaranteed tightness (minimum one damper of each type and size offered). In case such type test is already done, the reports of the same shall be submitted for review.
- iii) All dampers shall be checked for sealing dimensions to establish guaranteed tightness.

IV. Structure, Ducts, Hoppers

- i) All materials shall be tested for chemical and mechanical properties as per relevant standard. All plates above 40mm shall be 100% Ultrasonically tested.
- ii) Visual inspection of all welds shall be performed in accordance with AWS DI .1.
- iii) NDT requirements of structural steel welds shall be as under:
 - a) 100% RT/UT on butt-welds of plate thickness $\geq 32\text{mm}$.
 - b) For plates of $25\text{mm} \leq \text{thickness} < 32\text{mm}$: 10%RT and 100% MPI.
 - c) For plates of thickness $< 25\text{mm}$: 10% MPI/LPI.
- iv) Edge for shop and field weld shall be examined by MPI for plate thickness $\geq 32\text{mm}$.

Cladding material and its application on the ducts shall be tested as per applicable standard.

1.5.1.1.2 Absorber

I. Metal Structures

- i) Only material which has been identified against mill sheet or test certificates shall be used for construction. All plates above 40mm shall be 100% Ultrasonically tested.
- ii) Visual inspection of all welds shall be performed in accordance with AWS DI .1.
- iii) NDT requirements of structural steel welds shall be as under:
 - a) 100% RT/UT on butt-welds of plate thickness $\geq 32\text{mm}$.

- b) For plates of $25\text{mm} \leq \text{thickness} < 32\text{mm}$ -10%RT and 100% MPI.
- c) For plates of thickness $< 25\text{mm}$ -10% MPI/LPI.
- iv) Edge for shop and field weld shall be examined by MPI for plate thickness $\geq 32\text{mm}$.
- v) Tanks shall be water fill tested up to liquid level as per construction Code.
- vi) Vacuum test according to design code/standard as applicable.
- vii) Cladding/Bonding material and its application shall be tested as per applicable standard.
- viii) Corrosion test on Corrosion Resistant Alloy shall be performed as per ASTM G48/applicable relevant standard for C276 Gr material. For Ti Cladding, material shall be tested as per relevant applicable Indian/International standard.
- ix) Absorber fabricated and erected at site shall be subjected to all tests according to design code/ standard as applicable.

II. Spray Nozzles

- i) Spray nozzles shall be tested for physical properties.
- ii) Spray nozzles also shall be subjected to performance test.

III. Agitators

- i) Rubber lining shall be tested for hardness and spark test.
- ii) Impellers shall be tested for dimensional and balancing check.
- iii) Gear Boxes shall be tested for run test as per standard practice.

IV. Other Critical Equipment

Checks/ NDTs shall be done as per relevant Indian Standards or equivalent International Standards.

1.5.1.1.3 Limestone & Gypsum Handling System

I. Crushers

The details of the checks to be carried out for various components are to be submitted by the Contractor for Owner's approval. However, some indicative checks on different items are given below which should necessarily form part of the Quality Assurance Plan to be agreed with the Owner.

- i) All plates equal to or above 25mm thickness shall be ultrasonically tested.

- ii) Shaft forgings and suspension bars to be checked for ultrasonic testing in addition to check for chemical, mechanical, hardness, microstructure etc. as per applicable material specification.
- iii) Following minimum NDT requirements to be ensured for welds:
 - a) Butt welds - 10% UT/RT and 100% MPI/DPT.
 - b) Fillet Welds - 10% MPI/DPT.
- iv) Crusher rotor to be dynamically balanced. Procedure to be submitted for approval.
- v) No-load trial run test to be carried out at shop to check for speed(RPM), temperature rise, noise level and vibration.

II. Metal Detectors

- i) Functional test including sensitivity, Burn in test, operation of liquid spray marker, detection of smallest piece of different materials as specified.
- ii) Test report for Degree of protection test to be furnished.

III. Belt Conveyor System

The details of the checks to be carried out in the various equipment are to be submitted by the Contractor for Owner's approval. However, some indicative checks on different items are given below which should necessarily form a part of the Quality Assurance Plan to be agreed with the Owner.

IV. Idlers

- i) Check for run out and free movement shall be carried out on idlers. Run out shall be restricted as per IS:8598
- ii) Test for dust proofness, water proofness and dynamic friction factor of the Idlers shall be conducted at shop. The detailed procedures for the same shall be submitted for review and approval.

V. Belting

- i) Rubber cover of finished belt shall be checked for tensile strength and elongation at break before and after ageing. Rubber cover shall also be checked for abrasion, tear strength and hardness.
- ii) For finished belts, checks for elongation at 10% nominal tensile strength tensile and elongation at break in longitudinal (warp) direction and tensile in transverse (weft) direction shall be carried out.
- iii) Adhesion test between ply to ply and cover to ply shall be carried out.

- iv) Troughability test and Test for fire resistance shall be carried out.
- v) Test for procedure qualification for belt vulcanizing joint (at site) shall be done. Procedure for belt vulcanizing joint shall be discussed and finalized during FQP finalization.
- vi) There will be a limitation on the no. of repairs allowed on the belts. Following will be the acceptance norm for the cover repairs.
 - The maximum size of a repair shall be limited to a size equivalent to one fifth the belt width. No single dimension shall exceed one fifth (1/5) of belt width.
 - Small local repair by dough filling of size 25mm x 25mm to a limited extent shall not be counted of repairs. However, in case of cluster of repairs, same shall be counted as a patch repair.
 - The maximum number of patch repair shall not exceed 5 per 100 meter. However, the total number of patch and dough filling repairs shall not exceed 10 per 100 meters.
- vii) In addition to above, Steel Cord belt shall be tested for following also:
 - Cord dia and breaking strength
 - Finished belt shall be tested for cord pull-out strength before and after ageing, peeling resistance.
 - Dynamic cord pull out test
 - Cord dia, pitch and number of cords
- viii) In no case shall the cover thickness or the width of belt be less than that given in specification.
- ix) For testing purpose, belt sample shall be taken from anywhere of the belt roll length offered.

VI. Belt vulcanizing machine

- i) Check for tensile strength shall be carried out on a sample vulcanized belt joint for each type of belt in shop. However, if such test has been done earlier, the report for same shall be submitted for verification.
- ii) Complete assembly shall be tested at shop for temp. and pressure developed.

VII. Pulleys

- i) In addition to chemical, mechanical, hardness, microstructure as per applicable material specification, pulleys shaft forgings shall be subjected to ultrasonic testing.
- ii) 100% MPI/DPT on all welds shall be conducted and 10% RT/UT on butt welds shall be conducted.
- iii) Static balancing of pulleys shall be carried out after rubber lagging.
- iv) Checks on rubber lagging to include abrasion loss, shore hardness test, peel-off strength test and physical properties. Peel-off strength shall be 10 kg/cm, Abrasion loss shall be less than 250 cubic mm when tested as per DIN 53516.

VIII. Pull chord & belt sway switches

The following tests shall be carried out:

- i) Over all dimension and functional test.
- ii) HV & IR test
- iii) Degree of protection test report.

IX. Zero speed switch, under belt switch and chute blockage switch

The following tests shall be carried out:

- i) Burn in test at 50 deg. C for 48 hours shall be done for electronic switches.
- ii) Over all dimension and functional tests.
- iii) HV & IR
- iv) Degree of protection test

X. Drive Equipment

- i) Gear Boxes
 - In addition to checks for physical, chemical, hardness, microstructure as per relevant standard, the shaft and gear/pinion forgings shall be subjected to ultrasonic testing
 - MPI to be carried out on Gears/Pinions after machining. Case depth, hardness and MPI after hard facing shall be checked to ensure freedom from defects.
 - Gear reducer shall be checked for reduction ratio, backlash and contact pattern. No load shop trial run to be conducted on gear boxes to check for oil leakage, temperature rise, noise level and vibration.
- ii) Flexible Coupling

- Ultrasonic testing shall be conducted on forgings for gear sleeve and gear hub, if gear coupling is provided.
- MPI shall be carried out after machining to ensure freedom from cracks.

iii) Fluid Coupling

- Dynamic balancing shall be carried out for the rotating parts.
- Check for leak tightness of the coupling shall be carried out.
- Functional test on fusible plug for each type of coupling shall be conducted at shop.
- All couplings to be run tested at shop on no load.
- Check for temperature rise, torque-speed, torque-slip characteristics and over speed test shall be included during performance test of one coupling of each type preferably at full load.

XI. Belt Scales

The details of the checks to be carried out in the various equipment are to be submitted by the Contractor for Employer's approval. However, some indicative checks are given below which should necessarily form a part of the quality assurance plan to be agreed with the Employer.

- i) Mounting arrangement/Overall dimensional check shall be carried out on the Belt Scales.
- ii) Belt scale shall be calibrated with test weight/test chain in static at works and with test weight for dynamic condition at site.
- iii) All electronic modules shall be subjected to burn in test at 50 deg. C for 48 hours.
- iv) General check for load cell shall be carried out.
- v) Test report for degree of protection on enclosure shall be furnished.
- vi) Accuracy/performance check shall be demonstrated at site.

1.5.1.1.4 Dust Control System

The tests etc. shall be carried out various components of dust extraction system offered by the bidder in line with standard industry practice. The details of the checks to be carried out on the various equipment are to be submitted by the Contractor for Purchaser's approval.

1.5.1.1.5 Bucket Elevator

- i) All plates equal to or above 25 mm thickness shall be ultrasonically tested.
- ii) Castings and forgings, forged/rolled bar/section shall be subjected to ultrasonically test in addition to check for chemical, mechanical, hardness, microstructure etc. as per applicable material specification.
- iii) Machined and hard faced surface of casting/forging and other hardened, stellited parts shall be subjected to DPT/MPI in addition to check for case depth, hardness as applicable for chain/sprocket/gear reducer/ rollers/ wheel/ pan etc.
- iv) Following minimum NDT requirements shall be followed for welds:
 - a) Butt Welds in Tension - 100% UT/RT and 100% MPI/DPT.
 - b) Butt Welds in Compression - 10% UT/RT and 10% MPI/DPT.
 - c) Fillet Welds - 10% MPI/DPT.
- v) For other items like drive system, motor, pulley, belt etc. relevant portion of specification shall be applicable.
- vi) No load trial run test shall be carried out at shop on completely assembled Bucket Elevator to check for trouble free operation, temperature rise, Noise & vibration. The procedure for the No load trial run shall be submitted for approval.

1.5.1.1.6 Silos/ Bunkers

- i) All material shall be tested for Chemical & Mechanical properties as per relevant standard. MPI/DP test on welding shall be carried out. Fit up assembly checks shall be carried out at shop for all despatchable segments.
- ii) Bag Filters: Bag Leakage test shall be carried out for pressure parts. Pulsing and sequential test on bag filter shall be done.

1.5.1.1.7 Ball Mills

- i) Raw material for shaft, coupling, gears and pinions, top and bottom races and other rotating components shall be subjected to UT. MPI/LPI shall be carried out to check surface soundness.
- ii) Wear-resistant parts shall be UT/RT tested to check soundness after suitable heat treatment. Check for chemical composition, hardness and microstructure shall be carried out.
- iii) Butt welds in the tube/separator/body casing of the mill shall be tested by RT and MP'. All other welds in main tube/separator shall be tested by MPI/LPI for acceptance. The tube shall be statically balanced.
- iv) All gearboxes shall be run tested for adequate duration to check rise in oil temperature, noise level and vibration. Check for leak tightness of gear case also shall be performed.

No load run test of the assembly shall be demonstrated at shop/site depending upon its

design/feasibility

I. Gravimetric Feeders

- i) Any welds in the casing/pulley fabrication shall be checked with MPI.
- ii) Routine tests shall be done as per relevant Indian Standards or equivalent International Standards.
- iii) All major items like plates for casing, head pulley, tail pulley, pulley shaft and major castings shall be procured with respective material test certificates.
- iv) Calibration check shall be carried out on all feeders.

II. Lubricating oil systems

Complete lube oil system shall be checked suitably as per standard practice.

1.5.1.1.8 Vacuum Belt Filter System

- i) Impeller, casing and shaft of vacuum pumps shall be tested for chemical and mechanical properties as per relevant standard. All plates above 40mm shall be 100% Ultrasonically tested.
- ii) UT on shaft (if greater or equal to 40mm) and impeller shall be carried out.
- iii) All vacuum pumps shall be tested at shop for capacity, power, pressure, efficiency, noise and vibration etc.
- iv) Filter cloths and belts shall be tested for physical properties as per relevant standard
- v) Hydro cyclones shall be checked by visual, dimensional etc.

1.5.1.1.9 Monorail and Hoists

I. Hooks

- i) All tests including proof load test as per relevant IS/BS/DIN shall be carried out.
- ii) MPE or DPT shall be done after proof load test.

II. Steels castings

Steel castings shall be subjected to DPT on machined surface.

III. Forgings

- i) All forgings (wheel, gears, pinions, axles, hooks and hook trunnion) greater than or equal to 40mm diameter or thickness shall be subjected to ultrasonic testing.
- ii) DPT or MPE shall be done after hard facing and machining.

IV. Wire rope

Wire rope shall be tested as per relevant standard.

V. Electric hoists

All electric hoists shall be tested as per IS-3938 and chain pulley blocks shall be tested as per IS-3832.

1.5.1.1.10 Ventilation System

- i) Fans
 - a) 20% DPT of welding on fan hub, blades, casing and impeller as applicable shall be carried out.
 - b) DPT of fan shafts shall be carried out after machining.
 - c) UT of fan shafts (diameter greater than or equal to 40mm) shall be carried out.
 - d) Rotating components of all fans shall be statically and dynamically balanced to ISO-1940 Gr. 6.3.
 - e) All centrifugal fans shall be subjected to run test for 4 hour or till temperature stabilization is reached. Vibration, noise level, temperature rise and current drawn shall be measured during the run test.
 - f) One fan of each type and size will be performance tested as per corresponding IS code for airflow, static pressure, total pressure, speed, efficiency, power consumption, noise, vibration and temperature rise.

1.5.1.1.11 Piping Valves and fittings- Refer clause 1.4.6

1.5.1.1.12 Packaged, split and window air conditioners

- i) Compressor of packaged air conditioner (PAC) shall be tested as per relevant code/standard.
- ii) PAC shall be subjected to production routine test in accordance with IS: 8148 for the following.
 - a) General running test.
 - b) Pressure or leakage test of refrigerant.
 - c) Insulation resistance test.
 - d) High voltage test.
 - e) Performance test on one PAC of each type/size at ambient condition to check for following:
 - DBT and WBT of supply and return air.
 - Air flow
 - Current, voltage measurement and power consumption
 - Noise and vibration measurement

- iii) Manufacturer's standard test certificate or guarantee certificate shall be submitted for split and window air conditioners.

1.5.1.1.13 Elevators

- i) The details of the checks to be carried out in the various equipment are to be submitted by the contractor for owner's approval. However, some indicative checks on different items are given below which should necessarily form part of the quality assurance plan to be agreed with the owner.
- ii) All forgings shall be subjected to ultrasonic test to ensure free from internal defects in addition to check for chemical and mechanical properties.
- iii) 10% of the welds selected at random shall be subject to DP test.
- iv) All forged components shall be subjected to DPI/MPI after machining.
- v) Gear Reducer shall be checked for gear ratio, backlash, contact pattern. No load shop trial run shall be conducted on gear boxes to check for oil leakage, temperature rise, noise and vibration.
- vi) Buffer springs shall be subjected to load test as per relevant specifications. Material certificates for springs shall also be furnished
- vii) All components prior to assembly shall be checked for dimensions.
- viii) Car sling and car body in assembled condition shall be checked for position of all major components i.e. car sling, inside depth, width, height, positions of push box, indicator box lights, fans etc.
- ix) Function test of Elevator assembly shall be carried out.
- x) All electrical equipment shall be of proven quality.
- xi) Galvanized components/parts shall be checked for weight of Zn coating, thickness of coating, uniformity of coating and adhesion test and visual examination as per IS 2633 and IS 2629.

1.5.1.1.14 Fire Detection & Protection System

i) Hydrant System: Shop Tests

- a) Hydrant Valve:
 - All valves shall be hydro tested for body and seat.
 - Capacity test / flow test shall be done as per relevant standard.
- b) Water Monitor, Hoses, Branch Pipes, Couplings and Nozzles:

- All tests including hydraulic test shall be done as per relevant Indian/ International standard.
- ii) **High/ medium velocity water spray & sprinkler system: shop tests**
 - a) For Pipes, Fittings, Valves and specialties, requirements are indicated separately.
 - b) Deluge Valves, Alarm Valves and Spray Sprinkler Nozzles
 - All valves shall be hydro tested for body and seat.
 - Performance test / functional test of 'Deluge Valves' and 'Spray Nozzles' shall be carried out.
- iii) **Detectors**

All 'Detectors' shall be tested as per relevant Indian / International Standards.
Detectors shall also meet the requirements of UL / FM / LPC/VDS etc.
- iv) **Piping, valve and specialties**

Refer clause 1.4.6
- v) **Portable & mobile fire extinguishers**
 - a. All fire extinguishers shall be tested as per relevant standard.
 - b. Performance / function test shall be carried out on sampling basis as per relevant code / standard.
- vi) **Site tests.**
 - a) Fire Extinguishers: A performance demonstration test at site of five (5) percent or one (1) number, whichever is higher, of each type and capacity of the extinguisher shall be carried out by the contractor. All consumables and replaceable items require for the contractor without any extra cost to employer would supply this test would be supplied by the Contractor without any extra cost to employer.
 - b) Piping Protection:
 - Thickness, Holiday by spark test, Adhesion test shall be carried out as per relevant standard.
 - Complete piping shall be Hydro pressure tested, at 1.5 X DP or 2 X MWP whichever is higher, before protection.
 - c) Welding of Pipes:
 - ERW Black / rolled welded:
 - 100% DPT on root of butt and finish weld of butt and fillet.
 - RT on 10% randomly selected joints shall be carried out (for underground piping).

- GI Pipes
Welding on GI Pipes in general shall not be done. Welding of GI Pipes, if permitted by design, (butt / socket / fillet weld) shall be done strictly as per approved drawing and procedure approved by NTPC Engineering. For all such welds 100% DP test and random 1% RT shall be done.

1.5.1.1.15 Miscellaneous Equipment

I. Pressure Vessels

- i) NDT on weld joint shall be as per respective code requirements or the minimum as specified as below:
 - a) 100% DPT on root run of butt weld, nozzle welds and finished fillet welds.
 - b) 10% DPT on all finished butt welds.
 - c) 10% RT (covering all 'T'/cross joints) of butt welds.
- ii) Butt welds of dished ends shall be stress relieved and subjected to 100% RT.
- iii) Each finished vessels shall be hydraulically tested to 150% of the design pressure for a duration of 30 minutes.

II. Tanks

- i) Tests shall be carried out on the materials of the tanks to establish their mechanical properties, and chemical compositions
- ii) Plates shall be subject to 100% ultrasonic testing.
- iii) Welding
 - a) Root run of butt welds shall be examined by 100% DPT or MPE.
 - b) Butt welds, full penetration joints and nozzle welds shall be checked by 100% RT and 100% MPE or DPT.
 - c) Fillet welds shall be checked by 100% MPE or DPT.
- iv) Rubber lining shall be subject to hardness and spark test, as applicable.
- v) Visual check and dimensional measurement shall be carried out on the completed equipment.
- vi) The fabricated tank shall be subject to water fill test to check for the leakage.

1.5.1.1.16 Low Pressure Piping, Valves and Fittings etc. Refer clause 1.4.6

1.5.1.1.17 Pumps & Blowers

I. Dosing pumps/ metering pumps

- i) Tests shall be carried out on the materials of the pumps to establish their properties, and chemical compositions
- ii) Pump casings shall be subject to hydraulic test at 1.5 times the shut off pressure or 2 times the working pressure whichever is higher.
- iii) Pumps shall be performance tested as per HIS, USA.

II. Horizontal centrifugal pumps

- i) Tests shall be carried out on the materials of the pumps to establish their properties, and chemical compositions.
- ii) 100% DPT or MPE shall be carried out for the rotor and machined surfaces of casing and impellers.
- iii) UT on pump shaft (diameter greater than or equal to 40mm), MPI or DPT on pump shaft and impeller after machining shall be carried out.
- iv) All rotating components of the pumps shall be statically and dynamically balanced to ISO-1940 Gr. 6.3.
- v) Assembly fit up check, and dimensional check shall be carried out for the completed pump assembly.
- vii) Pump casings shall be subject to hydraulic test at 1.5 times the shut off pressure or 2 times the working pressure whichever is higher for a minimum duration of 30 minutes.
- viii) All pumps shall be tested at manufacturer's works for head, capacity, power and efficiency as per requirements of HIS, USA or equivalent standard. Pump shall be given running test over the entire operating range covering from the shut-off head to the maximum flow. Acceptance shall be as per approved datasheet and HIS only.
- ix) Pumps shall be subjected to strip down examination visually to check for mechanical damages after testing at shop in case abnormal noise level and/or excessive vibration are observed during the shop test.

III. Vertical sump pumps

- i) All materials should be of tested quality and test certificates to be provided.

- ii) Shaft forgings to be subjected to ultrasonic testing. iii) DPT to be done on machined shaft and impeller. iv) Impellers to be dynamically balanced to ISO 1940 Gr.6.3.
- v) All pressure parts shall be hydraulically tested at 150% of the shut-off head or 200% of rated head, whichever is higher for 30 minutes. No leakage is allowed.
- vi) All pumps to be performance tested as per Hydraulic Institute Standard/ Indian Standard. Performance test to include check for noise, vibration level and bearing temperature rise.

IV. Blowers

- i) Tests shall be carried out on the materials of the rotary blowers/ compressors to establish their properties, and chemical compositions.
- ii) 100% DPT or MPE shall be carried out for the rotor and machined surfaces of casing and impellers.
- iii) The shaft and impellers shall be dynamically balanced.
- iv) Assembly fit up check, and dimensional check shall be carried out for the completed blower assembly.
- v) The blower/ compressor casing shall be subject to hydraulic test at 1.5 times the shut off pressure or 2 times the working pressure whichever is higher.
- vi) The blowers/ compressor shall be performance tested as per relevant code/ standard.

V. Air Compressors

- i) All pressure parts shall be hydraulically tested at not less than 150% of design pressure for a duration of 30 minutes prior to painting.
- ii) All other parts including inter-connecting piping shall be hydraulically tested wherever possible, as per relevant codes.
- iii) Ultrasonic testing shall be carried out on all forgings and rotor for dia 40mm and above. MPI/DPT shall be done on machined area of the components.
- iv) During assembly all clearances and alignments shall also be checked and recorded
- v) Rotor shall be statically and dynamically balanced.
- vi) Performance test on the compressor shall be carried out in accordance with ISO: 1217/Eq. The test shall also include demonstration of loading and unloading mechanism (Capacity control) and operation of safety valve.

vii) Vibration and Noise level measurement shall be done during shop performance test.

viii) Air Receiver

a) Each finished vessel shall be hydraulically tested at 150% of the design pressure for a duration of 30 minutes

b) NDT on weld joints shall be as per respective code requirements or the minimum as specified below:

- 100% DPT on root run of butt welds
- 100% DPT on all finished butt welds and fillet welds
- 10% RT on butt welds which shall include all T-Joints

1.5.1.1.17 Hangers and Supports

i) All raw materials used shall have co-related mill test certificate meeting mandatory checks of material specification.

ii) Completed springs shall be tested for sagging test and load versus deflection test. For diameter more than 25mm, MPE shall also be carried out.

iii) Butt welds of thickness 32mm and above shall be tested for UT, and for butt welds of thickness less than 32mm MPE shall be done. Fillet welds shall be tested for MPE.

iv) Dampers with viscous fluids shall be checked for viscosity of liquid used, damping resistance of the damper, stiffness of the damper etc.

v) Turn-buckle, pipe clamps and hangers of thickness greater than 25mm shall be checked by MPE or DPT on bent portions.

vi) Assembled hanger, one number of each type and size shall be checked for variation in deflection and travel versus load test.

1.5.2 Sea water based FGD System:

The testing and inspection requirements of major equipment of FGD system to be followed by the bidder over and above the respective code/ standard requirements are given hereunder:

1.5.2.1 MECHANICAL SYSTEMS

1.5.2.1.1 Flue Gas System

I. Booster fans

i) Rotor components shall be subjected to ultrasonic test at mill and magnetic particle examination/dye penetration examination after rough machining.

ii) Butt welds in rotor components shall be subjected to 100% UT and all welds shall be subjected to MPE or DPT after stress relieving.

iii) The rotor of fans shall be dynamically balanced to quality grade 6.3 of ISO 1940.

- iv) Test for natural frequency and hardness of fan blades shall be carried out as per technical specification/ relevant standard.
- v) Full range performance test shall be carried out on one fan as per BS 848, Part-1/ equivalent standard.

II. Thermal insulation

- a) Pre-formed fibrous pipe insulation and LRB mattresses/ sections of rock wool/ mineral wool from approved manufacturing sources conforming to and tested as per relevant standards shall be used.
- b) For resin bonded mineral wool insulation, testing shall be carried out as per IS: 8183.
- c) For resin bonded rock wool insulation, testing shall be carried out as per IS: 9842.
- d) Type tests except thermal conductivity shall be regularly carried out once in three months.
- e) Type test for thermal conductivity shall be carried out by the manufacturer minimum once in six months. Thermal conductivity (K value) shall be measured in line with IS: 3346.

III. Dampers

- i) All the dampers shall be subjected to operational test/checks.
- ii) Gas tight dampers shall be subjected to shop leakage test to demonstrate the guaranteed tightness (minimum one damper of each type and size offered).
- iii) All dampers shall be checked for sealing dimensions to establish guaranteed tightness.

IV. Structure, Ducts, Hoppers

- i) All materials shall be of tested quality and test certificates for chemical and mechanical properties as per relevant standard shall be provided. All plates above 40mm shall be 100% Ultrasonically tested.
- ii) Visual inspection of all welds shall be performed in accordance with AWS DI .1.
- iii) NDT requirements of structural steel welds shall be as under:
 - a) 100% RT/UT on butt-welds of plate thickness $\geq 32\text{mm}$.
 - b) For plates of $25\text{mm} \leq \text{thickness} < 32\text{mm}$: 10%RT/UT and 100% MPI.
 - c) For plates of thickness $< 25\text{mm}$: 10% MPI/LPI.
- iv) Edge for shop and field weld shall be examined by MPI for plate thickness $\geq 32\text{mm}$.

1.5.2.1.2 Gas to gas heater (GGH)

- i) Forged shafts for GGH like stub shaft, main rotor forging, housing hub shall be subjected to 100% UT at mill and magnetic particle examination after machining.
- ii) Critical welds of rotor post shall be subjected to radiographic examination.
- iii) Trial run of GGH drive assembly (gear box+pinion+electric motor+ air motor) shall be carried out at shop.

1.5.2.1.3 Scrubber

I. Metal Structures

- i) Only material which has been identified against mill sheet or test certificates shall be used for construction. Check testing shall be carried out in the absence of MTC. All plates above 40mm shall be 100% Ultrasonically tested.
- ii) Visual inspection of all welds shall be performed in accordance with AWS D1.1.
- iii) NDT requirements of structural steel welds shall be as under:
 - a) 100% RT/UT on butt-welds of plate thickness ≥ 32 mm.
 - b) For plates of $25\text{mm} \leq \text{thickness} < 32\text{mm}$ - 10% RT and 100% MPI.
 - c) For plates of thickness $< 25\text{mm}$ - 10% MPI/LPI.
- iv) Edge for shop and field weld shall be examined by MPI for plate thickness ≥ 32 mm.

II. Spray Nozzles

- i) Spray nozzles shall be tested for physical properties.
- ii) Spray nozzles also shall be subjected to performance test.

III. Other Critical Equipment

Checks/ NDTs shall be done as per relevant Indian Standards or other applicable International Standards.

1.5.2.1.4 Sea water pumps

- i) All materials shall be of tested quality and test certificates for chemical and mechanical properties as per relevant standard shall be provided.
- ii) Shaft forgings and thrust bearing to be subjected to ultrasonic testing.
- iii) DPT/ MPI to be done on machined shaft, impeller, castings, column pipes, companion flanges and thrust bearing.
- iv) RT to be done on impeller, column pipe discharge head assembly and flanges.

- v) Impeller and shaft to be dynamically balanced to ISO 1940 Gr.6.3.
- vi) Casing shall be hydraulically tested at 150% of the shut-off head for 30 minutes. No leakage is allowed.
- vii) All pumps to be performance tested as per Hydraulic Institute Standard/ Indian Standard. The performance test shall be conducted to verify output (flow) against total head, power input, efficiency and to establish the characteristic curves of the pump. The performance test shall also include check for noise, vibration level and bearing temperature rise.

1.5.2.1.5 Monorail and Hoists

I. Hooks

- i) All tests including proof load test as per relevant IS shall be carried out.
- ii) MPE or DPT shall be done after proof load test.

II. Steels castings

Steel castings shall be subjected to DPT on machined surface.

III. Forgings

- i) All forgings (wheel, gears, pinions, axles, hooks and hook trunnion) greater than or equal to 40mm diameter or thickness shall be subjected to ultrasonic testing.
- ii) DPT or MPE shall be done after hard facing and machining.

IV. Wire rope

Wire rope shall be tested as per relevant standard.

V. Electric hoists

All electric hoists shall be tested as per IS-3938 and chain pulley blocks shall be tested as per IS-3832.

1.5.2.1.6 Ventilation System

- i) Fans
 - a) 20% DPT of welding on fan hub, blades, casing and impeller as applicable shall be carried out.
 - b) DPT of fan shafts shall be carried out after machining.
 - c) UT of fan shafts (diameter greater than or equal to 40mm) shall be carried out.
 - d) Rotating components of all fans shall be statically and dynamically balanced to ISO-1940 Gr. 6.3.
 - e) All centrifugal fans shall be subjected to run test as per manufacturer's standard practice.

- f) One fan of each type and size will be performance tested as per corresponding IS code for airflow, static pressure, total pressure, speed, efficiency, power consumption, noise, vibration and temperature rise.

ii) Piping, valves and fittings- Refer clause 1.4.6

1.5.2.1.7 Packaged, split and window air conditioners

- i) Compressor of packaged air conditioner (PAC) shall be tested as per relevant code/standard.
- ii) PAC shall be subjected to production routine test in accordance with IS: 8148 for the following.
 - a) General running test.
 - b) Pressure or leakage test of refrigerant.
 - c) Insulation resistance test.
 - d) High voltage test.
 - e) Performance test on one PAC of each type/size at ambient condition to check for following:
 - DBT and WBT of supply and return air.
 - Air flow
 - Current, voltage measurement and power consumption
 - Noise and vibration measurement
- iii) Manufacturer's standard test certificate or guarantee certificate shall be submitted for split and window air conditioners.

1.5.2.1.8 Elevators

- i) The details of the checks to be carried out in the various equipment are to be submitted by the contractor for owner's approval. However, some indicative checks on different items are given below which should necessarily form part of the quality assurance plan to be agreed with the owner.
- ii) All critical forgings shall be subjected to ultrasonic test to ensure free from internal defects in addition to check for chemical and mechanical properties.
- iii) 10% of the welds selected at random shall be subject to DP test.
- iv) All forged components shall be subjected to DPI/MPI after machining.
- v) Gear Reducer shall be checked for gear ratio, backlash, contact pattern. No load shop trial run shall be conducted on gear boxes to check for oil leakage, temperature rise, noise and vibration.
- vi) Buffer springs shall be subjected to load test as per relevant specifications. Material certificates for springs shall also be furnished
- vii) All components prior to assembly shall be checked for dimensions.

- viii) Car sling and car body in assembled condition shall be checked for position of all major components i.e. car sling, inside depth, width, height, positions of push box, indicator box lights, fans etc.
- ix) Function test of Elevator assembly shall be carried out.
- x) All electrical equipment shall be of proven quality.
- xi) Galvanized components/parts shall be checked for weight of Zn coating, thickness of coating, uniformity of coating and adhesion test and visual examination as per IS 2633 and IS 2629.

1.5.2.1.9 Fire Detection & Protection System

i) Hydrant System: Shop Tests

a) Hydrant Valve:

- All valves shall be hydro tested for body and seat.
- Capacity test / flow check shall be done as per relevant standard.

b) Water Monitor, Hoses, Branch Pipes, Couplings and Nozzles:

- All tests including hydraulic test shall be done as per relevant Indian/ International standard.

ii) High/ medium velocity water spray & sprinkler system: shop tests

a) For Pipes, Fittings, Valves and specialties, requirements are indicated separately.

b) Deluge Valves, Alarm Valves and Spray Sprinkler Nozzles

- All valves shall be hydro tested for body and seat.
- Performance test / functional test of 'Deluge Valves' and 'Spray Nozzles' shall be carried out.

iii) Detectors: All 'Detectors' shall be tested as per relevant Indian / International Standards. Detectors shall also meet the requirements of UL / FM / LPC/VDS etc.

iv) Piping, valve and specialties

Refer clause 1.4.6.

v) Portable & mobile fire extinguishers

- a. All fire extinguishers shall be tested as per relevant standard.
- b. Performance / function test shall be carried out on sampling basis as per relevant code / standard.

vi) **Site tests.**

- a) Fire Extinguishers: A performance demonstration test at site of five (5) percent or one (1) number, whichever is higher, of each type and capacity of the extinguisher shall be carried out by the contractor. All consumables and replaceable items require for the contractor without any extra cost to employer would supply this test would be supplied by the Contractor without any extra cost to employer.
- b) Piping Protection:
 - Thickness, Holiday by spark test, Adhesion test shall be carried out as per relevant standard.
 - Complete piping shall be Hydro pressure tested, at 1.5 X DP or 2 X MWP whichever is higher, before protection.
- c) Welding of Pipes:
 - ERW Black / rolled welded:
 - 100% DPT on root of butt and finish weld of butt and fillet.
 - RT on 10% randomly selected joints shall be carried out (for underground piping).
- d) GI Pipe: Welding on GI Pipes in general shall not be done. Welding of GI Pipes, if permitted by design, (butt / socket / fillet weld) shall be done strictly as per approved drawing and procedure approved by Owner. For all such welds 100% DP test and random 1% RT shall be done.

1.5.2.1.10 Miscellaneous Equipment

I. Pressure Vessels

- i) NDT on weld joint shall be as per respective code requirements or the minimum as specified below:
 - a) 100% DPT on root run of butt weld, nozzle welds and finished fillet welds.
 - b) 10% DPT on all finished butt welds.
 - c) 10% RT (covering all 'T'/cross joints) of butt welds.
- ii) Butt welds of dished ends shall be stress relieved and subjected to 100% RT.
- iii) Each finished vessels shall be hydraulically tested to 150% of the design pressure for a duration of 30 minutes.

II. Tanks

- i) All materials shall be of tested quality and test certificates for chemical and mechanical properties as per relevant standard shall be provided.
- ii) Plates above 40 mm thickness shall be subject to 100% ultrasonic testing.
- iii) Welding
 - a) Root run of butt welds shall be examined by 100% DPT or MPE.

- b) Butt welds, full penetration joints and nozzle welds shall be checked by 100% RT and 100% MPE or DPT.
- c) Fillet welds shall be checked by 100% MPE or DPT.
- iv) Rubber lining shall be subject to hardness and spark test, as applicable.
- v) Visual check and dimensional measurement shall be carried out on the completed equipment.
- vi) The fabricated tank shall be subject to water fill test to check for the leakage.

III. GRP/GRE Piping for Sea Water

- i) The testing and quality inspection for GRP/ GRE pipe and piping materials shall be carried at manufacturer's works as per the approved quality assurance plan. The tests to be carried out, as a minimum, shall cover the following:

- a) Raw materials

Raw materials for use in pipes covered in this specification shall be sampled and tested by the manufacturer prior to their use to ensure that they comply with the requirement of ASTM D 3517. The glass content in the GRP/GRE material used shall be determined in accordance with ASTM D2584 using ignition loss method.

- b) Pipe stiffness

Samples of pipe shall be tested for compliance with the pipe stiffness in accordance with ASTM D2412, with the stiffness value rated at 3% deflection.

- c) Load capacity (hoop and axial)

Tests will be performed in accordance with AWWA C950 and ASTM D3517 to check the load capacity (or strength) in both the hoop and axial directions.

- d) Barcol hardness test

Barcol hardness tests will be conducted in accordance with ASTM D 2583. Pipes failing to reach the required level of cure will be rejected.

- e) Critical dimensions

The pipes shall be measured for compliance with critical dimensions as specified in ASTM standards including diameter, thickness, end squareness and length. Measurements shall be made in accordance with ASTM D3567. Wall thickness should be measured as per ASTM D3567 once per every production lot.

f) *Visual acceptance*

Visual Inspection and classification of defects shall be carried out according to ASTM D2563 level III. The pipe should be free from all defects, including delamination, bubbles, pinholes, cracks, pits, blisters, foreign inclusions, and resin-starved areas. The pipe should be as uniform as commercially practicable in color, opacity, density and other physical properties.

ii) Field Hydro Testing

- a) Hydrostatic test of the GRP/ GRE pipe for the entire length or on a segment of it shall be carried out to verify the hydraulic sealing of the system at the testing pressure and its structural integrity.
- b) The hydro testing of the piping system shall be carried out before encasing or closing the trench covers. Permanent flanged spool pieces should be provided at the start and end point of the pipe to facilitate blinding of the line for hydro testing. The joints should be visible in order to allow for the inspection, unless otherwise stated by Owner & their representative.
- c) Hydrostatic testing shall be carried out at 1.5 times the design pressure for a duration of minimum 2 hours after stabilization of the test pressure. The contractor shall submit the detailed plan indicating the methodology for hydro testing, blinding the ends, air venting, supply of sea water, pressurizing and draining. The test procedure shall be subject to purchase's approval.
- d) The hydraulic testing shall be considered acceptable, if at the end of the 2 hours of testing, the following parameters are met:
 - During the period of the hydraulic test, the pressure stays stable.
 - No leakages are noticed in any point of the pipeline, or in the joints, or in the testing equipment.
- e) In case of failure of the test, the reason for leakage shall be examined. Any pipes found defective or leaking (except at the butt joints) shall be replaced with a new one. Repair or patch work on pipes shall not be acceptable. The pipeline shall then be tested again according to the approved test procedure.

IV. Low Pressure Piping, Valves and Fittings etc. Refer CI 1.4.6

V. Horizontal centrifugal pumps

- i) Tests shall be carried out on the materials of the pumps to establish their properties, and chemical compositions.
- ii) 100% DPT or MPE shall be carried out for the rotor and machined surfaces of casing and impellers.
- iii) UT on pump shaft (diameter greater than or equal to 40mm), MPI or DPT on pump shaft and impeller after machining shall be carried out.

- iv) All rotating components of the pumps shall be statically and dynamically balanced to ISO-1940 Gr. 6.3.
- v) Assembly fit up check, and dimensional check shall be carried out for the completed pump assembly.
- vi) Pump casings shall be subject to hydraulic test at 1.5 times the shut off pressure for a minimum duration of 30 minutes.
- vii) All pumps to be performance tested as per Hydraulic Institute Standard/ Indian Standard. The performance test shall be conducted to verify output (flow) against total head, power input, efficiency and to establish the characteristic curves of the pump. The performance test shall also include check for noise, vibration level and bearing temperature rise.
- viii) Pumps shall be subjected to strip down examination visually to check for mechanical damages after testing at shop in case abnormal noise level and/or excessive vibration are observed during the shop test.

VI. Vertical pumps

- i) All materials should be of tested quality and test certificates to be provided.
- ii) Shaft forgings to be subjected to ultrasonic testing.
- iii) DPT to be done on machined shaft and impeller.
- iv) Impellers to be dynamically balanced to ISO 1940 Gr.6.3.
- v) All pressure parts shall be hydraulically tested at 150% of the shut-off head for 30 minutes. No leakage is allowed.
- vi) All pumps to be performance tested as per Hydraulic Institute Standard/ Indian Standard. The performance test shall be conducted to verify output (flow) against total head, power input, efficiency and to establish the characteristic curves of the pump. The performance test shall also include check for noise, vibration level and bearing temperature rise.

VII. Blowers

- i) Tests shall be carried out on the materials of the rotary blowers/ compressors to establish their properties, and chemical compositions.
- ii) 100% DPT or MPE shall be carried out for the rotor and machined surfaces of casing and impellers.
- iii) The shaft and impellers shall be dynamically balanced.
- iv) Assembly fit up check, and dimensional check shall be carried out for the completed blower assembly.
- v) The casing shall be subject to hydraulic test at 1.5 times the shut off design pressure.
- vi) The blowers shall be performance tested as per relevant code/ standard.

VIII. Air Compressors

- i) All pressure parts shall be hydraulically tested at not less than 150% of design pressure for a duration of 30 minutes prior to painting.
- ii) All other parts including inter-connecting piping shall be hydraulically tested wherever possible, as per relevant codes.
- iii) Ultrasonic testing shall be carried out on all forgings and rotor for dia 40mm and above. MPI/DPT shall be done on machined area of the components.
- iv) During assembly all clearances and alignments shall also be checked and recorded.
- v) Rotor shall be statically and dynamically balanced.
- vi) Performance test on the compressor shall be carried out in accordance with ISO: 1217/Eq. The test shall also include demonstration of loading and unloading mechanism (Capacity control) and operation of safety valve. Power consumption at motor input terminal at rated capacity as well as at fully unloaded condition of all the compressor shall be measured.
- vii) Vibration and Noise level measurement shall be done during shop performance test.
- viii) Air Receiver
 - a) Each finished vessel shall be hydraulically tested at 150% of the design pressure for a duration of 30 minutes
 - b) NDT on weld joints shall be as per respective code requirements or the minimum as specified below:
 - 100% DPT on root run of butt welds
 - 100% DPT on all finished butt welds and fillet welds
 - 10% RT on butt welds which shall include all T-Joints

IX. Hangers and Supports

- i) All raw materials used shall have co-related mill test certificate meeting mandatory checks of material specification.
- ii) Completed springs shall be tested for sagging test and load versus deflection test. For diameter more than 25mm, MPE shall also be carried out.
- iii) Butt welds of thickness 32mm and above shall be tested for UT, and for butt welds of thickness less than 32mm MPE shall be done. Fillet welds shall be tested for MPE.
- iii) Dampers with viscous fluids shall be checked for viscosity of liquid used, damping resistance of the damper, stiffness of the damper etc.
- iv) Turn-buckle, pipe clamps and hangers of thickness greater than 25mm shall be checked by MPE or DPT on bent portions.

- v) One assembled hanger of each type and size in each lot shall be checked for variation in deflection and travel versus load test.

X. EOT Crane, Semi Gantry Crane and Hoists

i) Hooks

- a) All tests including proof load test as per relevant IS/BS/DIN shall be carried out.
b) MPI/DPT shall be carried out after proof load test.

ii) Steel casting

DPT on machined surface shall be carried out.

iii) Girders, end carriage, crab, gear box and rope drum

- a) The plates of thickness 25mm and above shall be ultrasonically tested.
b) NDT requirements on weldments shall be as follows:

- Butt welds in tension:-100% RT and 100% DPT
- Butt welds in compression:-10% RT and 100% DPT
- Butt welds in rope drum:-100% RT and 100% DPT
- Fillet welds:- 10% DPT

iv) Forging (wheel, gears, pinions, axle, hooks & hook trunnion)

- a) All forgings greater than or equal to 40 mm diameter or thickness shall be subjected to ultrasonic testing.

b) DPT/MPI shall be done after hard facing and machined surfaces.

- v) Wire rope shall be tested as per relevant standard.

- vi) Reduction gears shall be tested for reduction ratio, backlash & contact pattern. Gear box shall be subjected to no-load run test to check for oil leakage, temperature rise, noise and vibration.

- vii) The cranes shall be completely assembled at shop for final testing. All tests for dimension, deflection, load, overload, hoisting motion, cross travel etc. As per IS-3177 shall be carried out at shop.

- viii) All electric hoists shall be tested as per IS-3938 and chain pulley blocks shall be tested as per IS-3832.

1.5A GENERAL REQUIREMENTS FOR FIELD ERECTION CHECKS AND TESTS

The QA requirements for raw materials, in-process tests and NDTs indicated for shop manufacture shall be applicable for site fabrication/erection of the respective item.

(a) Hydraulic Test of Pressure Parts

On completion of erection of pressure parts of each steam generator, the unit with its fittings and mountings in position shall be subjected to hydraulic test pressure in accordance with requirement of Indian Boiler Regulations. Water used for hydraulic test shall be made alkaline by addition of suitable chemical. After the test, all parts shall be drained and suitably preserved.

(b) Turbine Assembly

Bidder shall clearly indicate the extent of assembly to be carried out at site for steam turbine and BFP drive turbine(s). Accordingly, bidder shall submit elaborate erection and assembly inspection programme of turbines for purchaser's approval.

(c) Condenser Assembly

- i) If the condenser sections calls for site assembly, care shall be taken in assembly of sections and correctness of alignment and fit up shall be checked. Site welding shall be carried out as per the procedure approved by the purchaser.
- ii) All weld seams shall be subjected to magnetic particle examination. At least 10% of butt welds shall be subjected to radiographic examination.
- iii) All welds between condenser neck and LP turbine shall be subjected to 100% radiographic and magnetic particle examination.
- iv) Condenser tubes shall be visually examined for dents, mechanical damages or any other defects prior to insertion. Both tube ends shall be thoroughly cleaned to a length of 100mm to remove oil, grease etc. and shall be checked for freedom from burrs prior to insertion.
- v) Tube expansion shall be carried out by electronic automatic torque control expanding unit, which shall be calibrated before use. Tube wall thinning and length of expansion shall be controlled and recorded.
- vi) Hydrostatic testing of condenser steam space shall be carried out after connecting all the pipes with the condenser along with condenser vacuum systems by filling the steam space with water up to the tip of the last stages of blades of LP cylinder.
- vi) Condenser water boxes shall be tested hydraulically at a minimum test pressure of 1.5 times the design pressure.

(d) General

- i) All rotating equipment shall be checked for their direction of rotation and free

movement after placing on the respective foundations.

- ii) Piping system shall be tested hydraulically or pneumatically as per application requirement.
- iii) All valves shall be checked for their direction of flow.
- iv) Insulation shall be carried out only after satisfactory inspection of leak test.
- v) After complete installation of air conditioning and ventilation systems, all ducting system shall be tested for air leakage test or smoke tightness test.

Sl. No .	Component & Operation	Characteristics	Class	Type of Check	Quantum of Check		Reference Document / Acceptance Norms	Format of Record	D	Agency			Remarks
					M	C/U				M	C	U	
LEGEND: M: MANUFACTURER/SUB-SUPPLIER, C: MAIN SUPPLIER, U: CUSTOMER, P: PERFORM, W: WITNESS, V: VERIFICATION, IR- INSPECTION REPORT, D: DOCUMENT REQUIRED													
Note:													
1) ‘ W ’ UNDER AGENCY COLUMN INDICATES THE CUSTOMER HOLD POINT (CHP), WHICH SHALL BE IDENTIFIED BY CUSTOMER.													
2) RECORDS, INDENTIFIED WITH “TICK’’ (√) SHALL BE ESSENTIALLY INCLUDED BY SUPPLIER UNDER COLUMN ‘D’													
MODEL QUALITY ASSURANCE PLAN (MQAP)													
1 STEAM GENERATOR													
1.1 PRESSURE PARTS													
1.0 Raw Material Inspection													
1.1	Drums Plates, pipes, bar and forging	Chemical properties	Critical	Chemical Test	100 %	Minimum requirements of BS: 5996 grade LC3/ ASTM or equivalent standards, Approved Drawing/ Datasheet	MTC	√	P	V	V	See Note 3	
		Mechanical properties	Critical	Mech. Tests	100 %		MTC	√	P	V	V		
		Elevated temp. tensile test on plate material	Critical	Tests	100 %	Approved Drawing/ Datasheet	MTC	√	P	V	V		
		UT (normal UT at Mill)	Critical	UT	100 %	SA 577/ SA 578 level B	MTC	√	P	V	V		
1.2	Headers (Plates & seamless pipes, bar, forging, Tubes)	Chemical properties, Mechanical Properties., Heat Treatment, Ultrasonic testing, Hardness test	Critical	Review of TC	100%	i) Ultrasonic Test ii) Transverse test on headers (greater than 8’’) iii) Flattening test on pipe from one end	MTC	√	P	V	V		

		Grade 22 and above				As per applicable material standard/code As per IBR or ASME E 213 or equivalent with the longitudinal calibration notch of depth 5% of wall thickness (0.3mm min. and 1.5mm max.)						
1.3	Tubes & Tube Elements	Chemical properties, Mechanical Properties., Heat Treatment, Ultrasonic testing, Hardness test Grade 22 and above	Critical	Tests	100%	As per applicable material standard/code As per IBR or ASME E 213 or equivalent with the longitudinal calibration notch of depth 5% of wall thickness (0.3mm min. and 1.5mm max.)	MTC	√	P	V	V	See Note 4
1.4	Piping	Chemical, Mechanical Properties, NDT	Critical	HT & UT	100%	Approved Drawing/Datasheet	MTC	√	P	V	V	
1.4.1	Fittings	Chemical, Mechanical Properties, NDT	Critical	Ultrasonic Test	100%	Approved Drawing/Datasheet	Test Report	√	P	V	V	
1.5	Valves	Chemical, Mechanical Properties, NDT	Critical	HT & UT	100%	Approved Drawing/Datasheet	MTC	√	P	V	V	NDT as per clause 1.2.1.6
1.6	Compensating Pads	Chemical, Mechanical properties	Critical	Tests	100%	Approved Drawing/Datasheet		√	P	V	V	
2.0	In Process Inspection											
2.1	Drum	Dimensional	Major	Measurement	100%	Approved Drawing/Datasheet	IR		P	-	-	

		Dished end pressing, Dished end HT after forming, Thickness,						√	P	V	V	
		Tensile test on cut out piece, WPS, PQR, Welder qualification records						√	P	V	V	
		Forming of Drum suspension Rod						√	P	V	V	
2.1.1		Visual	Major	Visual	100%	MANUFACTURING DRAWING		√	P	-	-	
2.1.2	a) Butt welds	NDT	Critical	UT/RT/PAUT+TOFD	100%	Approved Drawing/Datasheet	RT	√	P	V	V	UT/RT/PAUT+TOFD after stress relief. RT Film review
	b) Butt welds	NDT	Critical	UT/RT/PAUT+TOFD & MPE	100%	Approved Drawing/Datasheet	UT REPORT	√	P	W	W	Before stress relieving in the furnace of the entire drum
2.1.3	Full penetration Welds & Lifting lug	NDT	Critical	Ultrasonic testing	100%	Approved Drawing/Datasheet	UT REPORT	√	P	V	V	Before stress relief
2.1.4	Mechanical testing of long seam & C-seam weld joint test coupon	Mechanical Test	Critical	Tensile test	100%	Approved Drawing/Datasheet	TC	√	P	W	W	
2.1.5	Suspension rod weld joint UT	NDT	Critical	UT	100%	Approved Drawing/Datasheet	TC	√	P	W	W	

2.1.6	Heat treatment of Drum inside furnace	Heat Treatment	Critical	HT Chart	100%	Approved Drawing/Datasheet	HT Chart	√	P	V	V	
2.1.7	Dimension of drum, suspension rod, assembly of internal, nozzles	Visual & Dimensional	Critical	Measure ment	100%	Approved Drawing/Datasheet	TC	√	P	W	W	
2.2	Headers											
2.2.1	Forming, Heat Treatment	Manufacturing		Visual, HT Chart	100%	Approved Drawing/Datasheet		√	P	V	V	
2.2.2	Butt welds (Carbon steel and low alloy except for P91/92 and X20)	NDT	Critical	a)Radiographic test (RT) b) TOFD	100%	Approved Drawing/Datasheet	RT REPORT	√	P	V	V	Before stress relief. Review of RT film
2.2.3	Butt welds (P91/92 and X20)	NDT	Critical	a) MPE b)Radiographic test (RT) c) TOFD	100%	Approved Drawing/Datasheet		√	P	V	V	After stress relief.
2.2.4	Full penetration Nozzle (carbon and low alloy steels)	NDT	Critical	Ultrasonic testing & MPE/DPT	100%	Approved Drawing/Datasheet	UT REPORT	√	P	W	W	UT before PWHT
2.2.5	Full penetration welds (Gr91/92) & attachment welds	NDT	Critical	Ultrasonic testing & MPE	100%	Approved Drawing/Datasheet	UT REPORT	√	P	W	W	UT after PWHT
2.2.6	Production weld test for L seam if welded	Tensile test	Critical	Measure ment	100%	Approved Drawing/Datasheet	TC	√	P	W	W	
2.2.7	WPS, PQR & welder qualification test	WPS,PQR & WTQ	Critical	Review	100%	ASME Sec.IX, IBR	WPS/PQR	√	P	V	V	

2.2.8	Hydro test	Hydraulic Test	Critical	Measurement	100%	Approved Drawing/Datasheet	TC	√	P	W	W	
2.3	Tubes and Tube Elements	NDT	Critical	MPI/DPT	100%	Approved Drawing/Datasheet	TC	√	P	V	V	
		Bending, sizing, swaging	Major	Visual			TC	√	P	V	V	
		Heat treatment	Major	HT			TC	√	P	V	V	
		Bend- First off inspection	Major	Visual			TC	√	P	W	W	
		WPS, PQR, Welder qualification	Major	WPS/PQR/WTQ			TC	√	P	V	V	
		Hardness test	Critical	Hardness			TC	√	P	W	W	
		Hydro Test	Critical	Measurement			TC	√	P	W	W	
		Steel ball and sponge ball test	Critical	Measurement			TC	√	P	W	W	
2.3.1	Butt Welds	NDT	Critical	RT / UT	100% (in case of random sampling – min. 20%)	Approved Drawing/Datasheet	TC	√	P	W	W	Review of RT Films
2.3.2	Fillet welds (including fins if any)	NDT	Critical	MPE or dye penetration test (DPT)	100%	Approved Drawing/Datasheet	TC	√	P	W	W	
2.4	Piping and Fitting	NDT	Critical	MPI/ PT	100%	Approved Drawing/Datasheet	TC	√	P	V	V	
2.4.1	Valves	NDT	Critical	MPI/DPT	100%	Approved Drawing/Datasheet	TC	√	P	V	V	
2.4.2	Butt Welds	NDT All butt welds in alloy steel piping of P91, X20 and X 22 after stress relieving (SR).	Critical	RT or UT & MPE	100%	Approved Drawing/Datasheet	TC	√	P	V	V	Review of RT Films

		UT shall be of digital recordable type										
2.4.3	All Weld joints	NDT 100% (Alloy steel piping of P91, X20 and X 22) (3% hardness on welds of other alloy steel piping.)	Critical	Hardness	100%			√	P	V	V	For preheating and post weld heat treatment (PWHT) induction heating shall be deployed. However, PWHT can be done in furnace also.
2.4.4	HP Piping	NDT	Critical	Examination of welds	100%	Approved Drawing/Datasheet	TC	√	P	V	V	See Note 5
2.4.5	Any other pipe not covered	NDT	Critical	MPE/ DPT/ RT	100% / 10%	Approved Drawing/Datasheet	Test Reports	√	P	V	V	See Note 6
2.5	Valves	NDT	Critical	MPI / DPT	100%			√	P	V	V	
2.5.1	Pressure retaining parts	NDT	Critical	MPE/ RT	100%/ 105							See Note 7
2.5.2	Stem	NDT	Critical	UT	100%	Dia 500mm or more		√	P	V	V	
2.5.3	Body & Bonnet	NDT	Critical	a) UT/RT b) MPE/ DPT	100% 100%	a) 1500 class or above b) 600 class or above		√	P	V	V	
2.6	Compensating Pads											
3.0	Final Inspection											
3.1	Drum	Hydraulic Pressure Test, Dimension	Major	Measurement	100%	Approved Drawing/Datasheet	TC	√	P	W	W	See Note 8
3.1.1	Compensating Pads	Pneumatic test	Major	Measurement				√	P	W	W	See Note 9
3.2	Complete closed end headers	Hydraulic Pressure Test, dimension	Major	Measurement				√	P	W	W	See Note 8

3.2a	Drum Suspension Rod	Profile check, Thread matching, Length Measure	Major	Measurement	100%	Drawing/Datasheet	IR	√	P	W	W	
3.3	Tubes and Fabricated, panels, coils	Hydraulic Pressure Test, Steel Ball and sponge ball test, dimension, trail assembly of spiral wall assembly	Major	Measurement				√	P	W	W	See Note 8
3.4	Valves	a) Hydraulic Pressure Test b) Seat leak test c) Back Seat test	Critical	Measurement	100%	a) ASME/ ANSI 16.34 or IBR b) As per applicable standards or codes c) Approved drawing/ datasheet	TC	√	P	W	W	See Note 8
		Visual & Dimension	Critical	Measurement	100%	Approved drawing/datasheet	TC	√	P	W	W	
		Functional Test	Critical	Measurement	100%							
3.5	Non pressure bearing attachments	NDT, UT and MPE for Load bearing welds after stress relieving. MPE for Non load bearing welds after stress relief.	Major	UT & MPE/DPT MPE/ DPT	100%	Approved drawing/datasheet	TC	√	P	W	W	The toes of the welds adjoining the drum shall be ground smooth prior to stress relieving before carrying out this examination.

NOTE -

1	INSPECTION / NDT / TEST CERTIFICATE / REPORT FOR UT, MPI, BALANCING, PRESSURE TEST SHALL INDICATE REFERENCE OF ITEM SERIAL NUMBER & MODEL FOR CO-RELATION.
2	MATERIAL TEST CERTIFICATE / REPORT FOR TEST SUCH AS CHEMICAL & MECHANICAL TEST ETC. SHALL INDICATE REFERENCE OF ITEM SERIAL NUMBER & ENGINE MODEL FOR CO-RELATION AS PER SAMPLING NORMS.
3	ONLY THOSE MATERIALS SHALL BE USED IN THE MANUFACTURE OF PRESSURE PARTS WHICH CAN BE IDENTIFIED AGAINST MILL SHEET OR MANUFACTURER TEST CERTIFICATES. MATERIAL SHALL MEET ALL THE MANDATORY REQUIREMENTS (AND SUPPLEMENTARY CHECKS, IF ASKED FOR) OF SPECIFIED SPECIFICATION, INDIAN BOILER REGULATIONS (IBR), AND RELEVANT CODE/STANDARD. ALL NON-DESTRUCTIVE TESTING AS DETAILED AGAINST RELEVANT EQUIPMENT SHALL MEET THE REQUIREMENT OF ASTM SECTION 3 VOL. 3.03 OR EQUIVALENT BIS STANDARD

4	RAW MATERIAL FOR TUBES FOR WATER WALL, SUPERHEATER, REHEATER, ECONOMIZER, RISER, SUPPLY AND CONNECTING TUBES INCLUDING NOZZLE/STUBS, CONNECTIONS FOR DRUM, HEADERS, PIPE WORK ETC., SHALL BE SUBJECTED TO 100% UT PRIOR TO FABRICATION											
5	a) Temperature > 400°C and/ or pressure exceeding 71 bar • 100% RT or UT on butt welds and full penetration branch welds. • 100% MPE. b) Temperature > 175°C up to 400°C and/or pressure exceeding 17 bar adopt 71 bar • 100% RT or UT on butt welds and full penetration branch welds for pipe diameter more than 100NB. • 10% RT or UT on butt welds and full penetration branch welds for pipe diameter up to 100NB. • 100% MPE.											
6.	100% MPE or DPT for underground pipes and 10% MPE or DPT for above ground piping. RT for 10% of butt welds of underground piping.`											
7.	a. The drum and all components which are to be subjected to fluid pressure shall be tested to minimum of 150% of the design pressure. The duration of the pressure tests shall be sufficient, as approved by the purchaser, to show any leakage paths and to permit a thorough examination of the component whilst under pressure. b. The temperature of the fluid used for the pressure test shall be such as to avoid any possibility of brittle fracture at a low temperature and the same shall be modified and submitted to the purchaser for approval, before commencing the test. c. The fluid used shall be of a sufficient purity and where relevant, suitable inhibitors shall be used to avoid excessive corrosion and /or damage to temporary parts either during the test or prior to drying and cleaning.											
8.	All compensating pads shall be provided with two-threaded weep holes to test welds at 0.5kg/cm ² (g) with soap solution and no leakage shall be ensured.											
9.	Chemical requirements: As per ASME Sec II A											
10.	For supercritical unit The design of all pressure parts (tubes, headers, separators, vessels etc.) shall meet the requirements of Indian Boiler Regulations (IBR).											
1.2 BOILER WATER CIRCULATION PUMPS												
1.0 Raw Material Inspection												
1.1	Casing Pump , Discharge nozzle, Heat Barrier, Motor frame (casing), pipes, Thrust bearing case,	Chemical, Mechanical, Heat treatment, Hardness, High temperate physical properties, UT	Critical	Review of TC	100%	Apart from mandatory and supplementary check of material specification. Approved Drawing/Datasheet	MTC	√	P	V	V	
1.2	Shaft, Stud bolt	Mechanical Properties, Chemical, Heat	Critical	Review of TC	100%	Apart from mandatory and	MTC	√	P	V	V	

		treatment, Impact, High temp physical, Run out, Dimension, DPT, RT, Hardness				Supplementary check of material specification. Approved Drawing/Datasheet						
1.3	Impeller	Mechanical Properties, Chemical, Heat treatment, Impact, High temp physical, Dimension, DPT, Dynamic balancing	Critical	Review of TC	100%	Apart from mandatory and supplementary check of material specification. Approved Drawing/Datasheet	MTC	√	P	V	V	
1.4	Spherical Casing	PT Bevel edge surface for site joint and sealing surface	Major	NDT	100%	ASME SEC V / ASME SEC VIII DIV 1	TC	Y	P/V	V	V	
2.0	In Process Inspection											
2.1	Welding Documents	WPS, PQR, Welder qualification	Major	Review	100%	ASME SEC IX, IBR	WPS/PQR	√	P	V	V	
2.2	Butt welds on pressure parts for these pumps	NDT, Heat Treatment as applicable	Critical	UT or RT and MPE or DPT	100%	Approved Drawing/Datasheet	MTC	√	P	V	V	
3.0	Final Inspection											
3.1	Boiler water circulation pumps	a) Hydraulic Test b) Dimension	Critical	Measure ment	100%	Approved Drawing/Datasheet	TC	√	P	W	W	The pump casing to be tested to minimum of 150% of the design pressure
		Performance	Critical	Measure ment	100%	Approved Drawing/Datasheet	TC	√	P	W	W	
		Electrical checks like Winding Resistance, HV, IR,	Critical	Measure ment	100%	Approved Drawing/Datasheet	TC	√	P	W	W	

		Endurance test of motor windings, joints and terminal seals										
		NPSH test, Temp. rise test, under voltage test, function of pump at operating temp. & pressure and hot standstill & start test, ,No-load test, lock rotor test, performance test , proof test, noise, vibration, over speed test	Critical	Measure ment	One no.	Approved Drawing/Datasheet	TC	√	P	W	W	Documents will be submitted for review and approval of the purchaser
1.3 AIR PREHEATERS, STEAM COIL AIR PRE-HEATER AND FUEL OIL HEATERS												
Refer Appendix-B - Page No.: C-264												
1.4 SOOT BLOWERS												
1.0	Raw Material Inspection											
1.1	Casing, Tubes	Chemical & Mechanical, NDT	Major	UT	100%	Approved Drawing/Datasheet	MTC	√	P	V	V	
1.2	Welding Documents	WPS, PQR, Welder qualification	Major	Review	100%	ASME SECIX,IBR	WPS/PQR	√	P	V	V	
2.0	In-Process Inspection											
2.1	Soot blowers : Butt weld	NDT	Critical	MPI/LPI	20%	Approved Drawing/Datasheet	TC	√	P	V	V	
2.2	Leakage Test	Hydraulic test	Critical	Measure ment	100%	Approved Drawing/Datasheet	TC	√	P	W	W	
3.0	Final Inspection											
3.1	Soot blowers, total travel, Deflection, direction of rotation	Operational checks & Dimension	Critical	Measure ment	100%	Approved Drawing/Datasheet	TC	√	P	W	W	For Long tube travel, closing and opening time and current drawn

1.5 ID, FD AND PA FANS: ROTATING PARTS & STATIONARY PARTS

1.0	Raw Material Inspection										
1.1	Shaft, Hub, Impeller, Blades, Disc, Servo motor, Bearing Housing, Static Parts	Chemical & Mechanical Properties, Hardness, HT & NDT(UT), MPI	Major	Review of TC	100%	Approved Drawing/Datasheet	TC	√	P	V	V
2.0	In-Process Inspection										
2.1	Rotor components	NDT	Critical	MPI/PT	100%	Approved Drawing/Datasheet	TC	√	P	W	W
2.2	WPS, PQR, welder qualification PWHT Shaft dimension, Impeller assembly & bearing housing dimensions, Balancing, Blades moment sorting, blade natural frequency test	Visual, Dimensional, Balancing	Critical	Review/ Measurement	100%	Approved Drawing/Datasheet	TC	√	P	V	V
2.3	Function test – for shaft with bearing assembly for temp rise, Impeller assembly with servomotors for blade shaft movement, assembly of blades- no fouling at fully close condition, Static parts assembly- dimensions	Functional Test	Critical	Measurement	100%	Approved Drawing/Datasheet	TC	√	P	W	W
2.4	Butt welds in Rotor components	NDT	Major	UT/ MPE	100%	UT for Butt welds All welds MPE after stress relief.		√	P	W	W

2.5	Main Shaft	Check of dimension R/O & F/O	Major	Dimensional	100%	Drawing	Test Report	√	-	P	W	
2.6	Impeller Hub	Check of dimensions incl radial and axial run at machines conditions, checking of bore dimension	Major	Dimensional	100%	Drawing	Test Report	√	P	-	#W	# Bore dimension
		Dynamic balancing of Impeller without internal and without blades	Major	Dimensional	100%	Gr 2.5 of ISO 1940/1	Test Report	-	-	P	V	
3.0	Final Inspection											
3.1	Hydraulic coupling of ID fan	Operational checks String test for coupling to check temperature rise, smooth operation, vibration and noise level	Critical	Measure ment	One fan assembly	Approved Drawing/Datasheet	TC	√	P	W	W	Dry run test to be carried out during string test.
3.2	Performance test of assembled fan	Functional Test	critical	Measure ment	1/type	Approved Drawing/Datasheet	TC	√	P	W	W	
1.5.1 HYDRAULIC BLADE ADJUSTING UNIT												
1.0 Manufacturing of Hydraulic Blade Adjusting Unit												
1.1	Hydraulic Unit	Material Grade	Critical	Mecha nical	100%	DIN EN 10204	TC		P	V	-	
1.2	Hydraulic Unit	Test run on testing stand duration of test -Pressure Equalizing -Leakage oil volume at testing pressure	Critical	Mecha nical	100%	Manufacturing Standard	TC	√	P	V	V	

		-Temperature at Hydraulic housing -Testing of function										
1.3	Hydraulic Unit	Dynamic balancing of cylinder Piston	Critical	Mechanical	100%	Grade 2.5 of ISO-1940 (as min)	TC	√	P	V	V	
1.5.2 STATIC PARTS												
1.0	Welders & Procedures are qualified as per AWS D1.1 for static parts.											
2.0	Raw Materials											
2.1	Plates	Chemical & mechanical properties	Major	Inspection of documents test/review	Melt wide	IS 2062/ENGG DRG/SPECIFICATION	Test report		P	V *	-	*Verification of compliance TC. Round>50mm & plates more than 40 mm shall be UT tested.
2.2	Welder	Check of welder certificates	Major	Inspection of documents test/review	100%	AWSD1.1	Welder certificate		P	-	-	
2.3	Welding procedure specification	Check of welding procedures before manufacturing start up	Major	Inspection of documents test/review	100%	AWSD1.1	WPS		P	V	-	
2.4	Welding procedure qualification record	Check of PQR before manufacturing start up	Major	Inspection of documents test/review	100%	AWSD1.1	PQR		P	V	-	
2.5	Qualification of NDT operators	Check of NDT certificates	Major	Inspection of documents test/review	100%	ASNT SNT TC	NDT certificate		P	V	-	
3.0	In Process Inspection											

3.1	Fan Housing Assembly											
3.1.1	Fan Housing	Check of dimensions after machining	Major	Dimensional Measurement	100%	Drawing	Test report	√	P	W	W	
3.1.2	Fan Housing	Visual inspection of welds seams and test of gas tight weld seams by DPT	Major	Visual & dye penetrant test (DPT)	100%	for visual inspection, Drawing & Manufacturer Standard	Test report	√	P	W	W**	**10% witness by Owner
3.2	Suction Chamber Assembly											
3.2.1	Suction chamber	Check of dimensions	Major	Dimensional Measurement	100%	Drawing	Test report	√	P	W	W	Part processing from BHEL approved sub-contractor (refer note – 4)
3.2.2	Suction chamber	Visual inspection of welds seams and test of gas tight weld seams by DPT	Major	Visual & dye penetrant test (DPT)	100%	for visual inspection, Drawing & Manufacturer Standard	Test report	√	P	W	W**	**10% witness by Owner
3.3	Diffuser Assembly											
3.3.1	Diffuser	Check of dimensions	Major	Dimensional Measurement	100%	Drawing	Test report	√	P	W	W	
3.3.2	Diffuser	Visual inspection of welds seams and test of gas tight weld seams by DPT	Major	Visual & dye penetrant test (DPT)	100%	for visual inspection, Drawing & Manufacturer Standard	Test report	√	P	W	W**	**10% witness by Owner
3.4	Silencer Assembly											
3.4.1	Silencer	Visual inspection of welds	Major	Visual	100%	for visual inspection, Drawing & Manufacturer Standard	Test report	√	P	W	W	

3.4.2	Silencer	Measurement of surface protection coating thickness	Major	Visual & paint coating thickness	100%	AS per Owner approved painting scheme	Test report	√	P	W	V	
4.0	Final Inspection											
4.1	Painting & Preservation	Painting & Preservation	Major	Visual and paint coating thickness	100%	As per Owner approved painting scheme	Test report	√	P	W	V	
Notes: 1. Manufacturer Standard internal NDT norm and the acceptance norm is in line with AWS. 2. Manufacturer Standard internal procedure for visual inspection of non-pressure parts. 3. GA Drawing approved by Owner Engineering. Actual material shall be as per owner approved GA Drawing. If material is not indicated in owner approved GA drawing the same shall be as per manufacturer drawing.												
1.6 COAL MILLS												
1.0	Raw Material Inspection											
1.1	Shaft, coupling, gears and pinions, top and bottom races and other rotating components Plates, casting like bearing housing, forgings, gear box	Chemical & Mechanical, HT, NDT(UT), Impact test	Critical		100%	Approved Drawing/Datasheet	MTC	√	P	V	V	
1.2	Journal Pressure Spring	Chemical	Major	Lab Test	Test piece per Heat	Manufacturing Standard	MTC	√	P	V	V	1) Testing at raw material manufacturer/supplier works 2) Verification of MTC & correlation with material at manufacturer/Owners approved works
		Hardness Check	Major	Hardness	10%	Manufacturing Standard	MTC	√	P	V	V	
		Load Test	Major	Lab Test	10%	Manufacturing Standard	MTC	√	P	V	V	
		Compression Test	Major	Lab Test	As per manufacturer specification	Manufacturing Standard	MTC	√	P	V	V	

		Dimensional Exam	Major	Measurement	10 %	Manufacturing Standard	MTC	√	P	V	V	
2.0	In Process Inspection											
2.1	Butt welds in tube/body casing and separator of mill	NDT	Major	RT/ UT and MPE/ LPI	100%	RT/ UT for Butt welds All welds MPE or LPI		√	P	W	W	
2.2	WPs, PQR, Welder qualification Dimensions, NDT- RT on butt weld.	Visual, Dimensional, NDT, heat treatment	Critical	Review/ Measure ment	100%	Approved Drawing/Datasheet	TC	√	P	V	V	
2.3	Gear box body	Kerosene Leakage test	Critical	Measure ment	100%	OEM/Manufacturer's standard	TC/IR	√	P	V	V	
3.0	Final Inspection											
3.1	Gearboxes	Operational checks Check rise in oil temperature, noise level and vibration.	Critical	Measure ment	100%	Approved Drawing/Datasheet	TC	√	P	W	W	Check for leak tightness of gear case also to be done.
3.2	Burner Assembly	Assembly	Critical	Measure ment	100%	Approved Drawing/Datasheet	TC	√	P	W	W	
3.3	Trial assembly of mill & No load trial run	No Load Trial	Critical	Measure ment	At least one no.	Approved Drawing/Datasheet	TC	√	P	W	W	
1.6.1 ROLLS												
SL NO	COMPONENT / OPERATIONS & DESCRIPTION OF TEST	TYPE OF CHECK	Quantu m of Check / Sampling with basis	REFERENCE DOCUMENT	ACCEPTANCE NORMS	FORMAT OF RECORD		AGENCY			REMARKS	
								M	C	N		
1	2	3	4	5	6	7	D *	8			9	
1.0	Raw Materials											
1.1	Moulding sand	Physical	Each batch	Manufacturer's standard/ Customer Tech. Specification	Lab record	-	P	-	-			
1.2	Ferro alloys	Chemical	Each batch	Manufacturer's standard/ Customer Tech. Specification	Lab record	-	P	-	-			

1.3	Scrap	Chemical	Each batch	Manufacturer's standard/ Customer Tech. Specification	Lab record	-	P	-	-	
2	Inprocess Inspection – Inserts for Sintercast Grinding Rolls									
2.1	Melting	Temp. control	Each melt	Manufacturer's standard/ Customer Tech. Specification	Log sheet	-	P	-	-	
		Chemical test	Each melt	Manufacturer's standard/ Customer Tech. Specification	Test report	✓	P	V	V	
		Pouring	Each melt	Manufacturer's standard/ Customer Tech. Specification	Log sheet	-	P	-	-	
2.2	Insert casting	Visual	Each insert	Manufacturer's standard/ Customer Tech. Specification	Test report	-	P	-	-	
3.0	Inprocess Inspection – Sintercast Inserted Roll Castings									
3.1	Melting	Temp. control	Each melt	Manufacturer's standard/ Customer Tech. Specification	Log sheet	-	P	-	-	
		Chemical test	Each melt	Manufacturer's standard/ Customer Tech. Specification	Test report	✓	P	V	V	
		Pouring	Each melt	Manufacturer's standard/ Customer Tech. Specification	Log sheet	-	P	-	-	
3.2	Roll casting	Heat treatment	Each batch	Manufacturer's standard/ Customer Tech. Specification	H/T charts	✓	P	V	V	
		Visual	Each Roll	Manufacturer's standard/ Customer Tech. Specification	Test report	-	P	-	-	
		Microstructure	Manufacturer's standard	Manufacturer's standard/ Customer Tech. Specification	Test report	-	P	V	V	
		Hardness	Each H/T Batch	Manufacturer's standard/ Customer Tech. Specification	Test report	-	P	-	-	
4.0	Final Inspection - Finished Sintercast Inserted Roll									
4.1	Finished rolls	Visual	Each roll	Manufacturer's standard/ Customer Tech. Specification	Inspection report	✓	P	W	W	
		Dimensions	Each roll	Manufacturer's standard/ Customer Tech. Specification	Inspection report	✓	P	W	W	
		Hardness test	Each roll- (One	Manufacturer's standard/ Customer Tech. Specification	Inspection	✓	P	W	W	To be checked with Equotip

			insert/roll)		report					hardness tester
		Bore surface contact	Each roll	Min. 85% of area	Inspection report	✓	P	W	W	Not applicable for straight bore rollers
		Identification & Painting	Each roll	Manufacturer's standard/ Customer Tech. Specification	Inspection report	-	P	V	-	

Note : 1. For reference: Heat treatment: Rolls are air quenched from 980 to 1060°c after holding for 9 to 12 hours (Please refer Manufacturer's standard).

Note : 2. Roll serial number to be punched on the non wearing surface of the roll.

Note : 3. The sub vendor must provide and comply with the performance guarantee for operation life of these rollers as described in the contract.

Note 4: M- Manufacturer, C- Main Contractor, N- Customer

1.6.2 TABLE LINERS

1	Raw Materials									
1.1	Moulding sand	Physical	Each batch	Manufacturer's standard/ Customer Tech. Specification	Lab record	-	P	-	-	
1.2	Ferro alloys	Chemical	Each batch	Manufacturer's standard/ Customer Tech. Specification	Lab record	-	P	-	-	
1.3	Scrap	Chemical	Each batch	Manufacturer's standard/ Customer Tech. Specification	Lab record	-	P	-	-	
2	Inprocess Inspection									
2.1	Melting	Temp. control	Each melt	Manufacturer's standard/ Customer Tech. Specification	Log sheet	-	P	-	-	
		Chemical test	Each melt	Manufacturer's standard/ Customer Tech. Specification	Test report	✓	P	V	V	
		Pouring	Each melt	Manufacturer's standard/ Customer Tech. Specification	Log sheet	-	P	-	-	
2.2	Liner casting	Heat treatment	Each H/T batch	Manufacturer's standard/ Customer Tech. Specification	HT Chart	✓	P	V	V	
		Microstructure	Each H/T batch	Manufacturer's standard/ Customer Tech. Specification	Report	✓	P	V	V	
3.0	Final Inspection And Testing									
3.1	Liner casting	Visual	100%	Manufacturer's standard/ Customer Tech. Specification	Inspection report	✓	P	W	W	
		Dimensions and trial assembly	Assy. On test bowl-	Manufacturer's standard/ Customer Tech. Specification	Inspection report	✓	P	W	W	

			100%							
		Hardness test	100%	Manufacturer's standard/ Customer Tech. Specification	Inspection report	✓	P	W	W	Quantum per set to be given
3.2	Liner casting	Identification	Each Set	See notes	Test Certificate	✓	P	V	-	

Note : 1. For reference: Heat treatment: 980 to 1060°C after holding for 4 to 6 hours and air quench (Please refer Manufacturer's standard).

Note : 2. Serial number and set number will be painted on working face of each casting.

Note : 3. The sub vendor must provide and comply with the performance guarantee for operation life of these rollers as described in the contract.

Note 4: M- Manufacturer, C- Main Contractor, N- Customer

1.7 COAL FEEDERS

Sl. No .	Component & Operation	Characteristics	Class	Type of Check	Quantum of Check		Reference Document / Acceptance Norms	Format of Record	D	Agency			Remarks	
					M	C/U				M	C	U		
1.0	Raw Material													
1.1	Plates, Pipes, Forging etc.	Chemical & Mechanical, NDT, Properties	Major	Review of TC	100%		Approved Drawing/Datasheet	TC	√	P	V	V		
1.2	Welding Documents	WPS, PQR, Welder qualification	Major	Review	100%		ASME SEC IX	WPS/PQ R	√	P	V	V		
1.3	Gear box assembly including motor, Feeder main & clean out conveyor drives	Run test, Type and Routine test for motors	Critical	Review of documents	100%		Purchase Specification & IS :325	TC			P	V	V	Procured with TC from manufacturer approved sources
1.4	Remote and Local Cabinets	Overall dimension Functional Test H. V Test	Minor	Review of documents	100%		Owner Approved QP & CHP clearance reports	CHP			P	V	V	
2.0	In-Process Inspection													

2.1	Welds in the casing and pulley fabrication	NDT	Major	MPE/PT & RT on Butt Joints	100%	Approved Drawing/Datasheet	TC	√	P	V	V	Review of RT Films	
3.0	Final Inspection												
3.1	Feeder casing Leak tightness check, endurance test for Load,, Dimension, Feeder body hydro test, gear box run test, dimension, function performance & calibration test	Visual, Dimensional, Functional	Major	Measurement	100%	Approved Drawing/Datasheet	TC	√	P	W	W		
1.8 BOILER STRUCTURE, DUCTS, HOPPERS, DAMPERS ETC.													
Refer Appendix-A -Page No. – C-248													
1.9 ELECTRO- STATIC PRECIPITATORS (ESPs)													
SL. NO	COMPONENT & OPERATIONS	CHARACTERISTICS	CLASS	TYPE OF CHECK	QUANTUM OF CHECK		REFERENCE DOCUMENT	ACCEPTANCE NORMS	FORMAT OF RECORD	AGENCY			REMARKS
					M	C/N				M	C	N	
1.0	Raw Material: Raw material to be procured from OWNER-approved sources.												
1.1	Plates & Rolled Sections	a. Visual & Dimensional	Major	Visual & Measurement	100 %	100 %	Material spec./ OWNER Approved GA Drawings. / Mfg. Drawings	Inward Inspection Report	--	P	V	--	Refer Note 2 & 3
		b. Chemical Composition	Critical	Document Review	1 per Heat / Lot	1 per Heat / Lot		MTC	√	P	V	V	
		c. Mechanical Properties	Critical	Document Review	1 per Heat / Lot	1 per Heat / Lot		MTC	√	P	V	V	
		d. UT (Bars > 50 mm)	Critical	Document Review	100 %	100%	ASTM 388	MTC	√	P	V	V	
		e. UT (Plates Thk. > 40 mm)			100 %	100 %	ASTM A 578 Level B S2		√	P	V	V	
2.0	Welding / Fit-Up												

2.1	Welding	Welding Procedure – WPS & PQR	Major	Document Review	100 %	100 %	AWS D1.1 / ASME Sec. IX	WPS	√	P	V	V	- WPS & PQR.	
		Welder Performance Qualification – WPQ	Major		100 %	100 %		WPQ	√	P	V	V	-	
2.2	Welding consumable	Verification of Make, Type & Batch No.	Major	Visual	100 %	100 %	ASME Sec II Part C	BTC	√	P	V	-	As per the MAIN SUPPLIER-approved list	
2.3	Straightening Marking, Cutting, Edge Preparation, Drilling and Splicing	a. Shape b. Size c. Diagonal d. Straightness	Major	Visual & Measurement	100%	-----	Drawings / spec.	Job History Card	--	P	--	--		
2.4	Fit Up, Welding, Dimensional Control,	Orientation, Weld Visual, Dimensions, Offset, Straightness, Shape, Diagonal & etc.	Major	Visual & Measurement	100 %	--	Drawings / spec.	Job History Card	--	P	--	--		
2.5	Pre-heating (as applicable)	Preheating Temperature Control	Major	Measurement	100 %	100 %	As per WPS & Tech. Spec.	Register	--	P	V	V		
3.0	Non-Destructive Examination													
3.1	Edge Preparation for Plates THK ≥ 32mm	Lamination/ Flaws	Major	MPT/LPT	100 %	100 %/ 10%	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	V	V	
3.2	Butt weld Joints for Plates THK ≥ 32mm	Weld Soundness	Critical	*RT / UT	100 %	100 % / 10 %	AWS D1.1	AWS D1.1	RT / UT Report	√	P	W	W	*RT films review

3.3	Butt Weld Joints for Plates 25mm ≥ THK < 32mm	Weld Soundness	Critical	MPT / LPT	100 %	100 % / 10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
				*RT / UT	10 %	10 %	AWS D1.1	AWS D1.1	RT / UT Report	√	P	W	W	*RT films review
3.4	All Butt Weld Joints of Plates THK < 25mm	Weld Soundness	Critical	MPT / LPT	10 %	10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
3.5	All fillet Weld Joints	Weld Soundness	Major	MPT / LPT	10 %	10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
3.6	All fillet weld of Built-up Girders, including the Bracket	Weld Soundness	Major	MPT / LPT	100 %	100 % / 10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
3.7	Fillet weld at Base and Top Plate Welding in Columns	Weld Soundness	Major	MPT / LPT	100 %	100 % / 10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
3.8	Weld Visual	Completeness of welding, Surface Defects, Weld Size	Major	Visual	100 %	100 % / 10 %	Mfg. Drawings	AWS D1.1	Inspection Report	√	P	W	W	
3.9	Bend line – Hopper, Funnel & etc.	Bending area	Major	MPI / LPT	100 %	10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
4.0	Final Inspection													

4.1	Orientation and Dimensional Inspection	Verification of Orientation, Dimensions, hole & position. Straightness, Twist, Web verticality, web shift - as applicable	Major	Visual and Measurement	100 %	100 % / 10 %	Drawings / Applicable Standard	Inspection Report	√	P	W	W	If columns are to be dispatched in two or more parts, a mock-up assembly of the columns must be conducted at the shop. Ground Mock-up assembly of 1 st Hopper shall be performed at each Manufacturer.
4.2	Match Marking & Identification	Material Identification Mark No. with DU No	Major	Visual	100 %	100 %		Inspection Report	√	P	W	R	All Components except Loose materials shall be identified by a Hard Punch. The same shall be highlighted by silver Paint and stencilled by Weatherproof Paint.
5.0	Surface Preparation And Painting												
5.1	Surface Preparation & Painting	Removal of Scale, Rust, Foreign Materials, etc.	Major	Visual	100%	100%	Approved Painting Scheme/ Approved Drawing / Tech. Spec.	Inspection Report	√	P	R	-	
		Review of Paint Certificate	Major	Visual	Per Batch	Per Batch		Paint Mfr. TC	√	P	R	V	
		Complete Coverage of Paint	Major	Visual	100 %	Random		Inspection Report	√	P	W	V	
		DFT	Major	Measurement	100 %	Random		Inspection Report	√	P	W	V	
5.2	Completeness of Supply	Verification of Scope of Supply & Correctness of QA Documents	Major	Verification	100 %	100 %	Purchase Order / Quality Plan / Drawing	Inspection reports	--	P	V	--	

Notes:

1. M/s OWNER and M/s MAIN SUPPLIER inspectors shall have full access to take photos of the respective jobs without seeking prior permission from the manufacturer.
2. Raw materials must be sourced from OWNER-approved suppliers, and the MTC must be submitted.
3. The manufacturer shall provide appropriate storage for raw materials, ensuring colour coding and traceability.
4. The Heat Number or Internal Allocation Number (IAN) shall be provided on the main parts (Flange & Web), excluding attachments.

Qualified and continuously employed welders may be deployed without requalification. Records of their qualification and continuity must be available for verification. A newly qualified welder by the Main Contractor shall be deployed for welding.

1.9.1 COLLECTING ELECTRODES

1.0	Raw Material												
1.1	COIL	CHEMISTRY	MAJOR	CHEMICAL	EACH HEAT	EACH HEAT	AS PER DRAWING / GMS	TC	√	P	V	V	
		HARDNESS AND ERICHSEN CUPPING VALUE	MAJOR	HARDNESS AND ERICHSEN CUPPING TEST	EACH HEAT	EACH HEAT	MATL. AS PER SPECN.	TC		P	V	V	
2.0	In Process Control												
2.1	ROLLING AND DIMENSIONAL CONTROL	PROFILE, CAMBER, BEND (LONG), LENGTH & PROFILE WIDTH, TRANSVERSE BEND, TWIST & WAVINESS	MAJOR	MEASUREMENT	Daily Once	ONCE IN A MONTH ON SURVEILLANCE BY OWNER	DRAWING	R		P	W	W	
3.0	FINAL INSPECTION	FINISH, PRESERVATION AND STURDINESS OF PACKING	CRITICAL	VISUAL	ALL BUNDLES	RANDOM	RUST PREVENTIVE FLUID APPLICATION AND AS PER PACKING DRAWING	R	√	P	W	V	

1.9.2 EMITTING ELECTRODES

1.0	Raw Material												
		CHEMISTRY	MAJOR	ANALYSIS	EACH HEAT	EACH HEAT	AS PER DRG / GMS	MILL TC	√	P	V	V	MATERIALS TO BE,
		MECHANICAL (UTS)	MAJOR	TENSILE TEST	ALL COILS	ALL COILS	AS PER DRG / GMS	MILL TC		P	V	V	

1.1	WIRE	MICRO STRUCTURE (Longitudinal & transverse (250X))	MAJOR	MICRO EXAMINATION	ONE SAMPLE /BATCH OF 1000 KGS	ONE SAMPLE /BATCH OF 1000 KGS	AS PER DRG / GMS	MILL TC	✓	P	V	V	PROCU RED WITH MILL TC	
		SURFACE PURITY												
		FREEDOMFROM CHLORIDE IONS	MAJOR	CHLORIDE ION CHECK	EACH COIL	EACH COIL	AS PER DRG / GMS	MILL TC		P	V	V		
		FREEDOM FROM SURFACE DEFECTS	MAJOR	MICRO EXAM	EACH COIL	EACH COIL	AS PER DRG / GMS	MILL TC		P	V	V		
		COILING TEST	MAJOR	UNIFORMITY OF COIL	SAMPLE COILS	SAMPLE COILS	AS PER DRG / GMS	MILL TC		P	V	V		
1.2	SHROUD TUBE FOR EMITTING ELECTRODE	CHEMISTRY & MECHANICAL	MAJOR	ANALYSIS	EACH HEAT	EACH HEAT	AS PER DRG / GMS	TC		P	V	V	REFERE- NCE TC WILL BE FU ED AS THESE ARE STOCK ITEMS	
2.0	In Process Control													
	NOTE: DURING STORAGE AND HANDLING THE CONTACT OF EMITTING ELECTRODE WIRE WITH STEEL SURFACE SHALL BE AVOIDED. IF REQUIRED SURFACE LIKELY TO COME IN CONTACT MAY BE LINED WITH RUBBER.													
2.1	DIMENSION	NO OF TURNS & END CONTSRUCTION ETC	MAJOR	MEASUREMENT AND CHECKING WITH TEMPLATE	100% BY SUB- VENDOR	SINGLE SAMPLING BY BHEL AQL 1.0% IL -III, IS: 2500 PART – I	DRAWING	R		P	V	V		
2.2	SURFACE CONDITION	FREE FROM SURFACE DEFECTS AND NO JOINTS	MAJOR	VISUAL & MAGNIFYING GLASS	100%	10%	DRAWING / BHEL ACCEPTANCE STANDARD (Scratches not accepted)	R		P	V	V		
3.0	Final Inspection													
3.1	SPRING BACK, RETENTION FORCE & INITIAL TENSION	SPRING BACK TEST, RETENTION FORCE, INITIAL TENSION	MAJOR	MEASUREMENT	ONE ELECTRODE PER HEAT	ONE ELECTRODE PER HEAT	DRAWING / BHEL ACCEPTANCE STANDARD	R		P	W	V/ W @	@once in three months	
3.2	DIMENSIONS & SURFACE CONDITION	DIMESIONS & SURFACE CONDITIONS	MAJOR	MEASUREMENT & VISUAL	20 piece per lot	20 piece per lot	DRAWING/BHEL ACCEPTANCE STANDARD			P	W	V/ W @	@once in three months	

3.3	FINISH & PACKING	FINISH AND PACKING	MAJOR	VISUAL	100%		DRAWING/BHEL ACCEPTANCE STANDARD	R		P	W	V	
NOTE -													
1	INSPECTION / NDT / TEST CERTIFICATE / REPORT FOR UT, MPI, BALANCING, PRESSURE TEST SHALL INDICATE REFERENCE OF ITEM SERIAL NUMBER & MODEL FOR CO-RELATION.												
2	MATERIAL TEST CERTIFICATE / REPORT FOR TEST SUCH AS CHEMICAL & MECHANICAL TEST ETC. SHALL INDICATE REFERENCE OF ITEM SERIAL NUMBER & ENGINE MODEL FOR CO-RELATION AS PER SAMPLING NORMS.												
3	For Critical items of APH like Rotor post, module, bearings, lub oil system etc. Oil heater & SCAPH, separate MQPs shall be prepared with mutual agreement between manufacturer and purchaser.												
4	For coal mills, MQP for mill shell, grinding rolls, planetary gear box, bullring segment, Bowl assembly separator MQP shall be prepared with mutual agreement between manufacturer and purchaser.												
2. STEAM TURBINE GENERATOR AND AUXILIARIES													
2.1 STEAM TURBINE - HIGH PRESSURE (HP) AND INTERMEDIATE PRESSURE (IP) ENCLOSURES													
1.0 Raw Material Inspection													
1.1	Casting material	Chemical properties, Heat Treatment.	Critical	Chemical composition, Review of HT charts, Defectogram etc.	100%	OEM Standard	MTC	√	P	V	V	Creep data in case of New or Non-established vendor	
1.2	Test pieces of the material and condition of the casting	Chemical & Mechanical properties, Micro structure	Critical	Chemical & Mechanical testing	100%	OEM Standard	MTC	√	P	W	W		
2.0	In Process Inspection												
2.1	Casting material	Final NDT after HT	Critical	MPE	100%	OEM Standard & NDE plan	MTC	√	P	W	W	Both inside & outside surface	
			Critical	UT or RT	100%		MTC	√	P	W	W		
2.2	Welds of stub pipes and transition pieces to the main body of an enclosure	NDT	Critical	RT and MPE or DPT	100%	OEM Standard & NDE plan	MTC	√	P	V	V	After stress relieving	

				RT/UT/MPE of finished welds								
2.3	Casting material	Dimensional control, Hardness survey	Critical	Dimension check, UT, Hardness check	100%	OEM Standard	MTC	√	P	W	W	
2.4	Casting material-machining	Machining of Horizontal Joint surface, Groves & other parts	Critical	Dimension check	100%	OEM Standard	MTC	√	P	W	W	Random for Customer
3.0	Final Inspection											
3.1	Cast enclosure	Hydraulic Test	Critical	Measure ment	100%	OEM Standard	TC	√	P	W	W	
3.2	Cast enclosure	Box up of upper & lower half(Contact check)	Critical	Measure ment	100%	OEM Standard	TC	√	P	W	W	
2.2 STEAM TURBINE - LOW PRESSURE (LP) ENCLOSURES (FABRICATED)												
1.0	Raw Material Inspection											
1.1	Plates, Piping & Fittings, Casing Casting (if applicable)	Chemical, Mechanical & NDT	Major	TC Review	100%	Approved Drawing/Datasheet	MTC	√	P	V	V	Co-related MTC from approved sources, UT more than 25mm for Plates
2.0	In Process Inspection											
2.1	Welds of an enclosure	Stress Relieving after completion of all welding	Major	Heat Treatment	100%	OEM Standard	IR	√	P	V	V	
		Welding & NDT	Critical	RT and MPE or DPT RT/UT/MPE of finished welds	100%	OEM Standard	IR	√	P	V	V	See Note 3 for NDT, Welding as per approved WPS/PQR/WTQ

3.0	Final Inspection											
3.1	Fabricated enclosure	Hydraulic Test	Major	Mech. Tests /Measurement	100%	OEM Standard	IR/TC	√	P	W	W	After stress relieving
		Box up of upper & lower half (Contact check)	Critical	Measurement	100%	OEM Standard	IR/TC	√	P	W	W	
		Visual & Dimensional	Critical	Visual, Measurement	100%	Approved drawing/datasheet	IR/TC	√	P	W	W	
2.3 TURBINE ROTORS - FORGING AND COMPLETE ROTOR												
1.0	Raw Material Inspection											
1.1	Forging material	Chemical properties, Heat Treatment,	Major	Chemical composition, Review of HT charts etc.	100%	OEM Standard	MTC	√	P	V	V	Creep data in case of New or Non-established vendor
1.2	Test pieces of the material	Chemical & Mechanical Properties, Microstructure	Critical	Chemical & Mechanical testing	100%	OEM Standard	MTC	√	P	W	W	
1.3	Forging material	Overlay Welding & its NDT	Critical	WPS/PQR review	100%	OEM Standard	MTC	√	P	V	V	
1.4	Forging material	NDT	Critical	UT	100%	OEM Standard & NDE Plan	MTC	√	P	W	W	
2.0	In Process Inspection											
2.1	Rotor Welds, if applicable	NDT	Critical	RT and MPE or DPT	100%	OEM Standard & NDE Plan	MTC	√	P	W	W	Ultrasonic examination with normal and angular probes of the weld • Run out of rotor before and after welding

												• MPE on finish welds • Hardness survey on the welds • Stress relieve annealing
				RT/UT/MPE of finished welds				√	P	W	W	
2.2	Forging material	Residual Stress Measurement	Major	Residual Stress	100%	OEM Standard	TC/IR	√	P	W	W	
2.3	Forging material	Machining of Shaft	Critical	Measurement	100%	Approved drawing/datasheet	TC/IR	√	P	W	W	Witness of Critical dimensions
3.0	Final Inspection											
3.1	HP and IP rotor forgings	Thermal stability tests	Critical	Measurement	100%	OEM Standard	TC/IR	√	P	W	W	Thermal stability of the rotors in service and at over speed
3.2	Each Rotor	NDT on Journal areas	Critical	MPE	100%	OEM Standard	TC/IR	√	P	W	W	After final machining and before gashing on other areas
3.3	Each Rotor	Blades Assembly (Feeler check b/w blade to blade at shroud & root)	Critical	Measurement	100%	Approved drawing/datasheet	TC/IR	√	P	W	W	Random check by Customer
3.4	Complete Rotor	i) Axial & radial run-outs before and after blading and after over speed tests.	Critical	Measurement	100%	OEM Standard/Technical Specification		√	P	W	W	See Note 4

		ii) dynamically balancing at rated speed iii) An over speed test to withstand over speed of 120% for five continuous minutes or 125% for two continuous minutes. iv) Vibration tests to determine natural frequencies in various vibration modes to ensure that the ranges are outside operating frequencies v) Lock Blade Lift check										
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2.4 TURBINE ROTORS – STATOR AND ROTOR BLADES AND SHROUD BANDS

1.0	Raw Material Inspection											
1.1	Blades Material Each bar stock	Chemical & Mechanical Properties, Heat Treatment, Hardness	Critical	Measurement	100%	OEM Standard/Technical Specification	MTC	√	P	V	V	Creep data in case of New or Non-established vendor
		Mechanical properties, chemical composition, and microstructure	Critical	Measurement	Per Heat/Batch	OEM Standard	MTC	√	P	W	W	On softest & Hardest Bar

1.2	Each bar stock	NDT	Critical	UT	100%	OEM Standard	MTC	√	P	V	V	Random check by Customer
2.0	In Process Inspection											
2.1	Erosion shield and blade joint	NDT	Critical	Radiography, Visual check PT	100%	OEM Standard	IR/TC	√	P	W	W	Shall be done prior to fitting to the wheel and after over speed tests
2.2	Shroud bands after punching and after riveting	NDT	Critical	DPT	100%	OEM Standard	IR/TC	√	P	W	W	Random check by Customer
3.0	Final Inspection											
3.1	Dimensional Inspection after machining & polishing	Dimensions check (Total Length, Throat, Min. & Max. Thickness etc.)	Critical	Measurement	100%	OEM Standard	IR/TC	√	P	W	W	Random check by Customer
3.2	Machined Blades	NDT	Critical	MPI	100%	OEM Standard	IR/TC	√	P	W	W	Random check by Customer
3.3	Moving blades of over 225mm active length	Moment weighed	Critical	Measurement	100%	OEM Standard	IR/TC	√	P	W	W	Random check by Customer
3.4	LP turbine blades	Natural Frequency Test	Critical	Measurement	100%	OEM Standard	IR/TC	√	P	W	W	Random check by Customer
3.5	Cast blades (if provided)	Chemical analysis and mechanical properties, Heat Treatment, NDT	Critical	(RT/UT)	100%	OEM Standard	IR/TC	√	P	W	W	Random check by Customer, Note-5
2.5 DIAPHRAGMS												
1.0	Raw Material											
1.1	Blade material	Chemical & Mechanical Properties, Heat Treatment, UT, Hardness, PMI	Major	Review of CHP/TC & IR	As per separate blade bar QAP	Approved Drawing/Datasheet	CHP/MT C & IR	√	P	V	V	

1.2	Diaphragm Outer ring	Chemical, Mechanical Properties & UT	Major	Review of TC	Per Heat/Batch	Approved Drawing/Datasheet	MTC	√	P	V	V	
1.3	Diaphragm Flange (Inner Ring)	Chemical, Mechanical & NDT	Major	Review of TC, NDT	Per Heat/Batch & 100%	Approved Drawing/Datasheet	MTC & IR	√	P	V	V	
2.0	In-Process Inspection											
2.1	Welded and Fabricated diaphragms	NDT	Critical	UT/ MPE	100%	OEM Standard	MTC	√	P	V	V	
3.0	Final Inspection											
3.1	Guide blade, moving blades and port wall	Concentricity, flatness	Critical	Measure ment	100%	OEM Standard	IR/TC	√	P	V	V	
3.2	Colour Matching	Surface contact of mating parts	Major	Visual	100%	Manufacturer Standard	TC/IR	√	P	V	V	Min.80 % contact & no through passage
3.3	Visual & Dimensional		Major	Measure ment	100%		TC/IR	√	P	V	V	
2.6	EXHAUST HOOD											
1.0	Raw Material Inspection											
1.1	Castings, Plates (if fabricated construction)	Chemical analysis and mechanical properties	Major	Review of TC	100%	As per drawing/datasheet	MTC	√	P	V	V	
2.0	In-Process Inspection											
2.1	Butt welds of Fabricated construction (if provided)	NDT	Critical	RT/ DPT	10%/ 100%	OEM Standard	TC/IR	√	P	V	V	
3.0	Final Inspection											

3.1	Finished hood	Pressure test	Critical	Measure ment	100%	As per drawing/datasheet		√	P	W	W	
3.2	Exhaust hood parting planes	Blue matching & Dimensional check	Critical	Measure ment	100%	As per applicable standard		√	P	W	W	
2.7 STOP, CONTROL AND BYPASS VALVES, ACTUATORS/ SERVO-MOTORS AND STEAM STRAINERS												
1.0	Raw Material Inspection											
1.1	Raw material of valve body, bonnet, valve disc, seat & spindle	Chemical & Mechanical properties, Heat Treatment, NDT,IBR Forms	Major	Review of TC	100%	Approved drawing/ datasheet, OEM Standard	MTC	√	P	V	V	Creep data in case of New or Non- established vendor
2.0	In Process Inspection											
2.1	Stellited and nitrided areas of components and on stellite components in the finish ground or honed condition	NDT	Major	Visual check/ DPT	100%	OEM Standard	TC/IR	√	P	V	V	Random check for Customer
2.2	Valve body and bonnet castings or forgings	NDT	Critical	RT/ UT MPE	100%	Body and bonnet shall also be subjected to 100% MPE on entire surface/ OEM Standard		√	P	W	W	
2.3	All pressure containing welds in body and bonnet	NDT	Critical	RT/ UT, MPE	100%	OEM Standard	TC/IR	√	P	V	V	Random check for Customer
2.4	Bar stock of 40 mm and above size for valve stem	NDT	Critical	UT, MPE/ DPT	100%	OEM Standard	TC/IR	√	P	V	V	Random check for Customer
3.0	Final Inspection											
3.1	Each valve body and bonnet	Pressure test, Seat Leakage Test	Critical	Measure ment	100%	As per drawing/ datasheet/ IBR requirements (if applicable)	TC/IR	√	P	W	W	At min.1.5 times the max. working pressure after applying temp.

												corrections for min. 30 minutes
3.2	All the actuating cylinders and servomotors	Operational checks	Critical	Measurement	100%	Approved drawing/datasheet, Manufacturer Standard	TC/IR	√	P	W	W	
3.3	Functional Testing with Actuator	Operational checks	Critical	Measurement	100%	Approved drawing/datasheet, Manufacturer Standard	TC/IR	√	P	W	W	
2.8 CAST AND FORGED STEEL COMPONENTS												
1.0	Final Inspection											
1.1	LP casing- in case of cast design, inlet, extraction and exhaust connections, shaft seal covers and rings, governor shaft, breach nut, threaded ring, angle ring, U- ring, servomotor parts such as body, piston, cover, yokes; turning gear casing	Chemical analysis and mechanical	Major	Measurement	100%	Approved drawing/datasheet, Manufacturer Standard	MTC	√	P	V	V	Including other items
2.9 BOLTS AND NUTS FOR PRESSURE RETAINING ENCLOSURES AND ROTOR COUPLINGS												
1.0	Raw Material											
1.1	Bar stock for bolts	Chemical & Mechanical, NDT	Major	Review of TC	100%	Approved drawing/datasheet, Manufacturer Standard	MTC	√	P	V	V	
2.0	Final Inspection											
2.1	Finish machined bolts	NDT	Major	MPE or DPT	100%		TC/IR	√	P	V	V	

		Visual & Dimensional	Major	Measure ment	100%		TC/IR	√	P	V	V	
2.10 GOVERNING AND PROTECTION SYSTEM EQUIPMENT												
1.0	Final Inspection - Electro-hydraulic controller, hydraulic amplifier, hydraulic controller, electro hydraulic convertors, hydraulic convertors, hydraulic speed governor, trip devices etc.											
1.1	Pressure retaining parts	Hydraulic testing	Critical	Measure ment	100%	Approved drawing/ datasheet, Manufacturer Standard	TC/IR	√	P	W	W	
1.2	All the main assemblies and sub-assemblies	Functional test	Critical	Measure ment	100%		TC/IR	√	P	W	W	
1.3	All control equipment	Rig testing	Critical	Measure ment	100%		TC/IR	√	P	W	W	
1.4	Nitrided, stellite components in the finish ground	NDT, Hardness properties	Critical	DPT, Hardness	100%		TC/IR	√	P	W	W	
2.11 INSPECTION OF COMPLETED TURBINE												
2.11.1 HP, IP & LP TURBINE												
1.0	Final Inspection											
1.1	Complete turbine	Various checks and measurement: • Clearance between shaft seal casing and shaft seal ring • Clearance in anti- rotational device in shaft seal casing joint • Axial and radial alignments of inner and outer casings	Critical	Measure ment	100%	OEM Standard/ Technical Specification/approved drawing & datasheet	TC/IR	√	P	W	W	

		- For radial and axial blade clearances in blading section - For axial and radial clearances in shaft seal - Min. axial clearances and min. radial clearances in completely assembled steam turbine - Alignment of over speed governor - Axial distances for shroud bands for casings										
1.2	Steam turbine control and emergency control equipment	No load running test	Critical	Measurement	100%	OEM Standard/ Technical Specification	TC/IR	√	P	W	W	
1.3	Pre- dispatch inspections	Visual including clearance check for transportation device	Major	Measurement	100%	OEM Standard/ Technical Specification	TC/IR	√	P	V	V	
2.11 INSPECTION OF COMPLETED TURBINE												
2.11.1 HP, IP AND LP TURBINES												
1.0	Final Inspection											
1.1	Complete turbine	Various checks and measurement: radial and axial blade clearance in blading section	Critical	Measurement	100%	OEM Standard/ Technical Specification/approved drawing & datasheet	TC/IR	√	P	W	W	

		- min. axial clearances and min. radial clearances in completely assembled steam turbine - alignment of stationary blade carriers										
1.2	Steam turbine control and emergency control equipment	No load running test	Critical	Measurement	100%	OEM Standard/ Technical Specification	TC/IR	√	P	W	W	
1.3	Pre- dispatch inspections	Visual including clearance check for transportation device	Major	Measurement	100%	OEM Standard/ Technical Specification	TC/IR	√	P	V	V	
2.12 INTEGRAL AUXILIARIES OF STEAM TURBINE												
1.0	Raw Material - Bearing Pedestals, Housings And Bearings											
1.1	Bearing Pedestal, Housings & Bearings	Chemical & Mechanical properties, NDT	Major	Review of TC	100%	Approved Drawing/Datasheet	MTC	√	P	V	V	
2.0	In-Process Inspection											
2.1	Pedestals	Leakage test	Critical	Measurement	100%	OEM/Manufacturer's standard	TC/IR	√	P	V	V	
2.2	Butt Welds (for fabricated pedestals and housing)	NDT	Critical	RT or UT	10%	OEM/Manufacturer's standard	TC/IR	√	P	V	V	after stress relieving
2.3	Bearing shell	NDT	Critical	RT or UT	100%	OEM/Manufacturer's standard	TC/IR	√	P	V	V	
3.0	Final Inspection											
3.1	Bearing oil inlet	Hydraulic Test	Major	Measurement	100%	OEM/Manufacturer's standard	TC/IR	√	P	V	V	

	pipng and jacking oil piping											
3.2	UT & DPT of bonding of lining to Shell/Liner & Thrust Bearing Pads	NDT	Critical	UT & DPT	100%	OEM/Manufacturer's standard	TC/IR	√	P	W	W	
3.3	Parting Plane tightness check	Feeler Gauge tightness of parting planes of bearing shell	Major	Measure ment	100%	OEM/Manufacturer's standard	TC/IR	√	P	V	V	
3.4	Colour Contact	Colour contact of joint plane for Journal bearing/combined Thrust & Journal bearing	Major	Measure ment	100%	OEM/Manufacturer's standard	TC/IR	√	P	V	V	
3.5	Visual & Dimensional	Verification of Dimension & Visual inspection	Major	Measure ment	100%	OEM/Manufacturer's standard	TC/IR	√	P	V	V	
2.12.1 LUBRICATING OIL, JACKING OIL AND CONTROL OIL SYSTEMS												
2.12.1.1 PUMPS												
1.0	Raw Materials											
1.1	Pumps Impeller, Shaft, Casing	Chemical & Mechanical Properties, NDT	Major	Review of TC	100%	Approved drawing/datasheet	MTC	√	P	V	V	
2.0	In-process Inspection											
2.1	Oil pump shaft	NDT	Critical	UT	100%	OEM/Manufacturer's Std.	TC/IR	√	P	V	V	
2.2	Pump impeller	NDT	Critical	MPE or DPT	100%	OEM/Manufacturer's Std.	TC/IR	√	P	V	V	
		Over speed test, if applicable for Main Oil Pump	Critical		100%	OEM/Manufacturer's Std.	TC/IR	√	P	V	V	
		Impeller Balancing	Critical	Balancing	100%	OEM/Manufacturer's Std.	TC/IR	√	P	V	V	

2.3	Pump casing	Hydraulic pressure test	Major	Measure ment	100%	Approved drawing/datasheet	TC/IR	√	P	V	V	
3.0	Final Inspection											
3.1	Pumps	Performance Test	Critical	Measure ment	100%	Approved drawing/ datasheet, OEM/ Manufacturer's Std.	TC/IR	√	P	V	V	
3.2	Coolers and lubricating oil tanks	Pressure test(Tube & Shell side)	Critical	Measure ment	100%	Approved drawing/datasheet,	TC/IR	√	P	W	W	
3.3	Atmospheric tanks	Performance (leakage test) By water fill test for at least 12 hrs.	Critical	Measure ment	100%	Approved drawing/datasheet,	TC/IR	√	P	W	W	
3.4	Functional Test of Lube Oil Skid	Skid Functional Test for Pump discharge, Pressure, Noise, Vibration & current	Critical	Measure ment	100%	Approved drawing/datasheet, OEM/Manufacturer's Std.	TC/IR	√	P	W	W	
2.12.1.2 OIL PURIFIERS (CENTRIFUGAL TYPE)												
1.0	Raw Materials											
1.1	Major components like Bowl Body, Bowl Hood, Top Disc, Distributor, Shafts, Impeller etc.	Chemical & Mechanical properties	Major	Review of TC	Per Heat/Batch	Manufacturer 'Standard	TC/IR	√	P	V	V	
2.0	Final Inspection											
2.1	All pressure parts	Hydraulic Pressure test	Critical	Measure ment	100%	Approved drawing /datasheet, OEM/ Manufacturer's Std.	TC/IR	√	P	V	V	
2.2	Bowl Complete	Unbalance	Major	Balancing	100%	Manufacturer's Standard	TC/IR	√	P	V	V	

2.3	Complete purifier	Performance Test at manufacturer's works for capacity, mechanical running, sequential operation and interlocks, moisture content, vapour tightness, vibration, noise level, quality improvements etc.	Critical	Measurement	100%	Approved drawing/datasheet, OEM/Manufacturer's Std.	TC/IR	√	P	W	W	See Note 6
2.4		Particle Size at Inlet & Outlet of Centrifuge	Critical	Measurement	01/Type	Approved drawing/datasheet, OEM/Manufacturer's Std.	TC/IR	√	P	W	W	

2.12.1.3 OIL PURIFIERS (COALESCER TYPE)

Sl.No.	Component/Operation	Characteristics	Class	Type of Check	Quantum of Check	Reference document/Acceptance Norms	Format	Agency				Remarks
								D	M	C	U	
1.0	Raw Materials & Bought Out Items											
1.1	Major components like Filter Housing,Piping Assembly etc.	Chemical & Mechanical properties,NDT	Major	Review of TC	Per Heat/Batch	Manufacturer'Standard	TC/IR	√	P	V	V	
1.2	Pump,Motor, Catridges, Intrumentation (DP Gauge, Pressure Gauge etc.)	Receiving Inspection	Major	Review of TC	100%	Manufacturer'Standard	TC/IR	√	P	V	V	

2.0	In-Process Inspection											
2.1	WPS/PQR Check	Welding	Major	Documentation	100%	ASME Sec.IX	WPS/PQR	√	P	V	V	
2.2	Visual & NDT after Welding	Visual,DPT	Major	Visual,DPT	100%	As per relevant Std.	TC/IR	√	P	V	V	
3.0	Final Inspection of Final SKID Assembly											
3.1	Hydrostatic Test with Oil	Hydraulic Pressure test	Critical	Measure	100%	Approved drawing/datasheet, OEM/Manufacturer's Std.	TC/IR	√	P	W	V	
3.2	Visual & Dimensional	Visual, Measurement	Major	Visual, Measurement	100%	Approved drawing/datasheet	TC/IR	√	P	W	V	
3.3	Functionality Test(Flow,Discharge Pressure,Noise,	Function Test	Critical	Measure	100%	Approved drawing/datasheet, OEM/Manufacturer's Std.	TC/IR	√	P	W	W	
3.4	Water & Particle Removal Test	Type Test	Critical	Measure	01/Type	Approved drawing/datasheet, OEM/Manufacturer's Std./Technical Specification	TC/IR	√	P	W	W	
3.5	Painting, Packing & Preservation	Visual,Measurement	Major	Visual,Measurement	100%	Approved Drawing/Datasheet	IR/TC	√	P	V	V	

2.12.2 STEAM CONDENSER

1.0	Raw Material											
1.1	Plate,Pipe, Forgings,Tubes	Chemical & Mechanical Properties,Heat Treatment, NDT	Major	Review of TC	100%	Approved drawing/datasheet	MTC	√	P	V	V	
2.0	In-Process Inspection:-											
2.1	Welding Document	WPS/PQR/WTQ	Major	Verification	100%	ASME Sec.VIII/IX,Technical Specification	WPS & POR	√	P	V	V	
2.2	Weld Visual	Surface Defects	Major	Visual	100%	Approved Drawing,ASME	IR	√	P	V	V	

2.3	Forgings of the nozzles	NDT	Critical	UT & MPE	100%	Manufacturer's Std./Technical Specification	MTC	√	P	V	V	
2.4	Butt welds (including T- joints)	NDT For vacuum containing welds, RT on atleast 10% of each butt weld shall be carried out.	Critical	RT & MPE/ DPT, Review of RT Films	10%/ 100%	Manufacturer's Std./Technical Specification	TC/IR	√	P	V	V	Nozzle welds shall be subject to 100% MPE or DPT
2.5	Welds (of fabricated	NDT	Critical	RT/ UT & MPE	100%	Manufacturer's Std./Technical Specification	TC/IR	√	P	V	V	
2.6	Tube to tube sheet weld joints	NDT	Critical	DPT	100%	Manufacturer's Std./Technical Specification	TC/IR	√	P	V	V	
2.7	Tube sheet Visual & Dimensional	Hole Size,Pitch, Ligament etc.	Critical	Measure	100%	Approved drawing/datasheet	TC/IR	√	P	V	W	Random inspection by Customer
2.8	Condenser Tubes	1.Hydro Test 2. Eddy Current Test 3. Visual & Dimensional check (Length, OD, T hickness, Ovality etc.) In place of hydraulic test, air under water or Helium leak detection test can be carried out	Critical	Visual/ Measurement	100%	Approved drawing/datasheet, Mfg.Std.	TC/IR	√	P	V	W	Random inspection by Customer
3.0	Final Inspection (After assembly at site)											
3.1	Mock-Up for Tube Expansion	Dimensional Check	Major	Measure	100%	Approved Drawing, ASME, Mfg.Std.	IR/TC	√	P	V	V	Random check by Customer
3.2	Tube side of condenser	Hydraulic Pressure test	Critical	Measure	100%	Approved drawing/datasheet, Mfg.Std.	TC/IR	√	P	W	W	
3.3	Shell side of condenser	Water fill test										

3.4	Standpipe, Flash tank etc.	Water fill test	Major	Visual, Measurement	100%	Approved Drawing/Datasheet	IR/TC	√	P	V	V	
3.5	Compensating pads (RF pads)	Pneumatical test										
3.6	Painting, Packing &	Visual, Measurement										

2.12.3 CONDENSER AIR EVACUATION SYSTEM

1.0	Raw material(Bought Out Items)											
1.1	Pump Casing, Head, Impeller, Shaft, Cooler, Separator Tank	Chemical & Mechanical Properties, Heat Treatment, NDT	Major	Review of TC	100%	Approved drawing/datasheet	MTC	√	P	V	V	UT for shaft dia. more than 40mm
2.0	In-Process Inspection											
2.1	Rotor Assembly (Impeller & Shaft Assembly)	Static & Dynamic Balancing	Major	Balancing	100%	Approved drawing/datasheet	MTC	√	P	V	V	
3.0	Final Inspection											
3.1	Pump Casing	Hydraulic testing	Critical	Measure	100%	Approved drawing/datasheet	TC/IR	√	P	W	W	
3.2	Heat exchangers	Hydro test On both tube side and shell side	Critical	Measure	100%	Approved drawing/datasheet	TC/IR	√	P	W	W	After, the heat exchangers shall be suitably dried

3.3	Complete unit	Operation and performance testing including Noise & Vibration	Critical	Measure	100%	Approved drawing/datasheet, Mfg. Std.	TC/IR	√	P	W	W	The test shall be conducted with the respective motors to be supplied
3.4	Painting, Packing & Preservation	Saturated Air(Humid Air Test) & Cavitation Test	Critical	Type Test	1/Type	Approved drawing/datasheet, Mfg. Std.	TC/IR	√	P	W	W	
		Visual,Measurement	Major	Visual,Measurement	100%	Approved Drawing/Datasheet	IR/TC	√	P	V	V	

2.12.4 (A) DEAERATOR & FEED WATER STORAGE TANK

1.0	Raw Materials											
1.1	Plate,Pipe, Fittings,Spray Nozzles, Dished End & Manhole Cover (as applicable)	Chemical & Mechanical Properties,Heat Treatment, NDT	Major	Review of TC	100%	Approved drawing/datasheet	MTC	√	P	V	V	
2.0	In-process Inspection											
2.1	Welding Document	WPS/PQR/WTQ	Major	Verification	100%	ASME Sec.VIII/IX,Technical Specification,IBR as	WPS & POR	√	P	V	V	
2.2	Welding	Surface Defects/NDT	Major	Visual & PT/MT & RT/TOFD/ PAUT	100%	Approved Drawing, ASME,Mfg.Std., IBR as applicable	IR/TC	√	P	V	V	Random Check by Customer, Review of RT Films
2.3	PWHT(as applicable)	Heat Treatment	Major	Visual	100%	Mfg.Std.	IR/TC	√	P	V	V	
3.0	Final Inspection											

3.1	Deaerator & Storage Tank	Hydraulic Test	Critical	Measurement	100%	Drawing/Datasheet	IR/TC	√	P	W	W	
3.2	Visual & Dimensional	Visual,Dimensional	Critical	Measure	100%	Approved Drawing/Datasheet, IBR documents as applicable	TC/IR	√	P	W	W	
3.3	Painting, Packing & Preservation	Visual,Measurement	Major	Visual,Measurement	100%	Approved Drawing/Datasheet	IR/TC	√	P	V	V	

2.12.4 (B) FEED WATER HEATERS, DRAIN COOLERS, GLAND STEAM CONDENSER

1.0	Raw Materials											
1.1	Plate,Pipe, Forgings,Tubes, Tubesheet, Nozzles, Hemihead, Dish End & Manhole Cover (as applicable)	Chemical & Mechanical Properties, Heat Treatment, NDT	Major	Review of TC	100%	Approved drawing/datasheet	MTC	√	P	V	V	
2.0	In-process Inspection											
2.1	Welding Document	WPS/PQR/WTQ	Major	Verification	100%	ASME Sec.VIII/IX, Technical Specification, IBR as	WPS & POR	√	P	V	V	
2.2	Welding	Surface/Volumetric Defects/NDT	Major	Visual & PT/MT & RT/TOFD/ PAUT(as applicable)	100%	Approved Drawing, ASME, Mfg. Std.	IR/TC	√	P	V	V	Random Check by Customer, Review of RT Films
2.3	Dishend Forming	Visual, Dimensional, NDT, Heat	Major	Verification of TC	100%	Approved Drawing, ASME, Mfg. Std.	IR/TC	√	P	V	V	

2.4	Tube Sheet Fabrication	Weld Overlay & its NDT, Visual & Dimensional	Major	Verification of TC	100%	Approved Drawing, ASME, Mfg. Std.	IR/TC	√	P	V	V	Random check by Customer
2.5	Tubes	1. Hydro Test 2. Eddy Current Test 3. UT Test(as applicable) 4. Visual & Dimensional	Critical	Visual, Measurement	100%	Approved drawing/datasheet, Mfg. Std.	TC/IR	√	P	V	V	
2.6	Mock-Up for Tube Expansion	Dimensional Check	Major	Measure	100%	Approved Drawing, ASME, Mfg. Std.	IR/TC	√	P	V	V	
2.7	Tube Bundle Assembly	Dimensional & NDT	Major	Measure, PT	100%	Approved Drawing, ASME, Mfg. Std.	IR/TC	√	P	V	V	
2.8	Tube to tube sheet weld joints	NDT	Critical	DP	100%	Mfg. Std.	IR/TC	√	P	V	V	
2.9	PWHT(as applicable)	Heat Treatment	Major	Visual	100%	Mfg. Std.	IR/TC	√	P	V	V	
3.0	Production Test Coupon testing(as applicable)	Visual & Dimensional	Major	Visual	100%	Drawing, Mfg. Std., IBR as applicable	IR/TC	√	P	V	V	
3.0	Final Inspection											
3.1	Heat exchangers	Hydro test (On both tube side and shell side)	Critical	Measure	100%	Drawing/Datasheet, IBR as applicable	IR/TC	√	P	W	W	After, the heat exchanger s shall be suitably dried and Nitrogen capped
3.2	Complete Unit	Visual, Dimensional	Critical	Measure	100%	Approved Drawing/Datasheet, IBR as applicable	TC/IR	√	P	W	W	

3.3	Painting, Packing & Preservation	Visual, Measurement	Major	Visual, Measurement	100%	Approved Drawing/Datasheet	IR/TC	√	P	V	V	
2.13.4 BOILER FEED PUMPS												
1.0	Raw Material											
1.1	Barrel casing, casing cover, suction & discharge branches, shaft, Impeller, Diffusers ring sections, and wearing rings	Chemical & Mechanical Properties, Heat Treatment & NDT	Major	Review of TC	100%	Approved Drawing/ Datasheet	MTC	√	P	V	V	
2.0	In-Process Inspection											
2.1	On machined components	NDT	Critical	DPT	100%	Manufacturer Standard	MTC	√	P	V	V	
2.2	Rotor Assembly	Runout Checking Dynamic Balancing	Critical	Measurement	100%	Manufacturer Standard, Relevant Std.	TC/IR	√	P	W	W	
2.3	Pump Assembly	Radial & Axial Running Clearances	Critical	Measurement	100%	Drawing, Manufacturer Standard.	TC/IR	√	P	W	W	
3.0	Final Inspection											
3.1	On barrel casing, discharge cover, and suction and discharge branches	Hydraulic testing	Critical	Measurement	100%	Drawing/Datasheet	TC/IR	√	P	W	W	
3.2	Overall Visual & Dimensional Check	Visual & Dimensional	Critical	Measurement	100%	Drawing/Datasheet	TC/IR	√	P	W	W	

3.3	Complete unit	Operation and performance testing -Performance testing as per Hydraulic Institute Standards (HIS) to determine its characteristic curve at design speed and to ensure compliance with design requirement. - Vibration to be Measured in transverse, horizontal and vertical direction at all measuring points -Noise Level as per HIS. -Strip down test	Critical	Measurement	100%	Drawing/Datasheet/Technical Specification, Manufacturer Standard, HIS/ASME	TC/IR	√	P	W	W	Tests shall be carried out with loop water at specified design temperature. Soften quality water shall be used for the performance testing
2.13.4.2 BOOSTER PUMP												
1.0	Raw Materials											
1.1	Casing, Casing cover, Shaft, Impeller, and wearing rings	Chemical & Mechanical Properties, Heat Treatment & NDT	Major	Review of TC	100%	Approved Drawing/Datasheet	MTC	√	P	V	V	
2.0	In-process Inspections											
2.1	On machined components	NDT	Critical	DPT	100%	Manufacturer Standard	MTC	√	P	V	V	
2.2	Rotor Assembly	Runout Checking	Critical	Measurement	100%	Manufacturer Standard, Relevant Std.	TC/IR	√	P	W	W	

2.3	Pump Assembly	Radial & Axial Running Clearances, Dimensional	Critical	Measurement	100%	Drawing, Manufacturer Standard,	TC/IR	√	P	W	W	
3.0	Final Inspection											
3.1	On pump casing	Hydraulic testing	Critical	Measurement	100%	Drawing, Manufacturer Standard,		√	P	W	W	
3.2	Impeller and completed rotor assembly	dynamic balancing test	Critical	Measurement	100%	Manufacturer Standard, Relevant Std.		√	P	W	W	
3.3	Complete unit	Operation and performance testing Performance testing as per (HIS) to determine its characteristic curve at design speed and to ensure compliance with design requirement. Vibration to be Measured in transverse, horizontal and vertical direction at all measuring points Noise Level as per HIS. Strip down test	Critical	Measurement	100%	Drawing/Datasheet/Technical Specification, Mfg. Std., HIS/ASME	TC/IR	√	P	W	W	Tests shall be carried out with loop water at specified design temperature. Soften quality water shall be used for the performance testing
2.13.4.3 GEAR BOX AND HYDRAULIC COUPLING												
1.0	Raw Materials											

1.1	Gears, pinions, wheels and shafts	Chemical & Mechanical Properties, NDT, Hardenability Test	Major	Review of TC	100%	Drawing, Manufacturer Standard	MTC	√	P	V	V	
2.0	In-process Inspection											
2.1	Gears, pinions, wheels and shafts	NDT on machined components	Major	PT/MPT	100%	Manufacturer Standard	TC/IR	√	P	V	V	
2.2	Assembled rotating component	Dynamic balancing test	Critical	Measurement	100%	Manufacturer Standard	TC/IR	√	P	V	V	
3.0	Final Inspection											
3.1	Gear box Assembly	Full load speed and back to back locked rotor torque test	Critical	Measurement	01 Gear Box/ Model	Manufacturer Standard	TC/IR	√	P	W	W	
		Visual & Dimensional Static Tooth Contact, Backlash	Critical	Measurement	100%	Manufacturer Standard	TC/IR	√	P	W	W	
		Mechanical Run Test	Critical	Measurement	100%	Manufacturer Standard, AGMA Std.	TC/IR	√	P	W	W	
3.2	Hydraulic Coupling Assembly	Visual & Dimensional	Critical	Measurement	100%	Approved Drawing	TC/IR	√	P	W	W	
		Performance test at No-Load	Critical	Measurement	100%	Manufacturer Standard	TC/IR	√	P	W	W	
2.13.5 CONDENSATE EXTRACTION PUMPS (CEPS)												
2.13.5.1 PUMP												
1.0	Raw Materials											
1.0	Casings, Diffuser, Shafts, Impeller	Chemical & Mechanical Properties, Heat Treatment, NDT	Major	Review of TC	100%	Approved Drawing/Datasheet	MTC	√	P	V	V	

2.0	In-Process Components											
2.1	Machined components	NDT	Major	PT/MPI	100%	Manufacturer Standard	TC/IR	√	P	V	V	
2.2	Pump shaft	NDT Runout	Major	UT Measurement	100%	Manufacturer Standard	TC/IR	√	P	V	V	
2.3	Pump Impeller	NDT	Major	RT	100%	Manufacturer Standard	TC/IR	√	P	V	V	Review of RT Films
2.4	Butt welds on fabricated components of thickness more than 10 mm	NDT	Major	RT	10%	Manufacturer Standard	TC/IR	√	P	V	V	Review of RT Films
2.5	Welds on Casing, suction bell, shaft, impeller and fabricated components	NDT	Major	DPT	100%	Manufacturer Standard	TC/IR	√	P	V	V	
3.0	Final Inspection											
3.1	On pump casing and pressure containing fabricated parts	Hydraulic testing	Critical	Measurement	100%	Drawing/Datasheet, Manufacturer Standard	TC/IR	√	P	W	W	
3.2	Individual Impeller and completed rotor assembly	Dynamic balancing Test, Runout	Critical	Measurement	100%	Manufacturer Standard, Relevant Std.	TC/IR	√	P	W	W	
3.3	Complete unit	Operation and performance testing • Performance testing as per Hydraulic Institute Standards (HIS) to determine its characteristic	Critical	Measurement	100%	Drawing/Datasheet/Technical Specification, Manufacturer Standard, HIS	TC/IR	√	P	W	W	Tests shall be carried out using cold soften quality water.

		curve at design speed and to ensure compliance with design requirements Vibration to be Measured in transverse, horizontal and vertical direction at all measuring points Noise Level as per HIS. Strip down test										
2.13.6 CONDENSATE POLISHING UNIT (CPU)												
2.13.6.1 CPU SERVICE VESSELS												
1.0	Raw Material											
1.1	Plates for Shell & Dish Ends, Pipes	Chemical & Mechanical, HT & NDT	Major	Review of TC	100%	Approved drawing/datasheet	MTC	√	P	V	V	UT for plates thickness more than 25mm
1.2	Rubber	Chemical & Mechanical	Major	Review of TC	100%	Approved drawing/datasheet, Mfg. Std.	MTC	√	P	V	V	
2.0	In-Process Inspection											
2.1	Finished and formed dished ends and hemi heads	NDT Knuckle portion shall be checked by 100% MPE or DP test	Critical	DP/ MPE	100%	Manufacturer Standard	MTC	√	P	V	V	

2.2	Butt welds full penetration joints and nozzle welds	NDT	Critical	RT & MPE or DPT	100%	Manufacturer Standard	MTC	√	P	V	V	Review of RT Films
2.3	Rubber lining	Mechanical Properties (Tests as per IS-4682 part-I) • Tensile & Elongation, Specific Gravity, Ash content • Adhesion test • Shore hardness test • Ozone resistance test: (50 PPHM) / 40° C / 20 % Strain /70 hrs. • Water Absorption Test .Spark Test	Critical	Measure ment	100%	Technical Specification, Relevant std.	TC/IR	√	P	V	V	Water Absorp tion Test, After ageing test & Compr ession Set test
3.0	Final Inspection											
3.1	Fabricated vessel	Hydraulic testing	Critical	Measure ment	100%	Drawing/datasheet, Manufacturer Standard	TC/IR	√	P	W	W	
3.2	Visual & Dimensional	Dimensional Conformity	Critical	Measure ment	100%	Drawing/datasheet	TC/IR	√	P	W	W	
2.13.6.2 ACID/ ALKALI HANDLING TANKS												
1.0	Raw Materials											
1.1	Plates for Shell & Dish Ends, Pipes	Chemical & Mechanical, HT & NDT	Major	Review of TC	100%	Approved drawing/datasheet	MTC	√	P	V	V	UT for plates thickne ss more

												than 25mm
1.2	Rubber	Chemical & Mechanical	Major	Review of TC	100%	Approved drawing/datasheet, Mfg. Std.	MTC	√	P	V	V	
2.0	In-process Inspection											
2.1	Butt welds full penetration joints and nozzle welds	NDT	Critical	RT & MPE or DPT	100%	Manufacturer Standard	TC/IR	√	P	V	V	
2.2	Rubber lining	Mechanical Properties (Tests as per IS-4682 part-I)	Critical	Measurement	100%	Technical Specification, Relevant std.	TC/IR	√	P	V	V	Tests are same as applicable for CPU
3.0	Final Inspection											
3.1	Fabricated tank	Leakage testing	Critical	Measurement	100%	Drawing/Datasheet	TC/IR	√	P	V	V	Random check by Customer
2.13.6.3 DOSING PUMPS/ METERING PUMPS												
1.0	Raw Materials											
1.1	Casing, Shaft, Impeller, valve seat, diaphragms	Chemical & Mechanical Properties, NDT	Major	Review of TC	100%	Drawing/Datasheet	MTC	√	P	V	V	
2.0	In-process Inspection											
2.1	Machined parts	Dimension LPI	Major	Dimension	100%	ASTM E-165	IR	√	P	V	V	
3.0	Final Inspection											
3.1	Pump casing	Hydraulic testing (at 1.5 times the shut off pressure or 2 times the working pressure)	Critical	Measurement	100%	Drawing/Datasheet	TC/IR	√	P	V	V	Random check by Customer

		whichever is higher)										
3.2	Complete unit	Operation and performance testing	Critical	Measurement	100%	Relevant HIS, Mfg. Std.	TC/IR	√	P	V	V	Random check by Customer
2.13.7 CONDENSER ON-LINE TUBE CLEANING SYSTEM												
2.13.7.1 BALL RECIRCULATION PUMP												
1.0	Raw Material											
1.1	Casing, Impeller & Shaft	Chemical & Mechanical Properties, NDT	Major	Review of TC	100%	Drawing/Datasheet	MTC	√	P	V	V	
2.0	Final Inspection											
2.1	Pump casing	Hydraulic testing	Critical	Measurement	100%	Drawing/Datasheet, Mfg. Std.	TC/IR	√	P	V	V	
2.2	Complete pump	Performance test	Critical	Measurement	100%	Drawing/Datasheet, Mfg. Std.	TC/IR	√	P	V	V	
2.13.7.2 BALL SORTER/ FABRICATED BODY (HOUSING)												
1.0	Raw Materials											
1.1	Housing Shell, Nozzle Flanges, Pipes, Screen Shaft, Gear Boxes	Chemical & Mechanical Properties, NDT	Major	Review of TC	100%	Drawing/Datasheet	MTC	√	P	V	V	
2.0	Final Inspection											
2.1	Body	Hydraulic testing	Critical	Measurement	100%	Drawing/Datasheet	TC/IR	√	P	W	W	
2.2	Ball sorter assembly	Performance test	Critical	Measurement	100%	Drawing/Datasheet, Manufacturer Standard	TC/IR	√	P	W	W	
2.13.7.3 COATING/ LINING												
1.0	Final Inspection											

1.1	Rubber lining	Mechanical properties	Critical	Tests as per IS-4682 part-I	100%	- Tensile & Elongation, Specific Gravity, Ash content - Adhesion test - Shore hardness test - Ozone resistance test: (50 PPHM) / 40° C / 20 % Strain /70 Hrs - Water Absorption Test.		√	P	V	V	Water Absorption Test, After ageing test & Compression Set test
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2.13.7.4 DEBRIS FILTER

1.0	Raw Material											
1.1	Body, strainer mesh and other components	Chemical & Mechanical Properties	Major	Review of TC	100%	Drawing/Datasheet, Manufacturer Standard	MTC	√	P	V	V	
2.0	Final Inspection											
2.1	Body	Hydraulic testing	Critical	Measurement	100%	Drawing/Datasheet, Manufacturer Standard	TC/IR	√	P	W	W	
2.2	Filter assembly	Performance test	Critical	Measurement	100%	Drawing/Datasheet, Mfg.Std.	TC/IR	√	P	W	W	

2.13.8 DRIP PUMP

1.0	Raw Material											
1.0	Casings, Diffuser, Shafts, Impeller	Chemical & Mechanical Properties, Heat Treatment, NDT	Major	Review of TC	100%	Approved Drawing/Datasheet	MTC	√	P	V	V	
2.0	In-Process Components											
2.1	Machined components	NDT	Major	PT/MPI	100%	Mfg.Std.	TC/IR	√	P	V	V	
2.2	Pump shaft	NDT Runout	Major	UT Measure	100%	Mfg.Std.	TC/IR	√	P	V	V	
2.3	Pump Impeller	NDT	Major	RT	100%	Mfg.Std.	TC/IR	√	P	V	V	Review of RT Films

2.4	Butt welds on fabricated components of thickness more than 10 mm	NDT	Major	RT	10%	Mfg.Std.	TC/IR	√	P	V	V	Review of RT Films
2.5	Welds on Casing, suction bell, shaft, impeller and fabricated components	NDT	Major	DPT	100%	Mfg.Std.	TC/IR	√	P	V	V	
3.0	Final Inspection											
3.1	On pump casing and pressure containing fabricated	Hydraulic testing	Critical	Measure	100%	Drawing/Datasheet,Mfg.Std.	TC/IR	√	P	W	W	
3.2	Individual Impeller and completed rotor assembly	Dynamic balancing Test, Runout	Critical	Measure	100%	Mfg.Std.,Relevant Std.	TC/IR	√	P	W	W	

3.3	Complete unit	Operation and performance testing Performance testing as per Hydraulic Institute Standards (HIS) to determine its characteristic curve at design speed and to ensure compliance with design requirements Vibration to be measured in transverse, horizontal and vertical direction at all measuring points Noise Level as per	Critical	Measure	100%	Drawing/Datasheet/Technical Specification,Mfg.Std., HIS	TC/IR	√	P	W	W	Tests shall be carried out using cold soften quality water
3.4	Painting, Packing & Preservation	Visual,Measurement	Major	Visual,Measurement	100%	Approved Drawing/Datasheet	IR/TC	√	P	V	V	

NOTE -

1	INSPECTION / NDT / TEST CERTIFICATE / REPORT FOR UT, MPI, BALANCING, PRESSURE TEST SHALL INDICATE REFERENCE OF ITEM SERIAL NUMBER & MODEL FOR CO-RELATION.											
2	MATERIAL TEST CERTIFICATE / REPORT FOR TEST SUCH AS CHEMICAL & MECHANICAL TEST ETC. SHALL INDICATE REFERENCE OF ITEM SERIAL NUMBER & ENGINE MODEL FOR CO-RELATION AS PER SAMPLING NORMS.											
3	a) Butt welds & full penetration welds : 10% RT or UT and 10% MPE or DPT 00%) b) Fillet welds : 10% MPE or DPT (100%) c) Nozzle welds : 10% MPE or DPT (100%) d) Lifting lug and other load bearing : 100% MPE or DPT											

	fillet welds e) Site weld edge preparations : 10% MPE or DPT											
4	- Run out examination shall be carried out at blade shrouds also - In case, bidder's practice is different, the same shall be furnished for purchaser's approval. During over speed test, vibration measurement and analysis shall also be carried out. - In case impulse stage and or blade discs are fitted on the rotor, fit up between such disc and rotor shall be checked up before and after over speed test - Lock blade lift after the over speed test shall be checked and record for same shall be maintained.											
5.	Before starting mass productions, following technological tests shall be carried out on the first lot of 10 to 15 blades: 100% RT and 100% MPE on blades 100% hardness testing. - Mechanical testing and metallurgical testing. - Weld repair shall not be permitted											
6.	Sample shall be drawn from inlet and outlet of purifier after works test and tested for moisture content, chemical tests and particle size of impurities											
3. HIGH PRESSURE PIPING, VALVES, THERMAL NSULATION AND MISCELLANEOUS SYSTEMS/ EQUIPMENT												
3.1 HIGH PRESSURE PIPING AND FITTING												
1.0	Raw Material											
1.1	All pipe lengths, Fittings	Chemical & Mechanical Properties, Heat Treatment, NDT	Major	Review of TC	100%	Drawing/Datasheet, Technical Specification, IBR requirement	MTC	√	P	V	V	
1.2	Review of WPS,PQR & WTQ	Verification, Validity	Major	Review of documents	100%	Drawing, ASME Sec IX,IBR	WPS,PQR, IBR Cert./ WTQ	√	P	V	V	
2.0	In-Process Inspection											
2.1	All mother pipes used for fittings	Hydro/NDT	Critical	hydraulic test or ultrasonic test	100%	Drawing, Technical Specification,	IR/TC	√	P	V	V	
2.2	All alloy and carbon steel pipes	NDT	Critical	UT	100%	Drawing, Technical Specification,	IR/TC	√	P	V	V	Random Check by Customer

2.3	Longitudinal welds	Hydro/NDT	Critical	UT or hydraulic tests	100%	Drawing, Technical Specification,	IR/TC	√	P	V	V	Random Check by Customer
2.4	Pipe bends (cold/ hot formed)	Bend Angle, Radius, Thinning, ,Surface Defect, Hardness	Critical	Visual, Measurement, Hardness, PT/MT	100%	Drawing, Technical Specification, Mfg. Std.	IR/TC	√	P	V	V	Random Check by Customer
2.5	Welds	NDT, Visual (after HT, if applicable)	Critical	RT/UT & PT/MPI,	100%	Drawing, Technical Specification, Mfg. Std.	IR/TC, Review of RT Films	√	P	V	V	Random Check by Customer, See Note 3
2.6	Finished welding for alloy steel, BFP discharge, MS, HRH, CRH piping and associated piping	NDT, Visual (after HT, if applicable)	Critical	RT/UT & PT/MPI	100%	Drawing, Technical Specification, Mfg. Std.	IR/TC, Review of RT Films	√	P	V	V	Random Check by Customer, See Note 4
2.7	Finished welding for Carbon Steel Class- I piping	NDT, Visual (after HT, if applicable)	Critical	RT/UT & PT/MPI	100%	Drawing, Technical Specification, Mfg. Std.	IR/TC, Review of RT Films	√	P	V	V	Random Check by Customer, See Note 5,6 &7
2.8	Oil piping	NDT, Visual (after HT, if applicable)	Critical	RT/UT & PT/MPI	100%	Drawing, Technical Specification, Mfg. Std.	IR/TC, Review of RT Films	√	P	V	V	Random Check by Customer

												er, See Note 8
2.9	Hardness of Alloy Steel Butt Joint (Weld & HAZ)	Metal properties	Critical	Hardness	100% for P91 material only, 3% for Other Alloy steel	Drawing, Technical Specification, Manufacturer Standard	IR/TC,	√	P	V	V	Rando m Check by Custom er
3.0	Final Inspection											
3.1	Final Dimensional & Visual Inspection	Weld Visual, Dimensional, PMI	Critical	Visual, Measurem ent, Chemical	100%	Drawing, Technical Specification, Manufacturer Standard	IR/TC	√	P	V	V	Rando m Check by Custom er
3.2 POWER CYCLE VALVES												
3.2.1 VALVES INCLUDING EXTRACTION LINE VALVES												
1.0	Raw Materials											
1.1	Body, Bonnet, Disc, Seat	Chemical & Mechanical Properties, Heat Treatment, NDT	Major	Review of TC	100%	Drawing/Datasheet, IBR if applicable	MTC	√	P	V	V	
1.2	Actuators (if applicable)	Compliance with Specification	Major	Review of TC	100%	Drawing/Datasheet	TC/IR	√	P	V	V	
1.3	Gear Box	MTC, Run Test	Major	Review of TC	100%	Drawing/Datasheet	TC/IR	√	P	V	V	Run Test- Noise, Vibratio n, Gear Ratio, DOR
2.0	In-Process Inspection											
2.1	Weld Overlay Inspection	Surface Defect Overlay Hardness	Major	PT, Hardness	100%	Drawing/Datasheet, Manufacturer Standard	TC/IR	√	P	V	V	Perform ed on

												Test coupon
2.2	Pressure retaining parts of valves	NDT	Critical	RT/UT & MPI/PT	100%	Drawing/Datasheet, Technical Specification, Manufacturer Standard	TC/IR	√	P	V	V	Random Check by Customer, See Note 9
2.3	Colour Matching of Disc/Plug/Seat	Surface contact of mating parts	Major	Visual	100%	Manufacturer Standard	TC/IR	√	P	V	V	Min.80 % contact & no through passage
3.0	Final Inspection											
3.1	Assembled Valves	Hydraulic testing-Shell Seat leak test-Hydro Seat leak test-Air	Critical	Visual & Measurement	100%	Drawing/Datasheet, Technical Specification, Applicable. Std., IBR if applicable	TC/IR	√	V	V	V	Random check by Customer
3.2	Fully assembled valve	Functional testing with actuator • Valve travel, closing and opening time • Current drawn by actuators	Critical	Measurement	100%	Drawing/Datasheet	TC/IR	√	V	V	V	Random check by Customer
3.2.2 BUTTERFLY VALVE												
1.0	Raw Material Inspection											
1.1	Plates for body, disc and flanges (for fabricated valves),	Chemical & Mechanical Properties, Heat Treatment, NDT	Major	Review of TC	100%	Drawing/Datasheet	MTC	√	P	V	V	

	Body & Disc Castings for Casted Valve, Drive & Stub Shafts											
1.2	Actuators (if applicable)	Compliance with Specification	Major	Review of TC	100%	Drawing/Datasheet	TC/IR	√	P	V	V	
1.3	Gear Box	MTC, Run Test	Major	Review of TC	100%	Drawing/Datasheet	TC/IR	√	P	V	V	Run Test- Noise, Vibration, Gear Ratio, DOR
2.0	In-Process Inspection											
2.1	Austenitic stainless steel welds	Material property for sea water application valves	Critical	IGC (inter-Granular corrosion)test	100%	Manufacturer Standard	TC/IR	√	P	V	V	
2.2	Shafts, seat rings etc.	NDT	Critical	Dye Penetration test	100%	Manufacturer Standard	TC/IR	√	P	V	V	
2.3	Rubber seal	Mechanical Properties, Hardness, Hydraulic stability test (Bleed Resistance) and ozone crack resistance tests	Critical	tensile, elongation and hardness test for vulcanising	Sample from lot	Relevant Std., Technical Specification	MTC	√	P	V	V	
3.0	Final Inspection											
3.1	On each valve	Hydraulic testing • To check body and bonnet strength	Critical	Measurement	100%	As per AWWA- C504/C516/ BS-5155	TC/IR	√	P	W	W	

		• Air seat leakage test										
3.2	Disc Strength test	Hydro (Forward and Backward)	Critical	Measurement	100%	as per EN12266-1/ AWWA-C516	TC/IR	√	P	W	W	
3.3	Proof of Design (POD) Test	Type Test	Critical	Review	01 per design/rating	AWWA C-516	POD Report	√	P	V	V	
3.4	On each valve with actuator	Performance test Opening and closing the valve from fully closed to fully open position and the reverse, under no flow for at least 25 cycles to check the following : • Closing and opening time • Operation of tripping switch and position indicator	Critical	Measurement	100%	Approved drawing/datasheet	TC/IR	√	P	W	W	See Note 10
3.2.3 METALLIC EXPANSION BELLOW												
1.0	Raw Material											
1.1	Bellows/Pipe Ends/Plates	Chemical & Mechanical properties	Major	Review of TC	100%	Approved drawing/datasheet	MTC	√	P	V	V	
2.0	In-Process Inspection											
2.1	Longitudinal butt weld on bellow	NDT Before forming and after forming	Critical	MPE or DPT	100%	Manufacturer Standard	TC/IR	√	P	V	V	
2.2	Butt/Groove Welds	NDT	Critical	RT, MPE or DPT	100%	Technical Specification, Manufacturer Standard	TC/IR	√	P	V	V	Review of RT Films
3.0	Final Inspection											

3.1	Visual & Dimensional	Dimensions	Critical	Measurement	100%	Approved drawing/datasheet	TC/IR	√	P	W	W	
3.2	On each pipe and expansion bellow	Hydraulic testing & Vacuum Test (for vacuum applications)	Critical	Measurement	100%	Approved drawing/datasheet	TC/IR	√	P	W	W	
3.3	On prototype/ expansion bellow	Type Test	Critical	Life cycle test, Squirm Test, Yield & Rupture	01/Type	Approved drawing/datasheet	TC/IR	√	P	W	W	
3.4	Routine Test	Deflection Test, Spring Rate(Only Axial)	Critical	Measurement	01/Type	Mfg. Std.	TC/IR	√	P	W	W	
3.2.4 HANGERS AND SUPPORT												
1.0	Raw Materials											
1.1	Round Bar, Plate, Coil Spring	Chemical & Mechanical Load Deflection, HT, Hardness, scragging test for Spring	Major	Review of TC	100%	Drawing/Datasheet, Technical Specification	MTC	√	P	V	V	
2.0	In-Process Inspection											
2.1	Welds	NDT • Butt welds of thickness 32mm and above • Butt welds of thickness less than 32mm & Fillet welds	Critical	UT/RT & MPE/DPT	100%	Mfg. Std.	TC/IR	√	P	V	V	
2.2	Turn-buckle, pipe clamps and hangers of thickness greater than 25mm	NDT	Critical	MPE/ DPT	100%	Manufacturer Standard	TC/IR	√	P	V	V	On bent portions

3.0	Final Inspection											
3.1	Completed Springs, & Assembled hangers	Performance test (Sagging test and load versus deflection test)	Critical	MEASUREMENT	100%	Drawing/Datasheet, Manufacturer Standard	TC/IR	√	P	V	V	Random check by Customer
3.2.5 THERMAL INSULATION, REFRACTORY, LAGGING AND CLADDING												
3.2.5.1 THERMAL INSULATION												
1.0	Final Inspection											
1.1	Resin bonded mineral wool insulation	Performance test	Critical	Measurement	100%	As per IS: 8183, Technical Specification	MTC	√	P	V	V	Random check by Customer
1.2	Resin bonded rock wool insulation	Performance test	Critical	Measurement	100%	As per IS: 9842, Technical Specification	MTC	√	P	V	V	
1.3	Sprayed mineral wool	Performance test	Critical	Measurement	100%	As per IS: 9724, Technical Specification	MTC	√	P	V	V	
1.4	Ceramic fiber blankets and block insulation	Performance test	Critical	Measurement	100%	as per IS: 15402, Technical Specification	MTC	√	P	V	V	
3.2.5.2 REFRACTORY												
1.0	Final Inspection											
1.1	Fire bricks or castable refractory	Performance test Castable refractory shall have proper identification, supplier name, customer name, batch no., date, material name and net weight in kg with proper instructions for handling	Critical	Measurement	100%	Manufacturer Standard, Technical Specification	TC/IR	√	P	W	W	

3.2.5.3 L AGGING AND CLADDING

1.0	Final Inspection											
1.1	Aluminum sheeting	Chemical & Mechanical Properties	Major	Review of TC	100%	Manufacturer Standard, Technical Specification	MTC	√	P	V	V	

3.3 LOW PRESSURE PIPING, VALVES AND FITTINGS ETC.**3.3.1 PIPES, FITTINGS AND MITRE BENDS****3.3.1.1 MS PIPES (BLACK & GALVANISED)**

1.0	Raw Material Inspection											
1.1	Steel strip/sheet	Mechanical & Chemical Properties	Major	Review of TC	1 sample/heat	Approved Drg/ Data sheet/Tech Spec/ IS 1239/IS 3589	MTC	√	P	V	V	
2.0	In-Process Inspection											
2.1	Welds of pipes and fittings Including welds of rolled and welded pipes	NDT	Critical	Dye penetration test	100%	ASTM E 165	TC/IR	√	P	V	V	
3.0	Final Inspection											
3.1	Pipe (Black)	Surface defect, Finish & End connection Dimensions, Straightness, Ovality, Bend removal Tensile, Elongation, Bend & Flattening	Critical	Visual Measure ment Physical	Sampling as per IS 4711	Approved Drg/Data sheet/Tech Spec/ IS 1239/IS 3589	Inspection report	√	P	W	W	
3.2	Galvanizing (If applicable)	Surface defect, Finish	Critical	Visual	Sample as Per IS 4736	IS:2633	Inspection report	√	P	W	W	

		Uniformity of Zinc coating	Critical	Dip test	Sample as Per IS 4736	IS 4736	Inspection report	√	P	W	W	
		Thickness of coating, Thickness of Zn coating, Adhesion of Zn coating, Free Bore test	Critical	Measurement			Inspection report	√	P	W	W	

3.3.1.2 ROLLED & WELDED PIPES / MITRE BENDS >600 NB TO 2200 NB

1.0	Raw Material Inspection												
1.1	PLATES	MECHANICAL & CHEMICAL PROPERTIES	MAJOR	MECHANICAL & CHEMICAL	1 SAMPLE/HEAT		APPROVED DRAWING/DATA SHEET / RELEVANT MATERIAL SPEC AS PER DATA SHEET	TC	√	P	V	V	CORRELATED MILL TCs ARE REQUIRED
		SURFACE DEFECTS	MAJOR	VISUAL	100%	-	RELEVANT MATERIAL SPEC AS PER DATA SHEET / NO PITTING/CORROSION	IR		P	-	-	
		DIMENSIONS	MAJOR	MEASURE	100%	-	MFG.DRG.	IR		P	-	-	
2.0	In Process Inspection												

2.1	Welding	WPS/PQR/WQR	Major	Qualification/Verification	100%	-	ASME Sec-IX / ASME Sec-IX	WPS, PQR		P	-	-	Refer Note #1
		Marking, cutting, rolling,, joint set-up	Major	Visual & Measure	100%	-	MFG DRG / MFG DRG	IR		P	-	-	
	Welding	NDT on root run	Critical	DPT	100%	-	ASTM E 165 / No Indications	IR	√	P	V	V	Refer Note #2
		NDT on finished weld	Critical	DPT	100%	-	ASTM E 165 / No Indications	IR	√	P	W	W	During Final Inspection
		NDT on ‘T’ joints (if any)	Critical	RT	100%	-	ASME Sec-VIII / ASME Sec-VIII	IR	√	P	V	V	Review of radiographs in CHP
3.0	Final Inspection												
3.1	Complete Pipe / Mitre Bend	Dimensional (thickness, diameter, ovality, straightness)	Critical	Measure	100%	2%	IS 3589 / BS 534 / Tech sec	IR	√	P	W	W	CHP
		Flattenning test	Critical	Mechanical	1 Sample / Lot		IS:3589	IR	√	P	W	W	CHP
		Hydraulic test	Critical	Hydro test	100%	10%	IS:3589 / No leakage	IR	√	P	W	W	Refer Note 3
NOTE 1: ONLY QUALIFIED WELDER / WPS SHALL BE USED. NOTE 2: DPT AFTER BACKGAUGING SHALL BE DONE, INSTEAD OF DPT ON ROOT RUN, IN CASE BACK GAUGING IS POSSIBLE. NOTE 3: IN CASE HYDRO TEST IS NOT POSSIBLE, 100 % RT OF WELD JOINTS IS REQUIRED TO BE DONE.													
3.3.2 VALVES AND CAST & FABRICATED BUTTERFLY VALVE													
3.3.2.1 VALVES													
1.0	Raw Material Inspection												
1.1	Shaft/ spindle of size ≥ 40 mm diameter	Mechanical properties	Critical	• UT • MPE or DPT	100%		Machined surfaces of casing, disc and shaft for MPE or DPT to check surface soundness.	MTC	√	P	V	V	
2.0	Final Inspection												

2.1	All valves	Hydraulic testing	Critical	Measurement	100%	Body, seat and back seat (wherever provided) at 1.5 times the maximum pressure to which respective valves can be subjected during plant operation.		√	P	W	W	Check valves shall also be tested for leak tightness test at 25% of the specified seat test pressure. For rubber lined valves, hydraulic test shall be carried out before rubber lining
2.2	Butterfly valves	Hydraulic testing, seat and disc string test	Critical	AWWA-C-504 standard	100%			√	P	W	W	

3.3.2.2 CAST & FABRICATED BUTTERFLY VALVE

1.0	Raw Material Inspection												
1.1 A	BODY & DISC CASTING, BAR STOCK FOR DRIVE & STUB SHAFTS, SEAT RING, CLAMPIN G RING	CHEMICAL & MECHANICAL PROPERTIES	Major	CHEMIC AL & MECHAN IC AL	1 SAMPLE/ HEAT		RELEVANT MATERIAL SPECIFICATION AS PER APPROVED DATA SHEET/ APPROVED DRAWING	MTC	√	P	V	V	REFER NOTE-1
1.1 B		SURFACE DEFECT	Major	VISUAL	100 %	--	MSS SP-55	√		P	V	-	
1.1 C		HEAT TREATMENT (WHEREVER APPLICABLE)	Major	REVIEW OF H.T CHARTS	100 %	100%	-DO-	√		P	V	V	
1.1 D		IGC TEST(WHEREVER APPLICABLE)	Major	CHEMIC AL	1 SAMPLE / HT BATCH		ASTM A 262 / Practice – “E”		√	P	V	V	
1.1 E		INTERNAL DEFECTS	Major	UT	100%		ASTM A 388	REFER NOTE- 2	√	P	V	V	ON BAR DIAMETER ≥ 40 MM
1.1 F	MS PLATES FOR BODY & DISC	CHEMICAL & MECHANICAL PROPERTIES	Major	CHEMIC AL	1 SAMPLE/ HEAT		RELEVANT MATERIAL SPECIFICATION AS PER	MTC	√	P	V	V	REFER NOTE-1

				& MECHAN IC AL			APPROVED DATA SHEET/ APPROVED DRAWING						
1.2 A	Rubber Seals, Seat liners	DIMENSIONS	Major	MEASUR E	100%		MANUFACTURING DRAWING	IR	√	P	V	-	Type of Rubber as per approved drawing / data sheet Acceptance norms shall be as per APPRVD DRG / DATA sheet if not mentioned in that value given in QP is to be followed
1.2 B		SURFACE DEFECT	Major	VISUAL	100%		NO CUT MARKS	IR	√	P	V	V	
1.2 C		SHORE HARDNESS	Major	MEASUR E	100%		IS 3400-II/ SHORE A 75+/- 5 For Nitrile & 70+/-5 for EPDM	IR	√	P	V	V	
1.2 D		Tensile & Elongation	Major	Measure	1 Sample / Batch		IS 3400-II/ASTM D 412 150KG/CM2 Min &250% MIN	TC	√	P	V	V	
1.2 E		Ageing test	Major	Physical	1 sample/ batch		ASTM D 573 / Refer Note -3	IR	√	P	V	V	
1.2 F		Bleed resistance test	Major	Hydraulic Stability	1 sample/ batch		IS 3400 part 6 / ASTM D 471 Refer note-4	IR	√	P	V	V	
1.2 G		Ozone resistance test	Major	Chemical	1 sample/ batch		ASTM D 1149 / No crack at 50 PPM ozone for 72 hrs at 40deg C	IR	√	P	V	V	
1.3	Gear box, bearing hardware	Dimensions & Make	Major	Measure & verification	100%		Manufacturer catalogue / drawing	TC/COC	√	P	V	V	Refer note-5
1.4	ELECTRICAL ACTUATOR (IF APPLICABLE) - REFER SEPARATE MQP/ RQP OF ACTUATOR MANUFACTURER FOR INSPECTION. THE CHP REPORT OF ACTUATOR MANUFACTURER WILL BE SHOWN FOR CORRELATION & VERIFICATION												
1.5 A	PNEUMATIC ACTUATOR (AS APPLICABLE)	APPEARANCE	Major	VISUAL	100%	100%	APPROVED DRAWING/ DATA SHEET/ MFG STD /CATALOGUE/ AWWA C504	IR /TC	√	P	V	V	
1.5 B		PERFORMANCE	Major	OPERATI ON	100%	100%		IR	√	P	V	V	
1.5 C		LEAK TIGHTNESS	Major	PNEUMAT IC TEST	100%	100%	APPROVED DRAWING/ DATA SHEET/ MFG STD/ CATALOGUE NO LEAKAGE	IR	√	P	V	V	
2.0	In process inspection												
2.1 A	CAST BODY & DISC AFTER MACHINING	DIMENSION	Major	MEASUR E	100%	-	MANUFACTURER DRAWING	IR	-	P	-	-	

2.1 B		BODY SHELL STRENGTH TEST	Major	HYDRO TEST	100%		APPLICABLE DESIGN STANDARD / APPROVED DRAWING	IR	√	P	V	V	
2.1 C		NDT on change of sections & machined surfaces	Major	MPI	100%		ASTM E 709 REFER NOTE – 6	NDT REPORT	√	P	V	V	
2.1 D		NDT ON MACHINED SURFACES FOR STAINLESS STEEL CASTINGS	Major	DPT	100%		ASTM E 165 NO LINEAR INDICATIONS	NDT REPORT	√	P	V	V	
2.2 A	DRIVE & STUB SHAFTS	DIMENSIONS	Major	MEASURE	100 %	--	MANUFACTURER DRAWING	IR	-	P	-		
2.2 B		SURFACE FINISH	Major	VISUAL & MEASURE	100 %	--	MANUFACTURER'S DRAWING	-	-	P	-	-	
2.2 C		SURFACE DEFECTS	Major	DPT	100%		ASTM E 165 NO LINEAR INDICATIONS	NDT REPORT	√	P	V	V	
2.3 A	PLATES FOR FABRICAT ION(FOR FABRICAT ED VALVE ONLY)	CUTTING OF PLATES	Major	VISUAL & MEASURE	100%	-	CUTTING PLAN/MANUFAV TURING DRAWING	REGIS TER		P	-	-	
2.4 A	BODY AND DISC FABRICAT ION(FOR FABRICAT	EDGE PREPARATION AND FIT UP	Major	VISUAL & MEASURE	100%	-	APPROVED WPS/ FABRICATION DRG	REGIS TER		P	-	-	
2.4 B		WPS, PQR / WELDER QUALIFICATION	Major	VISUAL & MECHANICAL TEST /NDT	100%		ASME SEC IX	QW 482, 483, 484	√	P	V	V	Note-9
2.5 A	ED VALVE ONLY)	NDT ON BUTT WELD OF BODY & DISC AFTER BACK GAUGING	Major	DPT AFTER BACK GAUGING	100%		ASME SEC V/ ASME SEC VIII DIV- 1 APPENDIX 8	NDT REPORT	√	P	V	V	

2.5 B		& FINAL WELDING	Major	RT FINAL WELDING	100%		ASME SEC V/ ASME SEC VIII DIV- 1 UCS 56		RT REPORT	√	P	V	V	RT FILM SHALL BE REVIEWED BY CUSTOMER/OWNER
2.5 C		STRESS RELIEVING (AS APPLICABLE)	Major	TIME-TEMP MEASURE	100%		ASME SEC VIII DIV-1 UW-51		SR CHART	√	P	V	V	
2.5 D		NDT ON BUTT WELD AND FILLET WELD OF BODY & DISC ON FINISHED WELD	Major	DPT	100%	10%	ASME SEC V	√	NDT REPORT		P	W	W	CUSTOMER/OWNER WITNESS DURING FINAL INSPECTION
2.5 E		BODY SHELL STRENGTH CHECK	Major	HYDRO TEST	100%		APPLICABLE DESIGN STANDARD/APPROVED DRAWING/DATA SHEET		IR	√	P	V	V	
3.0	Final Inspection													
3.1 A	COMPLETE VALVE WITH JOB ACTUATOR	PROOF OF DESIGN TEST	Major	TYPE TEST	1 NO PER DESIGN / RATING		APPLICABLE DESIGN STANDARD / /POD PRECEDURE		POD REPORT	√	P	V / W	V / W	REFER NOTE-7
3.1 B		VISUAL & DIMENSION	Major	MEASURE	100%	10% OR MIN 02 NOS	APPLICABLE DESIGN STANDARD / APPROVED DRAWING / DATA SHEET		√	√	P	W	W	
3.1 C		SHELL TIGHTNESS TEST	Critical	HYDRO TEST	100 %	OF EACH SIZE WHICH EVER IS HIGHER	APPLICABLE DESIGN STANDARD /APPROVED DRAWING/DATA SHEET No leakage		√	√	P	W	W	
3.1 D		SEAT TIGHTNESS TEST	Critical	HYDRO TEST	100 %		APPLICABLE DESIGN STANDARD /APPROVED DRAWING/DATA SHEET No leakage		√	√	P	W	W	
3.1 E		DISC STRENGTH TEST	Critical	HYDRO TEST	01 NO. OF EACH SIZE & RATING		APPLICABLE DESIGN STANDARD		IR	√	P	W	W	

							/APPROVED DRAWING/DATA SHEET NO PERMANENT DEFORMATION & LEAKAGE PERMITTED.						
3.2	FUNCTIONAL TEST OF COMPLETE VALVE WITH JOB ACTUATOR/ OPERATOR	SMOOTH OPERATION CURRENT DRAWN LIMIT SETTING INTERLOCKS, OPEN & CLOSING TIME CHECK (IF APPLICABLE)	Critical	Operation	100%	10%	APPROVED DATA SHEET / DRAWING	IR	✓	P	W	W	
3.3	PAINTING	PAINTING APPEARANCE, THICKNESS & ADHESION	Major	VISUAL & MEASURE	100%	100%	TECHNICAL SPECIFICATION/ APPROVED DRAWING / PAINTING SCHEDULE	IR	✓	P	V	V	
3.4	MARKING	NAME PLATE/ FLOW DIRECTION INDICATION	Minor	VISUAL	100%	--	APPROVED DRAWING	LOG SHEET	-	P	-	-	
NOTES													
1	NO WELD REPAIR IS ALLOWED ON CASTINGS OF CAST IRON. ONLY MECHANICAL PROPERTIES REQUIRED FOR CI/SGI CASTINGS. AS CAST HEAT NO. MARKS FOR SIZE ABOVE 500 NB AND PUNCHED HEAT NO. MARKS FOR SIZE BELOW 500 NB IS REQUIRED FOR CORRELATION OF CASTINGS. MS PLATES SHALL BE OF CUSTOMER/OWNER ACCEPTABLE SOURCE WITH CORRELATED TC. #-IN CASE CASTING OF BODY IS OUTSOURCED V/V MFR TO WITNESS TESTS ONE PER HEAT AS PER RELEVANT STD.												
2	NORMAL PROBE OF 2MHZ FREQUENCY AND 20MM DIAMETER SHALL BE USED. WITH THIS PROBE THE BACK WALL ECHO (BWE) IN SOUND AREA SHALL BE SET AT 100% FSH (FULL SCREEN HEIGHT). AT THIS GAIN SETTING, ANY DEFECT ECHO $\geq$ 20% FSH IS NOT ACCEPTABLE. ALSO, LOSS OF BWE > 20% FSH IS NOT ACCEPTABLE.												
3	AGEING TEST SAMPLE SHALL BE KEPT AT 100° C(FOR NITRILE RUBBER) & AT 70° C(FOR EPDM RUBBER) FOR 72 HOURS. ACCEPTANCE CRITERIA SHALL BE AS PER APPVD DRG/DATA SHEET OR AS BELOW IF NOT MENTIONED IN APPVD DRG/DATA SHEET: A) PERMISSIBLE VARIATION IN HARDNESS $\pm$ 5 % OF ORIGINAL VALUE.												

	B) PERMISSIBLE VARIATION IN TENSILE STRENGTH ±30% OF ORIGINAL VALUE. C) PERMISSIBLE VARIATION IN % ELONGATION ±50 % OF ORIGINAL VALUE.											
4	FOR BLEED RESISTANCE TEST, ONE SAMPLE EACH SHALL BE KEPT IN 33% HCL, 48% NA OH AND DM WATER SEPARATELY FOR 72 HOURS AT 27 DEG. ACCEPTANCE CRITERIA WILL BE;											
	a. NO DISCOLORATION					b. PERMISSIBLE CHANGE IN MASS SHALL BE 0% TO +2%						
5	GEAR BOX MANUFACTURERS TEST CERTIFICATE WITH GEAR RATIO, TORQUE RATING AND RUN TEST. FOR BEARINGS AND OTHER HARDWARE COC TO BE SUBMITTED.											
6	ACCEPTANCE NORMS FOR MPI TEST (AS PER ASTM E 125).											
	TYPE	DISCONTINUITY	ACCEPTABLE LEVEL		TYPE	DISCONTINUITY	ACCEPTABLE LEVEL					
	I	HOT TEARS & CRACKS	NOT ACCEPTABLE		IV	INTERNAL CHILLS & UN FUSED CHAPLETS	1					
	II	SHRINKAGE	3		V	POROSITY	1					
	III	INCLUSION	3									
7	POD CLEARANCE SHALL BE TAKEN FROM CUSTOMER/OWNER-ENGG (WHEREVER APPLICABLE AS PER CONTRACTUAL REQUIREMENT). IN CASE POD TEST HAS BEEN CARRIED OUT ALREADY FOR VALVE OF THE SAME SEAT DESIGN WITH THE SAME MOC & SIZE GROUP AND WITNESSED BY CUSTOMER, REPORT FOR THE SAME NEED TO BE REVIEWED ONLY. IF POD IS ENVISAGED IN TECHNICAL SPECIFICATION AND NOT CARRIED OUT EARLIER AS ABOVE, THE SAME SHALL BE OFFRED FOR WITNESS OF CUSTOMER/OWNER.											
8	POSITIVE MATERIAL IDENTIFICATION FOR ALLOY/SS MATERIALS LIKE BODY, SEAT, FASTENERS ETC TO BE CARRIED OUT. THE QUANTUM OF CHECK SHALL BE 10 % OF THE OFFERED QUANTITY (EACH SIZE) OR MIN 2 NUMBERS OF VALVE OF EACH SIZE WHICHEVER IS HIGHER											
9	MAIN CONTRTRACTOR/CUSTOMER/OWNER APPROVED WPS & PQR TO BE USED. ONLY QUALIFIED WELDERS TO BE ENGAGED FOR WELDING.											
3.3.3 RUBBER LINING OF PIPES AND VALVES												
1.0	Final Inspection											
1.1	Rubber lining of pipes and valves	Mechanical properties	Critical	Tests as per IS-4682 part-I	100%	• Tensile & Elongation, Specific Gravity, Ash content • Adhesion test • Shore hardness test		✓	P	V	V	Water Absorption Test, After ageing test & Compression Set test

						• Ozone resistance test: (50 PPHM) / 40° C / 20 % Strain /70 hrs. • Water Absorption Test • Resistance to bleeding • Dimensions • Cleanliness & Uniformity • Spark Test • Control of Pr./Temp/ & Time • Thickness • Adhesion • Surface Defects • Completeness						
3.3.4 COATING AND WRAPPING MATERIAL												
1.0	Raw Material											
1.1	Coal tar based synthetic Primer	Viscosity Drying time Flash Point Volatile matter	Major	Chemical & Physical test	IS 15337/AWWA C203	Approved Data sheet/ IS 15337/AWWA C203	TC	√	P	V	V	
1.2	Fabric	Pliability Heat Resistance Breaking load in longitudinal direction Porosity	Major	Physical test	IS 15337/AWWA C203/ASTM D 146	Approved Data sheet/ IS 15337/AWWA C203/ASTM D 146	TC	√	P	V	V	
2.0	In process Inspection											
2.1	COAL TAR TAPE	A) UNIFORMITY OF COATING (ON BOTH SIDES)	MAJOR	VISUAL	AS PER IS:15337/AWWA C203	APPD DATA SHEET/AWWA C203/IS:15337	IR	√	P	V	V	

		B) PLIABILIT Y										
3.0	Final Inspection											
3.1	Coal tar tape	Thickness Width Softening point Penetration Filler content Service temperature Breaking load in longitudinal direction Adhesion test Holiday test Direct Impact test Petrol solubility test	Critical	Measurement & Physical test	IS 15337/AWWA C203	Approved Data sheet/ IS 15337/AWWA C203	Inspection report	√	P	W	V	Note: Cathodic disbondment test to be carried out by Mfrs as per IS 15337 and report to be submitted for review.
3.3.5 RUBBER EXPANSION JOINTS												
1.0	Raw Materials											
1.1	Neoprene Rubber	Chemical & Mechanical, Shore Hardness, Ozone resistance test, Hydro stability test, Flame resistance test, Kerosene Dip Test	Major	Review of TC	01 Sample/Batch	As per relevant IS	MTC	√	P	V	V	
1.2	Rubber compound test slab after vulcanising	Tensile strength, elongation, shore hardness	Major	Review of TC	01 Sample/Batch	As per relevant IS	MTC	√	P	V	V	
1.3	Fabric strength of synthetic fiber for reinforcement	Rubber to fabric adhesion Rubber to metal adhesion	Major	Review of TC	01 Sample/Batch	As per relevant IS	MTC	√	P	V	V	

2.0	Final Inspection											
2.1	Bare Bellows/ Expansion joints in assembled condition	1. Visual & Dimensional 2. Hydro Test 3. Vacuum Test 4. Combined Movement Test 5. Deformation Check	Critical	Measure ment	100%	Drawing/Datasheet, Technical Specification	TC/IR	√	P	V	V	Random check by Customer Proof of design (POD) test documents verification.
3.4 EQUIPMENT COOLING WATER (ECW) SYSTEM												
3.4.1 PRIMARY SIDE AND SECONDARY SIDE PUMPS												
1.0	Final Inspection											
1.1	Pump casing and pressure containing fabricated parts	Hydraulic testing	Critical	Measure ment	100%	At 1.5 times the shut off head or 2 times the working pressure whichever is higher		√	P	W	W	
1.2	Complete unit	Operation and Performance testing as per Hydraulic Institute Standards (HIS) to determine its characteristic curve at design speed and to ensure compliance with design requirements Vibration to be measured in transverse, horizontal and vertical direction	Critical	Measure ment	100%	Drawing/Datasheet		√	P	W	W	Other tests shall also be conducted in line with details given above for other pumps

		at all measuring points.										
		Noise Level as per HIS.										
		Strip down test										

3.4.2 PLATE HEAT EXCHANGERS

SL. NO	COMPONENT & OPERATIONS	CHARACTERISTICS	CLASS	TYPE OF CHECK	QUANTUM OF CHECK		REFERENCE DOCUMENT	ACCEPTANCE NORMS/	FORMAT OF RECORD/		AGENCY			REMARKS
					M	C/ O					M	C	O	
1.	2.	3.	4.	5.	6.		7.	8.	9.	D*	** 10			11

1	Raw Material Inspection													
1.1	FRAME PLATE & PRESSURE PLATE, CONNECTION LINING MATERIAL. TOP AND BOTTOM CARRYING BAR, COUNTER FLANGE	CHEMICAL & MECHANICAL	MAJOR	CHEMICAL & MECH	1SAMPLE / HEAT		APPROVED DRAWING / DATA SHEET	MTC	√	P	V	V	REFER NOTE-1	
1.2		HEAT TREATMENT (AS APPLICABLE)	MAJOR	MEASURE	100%			HT Chart		P	V	V		
1.3		INTERNAL SOUNDNESS OF PLATES (≥ 25 MM THICK)	MAJOR	UT	100%		ASME SEC VIII DIV 1 SA 578 LEVEL B /ASTM A 435 (NOTE-2)	MTC/ IR	√	P	V	V	(APPLICABLE FOR FRAME AND PRESSURE PLATE ONLY)	
1.4		INTERNAL SOUNDNESS BARS (≥ 40 MM Dia)	MAJOR	UT	100%		ASTM A 388 (NOTE-2)	MTC/ IR	√	P	V	V	REFER NOTE-2	
1.5	HEAT TRANSFER PLATES	CHEMICAL PROPERTIES	MAJOR	CHEMICAL TEST.	1SAMPLE / HEAT		APPROVED DRAWING / DATA SHEET	MTC	√	P	V	V	REFER NOTE- 53	
1.6		MECHANICAL PROPERTIES	MAJOR	MECHANIC AL TEST.	1SAMPLE / HEAT			MTC	√	P	V	V		
1.7	PIPES/NOZZLES	REVIEW OF MILL TC	MAJOR	CHEM, MECH & HYDRO	1/LOT		APPROVED DRAWING / DATA SHEET	TC	√	P	V	V	REFER NOTE-4	
1.8	GASKET	DIMENSIONS	MAJOR	MEASUREM ENT	100 %	100 %	APPROVED DRAWING	I.R.	√	P	V	V	REFER NOTE-5	
1.9		WORKMANSHIP AND FINISH	MAJOR	VISUAL.	100 %	100 %	NO DAMAGE. NO SURFACE DEFECTS	I.R.	√	P	V	V		
1.10		CONTOUR	MAJOR	VISUAL	100 %	100 %	MFG DRGS/ SPECIFICATION	I.R.	√	P	V	V		

1.11		HARDNESS	CRITICAL	MEASUREMENT	100 %	100 %	APPROVED DRAWING	I.R.	√	P	V	V	
1.12	TIGHTENING BOLT / STUD & NUT/TIE ROD	CHEMICAL & MECHANICAL	MAJOR	CHEMICAL & MECH	1SAMPLE / HEAT		APPROVED DRAWING / DATA SHEET	MTC	√	P	V	V	
1.13		WORKMANSHIP AND FINISH.	MAJOR	VISUAL.	100 %	100 %	APPROVED DRAWING	I.R.	√	P	V	V	
1.14		INTERNAL SOUNDNESS (FOR DIA >= 40 MM	CRITICAL	UT	100 %	100%	ASTM A 388 (NOTE-2)	MTC/IR	√	P	V	V	
2.0	Inprocess Inspection												

2.1	WELDING (AS APPLICABLE)	WPS, PQR, WPQ	CRITICAL	VERIFICATION	100%		ASME-SEC.IX		QW-482, 483, 484	√	P	V	V	REFER NOTE-6
2.2	HEAT TRANSFER PLATES	PLATES PRESSED IN SINGLE OPERATION	MAJOR	VERIFICATION	100 %	100 %	MANUFACTURER'S STANDARD PROCEDURE	UNIFORM CORRUGATION THROUGHOUT PLATE PRESSING CAPACITY	COMPLIANCE REPORT	√	P	V	V	REFER NOTE-7
2.3		HT AREA MEASUREMENT	MAJOR	WHITE LIGHT SCANNING	1/ MODEL TYPE	1/ MODEL TYPE	MANUFACTURER'S STANDARD PROCEDURE	APPROVED DRAWING / DATASHEET	I.R.	√	P	V	V	REFER NOTE-8
2.4		GASKET GROOVE DEPTH ON EACH PLATE	MAJOR	VISUAL & MEASUREMENT	100 %	100 %	MFG DRGS/ DATA SHEET	GASKET SHOULD NOT PROTRUDE OUTSIDE THE GASKET GROOVE FOR MORE THAN	IR	√	P	V	V	GASKET GROOVE IN EACH PLATE SHOULD BE ABLE TO SUPPORT THE GASKET

							20 % OF GASKET THICKNESS						FROM THREE SIDES.
2.5		WORKMANSHIP AND FINISH	MAJOR	VISUAL	100 %	100 %	NO SCRATCHES, CRACKS ETC.	IR		P	V	V	
2.6		SURFACE DEFECTS AND CRACKS	CRITICAL	LIGHT BOX /VACUUM/ AIR CHAMBER TEST	100 %	10 %	MANUFACTURER'S LIGHT BOX TEST PROCEDURE NO	IR	√	P	W	W	REFER NOTE-9
2.7				DP TEST	10 %	2% / 1%	MANUFACTURER'S DP TEST PROCEDURE/ ASTM E 165	IR	√ P		W	W	REFER NOTE-10
2.8	WELDING OF PIPE AND PIPE TO INNER & OUTER RINGS(NOZZLE CONNECTION).	WELDING JOINT SOUNDNESS	MAJOR	DP TEST	100 %	100 %	MANUFACTURER'S DP TEST PROCEDURE/ ASTM E 165	IR	√	P	W	V	
2.9		VISUAL INSPECTION OF WELD	MAJOR	VISUAL	100 %	100 %	APPROVED DRAWING	IR	√	P	V	V	
2.10	PHE STRUCTURE	WORKMANSHIP	MAJOR	MEASURE &	100	100	APPROVED DRAWING	IR	√	P	V	V	

		AND FINISH		VISUAL	%	%							
2.11	PLATE GASKETS	PRESENCE OF GASKET	MAJOR	VISUAL	100 %	100 %	MANUFACTURES SPECIFICATION	IR	√	P	V	V	
2.12	PLATE ARRANGEMENT TO FLOW DIAGRAM	CORRECTNESS	MAJOR	VISUAL AS PER FLOW DIAGRAM	100 %	100 %	APPROVED DRAWING	IR	√	P	V	V	
2.13	ASSEMBLY OF TIGHTENING BOLTS	SQUEEZING OF	MAJOR	VISUAL	100 %	100 %	APPROVED DRAWING / DATA SHEET	IR		P	V	-	

	AND NUTS	THREADS ON TIGHTENING BOLTS.											
2.14	PLATE PACK	DIMENSION	MAJOR	MEASURE	100 %	100 %	APPROVED DRAWING	IR		P	V	-	
2.15	PHE ASSEMBLY	LEAKAGE/ STRENGTH OF THE STRUCTURE	MAJOR	HYDRO TEST	100 %	100 %	MANUFACTURER'S HYDRO TEST PROCEDURE NO.	IR	√	P	V	V	
3.0	Final Inspection & Testing												
3.1	COMPLETE ASSEMBLY	DIMENSION, NO OF HEAT TRANSFER PLATES, WORKMANSHIP & FINISH.	MAJOR	MEASURE & VISUAL	100 %	100 %	APPROVED DRAWING/DATA SHEET	IR	√	P	W	W	
3.2		LEAKAGE/ STRENGTH OF THE STRUCTURE	MAJOR	HYDRO TEST	100 %	100 %	MANUFACTURER'S HYDRO TEST PROCEDURE NO.	IR	√	P	W	W	UNBALANCED HYDROSTATIC PRESSURE (BOTH PRIMARY & SECONDARY SIDE)
3.3	HT PLATE MATERIAL GRADE VERIFICATION	PMI WITH XRF ANALYZER	MAJOR	PMI	100 %	10 %	MATERIAL SPECIFICATION AS PER APPROVED DRAWING / APPROVED DATA SHEET	IR	√	P	W	W	NOTE-11
3.4	SURFACE PREPARATION & PAINTING	SHADE, DFT	MAJOR	VISUAL / MEASURE	100 %	100 %	APPROVED DRAWING/ APPROVED PAINTING PROCEDURE	IR	√	P	W	V	
4.0	Preservation & Packing												
4.0	PACKING & PRESERVATION	SOUNDNESS AND COMPLETENESS	MAJOR	VISUAL	100 %	100 %	TECH SPEC/ APPROVED QAP/ APPROVED DRAWING	COC	-	P	V	-	

Notes:

NOTE-1 – IN ABSENCE OF CO-RELATED MILL TCS, MANUFACTURER SHALL ARRANGE CHECK TESTING AND TC SHALL BE SUBMITTED FOR CUTOMER'S REVIEW.

NOTE-2 – ULTRASONIC TEST SHALL BE CARRIED OUT USING 10 MM/20 MM SIZE NORMAL BEAM PROBE OF FREQUENCY 2 MHZ. UT ACCEPTANCE: BACK WALL ECHO (BWE) IN SOUND AREA SHALL BE SET AT 100% OF FULL-SCALE HEIGHT (FSH). THE WHOLE BAR SHALL BE SCANNED UNDER THIS SENSITIVITY SETTING. AT THIS SENSITIVITY SETTING ANY DEFECT ECHO EXCEEDING 20% OF FSH IS NOT ACCEPTABLE. ALSO, AT ANY PLACE LOSS IN BACK WALL ECHO MORE THAN 20% OF FSH IS NOT ACCEPTABLE.

NOTE-3 ALL THE SOURCE OF RAW MATERIAL FOR COIL / CUT TO SHEET SHOULD BE FROM CUTOMER'S APPROVED VENDORS (e.g. JINDAL / SAIL/ARCELLORMITAL / OUTOKUMPU). MILL TCS FOR HT PLATES FROM CUTOMER'S APPROVED VENDORS SHALL BE SUBMITTED TO CUTOMER'S FOR REVIEW WITH PROPER CORRELATION AS PER MFRS HEAT NUMBER.

NOTE-4 – TC REVIEW FOR CHEMICAL, MECHANICAL PROPERTIES. & HYDRO TEST.

NOTE-5 TCS FOR GASKETS FROM CUTOMER'S ACCEPTED VENDORS WILL BE SUBMITTED TO CUTOMER'S FOR REVIEW WITH PROPER CORRELATION AS PER, MFRS BATCH NUMBER.

NOTE-6 MAIN CONTRACTOR/ LLOYDS/TUV/BV/CUTOMER'S APPROVED WPS & PQR SHALL BE ACCEPTABLE. IF WELDERS ARE ALREADY QUALIFIED BY LLOYDS/TUV/BV/CUTOMER'S & DOING SIMILAR JOB REGULARLY, REQUALIFICATION IS NOT REQUIRED.

NOTE-7: SINGLE STEP PRESSING OPERATION - ONLY SINGLE STROKE (UNLESS UNTILL ALLOWED IN CONTRACT) SHALL BE USED TO FORM WHOLE CORRUGATIONS ON EACH WHOLE PLATE BY USING A SINGLE PRESS AND NOT IN MULTIPLE STROKE OR IN INCREMENTAL MANNER. CERTIFICATE OF COMPLIANCE FOR HT PLATE WITH WITNESS OF TPI AGENCY LIKE TUV/LLOYDS/BV WILL BE SUBMITTED FOR MAIN CONTRACTOR AND CUTOMER'S REVIEW.

NOTE-8: INSPECTION OF HT PLATE AREA MEASUREMENT SHALL BE BY WHITE SCANNING METHOD FROM THIRD PARTY LIKE TUV/ LLOYD/BV & SHALL BE WITNESS BY MAIN CONTRACTOR & CUTOMER'S. HOWEVER, IN CASE SAME HAS ALREADY BEEN WITNESSED BY CUTOMER'S IN SOME EARLIER PROJECTS, THE INSPECTION REPORT FOR THE SAME SHALL BE FURNISHED FOR CUTOMER'S REVIEW.

NOTE-9: 10% LIGHT BOX / VACUUM / AIR CHAMBER TEST SHALL BE RANDOMLY WITNESS BY MAIN CONTRACTOR & CUTOMER'S AT VENDORS WORKS, IN CASE ANY DEFECT IS FOUND IN ANY OF THESE 10% PLATES, THE WHOLE LOT SHALL BE TESTED IN PRESENCE OF MAIN CONTRACTOR & CUTOMER'S. H.T. PLATES WITHOUT DEFECT ONLY SHALL BE ACCEPTED.

NOTE-10: DPT 1% OR MINIMUM 50 NOS. OF PLATES WHICHEVER IS MAXIMUM SHALL BE RANDOMLY WITNESS BY CUTOMER'S AT VENDORS WORKS. MAIN CONTRACTOR TO WITNESS DP TEST ON 2 % OR

MINIMUM 100 NOS. OF PLATES. IN CASE ANY DEFECT IS FOUND IN ANY OF THESE PLATES DURING ABOVE

D.P. TEST , MANUFACTURER SHALL CONDUCT D.P.TEST ON ENTIRE LOT OF PLATES. CUTOMER'S SHALL WITNESS TWICE THE SAMPLE AS EARLIER VIZ. 2% OF TOTAL NOS. OF PLATES RANDOMLY SELECTED FROM THE ENTIRE LOT.

MAIN CONTRACTOR TO WITNESS D.P. TEST ON 4 % TO TOTAL NOS. OF PLATES RANDOMLY SELECTED FROM THE ENTIRE LOT. VENDOR SHALL FIRST CARRYOUT DPT AS STATED ABOVE THEMSELVES AND ONLY AFTER SATISFACTORY TESTING, SHALL OFFER THE DPT FOR CUTOMER'S WITNESS.

NOTE-11: 100% PMI INSPECTION FOR MATERIAL GRADE VERIFICATION OF HT PLATES OF PHE SHALL BE WITNESSED BY REPUTED TPIA LIKE TUV/ LLOYD/BV AND CERTIFICATE SHALL BE SUBMITTED FOR REVIEW OF MAIN CONTRACTOR & CUTOMER'S. RANDOM 10% PMI INSPECTION FOR MATERIAL GRADE VERIFICATION OF HT PLATES OF PHE SHALL BE WITNESSED BY MAIN CONTRACTOR & CUTOMER'S.

3.5	EOT CRANES AND HOISTS														
SL. No.	COMPONENT & OPERATION S	CHARACTERISTICS	CLASS	TYPE OF CHECK	QUANTUM OF CHECK		REFERENCE DOCUMENT	ACCEPTANCE NORMS	FORMAT OF RECORD	AGENCY			REMARKS		
A. Raw Material Inspection															
A.1	MS PLATES FOR BRIDGE GIRDER, END CARRIAGES, CRAB FRAME, ROPE DRUM & PULLEYS (FABRICATED)	VISUAL& DIMENSIONAL	MAJOR	VISUAL& MEASUREMENT	100%	100%	RELEVANT MATERIAL SPECIFICATION AS PER APPROVED DATA SHEETS /DRAWINGS.		IR		P	V	-		
		CHEMICAL COMPOSITION	MAJOR	CHEMICAL ANALYSIS	1 SAMPLE / HEAT				MTC	√	P	V	V	In the absence of correlated TC check testing shall be done.	
		MECHANICAL PROPERTIES	MAJOR	TENSILE,% ELONGATION & YIELD STRESS	1 SAMPLE / HEAT					√	P	V	V		
		ULTRASONIC TEST OF PLATES (THICKNESSES ≥ 25 MM)	MAJOR	NDT	100%	100%	ASTM A 435	ASTM A 435	IR	√	P	V	V		
				CHEMICAL COMPOSITION	MAJOR	CHEMICAL ANALYSIS	1 SAMPLE / HEAT/SIZE		RELEVANT			√	P	V	V

A.2	ROPE DRUM (IF SEAM LESS PIPE USED)..	MECHANICAL PROPERTIES	MAJOR	TENSILE,% ELONGATION , YIELD STRESS, FLATTENING & MACRO ETCHING TEST	1 SAMPLE / HEAT/SIZE		MATERIAL SPECIFICATION AS PER APPROVED DATA SHEETS /DRAWINGS.		IR	√	P	V	V		
		ULTRASONIC TEST FOR SEAMLESS Pipe	MAJOR	UT	100 %	100 %	ASTM E-213	ASTM E-213	UT REPORT	√	P	V	V		

A.3	FORGINGS / ROLLED BARS FOR GEARS, PINIONS WHEELS AXLES, SHAFTS AND BRAKE	CHEMICAL PROPERTIES	MAJOR	CHEMICAL ANALYSIS	1 SAMPLE / HEAT/SIZE		RELEVANT MATERIAL SPECIFICATION AS PER APPROVED DATA SHEETS /DRAWINGS.		IR	√	P	V	V	in the absence of correlated TC check testing shall be done
	DRUM OF OUTER DIAMETER 200 MM	MECHANICAL PROPERTIES	MAJOR	TENSILE, % ELONGATION, YIELD STRESS	1 SAMPLE / HEAT/SIZE				IR	√	P	V	V	
		ULTRASONIC TEST	MAJOR	NDT	100%	100%	ASTM A 388	REFER NOTE-9	UT REPORT	√	P	V	V	
A.4	STEEL CASTING FOR PULLEY BREAK DRUM OF OUTER DIAMETER ≥200 MM	CHEMICAL COMPOSITION	MAJOR	CHEMICAL ANALYSIS	1 SAMPLE / HEAT/HT BATCH		RELEVANT MATERIAL SPECIFICATION AS PER APPROVED DATA SHEETS /DRAWINGS.		IR	√	P	V	V	
		MECHANICAL PROPERTIES	MAJOR	TENSILE, % ELONGATION, YIELD STRESS	1 SAMPLE/ HEAT/HT BATCH					√	P	V	V	
		ULTRASONIC TEST	MAJOR	NDT	100%	100%	ASTM A 609	ASTM A 609	UT REPORT	√	P	V	V	On accessible Area
A.5	CI CASTING FOR GEAR BOX HOUSING	CHEMICAL & MECHANICAL PROPERTIES	MAJOR	CHEMICAL, TENSILE, % ELONGATION, YIELD	1 SAMPLE/ HEAT/HT BATCH		SPECIFICATION/ DATA SHEETS /DRAWINGS		MTC	√	P	V	V	in the absence of correlated TC check testing shall be done

		S		STRESS										
B. Bought out items														
B.1	RAILS (IF APPLICABLE)	CHEMICAL & MECHANICAL PROPERTIES	MAJOR	CHEMICAL ANALYSIS, MECHANICAL PROPERTIES	1 SAMPLE / HEAT		IS 3443 / SAIL PRODUCT/ APPROVED DATA SHEETS /DRAWINGS.		MTC / CHECK TEST REPORT	√	P	V	V	Test to be carried out in absence of Mill T.C
		VISUAL & DIMENSIONAL	MAJOR	VISUAL & MEASUREMENT	RANDOM 25% (EACH SIZE)				IR	√	P	V	V	
B.2	HOOKS	CHEMICAL & MECHANICAL PROPERTIES	MAJOR	CHEMICAL ANALYSIS, TENSILE STRENGTH, % ELONGATION	1 SAMPLE/ HEAT/HT BATCH		MATERIAL AS PER APPROVED DATA SHEETS /DRAWINGS		MTC	√	P	V	V	
		PROOF LOAD TEST	MAJOR	MECHANICAL	100%	100%			IS 15560/IS:57 49	IS 15560/IS:57 49	TEST Report	√	P	V

		UT BEFORE & AFTER PROOF LOAD TEST	CRITICAL	NDT	100 %	100 %	ASTM A 388	REFER NOTE-9	UT REPORT	√	P	V	V			
		DPT TEST BEFORE & AFTER PROOF LOAD TEST	CRITICAL	NDT ON MACHINE D SURFACE	100%	100 %	ASTM E 165	ASTM E 165	PT REPORT	√	P	v	V			
		DIMENSIONS	MAJOR	MEASUREMENT	100%	100 %	MANUFACTURE'S DRAWING		IR	√	P	V	V			
B.3	WIRE ROPES	IDENTIFICATION DIMENSIONAL CONFORMITY & BREAKING STRENGTH	MAJOR	CONSTRUCTION, DIAMETER, TENSILE, DESIGNATION, BREAKING LOAD	100% (EACH BATCH)		IS-2266 / APPROVED DATA SHEET		MTC	√	P	V	V			
B.4	GEAR COUPLINGS	CHEMICAL COMPOSITION	MAJOR	CHEMICAL ANALYSIS,	1 SAMPLE / HEAT		IS-3177 / APPROVED DRAWING / APPROVED DATA SHEET		MTC / CHECK TEST REPORT	√	P	V	V			
		MECHANICAL PROPERTIES	MAJOR	MECHANICAL TESTS	1 SAMPLE / HEAT					√	P	V	V			

		VISUAL & DIMENSIONAL	MAJOR	VISUAL & MEASURE	100%	100%	PRODUCT CATALOGUE / COMPONENT DRAWING	IR	√	P	V	V				
B.5	BEARINGS	MAKE, & TYPE,	MAJOR	VERIFICATION	100%	100%	APPROVED DATA SHEET / BEARING CATALOGUE	IR		P	V	V				
B.6	MOTOR	MAKE / TYPE / RATING	MAJOR	VISUAL,	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / DATA SHEET / CUSTOMER/OWNER SPECIFICATION / IS 325	IR,		P	V	V				
		ROUTINE TESTS	MAJOR	REVIEW	100%	100%		MTC / COC	√	V	V	V				
B.7	SHROUDED DSL (DOWN SHOP LEAD) & CURRENT COLLECTOR	MAKE / TYPE / RATING HV & IR Test	MAJOR	VISUAL	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / CUSTOMER/OWNER SPECIFICATION / IS 3177	IR / MTC	√	P	V	V				
B.8	BRAKES –EM / EHT	MAKE / TYPE / RATING	MAJOR	VERIFICATION	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / CUSTOMER/OWNER SPECIFICATION / IS 3177	IR	√	P	V	V				
		FUNCTIONAL CHECK INCLUDING	MAJOR	REVIEW	100%	100%		MTC	√	V	V	V				

		IR-HV													
B.9	LIMIT SWITCH	MAKE / TYPE / RATING, OPERATION	MAJOR	VISUAL	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / CUSTOMER/OWNER SPECIFICATION / IS 3177	IR	√	P	V	V			
B.10	DBR (DYNAMIC BRAKING RESISTOR), LINE CHOKE	RATING, IR / HV,	MAJOR	VISUAL /REVIEW	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / CUSTOMER/OWNER SPECIFICATION / IS 3177 / MANUFACTURER STANDARD	IR / MTC		P	V	-			
B.11	CABLES - LT POWER / CONTROL / EPR FLEXIBLE TRAILING	MAKE / TYPE / RATING	MAJOR	VERIFICATION	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / CUSTOMER/OWNER SPECIFICATION / IS 1554 /IS 694/ IS 9968 / IS 4289 / IS 7098	IR		P	V	V			
		ROUTINE & ACCEPTANCE TESTS	MAJOR	REVIEW	100%	100%		MTC	√	V	V	V			
B.12	LOAD CELL (AS APPLICABLE)	MAKE / TYPE / RATING/FUNCTI ONAL CHECK	MAJOR	REVIEW	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / CUSTOMER/OWNER SPECIFICATION / IS 3177	IR		P	V	V			
B.13	MCCB / MCB / MPCB/ OVERLOAD RELAY/ SFU / CONTACTOR / TIME DELAY RELAY / AUX(CONTROL) RELAY/INDICATING METER, CONTROL	MAKE / TYPE / RATING IR & FUNCTIONAL CHECK	MAJOR	VERIFICATIO N	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / CUSTOMER/OWNER SPECIFICATION	IR		P	V	V	SEE NOTE - 8		

	TRANSFORMER/CURRENT TRANSFORMER														
B.14	PUSH BUTTON/INDICATING LAMP/SELETOR SWITCH/TERMINAL BLOCK	MAKE / TYPE / RATING	MAJOR	VISUAL & MEASU RE	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / CUSTOMER/OWNER SPECIFICATION /IS 3177	IR		P	V	V	SEE NOTE - 8		
B.15. 1	PANEL & PENDENT STATION HOUSING / BOX.	SHEET/GLAND PLATE THICKNESS/MAT ERIAL PAINT SHADE/THICKNE SS (MIN 70 MICRONES)/ADH ESION ,DOP BY PAPER INSERTION	MAJOR	-VISUAL & REVIEW	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / CUSTOMER/OWNER SPECIFICATION	IR		P	V	V	SEE NOTE - 8		
B.15.2	ASSEMBLED PANEL AND PENDENT STATION (EXCLUDING VVVF DRIVE)	-MAKE / TYPE / RATING OF BOIs ACCESSIBLE MOUNTING & LABELIN G OF COMPONENTS, WIRE DRESSING & FRRULING , CRIMPING CHECK BY GENTLE PULLING , EARTHING OF	MAJO R	-VISUAL & MEASU RE	100%	100%	CUSTOMER/OWNER APPROVED DRAWING / CUSTOMER/OWNER SPECIFICATION / IS-8623, LIST OF MAKES FOR BOI	IR		P	V	V			

		DOOR & PANEL ,IR-HV TEST													
B.15.3	VVVV DRIVE (AS APPLICABLE)	MAKE, TYPE, RATINGS & ROUTINE TEST.	MAJOR	VISUAL	100%	100%	BOI LIST & CUSTOMER/OWNER APPROVED DRAWING/DATA SHEET		IR		P	V	V		
C	IN PROCESS INSPECTION														
C.1	GEARS, PINIONS & WHEELS	DP TEST ON TOOTH AREA OF GEAR & PINION & TREAD OD OF WHEELS	MAJOR	NDT – DP TEST	100%	100%	ASTM E 165	NO RELEVAN T INDICATI ON	IR	√	P	V	V		
		HARDNESS TEST	MAJOR	HARDNE SS TEST	100%	10% RAND O M	IS-3177 / APPROVED DATA SHEET / APPROVED DRAWING.		IR	√	P	V	V		
		VISUAL & DIMENSIONA L	MAJOR	VISUAL & MEASURE	100%	100%	MANUFACTURER'S DRAWINGS		IR		P	V	-		
C.2	BRAKE DRUM, PULLEYS	VISUAL & DIMENSIONA L	MAJOR	VISUAL & MEASU RE	100%	100%	MANUFACTURER'S DRAWINGS		IR		P	V	-		
		DP TEST ON MACHINED AREA	MAJOR	NDT	100%	100%	ASTM E 165	NO RELEVAN T INDICATI O N	DPT REPOR T	√	P	V	V		

C.3	WELDING OF BOX GIRDER, CRAB, END CARRIAGES, ROPE DRUM (IF	WPS, PQR & WPQ	MAJOR	VISUAL & MECHANICAL TESTS	100%	100%	APPROVED WPS, ASME SEC-IX	ASME SEC-IX	QW-481-484	√	P	V	V		

	FABRICATED) GEAR BOX CASINGS, ETC	MARKING & CUTTING OF MS PLATES	MAJOR	VISUAL & MEASURE	100%	100%	MANUFACTURING DRAWING		IR		P	V	-			
		EDGE PREPARATION	MAJOR	VISUAL & MEASURE	100%	100%	MANUFACTURING DRAWING / APPROVED WPS		IR		P	V	-			
		WELD JOINT SIZE / LOCATION / WELDER NO. CO-RELATION	MAJOR	MARKING, VERIFICATION	100%	100%	MANUFACTURING DRAWING		IR		P	V	V			
		DP TEST ON WELD	MAJOR	NDT	100% FOR BW JOINTS & CRITICAL FILL ET WELDED JTS	100% FOR BW JOINTS & 10% FOR FW	ASTM E 165	NO RELEVANT INDICATION	DPT REPORT	√	P	V	V			
		RADIOGRAPHY OF BUTT WELD JOINTS	CRITICAL	NDT TEST	100% IN TENSION ZONE & 25% IN		ASME – SEC-V	ASME –SEC-VIII DIVISION – I, UW 51	RT FILMS & REPORT	√	P	V	V			

					COMPRESSION ZONE 100% FOR ROPE DRUM JOINT			& UW-52								
		DIMENSIONAL CHECK AFTER FINAL FABRICATION / MACHINING	MAJOR	MEASURE	100%	100%	MANUFACTURING DRAWING		IR	√	P	V	—			
C.4	GEAR BOXES	BACK LASH, TOOTH CONTACT REDUCTION RATIO	MAJOR	MEASUREMENT	1 PER STAGE		IS:4460 , APPROVED DRAWING / APPROVED / DATA SHEET		IR	√	P	V	V			
		NO LOAD RUN OF GEAR BOXES (Duration -4 hours)	MAJOR	SOUND LEVEL,TEMP Rise & Oil Leakage	100% EACH		APPROVED DRAWING /		IR	√	P	V	V			
D	FINAL INSPECTION															

D.1	COMPLETE ASSEMBLED CRANE AT SHOP FLOOR FOR NO LOAD / FULL LOAD/125% OVER LOAD TEST ALONG WITH ACTUAL CONTROL PANEL, VVFD, PUSH BUTTON PENDENT/RADIO REMOTE CONTROL CONNECTED(AS APPLICABLE)	OVERALL DIMENSIONS, SPEED,CURRENT, DURING NO LOAD TEST & FULL LOAD, DEFLECTION, OPERATION OF BRAKES ON FULL LOAD TEST, LIMIT SWITCH OPERATION OF ALL MOTIONS WITH PENDENT AS WELL AS RRC. CHECKING OF SAFETY INTERLOCKS, FUNCTIONAL TEST, DOP CHK BY PAPER INSERTION	MAJOR	VISUAL & MEASURE (SEE NOTE – 6)	100%	100%	IS-3177/APPROVED DRAWING /APPROVED DATA SHEET/ APPROVED SIZING CALCULATION	IR	√	P	W	W		
D.2	PAINTING OF CRANE	ADHESION ,VISUAL & DFT MEASUREMENT	MAJOR	VISUAL,SHADE VERIFICATION ,DFT & MEASUREMENT	100%	100%	AS PER CUSTOMER/OWNER SPECIFICATION	IR	√	P	V	V		
NOTES:-														

1.	MOTOR POWER RATING LESS THAN 30KW: ACCEPTANCE OF MOTOR LESS THAN 30KW IS BASED ON COC OF THE MANUFACTURER & THE CONTRACTOR CONFIRMING AS FOLLOWS: "IT IS HEREBY CONFIRMED THAT THE ABOVE MENTIONED MOTOR /MOTORS WAS/ WERE MANUFACTURED TAKING CARE OF CUSTOMER/OWNER SPECIFIC REQUIREMENTS REGARDING AMBIENT TEMP., VOLTAGE & FREQUENCY VARIATION, HOT STARTS, PULL OUT TORQUE, STARTING KVA/KW, TEMP. RISE, DISTANCE BETWEEN CENTRE OF STUD & GLAND PLATE AND TESTED IN ACCORDANCE WITH APPROVED DRAWING /DATA SHEETS." MOTOR POWER RATING MORE $\geq$ 30 KW BUT 50 KW ACCEPTANCE OF MOTOR RATING BETWEEN 30 KW & 50 KW IS BASED ON CUSTOMER/OWNER REVIEW OF ROUTINE TEST INSPECTION REPORT AS PER IS 325 WITNESSED BY MAIN CONTRACTOR ALONG WITH COC OF THE MANUFACTURER & THE CONTRACTOR CONFIRMING AS FOLLOWS: "IT IS HEREBY CONFIRMED THAT THE ABOVE MENTIONED MOTOR /MOTORS WAS/ WERE MANUFACTURED TAKING CARE OF CUSTOMER/OWNER SPECIFIC REQUIREMENTS REGARDING AMBIENT TEMP., VOLTAGE & FREQUENCY VARIATION, HOT STARTS, PULL OUT TORQUE, STARTING KVA/KW, TEMP. RISE, DISTANCE BETWEEN CENTRE OF STUD & GLAND PLATE, SPACE HEATER AND TESTED IN ACCORDANCE WITH APPROVED DRAWING /DATA SHEETS." MOTOR POWER RATING MORE $\geq$ 50 KW SEPARATE RQP / MQP SHALL BE SUBMITTED FOR MOTOR > 50KW WHERE CUSTOMER/OWNER SHALL WITNESS THE TESTS.	
2.	CUSTOMER/OWNER/BHEL/LLOYD/BVQI/TPL/NPCIL WPS SHALL BE USED FOR FABRICATION. ONLY CUSTOMER/OWNER/BHEL/LLOYD/BVQI/TPL/NPCIL APPROVED WELDERS SHALL BE ENGAGED FOR WELDING. (NO WELDING IS ALLOWED FOR CI MATERIAL)	
3.	THE HEAT NUMBER / PLATE IDENTIFICATION NUMBER SHALL BE TRANSFERRED ON ALL MAJOR CUT PIECES OF THE MS PLATE FOR PROPER CORRELATION. CUTTING PLAN OF EACH PLATE SHALL BE MAINTAINED FOR PROPER TRACEABILITY.	
4.	WELDER'S NUMBER SHALL BE PUNCHED NEAR BUTT WELD JOINTS. THE WELDING PLAN OF EACH FABRICATED ITEM SHALL BE MAINTAINED.	
5.	HV TEST ON CONTROL PANELS WILL BE DONE ISOLATING VVVF DRIVE AND OTHER ELECTRONIC PARTS. HV TEST AT 2 KV FOR POWER & 1.5 KV FOR CONTROL CIRCUIT.	
	ALL CRITICAL DIMENSIONS SHALL BE MEASURED. DEFLECTION SHALL BE MEASURED AT SAFE WORKING LOAD TEST THE CURRENT AND SPEED FOR HOISTING MOTION AND CROSS TRAVEL	
6.	MOTION SHALL BE MEASURED DURING NO LOAD TEST AND SAFE WORKING LOAD (SWL) TEST. THE SPEED AND CURRENT FOR LONG TRAVEL MOTION SHALL BE MEASURED DURING FREE RUNNING TEST. ALL INTERLOCKS, BRAKES CONTROL PANEL/PUSH BUTTON PENDANT, RADIO REMOTE CONTROL AND SAFETY MEASURES SHALL BE CHECKED DURING FINAL INSPECTION.	
7.	SAFE WORKING LOAD TEST & 125% OVERLOAD TEST SHALL BE CONDUCTED AT SHOP WITH ACTUAL VVFD PANELS, SHOP WIRE ROPES & TEMPORARY CABLES.	
8.	LIST OF ALL MAJOR BOUGHT OUT ITEMS AND THEIR ACCEPTABLE MAKE IS ATTACHED.	
9.	1) ASTM A -388/A-388M, APPENDIX 12.12.3	

- ii) ANT DEFECT ECHO HEIGHT MORE THAN 20% OF FULL SCREEN HEIGHT (WHEN SENSITIVITY LEVEL IS AT PRIMARY REFERENCE LEVEL) IS NOT ACCEPTABLE
- iii) AT ANY LOCATION, IF LOSS IN FIRST BACK WALL ECHO IS OBSERVED TO BE MORE THAN 20% (I.E FIRST BACK WALL ECHO HEIGHT IS LESS THAN 80% OF FULL SCREEN HEIGHT) WHEN SENSITIVITY SETTING IS AT PRIMARY REFERENCE LEVEL THE SAME IS ALSO NOT ACCEPTABLE.

Sl. No.	Component & Operation	Characteristics	Class	Type of Check	Quantum of Check		Reference Document / Acceptance Norms	Format of Record	D	Agency			Remarks
					M	C/U				M	C	U	
3.6 AIR CONDITIONING SYSTEM													
3.6.1 REFRIGERANT COMPRESSOR (RECIPROCATING/ SCREW/ CENTRIFUGAL)													
1.0	Raw Material												
1.1	Raw Materials	Chemical & Mechanical	Major	Chemical & Mechanical	1/heat	Approved Drg/ Data sheet	TC	√	P	V	V		
2.0	Final Inspection												
2.1	Castings for cylinder block, crank case and casings	Operation and performance testing	Critical	Hydraulic / pneumatic test	100%	Approved Drg/ Data sheet	Inspection Report	√	P	V	V		
2.2	Oil pump for reciprocating compressor			Capacity test	100%	Approved Drg/ Data sheet	Inspection Report	√	P	V	V		
	Assembled compressor :												
2.3	Compressors and chilling units	Performance testing to check FAD (free air delivery), noise, vibration and temperature rise of bearing and body	Critical	No load air run (free run) test	100%	Approved Drg/ Data sheet	Inspection Report	√	P	V	V		
2.4	Reciprocating compressor	Performance testing	Critical	Hydraulic/	100%	Approved Drg/ Data sheet	Inspection Report	√	P	V	V		

				leakage test								
2.5	Compressors	Performance testing	Critical	Functional run test	100%	Approved Drg/ Data sheet	Inspection Report	✓	P	V	V	Also capacity control (for part load performance) check
3.6.2 CONDENSER AND EVAPORATOR												
1.0	Final Inspection											
1.1	Shell side and tube side of condenser and evaporator	Operation and performance testing	Critical	Hydraulic / pneumatic test	100%	No leakage shall be permitted		✓	P	V	V	
3.6.3 VAPOUR ABSORPTION MACHINE (VAM)												
1.0	Final Inspection											
1.1	Complete assembled unit	Operation and performance testing	Critical	Capacity (TR) and steam consumption	100%	Approved Drg / Data Sheet	Inspection report	✓	P	W	W	All the controls shall be tested for proper functioning during the above test
1.2	N2 testing	leakage	Critical	Leakage	100%	Mfrs std	Inspection report	✓	P	W	V	
1.3	Helium spray test	leakage	Critical	Leakage	100%	Mfrs std	Inspection report	✓	P	W	V	
1.4	Helium shroud test	leakage	Critical	Leakage	100%	Mfrs std	Inspection report	✓	P	W	W	
1.5	Hydro testing	leakage	Critical	Leakage	100%	Mfrs std/ Approved Drg / Data Sheet	Inspection report	✓	P	W	V	
3.6.4 AIR HANDLING UNIT (AHU)												
1.0	Final Inspection											
1.1	Fan	Operation and performance testing	Critical	As per AMCA / IS	One of each type and size	for air flow, static pressure, speed, efficiency,		✓	P	V	V	

						power consumption, noise and vibration						
1.2	Assembled AHU (AHU casing and fan assembly)	Operation and performance testing	Critical	As per IS	One per type	free run test. Noise, vibration and temperature rise of bearing		√	P	V	V	
3.6.5 COOLING TOWER (AIR CONDITIONING)												
1.0	Raw Material											
1.1	Hub Plates, Gear Boxes, Fan Shaft	Chemical & Mechanical Properties, NDT	Major	Review of TC	100%	Approved drawing/datasheet	MTC	√	P	V	V	
2.0	In-Process Inspection											
2.1	Fan Static Balancing, Drive Shaft Dynamic Balancing Gear Assembly Run Test	Balancing Run Test	Critical	Visual/ Measurement	100%	Approved drawing/datasheet, Manufacturer Standard	TC/IR	√	P	V	V	
3.0	Final Inspection											
3.1	Assembled Cooling towers	Operation and performance testing Free run test at shop to measure FAD, noise and vibration Visual & Dimensional	Critical	Measurement	100%	Approved drawing/datasheet, Manufacturer Standard	TC/IR	√	P	W	W	For cooling towers being supplied in knocked- down condition, these tests shall be done at site
3.6.6 LOW PRESSURE AIR DISTRIBUTION SYSTEM												
1.0	Final Inspection											
1.1	Ducting system	Leakage	Major	Visual	100%	Smoke Test		√	P	W	W	

1.2	Air conditioning	Temperature and humidity	Major	Measurement	100%	Approved drawing/datasheet, Manufacturer Standard		✓	P	W	W	
3.7 VACUUM PUMP												
1.0	Raw Material Inspection											
1.1	Pump components	Chemical, Physical properties & Hardness	Major	Chemical, Physical & Hardness	1/heat	Approved Drg/Data sheet	TC	✓	P	V	V	
1.2	Separator	UT of forgings like shaft, couplers etc.	Major	UT	100%	ASTM A 388	UT Report	✓	P	V	V	Note-1
1.3	Heat Exchanger	Chemical, Physical properties	Critical	Chemical, Physical	1/heat	Approved Drg/Data sheet	TC	✓	P	V	V	
		Hydraulic test	Critical	Leakage test	100%	Approved Drg/Data sheet	TC	✓	P	V	V	
1.4	Gear Box	Chemical, Physical properties	Critical	Chemical, Physical	1/heat	Approved Drg/Data sheet	TC	✓	P	V	V	
1.5	Valves	Hydraulic test of tube side & shell side	Critical	Leakage test	100%	Approved Drg/Data sheet	TC	✓	P	V	V	
1.6	Instruments	Performance test	Critical	Performance test	100%	Approved Drg/Data sheet	TC	✓	P	V	V	No load condition, backlash, no. of gear train etc.
2.0	In Process Inspection											
2.1	Rotor	NDT Of machined portion	Critical	DPT	100%	ASTM E 165/no relevant indication	Inspection report	✓	P	V	V	
		Dynamic Balancing, Run out check	Critical		100%	ISO 1940 Gr.6.3 or better	Inspection report	✓	P	V	V	

3.0	Final Inspection											
3.1	Pump Assembly	Major dimensions	Critical	Measure ment	100%	Approved Drg/Data sheet	Inspection report	√	P	W	W	
		Hydrostatic test	Critical	Leakage Test	100%	Approved Drg/Data sheet	Inspection report	√	P	W	W	
		Performance test	Critical	Measure ment	100%	Approved Drg/Data sheet	Inspection report	√	P	W	W	
		Noise level check, Vibration check, Bearing temp. rise check	Critical	Measure ment	100%	Approved Drg/Data sheet	Inspection report	√	P	W	W	
3.8 OXIDATION BLOWER												
1.0	Raw Material Inspection											
1.1	Body/ casing, End plate/ Cover, Rotor	Chemical, Physical properties & Hardness	Major	Review of TC	100%	Approved Drawing/datasheet	MTC	√	P	V	V	
1.2	Shaft	Chemical, Physical properties	Major	Review of TC	100%	Approved Drawing/datasheet	MTC	√	P	V	V	
		NDT	Major	UT				√	P	V	V	
1.3	Timing gears	Chemical, Physical properties	Major	Review of TC		√		P	V	V		
		Hardness	Major			√		P	V	V		
		Mechanical properties	Major	UT after proof machinin g		√		P	V	V		
1.4	Base frame, Pipe & fitting, Foundation bolts	Chemical, Physical properties	Major	Review of TC	√	P		V	V			

1.5	Coupling/ Gear box, Valves, Filter, Silencer, Bearings etc.	Functional & Performance check	Major	Review of TC				√	P	V	V	
1.6	Actuators & Motor	Functional & Performance check	Major	Review of TC				√	P	V	V	
2.0	In-Process Inspection											
2.1	Body/ Casing	Hydro test	Critical	Measure ment	100%	Approved Drawing/datasheet	TC	√	P	V	V	
2.2	Shaft, Gears, End plate after machining	NDT	Critical	DPT	100%	Mfg.Std,	TC	√	P	V	V	
2.3	Rotor assembly (shaft with lobe)	Balancing	Critical	Visual	100%	Mfg.Std,	TC	√	P	V	V	
3.0	Final Inspection											
3.1	Complete blower assembly with motor	Performance Run • Overall major dimensions • Free air delivery • Head Power • Efficiency, speed • Discharge Pressure & temp. • Noise & Vibration • Oil temp. rise test	Critical	Measure ment	100%	Approved Drawing/datasheet	TC	√	P	W	W	
NOTE -												
1	INSPECTION / NDT / TEST CERTIFICATE / REPORT FOR UT, MPI, BALANCING, PRESSURE TEST SHALL INDICATE REFERENCE OF ITEM SERIAL NUMBER & MODEL FOR CO-RELATION.											

2	MATERIAL TEST CERTIFICATE / REPORT FOR TEST SUCH AS CHEMICAL & MECHANICAL TEST ETC. SHALL INDICATE REFERENCE OF ITEM SERIAL NUMBER & ENGINE MODEL FOR CO-RELATION AS PER SAMPLING NORMS.		
3	However, as a minimum, the following requirements shall be met (except for oil piping) : - For temperature > 400 °C and/or pressure exceeding 71 bar, the butt welds and full penetration branch welds shall be subject to 100% RT or UT and 100% MPE - For temperature > 1750C up to 400 0C and/or pressure exceeding 17 bar and up to 71 bar, the butt welds & full penetration branch welds shall be subject to NDTs as below: - pipe diameter > 100 NB100% RT or UT and 100% MPE - pipe diameter ≤ 100 NB10% RT or UT and 100 % MPE - In addition to above, statutory requirement, wherever applicable, shall also be complied with		
4	- Butt welds 100% RT and 100% MT or DPT - Welds for nozzle, branch 100% RT and 100% MT or DPT connections of size > 100NB - Welds for all other attachments 100% MT or DPT - Removal of weld defects 100% MT or DPT		
5.	i)	Butt welds: pipe size > 100 NB pipe size ≤ 100 NB	100% RT and 100% MPE or DPT 10% RT and 10% MPE or DPT
	ii)	Welds for nozzle, branch and thickness > 19 mm	100% RT and 100% MPE or DPT connections of size > 100NB
	iii)	Welds for all other attachments	10% MPE or DPT
	iv)	Removal of weld defects/weld repair	100% MPE or DPT
6.	a)	Butt welds: pipe size > 100 NB pipe size ≤ 100 NB	100% RT and 10% MPE or DPT 10% MPE or DPT
	b)	Welds for nozzle, branch	10% MPE or DPT connections/ attachments
	c)	Removal of weld defects	100% MPE or DPT
7.	All other pipes not covered above (except oil piping) shall be subjected 100% MPE or DPT in case of underground piping and 10% MPE or DPT in case of over ground piping. 10% of butt welds of underground piping shall be subjected to RT		
8.	a) Butt welds of Oil piping shall be subjected to 10% RT and 10% DP Test. For Jacking oil lines 100% RT and 100% DPT shall be carried out on butt welds. b) Fillet welds with load transfer shall be subjected to 100% MPE or DPT and fillet welds without load transfer shall be subjected PE		
9.	a) Valve size < 50 NB - Visual examination for rating below ANSI Class 900 - MPE for rating above ANSI Class above 900		

	b) Valve size ≥ 50 NB and < 100 NB - Visual examination for rating up to ANSI Class 600 - MPE for rating above ANSI Class above 600 and below 900 100% MPE and 100% RT on 10% of valves for rating above ANSI Class above 900 and below 4500 c) Valve size ≥ 100 NB and < 300 NB - Visual examination for rating up to ANSI Class 300 - MPE for rating above ANSI Class above 300 and below 600 100% MPE and 100% RT on 10% of valves on change of section and weld ends for rating above ANSI Class above 600 and below 900 100% MPE and 100% RT on valves for rating above ANSI Class 900 and below 4500 d) Valve size ≥ 300 NB - MPE for rating up to ANSI Class 600 100% MPE and 100% RT on all valves in the areas of change of section and weld ends for rating above ANSI Class 600 and below Class 900 100% MPE and 100% RT on valves for rating above ANSI Class 900 and below Class 4500 Note : In the above NDTs, MPE may be replaced by DPT for austenitic steels											
10.	After assembly, one valve of each size with respective actuator shall be shop operated over the full range of movement in both the directions, with the body subjected to the full hydrostatic pressure conditions. During the test, hand wheel operation, opening and closing time and current drawn shall also be checked. The test shall be conducted for three consecutive cycles with valve shaft both in vertical and horizontal planes											
3.9 COOLING WATER PUMP												
1	Raw Material											
1.1	Bell Mouth	Chemical Properties	Major	Chem. Analysis	1 Sample/Heat	Relevant Material Specification as per	MTC	√	P	V	V	
		Mech. Properties	Major	Mech.	1 Sample/Heat	Approved Drawing and Approved Data sheet	MTC	√	P	V	V	
		Visual Inspection	Major	Visual	100%	Manufacturer's Practice	IR	√	P	V	V	
1.2	Diffuser Casing, Impeller casing and Casing Cap (Bowl)	Chemical Properties	Major	Chem.	1 Sample/Heat	Relevant Material Specification as per	MTC	√	P	V	V	As Cast Heat Nos required for Co-Relation. Also Refer Note – 1.
		Mech. Properties	Major	Mech.	Each melt & HT batch	Approved Drawing and Approved Data sheet	MTC	√	P	V	V	
		Visual Inspection	Major	Visual	100%	Manufacturer's Practice	IR	√	P	V	V	
1.3	Impeller (Casting), Impeller Nut	Chemical Properties	Major	Chem. Analysis	1 Sample/Heat	Relevant Material Specification as per	MTC	√	P	V	V	As Hard Punch Heat Nos required for Co-relation.

		Mech. Properties		Mech. test	1 Sample/Heat	Approved Drawing and Approved Data sheet	MTC	√	P	V	V	Mech check carried out after HT. Also Refer Note – 1.
		Visual Inspection		Visual	100%	Manufacturer's Practice	IR	√	P	V	V	
1.4	Forging for Muff coupling	Chemical Properties	Major	Chem. Analysis	1 Sample/Heat	Relevant Material Specification as per Approved Drawing and Approved Data sheet	MTC	√	P	V	V	Refer Note-2
		Mech. Properties		Mech. test	1 Sample/Heat		MTC	√	P	V	V	
		Internal Defects		UT (on rough Machined)	100%	App. Drg/ Data sheet/ specification	UT Certificate	√	P	W	V	
1.5	Forging for Rigid coupling	Chemical Properties	Major	Chem. Analysis	1 Sample/Heat	Relevant Material Specification as per Approved Drawing and Approved Data sheet	MTC	√	P	V	V	
		Mech. Properties		Mech. test	1 Sample/Heat/ HT Batch		MTC	√	P	V	V	
		Internal Defects		NDT-UT	100%	App. Drg/ Data sheet/ Relevant Standard /Manufacturer's Standard	UT Certificate	√	P	W	V	
1.6	PUMP SHAFT	Chemical Properties	Major	Chem. Analysis	1 Sample/Heat	Relevant Material Specification as per Approved Drawing and Approved Data sheet	MTC	√	P	W	V	Refer Note-2
		Mech. Properties	Major	Mech. test	1 Sample/Heat/ Heat batch		MTC	√	P	W	V	
		Heat Treatment	Major	Measure (Time-Temp chart)	100%		HT Chart	√	P	V	V	
		Sub Surface Defects	Major	MPI	100%	Relevant Standard	MTC	√	P	W	V	

		Internal Defects	Major	NDT-UT	100%		App. Drg/ Data sheet/ Relevant Standard /Manufacturer's Standard		UT Certificate	√	P	W	V	
1.7	Forging for shaft sleeves	Chemical Properties	Major	Chem. Analysis	1 Sample/Heat		Relevant Material Specification as per Approved Drawing and Approved Data sheet		MTC	√	P	V	V	
		Mech. Properties	Major	Mech. test	1 Sample/Heat/ Heat batch	MTC			√	P	V	V		
		Internal Defects		NDT-UT	100%		App. Drg/ Data sheet/ Relevant Standard /Manufacturer's Standard		UT Certificate	√	P	W	V	
1.8	Taper Column & Discharge Head, Sole plate, Motor Stool & Flanges	Chemical & Mech. Properties	Critical	Chemical & Mech Check	1 Sample/Lot		Relevant Material Specification as per Approved Drawing and Approved Data sheet		MTC	√	P	V	V	
		Internal Defects	Major	UT*`	100%		ASTM A 435		UT Certificate	√	P	W	V	*For plate t >=25 mm
2	In Process Inspection													
2.1	Diffuser Casing, Impeller casing (Bowl)	Dimensions	Major	Measure	100%	-	Manufacturing Drawing		IR		P	V		
		DPT on Machined Area		NDT-DPT	100%		ASME Sec V article 6	ASME Sec VIII DIV I, App 7	TC	√	P	V	V	
		Hydraulic Test of Bowl	Major	Leak tightness Test	100%	100%	App. Drg/ Data sheet	NO LEAKAGE	TC	√	P	V	V	Test duration 30min
2.2	Impeller	Dimensions	Major	Measure	100%	-	Manufacturing Drawing		IR		P	V		
		RT	Critical	NDT	100%		ASME Sec V	ASTM E-446		√	P	V	V	Review of RT film reports by Owner

							article 2		RT FILM & REPORT					
	Impeller (after clearance of RT)	Solution Annealing (SS – CF8M Material)	Major	Measure (Time-Temp chart)	100%		App. Drg/ Data sheet	App. Drg/ Data sheet	HT Chart	√	P	V	V	
		IGC Test	Major	IGC Test	1 Sample/Heat/ Heat batch		ASTM 262 Practice E		TC	√	P	V	V	
		Dimensions	Major	Measurement	100%		Manufacturing Drawing		IR	√	P	W	V	
		DPT on Machined Area	Major	NDT/DPT	100%		ASME Sec V article 6	ASME Sec VIII DIV I, App 7	TC	√	P	W	V	
		Grinding/ Finishing	Major	Visual	100%	-	Manufacturing Drawing		-		P	-	-	
		Dynamic Balancing	Major	Balancing	100%		ISO 21940, GR, 6.3		IR	√	P	W	V	
2.3	Bell Mouth	DPT on Machined Area	Major	NDT - DPT	100%		ASTM 165		TC	√	P	W	V	
		Dimensions	Major	Measurement	100%	-	Manufacturing Drawing		IR		P	V		
2.4	Shaft & Sleeves, Muff Coupling After Machining	Dimensions	Major	Measurement	100%	-	Manufacturing Drawing		IR		P			
		Surface Finish	Major	Visual	100%	-	Manufacturing Drawing		-		P	-	-	
		DPT on Machined Area	Major	NDT - DPT	100%		ASTM 165		TC	√	P	W	V	
2.5	Pump & Motor Coupling	Dimensions	Major	Measurement	100%	-	Manufacturing Drawing		IR		P	V		
		Dynamic Balancing	Major	Balancing	100%		ISO 21940, GR 6.3		IR	√	P	W	V	

2.6	Fabricated Parts													
2.6.1	Fabricated Items mentioned above & Segmental Flanges (If made for components Mentioned in 1.7)	WPS, PQR / Welder Qualification	Major	Visual & Mech test	100%		ASME SEC- IX/Approved WPS		QW 483 484	√	P	V	V	See Note-3
		Joint Fit-up	Major	Measure	100%	-	Manufacturing Drawing		-		P	-	-	
		Root run welding	Major	Visual	100%	-	Manufacturing Drawing		-		P	-	-	
		DPT after Back gauging (In Case of Butt Weld)	Major	NDT	100%		ASME Sec VIII DIV I, App 8		IR	√	P	V	V	
		Spot RT on Full Penetration Butt Weld Joint for thickness > 10mm	Critical	NDT	10% of Butt Weld Length		ASME SEC-V	ASME SEC-VIII DIV-I	RT REPORT & RT FILM	√	P	V	V	Review of RT film reports by owner
		RT Test on Weld Joint for flange thickness <=56mm	Critical	RT	100%		ASME SEC-V	ASME SEC-VIII DIV-I	RT REPORT & RT FILM	√	P	V	V	Review of RT film reports by owner
		UT Test on Weld Joint for Plate thickness > 56mm	Critical	UT	100%	100/20%	ASME SEC-V	ASME SEC-VIII DIV-I, APP 12	UT REPORT	√	P	W	W	
		DPT on finished weld	Major	DPT Test	10% of Butt And 10% on Fillet weld		ASME SEC-V Art 6	ASME SEC-VIII DIV-I, App 8	IR	√	P	W	V	
		Stress Relieving (segmental flanges if made for these parts and thickness >=	Major	Measure	100%		ASME SEC-VIII	ASME SEC-VIII	HT Chart	√	P	V	V	

		37.5mm												
2.6.2	Finished Column Pipe & Discharge Head	Dimensions	Major	Measure	100%	-	Manufacturing Drawing		IR	-	P	-	-	
		Hydraulic test of column pipe &		Leak Tightness Test	100%	100%	App. Drg/ Data sheet	NO LEAKAGE	IR	√	P	W	V	Test duration 30min
		Discharge Head			100%	100%								
3.0	Final Inspection													
3.1	Pump Assembly/ Bowl Assembly	Assembly Checks, Free rotation of assembly	Major	Measurem ent	100%		App. Drg	App. Drg	Log Sheet	√	P	W	W	
		Performance Testing, Noise Level, Vibration, Bearing Temp	Critical	Testing	100%		Approved Test Procedure	Approved Test Procedure	Test Repot	√	P	W	W	
		Post test inspection(Up to Journal & Thrust bearings only)	Major	Visual	100%		App. Drg	No Damages	Inspection Report	√	P	W	W	
3.2	Painting & Packing	Verification of completeness & Packing	Major	Visual	100%		App. Drg/ Data sheet	App. Drg/ Data sheet	Inspection Report	√	P	W	W	
3.3	QA Documentation	All Reports	Major	Review	100%		Approved QAP	Approved QAP	Report	√	P	V	V	
NOTE- 1. NO WELDING IS PERMISSIBLE ON CAST IRON CASTINGS. FOR CI MATERIAL – "S" & "P" CONTENT SHALL BE S = 0.1 % MAX & P = 0.15 % MAXIMUM, NI % AS PER APPROVED DATA SHEET. 2. IN CASE, CORRELATED MTC ARE NOT AVAILABLE, THEN QUANTUM OF CHECK SHALL BE ONE SAMPLE FROM EACH BAR. 3. FABRICATION SHALL BE CARRIED OUT IN MANUFACTURER’S OR ITS APPROVED SOURCES ONLY.														
3.10 ASH HANDLING PLANT														
3.10.1 BAG FILTER FOR BUFFER HOPPER & VENT FILTER OF FA SILO														
1	Raw Material													

1.1	Dish end, Plenum casing flanges & Tube, MS Sheets/plates and Std section	Visual & Dimension	Major	Visual/ measurement	20% random	IS:2062 & IS 1079/ Approved Drg/ DS	Mfg. TC/QCR	√	P	V	V	
		Chemical & Mechanical	Major	Chemical & Mechanical	Sample/Lot	IS:2062 & IS 1079/ Approved Drg/DS	Mfg. TC/QCR	√	P	V	V	
1.2	Pipe for air manifold and pulsing pipe	Chemical & Mechanical	Major	Chemical & Mechanical	Sample/Lot	IS:1239 & IS 3589/ Approved Drg/DS	Mfg. TC/QCR	√	P	V	V	
1.3	Wire cages	Visual & Dimension	Major	Visual/ measurement	5%	Approved Drg/DS	Mfg. TC/QCR	√	P	V	V	
1.4	Venture	Visual & Dimension Tensile & Hardness	Major	Visual/ Measurement & Mechanical Test	10% random	Approved Drg/DS	Inspection report	√	P	V	V	
1.5	Sequential controller, gauges as applicable	Visual & review of MTC, review of calibration certificate & Functional test	Major	Verification	100%	Approved Drg/DS	Test & guarantee certificate from mfc	√	P	V	V	
1.6	Pulse valve & solenoid Valves	Visual & review of MTC and Functional Test	Major	Visual/ Measurement & Testing	100%	Approved Drg/DS	Test & guarantee certificate from mfc	√	P	V	V	
1.7	Filter bags	Visual review of MTC	Major	Visual/ measurement	5%	Approved Drg/DS	Test & guarantee	√	P	V	V	

							certificate from mfc					
2	In Process Control											
2.1	Marking and Cutting	Visual & Dimension	Major	Visual/ measureme nt	100%	Approved Drg./DS	IR	√	P	V	V	
2.2	Set up & truck welding	Visual & Dimension	Major	Visual/ measureme nt	100%	Approved Drg./DS	IR		P	V	V	
2.3	Welding Qualification	WPS & PQR	Major	Verificatio n	100%	ASME SEC IX	WPS/PQR		P	V	V	
2.4	Welding (by approved welder)	Weld quality	Major	Visual & DP	100%	Approved Drg/ASTM E-165	IR	√	P	V	V	
2.5	Air Manifold/ reservoir	Leak tightness	Major	Hydro	100%	Approved Drg./DS	IR	√	P	V	V	
3	Final Inspection											
3.1	Assembly surface, Cleaning & painting	Visual	Major	Visual	100%	Approved GA DRG	IR	√	P	W	W	
3.2		Dimensional, quality	Major	Measurem ent	100%	10%	Approved GA DRG	IR	√	P	W	W
3.3		Leakage of Casing	Major	Pneumatic Test	100%	Approved GA DRG / Data Sheet / No leakage	IR	√	P	V	V	
3.4		Functional test	Major	Pulsating test	100%	Approved GA DRG/Data sheet/approved test procedure	IR	√	P	V	V	
3.5		Painting	Minor	Visual	random	Approved painting Schedule/ Mfd Std.	IR	√	P	V	V	
3.2	QA Documentation	Visual	Major	Review	100%	Approved QAP	Report	√	P	V	V	
3.10.2 BAH FEED GATE												
1	Raw Material											

1.1	Housing	Chemical Composition	Major	Chemical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	
		Mechanical Test	Major	Mechanical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	
		Visual and Dimension	Major	Visual & Measurement	100%	Approved DRG/Approved Specification	IR	√	P	V	V	
1.2	Wear Strip/Liner	Chemical Composition	Major	Chemical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	
		Mechanical Test	Major	Mechanical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	
		Visual and Dimension	Major	Visual & Measurement	100%	Approved DRG/Approved Specification	IR	√	P	V	V	
1.3.1	Rollers	Mechanical Test/HT Chart	Major	Mechanical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	
		Chemical Composition	Major	Chemical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	
		Hardness	Major	Mechanical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	
		Visual and Dimension	Major	Visual & Measurement	100%	Approved DRG/Approved Specification	IR	√	P	V	V	
1.3.2	Roller wedges	Chemical Composition	Major	Chemical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	
		Mechanical Test	Major	Mechanical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	
		Visual and Dimension	Major	Visual & Measurement	100%	Approved DRG/Approved Specification	IR	√	P	V	V	

1.4	Door and Door Frame	Chemical	Major	Chemical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	
		Mechanical Test	Major	Mechanical	1 /heat	Approved DRG/Approved Specification	MTC/Lab TC	√	P	V	V	All bolts and nuts are Galvanized
		Visual and Dimension	Major	Visual & Measurement	100%	Approved DRG/Approved Specification	IR	√	P	V	V	
1.5	Pneumatic Cylinder	Visual and Dimension & Operation Test	Major	Documents Review	100%	As per Approved Drg./ DS	MTC/Lab TC	√	P	V	V	
1.6	Shafts	Chemical Composition	Major	Chemical	1 /heat	Approved DRG./ Specification	MTC/Lab TC	√	P	V	V	
		Visual and Dimension	Minor	Visual & Measurement	100%	Approved DRG./ Specification	IR	√	P	V	V	
		Mechanical Properties	Major	Mechanical	One sample/lot	Approved DRG./ Specification	MTC/Lab TC	√	P	V	V	
		Sub surface Defect (above 40min dia)	Major	UT	100%	Approved DRG./ Specification/ ASTM A 388	IR	√	P	V	V	NOTE-1
		Surface Defect after machining	Major	OPT	100% on final weld	ASME SEC V	IR	√	P	V	V	
2.0	In Process Inspection											
2.1	Gate & Flame, Cylinder Mounting pedestal, Follower, Wear Plates, Body for access door & inspection window	Casting Defects	Major	Visual	100%	Approved DRG/ Specification	IR	√	P	V	V	

		Visual & Dimensional	Major	Visual & Measurement	100%	Approved DRG./ Specification	IR	√	P	V	V	
2.2	Marking, Cutting, Edge Preparation, tacking, Drilling, Assembly of liner plates for Fabrication of Housing	Visual & Dimensional	Major	Visual & Measurement	100%	Approved DRG/ Specification	Shop Record	√	P	V	V	
		DP Test	Major	DPT	100%	ASME SEC V	IR	√	P	V	V	
	Welding Documents for Fabrication of Housing.	WPS & PQR, WPQ	Major	Review	100%	ASME Sec. IX / ASME SEC. VIII	WPS & PQR, WPQ	√	P	V	V	
	Welding (Final) Housing Joints only for Fabrication of Housing.	Weld Quality	Major	Visual & DPT	100%	ASME SEC V	IR	√	P	V	V	
2.3	Wear Strip, Roller Wedge, Rollers & Roller Axial	Visual & Dimensional	Major	Visual & Measurement	100%	Manufacturing Drg.	IR	√	P	V	V	
2.4	Assy. With Pneumatic Cylinder	Matching of Door 7 Frame movement of door clearance	Major	Operation, Measurement, Hydro test	100%	Approved DRG. Approved Specification	IR	√	P	V	V	
3.0	Final Inspection											
3.1	Gate way along with Pneumatic Cylinder	Fitment Dimensions Completeness	Major	Visual & Measurement	100%	As per Approved Drg.	IR	√	P	W	W	PMI test to be done for alloy materials.
3.2		Leakage test of Sealing area of feed gate	Major	Water Fill test	100%	As per Approved Drg./DS	IR	√	P	W	W	

3.3		Functional test	Major	Open and close operations for free movement	100%	As per Approved Drg./DS	IR	√	P	W	W	
3.4	Painting	Final Finish & DFT	Minor	Visual & Measurement	100%	Approved Painting Schedule/Data sheet.	IR	√	P	V	V	
3.5	QA Documentation	Visual	Minor	Review	100%	Approved QAP	IR	√	P	V	V	

NOTE-1 : UT SHALL BE CARRIED OUT BY USING SUITABLE FREQUENCY PROBE. USING THIS PROBE, THE BACK WALL ECHO SHALL BE SET AT 100% OF FULL SCALE HEIGHT (FSH) IN SOUND AREA. AT THIS SENSITIVITY LEVEL, ANY DEFECT ECHO EXCEEDING 20% OF FSH IS NOT ACCEPTABLE. ALSO LOSS OF BACK WALL ECHO MORE THAN 20% AT ANY LOCATION IS NOT ACCEPTABLE

3.10.3 CLINKER GRINDER

1.0	Raw Material Inspection											
1.1	Housing	Chemical Composition	Major	Chemical	1 Sample /heat	Approved DRG. Specification	MTC/Lab TC	√	P	V	V	NOTE-1
		Physical properties	Major	Mechanical	1 Sample /heat	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
		UT OF PLATES >25 MM THK	Major	Mechanical	1 Sample /heat	ASME SEC VIII DIV 1 SA 578 LEVEL B /ASTM A 435	MTC/UT REPORT	√	P	V	V	
		Dimensional conformity	Major	Measurement	100%	Approved DRG. Specification	IR	√	P	V	V	
1.2	Driving Gear	Chemical Composition	Major	Chemical	As per Approved DRG/Specification	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
		Mechanical Test	Major	Mechanical	Approved DRG/ Specification	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
1.3	Grinder Rolls	Chemical Composition & Mechanical properties	Major	Chemical & Mechanical	1 Sample/heat	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	

		Dimensional conformity	Major	Measurement	100%	Approved DRG/ Specification	IR	√	P	V	V	
		Heat Treatment	Major	Review of HT chart	100%	Approved DRG/ Specification	IR	√	P	V	V	
		Bend Test	Major	Mechanical	1 Sample/heat	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
		Hardness	Major	Measurement	1 Sample/heat	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
		Casting Defects	Major	Visual	100%	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
1.4	Shaft	Chemical Composition	Major	Chemical	1 Sample /heat	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
		Dimensional conformity	Minor	Measurement	100%	Approved DRG/ Specification	IR	√	P	V	V	
		Mechanical Properties	Major	Mechanical	1 Sample /heat	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
		Sub surface Defect (above 40mm dia)	Major	UT	100%	Approved DRG/ Specification/ASTM A 388	IR	√	P	v	v	NOTE-1
		Surface Defect after Ma	Major	DPT	100% on final weld	ASME SEC V	IR	√	p	V	V	
1.5	Shaft Sleeve	Chemical Composition & Mechanical properties	Major	Chemical & Mechanical	On 10% sample	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
		Hardness	Minor	Measurement	100%	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
		Dimensional conformity	Minor	Measurement	100%	Approved DRG/ Specification	IR	-	P	V	V	
1.6	Wear Plate	Chemical Composition	Major	Chemical	1 Sample /heat	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	

		Hardness	Major	Measurement	1 Sample /heat	Approved DRG/ Specification	IR	√	P	V	V	
1.7	Bearing, Screws, Flexible hose pipes and other items	Verification of Make/Model no.	Major	Visual	100%	Approved DRG/ Specification	MTC/Lab TC	√	P	V	V	
2.0	In Process Inspection											
2.1	Big Support	Casting Defect	Major	Visual	100%	Approved DRG/ Specification	IR	√	P	V	V	
			Major	DPT	100% on final weld	ASME SEC V	IR	√	P	V	V	
		Dimensional Conformity	Major	Measurem ent	Critical dimensions	Approved Drg.	IR	√	P	V	V	
2.2	Fabrication of Housing	Dimensional conformity	Major	Measurem ent	Critical dimensions	Approved Drg.	IR	√	P	V	V	
2.3	Welding Qualification	WPS, PQR, WQR	Major	Verificatio n	100%	ASME SEC IX	IR	√	P	V	V	
2.4	Welding of Housing, Shaft & Rolls	DPT on final weld	Major	DPT	100% on final weld	ASME SEC V	IR	√	P	V	V	
2.5	Alignment of Clinker grinder	Alignment of fire movement	Major	Visual & by manual operation	100%	Approved Drg.	IR	√	P	V	V	
3.0	Final Inspection											
3.1	Check for Completeness, appearance, drilled holes, straightness, diameter, no. of holes verification,	Fitment Dimension	Major	Measurem ent/Visual	100%	Approved Drg./Data sheet	IR	√	P	W	W	

	Cleanliness & dimension											
3.2	No Load running test at least for 2hours with shop drive	Free Rotation	Critical	Visual	100%	Approved Drg	IR	√	P	W	W	
		Bearing Temp rise & Vibration	Major	Measurement	100%	Approved Drg ./DATA SHEET	IR	√	P	W	W	Temp rise ambient +30° Vibration below B5 Micron
3.3	Painting	Paint shade, DFT	Minor	Visual & Measurement	100%	Approved painting Schedule	IR	√	P	V	V	
3.4	QA Documentation	Visual	Major	Review	100%	Approved QAP	Report	√	P	V	V	

NOTE-1: UT SHALL BE CARRIED OUT BY USING SUITABLE FREQUENCY PROBE. USING THIS PROBE, THE BACK WALL ECHO SHALL BE SET AT 100% OF FULL SCALE HEIGHT (FSH) IN SOUND AREA. AT THIS SENSITIVITY LEVEL, ANY DEFECT ECHO EXCEEDING 20% OF FSH IS NOT ACCEPTABLE. ALSO LOSS OF BACK WALL ECHO MORE THAN 20% AT ANY LOCATION IS NOT ACCEPTABLE

3.10.4 AIR OIL CONVERTER TANK

1.0	Raw Material Inspection											
1.1	Body & Baffle plate, Structural supports	Chemical Composition	Major	Chemical	1 Sample/heat	Approved DRG/Specification	MTC/Lab TC	√	P	V	V	
		Mechanical Test	Major	Mechanical	1 Sample/heat	Approved DRG/Specification	MTC/Lab TC	√	P	V	V	
		Surface Defect	Major	Visual & DP T	100%	Approved DRG/Specification	IR	√	P	V	V	
2.0	In Process Inspection											
2.1	Body & Structural, Dish end											

	a) Marking, cutting edge preparation, Cleaning, tacking, straightening etc.	a) Fit up check (ovality / misalignment)	Major	Dimensional/ Visual	100%	Approved DRG/Specification	IR	√	P	V	V	
	b) Welder Qualification and welding procedure	Welding Documents	Major	Review	100%	ASME SEC IX	IR	√	P	V	V	
	c) NDT	Surface Defect	Major	Visual of DP Test	100%	Approved DRG/Specification	IR	√	P	V	V	
		Weld Quality	Major	DP Test	100% Butt 10% Fillet	ASME SEC V / ASME SEC VIII Appendix-B	IR	√	P	V	V	
3.0	Final Inspection											
3.1	Check for critical dimensions, straightness, Distortion, alignments etc.	Completeness and visual/ dimensional examination	Major	Measurement/Visual	Overall Dimensions 100%	Approved DRG/Specification	IR	√	P	W	W	Witness 100% of offered lot
3.2	Hydrostatic Pr. Testing of shell	Hydraulic pressure for 5 min.	Major	Pressure Test	100%	Test Pressure as per Approved Drg/DS/Relevant code & standard / No leakage/Pressure Drop	IR	√	P	W	W	
3.3	Painting	Paint shade, DFT	Major	Visual & Measurement	100%	Approved painting Schedule	IR	√	P	V	V	
3.4	QA Documentation	Visual	Major	Review	100%	Approved QAP-	Report	√	P	V	V	
3.10.5 CAST IRON PIPE												
1.0	Raw Material Inspection											
1.1	Raw Material Pig Iron & Scrap	Material Compliance	Major	Chemical Check	One Sample / Lot	AS per Mfg. Std	MTC	√	P	V	V	

1.2	RUBBER GASKETS FOR SOCKET & SPIGOT ENDS PIPES**	CONFORMANCE TO IS 1536 REQUIREMENTS	MINOR	VISUAL	ONE/ LOT	IS 5382 & IS 12820	IR		P	V		** IF APPLICABLE
2.0	In Process Inspection											
2.1	Pipe Spinning	Surface Condition	Major	Verification of Lot Sheet	100%	IS:1536/BS:1211	IR	√	P	V	V	
		Thickness	Major	Visual	100%	IS:1536/BS:1211	IR	√	P	V	V	
2.2	Heat Treatment	Verification of Heat Treatment	Critical	Review	100%	IS 1536/APPVD DRG & DS/MFRS STD	IR	√	P	V	V	
2.3	Pipe after heat Treatment	Major Dimensions	Major	By Vernier Calliper	Every Hour	IS 1536/APPVD DRG & DS/MFRS STD	IR	√	P	V	V	
3.0	Final Inspection											
3.1	Physical Testing	Mechanical Test	Critical	Modulus of rupture/ tensile test	One sample / Batch	IS:1536/BS:1211	IR	√	P	W	W	
		Brinell Hardness	Critical	Hardness Test	One sample / Batch	IS:1536/BS:1211	IR	√	P	W	W	
3.2	Dimension	OD, THICKNESS, OVALITY & LENGTH, QUALITY, STRAIGHTNES S, LENGTH AND SOCKET	Major	BY GO GAUGES, VERNIER CALIPER, MEASURING TAPE	100%	IS:1536/BS:1211	IR	√	P	W	W	10% from the offer lot (Rev 01)

		& SPIGOT DIMENSIONS										
3.3	Hydrostatic Test	Leak Test	Major	Hydrostatic Test	100%	IS:1536/BS:1211	IR	√	P	W	W	10% from the offer lot
3.4	Bitumeneous Black Paint / Tar	Coating Adherence	Major	Adherence Test	100%	IS:1536/BS:1211	IR	√	P	V	V	
3.5	Marking	Compliance	Major	Verification	100%	IS:1536/BS:1211	IR	√	P	V	V	
3.10.6 EXPANSION BELLOW, BELLOW APH, DUCT & SCR HOPPER												
1.0	Raw Material Inspection											
1.1	Sheet for Bellows	Mechanical & Chemical Properties	Major	Mechanical & Chemical	Sample Test Report or Mill TC	Approved Drawing/ Specification	TC	√	P	V	V	
1.2	CS Flange and plate for socket/centre pipe	Mechanical & Chemical Properties	Major	Mechanical & Chemical	Sample Test Report or Mill TC	Approved Drawing/ Specification	TC	√	P	V	V	
2.0	In Process Inspection											
2.1	Marking, Cutting & machining	Measurement	Major	Dimension	100%	Approved Drawing	IR	√	P	V	V	
2.2	Fabrication before welding	Edge preparation & Fit up checking	Major	Visual & Dimension	100%	Approved Drawing	IR	√	P	V	V	
2.3	After welding of bellows Long scam (plies)	Soundness of weld joints	Major	Visual & DP Test	100%	ASTM E 165 / As per std. Practice	IR	√	P	V	V	
3.0	Final Inspection											
3.1	After welding of bellows with socket & Flange with socket full assembly	Surface defect, Dimension	Major	Visual, Measurement	100%	Approved Drawing	IR	√	P	W	W	

3.2	Deflection Test (Routine Test)	Deflection Test	Critical	Axial & Lateral Movement Measurement	100%	Approved Drawing	IR	√	P	W	W	
3.3	Pneumatic Test (Routine Test)	Leakage Test at Specified Test Pressure with air and soap solution	Major	Pneumatic Leakage Test	100%	Approved Drawing/ Specification	IR	√	P	W	W	
3.4	Painting on all MS Paints	Surface cleaning and painting	Minor	Visual Checking	100%	As per approved paint specification	IR	√	P	V	W	
3.5	QA Documentation	Visual	Major	Review	100%	Approved Drawing & QAP	IR	√	-P	V	V	
3.10.7 FLUIDISING BLOWER												
1.0	Raw Material Inspection											
1.1	Casing, Rotary Lobi (Rotor), end cover/end plate/side plate	Physical & Chemical	Major	Physical & Chemical	1/Heat	IS:210/ Approved Drg/ DS	MTC	√	P	V	V	
1.2	Shaft	Physical & Chemical	Major	Physical & Chemical	1/Heat	approved Drg/Data sheet	MTC	√	P	V	V	
		NDT	Major	UT	100%	ASTM – A 388	IR					For Dia > 40MM, NOTE-1
1.3	Gears	Chemical	Major	Chemical	1/Heat	approved Drg/Data sheet	MTC	√	P	V	V	
		Internal Defects	Major	UT	100%	ASTM – A 388	IR	√	P	V	V	For Dia > 40MM, , NOTE-1
		Hardness 54 to 60 HRC	Major	Mechanical	100%	Approved Drg/DS	Test Report	√	P	V	V	
2.0	In Process Inspection											
2.1	Impeller (Lobe with shaft assembly) after machining	Surface Defects in machining area	Major	DPT	100%	ASTM – 165 / No Linear Indication	IR	√	P	V	V	

		Dynamic Balancing	Major	Balancing	100%	ISO:1940 GR 6.3	IR	√	P	V	V	
2.2	Casing and end cover assembly	Pressure Test	Major	Hydro Test	100%	1.5 Times of Design pressure or 2 times of working pressure whichever is higher for 30 minutes / No leakage	IR	√	P	V	V	
3.0	Final Inspection											
3.1	Complete Blower Assembly	Overall Dimensions, completeness	Major	Visual & Measure	100%	Approved DRG	IR	√	P	W	W	
3.2		Performance Test with Shop/JOB motor	Critical	Capacity, Head, speed Power, efficiency	100%	BS-1571 Part II/Approved Drg/Data Sheet	IR	√	P	W	W	
3.3		Noise & Vibration	Critical	Measurement	100%	Approved Drg/Data Sheet	IR	√	P	V	W	
3.4		Brg Temp Rise	Critical	Measure	100%	Approved Drg/Data Sheet/Mfrs std	IR	√	P	V	V	
3.5		Painting	Minor	Visual & Measurement	100%	As per Approved Painting Specification	IR	√	P	V	V	
3.6	QA Documentation	Visual	Major	Review	100%	Approved QAP	Report	√	P	V	V	
NOTE-1: UT shall be carried out by using suitable frequency PROBE. Using this probe, the Back Wall Echo shall be set at 100% of Full Scale Height (FSH) in sound area. At this sensitivity level, any defect echo exceeding 20% of FSH is not acceptable. Also loss of back wall echo more than 20% at any location is not acceptable												
3.10.8 FLY ASH FEED VALVE												
1.0	Raw Material Inspection											

1.1	Body Intake, Tee Intake	Mechanical Properties	Major	Hardness	1/Heat	Approved Drg./TDS	MTC	√	P	V	V	
		Chemical		Chem. Compo.	1/Heat		MTC	√	P	V	V	
		Dimensions		Measurem ent	100%		IR	√	P	V	V	
		Surface Defects		Visual	100%		IR	√	P	V	V	
1.2	Disc & Seat	Mechanical Properties	Major	Hardness	1/Heat	Approved Drg./TDS	MTC	√	P	V	V	
		Chemical		Chem. Compo.	1/Heat		MTC	√	P	V	V	
		Dimensions		Measurem ent	100%		IR	√	P	V	V	
		Surface Defects		Visual	100%		IR	√	P	V	V	
1.3	Shafts	Chemical & Mech. Properties	Major	Chem. Analysis & Mech. Prop	1/Batch/Lot	Approved Drg./T DS	MTC	√	P	V	V	
		Internal soundness	Major	UT (DIA ≥40 MM)	100%	ASTM A 388 / Note-1	UT Report	√	P	V	V	
1.4	Hopper Inlet	Physical Properties	Major	UTs & Hardness	1/Heat	Approved Drg./TDS	MTC	√	P	V	V	

		Surface Defects		Visual	100%		IR	√	P	V	V	
		Dimensions		Measurement	100%		IR	√	P	V	V	
1.5	Spring (Spring Steel)	Chemical Properties	Major	Chem. Analysis stiffness measurement	1/Batch	Approved Drg./TDS	MTC	√	P	V	V	
		Physical Properties			25%		IR	√	P	V	V	
		Dimensions			100%		IR	√	P	V	V	
1.6	Diffuser Mounting plenum	Review Document & Operation	Major	Verification	100%	Approved Drg./TDS	IR & TC	√	P	V	V	
								√	P	V	V	
								√	P	V	V	
								√	P	V	V	
1.7	Porus Tile	Review Document & Operation	Major	Verification	100%	Approved Drg./TDS	IR & TC	√	P	V	V	
1.8	Pneumatic Cylinder	Review Document & operation	Major	Verification	100%	Approved Drg./TDS	IR & TC	√	P	V	V	
2.0	In Process Inspection											
2.1	Machining of components	Dimensions Surface Defects	Major	Measurement Visual	100%	Approved/Mfg. Drg	IR IR	√	P	V	V	
								√	P	V	V	

2.2	Machining of valve seat & Disc.	Machining Grinding Lapping	Minor	Measurement Blue Matching of Disc & Seat	100%	Mfr. Drg / Full circle matching (at least 80%of area)	IR IR IR	√	P	V	V	
								√	P	V	V	
								√	P	V	V	
2.3	Pressure testing of Hopper Inlet & Tee Intake	Leak Tightness	Major	Hydro Test	100%	Approved Drg./TDS	IR	√	P	V	V	
3.0	Final Inspection											
3.1	Complete Fly ash Feed Valve Assembly along with Air Cylinder	Dimensions & Completeness Operational Test(No Load) 10 cycles Seat vacuum Test	Major Major Major	Measurem ent Visual By Comp. Air -Do-	100% 100% 100%	Approved Drg./TDS Smooth operation	IR IR IR	√	P	W	W	
3.2	Painting	Appearance & Cleanliness	Major	Visual	100%	Approved painting schedule	IR	√	P	V	V	
3.3	QA Documentation	Visual	Major	Review	100%	Approved QAP	Report	√	P	V	V	
3.10.9 AIR WASHER												
1.0	Raw Material Inspection											
1.1	Body Steel plates, structural steel members	Chemical Composition	Major	Chemical	100%	Approved DRG/DATA SHEET	MTC/IR	√	P	V	V	UT of plate thickness more than 25mm
		Mechanical Test	Major	Mechanical	100%	Approved DRG/ DATA SHEET	IR	√	P	V	V	

2	In Process Inspection											
2.1	Body a) Marking, Cutting, edge preparation, Cleaning, tacking, straightness	Dimensional conformity	Major	Measurement/Visual	100%	AS per shop Drg	IR	√	P	V	V	
	Weld joint & Welding	Welder Qualification and procedure welding	Major	Mechanical	100%	ASME Sec IX	PQR/WPS Certificate	√	P	V	V	
		Weld Quality	Major	DPT	100%	ASTME-165	IR	√	P	V	V	
2.2	Pipe connection and fittings	Dimensional conformity Alignment	Major	Measurement/Visual	100%	AS per shop Drg	IR	√	P	V	V	
2.3	Structural Welding	Weld Quality	Major	DPT on root run	100%	ASTME-165	IR	√	P	V	V	
				DPT on final & fillet weld	100%	ASTME-165	IR	√	P	V	V	
3	Final Inspection											
3.1	Check for critical dimensions, completeness	Dimensional conformity	Major	Measurement/Visual	100%	As per Approved DRG	IR	√	P	W	W	
3.2	Leak Test	Water Fill Test for 1 hour duration	Major	Fill Test	100%	As per Approved DRG	IR	√	P	W	W	
3.3	Painting	Paint shade, DFT	Major	Visual & Measurement	100%	Approved painting Schedule	IR	√	P	V	V	

3.4	QA Documentation	Visual	Major	Review	100%	Approved QAP-	Report	√	P	V	V	
3.10.10 ASH SLURRY PUMPS												
1.0	Raw Material Inspection											
1.1	Casing, impeller, Wear Ring/Wear Plate	Visual, Physical & Chemical	Major	Visual Measurement	1/Heat	App. Drg/ Data sheet	MTC	√	P	V	V	
1.2	Shaft	Physical, Chemical	Major	Visual Measurement	1/Lot	App. Drg/ Data sheet	MTC	√	P	V	V	
1.3		Internal Soundness	Major	UT(DIA ≥40 MM)	100%	ASTM A 388 / Note-1	UT Report	√	P	V	V	
2	In Process Inspection											
2.1	Machining of shaft, Bearing Housing Sleeve, Impeller Casing etc.,	Compliance Report	Major	Visual Measurement	100%	Mfr. Drawing	Inspection Report	√	P	V	V	
2.2	Casing Hydrostatic Test	Hydrostatic	Critical	Hydrostatic	As per Std	App. Drg/ Data sheet/ / As per IS 5120	Inspection Report	√	P	V	V	
2.3	Impeller Dynamic Balancing	Dynamic Balancing	Critical	Measurement	AS per Std	As per Std / ISO 1940-1973 Gr.6.3	Inspection Report	√	P	V	V	
3	Final Inspection											
3.1	Dimensional	Overall Dimensional Conformity	Major	Visual Measurement	One/Type/Size	Approved Drawing/Data Sheet	Inspection Report	√	P	W	W	
3.2	Performance Test with Calibrated test lab motor	Q Vs H, P & Noise, Vibration	Critical	Performance	One/Type/Size	Approved Drawing/Data Sheet/IS 5120/BIS	Inspection Report	√	P	W	W	

3.3	Painting	Visual & DFT	Major	Visual & measure	Random	Approved Painting Schedule	Inspection Report	√	P	V	V	
3.10.11 WETTING HEAD												
1.0	Raw Material Inspection											
1.1	Casting (ejector head)	Chemical Composition	Major	Chemical	1 Sample /heat	Approved DRG/ Specification/DS	MTC/Lab TC	√	P	V	V	
		Mechanical Properties	Major	Mechanical	1 Sample /heat	Approved DRG/ Specification/DS	MTC/Lab TC	√	P	V	V	
		Casting Defects	Major	Visual Checking	100%	Approved DRG/ Specification/DS	IR	√	P	V	V	
		Hydro Test (Water jacket only) for 5 Min.	Major	Hydro Test	100%	Approved DRG/ Specification/DS / No Leakage/Sewage & Drop	IR	√	P	V	V	Hydro Test to be done before drilling of Nozzle hole.
		Hardness	Major	Measurement	100%	Approved DRG/ Specification/DS	IR	√	P	V	V	
1.2	Nozzle Tip	Chemical Composition	Major	Chemical	1 Sample /heat	Approved DRG/ Specification/DS	MTC/Lab TC	√	P	V	V	
		Mechanical Properties	Minor	Mechanical	1 Sample /heat	Approved DRG/ Specification/DS	MTC/Lab TC	√	P	V	V	
		Dimensional conformity	Major	Measurement	100%	Plant Drg.	IR	√	P	V	V	
		Hardness	Major	Measurement	100%	Approved DRG/ Specification/DS	IR	√	P	V	V	
2.0	In Process Inspection											
2.1	Drilling Machining, Visual & Dimensions	Accuracy/ Appearance & conformity to the	Major	Visual & Measurement	100%	Approval/Details Drg.	IR	√	P	V	V	

		drawing shape, size, orientation.										
		All machined area DPT to be done before & after machining	Major	DP Test	100%	ASME SEC V / ASME SEC VIII Appendix & (cl 8-4)	IR	√	P	V	V	
3.0	Final Inspection											
3.1	Assembly	Dimension	Major	Measurement	100%	Approved DRG/ Specification/DS	IR	√	P	W	W	
		Completeness	Major	Visual Checking	100%	Approved DRG/ Specification/DS	IR	√	P	W	W	
3.2	Leakage Test	Water fill Test (2hr)	Major	Water fill Test	100%	Approved DRG/ Specification/DS / No Leakage/ Sewage & Drop	IR	√	P	W	W	
3.3	Painting	Paint shade, DFT	Minor	Measurement/Visual	100%	Approved painting Schedule	IR	√	P	V	V	
3.4	QA Documentation	Visual	Major	Review	100%	Approved QAP	Report	√	P	V	V	
3.11 COAL HANDLING PLANT												
3.11.1 FEED EJECTOR												
1.0	Raw Material Inspection											
1.1	Inlet liner, discharge throat, Tall pipe, Elbow, Plate, Flange	Chemical	Major	Chemical	1 Sample /heat	Approved /Mfg. DRG/ Specification	MTC/Lab TC	√	P	V	V	
		Mechanical Test	Major	Hardness Measurement	1 Sample /heat	Approved /Mfg. Drg/ Specification	MTC/Lab TC	√	P	V	V	
1.2	Ejector head, Nozzle, Nozzle Tip	Chemical	Major	Chemical	1 Sample /heat	Approved /Mfg. DRG	MTC/Lab TC	√	P	V	V	

		Mechanical Test	Major	Mechanical	1 Sample /heat	Approved /Mfg. DRG	MTC/Lab TC	√	P	V	V	
1.3	NDT for Forged Flanges	NDT	Major	UT	1 Sample/heat	Approved /Mfg. DRG/REL STD	MTC/Lab TC	√	P	V	V	
2.0	In Process Inspection											
2.1	Inlet liner, Discharge throat, Tall pipe	Visual & Dimensional conformity	Major	Measurement/Visual	100%	Mfg. Drg/ Specification	IR	-	P	W	V	
2.2	Ejector Head	Visual & Dimensional conformity	Major	Measurement/Visual	100%	Mfg. Drg/ Specification	IR	√	P	W	V	
		Hydro Test (30 min)	Major	Hydro Test	100%	Approved Drg./ Specification/ No Leakage/ Seepage	IR	√	P	V	V	
2.3	Nozzle	Hardness	Major	Measurement	100%	Mfg. Drg/ Specification	IR	-	P	V	V	
3.0	Final Inspection											
3.1	Completeness and final dimensional examination for overall assembly	Fitment Dimension	Major	Measurement/Visual	100%	Approved Drg./ Specification	IR	√	P	W	W	
3.2	Leak Test	Water fill Test (1hr)	Major	Water fill Test	100%	Approved Drg./ Specification No Leakage/ Seepage /	IR	√	P	W	V	
3.3	Painting	Paint shade, DFT	Minor	Measurement/Visual	100%	Approved painting Schedule	IR	√	P	V	V	
3.4	QA Documentation	Visual	Major	Review	100%	Approved QAP	Report	√	P	V	V	
3.11.2 BELT WEIGHER												

1.0	Raw Material Inspection												
1.1	Steel	i) Dimension ii) Chemical analysis iii) Mechanical Properties	Major	Visual Chemical Mechanical	100%	100%	As per ISO/BIS	As per ISO/BIS/ MTC	√	P	V	V	
1.2	Load cell Blank	i) Visual dimension ii) Chemical test iii) Ultrasonic test iv) Hardness Test v) Planting DFT	Major	Visual, Chemical, Measurement	100%		Manufacturer Standard	Manufacturer Standard/ Test report/Lab report	√	P P	V W	V V	Witness of final plating DFT during Final Inspection
1.3	Speed Sensor	Visual & Functional	Major	Visual, Functional	100%		Relevant Standard/ Specification	Relevant Standard/ Specification/ Internal/ External Test Report	√	P	V	V	
1.4	Cable	i) Visual, Dimensional ii) Insulation, high voltage test	Major	Visual, Measurement	100%		Relevant Standard/ Specification	Relevant Standard/ Specification/ Internal/ External Test Report	√	P	V	V	
1.5	Junction Box	i) Visual, Dimensional ii) ingress test	Major	Visual	100%		Relevant Standard/ Specification	Relevant Standard/	√	P	V	V	

							Specification/ Internal/ External Test Report					
1.6	Enclosure	i) Visual, Dimensional ii) Paints & Shade	Major	Visual Dimensional DFT	100%	Relevant Standard/ Specification	Relevant Standard/ Specification/ Internal/ External Test Report	√	P	V	V	
2.0	In Process Inspection											
2.1	Indicated/ Controller assembly and testing	i) Calibration ii) Functional check iii) Surface finish iv) NDT	Major	Visual, Dimensional	100%	Approved Specification	Approved Specification/ Internal/ External Test Report	√	P	V	V	
2.2	Fabricated & machined item	i) Visual, Dimension	Major	Visual, Dimensional, DP	100%	Approved DRG/DS/MFRS DRG	Mfr. Procedure Internal/ External Test Report /	√	P	V	V	
3.0	Final Inspection											
3.1	Load Cell	i) Visual, Dimensional ii) Linearly, Accuracy	Major	Visual, Dimensional, Functional	100%	Design Documents, test procedure, standard operating procedure(SOP)	IPA Procedure Internal/	√	P P P	W V V	W W W	

		iii) Ingress & temp test iv) capacity, make & model						External Test Report		P	V	-	
3.2	Electronic testing under simulated conditions	i) Visual, Dimensional ii) Linearly, Repeatability, Calibration, functional, accuracy iii) Vibration, Ingress protection, temperature, sock test.	Major	Visual, Dimensional, Functional	100% Sampling 100%		Design Documents, test procedure, standard operating procedure(SOP)	Mfr. Procedure Internal/ External Test Report	√	P	W	W	
										P	W	W	
										P	V	W	
3.3	Overall Assembly	i) Visual, Dimensional ii) Paint & Shade	Major	Visual, Dimensional, DFT	100%	10%	Design Documents, standard operating procedure(SOP)	Mfr. Procedure Internal/ External Test Report	√	P	W	W	Witness of final paint, DFT during Final Inspection.
3.4		Painting	Major	Visual & Measurement	100%	100%	Approved Painting Schedule / Data Sheet	IR	√	P	V	V	
3.11.3 FR GRADE CONVEYOR BELT													
1.0	Raw Material Inspection												

1.1	Raw Rubber	A) Volatile Matter B) Ash Content C) Mooney Viscosity D) Dirt Content E) Plasticity F) Retention Index	Major	Physical Chemical Physical Physical Chemical	Random One Sample Per Lot	-	IS:4588 & Is:3660/ Mfr. Std.	Raw Material Analysis Sheet		V	-	-	
1.2	Zinc Oxide	A) Moisture Content B) Ignition Test C) Purity D) Particle Size E) Bulk Density F) Acid Insolubility	Major	Physical Chemical	Random One Sample Per Lot	-	IS:3399/ Mfr. Std.	Raw Material Analysis Sheet		V	-	-	
1.3	Stearic Acid	A) Ash Content B) Melting Point C) Acid Value	Major	Chemical Physical	Random One Sample Per Lot	-	IS:1675 / Mfr. Std.	Raw Material Analysis Sheet		V	-	-	
1.4	Carbon Black	A) Moisture Content B) Ash Content C) Iodine Absorption No. D) BDP Absorption No. E) pH	Major	Physical Chemical	Random One Sample Per Lot	-	IS:7497 & Is:7498/ Mfr. Std.	Raw Material Analysis Sheet	-	V	-	-	
1.5	White Filter	A) Moisture Content B) Ignition Test	Major	Physical Chemical Chemical	Random One	-	IS:1685/ Mfr. Std.	Raw Material	-	V	-	-	

		C) Insolubility in HCl D) Particle Size		Physical	Sample Per Lot			Analysis Sheet					
1.6	Protective Agent	A) Moisture Content B) Ash Content C) Softening Point/Melting Point	Major	Physical Chemical Physical	Random One Sample Per Lot	-	Mfr. Standard	Raw Material Analysis Sheet	-	V	-	-	
1.7	Rubber/Chemicals/Accelerators	A) Moisture Content B) Ash Content C) Meeting Point D) Solubility	Major	Physical Chemical Physical Physical	Random One Sample Per Lot	-	Mfr. Standard	Raw Material Analysis Sheet	-	V	-	-	
1.8	Sulphur	A) Moisture Content B) Ash Content C) Purity	Major	Physical Chemical Chemical	Random One Sample Per Lot	-	IS:8851/Mfr. Std.	Raw Material Analysis Sheet	-	V	-	-	
1.9	Plasticizer	A) Relative Density B) Aniline Point C) Viscosity	Major	Physical	Random One Sample Per Lot	-	Mfr. Standard	Raw Material Analysis Sheet	-	V	-	-	
1.10	Textile Fabric	A) Thickness B) Width C) Weight (GSM)	Major	Physical	Random One Sample Per Roll	Random One Sample Per Roll	Mfr. Standard	Textile Laboratory Testing Report	√	V	V	V	
1.11	Dipped Textile Fabric	A) Breaking Strength (Warp & Weft) B) Elongation At Break (Warp & Weft)	Major	Physical	Random One Sample Per Batch	Random One Sample Per Batch	Mfr. Standard	Textile Laboratory	√	V	V	V	

		C)Elongation T Ref. Load (Warp & Weft) D) Rubber Adhesio Cover To Ply And Ply To Ply E) Shrinkage (Warp & Weft) F) % Cramp (Warp & Weft)						Testing Report					
1.12	Rubber Compound	A) Rheometric Analysis B) Specific Gravity C)Hardness D)Tensile Strength E) Elongation At Break F) Adhesion	Major	Physical	Random One Sample Per Batch	Random One Sample Per Batch	Mfr. Standard	√Lab Report	√	P	V	V	
2.0	In Process Inspection												
2.1	Rubber Coating Of Fabrics	A)Ply Thickness B)Ply Width C)Ply Length D)Fabric Roll No. And Type E)Compound Code And Batch No.	Major	Physical	Each Ply	Each Ply	Mfr. Standard	Production Log Sheet/Inter nal Record	√	P	V	V	
2.2	Rubber Cover Sheeting/Calenderin g	A)Thickness B)Width C)Length D)Compound Code And Batch No.	Major	Physical	Each Sheet	Each Sheet	Mfr. Standard	Production Log Sheet/Inter nal Record	√	P	V	V	

2.3	Belt Building	A)Thickness B)Width C)Length D)Cover And Ply Position	Major	Physical	Each Belt	Each Belt	Mfr. Standard	Production Log Sheet/Internal Record√	√	P	V	V	
2.4	Moulding (Curing)	A)Curing Temperature B)Curing Time C) Hydraulic Pressure C) Cured Belt Width, Length And Thickness	Major	Physical	Each Belt	Each Belt	Mfr. Standard	Production Log Sheet/Internal Record	√	P	V	V	
2.5	Dressing & Sizing	A)Finish B)Edge (Mould Cut)	Major	Visual	100%	100%	Approved Drg/Ds/ Tech. Specs	Production Log sheet/Internal Record		P	V	-	
2.6	Cured Belt Inspection	Mapping Of Surface Defects Their Type And Repairs	Major	Visual	Each Belt	Each Belt	IS:1891/ Specification	Production Log sheet/Internal Record	√	P	V	V	Refer Not-01
3.0	Final Inspection												
3.1	Dimensional and Visual	A) Visual Exam For Surface Finish B) Mapping Of Surface Defects Their Type And Repairs C)Length D)Edge (Mould/Cut)	Critical	Physical	100% 100% 100% 100% Random	Sample As Per Is:1891-1	Approved Drg/DS/ Tech. Specs.	IR	√	P	W	W	Refer Note-01 & 02 & 05

		E) Width F) Shore Hardness G) Thickness Of Full Belt H) No. Of Plies I) Top & Bottom Cover Thickness			Location In Each Roll Random Spots In Each Roll Each Roll Each Roll								
3.2	Tensile Strength Of Full Belt Thickness	A) Breaking Strength (Warp & Weft) B) Elongation At Ref. Load (Warp) C) Elongation At Break (Warp)	Critical	Physical	Each Roll	Sample As Per Is:1891-I	IS:1891 Part-I / Specification / Approved Data Sheet	Laboratory Test Report	√	P	W	W	
3.3	Adhesion	A) Top Cover To Ply B) Ply To Ply C) Ply To Bottom Cover	Critical	Physical	Each Roll	Sample As Per Is:1891-I	IS:1891 Part-I / Specification / Approved Data Sheet	Laboratory Test Report	√	P	W	W	
3.4	Troughability	Troughability	Critical	Physical	Each Roll	Sample As Per Is:1891-I	IS:1891 Part-I	Laboratory Test Report	√	P	W	W	Refer Note -02
3.5	Cover Rubber Properties	A) Cover Tensile Strength (Before & After Ageing) B) Elongation At Break (Before & After Ageing), thickness	Critical	Physical	Each Roll	Sample As Per Is:1891-I	IS:1891 (Part-I & V) / Specification / Approved Data Sheet	Laboratory Test Report	√	P	W	W	

		C)Angular Tear Strength	Critical	Physical	Each Roll	Sample As Per Is:1891-I	ASTM D 624 Type-C / Specification/ Approved Data Sheet		√	P	W	W	
		D)Abrasion Loss	Critical	Physical	Each Roll	Sample As Per Is:1891-I	Din:53516 / Specification/ Approved Data Sheet	Laboratory Test Report	√	P	W	W	
3.6	Fire Resistivity Test	A) Drum Friction Test	Critical	Physical	Each Roll	Sample As Per Is:1891-I	CAN/CSA/M422-M87 Type C	Laboratory Test Report	√	P	W	W	
		B) Flame Test	Critical	Physical	Each Roll	Sample As Per Is:1891-I	ISO 340	Laboratory Test Report	√	P	W	W	
		C)Electrical Surface Resistance Test	Critical	Physical	Each Roll	Sample As Per Is:1891-I	CAN/CSA /M422-M87 Type C	Laboratory Test Report	√	P	W	W	
4.0	Identification & Marking	Belt number, manufacture's name, logo, belt ratting, grade, fabric type, and year of manufacture etc. shall be embossed by the manufacture on every valcanized length of belt or as per PO condition. Belts nos. Offered for inspection & belt nos. From which samples are drawn for testing shall be indicated in the CHP. Accepted belt nos. Shall be identified with signature and date using permanent marker as well as hard punch in white paint. The identification marks shall be preserved by the manufacturer.											
5.0	Dispatch	Packing And Marking	Major	Visual	Each Roll	Each Roll	Mfr. Standard Specs.	Finished Product Department Record		P	V	-	

NOTE-01: Defectogram shall be prepared by the belt manufacturer and shall be submitted for verification by NTPC representative/Inspection Engineer during Final Inspection. Following repair norms shall be applicable –

i) Patch repair: Localized rectification of surface blemishes/defect in cured belt by using rubber compound similar to the mother compound up to top carcass may be done followed by local vulcanization.

ii) Buffing/dough: Entrapment of foreign matters may be buffed. Depth of buffing should not exceed the difference in thickness of the rubber (as measured in test sample for the purpose of acceptance of cover rubber thickness) and the specified minimum cover thickness. Where the indentation depth is more, the same may be filled with rubber compound followed by vulcanization locally. The repairs of size up to and including (25X25 mm) (625 sq mm) shall not be considered as patch repair.

iii) a) Maximum number of repairs as per (i) as indicated above shall be limited to 5 per 100 sq . meters of belt area (rounded up to higher unit).

b) Total number of repairs as per (i) and (ii) indicated above shall not exceed more than 10 per 100 sq. meter of belt.

c) In case of patch repair as indicated in i above, the maximum size/area of each repair shall be limited to 1/5 W x1/5 W, with one-dimension Max, 1/5W, where ‘W’ is width of the belt.

Note-02: Sample shall be taken randomly from anywhere of belt roll/length offered for inspection. Belt to be supplied in 02 pieces and the lengths shall be indicated on packing drum.

Note-03: Latest edition of all the standards mentioned in the quality plan is to be used.

Note- 04 : Two Copies of MTC (material test certificate) and IR (Inspection report) to be submitted in hard form or soft form.

Note-05: Visual Inspection (including random check of belt width, hardness and total belt thickness) on both sides shall be done on the complete belts from which sample shall be taken as per IS-1891 part-I.

3.11.4 FABRICATED GEAR CASE:

1.0	Raw Material inspection											
A1		Chemical Composition	Critical	Check Test	1 per heat	1 Per Heat	Relevant Material Specification as per Approved Drawing / DS.	TC	√	P	V	V
A2		Mechanical Properties	Critical	Tensile Test	1 per heat	1 Per Heat	Relevant Material Specification as per Approved Drawing / DS.	TC	√	P	V	V
A3		Internal Defect	Major	UT	100%	100%	ASTM-A-435	TC	√	P	V	V
II	Pinions, Wheels & Shafts											
(A)	Raw Material (as per drawing)											
A1	(Forgings & Round Bars of Carbon Steel / Alloy Steel)	Surface Condition	Major	Visual	100%	100%	Drawing/Relevant Standard	I.R./TC	√	P	V	V
A2		Internal Defect	Critical	Ultrasonic	100%	100%	Relevant Standard	I.R./TC	√	P	V	V

A3		Chemical Composition	Critical	Check testing	1/heat	1/heat	Relevant Material Specification as per Approved Drawing / DS.	TC	√	P	V	V	
A4		Mechanical Properties	Critical	Tensile test	1/heat	1/heat	Relevant Material Specification as per Approved Drawing / DS.	TC	√	P	V	V	
A5		Microstructure	Critical	Visual under microscope	1/heat	1/heat	Relevant Material Specification as per Approved Drawing / DS.	TC	√	P	V	V	
	In Process Inspection												
III.A													
1)	Gear Casing (Fabricated)	WPS/PQR/WPQ	Major	Verification	100%	100%	ASME Sec-IX	WPS/PQR/WPQ	√	P	V	V	Refer Not-05
2)		Stress Relieving	Major	Visual	100%	100%	Mfr. S.R. Procedure	H.T. Chart	√	P	V	V	
3)		Surface Defect (Welding)	Major	DP Test on Weld	100%	100%	ASTM E 165 / No Linear Indication	IR	√	P	V	V	
III.B													
1)	Pinion and Gear (After Machining & Case Hardening)	Case Depth	Major	Measurement	1/HT Batch	1/HT Batch	Mfr. Standard	IR	√	P	V	V	
2)		Hardness	Major	Measurement	1/HT Batch	1/HT Batch	NTPC Approved Drawing / DS	IR	√	P	V	V	
3)		Surface Defect	Major	DP Test	100%	100%	ASTM E 165 / No linear Indication	IR	√	P	V	V	
4)		Dimension	Major	Measurement	100%	-	Mfr. Drawing	IR		P	-	-	
III.C													
1)	Input and Out Put Shaft after Machining	Dimension	Major	Measurement	100%	-	Mfr. Standard	IR	√	P	--	-	

2)		Surface Defect	Major	DP Test	100%	100%	ASTM E 165 / No linear Indication	IR	√	P	V	V	
3)		Hardness	Major	Measureme nt	1/HT Batch	1/HT Batch	NTPC Approved Drawing / DS	IR	√	P	V	V	
IV	Final Inspection of assembled gear box												
1)		Arrangement Verification	Major	Visual	100%	10%	Approved Drawing.	IR	√	P	W	W	
2)		Overall Dimensions & Completeness	Major	Measureme nt	100%	10%	Approved Drawing.	IR	√	P	W	W	
3)		Backlash	Critical	Measureme nt	100%	10%	Mfr. Standard / Approved Drawing	IR/TC	√	P	W	W	
4)		Contact Pattern	Critical	Blue matching	100%	10%	Mfr. Standard / Approved Drawing	IR/TC	√	P	W	W	
II	No-Load Running Trial Test												
1)		Reduction Ratio	Critical	Visual/ Measureme nt	100%	10%	Approved Drawing / Datasheet	IR	√	P	W V	W	
2)		Oil Leakage	Critical	Visual	100%	10%	Relevant Standard / No Leakage	IR	√	P	W V	W	
3)		Temp rise of Oil	Critical	Measureme nt	100%	10%	Relevant / Standard	IR	√	P	W V	W	
4)		Noise Level	Critical	Measureme nt	100%	10%	Max 85 DBA at 1 Mtr in horizontal direction and 1.5 M in elevation.	IR	√	P	W	W	

5)		Vibration	Critical	Measurement	100%	10%	ISO 10816 / 5mm/sec RMS MAX (Group B based on the class)	IR	√	P	W	W	
II)	Painting, packing & identification												
1)		Uniformity of Coating & Finishing, DFT Measurement	Major	Visual	100%	100%	As per Approved Data Sheet.	IR	√	P	V	V	
3.11.5 IN LINE MAGNETIC SEPARATOR													
1.0	Raw Material inspection												
1.1	Top & Side plates	Visual/ dimensional/ Chemical	Minor	Visual/ physical measurement	100%	100%	100%	G.A. Drg/ Approved Drawing	MTC	√	P	V	V
1.2	Nonmagnetic Sheet (S.S)	Visual/ Dimensional /Chemical	Minor	Visual/ physical/ chemical	100%	100%	100%	G.A. Drg/ Approved Drawing	MTC	√	P	V	V
1.3	Coil material – Anodized Annealed Aluminum strips	MTC	Critical	MTC	100%	100%	100%	Relevant Standard/ Specification. Approved Drawing	MTC	√	P	V	V
1.4	Switch gear/Geared Motor, Belting Pulley, Rectifier Transformer	Visual/Electrical/Type/ Rating Verify/ Continuity Check	Major	Visual	100%	100%	100%	G.A. Drg/ App. B.O.M	IR	√	P	V	V
1.5	In Process Fabrication Marking, Cutting, Drill	Workmanship, Dimensions & DP Test of Welding leakage test on tank	Major	Visual/ physical measurement	100%	100%	100%	G.A. Drg/ Approved Drawing	IR	√	P	V	V
1.6	Silicon Oil	Electrical & Chemical	Major	Visual, Measure	1/Lot	1/Lot	1/Lot	G.A. Drg/Relevant Std./ Approved Drawing	IR	√	P	V	V
2.0	In process inspection												
2.1	Magnet housing fabricator	Visual/ dimensional	Major	Visual/ physical measurement	100%	100%	100%	App.G.A. Drg	ITC	√	P	V	V
													Degree of protection (DOP) certificate to verification shall be produced.

2.2	Control Panel manufacturing	BOM/make ferruling/wire size/terminal/ marking interlock/ functional Check/H.V. Test /Insulation Resistance Test/Painting shade thickness	Major	Visual/ dimensional / electrical/ Megger test/HV test	100%	100%	100%	App.G.A. Drg	ITC	√	P	W	W	
3.0 Final Inspection														
3.1	In Line Magnetic Separator (Electric Magnetic Type)	Visual/ Dimensional/ operational check with control panel	Critical	Visual physical measurement/HV test & Insulation Resistance test current & voltage measurement/Temp rise test, load test in hot & cold condition, force index, calculation at hot steady state, Noise level.	100%	100%	100%	App.G.A. Drg	ITC	√	P	W	W	
3.2	Random lifting of tramp iron pieces as per App. Drg & Lifting force	performance	Major	Functional	100%	100%	100%	App.G.A. Drg	ITC	√	P	W	W	
3.3	Gauss Valve	Electrical	Major	Gauss valve	100%	100%	100%	App.G.A. Drg	ITC	√	P	W	W	

3.4	Painting	Painting shade/DFT	Major	Visual/ Thickness of paint	100%	100%	100%	As per Approved Painting Specs.	ITC	√	P	V	V	
Note: 1. Statutory requirement will be complied. 2. Instruments used for test shall have valid calibration certificate with tractability to national level. 3. Bought out items shall be from approved vendor by Customer. 4. All inspection/verification/NDT report along with material certificate shall be reviewed at the time of witness point. 5. W at customer column shall be considered as hold point/Witness.														
3.11.10 TOOTHED RING HAMMERS														
1.0	Raw Material Inspection													
1.1	Casting Hammers and Test bar Elements of C, Si, Mn, P, S, Cr	Chemical Properties	Major	Chemical analysis	One Sample/Heat		As per Approved Drawing & IS 276 Grade 3/ Clause 8, Table-1	MTC	√	P	V	V		
2.0	In Process Inspection													
2.1	Heat treatment	Mechanical	Major	Heat Treatment	100%		As Per IS 276 Grade 3	HT Chart	√	P	V	V		
2.2	Brinells Hardness Test	Mechanical	Major	Harness check Physical	Min 10% Heat & HT Batch		As Per IS 276 Grade 3	TC	√	P	V	V	Max Hardness HB 229 BHN Class II	
2.3	Bend Test	Mechanical	Major	Mechanica l Test Physical	IS 6907		As Per IS 276 Grade 3/ Clause 14.3	TC	√	P	V	V		
2.4	Liquid penetrant Test	Mechanical	Major	NDE	Min 10% of offer Qty		IS 3658/ IS 11732	TC	√	P	-	V		
3.0	Final Inspection													
3.1	Finished Hammer for Coal Crushers	Surface Defects, Marking and Dimensional (OD, ID, Height, Width,	Major	Measurem ent	100% / Random 5%		As per Approved Drawing	IR	√	P	W	W	See Note 1&2	

		Thickness, Edge Preparation)										
3.2		Chemical Composition	Major	Chemical Analysis	One Sample/Heat	Approved Drawing/DS / IS 276 Grade 3, Clause 8, Table-1	IR	√	P	W	W	Testing with PMI
3.3		Bend Test on Test bar	Critical	Bend Test	One Sample/Heat	Approved drawing/datasheet	IR	√	P	W	W	Max Hardness HB229BHR: and see Note 2 & 3
3.4		Hardness	Major	Mechanical	Random 10% of offer Qty/ 1 per Heat per Batch	Approved drawing/datasheet	IR	√	P	W	W	Approximate weightment of each hammer 39.5 kg Tolerance +/-0.50kg
3.5		Weight	Critical	Measurement	100% offer Qty / Random 5% offer Qty	As per Approved Drawing/Data Sheet	IR	√	P	W	W	
3.6		Drop Test	Major	Mechanical	Random 1% offer Qty	See Note 4	IR		P	W	W	
3.7	Finished Hammer for Coal Crushers	Painting	Major	Visual	100%	IS 276 Clause 18	IR	√	P	W	W	Painting with Red oxide and weightment marking on each hammer with yellow paint
Note:- 1. Casting should be properly fitted and dressed before inspection. Risers/Runners shall not be removed by gas cutting. Surface cracks and pitting visible to naked eyes, are not acceptable. 2. Dimensions and weights shall be as per approved drawing/data sheet/technical specifications with a max. Variation of +/-1mm in dimensions and +/-0.5 Kgs in Weights. 3. Weight of hammer casting should be punched on each casting and embossing of plant name, Heat batch number and firm name. 4. Drop test: The hammer casting shall be keep with its axis in horizontal position, over a steel plate. A dead weight of not less than 100Kgs shall be dropped on the casting, from not less than 03 Mtrs height. Any crack or chip off is not acceptable. Further any significant variation in dimensions after drop test is not acceptable. In case of failure of the first sample, two more samples, for each failed sample, from the same lot, shall be subjected to the drop test. In case of failure in retest, the entire lot shall be rejected and hammer castings used in drop test are not to be supplied and are to be destroyed in presence of purchaser. 5. Hardness shall be measured by digital hardness tester. 6. Weld repair on the hammer castings is not acceptable. 7. Manufacturer shall ensure proper surface protection on the castings, prior to dispatch.												

3.11.11 VIBRATING FEEDERS														
1.0	Raw Material Inspection													
1.1	Plates for Feeder/Screen body	Chemical & Mechanical Properties	Major	MTC	1/Heat	1/Heat	1/Heat	Approved Drawings/ Relevant STD	Mill TC	√	P	V	V	
		NDE (UT) for plate material soundness for 40mm thickness and above.	Major	Testing	100%	100%	100%	ASTM A 435	NDE Report	√	P	V	V	
1.2	Stiffness Angles/Flat/Channel /Rectangular Section/Pipe Beams/Exciter Shaft	Chemical & Mechanical Properties	Major	MTC	1/Heat	1/Heat	1/Heat	Approved Drawings/Relevant STD	Mill TC	√	P	V	V	
1.3	Liner Plates	Chemical & Mechanical Properties	Major	MTC	100%	100%	100%	Approved Drawing/ Relevant Standard	MTC	√	P	V	V	
2.0	Bought Out Items													
2.1	Bearings	Visual, Identification & Review of warranty certificate	Major	Warranty Cert.	100%	100%	100%	Tech. Specification	Warranty Certificate	√	P	V	V	Bearings shall be procured from original manufacturer
2.2.	Coil spring	Visual, Dimensions & Review of MTC	Major	MTC	100%	100%	100%	Approved Drawings/ Tech. Specification	MTC	√	P	V	V	
2.3	Fasteners of vibrating Parts	Chemical & Mechanical Properties, Visual and Grade	Major	Visual, Measurement & Verification	Random	Random	Random	Tech. Specification/ Mfg Drgs	MTC	√	P	V	V	

3.0	WELDING CONTROL													
3.1	Procedure & performance Qualification	Review of established WPS, PQR & WPQ	Major	Verification	100%	100%	100%	ASME Sec-IX	WPS, PQR & WPQ	√	P	V	V	
3.2	Welding electrodes	Baking & Holding of electrodes for low hydrogen electrodes (if applicable)	Major	Visual & Measurement	100%	100%	100%	Mfg .Recommendation /Standard Procedure	Register		P	V	V	
4.0	In Process Inspection													
4.1	Material Preparation	Edge preparation (Bevel angle and Root face)	Major	Visual & Measurement	100%	100%	100%	As per /Drawings	I/R	√	P	V	V	Verification at Final Inspection
	Weld Joints	Fitment and alignment (Root gap & bevel angle)	Major	Visual & Measurement	100%	100%	100%	As per /Drawings	I/R	√	P	V	V	Verification at Final Inspection
4.2	Weld soundness	NDE for filler welds	Major	DPT	10%	10%	10%	ASME Sec-VIII Div-1 app.8/ASTM SE 165	NDE Report	√	P	V	V	
5.0	Final Inspection													
5.1	Sub assembly of Feeder/Screen other accessories	Completeness & Dimensions	Major	Visual & Measurement	100%	100%	100%	Approved Drawings/ Tech. Specifications	IR	√	P	W	W	
5.2	UB Motor Exciter Assembly	Completeness & Dimensions of final assembly, Motor Data (Resistance Value, HV Test & Reduce voltage)	Major	RTC	100%	100%	100%	Manufacturing Drawing/ Tech. Specification	RTC	√	P	W	W	

5.3	Final Assembly	Completeness & Dimensions of final assembly, no load running test speed (RPM), amplitude (stroke), Temperature rise & Noise Level	Major	Visual & Measurement	100%	100%	100%	Mfg. Drawing & Standard Procedures	IR	√	P	W	W	No Load running test shall be conducted for 4-hours at available test bed in works. Amplitude of vibration, Bearing temperature rise and Noise Level is measured. Vibration will be recorded in vibograph, Noise Level shall be 85 dBA at 1.0m in horizontal and 1.5m in vertical.
5.4	Surface preparation & Painting	Roughness	Major	Visual & Measurement	100%	Random	Random	Approved Drawing/ Painting Specification	IR	√	P	V	V	
		Visual, Shade, DFT	Major	Visual & Measurement	100%	Random	Random	Approved Drawing/ Painting Specification		√	P	W	W	
5.5	Final Documentation	Complete Inspection documents	Major	Verification	100%	100%	100%	AS per approved MQP	IR	√	P	V	V	

Note: No load test procedure shall be tied up during QAP Approval.

3.11.12 VIBRATION ISOLATION SYSTEM

1.0 Raw Material Inspection

1.1	For Boxes	Chemical / Mechanical	Minor	Verify	One / Heat	IS 2062/Approved Diagram/ Approved Specification	MTC	√	V	V	V	
1.2	For Springs	) Chemical	Major	Verify	One / Lot	IS 2062/Approved Diagram/ Approved Specification	MTC	√	V	V	V	

		) Grain Size	Major	Verify	One / Lot	ASTM E 112/Approved Specification	MTC	√	V	V	V	
		) Micro Structure	Major	Verify	One / Lot	Tempered Martensite / Approved Specification	MTC	√	V	V	V	
1.3	Springs at Manufacturer's end	Hardness	Major	Test	One / Lot	IS 1500 / 415-460 BHN	MTC	√	V	V	V	
1.4	Springs at Manufacturer's end	Decarburization	Major	Test	One / Lot	IS 6396 / Max. depth 0.5% bar dia – partial	MTC	√	V	V	V	
1.5	Springs at Manufacturer's end	NDE after compression	Major	MPI	100%	IS 3703 / No cracks	MTC	√	V	V	V	
1.6	Adhesive pads (jute) 4mm thick, Steel shims	Dimensions (mm)	Minor	Measurement/ Visual	5%	Approved Drawing	Internal Record	√	P	V	V	
1.7	Studs & Nuts	Chemical/ Mechanical	Minor	Verify	One / Lot	Approved Specification	Lab Test Report	√	V	V	V	
		Dimensions	Minor	Measurement	5%	Approved Specification	Internal Record	√	P	V	V	
2.0	In Process Inspection											
2.1	Welding	Visual/Surface exam (mm)	Major	Visual/ Measurement	10% on welds	Approved Drawing/Data Sheet	Certificate	√	P	V	V	
		NDE	Major	DPT	10% Random	ASTM/E 165	Certificate	√	P	V	V	
2.2	Boxes	Dimensions (mm)	Major	Measurement	10%	Drawing/Data sheet	Internal Record	√	P	V	V	
3.0	Final Inspection											
3.1	Shot Blasting	Picture	Major	Compare	10%	Approved Drawing/Data Sheet	IR	√	P	V	V	
3.2	Painting	Thickness (180 microns)	Major	Measurement/ Visual	10%	Approved Drawing/Data Sheet	IR	√	P	V	W	

3.3	Spring Unit	Dimensions (Except HF&HV)	Minor	Measurement	10%	Approved Drawing/Data Sheet	Internal Record	√	P	W	W	
3.4	Constant of Spring Units (Vertical Kv)	Load Vs Displacement (kN/mm)	Critical	Test	10%	Approved Drawing/Data Sheet	Certificate	√	P	W	W	
3.5	Document Control	Verification of above documents (TC/IR)	Major	Compare	All	Approved MQP	Certificate	√	P	V	W	
3.6	Packing / Marking	visual	Major	Visual/ Compare	100%	Approved Specification	IR	√	P		V	

3.11.13 PULLEY

1.0	Raw Material & Bought out Inspection													
1.1	Shell Plate(MS)	Chemical, Mechanical, Physical	Major	Chemical Mechanical	1/ Heat	1/ Heat	1/ Heat	Approved drawing	Mill TC/Lab TC	√	P	V	V	UT fore plate above 25mm thickness
		Ultrasonic test*		NDT	100%	100%	100%	ASTM A435	UT Report	√	P	V	V	
1.2	SS Plate for Non-magnetic pulley	Chemical, Mechanical	Major	Chemical Mechanical	1/ Heat	1/ Heat	1/ Heat	Approved drawing	Mill TC/Lab TC	√	P	V	V	
1.3	Shaft(forged)	Chemical, Mechanical, Physical, Heat Treatment, Microstructure & Hardness	Major	Chemical Mechanical HT	1/ Heat	1/ Heat	1/ Heat	Approved drawing	Mill TC/Lab TC	√	P	V	V	UT Applicable on above 40mm Dia shaft
		Ultrasonic test*		NDT	100%	100%	100%	ASTM A 388	UT Report	√	P	V	V	
1.4	Hub	Chemical, Mechanical	Major	Chemical Mechanical	1/Heat	1/Heat	1/Heat	Approved drawing	Mill TC/Lab TC	√	P	V	V	UT Applicable on above 40mm Dia hub
		Ultrasonic test*		NDT	100%	100%	100%	ASTM A 388	UT Report	√	P	V	V	

1.5	SS Plate for Non-magnetic Pulley	Chemical, Mechanical	Major	Chemical Mechanical	1/Heat	1/Heat	1/Heat	Approved drawing	Mill TC/Lab TC	√	P	V	V	
1.6	Diaphragm	Chemical, Mechanical, Physical	Major	Chemical Mechanical	1/Heat	1/Heat	1/Heat	Approved drawing	Mill TC/Lab TC	√	P	V	V	UT fore plate above 25mm thickness
		Ultrasonic test*		NDT	100%	100%	100%	ASTM A435	UT Report	√	P	V	V	
1.7	Rib Plate	Chemical, Mechanical	Major	Chemical Mechanical	1/Heat	1/Heat	1/Heat	Approved drawing	TC/GC	√	P	V	V	
1.8	Plummer Block & Bearing	Verification of make & Number	Major	Visual	Random	Random		Approved drawing	TC	√	P	V	V	
2.0	In process Inspection													
2.1	Fitup & Welding of shell plate & Diaphragm Plate	Mismatch/ Dimension /Weld Defects	Major	Visual & Measurement	100%	100%	100%	Approved drawing	IR	√	P	V	V	
2.2	NDT	(*)Pulley Shell L-seam	Major	RT/UT&MPI	100%	10% from each offered lot		ASME SEC – V / ASME Sec VIII DIV.1	NDT Report	√	P	W	V	10% Witness & 100% verification
		Full penetration weld (End Disc to shell)		UT&DPT	100%	10% from each offered lot		ASME SEC – V / ASME Sec VIII DIV.1	NDT Report	√	P	W	V	
		Filler weld (End Disc to Hub)		DPT/MPI	100%	10% from each offered lot		ASME SEC – V / ASME Sec VIII DIV.1	NDT Report	√	P	W	V	
2.3	Machining of Pulley Shaft	Measurement	Major	Dimension & run out	100%	100%	100%	Approved drawing	IR	√	P	W	V	
		DPT on machined Shaft	Major	NDT	100%	100%	100%	ASME SEC – V / ASME Sec VIII DIV.1	DPT Report	√	P	V	V	

3.0	Final Inspection													
3.1	Visual, Dimensional, Out of roundness, Eccentricity & Static balancing before rubber lagging, Tensile, Elongation, Peel Off	Accuracy/appearance & conformity to the approved Drg.	Major	Visual, Measurement	100%	10% from each offered lot		Approved Drg, Spec & IS: 8531	IR	√	P	W	V	
3.2	Rubber Lagging	Visual, Thickness, Shore hardness, Spark Test, Strength, EI & Abrasion loss, adhesion strength, specific gravity	Major	Visual, Measurement	100%	10% from each offered lot		Approved drawing	IR	√	P	W	V	
		Ash content	Major	Test	100%	10%	10%	Approved drawing / 12% Max	IR	√	P	W	V	
3.3	Static balancing after lagging	Accuracy/appearance & conformity to the approved Drg.	Major	Measurement	100%	10% from each offered lot		Approved Drg, Spec & IS: 8531	IR	√	P	W	V	
3.4	Painting	Surface preparation, Painting thickness & Shade	Major	Measurement	100%	Random		AS per painting specification	IR	√	P	V	V	
3.5	QA documentation Marking, Packing	Checking of all QA point	Major	Visual	100%	-	-	Approved Diagram	IR	√	P	V	V	
3.6	Visual, Dimensional, Out of roundness,	Accuracy/appearance & conformity to the approved Drg.	Major	Visual, Measurement	100%	10% from each offered lot		Approved Drg, Spec & IS: 8531	IR	√	P	V	V	

	Eccentricity & Static balancing before rubber lagging.												
3.7	Rubber Lagging	Visual, Thickness, Shore hardness, Spark Test, Strength, EI & Abrasion loss, adhesion strength, specific gravity	Major	Visual, Measurement	100%	10% from each offered lot	Approved drawing	IR	√	P	V	V	
		Ash content	Major	Test	100%	10% 10%	Approved drawing / 12% Max	IR	√	P	V	V	
3.8	Static balancing after lagging	Accuracy/appearance & conformity to the approved Drg.	Major	Measurement	100%	10% from each offered lot	Approved Drg, Spec & IS: 8531	IR	√	P	V	V	
3.9	Painting	Surface preparation, Painting thickness & Shade	Major	Measurement	100%	Random	AS per painting specification	IR	√	P	V	V	
3.10	QA documentation Marking, Packing	Checking of all QA point	Major	Visual	100%	- -	Approved Diagram	IR	√	P	V	V	

Sl. No	Component & operations	Characteristics	Class	Type of check	Quantum Of check		Reference document	Acceptance norms	Format of record		Agency			Remarks
					M	C/N					M	C	N	
1.	2.	3.	4.	5.	6.		7.	8.	9.	D*	** 10			11.

ITEM /EQUIPMENT:

BOILER STRUCTURE:- CEILING GIRDERS, COLUMNS, HORIZONTAL & BUILT-UP BEAMS

SUB-SYSTEM : SG

1.0	Raw Material : The material of construction shall be as per drawing/approved data sheet. Raw Material to be sourced from OWNER agreed subvendors.													
1.1	Plates & Rolled sections Plates of IS 2062 E 250 Gr BR & E350 Gr BO shall be: i. Fully killed and impact tested ii. Plates of thickness lesser than and including 12mm- as -rolled. iii. Plates of thickness beyond 12 mm to and including 40 mm- Normalizing Rolling iv. Plates of thickness greater than 40 mm- Furnace normalized In addition, for plates of t > 40 mm: Vacuum degassing shall be done.	Chemical and Mechanical Properties	Critical	T.C.Verification	100%	100%	As per Material Specification / Drg.	Mfg TC	√	P	V	V*	Raw.matl. correlated to TC on receipt. Further correlation to W.O. number & Material Grade. *Verification of Material TC by OWNER for plates t >40 mm. TC Verification not required for rolled products like angles, channels etc. (* Refer to Note-4)	
		Soundness (UT***)	Critical	T.C.Verification	100%	100%	ASTM A578 Level B-S2	Mfg TC	√	P	V	V	*** For plates of t >40 mm only.	

2.0	Inprocess Controls : Refer QCP:002/ Rev. 04												
2.1	Preparation												
2.1.1	Edge preparation (Shop welds)	NDE	M in or	LPI/MPI	100%*	--	As per Material Specification	R	-	P	V		* Flame cut edges shall
2.1.2	Edge preparation (Site welds)	NDE	M in or	LPI/MPI	100%\$	100% / 10%	/ Drg/ASME/ AWS D1.1 Drg/ASME/, AWS D1.1	R	√	P	V	V	be ground before welding. & Note-2 \$ For t ≥32 mm
2.1.3	Edge preparation for Ceiling Girders (Site welds)	NDE	M aj or	UT	**	**	ASTM A 435 & AWS	R		P	V	V	** for a width of
2.2	Welding	Welding & Procedure Qualification	M aj or	Review of Documents	100%	100%	AWS D 1.1 & SIP:NP:07	PQR/ WPS		P	V	V	100mm from the edge, before edge preparation
		Personnel Qualification	M aj or	Review of Documents	100%	100%	AWS D 1.1 & SIP:NP:07	R		P	V	V	
2.3	Fit up	Dimension	M aj or	Measurement	100%	100%	AWS	R		P	W		
2.4	Pre-heat	Measurement	M aj or	Measurement	100%	100%	WPS	R		P	V		
2.3.1	Butt welds	Refer Note-1											

2.3.1.1	Back gouged weld grooves	Surface check	M in or	LPI	100%	100/10%	Drg/ASME/ AWS	R		P	W	V	
2.3.1.2	Finished butt welds												
i	Thickness ≥ 32mm	Weld quality	Maj	RT/ UT	100%	100%	Drg/ASME/ AWS	R	√	P	W	V*	*Review of films for RT.
ii	Thickness from 25 up to <32mm	Weld quality	Maj	RT MPI	10%# 100%	10% 100% / 10%	Drg/ASME/ AWS	R	√	P	W	V	# 10% of Joints per PGMA per Vendor.
iii	Thickness below 25mm	Weld quality	Maj	MPI/LPI	100%	10%	Drg/ASME/ AWS	R	√	P	W	W	Review of films for RT
2.3.1.3	Production test coupons for Columns & ceiling girders	Weld soundness	Maj	RT & Mech	**	**	Drg/ASME/ AWS	R		P	W	W	
	For Butt Welds **	** Quantum of check: 1. For ‘+’ and ‘H’ columns, one test plate per column (Main columns). 2. For Ceiling girder: One test plate per Ceiling Girder for each Sub contractor/ vendor. 3. For other Main columns: One test plate for each Sub contractor/ vendor.										** Only for Main Columns & Ceiling Girders.	
2.3.2	For Fillet welds												
2.3.2.1	Production Test coupons for “+”and “H” Columns (Main columns) & ceiling girders	Weld fusion	Maj	Macroetch	\$\$	\$\$	AWS D1.1, ASTM E 340/CIP:033	R		P	W	W	
		\$\$ Quantum of check: In case of Main columns, TWO Macros for each “+” and “H” columns out of which the first macro from the first column piece and the second from the last column piece welded – as manufactured.											
2.3.2.2	Groove & Fillet Welds quality	NDE	Maj	LPI / MPI	100	100/10%	Drg/ASME/ AWS	R	√	P	W	W	

2.4	Heat Treatment	Time & Temp	Maj	Review of documents	100%	100%	Drawing/TECH SPEC	R	-	P	V	V	
3.0	Final Inspection												
3.1.1	Dimensions	Straightness, web shift, Web verticality, twist, Squareness	Maj	Visual and Measurement	100%	100%	Drg/ASME/ AWS	R		P	W		
3.2	Individual Pieces(Welded Structure): Columns, Ceiling Girders & Beams & Vertical bracings	Shape,Size & Overall Dimensions Length Orientation ,Weld size and quality	Maj	Visual, Verification& Measurement	100%	100%	Drawing/TECH SPEC	R	√	P	W	W	*For columns with welded gussets, match marking with splice plates to be ensured. ** Ensure the following dimensions of columns gussets and vertical bracings, as per the relevant drawing- a. Hole size& Pitch b. Location of holes
3.2.1	Individual Pieces(Bolted Structure): Columns*, Ceiling Girders, Beams & Vertical Bracings**	Shape,Size & Overall Dimensions Length Orientation ,Weld size and quality	Maj	Visual,Verification& Measurement	100%	100%	Drawing/TECH SPEC	R	√	P	W	W	
3.3	Trial Assy: i. Columns\$ ii. Ceiling girders \$ along with cross members	Dimensions. Alignment, Match marking as per Drg orientation	Maj	Visual & Measurment	100%	100%	Drawing/TECH SPEC	R	√	P	W	W	
							Drawing/TECH SPEC	R	√	P	W	W	\$ At least two consecutive girders/columns to be assembled at a time. # Applicable only for vertical bracings of welded structure
3.4	iii. Vertical bracings of Welded Structures iv. Split type Ceiling Girder- Individual Top & Bottom piece. v. Split type -Ceiling Girders Compartmental Assy - Bottom pieces of two consecutive Girders with cross members. Verification of Completion	Lay-out # Dimensions. Alignment, Match marking Dimensions Orientations Data folder Compilations for Ceiling Girders & Main columns	Maj	Visual & Measurment Visual & Measurment Verification Visual	100%	100%	Drawing/TECH SPEC As per Drawing NDT reports, HT chart extracts, Trial assembly reports & Final CHP clearance	R	√	P	W	W	
3.5	Painting ,Preservation&	Finish &	Mi		100%	--	Drawing & Painting	-		P	V	--	*Approved by OWNER

3.6	protection Identification	coat thickness, Surface preparation W.O NO, DU project, Match Marking, Elevation marking.	n Min	Visual Measurement Visual	Rand om 100%	100%	Scheme * Drawing	R	√	P	V		Engg.
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NOTE:1 NDE FOR WELDS shall be carried out by qualified personnel and records to be maintained for verification.

NOTE:2 HEAT TREATMENT REQUIREMENTS (AWS D1.1)
 1) All butt welds when t> 50mm and all welds in a tension flange of ceiling Girder other than web-to- flange fillet welds and web stiffener welds at end bearing locations as per WPS. 2) All gas cut edges of load bearing members above 50 mm thickness shall be heat treated for minimum 30 minutes (600-650 °C)- .
 Alternatively, the gas cut edges shall be ground / machined to 3mm.

NOTE:3 VISUAL INSPECTION;
 a) Visual inspection of welds shall be performed after completion of welding and subsequent cooling to room temperature.
 b) All welds shall be cleaned to remove the slag, spatter etc. and visually examined for defects like crack, undercut, porosity, lack of fusion etc.
 c) The welds shall be examined as below for size, shape and reinforcements.

Nature of defects	Accepted norms.	Disposition
i) Crack, Lack of fusion & overlap	Not accepted	To be repaired & retested by LPI/MPI

ii) Crater	Not accepted.	To be repaired
iii) Under cut, Porosity Undersize & weld reinforcement	AWS D1.1	
v) WELD CONTOUR Face of fillet Flat or concave accepted. Convexity up to < 0.1S + 1.5mm is acceptable where S is the fillet size (minimum) as per drawing. Under size up to 1.6mm for a cumulative length of 10% of total length permitted		
vi) For all SQP, SIP, NDT & other MANUFACTURER procedures referred in CQP 1476, MANUFACTURER ‘s latest revisions are applicable.		
In case of outsourcing of PWHT and NDT, MANUFACTURER approved agency to be deployed. MANUFACTURER approved procedures to be followed. Physical Surveillance at vendor works shall be done by MANUFACTURER as per internal schedule.		

NOTE :4
 Materials identified against mill sheet or test certificate only shall be used for construction. Strict compliance to the same shall be ensured by the sub-contractors/vendors. MANUFACTURER will verify MTC for plates in B2B system. OWNER will carry out random surveillance checking of MTC for plates of thickness t ≤ 40 mm and in case of any non-conformance, suitable action as deemed fit will be taken by MANUFACTURER & OWNER on the sub- contractor/vendor concerned.

		MANUFACTURER NAME & ADDRESS			MANUFACTURING QUALITY PLAN								
					ITEM: DUCT				PACKAGE: Main Plant Package (Steam Generator & Auxiliary)				
									SUBSYSTEM : Ducting				

1.0 Raw Material														
1.0	Raw Material	Chemical & physical properties	Major	Review of MTC	100%	100%	Drawing/ Specification	Specification	MTC	√	P	R	--	
		Surface Finish & Dimensions	Major	Visual & Measurement	100%	100%	Free from surface defects & Drawing / Note	No Surface Defects	--	--	P	R	--	
2.0 Welding & NDT														
2.1	Procedure Qualification	WPS, PQR and WPQ	Major	Review	100%	100%	Drawing / ASME Sec IX	ASME Sec IX	WPS, PQR and WPQ	√	P	*V	--	*Witness for all new qualification
2.2	Welding Inspection	Weld completion	Major	Size and Visual	100%	100%	Drawing	Drawing	IR	√	P	V	--	
2.2.1		Weld Surface Quality	Major	DPT	10%	10%	ASME Section VIII Div I	ASME Section VIII Div I	IR	√	P	V	--	
3.0 In Process														
3.1	Marking, cutting & edge preparation	Dimensions & Straightness	Minor	Measurement	100%	--	Drawing	Drawing	--	--	P	--	--	
3.2	Joint setup	Set up Dimensions, offset, Matching	Major	Visual & Measurement	100%	--	Drawing	Drawing	--	--	P	--	--	
3.3	Bending	Shape / Dimensions	Major	Visual & Measurement	100%	Random	Drawing / Welding is not acceptable in place of bending	Drawing	--	--	P	V	--	
3.4	Hole Drilling	Hole Centers and diameter	Major	Measurement	100%		Drawing	Drawing	--	--	P	--	--	
4.0 Ducts														
4.1	Ducts*	Overall Dimension & Orientations	Major	Visual & Measurement / layout matching	100%	100%	Drawing	Drawing	IR	√	P	W	V	*Bends, Transition Ducts to be trial assembled with adjacent member
		Diagonals	Major	Measurement	100%	100%	Drawing	Drawing	IR	√	P	W	V	
		Stiffeners Location	Major	Visual & Measurement	100%	100%	Drawing	Drawing	IR	√	P	W	V	
		Guide vane/ Baffle plate locations	Major	Visual & Measurement	100%	100%	Drawing	Drawing	IR	√	P	W	V	

4.2	Ducts *	Free of Bulging / Waviness	Major	Visual	100%	100%	Bulging/ Waviness	5 mm Max	IR	√	P	W	V	
		Match Marking	Major	Visual	100%	100%	Drawing / Hard Punching	Drawing	IR	√	P	W	V	

5.0 Final Inspection														
5.1	Surface Preparation	Surface Finish	Major	Visual	100%	100%	Drawing / PO	Drawing / PO		--	P	V	--	
5.2	Painting	Paint shade , type , DFT and Finish	Major	Visual & Measurement	100%	Random	Drawing / PO	Drawing / PO	IR	√	P	W	--	
5.3	Date Folder	QAP Compliance & Inspection Clearance	Major	Review	100%	100%	Raw Material Test Certificates, Inspection Reports; NDT Reports;	Final Dossier	MICN	√	P	H	--	
5.4	Pre-Shipment	Packing	Major	Visual	100%	100%	Packing Procedure , MICN & PO	Packing Procedure	Packing list	√	P	V	--	

Note: The thickness mentioned in the order/drawing for various sections shall not vary more than +/- 2.5% & the mentioned thickness shall be available in at least 75% of the material used.

			<u>Manufacturer's Name & Address</u>			REFERENCE QUALITY PLAN											
						ITEM: HOPPER											
												SUB-SYSTEM:					
Sl No	Component & Operation	Characteristics	Class	Type of Check	Quantum of Check		Reference Documents	Acceptance Norms	Format of Record		Agency			Remarks			
					M	C/N					M	C	N				
1	2	3	4	5	6		7	8	9	D*	**10			11			

1.0	Raw Material: Raw Material to be procured from OWNER-approved sources.												
1.1	Plates & Rolled Sections	a. Visual & Dimensional	Major	Visual & Measurement	100 %	100 %	Material spec./ OWNER Approved GA Drawings. / Mfg. Drawings	Inward Inspection Report	--	P	V	--	Refer Note 2 & 3
		b. Chemical Composition	Critical	Document Review	1 per Heat / Lot	1 per Heat / Lot		MTC	√	P	V	V	
		c. Mechanical Properties	Critical	Document Review	1 per Heat / Lot	1 per Heat / Lot		MTC	√	P	V	V	
		d. UT (Bars > 50 mm)	Critical	Document Review	100 %	100%	ASTM 388	MTC	√	P	V	V	
		e. UT (Plates Thk. > 40 mm)			100 %	100 %	ASTM A 578 Level B S2		√	P	V	V	
2.0	Welding / Fit-Up												

2.1	Welding	Welding Procedure – WPS & PQR	Major	Document Review	100 %	100 %	AWS D1.1 / ASME Sec. IX	WPS	√	P	V	V	- WPS & PQR. -
		Welder Performance Qualification – WPQ	Major		100 %	100 %		WPQ	√	P	V	V	
2.2	Welding consumable	Verification of Make, Type & Batch No.	Major	Visual	100 %	100 %	ASME Sec II Part C	BTC	√	P	V	-	As per the MAIN SUPPLIER-approved list
2.3	Straightening Marking, Cutting, Edge Preparation, Drilling and Splicing	a. Shape b. Size c. Diagonal d. Straightness	Major	Visual & Measurement	100%	----	Drawings / spec.	Job History Card	--	P	--	--	
2.4	Fit Up, Welding, Dimensional Control,	Orientation, Weld Visual, Dimensions, Offset, Straightness, Shape, Diagonal & etc.	Major	Visual & Measurement	100 %	--	Drawings / spec.	Job History Card	--	P	--	--	
2.5	Pre-heating (as applicable)	Preheating Temperature Control	Major	Measurement	100 %	100 %	As per WPS & Tech. Spec.	Register	--	P	V	V	
3.0	Non-Destructive Examination												

3.1	Edge Preparation for Plates THK $\geq$ 32mm	Lamination/ Flaws	Major	MPT/LPT	100 %	100 %/ 10%	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	V	V	
3.2	Butt weld Joints for Plates THK $\geq$ 32mm	Weld Soundness	Critical	*RT / UT	100 %	100 % / 10 %	AWS D1.1	AWS D1.1	RT / UT Report	√	P	W	W	*RT films review
3.3	Butt Weld Joints for Plates 25mm $\geq$ THK < 32mm	Weld Soundness	Critical	MPT / LPT	100 %	100 % / 10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
				*RT / UT	10 %	10 %	AWS D1.1	AWS D1.1	RT / UT Report	√	P	W	W	*RT films review
3.4	All Butt Weld Joints of Plates THK < 25mm	Weld Soundness	Critical	MPT / LPT	10 %	10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
3.5	All fillet Weld Joints	Weld Soundness	Major	MPT / LPT	10 %	10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
3.6	All fillet weld of Built-up Girders, including the Bracket	Weld Soundness	Major	MPT / LPT	100 %	100 % / 10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	

3.7	Fillet weld at Base and Top Plate Welding in Columns	Weld Soundness	Major	MPT / LPT	100 %	100 % / 10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
3.8	Weld Visual	Completeness of welding, Surface Defects, Weld Size	Major	Visual	100 %	100 % / 10 %	Mfg. Drawings	AWS D1.1	Inspection Report	√	P	W	W	
3.9	Bend line – Hopper, Funnel & etc.	Bending area	Major	MPI / LPT	100 %	10 %	AWS D1.1	AWS D1.1	MPT / LPT Report	√	P	W	W	
4.0	Final Inspection													
4.1	Orientation and Dimensional Inspection	Verification of Orientation, Dimensions, hole & position. Straightness, Twist, Web verticality, web shift - as applicable	Major	Visual and Measurement	100 %	100 % / 10 %	Drawings / Applicable Standard	Inspection Report	√	P	W	W	If columns are to be dispatched in two or more parts, a mock-up assembly of the columns must be conducted at the shop. Ground Mock-up assembly of 1 st Hopper shall be performed at each Manufacturer.	
4.2	Match Marking & Identification	Material Identification Mark No. with DU No	Major	Visual	100 %	100 %		Inspection Report	√	P	W	R	All Components except Loose materials shall be identified by a Hard Punch. The same shall be highlighted by silver Paint and stencilled by Weatherproof Paint.	
5.0	Surface Preparation And Painting													

5.1	Surface Preparation & Painting	Removal of Scale, Rust, Foreign Materials, etc.	Major	Visual	100%	100%	Approved Painting Scheme/ Approved Drawing / Tech. Spec.	Inspection Report	√	P	R	-	
		Review of Paint Certificate	Major	Visual	Per Batch	Per Batch		Paint Mfr. TC	√	P	R	V	
		Complete Coverage of Paint	Major	Visual	100 %	Random		Inspection Report	√	P	W	V	
		DFT	Major	Measurement	100 %	Random		Inspection Report	√	P	W	V	
5.2	Completeness of Supply	Verification of Scope of Supply & Correctness of QA Documents	Major	Verification	100 %	100 %	Purchase Order / Quality Plan / Drawing	Inspection reports	--	P	V	--	

Notes:

1. M/s OWNER and M/s MAIN SUPPLIER inspectors shall have full access to take photos of the respective jobs without seeking prior permission from the manufacturer.
2. Raw Materials must be sourced from OWNER-approved suppliers, and the MTC must be submitted.
3. The manufacturer shall provide appropriate storage for Raw Materials, ensuring colour coding and traceability.
4. The Heat Number or Internal Allocation Number (IAN) shall be provided on the main parts (Flange & Web), excluding attachments.
5. Qualified and continuously employed welders may be deployed without requalification. Records of their qualification and continuity must be available for verification. A newly qualified welder by the Main Contractor shall be deployed for welding.

		MANUFACTURER'S NAME AND ADDRESS		ITEM / EQUIPMENT: DAMPERS		QP NO		SIGN. OF MFGR'S								
						PAGE NO										
SL. NO	COMPONENT & OPERATIONS		CHARACTERIS TICS	CLASS	TYPE OF CHECK	QUANTUM OF CHECK		REFERENCE DOCUMENT#	ACCEPTANCE NORMS		FORMAT OF RECORD		AGENCY			REMARKS
					M	C/N	M						C	N		
1.	2.		3.	4.	5.	6.		7.	8.			D*	** 10.			11.

1.0	Raw Material												
1.1	Plates, Rolled sections	Chemical & Mechanical Properties	Major	TC Verification	100%	100%	As per drawing/GMS	TC		P	V	V	
		Soundness (UT)#	Critical	TC Verification	100%	100%	ASTM A 435	TC		P	V	V	
1.2	Rounds	Chemical & Mechanical properties	Major	TC Verification	100%	100%	As per drawing / GMS	TC		P	V	V	
		Soundness (UT) #	Critical	TC Verification	100%	100%	ASTM A 388 / ASME Sec VIII Div-2	TC		P	V	V	
1.3	Seals	Chemical & Mechanical properties	major	TC Verification	100%	100%	As per drawing/GMS	TC		P	V	V	
1.4	Sub deliveries	Conformance to PO & as per drawing/GMS	Major	TC Verification	100%	100%	Note 3	TC		P	V	V	
TC - Test Certificate ➤ Review of material TCs shall be for plates above 40 mm thickness only. For plates of thickness less than or equal to 40 mm, COC will be submitted by MAIN SUPPLIER to OWNER ➤ TC verification of channels of size 400 mm and above (if applicable) and Beams of Size 450,500 and 600 mm. ➤ # Plates of thickness more than 40 mm and Rounds of Dia. more than 40 mm shall be UT tested. ➤ ## Surveillance checks of TC verification will be carried out at Raw Materials.													
2.0	In process Control												

	Welders are Qualified to AWS D1.1. Procedures are prequalified & OWNER will review procedures and welder's qualification records												
2.1	Weld Inspection	Weld Quality & Size	Major	Visual & Measurement	100%	100%(C) 10% (N)	Note – 1	Report		P	W	V	
2.2	Welds of studs with gland plate & butt welds of frame	Weld Soundness	Major	MPI/LPI	100%	100%	Drawing & Tech Spec/ ASME/ AWS	Report		P	W	W	
2.3	Welds of blade block, mounting bracket and all other welds	Weld Soundness	Major	MPI / LPI	100%	10%	Drawing & Tech Spec/ ASME/ AWS	Report		P	W	V	
2.4	Assembly	Overall Dimensions & Match marks	Major	Measurement	100%	100%	Drawing	Report		P	W	V	
3.0	Final Inspection												
3.1	Trial operation	Free operation.	Major	Open and close Movement	100%	100%	Drawing & Tech Spec & Main supplier procedure	Report	√	P	W	W	
3.2	Trial operation	Seal Tightness.	Major	Check with Feeler Gauge	100%	100%	Drawing/ Tech Spec / Main supplier procedure/ OWNER Engg approved test procedure and Damper testing schedule	Report	√	P	W	W	
3.3	Performance Leak Tightness Test (For Bi plane Damper only)	Leak Tightness Test (as per contract)	Major	Performance Testing	Refer Test Schedule	Refer Test Schedule	OWNER Engg approved test procedure and Damper testing schedule	Report	√	P	W	W	Witness by OWNER Engg

3.4	Verification of completion	Over all dimension, Orientation	Major	Verification / Visual	100%	100%	As per drawing, All identified by mark under D in QP	Report		P	W	V	
3.5	Painting, preservation & protection.	Finish	Minor	Visual	100%	100%	Drawing	Report		P	W	V	
		Paint DFT	Major	Measurement	10%	10%	As per OWNER approved Painting scheme	Report		P	W	V	

NOTE 1: VISUAL INSPECTIONS

- a) Visual inspection of welds shall be performed after completion of welding on shaft, Blower air mounting brackets welds to be witness by MAIN SUPPLIER & verification of reports by OWNER
- b) All welds shall be cleaned to remove slag, spatter etc and visually examined for defects like crack, undercut, porosity, lack of fusion etc.
- c) The welds shall also be examined for size, shape and re-inforcement.

Nature of defects	Acceptance Norms	Disposition
Crack, lack of fusion, overlap	Not accepted repair and retest.	Confirm by LPI/MPI.
Crater	Not accepted	Fill by weld deposit
Undercut	Up to 0.8 mm accepted. >0.8 mm not accepted	----- Fill and grind smooth
Porosity for butt/fillet welds	One pore of dia < 2.4 mm every 100 mm length is permitted. However, pores of dia > 2.4 mm not accepted.	----- To be repaired.

Weld contour

1. Face of fillet	Flat or concave accepted. Slightly convex also accepted provided convexity < 0.1 S + 1.5 mm where S is fillet size.
2. Size (Minimum)	As per drawing. Under size permitted up to 1.6 mm for a cumulative length of 10% of total length.
3. Reinforcement	Max 3 mm up to 19 mm., Max 5 mm above 19 mm.

ITEM / EQUIPMENT: ROTOR POST ASSY SUB-SYSTEM: APH													
SL. NO	COMPONENT & OPERATION	CHARACTERISTICS	CLASS	TYPE OF CHECK	QUANTUM OF CHECK		REFERENCE DOCUMENT#	ACCEPTANCE NORMS	FORMAT OF RECORD	AGENCY			REMARKS
					M	C/N				M	C	N	
1.	2.	3.	4.	5.	6.		7.	8.		D *	** 10.		11.

1.0	Raw Materials													
1.1	Plates	Chemical & Mechanical properties.	Major	Chemical & Tensile Bend	Each Heat	Refer Notes- 1 & 2	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	**** When back wall Echo is set to 100 % of FSH in Sound area a) Defect Echo not to exceed 20 % of FSH during scan. b) Back wall Echo loss permissible is 20 % of FSH.
		Soundness	Major	UT for Plates > 40 mm	100%	Refer Notes- 1 & 2	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
1.2	Trunnion Forging	Chemical & Mechanical properties.	Major	Chemical, Tensile	100%	Refer Notes- 1 & 2	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
		Soundness	Major	UT	100%	Refer Notes- 1 & 2	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
1.3	Header Plate	Chemical & Mechanical properties.	Major	Chemical, Tensile	100%	Refer Notes- 1 & 2	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
		Soundness	Major	UT for Thick ³ 25mm	100%	100%	ASTM-A 435/A 388	ASTM-A-435/ A 388 (****) (No laminations)	TC		P	V	V	
1.4	Lower Post Pins	Chemical & Mechanical Properties	Major	Chemical, Tensile, Hardness	Each heat	Refer Notes- 1 & 2	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
		Soundness	Major	UT for > 50mm	100%	100%	ASTM A388	ASTM A388****	DR		P	V	V	
2.0	In Process Control													
	WELDERS ARE QUALIFIED TO AWS D1.1 & PROCEDURES ARE PREQUALIFIED. OWNER WILL REVIEW PROCEDURES AND WELDERS QUALIFICATION RECORDS													
2.1	Welding	Weld Size	Major	Measurement	100%	100%	Drawing	Drawing	DR		P	W	-	
		Finish	Major	Visual	100%	10%	--	--	--		P	W	-	

2.2	Rotor post assy. after welding	Dimensions	Major	Measurement	100%	100%	Drawing	HC		P	W	V	
2.3	Support Trunnion Shell Butt Weld	Weld soundness	Major	RT	100 %	100%	BHE:NDT:RP:RT:01 (Latest)	NDT Report	Ö	P	V	V	CHP RT Film Review
	Support Trunnion Shell to common Header Plate	Weld soundness	Major	UT/MT (Refer Note – 3)	100 %	100%	BHE:NDT:RP:UT:04 (Latest) BHE:NDT:RP:MT:01 (Latest)	NDT Report	Ö	P	W	V	
	Rotor Post Butt Weld	Weld soundness	Major	PT/ MT	100 %	100%	BHE:NDT:RP:PT:01 (Latest) BHE:NDT:RP:MT:01 (Latest)	NDT Report	Ö	P	W	V	
	Header Plate to Rotor Post Shell	Weld soundness	Major	PT/ MT	100 %	100%	BHE:NDT:RP:PT:01 (Latest) BHE:NDT:RP:MT:01 (Latest)	NDT Report	Ö	P	W	V	
	Rotor Post Shell to Common Header Plate	Weld soundness	Major	PT/ MT	100 %	100%	BHE:NDT:RP:PT:01 (Latest) BHE:NDT:RP:MT:01 (Latest)	NDT Report	Ö	P	W	V	
	Support trunnion shell to Header plate	Weld soundness	Major	PT/ MT	100 %	100%	BHE:NDT:RP:PT:01 (Latest) BHE:NDT:RP:MT:01 (Latest)	NDT Report	Ö	P	W	V	
	Lifting Lug Welding	Weld soundness	Major	PT/ MT	100 %	100%	BHE:NDT:RP:PT:01 (Latest) BHE:NDT:RP:MT:01 (Latest)	NDT Report	Ö	P	W	V	
2.4	Rotor post Dimensions after Machining.	Critical Dim. Drilling & Tapping	Major	Measurement	100%	100%	Drawing	HC	Ö	P	W	W	
2.5	Guide Trunnion to Adaptor Sleeve	Pressure Test for Taper Match with Adapter Sleeve	Major	Pressure Test	100%	See remarks	No Leakage and PRQA:403 (latest revision)	HC	Ö	P	W	W	100% by MAIN SUPPLIER 10% (minimum 1 No. per unit) by OWNER
3.0	Final Inspection												
3.1	Rotor post assembly and Trunnion spool assembly	Overall Dimensions	Major	Measurement	100%	100%	Drawing	HC		P	W	V	
3.2	Painting and Preservation	Painting DFT	Major	Visual & Measurement	100%	10%	OWNER Engineering approved painting scheme.	Report		P	W	V	

Notes:

1. Review of material TCs shall be for plates above 40 mm thickness only. For plates of thickness less than or equal to 40 mm, COC will be submitted by MAIN SUPPLIER to OWNER.
2. TC verification for trunnion forgings, header plate forgings, adopter plate forgings, adopter sleeve forgings.
3. UT to be carried out in case of design without ring weld and in other cases where design is with ring weld, layer by layer MPI to be performed after every 5 mm of welding.

ITEM / EQUIPMENT: **Module Assembly**

SUB-SYSTEM: **APH**

1.0	Raw Materials													
1.1	Lug plates	Chemistry & Mechanical properties	Major	Chemical & Mechanical	Each Heat	Note -1	As per Drawing/GMS	As per Drawing/GMS	TC	Ö	P	V	V	
		Soundness	Major	UT for Plates > 40 mm	100%	Note -1	ASTM A 435	ASTM A 435	TC	Ö	P	V	V	
1.2	Heating element (Hot /Hot intermediate)	Chemistry & Hardness, and Erichsen Cupping value	Major	Chemical & Hardness, and Erichsen Cupping Test.	Each Heat	Note -1	As per Drawing/GMS	As per Drawing/GMS	TC	Ö	P	V	V	
1.3	Heating element (cold)	Chemical & Mechanical	Major	Chemical & Mechanical.	Each Heat	Note -1	As per Drawing/GMS	As per Drawing/GMS	TC	Ö	P	V	V	
1.4	Diaphragm plates	Chemical & Mechanical	Major	Chemical & Mechanical.	Each Heat	Note -1	As per Drawing/GMS	As per Drawing/GMS	TC	Ö	P	V	V	
1.5	Stay plates, Basket support bar	Chemical & Mechanical	Major	Chemical & Mechanical	Each Heat	Note -1	As per Drawing/GMS	As per Drawing/GMS	TC	Ö	P	V	V	
1.6	Grating flats, plates	Chemical & Mechanical	Major	Chemical & Mechanical.	Each Heat	Note -1	As per Drawing/GMS	As per Drawing/GMS	TC	Ö	P	V	V	
2.0	In process control													
	WELDERS ARE QUALIFIED TO AWS D1.1 PROCEDURES ARE PREQUALIFIED & OWNER WILL REVIEW PROCEDURES AND WELDERS QUALIFICATION RECORDS.													
2.1	Lug plate cutting	Dimensions	Major	Check with template	100%	100%	Drawing	Drawing	DR		P	W	V	

2.2	Stress Relief Heat Treatment of Lug Plate	Compliance to SR cycle.	Major	Verification of HT chart	100%	100%	BAP: HT: 001 (Latest)	BAP: HT: 001 (Latest)	Chart		P	V	V		
2.3	Welding	Weld size	Major	Measurement	100%	10%	Drawing	Drawing	DR		P	W	V		
		Finish	Major	Visual	100%	10%	AWS D1.1	AWS D1.1	---		P	W	V		
2.4	NDE	Soundness	Major	PT	10%	10%	BHE: NDT: RP: PT: 01 (Latest)	BHE: NDT: RP: PT:01 (Latest)	NDE Report	Ö	P	W	W		
2.5	Lug Assembly Machining	Dimensions	Critical	Measurement	100%	Note-2	Drawing / MAIN SUPPLIER Procedure PRQA: 417	Drawing / MAIN SUPPLIER Procedure PRQA: 417	HC		P	W	W		
3.0	Final Inspection														
3.1	Module Assy (Note-3) (In Fixture)	Over All Dimensions.	Major	Measurement	100%	Note-3	Drawing	Drawing	HC		P	W	W		
		Weld size Weld finish	Major	Visual	100%	10%	AWS D 1.1	AWS D 1.1	--		P	W	V		
3.2	Painting & Preservation.	Painting DFT	Major	Visual / Measurement.	100%	10%	OWNER Engineering approved painting scheme	HC		P	W	V			
Note -1: ➤ Review of material TCs shall be for plates above 40 mm thickness only. For plates thickness more than or equal to 40 mm, COC will be submitted by MAIN SUPPLIER to OWNER ➤ TC verification for lug plates, HE coils. Note-2: 100% Inspection by MAIN SUPPLIER. Quantum of inspection for OWNER is 3 Nos for rotor of 12 modules and 6 Nos for rotor of 24 modules. Note-3: Lug assembly and module diaphragm plates are placed in fixture for fit up and dimension check by MAIN SUPPLIER. FOT is offered for OWNER in fixture. For other modules inspection in fixture is done by MAIN SUPPLIER and full welding is carried out after removal from fixture and offered for OWNER inspection in all cases.															

ITEM / EQUIPMENT: Connecting Plate Assembly SUB-SYSTEM: APH														
1.0	Raw Materials													
1.1	Structural Plates, Sheets, Angles	Chemical & Mechanical Properties	Major	Analysis Tensile Test	Each Heat	Refer Notes	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	## When back wall Echo is set to 100 % of FSH in Sound area a) Defect Echo not to exceed 20 % of FSH during scan. b) Back wall Echo loss permissible is 20 % of FSH.
		Soundness	Major	UT for Plates > 40 mm	100%	Refer Notes	ASTM A 435	ASTM A435	TC		P	V	V	
1.2	Pipes	Chemical & Mechanical Properties	Major	Analysis Tensile Test	Each Heat	Refer Notes	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
1.3	Rods	Chemical & Mechanical Properties	Major	Analysis Tensile Test	Each Heat	Refer Notes	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
		Soundness	Major	UT for Rod Dia > 50 mm	100%	100%	ASTM A 388	ASTM A388 ##	TC		P	V	V	
2.0	In Process Control													
	Welders are qualified to AWS D1.1. Procedures are Pre-qualified & OWNER will review procedures and welder's qualification records													
2.1	Welding	Weld Size	Major	Measurement	100%	10%	Drawing	Drawing	DR		P	W	V	
		Finish	Major	Visual	100%	10%	--	--	--		P	W	-	
2.2	a) Butt weld of CE center section base or bottom plate	Soundness	Major	UT	100%	10%	BHE: NDT: RP: UT/04 (Latest)	BHE: NDT: RPT: UT/04 (Latest)	NDT Report	Ö	P	W	W	
	b) Butt weld of CE center section H- Frame	Soundness	Major	UT	100%	10%	BHE: NDT: RP: UT/04 (Latest)	BHE: NDT: RPT: UT/04 (Latest)	NDT Report	Ö	P	W	W	
	c) CE sidewall butt weld >25 mm	Soundness	Major	UT	100%	10%	BHE: NDT: RP: UT/04 (Latest)	BHE: NDT: RPT: UT/04 (Latest)	NDT Report	Ö	P	W	W	

2.2	d) CE sidewall butt weld <25 mm & other butt welds of CE.	Soundness	Major	MPI (both sides) with Root LPI	100%	100%	BHE: NDT: RP: PT:01 (Latest), MT:01 (Latest)	BHE: NDT: RP: PT:01 (Latest) MT:01 (Latest)	NDT Report	Ö	P	W	V		
	e) HE Center section Butt weld.	Soundness	Major	PT	100%	100%	BHE: NDT: RP: PT:01 (Latest)	BHE: NDT: RP: PT :01 (Latest)	NDT Report	Ö	P	W	V		
	f) other welds of HE & CE	Soundness	Major	PT	10%	10%	BHE: NDT: RP: PT:01 (Latest)	BHE: NDT: RP: PT :01 (Latest)	NDT Report	Ö	P	W	V		
2.3	Sector plate after welding	Dimensions	Major	Measurement	100%	100%	Drawing	HC		P	W	V			
2.4	Stress relief of sector plate.	Compliance to stress relief cycle	Major	Verification of HT Chart	100%	100%	BAP: HT:001 (Latest)	HT Chart	Ö	P	V	V			
2.5	Sector plate after machining.	Dimensions	Major	Measurement	100%	100%	Drawing	HC		P	W	V			
		Finish	Major	Visual	100%	10%	Drawing	HC		P	W	-			
3.0	Final Inspection														
3.1	Trial assembly of CPA	Over all dimension	Major	Measurement	100%	100%	Drawing	HC	Ö	P	W	W			
3.2	Sector plate fixing and checking on center section and primary center section	Dimensions	Major	Measurement	100%	100%	Drawing	HC	Ö	P	W	V			
3.3	Checking movement of sector plate	Movement	Major	Measurement	100%	100%	Drawing	HC	Ö	P	W	W			
3.4	Painting and preservation	Painting DFT	Major	Visual / measurement	100%	10%	OWNER Engineering Approved painting scheme	Report		P	W	V			
Note: 1. Review of material TCs shall be for plates above 40 mm thickness only. For plates of thickness less than or equal to 40 mm, COC will be submitted by MAIN SUPPLIER to OWNER. 2. TC verification of channels of size 400 mm and above (if applicable) and Beams of Size 450, 500 and 600 mm. 3. TC verification for Bracing pipe.															

ITEM / EQUIPMENT: GUIDE BEARING ASSY														
SUB-SYSTEM:APH														
1.0	Raw Material													
1.1	Plates, Sheets	Chem. & Mech Properties	Major	Chem.& tensile Test	Each Heat	Refer Notes–1&2	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	**When back wall echo is set to 100% of FSH in sound area a) Defect echo not to exceed 20% of FSH during scanning. b) Back wall echo loss permissible 20% of FSH.
1.2	Rods	Chem. & Mech Properties	Major	Chem.& tensile Test	Each Heat	Refer Notes–1&2	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
		Soundness	Major	UT > 50 mm	100%	100%	ASTM A 388	ASTM A 388**	TC		P	V	V	
1.3	Tubes	Chem. & Mech Properties	Major	Chem.& Tensile Test	Each Heat	Each heat	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
1.4	Adapter sleeve forgings	Chem. & Mech Properties	Major	Chem.& Tensile Test	Each Heat	Each heat	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
		Soundness	Major	UT	100 %	Refer Notes–1&2	ASTM A 388	ASTM A 388**	TC		P	V	V	
1.5	Housing Sleeve Castings	Chem. & Mech Properties	Major	Chem. & tensile test	Each Heat	Refer Notes–1&2	As per Drawing / GMS	As per Drawing / GMS	TC		P	V	V	
		Soundness	Major	UT	100%	Refer Notes–1&2	ASTM A609	ASTM 609 Level II	TC		P	V	V	
1.6	Bearings	Compliance to Drawing	Major	Verification	100%	Refer Notes–1&2	As per Drawing / GMS	As per Drawing / GMS	CC		P	V	V	
2.0	In Process Control													
Welders are qualified to AWS D1.1 & procedures are pre-qualified. OWNER will review procedures & welders qualification records														
2.1	Welding	Weld size	Major	Measurement	100%	100%	Drawing	Drawing	DR		P	W	V	
		Soundness	Major	MT/PT	10%	10%	BHE: NDT: RP: MT:01 (Latest) BHE: NDT: RP: PT:01 (Latest)	NDT Report		P	W	V		

2.3	Machining of Guide bearing Housing	Dimensions	Critical	Measurement	100 %	100%	Drawing	Drawing	HC	Ö	P	W	W		
2.4	Kerosene Leak Test after machining	Soundness	Major	Visual	100 %	100%	No leakage	HC	Ö	P	W	V			
3.0	Final Inspection														
3.1	Assembly of Guide Bearing Housing with Bearing, Adapter Sleeve, Housing Sleeve & Housing Cover	Assembly Clearances of Bearing with Housing Sleeve / Adapter Sleeve Free rotation	Critical	Measurement	100 %	100%	Drawing	Drawing	DR	Ö	P	W	W	\$	\$ Free rotation witness by OWNER. Other checks in column 3 against clause 3.1 verification by OWNER
3.2	Painting and Preservation.	Painting DFT	Major	Visual	100 %	10%	OWNER Engineering approved painting scheme	Report		P	W	V			
Notes: 1. Review of material TCs shall be for plates above 40 mm thickness only. For plates of thickness less than or equal to 40 mm, COC will be submitted by MAIN SUPPLIER to OWNER 2. TC verification for adopter plate forgings, adopter sleeve forgings, adopter sleeve forgings. LEGEND: * RECORDS IDENTIFIED WITH “TICK” (Ö) SHALL BE ESSENTIALLY INCLUDED BY SUPPLIER IN QA															
ITEM / EQUIPMENT: SUPPORT BEARING ASSY SUB-SYSTEM: APH															
1.0	Raw Materials														
1.1	Plates and structural	Chem.& Mech. Properties	Major	Chem. &Tensile Test.	Each Heat	Note-1	As per Drawing/GMS	As per Drawing/GMS	TC		P	V	V	CC: compliance certificate	
		Soundness	Major	UT for plates > 40 mm	100%	Note-1	ASTM A 435	ASTM A 435	TC		P	V	V		
1.2	Adapter plate	Chem.& Mech. Properties	Major	Chem. &Tensile Test.	100%	Note-1	As per Drawing/GMS	As per Drawing/GMS	TC	Ö	P	V	V		
		Soundness	Major	UT	100%	100%	ASTM A 435	ASTM A 435/ No Lamination	TC	Ö	P	V	V		

1.3	Bearings	Compliance to Drawing/GMS	Major	Verification	100%	100%	As per Drawing/GMS	As per Drawing/GMS	CC	Ö	P	V	V		
2.0	Inprocess Control														
	WELDERS ARE QUALIFIED TO AWS D1.1 & PROCEDURES ARE PREQUAIFIED. OWNER WILL REVIEW PROCEDURES AND WELDERS QUALIFICATION RECORDS														
2.1	Welding of Support Bearing Housing	Weld Size & Finish	Major	Measurement	100%	10%	DRAWING	DRAWING	DR		P	W	V		
	Weld between Housing Ring & Bottom Plate	Soundness	Major	MPI	100%	100%	BHE: NDT:RP: MT:01 (Latest)	BHE: NDT: RP MT:01 (Latest)	REP	Ö	P	W	V		
	Other areas	Soundness	Major	LPI	10%	10%	BHE: NDT: RP PT:01 (Latest)	BHE: NDT: RP: PT:01 (Latest)	REP	Ö	P	W	V		
2.2	Machining of support bearing Housing	Bearing. Seating diameter	Major	Measurement	100%	100%	Drawing	Drawing	DR		P	W	V		
		Kerosene leak test for soundness	Major	Visual	100%	100%	No Leakage	No Leakage	HC	Ö	P	W	V		
3.0	Final Inspection														
3.1	Assembly checks	A) Clearance of bearing seating area	Critical	Measurement	100%	100%	Drawing	Drawing	DR	Ö	P	W	W		
		B) Free rotation of bearing inside housing	Critical	Visual	100%	100%	Drawing	Drawing	DR	Ö	P	W	W		
4.0	Painting & preservation	Painting DFT	Major	Visual/ Measurement.	100%	10%	OWNER Engineering approved painting scheme	REP		P	W	V			
Note-1: Review of material TCs shall be for plates above 40 mm thickness only. For plates of thickness less than or equal to 40 mm, COC will be submitted by MAIN SUPPLIER to OWNER.															
ITEM/ EQUIPMENT: ROTOR DRIVE ASSY															
SUB-SYSTEM: APH															

1.0	Raw Material													
1.1	Plates, sheets, Structural	Chemistry & Mech. Properties	Major	Chemical & Tensile Test	Each heat	See notes below	As per Drawing /GMS	TC		P	V	V		
1.2	Sub Deliveries (BOI)													
1.2.1	Pinion	Compliance to Drawing	Major	Verification	100%	100%	As per Drawing /GMS	TC	Ö	P	V	V		
1.2.2	Air Motor	Compliance to Drawing	Major	Verification	100%	100%	As per Drawing /GMS	TC	Ö	P	V	V		
1.2.3	Gear Box	Compliance to Drawing	Major	Verification	100%	100%	As per Drawing /GMS	TC	Ö	P	V	V		
1.2.4	Motor a) Make, type & rating	Compliance to Drawing	Major	Verification	100%	100%	OWNER Approved Drawing / Data sheet / Makes as per LOA	TC	Ö	P	V	V		
	b) COC for motor	Compliance to Drawing	Major	Verification	100%	100%	Note-1	TC	Ö	P	V	V		
2.0	In process Control													
	WELDERS ARE QUALIFIED TO AWS D1.1 & PROCEDURES ARE PREQUALIFIED. OWNER WILL REVIEW PROCEDURES AND WELDERS QUALIFICATION RECORDS													
2.1	Welding	Weld size	Major	Measurement	100%	10%	Drawing	Drawing	DR		P	W	V	
		Finish	Major	Visual	100%	10%	--		DR		P	W	V	
2.2	NDE	Soundness	Major	PT	100%	100%	BHE: NDT: RP: PT:01 (Latest)	NDT Report		P	W	V		

2.3	Dimensional checks of Rotor Drive base assy, Motor Bracket and pinion cover assy.	Overall Dimension	Major	Measurement	100%	100%	Drawing	DR	Ö	P	W	V	
ITEM / EQUIPMENT: ROTOR DRIVE ASSY SUB-SYSTEM: APH													
3.0	Final Inspection												
3.1	No Load Run Test	1) Noise level of electric & air motors	Major	Measurement	100%	100%	Noise level of Air motor shall be checked at a distance of 1M from shaft horizontally & at 1.5 M elevation from shaft axis and acceptance norms shall be as per OWNER technical specification	DR	Ö	P	W	W	*-No load run test power consumption requirement as indicated in the QP is for record purpose only.
		2) Power Consumption*	Major	Measurement	100%	100%	IS 325	DR	Ö	P	W	W	
		3) Wobble	Major	Measurement	100%	100%	Drawing	DR	Ö	P	W	W	
3.2	Painting and preservation	Painting DFT	Major	Visual / Measurement	100%	10%	OWNER approved painting scheme.	--		P	W	V	
Notes: ➤ Review of material TCs shall be for plates above 40 mm thickness only. For plates of thickness less than or equal to 40 mm, COC will be submitted by MAIN SUPPLIER to OWNER. Note 1: a) Less than 50 KW Acceptance of Motor less than 50 KW is based on COC of the manufacturer & the contractor confirming as follows: It is hereby confirmed that the above-mentioned motor /motors was/ were manufactured taking care of OWNER specific requirements regarding ambient temp., voltage & frequency variation, hot starts, pull out torque, starting KVA/KW, temp. rise, distance between centre of stud & gland plate and tested in accordance with approved drawing /data sheets.													
ITEM / EQUIPMENT: UPPER ROTOR POST PIN SUB-SYSTEM: APH													
1.0	Raw Materials												
1.1	Plates	Chemistry Mech. Properties	Major	Chemical & Mechanical	Each heat	Each heat	As per Drawing/GMS	As per Drawing/GMS	TC		P	V	V #When back wall Echo is set to

1.2	Rods	Chemistry Mech. Properties	Major	Chemical & Mechanical	Each heat	Each heat	As per Drawing/GMS	As per Drawing/GMS	TC		P	V	V	100 % of FSH in Sound area a) Defect echo not to exceed 20 % of FSH during scanning. b) Back wall echo loss permissible 20 % of FSH.
		Soundness	Major	UT for Rod > 50mm	100%	100%	ASTM A388	ASTM A388 #	TC		P	V	V	
2.0	In Process Control													
2.1	Welding	Weld size	Major	Measurement	100%	100%	DRAWING	DRAWING	DR	Ö	P	W	V	
3.0	Final Inspection													
3.1	Dimensional Check	Overall dimensions	Major	Measurement	100%	10% of lot	DRAWING	DRAWING	DR	Ö	P	W	V	
3.2	Painting and preservation	Painting DFT	Major	Visual / Measurement	100%	10% of lot	OWNER Engineering approved painting scheme		REP	Ö	P	W	V	
ITEM / EQUIPMENT: ROTOR HOUSING ASSY														
SUB-SYSTEM: APH														
1.0	Raw Material													
1.1	Plates & structural	Chemistry & Mech. Properties	Major	Chemical & Tensile Test	Each heat	Each heat	As per Drawing/GMS	As per Drawing /GMS	TC	Ö	P	V	V	
		Soundness	Major	UT for plates >40mm	100%	Note-1	ASTM A435	ASTM A435	TC		P	V	V	
2.0	In process control													
	Welders are qualified to AWS D1.1 & Procedures are prequalified. OWNER will review procedures and welder’s qualification Records													

2.1	Welding	Weld size	Major	Measurement	100%	10%	Drawing	Drawing	DR		P	W	V	V/W * W for Main pedestal. V for Other pedestal & Panels			
2.2	NDE	Soundness	Major	LPI/MPI	10%	10%	BHE: NDT: RP: PT: 01 (Latest) BHE: NDT: RP: MT: 01 (Latest)		NDT Report	Ö	P	W	V/ W *				
3.0	Final Inspection																
3.1	Dimensional Inspection	Over all dimension	Major	Measurement	100%	100%	Drawing	Drawing	DR	Ö	P	W	V/ W *				
3.2	Painting and Preservation.	Painting DFT	Major	Visual /Measurement	100%	10%	OWNER Engineering approved painting scheme		DR		P	W	V				
Note-1:																	
➤ Review of material TCs shall be for plates above 40 mm thickness only. For plates of thickness less than or equal to 40 mm, COC will be submitted by MAIN SUPPLIER to OWNER.																	
➤ TC verification of channels of size 400 mm and above (if applicable) and Beams of Size 450mm and above																	

1.6 ELECTRICAL SYSTEMS

1.6.1 Generator and auxiliary systems													
1.6.1.1 Process check for static parts of generator/ exciter													
Items/ Components/ Process	Tests/ Checks												
	Visual, dimension	Chemical property (Raw Material)	Heat treatment	Mechanical property (Raw Material)	Impact (Raw Material)	Hydraulic test	Pneumatic test	RT or UT (10% for butt weld)	MP or DPT (All welds)	Relative permeability	Ferrite content	DIN 43760, IS:2848,7358	DIN 48124
Sheet and Fabrication :													
- END shield	Y	Y	Y	Y	Y	Y	Y	Y	Y				
- Stator casing	Y	Y	Y	Y	Y	Y	Y	Y	Y				
- Bushing boxes	Y	Y	Y	Y	Y	Y	Y	Y	Y				
- Terminal plates	Y	Y	Y	Y	Y	Y	Y		Y				
- Manhole and covers	Y	Y	Y	Y	Y	Y	Y		Y				
- Trunnions	Y	Y	Y	Y	Y			Y	Y				
- Core bar	Y	Y		Y									
- Press ring	Y	Y		Y					Y				
- Core bolt (insulated)	Y	Y		Y				Y	Y				
- Gaskets	Y			Y									
- Bearing seals and hydrogen seals	Y	Y		Y				Y1					
- Terminal bushing													Y
- RTD/ Thermocouple												Y	
Additional check for nonmagnetic components										Y			
Non magnetic components welding											Y		
Y=Test applicable, Y1=UT on babbit for bearing													

1.6.1.2 Process check for core of generator/ exciter (Table 1/2)								
Items/ Components/ Process	Tests/ Checks							
	Specific loss before and after ageing	Magnetisation	Anotrophy of losses	Stacking factor	Burr level	Check for varnish, insulation (chemical, electrical, viscosity, cure time, solid content, dielectric properties)	Dimension and surface (uniformity of varnish coat	Spot weld check
Core lamination	Y	Y	Y	Y		Y	Y	
After punching insulated core laminations					Y	Y	Y	
Ventilation stamping								Y
Core assembly	Y	Y					Y1	
Y = Test applicable, Y1 = Visual checks								

1.6.1.2 Process check for core of generator/ exciter (Table 2/2)						
Items/ Components/ Process	Tests/ Checks					
	Process check including heating and	Insulation test of core tension bolt and core bar	Functional check of ventilation ducts	Hot spot by infra red camera, ELCID pressure application	Location of temperature detectors	Iron loss at rated flux density
Core assembly (additional checks for generator)	Y	Y	Y	Y	Y	
Y = Test applicable, Y1 = Visual check						

1.6.1.3 Process check for stator conductor and winding of generator/ exciter (Table 1/3)												
	Tests/ Checks											
	Mechanical Properties (sample)	Chemical Properties (sample)	Resistivity/ Resistance	Metallography properties	Eddy current and pressure test	Insulation adhesion	Flexibility of bending	Dielectric test	Dimension/ visual	Electric test	Physical properties	Brazing procedure
Winding copper and connecting busbars	Y	Y	Y		Y1				Y			
Insulated conductor						Y	Y	Y	Y	Y		
Insulation material	Y	Y							Y	Y	Y	
Manufacturing winding bar and phase bar								Y	Y	Y		Y
Winding laying								Y	Y	Y		
Water supply hoses	Y	Y						Y	Y			
Winding supporting		Y							Y		Y	
Connection between bars												Y
Wound stator												
Y = Test applicable, Y1 = Visual checks												

1.6.1.3 Process check for stator conductor and winding of generator/ exciter					
(Table 2/3)					
Items/ Components / Process	Tests/ Checks				
	X-Ray	Process check	Flow test	Helium leak test and pressure test	Check on RTD and location
Winding copper and connecting busbars					
Insulated conductor					
Insulation material					
Manufacturing winding bar and phase bar		Y	Y1	Y	
Winding laying		Y			Y
Water supply hoses				Y	
Winding support ring					
Connection between bars					
Wound stator				Y1	
Y = Test applicable, Y1 = Visual checks					

1.6.1.3 Process check for stator conductor and winding of generator/ exciter (Table 3/3)												
Items/ Components/ Process	Tests/ Checks											
	Tan delta and delta (tan delta upto 1.2 Un)	Corona protection resistance	Reactance of stator winding	Magnetic permeability of metallic parts	Magnetic test and Vibration fatigue	Dielectric test at elevated, room temperature	Inter strand Insulation test	Thermal shock and boroscopic examination of brazed water box	Slot wedge tightness and radial movement	Type test on two bars for heating cycle test, Thermal stability test, Voltage endurance test	Support arrangement	Stator-NFT Test
Winding copper and connecting bus bars												
Insulated conductor						Y						
Insulation material												
Manufacturing winding bar and phase bar	Y	Y					Y	Y1		Y		
Winding laying	Y										Y	
Water supply hoses				Y	Y						Y	
Winding support ring												
Connection between bars							Y					
Wound stator	Y		Y						Y			Y
Y = Test applicable, Y1= Applicable for hollow conductor												

1.6.1.4 Process check for rotor and assembly of generator / exciter
(Table 1/4)

Items/ Components/ Process	Tests/ Checks											
	Sample tensile stress	Sample 0.2 limit	Sample elongation	Hardness on sample	Impact and stress rupture propertiescheck on sample	Sample chemical properties	NDTT, FATT	Process check including heat treatment (as applicable)	Ultrasonic test/ RT (at suppliers works and after preliminary machining)	Sulphur prints check flux carrying capacity/ magnetic	Flux carrying capacity/ magneticproperties	Boroscopic examination
Rotor forging and slip ring shaft	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Rotor end retaining ring and cover, locking ring and slip ring forgings, diode wheel	Y	Y	Y		Y	Y		Y	Y			
Rotor winding copper rotor wedges, damper wedges, CC-bolts and D-leads	Y		Y			Y		Y	Y			
Rotor slot boxes/ insulating material						Y						
Coil manufacture												
Rotor winding								Y				
Winding connection studsand assembly												
Complete rotor								Y				
Test on completed rotor at various speed upto rated speed												
Test on completed rotor before and after overspeed												
Fan hubs/ blades						Y		Y	Y			
Generator assembly												
Rectifier wheel	Y		Y					Y	Y			
Permanent magnet					Y						Y	
Exciter assembly												
Y = Test applicable												

1.6.1.4 Process check for rotor and assembly of generator / exciter
(Table 2/4)

Items/ Components/ Process	Tests/ Checks											
	MPI or DP, NDT	Visual, dimension, cleanliness	Adhesion, thickness of coat on silver plating	Electrical conductivity and oxygen content	Mechanical test on sample	Electrical test	Resistance measurement	Purge test on vents	Helium leak test for H ₂ cooled machine	Inter-turn test	Dielectric test	Gas tightness
Rotor forging and slip ring shaft	Y	Y										
Rotor end retaining ring and cover, locking ring and slip ring forgings, diode wheel	Y		Y									
Rotor winding copper rotor wedges, damper wedges, CC-bolts and D-leads	Y		Y	Y	Y	Y						
Rotor slot boxes/ insulating material					Y	Y						
Coil manufacture		Y										
Rotor winding	Y	Y				Y	Y	Y	Y	Y	Y	
Winding connection studs and assembly	Y				Y						Y	Y
Complete rotor							Y				Y	
Test on completed rotor at various speed upto rated speed							Y			Y	Y1	
Test on completed rotor before and after over-speed		Y								Y	Y	
Fan hubs/ blades	Y	Y										
Generator assembly		Y										
Rectifier wheel		Y				Y						
Permanent magnet		Y			Y							
Exciter assembly		Y				Y						
Y=Test applicable, Y1= at rated speed only												

1.6.1.4 Process check for rotor and assembly of generator / exciter
(Table 3/4)

Items/ Components/ Process	Tests/ Checks											
	Insulation Resistance	PI at 5 kV	Radial run out/ alignment	Impedance measurement/ Repetitive Surge Oscillograph	Dynamic balancing ISO 5393, 5406, 2372, 1940 including Air run test	Over speed test (120%) for 2 minute	Functional test	Axial run out, seal ring holder	Metallography examination	Torque on joint bolts	Fitting and locking of balancing weights	Brazer and brazing procedure
Rotor forging and slip ring shaft									Y			
Rotor end retaining ring and cover, locking ring and slip ring forgings, diode wheel												
Rotor winding copper rotor wedges, damper wedges, CC-bolts and D-leads										Y		
Rotor slot boxes/ insulating material												
Coil manufacture												Y
Rotor winding												Y
Winding connection studs and assembly	Y											
Complete rotor	Y		Y	Y	Y	Y				Y		
Test on completed rotor at various speed upto rated speed				Y								
Test on completed rotor before and after over-speed	Y		Y	Y								
Fan hubs/ blades											Y	
Generator assembly	Y	Y	Y				Y		Y	Y	Y	Y
Rectifier wheel			Y					Y		Y	Y	
Permanent magnet												
Exciter assembly			Y				Y			Y	Y	
Y=Test applicable												

1.6.1.4 Process check for rotor and assembly of generator/ exciter (Table 4/4) (additional checks for exciter)									
Item/ Components/ Process	Tests/ Checks								
	As per IEC-146	As per IEC-76	Pole parallelism and polarity	Mechanical chemical and Magnetic properties	Functional check	Insulation resistance	IEEE/ ANSI-C37.18	As per specification	Dimensional and visual
Fuse diode and filter Circuit	Y								Y
PMG and exciter stator			Y	Y		Y			
Banding wire				Y					
Exciter field Breaker, field discharge resistor					Y				
Bearing, exciter armature field, axis coil RTD						Y			
Voltage Regulator								Y	
Y = Test applicable									

1.6.1.5 Final acceptance tests for generator/ exciter (Table 1/3)										
Item/ Process	Tests/ Checks									
	Gas tightness for H ₂ cooled	Resistance measurement	Rotor impedance at various speeds	Heat run test	Function check	Voltage regulation	OCC	SCC	Record auxiliary parameters	Steady state reactance
Works tests on running generator	Y	Y	Y	Y			Y	Y	Y	Y
Without Excitation, OC and SC with rated voltage and current for generator				Y						
On total winding/ phases at interval 0.2Un for generator										
Condition after dismantling										
Works tests on brushless exciter		Y		Y			Y			
PMG works tests		Y		Y		Y				
Full load for PMG and converter assembly				Y						
Converter assembly				Y						
Y = Test applicable										

1.6.1.5 Final acceptance tests for generator / exciter (Table 2/3)										
Item/ Process	Tests/ Checks									
	Insulation resistance at 5 kV	Polarization index at 5 kV	Phase sequence voltage balance	Shaft voltage and current	HV test (except electronic circuit)	RTD, BTD Check	Capacitance measurement	Tan delta, delta tan-delta	Rotor journal	Bearing oil catcher
Works tests on running generator	Y	Y	Y	Y	Y	Y	Y	Y		
Without Excitation, OC and SC with rated voltage and current for generator										
On total winding/ phases at interval 0.2Un for generator							Y	Y		
Condition after dismantling									Y	Y
Works tests on brushless exciter	Y				Y					
PMG works tests	Y		Y		Y					
Full load for PMG and converter assembly										
Converter assembly	Y				Y					
Y = Test applicable										

1.6.1.5 Final acceptance tests for generator/ exciter (Table 3/3)

Item/ Components/ Process	Tests/ Checks									
	Seal rings, liners	Winding overhang	Vibration measurement	Reduced voltage running and No load	Load characteristics	Characteristics of search coil, quadrature	Ripple content	Visual and dimension	Partial discharge, DLA	Routine tests as per IS/ IEC
Works tests on running generator			Y					Y	Y	Y
Without Excitation, OC and SC with rated voltage and current for generator										
On total winding/ phases at interval 0.2Un for generator										
Condition after dismantling	Y	Y								
Works tests on brushless exciter			Y	Y	Y	Y		Y		
PMG works tests				Y	Y					
Full load for PMG and converter assembly										
Converter assembly										
Y =Test applicable										

Note:

1. All generators shall be assembled at works and shall be tested to verify/ ensure design and workmanship in accordance with IEC-34, VDE-0530, IEEE-115, IEEE 43. The manufacturer shall submit detailed test procedure which clearly specify test set up, instruments to be used, acceptance norms (wherever applicable), recording of different parameter, interval of recording, precautions, etc.
2. MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.

1.6.1.6 Type tests

1.6.1.6.1 Type tests (Category-I)

i) Generator

One number assembled generator shall be tested at works as per IEC-34. The tests shall be carried out keeping all conditions/ parameters as close as possible to site conditions with all the built-in instrumentation (like RTD etc.) suitably wired and the readings recorded. During various tests, bearing and shaft overhang vibrations shall also be measured with and without excitation. Recording of various parameters of bearing, seal oil system, gas system, stator water cooling system and environmental conditions (like temperature etc.) shall also be done. The following tests shall be conducted:

- a) Instantaneous short circuit test to determine transient and sub-transient reactance parameters and to ensure stability of winding during sudden short circuit condition
- b) Negative sequence and zero sequence impedance
- c) Voltage waveform factor and Total Harmonic Factor
- d) Short circuit heat run test at rated pressure and cooling parameters with one cooler out of circuit at two third of rated stator current. In case of unsymmetrical cooler configuration, test with all possible variants of one cooler out of circuit shall be carried out.
- e) Vibration measurement on all planes on stator overhang winding at suitable locations on each end and at other critical locations to be decided by purchaser for the following conditions:
 - Open circuit operation
 - Short circuit operation
 - Sudden short circuit conditions
 - Stand still condition - with Hammer test

ii) Brushless excitation system

- a) Exciter - Temperature rise test at peak rating of excitation system. Ceiling duty condition shall also be demonstrated
- b) Permanent magnet generator - Temperature rise test at peak rating of excitation system and ceiling duty condition shall also be demonstrated
- c) Converter assembly (of the exciter field) - Temperature rise test at peak rating of excitation system. Ceiling duty condition shall also be demonstrated

1.6.1.6.2 Type test (Category-II)

Brushless excitation system

- a) Converter assembly of the exciter field
 - Input and output surge withstand capability test
 - Soak test for Electronic module
- b) Degree of protection test for Excitation system panels

1.6.1.7 Site tests

The tests to be conducted on each generator at site shall include but not be limited to those listed below. Any other test considered necessary by the contractor shall also be carried out :

- a) Electrical
 - i) Measurement of the insulation resistance of the stator and rotor windings to the frame and between phases, after drying out the machine and measurement of the polarisation index
 - ii) Measurement of the DC resistance of all windings and embedded temperature detectors
 - iii) Measurement of the insulation resistance of bearings
 - iv) Capacitance measurement and dissipation factor between the winding and body at rated voltage
 - v) Stator -Partial Discharge test @ 1.3 Phase to Neutral voltage for PRPD & Extinction & inception voltages (as per manufacturer's practice).
 - vi) Open circuit and short circuit tests
 - vii) Performance capability of the machine
 - viii) Line charging capacity
 - ix) Short circuit tests on the generator HV end and the generator transformer HV end to check the stability and operation of the generator and the overall (i.e. generator and generator transformer) differential protections and negative phase sequence protection.
 - x) Rotor-IR, Winding resistance, Impedance & RSO test
 - xi) Shaft voltage
- b) Mechanical
 - i) Hydrogen leakage test
 - ii) Vibration test
 - iii) Over-speed test
 - iv) Hydraulic tests on coolers
 - v) Bearing and shaft current test
- c) Load throw-off tests.

1.6.2 Generator Isolated Phase Bus ducts and Neutral Grounding equipment										
1.6.2.1 Generator Isolated Phase Bus duct						(Table 1/2)				
Items/ Components/ Sub-systems	Tests/ Checks									
	Visual and Dimensional Checks	Electrical/ Mechanical/ Chemical Properties	WPS and PQR	NDT, DP or MPI, RT	Painting shade and adhesion test	Galvanizing test as per IS: 2629/ 2633/ 6745	Electrical clearance and Creepage	Functional/ Operational check	Make, Type, Rating, Model, TC, General physical inspection	Routine test as per relevant standard
	Enclosure/ cubicle	Y	Y		Y	Y		Y		
	Busbar flexible connector and dis-connector link	Y	Y							
	Galvanized steel structure and plate	Y					Y			
	Seal of bushing and post insulator IS:5621, 2544	Y	Y					Y		Y
	Welding of enclosure and conductor	Y		Y	Y					
	Gasket, silicagel breather, CT, VT, surge capacitor and arrestor, NGT, NGR, elastomer spring head								Y	Y
	Busbar pressurisation system	Y							Y	
	Complete busduct IS:8084	Y				Y				Y
Y =Test applicable										

1.6.2 Generator Isolated Phase Bus ducts and Neutral Grounding equipment					
1.6.2.1 Generator Isolated Phase Bus duct				(Table 2/2)	
Items/ Components/ Sub-systems	Tests/ Checks				
	Trial assembly at works : Heat run test and Milli volt drop measurement acrossbolted flexible joints	Trial assembly at works : Short circuit withstand test	Trial assembly at works: Impulse withstand test	Trial assembly at works: one minute high voltage power frequency	Trial assembly at works: Air leakage rateand water tightness test
Enclosure/ cubicle					
Busbar flexible connector and disconnector link					
Galvanized steel structure and plate					
Seal of bushing and post insulator IS:5621, 2544					
Welding of enclosure and conductor					
Gasket, silicagel breather, CT, VT, surge capacitor and arrestor, NGT, NGR, elastomer spring head					
Busbar pressurisation system					
Complete busduct IS:8084	Y	Y	Y	Y	Y
Y =Test applicable					
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.					

1.6.2 Generator Isolated Phase Bus ducts and Neutral Grounding equipment											
1.6.2.2 Neutral Grounding Resistor											
Items/ Components/ Sub-systems	Tests/ Checks										
	Visual and Dimensional check	Mechanical properties	Electrical strength	Chemical Composition	Make, Type, Rating, Model, TC, General physical inspection.	Insulation resistance measurement before and after HV Test	HV Test	Degree of protection test	Routine test as per relevant standard	Galvanizing test	Routine test
Resistor	Y	Y	Y	Y		Y					
Cubicle	Y		Y					Y			
Galvanized steel structures (IS:2633/ 2629/ 6745/ 2062)	Y	Y								Y	
Bushing/ Post and support Insulator (IS:2544/ 5621)	Y	Y	Y		Y	Y			Y		Y
Complete NGR (IEEE-32)	Y					Y	Y	Y	Y		Y
Y =Test applicable											
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.											

1.6.2.3 Type Tests

1.6.2.3.1 Type test (Category-I)

The following type tests shall be conducted on one bus-ducts of each rating:

- Heat run test on an assembly of representative sections and fittings with all types of flexibles, CTs etc
- Short circuit withstand test (set up same as for heat run)
- Impulse withstand test (set up shall include typical X-section with flexible connections, 90° bend, CT's in position, seal off bushing, inspection cover and bellows)

- d) One minute high voltage power frequency withstand test (set up as for short circuit test)
- e) Air leakage rate and water tightness test (set up shall include inspection cover, flanged joint and bellow)

1.6.2.3.2 Type test (Category-II)

- Panels, cubicles and marshalling boxes shall be type tested for the degree of protection provided by the enclosure as given below :
 - For 5X-it shall not be possible to insert a thin sheet of paper under gaskets and through enclosure joints.
 - For 4X-It shall not be possible to insert a one mm diameter steel wire into enclosure from any direction without force.
 - For 2X-It shall not be possible to insert a twelve (12) mm dia steel wire into the enclosure from any direction without any force.
 - Test for second digit shall be in line with IS:13947 part-1
- For the equipment and materials such a current transformers, voltage transformers, lightning arresters, grounding transformers, loading resistors, bushings and surge capacitors for which type test are not specified in this section,successful type tests certificate shall be submitted to the purchaser.

1.6.2.4 Site tests

- i) 10% radiography and 100% DP test on all site welded joints of busbar and enclosure (for Root & Final welding).
- ii) Milli-volt drop test
- iii) Ratio and polarity tests on current and voltage transformers
- iv) Insulation measurement of equipment and all wiring
- v) Functional test on pressurization (Tightness test as per ANSI 37.20)
- vi) One minute high potential power frequency withstand test at 75% of rated test voltage
- vii) Enclosure Isolation test with support frame.
- viii) Air pressurisation test.

1.6.3 Power Transformers
1.6.3.1 Generator Transformers/ Unit auxiliary transformers/ Station transformers)
Please refer QAP for transformer mentioned in Annexure – A of Model Quality Assurance Plan (MQAP) For Equipment Of Substation, Switchyards And Transmission Lines

1.6.4 HT BUSDUCT										
1.6.4.1 11kV, 6.6kV & 3.3kV Segregated Phase Bus ducts										
Items/ Components/ Sub-systems	Tests/ Checks									
	Visual and Dimensional Checks	Electrical, Mechanical, Chemical properties	WPS and PQR	NDT, DP or MPI, RT	Paint shade and Adhesion test	Galvanizing Test as per IS: 2629/ 2633/ 6745	Electrical clearance and creepage	Functional/ Operational check	Make, Type, Rating, Model, TC, General physical inspection	Routine tests on complete busduct
Enclosure/ cubicle	Y	Y		Y	Y					
Arc test at rated short circuit current										
Busbar flexible connector and disconnector link	Y	Y								
Steel structure and plate IS:2062		Y				Y				
Bushing, post and support insulator (IS:9431 and 2544)	Y						Y			Y
Welding of enclosure and conductor	Y		Y							
Gasket, silicagel breather								Y	Y	Y
Complete busduct IS:8084	Y				Y					Y
Y =Test applicable										
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.										

1.6.4.2 Type test (Category-II)

- (a) All bus-ducts supplied shall be of type tested quality.
- (b) Type test reports for the following shall be submitted:
 - One minute power frequency voltage withstand test
 - Temperature rise test. Milli volt drop shall also be measured across boltedflexible joint
 - Impulse voltage withstand test

- Short time current test
- Water tightness test (as per IS:8084)
- Arc test at rated short circuit for rated time
- Air leakage test

1.6.4.3 Site tests

The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out

- Power frequency voltage withstand test
- Air leakage test
- Insulation resistance measurement of equipment and all wiring
- Milli-volt drop
- Tightness test for bus bar joints.

1.6.5 Auxiliary Service Transformers

1.6.5.1 Oil-filled Outdoor Transformers

	Tests/ Checks													
Items/ Components/ Sub-systems	Visual and Dimensional checks	Mechanical properties	Electrical strength	Thermal properties	Chemical Composition	Compatibility with oil	Core Loss (on first Job) Hot Spot	NDT, DP or MPI, UT	Ageing Test	Voltage Ratio, Vector Group andPolarity, Magnetic balance test	Make, Type, Rating, Model, TC, General physical inspection.	WPS and PQR	Vacuum and Pressure Test	Routine Tests
Tank, HV and LV Cable Box/ Flange throat	Y	Y						Y				Y	Y	
Conservator/ Radiator/ Cooler/ Pipes	Y	Y						Y				Y		
Copper Conductor (IS:191)	Y	Y	Y		Y									
Insulating Material	Y	Y	Y	Y	Y	Y								
CRGO Lamination and Built Core	Y	Y	Y		Y	Y	Y							
Bushing / Insulator (IS:2544/ 5621)	Y	Y									Y			Y
Gasket	Y				Y	Y			Y					
Transformer oil (IS:335)														Y
Off-circuit tap changer	Y										Y			Y
Core coil assembly and pre-tanking	Y									Y				
Marshalling box	Y	Y						Y						Y
WTI, OTI, MOG, PRD, Breather, Terminal Connector, Bucholz Relay, Globe and Gate Valve,	Y										Y			
Welding (ASME Sect-IX)	Y											Y		
Complete Transformer (IS:2026)	Y									Y				Y

Y =Test applicable

Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.

1.6.5.2 Dry Type Indoor Transformers												
Items/ Components/ Sub-systems	Tests/ Checks											
	Visual and dimensional check	Mechanical properties	Electrical strength	Thermal properties	Chemical properties	Core loss (on first job), Hot	NDT, DP or MPI	Voltage ratio, vector group and polarity	Make, Type, Rating, Model, TC, General physical inspection	WPS and PQR	Measurement of capacitance and tan delta between winding	Routine tests
Enclosure door, HV and LV Cable box/ Flange throat	Y	Y							Y			
Copper conductor	Y	Y	Y		Y							
Insulating material	Y			Y	Y							
CRGO lamination and built core	Y					Y						
Bushing/ Insulator (IS:2544/ 5621)	Y								Y			Y
Gasket	Y								Y			Y
Off-circuit tap changer	Y								Y			
Core coil assembly	Y							Y				
Marshalling box	Y											
WTI, Thermister, Terminal connector	Y								Y			
Welding										Y		
Complete transformer (IS:11171)	Y							Y			Y	Y
Y =Test applicable												
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.												

1.6.5.3 Type tests

1.6.5.3.1 Type tests (Category-I)

The transformer shall be subjected to the following type tests for rating above 2 MVA.

- Short Circuit test – This test shall be carried out after conducting the routine tests. Rest of the type tests shall be conducted after successful short circuit testing.
- Noise level measurement
- Enclosure IP Degree test (for Dry Type Transformers only)
- Measurement of zero phase sequence impedance
- Measurement of the harmonics of no load current
- Temperature rise
- Lightning impulse voltage test on all the three limbs as per Cl. 13 of IS:2026 Part-III, 2018

1.6.5.3.2 Additional Type tests (Category-I) (for Oil Filled Transformers only)

- I. Vacuum Test on Transformer Tank - One Transformer tank of each rating shall be subjected to the specified vacuum. The tank designed for full vacuum shall be tested at an internal pressure of 0.35 kg/ cm² (absolute, 250 mm of Hg) for one hour. The permanent deflection of flat plates after the vacuum has been released shall not exceed the values specified below :

Horizontal Length of Flat Plate (mm)	Permanent deflection (mm)
Upto and including 750	5.0
751 to 1250	6.5
1251 to 1750	8.0
1751 to 2000	9.5
2001 to 2250	11.0
2251 to 2500	12.0
2501 to 3000	16.0
Above 3000	19.0

- II. Pressure Test on Transformer Tank - One transformer tank of each rating shall be subjected to a pressure corresponding to twice the normal pressure of normal pressure plus 0.35 kg/ cm² whichever is lower measured at the base of the tank and maintained for an hour. The permanent deflection of flat plates after the excess pressure has released shall not exceed the figure for vacuum test.
- III. Oil Leakage Test - All tanks and oil filled compartments shall be tested for oil tightness by oil of a viscosity not greater than that of insulating oil to IS:335, at the specified ambient temperature and subjected to a pressure equal to the normal pressure plus 35 KN/ m² (51 lb/ inch²) measured at the base of the tank. This pressure shall be maintained for a period of not less than 12 hours, during which time no leakage shall occur.
- IV. Tests on associated equipment - Porcelain bushings, bushing current transformers, winding temperature indicating devices, dial thermometers, buchholz relays, ON/OFF load tap changer, auxiliary motors and motor starting contactors, coolers, control device, Insulating oil and other associated equipment shall be tested in accordance with relevant IS. The following checks shall also be made before dispatch:
- ii) Check for proper packing and preservation of accessories like radiators, bushings, explosion vent, dehydrating breather, Buchholz relay, conservator etc.
 - iii) Check for proper provision of bracing to arrest the movement of core and winding assembly inside the tank.
 - iv) Test for Gas tightness and derivation of leakage rate to ensure adequate reserve gas capacity during transit and storage.

a. Type tests (Category-II)

All indoor transformers below and equal to 2 MVA rating to be supplied shall be of type tested quality. The test reports of the following type tests shall be submitted.

- v) Short Circuit test – This test shall be carried out after conducting the routine tests. Rest of the type tests shall be conducted after successful short circuit testing.
- vi) Noise level measurement
- vii) Measurement of zero phase sequence impedance
- viii) Measurement of the harmonics of no load current
- ix) Temperature rise
- x) Lightning impulse voltage test on all the three limbs as per Clause 13 of IS:2026 Part-III, 2018

1. Site tests

- i) The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out
 - a) Dry out test
 - b) Resistance measurement of windings
 - c) Ratio test
 - d) Vector group test
 - e) Tap changer test
 - f) Buchholz relay test
 - g) Low oil level alarm.
 - h) Temperature indicators
 - i) Marshalling kiosk
 - j) Protective relays
 - k) Magnetizing current
 - l) Door interlock tests
- ii) The following additional checks shall be made to see the following:
 - a) All oil valves are in correct position closed or opened as required
 - b) All air pockets are cleared
 - c) Thermometer pockets are filled with oil
 - d) Oil is at correct level in the bushing, conservator, divertor switch, tank etc.
 - e) Earthing connections are made
 - f) Colour of silica gel is blue
 - g) Bushing arcing horn is set correctly
 - h) CT polarity is correct (when bushing mounted CTs are provided)

1.6.6 Motors													
Item/ Components/ Sub-system	Tests/ Checks												
	Visual	Dimensional	Make, Type, Rating, TC, General	physical	Mechanical, Chemical properties	NDT, DP or MPI, UT	Metallography	Electrical characteristics	Welding/ Brazing (WPS/ PQR)	Heat treatment	Magnetic characteristics	Hydraulic, Leak, Pressure test	Thermal characteristics
Plates for stator frame, end shield, spider etc.	Y	Y	Y	Y					Y				
Shaft	Y	Y	Y	Y	Y	Y			Y				
Magnetic material	Y	Y	Y	Y	Y		Y			Y		Y	
Rotor copper/ Aluminium	Y	Y	Y	Y		Y	Y		Y				
Stator copper	Y	Y	Y	Y			Y		Y			Y	
SC ring	Y	Y	Y	Y	Y	Y	Y	Y	Y				
Insulating material	Y		Y	Y			Y					Y	
Tubes for cooler	Y	Y	Y	Y	Y				Y		Y		
Sleeve bearing	Y	Y	Y	Y	Y				Y		Y		
Stator, Rotor coils	Y	Y	Y				Y	Y					Y
Castings, stator frame, terminal box and bearing housing etc.	Y	Y	Y	Y	Y			Y					
Fabrication and machining of stator, rotor, terminal box	Y	Y			Y				Y				
Wound stator	Y	Y					Y	Y					Y

Rotor complete	Y	Y					Y							Y	Y				
Stator, Rotor, Terminal Box assembly	Y	Y					Y												
Accessories, RTD, BTD,CT, Brushes, Diodes, space heater, antifriction bearing, cable glands, lugs, gaskets etc.	Y	Y	Y																
Complete motor (IS: 325/ IS:4722/ IS:9283)	Y	Y	Y				Y									Y	Y	Y	Y 1
Y =Test applicable, Y1 = for 11kV, 6.6 kV and 3.3kV motors only																			
Notes																			
1)	This is an indicative list of tests/ checks. The manufacture is to furnish the detailed Quality Plan indicating the practices and procedure followed along with relevant supporting documents during QP finalization. However QP approval is not envisaged for 415V motors upto 50 KW.																		
2)	Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.																		

1.6.6.1 Type tests

1.6.6.1.1 Type tests (Category-I)

11kV, 6.6kV and 3.3kV motors

The following type tests shall be conducted on each type and rating of 11kV, 6.6kV 3.3kV motor:

- No load saturation and loss curves upto approximately 115% of rated voltage
- Measurement of noise at no load
- Momentary overload test (subject to test bed constraint)
- Full load test (subject to test bed constraint)
- Temperature rise test at rated conditions (During heat run test, bearing temperature, winding temperature, core temperature, coolant flow and its temperature shall also be measured. In case the temperature rise test is carried at load other than rated load, specific approval for the test method and procedure is required to be obtained. Wherever ETD's are provided, the temperature shall be measured by ETD's also for the record purpose).

1.6.6.1.2 Type tests (Category-II)

- i) 11kV, 6.6kV and 3.3kV motors

The following type test reports shall be submitted for each type and rating of motor.

- Degree of protection test for the enclosure followed by IR, HV and no load run test
- Terminal box-fault level withstand test for each type of terminal box
- Type test on Elastimold termination kit as per relevant standard

- ii) 415V motors

415V motors shall be of type tested quality. The type test reports for type tests as per relevant standards shall be submitted for each type and rating of 415V motors (above 100KW only)

- iii) Impulse Voltage test (on the sample coil after placing it in stator core at $(4U+5kV)$ and with at least five impulse of 1.2/50 micro-second wave, for MV motors only, where U is the line to line voltage in kV).

1.6.6.2 Site Tests

The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out

- i) Measurement of vibration.
- ii) Measurement of insulation resistance and polarization index.
- iii) Measurement of winding resistance of stator winding
- iv) HV Power frequency test for 1 minute preceded and followed by measurement of IR values.
- v) Capacitance & Tan Delta & Tip Up till rated voltage
- vi) Measurement of full load current.
- vii) Test running of the motors, checking the temperature rise and identifying the hot spot etc.

1.6.7 11kV, 6.6kV and 3.3kV Switchgear (Table 1/2)

Items/ Components/ Sub-systems	Tests/ Checks													
	Make, Type, Model, Rating and TC	Electrical Properties	Mechanical properties	Chemical properties	Dimensions and Finish	Functional and operational features	Item to conform to relevant Standards	Pretreatment as per IS 6005	Paint shade, thickness, adhesion and finish	Functional Checks	HV and IR Test	Degree of Protection - Routine test	CB Operation timing check	All Routine Tests as per relevant standards
Aluminum busbar material (IS: 5082)	Y	Y	Y	Y	Y		Y							
Copper busbar material (IS:613)	Y	Y	Y	Y	Y		Y							
Bus bar support insulation (IS:9431)	Y	Y	Y		Y		Y				Y			
HT Circuit Breaker (IEC:56)	Y				Y	Y	Y			Y			Y	Y
HT Contactors (IS:9046)	Y				Y	Y	Y			Y				Y
Protection and auxiliary relays (IS:3231/ 8686)	Y				Y	Y	Y			Y				Y
HT CT's and PT's (IS:2705/ 3156)	Y				Y		Y							Y
HT Fuses (IS:9385)	Y				Y	Y	Y							
Surge arrester (IEC:99-4)	Y				Y		Y							Y
LT Contactors (IS:13947)	Y				Y	Y	Y			Y				
Control and selector switches (IS:6875)	Y				Y	Y	Y			Y				
Indicating meters (IS:1248)	Y				Y	Y	Y			Y				Y
Indicating lamps (IS:13947)	Y				Y	Y	Y			Y				
Push buttons (IS: 4794)	Y				Y	Y	Y			Y				
Control transformer (IS:12021)	Y				Y	Y	Y							Y
LT fuses (IS:13703)	Y				Y	Y	Y							
Energy meters (IS:722)	Y				Y	Y	Y							Y
Transducers (IEC:60688)	Y				Y	Y	Y							Y
Y =Test applicable														
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.														

1.6.7 11kV, 6.6kV and 3.3kV Switchgear													(Table 2/2)			
Items/ system	Components, Sub-	system	Tests/ Checks													
			Make, Type, Model, Rating and TC	Electrical properties	Mechanical properties	Chemical properties	Dimensions and Finish	Functional and Operational features	Item to conform to relevant standards	Pretreatment as per IS: 6005	Paint shade, thickness, adhesion and	Functional checks	HV and IR Test	Degree of Protection - Routine test	CB Operation timing check	All Routine Tests as per relevant standard
Diodes			Y	Y				Y	Y			Y				
Terminal Blocks			Y	Y				Y	Y							
Synthetic Rubber Gasket (IS:11149/ 3400)			Y	Y			Y		Y							
Breaker Handling Trolley			Y				Y	Y			Y	Y				
HT Switchgear Panel (IS:3427)			Y				Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Y =Test applicable																
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.																

1.6.7.1 Type test (Category-II)

All 11kV, 6.6kV and 3.3kV Switchgears supplied shall be of type tested quality.

- a) Type test reports for the following shall be submitted:
 - i) Circuit breaker/ circuit breaker panels, of each voltage class and current rating:
 - Short circuit duty test on circuit breaker, mounted inside the panel offered alongwith CTs, bushing and separators
 - Short time withstand test on circuit breaker, mounted inside panel offered together with CTs, bushings and separators
 - Power frequency withstand test on breaker mounted in side panel

- Lightning impulse withstand test on breaker mounted in side panel
- Temperature rise test on breaker and panel together. For this test, the test set up shall include three panels with breakers, the test breaker and panel being placed in the Center.

The adjacent panels shall also be loaded to their rated current capacity. Alternatively the test panel may be suitably insulated at the sides, which will be adjoining to other panels in actual site configuration

- Test to verify pressure relief devices operation of the panel. This shall be done on one panel of each voltage class
- measurement of resistance of main circuit
- Mechanical endurance test on breaker
- Mechanical operation test

ii) Contactor and contactor panels of each type and rating:

- Verification of rated making and breaking capacities of the contactor
- Short time withstand test of panel
- Power frequency test on the contactor mounted in side panel
- Lightning impulse voltage withstand test of the contactor mounted inside panel
- Measurement of resistance of main circuit
- Test to confirm coordination between fuse and contactor

iii) Surge arrestor/ lightning arrestor (as applicable) of each type:

- Standard lightning impulse voltage spark-over test
- Front of wave sparkover test (For surge arrestor used along with motor feeder, this test shall be carried out with a voltage wave having a rate of rise of not less than 142kV/ micro-second for arrestor used in 11kV system, 85kV/ micro-second for 6.6kV system and 44kV/ micro-second for arrestor used in 3.3kV system).
- Power frequency sparkover and temporary overvoltage test
- Residual voltage test
- Operating duty test
- Current impulse withstand test (long duration and high current impulse)

- Pressure relief test

iv) Short circuit withstand test of earthing device (truck/ switch)

b) Shop testing of components and bought out items

For various bought out items like CT, VT, relays, meters, surge arrestors etc. for which identification can be physically verified and test certificate from manufacturer can be co-related, following shall be indicated in the QP.

- Review of manufacturer's Test Certificate (TC) as per relevant standard and compliance to requirements of this specification
- Physical check and functional/ operational check to ensure that item is fit for assembly on the switch board cubicle.

For all other items where identification has not been envisaged by manufacturers, a certificate of conformance to be obtained from manufacturer, and component identification shall be tagged as per approved drawing/ schematic. Either of the following or both shall be adopted .

- Random sample to be tested either in-house or at an independent laboratory for all tests envisaged in relevant standard.
- Physical check and functional/ operation check to ensure the item is fit for assembly on the switch board cubicle.

1.6.7.2 Site tests

The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out

a) S.N. General

- i) Check name plate details according to specification
- ii) Check for physical damage
- iii) Check tightness of all bolts, clamps and connecting terminals
- iv) Check earth connections
- v) Check cleanliness of insulators and bushings
- vi) Check heaters are provided
- vii) HV test on complete switchboard with CT and breaker/ contactor in position
- viii) Check all moving parts are properly lubricated
- ix) Check for alignment of busbars with the insulators to ensure alignment and fitness of insulators
- x) Check for interchangeability of breakers/ contactors
- xi) Check continuity and IR value of space heater
- xii) Check earth continuity for the complete switchgear board

- b) S.N. Circuit Breaker/ Contactors
- i) Check alignment of trucks for free movement
 - ii) Check correct operation of shutters
 - iii) Check slow closing operation (if provided)
 - iv) Check control wiring for correctness of connections, continuity and IR values
 - v) Manual operation of breakers completely assembled
 - vi) Power closing/ opening operation, manually and electrically at extreme condition of control supply voltage
 - vii) Closing and tripping time
 - viii) Trip free and anti-pumping operation
 - ix) IR values, resistance and minimum pick up voltage of coils
 - x) Simultaneous closing of all the three phases
 - xi) Check electrical and mechanical interlocks provided
 - xii) Checks on spring charging motor, correct operation of limit switches and time of charging
 - xiii) Check SF₆ pressure/ Leakage current in case of VCB
 - xiv) All functional checks
- c) S.N. Current Transformers
- i) Insulation resistance between windings and winding terminals to body
 - ii) Polarity tests
 - iii) Ratio identification checking of all ratios on all cores by primary injection of current
 - iv) Magnetisation characteristics (Knee point) and secondary winding resistance
 - v) Spare CT cores to be shorted to earth
- d) S.N. Voltage Transformers
- i) Insulation resistance test
 - ii) Ratio test on all cores
 - iii) Polarity test
 - iv) Line connections as per connection diagram
- e) S.N. Cubicle Wiring
- i) Check all switch developments
 - ii) It should be made sure that the wiring is as per relevant drawings. All interconnections between panels shall similarly be checked
 - iii) All the wires shall be meggered to earth
 - iv) Functional checking of all control circuit e.g. closing, tripping interlock, supervision and alarm circuit including proper functioning of component/equipment
 - v) Check terminations and connections

- vi) Wire ducting
- vii) Gap sealing and cable bunching
- f) S.N. Relays
 - i) Check internal wiring
 - ii) Insulation resistance between all terminals and body
 - iii) Insulation resistance between AC to DC terminals
 - iv) Check operating characteristics by secondary injection
 - v) Check minimum pick up voltage of DC coils
 - vi) Check operation of electrical/ mechanical targets
 - vii) Check CT connections with particular reference to their polarities for differential type relays
 - viii) Relay pickup & timing test as per Relay settings
- g) S.N. Meters
 - i) Insulation resistance of all insulated portions
 - ii) Check CT and VT connections with particular reference to their polarities for power type meter

1.6.8 415V Switchgear and Bus duct													
1.6.8.1 415V Switchgear (MCC, PCC, ACDB, DCDB, Local push button station, Local motor starters)													(Table 1/2)
Items/ Components/ Sub-system	Tests/ Checks												
	Make, Model, Type, Rating and TC	Dimensions and Finish	Electrical properties	Mechanical properties	Chemical properties	Functional and Operational features	Item to conform to relevant standards	Pretreatment as per IS 6005	Paint shade, adhesion, thickness and finish	Functional checks	Milli-volt drop Test	IR – HV – IR Test	Degree of protection Routine test
													All Routine tests as per relevant standards
Sheet steel (IS:513)	Y	Y		Y	Y		Y						
Aluminum bus bar material (IS:5082)	Y	Y	Y	Y	Y		Y						
Copper bus bar material (IS:613)	Y	Y	Y	Y	Y		Y						
Support insulator (IS:943, IS:10912, IEC: 660)	Y	Y	Y	Y			Y						
Air circuit breaker(IS:13947)	Y	Y				Y	Y			Y	Y		Y
Energy meters (IS:722)	Y	Y				Y	Y			Y			Y
Power and auxiliary contactor (IS:13947)	Y	Y				Y	Y			Y			
Protection and auxiliary relays (IS:3231)	Y	Y				Y	Y			Y			Y
Control and selector switches (IS:6875)	Y	Y				Y	Y			Y			
CT and VT (IS 2705/ 3156)	Y	Y					Y						Y
MCCB (IS:13947)	Y	Y					Y			Y			
Indicating meters (IS:1248)	Y	Y				Y	Y			Y			Y
Indicating lamps (IS:13947)	Y	Y				Y	Y			Y			
Air break switches (IS:13947)	Y	Y				Y	Y			Y			
Control terminal blocks	Y	Y				Y	Y						
Y =Test applicable													
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.													

1.6.8.1 415V Switchgear (Table 2/2)
(MCC, PCC, ACDB, DCDB, Local push button station, Local motor starters)

Items/ Components/ Sub-system	Tests/ Checks													
	Make, Model, Type, Rating and TC	Dimensions and Finish	Electrical properties	Mechanical properties	Chemical properties	Functional and Operational Features	Item to conform to relevant standards	Pretreatment as per IS: 6005	Paint Shade, Adhesion, Thickness and	Functional checks	Milli-volt drop Test	IR – HV – IR Test	Degree of Protection - Routine test	All Routine tests as per relevant standards and
Fuse (IS:13703)	Y	Y				Y	Y			Y				
Control transformer (IS:12021)	Y	Y				Y	Y			Y				Y
Push Buttons (IS:4794)	Y	Y				Y	Y			Y				
Transducer (IEC:60688)	Y	Y				Y	Y			Y				Y
MCB (IS:8828)	Y	Y				Y	Y			Y				
Breaker handling trolley	Y	Y				Y			Y	Y				Y
Synthetic rubber gasket (IS:11149)	Y	Y		Y	Y		Y							
415V Switchgear (IS:8623)	Y	Y				Y	Y	Y	Y	Y	Y	Y	Y	Y
Y =Test applicable														
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.														

1.6.8.2 415V Bus duct

Items/ Components	Tests/ Checks														
	Dimension and surface finish	Make, Type, Rating and Test certificate (TC)	Electrical properties	Mechanical properties	Chemical properties	Item to conform to relevant IS	WPS approval, Welder qualification	Weld quality check (DP test and X-ray)	Paint shade, Thickness, Adhesion and finish	Tightness by torque measurement	Electrical clearances	Galvanizing test as per IS 2629/ 2633	IR – HV – IR test	Phase sequence check	Degree of protection routine test
Aluminum sheets/ plates/ strips/ flexibles/ tubes (IS:5082/ 737)	Y	Y		Y	Y	Y	Y	Y							
CRCA flats/ ISMC (IS:2062)	Y	Y		Y	Y	Y									
Neoprene/ synthetic rubber gaskets (IS:11149/ 3400)	Y	Y		Y	Y										
Rubber bellows (IS:3400)	Y	Y		Y	Y										
Support insulator (BS:2782, IEC:660, IS:10912)	Y	Y	Y	Y											
Galvanized structure and GI earthing flat (IS:2629/ 2633/ 4749)	Y	Y				Y						Y			
Space heater and Thermostat		Y	Y												
LT busduct (IS:8623 Part 2)	Y	Y				Y	Y	Y	Y	Y	Y		Y	Y	Y

Y = Test applicable

Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.

1.6.8.3 Type test (Category-II)

All 415V Switchgears and bus duct supplied shall be of type tested quality.

- a) Type test reports for the following tests shall be submitted on each type and rating of 415V switchgear and AC, DC distribution boards:
 - i) Short time withstand test with circuit breaker mounted inside the switchgear panel
 - ii) Temperature rise test
 - iii) Arc test at rated short circuit capacity
 - iv) Type II - Short circuit co-ordination test for any three ratings of MCC module
 - v) Test sequence-1 and combined test sequence shall be carried out on each rating of circuit breaker mounted inside the panel
 - vi) Degree of protection test
- b) Type test reports for the following tests shall be submitted on each type and rating of 415V bus duct:
 - i) Heat run test on an assembly of representative sections and fittings
 - ii) Short circuit test for a duration and current on an assembly of representative section
 - iii) One minute high potential power frequency voltage withstand test
 - iv) Air and water tightness test on a typical section

1.6.8.4 Site tests

The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out

- a) S.N. General
 - i) Check name plate details according to specification
 - ii) Check for physical damage
 - iii) Check tightness of all bolts, clamps and connecting terminals
 - iv) Check earth connections
 - v) Check cleanliness of insulators and bushings
 - vi) Check heaters are provided
 - vii) HV test on complete switchboard with CT and breaker/ contactor in position
 - viii) Check all moving parts are properly lubricated

- ix) Check for alignment of busbars with the insulators to ensure alignment and fitness of insulators
 - x) Check for interchange ability of breakers/ contactors
 - xi) Check continuity and IR value of space heater
 - xii) Check earth continuity for the complete switchgear board.
 - xiii) Check Closed door operation - service, test & isolated positions, if applicable.
- b) S.N. Circuit Breaker/ Contactors
- i) Check alignment of trucks for free movement
 - ii) Check correct operation of shutters
 - iii) Check slow closing operation (if provided)
 - iv) Check control wiring for correctness of connections, continuity and IR values
 - v) Manual operation of breakers completely assembled
 - vi) Power closing/ opening operation, manually and electrically at extreme condition of control supply voltage
 - vii) Closing and tripping time
 - viii) Trip free and anti-pumping operation
 - ix) IR values, resistance and minimum pick up voltage of coils
 - x) Simultaneous closing of all the three phases
 - xi) Check electrical and mechanical interlocks provided
 - xii) Checks on spring charging motor, correct operation of limit switches and time of charging
 - xiii) Check SF₆ pressure/ vacuum (as applicable)
 - xiv) All functional checks
- c) S.N. Current Transformers
- i) Insulation resistance between windings and winding terminals to body
 - ii) Polarity tests
 - iii) Ratio identification checking of all ratios on all cores by primary injection of current
 - iv) Magnetisation characteristics (knee Point) and secondary winding resistance
 - v) Spare CT cores, if any to be shorted and earthed
- d) S.N. Voltage Transformers
- i) Insulation resistance test
 - ii) Ratio test on all cores
 - iii) Polarity test
 - iv) Line connections as per connection diagram
- e) S.N. Cubicle Wiring
- i) Check all switch developments
 - ii) It should be made sure that the wiring is as per relevant

- drawings. All interconnections between panels shall similarly be checked
- iii) Insulation resistance of all wires with respect to earth
 - iv) Functional checking of all control circuit e.g. closing, tripping interlock, supervision and alarm circuit including proper functioning of component/equipment
 - v) Check terminations and connections
 - vi) Wire ducting
 - vii) Gap sealing and cable bunching
- f) S.N. Relays
- i) Check internal wiring
 - ii) Insulation resistance between all terminals and body
 - iii) Insulation resistance between AC and DC terminals
 - iv) Check operating characteristics by secondary injection
 - v) Check minimum pick up voltage of DC coils
 - vi) Check operation of electrical/ mechanical targets
 - vii) Check CT connections with particular reference to their polarities for differential type relays
 - viii) Relay pickup & timing as per Relay settings
- g) S.N. Meters
- i) Insulation resistance of all insulated portions
 - ii) Check CT and VT connections with particular reference to their polarities for power type meter
- h) S.N. Bus duct
- i) Visual inspection
 - ii) Power frequency voltage withstand test
 - iii) Insulation resistance measurement of equipment and all wiring
 - iv) Milli-volt drop

1.6.9 Power and Control Cables :														
1.6.9.1 11kV, 6.6kV and 3.3kV Power Cables													(Table 1/2)	
Item/ Components/ Sub-system	Tests/ Checks													
	Make, Type, Rating and TC	Dimension/ surface finish	Mechanical properties	Chemical composition	Spark test	Curing properties	Electrical properties	Hot set test, Eccentricity and Ovality	Lay length and sequence	Armour coverage, cross over, looseness, gap between two wire	Sequential marking/ surface finish/ cable length	Tensile strength and elongation before and after ageing on outer sheath and insulation	Thermal stability on outer sheath	Metallic screening (if applicable)
Aluminium (IS-8130)	Y	Y	Y	Y			Y							
Semi conducting compound	Y		Y			Y	Y							
XLPE compound(IS-7098-Part-II)	Y		Y			Y	Y					Y		
FRLS HR PVC compound (IS- 5831, ASTM-D2843, ASTM-2863, IEC-754 Part-1)	Y		Y									Y	Y	
Triple extrusion and curing/ Manufacturing of core		Y			Y			Y	Y					
Copper tape	Y	Y	Y				Y							
Polyster tape	Y	Y												
Armour wire/strip	Y	Y	Y											
Copper tapping	Y	Y						Y						
Inner sheath	Y	Y												
Armouring										Y				Y
Outer sheathing		Y									Y	Y	Y	
Power cable (Finished) (IS :7098 Part-II) IEC:332, IS-5831, ASTM-D2843, ASTM-2863, IEC-754 Part-1) Cable Drum Rewinding	Y								Y	Y	Y	Y	Y	
Wooden drum (IS:10418)/ Steel drum		Y												

Y =Test applicable

Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.

1.6.9.1 11kV, 6.6kV and 3.3kV Power cables (Table 2/2)				
Item/ Components/ Sub-system	Tests/ Checks			
	Anti termite coating on wooden drums	Constructional requirements feature as per specification	Routine and acceptance test as per relevant standard and specification	FRLS tests
Aluminium (IS-8130)				
Semi conducting compound				
XLPE compound (IS-7098- Part-II)				
FRLS HR PVC compound (IS-5831, ASTM-D2843, ASTM-2863, IEC-754 Part-1)				Y
Triple extrusion and curing/ Manufacturing of core				
Copper tape				
Polyster tape				
Armour wire/strip				
Copper tapping				
Inner sheath				
Armouring				
Outer sheathing		Y		Y
Power cable (Finished) (IS : 7098 Part-II) IEC: 332, IS-5831, ASTM-D2843, ASTM-2863,IEC-754 Part-1)		Y	Y	Y
Cable Drum Rewinding				
Wooden drum(IS-10418) /Steel Drum	Y			
Y =Test applicable				
Notes				
1)	Additional acceptance tests like FRLS, thermal stability, tensile strength and elongation after ageing shall be done on one sample/ lot			
2)	Length measurement/ surface finish/ eccentricity/ ovality shall be checked on one length/ size/ lot			
3)	Routine test shall be carried out on each drum length as per specifications.			
4)	MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.			

1.6.9.2 1.1 kV PVC and XLPE cables			(Table1/2)									
Item/ Components/ Sub-system	Tests/ Checks											
	Make, Rating, Type and TC	Dimension/surface finish	Mechanical properties	Chemical composition	Electrical properties	Spark test	Hot set test (XLPE)	Lay length/ sequence	Armour coverage, cross over, looseness, Gap between two armour wire/ strip	Sequential marking/ surface finish/ cable length	Tensile strength, elongation before and after ageing of insulation and outer sheath	Thermal stability of insulation and outer sheath *
Aluminum (IS-8130)	Y	Y	Y	Y	Y							
PVC compound (IS-5831)	Y		Y		Y						Y	
XLPE compound (IS-7098 Pt-I)	Y		Y		Y		Y				Y	
FRLS HR PVC compound (IS-5831)ASTM-D-2843/ ASTM-D-2863 IEC-754 Part-I	Y		Y								Y	
Armour wire/strip (IS-3975)	Y	Y	Y									
Insulated core		Y				Y	Y					Y
Laid up core		Y						Y				
PVC Inner sheath		Y										
Armouring		Y							Y			
Outer sheath		Y								Y	Y	Y
Finish cable (IS-1554 and 7098 – Part-1) ASTM-D-2843/ ASTM-D-2863 IEC-754 Part-I Swedish Chimney SS 4241475 for (F3 category) Flammability test IEC-332 Part –3 Cat-B Cable Drum Rewinding	Y	Y							Y	Y	Y	Y
Wooden drum (IS-10418) / Steeldrum		Y										
Y =Test applicable												
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.												

1.6.9.2 1.1 kV PVC and XLPE CABLES		(Table 2/2)			
Item/ Components/ Sub-system	Tests/ Checks				
	Anti termite treatment on wooden drums	Constructional/ requirement as per specification	Routine and acceptance test as per relevant standard and specification	FRLS Tests	
Aluminum (IS-8130)					
PVC Compound (IS-5831)					
XLPE Compound (IS-7098 Pt-I)					
FRLS PVC Compound (IS-5831) ASTM-D-2843/ASTM-D-2863 IEC-754 Part-I					Y
Armour wire/strip (IS-3975)					
Insulated Core					
Laid up core					
PVC Inner sheath					
Armouring					
Outer sheath					Y
Finish cable (IS-1554 & 7098 – Part-1) ASTM-D-2843/ASTM- D-2863 IEC-754 Part-I Swedish Chimney SS 4241475 for (F3 category) Flammability test IEC-332 Part –3 Cat-B Cable Drum Rewinding		Y	Y		Y
Wooden drum (IS-10418) / Steel drum	Y				
Y =Test applicable					
Notes					
1)	(*) Not applicable for XLPE insulation				
2)	Additional acceptance tests like FRLS, thermal stability, tensile strength and elongation after ageing shall be done on one sample / lot.				
3)	Length measurement / surface finish shall be checked on one length/ size/ lot				
4)	Routine test shall be carried out on each drum length as per specifications.				
5)	MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.				

1.6.9.3 1.1 kV PVC Control cables		(Table 1/2)									
Item/ Components/ Sub-system	Tests/ Checks										
	Make, Type, Rating, Test certificate (TC)	Dimension/ surface finish	Mechanical properties	Chemical composition	Electrical properties	Spark test	Lay length/ sequence	Armour coverage, cross over, looseness, gap between two armour wire	Sequential marking/ surface finish/ cable length	Tensile strength, elongation before and after ageing of insulation and outer sheath	Thermal stability of insulation and outer sheath
Copper conductor (IS-8130)	Y	Y	Y	Y	Y						
PVC compound (IS-5831)	Y		Y		Y					Y	
FRLS PVC compound IS-5831 ASTM-D-2843/ ASTM-D-2863 IEC-754 Part-1	Y		Y							Y	
Armour wire/strip (IS-3975)	Y	Y	Y								
Insulated core		Y				Y	Y				
Laid up core		Y					Y				
PVC Inner sheath		Y									
Armouring		Y						Y			
Outer sheath		Y							Y	Y	Y
Finished cable (IS-1554) ASTM-D-2843/ ASTM-D-2863 IEC-754 Part-1 Swedish Chimney: SEN SS 424-1475(F3 category) Flammability test IEC-332 Part-3 Cat-B Cable Drum Rewinding	Y	Y						Y	Y	Y	Y
Wooden drum(IS:10418)/ Steel drum		Y									
Y =Test applicable											
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.											

1.6.9.3 1.1 kV PVC Control cables		(Table 2/2)			
Item/ Components/ Sub-system	Tests/ Checks				
	Anti termite treatment on wooden drums	Constructional feature as per specification	Routine and Acceptance test as per relevant standard and specification	FRLS test	
Copper conductor (IS-8130)					
PVC compound (IS-5831)					
FRLS PVC compound IS-5831 ASTM-D-2843/ ASTM-D-2863 IEC-754 Part-1					Y
Armour wire/strip (IS-3975)					
Insulated core					
Laid up core					
PVC Inner sheath					
Armouring					
Outer sheath					Y
Finished cable (IS-1554) ASTM-D-2843/ ASTM-D-2863 IEC-754 Part-1 Swedish Chimney: SEN SS 4241475 (F3 category) Flammability test IEC-332 Part-3 Cat-B Cable Drum Rewinding		Y	Y		Y
Wooden drum(IS : 10418) / Steel drum	Y				
Y =Test applicable					
Notes					
1)	Additional acceptance tests like FRLS, thermal stability, tensile strength and elongation after ageing shall be done on one sample/ lot				
2)	Length measurement / surface finish shall be checked on one length/ size/ lot				
3)	Routine test shall be carried out on each drum length as per specifications.				
4)	MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.				

1.6.9.4 Type tests (Category-I)

The type tests on one drum for each type and size of cables shall be conducted.

- Annealing test (for copper) as per IS:8130
- Tensile test (for Aluminium) as per IS:8130

- c) Wrapping test (for Aluminum) as per IS:8130
- d) Conductor resistance test as per IS:8130
- e) Test for armour wires/ strips as per IS:3975 - The hard drawn aluminium wires armour (in case of single core cables) shall also comply with the tensile test and wrapping test as per IS:8130.
- f) Test for thickness of insulation and sheath - Requirement and methods of test for the thickness of insulation and sheaths shall be as per relevant IS. The calculated diameter over stranded core and "Calculated nominal diameter under outer sheath" shall, however, be determined by the method given in Appendix "A" of IEC-60502.
- g) Tensile strength and elongation test for insulation and sheath - The value of tensile strength and elongation at break-point shall not be less than 125kg/cm² and 150% respectively for the type 'A' and type ST1 PVC compounds. For XLPE insulation these tests shall be carried out in accordance with IS:7098 (Part-III). The tensile strength and elongation at a break shall not be less than 125kg/cm² and 200% respectively.
- h) Ageing test for insulation and sheath - The ageing test shall be carried out as per the procedures laid down in relevant IS, however the period of test shall be 168 hours, instead of 120 hours. The test value obtained for the tensile strength and elongation must not differ from the corresponding values obtained before ageing by more than $\pm 25\%$ in case of type ST2 PVC and XLPE compounds and by more than $\pm 20\%$ in case of type A, and type STI PVC compound.
- i) Loss of mass test - This test is to be carried out on the PVC insulation and sheaths as per IEC-60540. The maximum permissible loss of mass shall be 2mg/cm².
- j) Shrinkage test - This test shall be carried out on the PVC insulation and sheath. The test procedures and test values shall comply with IS:5831.
- k) Cold bend test - This test is to be carried out upto 6mm² cables size in accordance with IS:5831.
- l) Cold impact test - This test shall be carried out on the PVC insulation and sheath in accordance with IS-5831.
- m) Heat shock test for PVC insulation and sheath - This test shall be carried out as per IEC-60540 and IEC-60502.
- n) Thermal stability test for PVC insulation and sheath - This test shall be carried out in accordance with IEC-60540/60540A and shall pass the minimum requirement of 100minutes.

- o) Test for bleeding and blooming of pigments for PVC - This test shall be carried out as per IS :5831.
- p) Fire resistance test - This test shall be carried out in accordance with relevant clause of IS:5831.
- q) Measurement of insulation resistance - This test shall be carried out as per IS:5831. The volume resistivity of the PVC insulated cables shall not be less than 1×10^{14} ohm-cm at 27°C and 1×10^{11} ohm-cm at 70°C.
- r) High voltage test - This test shall be performed as per IS:1554 (Part-I).

The cable with insulation/ sheath material other than PVC shall be subject to all the tests mentioned in their respective Indian or International Standards to which they are conforming. In that case if any of the above test is not applicable, the same will not be carried out.
- s) Hot set test for insulation - This test shall be carried out as per IEC-60540 and shall pass the requirement given in IEC-60502.
- t) Partial discharge test - This test shall be carried out in accordance with IEC-60540 and IEC-60502. The test requirements are as given in IEC-60502.
- u) Bending test - The method and test requirement shall be as per IEC-60502 and on completion of this test, the sample be subject to partial discharge measurement and comply with the requirements.
- v) Tan delta measurement as a function of voltage and capacitance measurement - This test shall comply with clause 16.1.6 of IEC-60502.
- w) Tan delta measurement as a function of temperature - This test shall comply with clause 16.1.7 of IEC-60502.
- x) Heating cycle test plus partial discharge test - This test shall be carried out as per IEC-540 and IEC-502.
- y) Impulse withstand test - This test shall be carried out as per IS:7098 (Part-II). The impulse withstand voltage shall be 75 kV.
- z) High voltage test - This test shall be performed as per IS:7098 (Part –II). The normal sequence of electrical test shall be as per clause 16.1.1 of IEC-60502.

1.6.9.5 Type test (Category-I) - Additional type tests on flammability

- a) The oxygen index test shall be carried out as per ASTM-D-2863. The minimum oxygen index shall be 29.
- b) Accelerated water absorption test and dielectric retention test - This test shall be carried out in accordance with NEMA-WC-5 irrespective of thickness of insulation (applicable for the thermoplastic material only).
- c) Temperature index test for sheath - As per ASTM-D-2863 at different temperature upto 250°C. However, the test shall be carried out by extrapolation method beyond temperature at which the material of sheath may start deformation as per BICC Hand Book Chapter-6 on cables in fire. The number of measurements up to 200°C (room temperature, 50°C, 100°C, 150°C and 200°C) plotted on a graph and then extrapolated. The minimum temperature index shall be 250°C.
- d) Flammability of finished cables - The test shall be carried out as per requirement of Swedish Standards SS-4241475 class-F3, IEEE-383, IEC-60332 and IEC- 60331 (for fire survival cables only) and cables should meet the requirement of all the above standards as mentioned in the respective category of cables.
- e) Acid gas generation during fire of sheath - The test shall be carried out as per IEC-60754-1 as well as per BS or other standard applicable for evaluation of halogen gases and the requirement of maximum halogen gases liberated shall be less than as mentioned elsewhere in the respective category of cables.
- f) Smoke generation by sheath under fire - The test shall be carried out as per ASTM D-2843.
- g) Test for rodent and termite repulsion property of sheath - The details to be given by the manufacturer.

1.6.9.6 Site tests

The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out

1.4.6.5.1.1 Insulation resistance test

1.4.6.5.1.2 High voltage test

1.6.10 Installation of Cables, Earthing system and Lightning Protection system

Items/ Components/ Sub-systems	Tests/ Checks													
	Make, Type, Model, Rating and TC	Dimension	Pre-treatment of sheet	Paint shade, paint thickness, adhesion	IP protection	Bought out items/ Bill of material	HV and IR	Galvanize test	Functional test as per specification	Proof load	Deflection test	Constructional feature as per specification	All Routine and acceptance tests	Item conform to relevant standard
Switch box/ Junction box/ Receptacles (IS-513, IS:5, IS:2629, IS:2633, IS:6745)	Y	Y	Y	Y	Y	Y	Y	Y	Y			Y	Y	
Cable glands (BS:6121)	Y	Y												Y
Cable lug (IS:8309)	Y	Y												Y
Lighting wire (IS:694)	Y	Y										Y		Y
Flexible conduits	Y	Y										Y		Y
Conduits (Galvanize and Epoxy) IS:9537, IS:2629, IS:2633 and IS:6745	Y	Y	Y					Y					Y	
RCC hume pipe (IS:458)	Y	Y												Y
Cable termination and Straight through joint (VDE-0278)	Y	Y			Y	Y						Y	Y	
Cable Trays, Flexible supports system and accessories IS:513, IS:2629, IS:2633 and IS:6745	Y	Y	Y			Y		Y		Y	Y	Y	Y	
Trefoil clamp	Y	Y												Y
GI flats for earthing and lightning protection (IS:2062, IS:2629, IS:6745 and IS:2633)	Y	Y	Y					Y						Y
GI wire (IS-280)	Y	Y	Y					Y						Y
Y =Test applicable														
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.														

1.6.10.1 Type test (Category-II)

- i) All cable trays, supports and accessories etc supplied shall be of type tested quality.
- ii) Cable termination kit and straight through joints shall be tested as per IS:3573 for 3.3kV and 11kV class.
- iii) Fire proof cable penetration system shall be tested for the following tests:
 - Accelerated ageing test
 - Water absorption test
 - Fire rating test
 - Hose stream test
 - Vibration test followed by fire rating test

iv) Galvanizing tests

The quality of galvanizing shall be inspected visually and shall be smooth, continuous, free from flux stains.

In addition following tests shall be conducted :

- Uniformity of coating – The coating of any article shall withstand for 1 minute dips in standard copper sulphate solution without the formation of an adherent red spot of metallic copper upon the basic metal.
- The quality of cadmium/ zinc plating on items with screw threads shall be inspected visually and shall be free from visible defects such as unplated areas, blisters and modules.
- In addition, the plating thickness shall be determined microscopically/ chemically or electronically.

v) Welding

The quality of welding shall be visually inspected, particular attention being paid to the following points.

- The welded joints shall be continuous along its length on both sides and of uniform width and thickness. It should be free from blow holes.
- The weld metal shall be properly fused with the parent metal without undercutting.
- The outside surface of the weld shall be clean. All slag shall have been removed. All welding shall be regularly checked for cracking using magnetic particle inspection or their equivalent technique.

- vi) Physical and dimensional checks for all items
- vii) Deflection test for cable trays
- viii) Following tests shall also be carried out on each type of equipment, devices and materials/ items supplied:
 - Physical and Dimensional checks
 - Check/ measurement of thickness for Nickel chrome plating for cable glands and tinning for cable lugs.
 - Check chemical composition of brass parts for cable glands
 - Hardness check on gaskets
 - Test for uniformity of galvanization

1.6.10.2 Site tests

The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out

- i) Cables
 - a) Check for physical damage.
 - b) Check for insulation resistance before and after termination/ jointing.
 - c) HT cables shall be pressure tested (test voltage as per IS:7098) before commissioning.
 - d) Check of continuity of all cores of the cables.
 - e) Check for correctness of all connections as per relevant wiring diagrams. Any minor modification to the panel wiring like removing/ inserting, shorting, change in terminal connections, etc shall be carried out.
 - f) Check for correct polarity and phasing of cable connections.
 - g) Check for proper earth connections for cable glands, cable boxes, cable armour, screens etc.
 - h) Check for provision of correct cable tags, core ferrules, tightness of connections.

- ii) Cable trays/ supports and accessories
 - a) Check for proper galvanizing/ painting and identification number of the cable trays/ supports and accessories.
 - b) Check for continuity of cable trays over the entire route.
 - c) Check that all sharp corners, burrs, and waste materials have been removed from the trays supports.
 - d) Check for earth continuity and earth connection of cable trays.
- iv) Earthing and Lightning protection system
 - a) Earth continuity checks.
 - b) Earth resistance of the complete system as well as sub-system.

1.6.11 DC Storage Battery												
1.6.11.1 Lead-Acid type battery												
Item/ Components/ Sub-system	Tests/ Checks											
	Dimensions and finish	Conformance to relevant drawing	Chemical composition	Lead coating thickness (minimum 25 microns, IS:6848 and Adhesion check	Conformance to CPWD Specification for teak wood	Paint process checks, paint shade, thickness, adhesion and finish	Constructional requirements as per specification	Insulation Resistance	Marking (Routine and Acceptance test)	Checking of polarity and absence of short circuit (Routine and Acceptance test)	Test for capacities for 10 hours discharge rate along with the test for voltage during discharge (Acceptance test)	
Container and Lids (IS:1146)	Y	Y										
Vent plugs	Y	Y										
Sealing compound (IS:3116)		Y	Y									
Positive and Negative plates		Y	Y									
Separators (IS:6071)	Y	Y										
Electrolyte (water/ sulphuric acid) (IS:1069/ 266)		Y	Y									
Inter-cell connectors and fasteners	Y	Y		Y								
Battery stand	Y	Y			Y	Y						
Cell insulators	Y	Y						Y				
Stack assembly	Y	Y										
Lead-Acid battery (IS:1652)	Y						Y		Y	Y	Y	
Y =Test applicable												
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.												

1.6.11.2 Ni - Cd type battery		(Table 1/2)									
Items/ Components/ Sub-system assembly	Tests/ Checks										
	Dimensions and finish	Impact strength	Conformance to relevant part drawing and Manufacturer' s standards	Resistance to alkali	Chemical composition	Nickle plating thickness	Paint shade, thickness, adhesion and finish	Air pressure test after heat sealing	Marking and mass (Routine and Acceptance test)	Air pressure test (Acceptance test)	
Container and Lids (IS:1146)	Y	Y	Y	Y							
Vent Plugs	Y		Y	Y							
Perforated steel strips	Y		Y	Y		Y					
Active material for positive and negative Plates			Y		Y						
Separators	Y		Y	Y							
Electrolyte			Y		Y						
Inter-cell connectors and fasteners	Y		Y	Y		Y					
Battery stand	Y			Y			Y				
Cell Insulators	Y		Y	Y							
Stack assembly	Y		Y					Y	Y		
Ni-Cd Battery (IS:10918)	Y								Y	Y	
Y =Test applicable											
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.											

1.6.11.2 Ni- Cd type battery (Table 2/2)				
Items/ Components/ Sub-system assembly	Tests/ Checks			
	Retention of charge test (Acceptance test)	AH test (Acceptance test)	Insulation Resistance (Acceptance test)	Polarity and absence of short circuit (Routine and Acceptance test)
Container and Lids (IS:1146)				
Vent Plugs				
Perforated steel strips				
Active material for positive and negative Plates				
Separators				
Electrolyte				
Inter-cell connectors and fasteners				
Battery stand				
Cell Insulators				
Stack assembly		Y		Y
Ni-Cd Battery (IS:10918)	Y	Y	Y	Y
Y =Test applicable				
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.				

1.6.11.3 Type test (Category-II)

All batteries supplied shall be of type tested quality.

1.6.11.3.1 Lead -Acid Plante battery

- i) Type test reports for the following shall be submitted for purchaser's approval :
 - a) Container (Rubber and Plastic containers)

- High voltage test
- Drop ball test
- Plastic Yield test
- Acid resistance test
- Hydraulic thrust endurance test.

b) Cells and batteries

- Tests for capacities for 10 hr and 30 minutes discharge rates and test for voltage during discharge.
- Ampere hours and watt-hour efficiency test.
- Tests for retention of charge.
- Endurance tests.

ii) Acceptance Tests - All acceptance tests as listed below shall be carried out on sample cell selected at random by the purchaser before dispatch and at site after completion of installation.

- Verification of markings.
- Verification of dimensions.
- Test for capacities for 10 hrs discharge rate along with the test for voltage during discharge.

If a battery fails to meet the guaranteed requirements, the purchaser shall have the option of asking to replace the same.

1.6.11.3.2 Nickel-Cadmium battery

i) Type test reports for the following shall be submitted for purchaser's approval :

- Life cycle test
- Tests for capacities for 5 hr. and ½ hr. discharge rates and test for voltage during discharge
- Ampere hours and watt-hour efficiency test
- Tests for retention of charge
- Endurance tests
- Discharge performance at low temperature

ii) Acceptance Tests - All acceptance tests as listed below shall be carried out on sample cell selected at random by the purchaser before dispatch and at site after completion of installation.

- a) Physical examination
- b) Dimensions, mass and layout
- c) Marking
- d) Polarity and absence of short circuit.
- e) Air pressure test
- f) Ampere - hour capacity
- g) Retention of charge

- h) Insulation resistance
- i) Test for capacities for 5 hrs. and ½ hr. discharge rates and test for voltage during discharge

If a battery fails to meet the guaranteed requirements, the purchaser shall have the option of asking to replace the same.

1.6.11.4 Site tests

The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out

- Physical examination
- Dimensions, mass and layout
- Marking
- Polarity
- Insulation resistance
- Impedance measurement

1.6.12 Battery Charger													
Item/ Components/ Sub-system	Tests/ Checks												
	Make, Model, Type, Rating and Finish	Chemical and Mechanical tests	Sheet steel pretreatment and painting process	Conform to relevant standard	Dimensional check and paint shade, thickness, adhesion and finish checks	Complete physical examination for constructional features of battery charger as per specification	Temperature rise test	Dynamic response test	Ripple content test, load limiter and Annunciator and AVR operation test	Operational and functional checks	HV and IR test	Burn-in test at 50°C for 48 hrs.	Degree of protection test .
Rectifier transformer (IS:2026)	Y			Y			Y				Y		
Electronic components including potentiometer (vernier type)	Y			Y									
PCB and electronic cards	Y			Y									
19"standard racks for electronic card	Y					Y							
Control and selector switches (IS:6875)	Y			Y						Y			
Indicating meters (IS:1248)	Y			Y						Y			
Indicating lamps (IS:13947)	Y			Y						Y			
Air break switches/ Fuses(IS:13947/ 13703)	Y			Y						Y			
Control terminal blocks (IS:13947)	Y			Y									
Control transformer (IS:12021)	Y			Y						Y			
Push buttons (IS:4794)	Y			Y						Y			
MCB (IS:8828)	Y			Y						Y			
PVC insulated copper control wires (IS:694)	Y			Y									
Sheet steel (IS:513)	Y	Y	Y	Y									
Synthetic rubber gaskets	Y	Y		Y									
Annunciator	Y									Y		Y	
Battery charger	Y				Y	Y	Y	Y	Y		Y	Y	Y
Y =Test applicable													
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.													

1.6.12.1 Type test (Category-II)

All battery chargers supplied shall be of type tested quality.

Type test reports for the following shall be submitted for purchaser's approval :

- i) Complete physical examination
- ii) Temperature rise test at full load.
- iii) Temperature rise test of rectifier assembly at 200% of full load (for chargers up to 400 A rating).
- iv) Insulation resistance test.
- v) High voltage (power frequency) test on power and control circuits except low voltage electronic circuits.
- vi) Ripple content test at no load, half load and full load
- vii) Automatic voltage regulator operation test at specified AC supply variations at no load, half load and full load
- viii) Load limiter operation test
- ix) Efficiency and power factor measurement.
- x) Input and output surge withstand capability test. Surge voltage as per ANSI-C37.90a shall be applied for period not less than 2 second at the following points of the charger operating at 50°C at full load :

- Across each AC input phase
- Across AC input line to ground.
- Across DC output terminals.
- Across each DC output terminal to ground

The charger shall not exhibit any component damage and there shall be no change in performance as per tests at (vii) and (viii).

- xi) Environmental tests - Steady state performance tests (vii) and (viii) shall be carried out before and after each of the following tests.
 - a) Soak Test - The electronic modules shall be subjected to continuous operation for a minimum period of 72 hours. During last 48 hours, the ambient temperature shall be maintained at 50°C. The 48 hour test period shall be divided into four equal 12 hour segments. The input voltage during each 12 hours shall be nominal voltage for 11 hours followed by 110% of nominal voltage for 30 minutes, followed by 90% of nominal voltage for 30 minutes.
 - b) Degree of protection test.

1.6.12.2 Site tests

The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out

- j) Complete physical examination
- ii) Checking of proper operation of annunciation system

- iii) Temperature rise test at full load for 30 min
- v) Insulation resistance test
- vi) Automatic voltage regulator operation
- vii) Load limiter operation.
- viii) Dynamic response test - overshoot/ undershoot in output voltage of the charger as a result of sudden change in load from 100% to 20% and 20% to 100% shall be measured with the batteries connected and disconnected. Output voltage of the charger connected with battery shall be within 94% to 106% of the voltage setting in above conditions and shall return to, and remain, within the limits specified as mentioned elsewhere, in less than 2 seconds (as applicable).
- ix) Regenerative / pumping back to grid test.

1.6.13 Protection and relay panel for generator, generator transformer and UATs

Items/ Components/ Sub-systems	Tests/ Checks												
	Make, Type, Model, Rating and TC	Electrical Properties			Dimensions and Finish	Functional and operational features	Item to conform to relevant Standards	Pretreatment as per IS 6005	Paint shade, thickness, adhesion and finish	Functional checks	HV and IR test	Degree of Protection - Routine test	All Routine & acceptance tests as per relevant IS / IEC and specification
Protective relays (IS:3231/ EC:60255 / IEC:61850	Y				Y	Y	Y			Y	Y	Y	Y
Auxiliary relays (IS:3231/ / IS:13947)	Y				Y	Y	Y			Y	Y	Y	Y
Control and selector switches (IS:6875)	Y				Y	Y	Y			Y			
Indicating meters (IS:1248)	Y				Y	Y	Y			Y			Y
Indicating lamps (IS:13947)	Y				Y	Y	Y			Y			
Push buttons (IS: 4794)	Y				Y	Y	Y			Y			
Control transformer (IS:12021)	Y				Y	Y	Y				Y		Y
LT fuses (IS:13703)	Y				Y	Y	Y						
Energy meters (IS:722, IS:14697, IEC 62052, IEC:62053	Y				Y	Y	Y						Y
Transducers (IEC:60688)	Y				Y	Y	Y						Y
Diodes	Y	Y				Y	Y			Y			
Terminal Blocks	Y	Y				Y	Y						
Synthetic Rubber Gasket (IS:11149/ 3400)	Y	Y			Y		Y						
Complete Panel (IS:3427)	Y				Y	Y	Y	Y	Y	Y	Y	Y	Y
Test realy kit	Y									Y			Y
Y =Test applicable													
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.													

1.6.13.1 Type test (Category-II)

The Protection and Relay panels shall be of type tested quality.

- i) For various bought out items like CT, VT, relays etc. for which identification can be physically verified and test certificate from manufacturer can be co-related, following shall be indicated in the QP.
 - Review of manufacturer's TC as per relevant standard and compliance to requirements of this specification
 - Physical check and functional/ operational check to ensure that item is fit for assembly on the switch board cubicle.
- ii) For all other items where identification has not been envisaged by manufacturers, a certificate of conformance to be obtained from manufacturer, and component identification shall be tagged as per approved drawing/ schematic. Either of the following or both shall be adopted .
 - Random sample to be tested either in-house or at an independent laboratory for all tests envisaged in relevant standard.
 - Physical check and functional/ operation check to ensure the item is fit for assembly on the panel.
- iii) Type Test of Protective relays / Numerical relays : Type test reports for the following shall be submitted for purchaser's approval as per IEC:60255. The Reports / Certificates of tests conducted in accredited Laboratories (accredited by the national accrediting body of the country where the lab is located) is acceptable
 - a) Insulations tests,
 - b) Electrical environment tests
 - c) EMC tests
 - d) Atmospheric environment tests
 - e) Mechanical Stress tests
 - f) Conformance tests
- iii) Factory Acceptance Tests (FAT) shall include all tests required to fully demonstrate to purchaser's satisfaction that each equipment/sub-system/system as software modules furnished as per this specification. The following minimum tests/ checks shall be conducted as FAT. Any other tests/ checks as per the manufacturer's recommendation and as agreed between Manufacturer and Purchasers shall also be carried out .The Manufacturer shall submit a detailed FAT procedure for Purchaser's approval
 - a) Pre power on checks
 - b) Power on checks
 - c) Hardware tests
 - d) Functional tests
 - e) Parametric tests
 - f) Specific tests on electronic hardware
 - g) Power failure auto-restart tests

h) Testing of interlocking

1.6.13.2 Site tests

The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out

a) Cubicle Wiring

- i) It should be made sure that the wiring is as per relevant drawings. All interconnections between panels shall similarly be checked
- ii) Insulation resistance of all the wires with respect to earth
- iii) Functional checking of all control circuit e.g. closing, tripping interlock, supervision and alarm circuit including proper functioning of component/ equipment
- iv) Check terminations and connections
- v) Wire ducting
- vi) Gap sealing and cable bunching

b) Relays

- i) Check internal wiring
- ii) Insulation resistance between all terminals and body
- iii) Insulation resistance between AC and DC terminals
- iv) Check operating characteristics by secondary injection
- v) Check minimum pick up voltage of DC coils
- vi) Check operation of electrical/ mechanical targets
- vii) Check CT connections with particular reference to their polarities for differential type relays
- viii) Pickup & timing test as per Relay settings
- ix) Configuration checks & communication checks.

c) Meters

- i) Insulation resistance of all insulated portions
- ii) Check CT and VT connections with particular reference to their polarities for power type meter

d) General

- i) Wiring check in conformity with schematics.
- ii) Insulation check.
- iii) Primary injection test.
- iv) Secondary injection test.
- v) On load test.
- vi) DC sequential test.

- e) The following minimum tests/ checks shall be conducted at site. as commissioning tests Any other tests/ checks as per the manufacturer's recommendation shall also be carried out .

- i) Start-up tests
- ii) Calibration and configuration checks
- iii) Pre-commissioning tests
- iv) Point-to-point testing of all the I/O signals
- v) Verification of all status indications by operating the plant
- vi) Verification of event / alarm indications by simulating alarm conditions
- vii) Verification of all analogue indications by injection testing
- viii) Verification of all controls by operating the plant

1.6.14 Emergency Diesel Generating Set											
1.6.14.1 Diesel Engine											
Items/ Components	Tests/ Checks										
	Material test	DP or MPI	UT (on forging and piston bonding)	Balancing	Hydraulic, water fill test	Assembly, Fit up	Dimension	Functional/ Operation test	Performance test as per BS-5514/ or equivalent IS/ ISO Standard including Governing test for 3 hours at full load and one hour at 110%	Fuel consumption (lit/ Kwh), rated power measurement, rated speed	All other tests (if applicable) as per specification and standard
Crank shaft	Y	Y	Y	Y							
Cylinder blocks/ heads	Y				Y						
Liner radiator	Y				Y						
Rotating, moving parts other than crank shaft	Y	Y									
Piston	Y	Y	Y								
Diesel Engine						Y	Y	Y	Y	Y	Y
Y =Test applicable											
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.											

1.6.14.2 Alternator		(Table 1/2)												
Items/ Components	Tests/ Checks													
	Visual	Dimension	Make, Type, Rating, TC, General physical inspection	Mechanical, Chemical properties	NDT, DP or MPI, UT	Metallography	Electrical characteristics	Welding/ Brazing (WPS/ PQR)	Heat treatment	Magnetic characteristics.	Hydraulic, Leak, Pressure test	Thermal characteristics	Run out	Dynamic balancing
Plates for stator frame, end shield, spider etc.	Y	Y	Y	Y					Y					
Shaft	Y	Y	Y	Y	Y	Y			Y					
Magnetic material	Y	Y	Y	Y	Y		Y			Y		Y		
Rotor copper/ aluminium	Y	Y	Y	Y		Y	Y		Y					
Stator copper	Y	Y	Y	Y			Y		Y			Y		
SC ring	Y	Y	Y	Y	Y	Y	Y	Y	Y					
Insulating material	Y		Y	Y			Y					Y		
Tubes for cooler	Y	Y	Y	Y	Y				Y		Y			
Sleeve bearing	Y	Y	Y	Y	Y				Y		Y			
Stator, Rotor, Exciter coils	Y	Y	Y				Y	Y						
Castings, stator frame, terminal box and bearing housing etc.	Y	Y	Y	Y	Y			Y						
Fabrication and machining of stator, rotor, terminal box	Y	Y			Y				Y					
Wound stator	Y	Y					Y	Y						
Wound exciter	Y	Y					Y	Y						
Rotor complete	Y	Y					Y						Y	Y
Exciter, Stator, Rotor, Terminal box	Y	Y					Y							
Accessories, RTD, BTM, CT, AVR. Brushes, Diodes, Space heater, antifriction, bearing, cable glands, lugs, gaskets etc.	Y	Y	Y											
Alternator (IS:4722)	Y	Y	Y				Y							
Y =Test applicable Y1 = for HT Machines only														
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.														

1.6.14.2 Alternator		(Table 2/2)			
Items/ Components	Tests/ Checks				
	All tests as per IS:4722	Vibration	Over speed	Tan delta, shaft voltage and polarization index	
Plates for stator frame, end shield, spider etc.					
Shaft					
Magnetic Material					
Rotor copper/ aluminum					
Stator copper					
SC Ring					
Insulating material					
Tubes for cooler					
Sleeve bearing					
Stator/ Rotor, Exciter coils					
Castings, stator frame, terminal box and bearing housing etc.					
Fabrication and machining of stator, rotor, terminal box					
Wound stator					
Wound exciter					
Rotor complete					
Exciter, Stator, Rotor, Terminal box					
Accessories, RTD, BTM, CT, AVR. Brushes, Diodes, Space heater, antifriction, bearing, cable glands, lugs, gaskets etc.					
Alternator (IS:4722)	Y	Y	Y	Y1	
Y =Test applicable Y1 = for HT Machines only					
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.					

1.6.14.3 Final assembly										
Items/ Components	Tests/ Checks									
	Material test	Dimension	WPS, PQR, Welding	NDT, DP or MPI, UT	Check completeness	Hydraulic, Leak, Pressure test	Functional test	All Routine test as per Specification /IS	No load test for one hour of the DG set assembly	Clearances and alignment
Base frame	Y	Y	Y	Y	Y					
Fuel tank	Y	Y	Y	Y	Y	Y				
Battery (IS:1691)								Y		
Battery charger								Y		
Control panel								Y		
Assembled DG set		Y			Y		Y	Y	Y	Y
Y =Test applicable										
Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.										

1.6.14.4 Type test (Category-II)

All DG sets and accessories supplied shall be of type tested quality.

Type test reports for the following shall be submitted for purchaser's approval :

- Measurement of resistance and air gap
- Phase sequence test
- Regulation test
- Measurement of open circuit and short circuit characteristics
- Efficiency test
- Temperature rise test
- Momentary overload test
- Over speed test
- High Voltage test
- Insulation resistance test (both before and after high voltage test)
- Noise level as per IS:12065

- l) Vibration as per IS:12075
- m) Determination of deviation of voltage waveform from sinusoidal
- n) Degree of protection test on control panel for IP-52

1.6.14.5 Site Tests

The following minimum tests/ checks shall be conducted at site. Any other tests/ checks as per the manufacturer's recommendation shall also be carried out

- a) Visual Test
- b) Electrical Tests
 - i) Calibration of instruments, relays etc.
 - ii) Primary and secondary injections tests of protective devices.
 - iii) Function, interlock test of control panel
 - iv) Insulation resistance measurement of power and control circuits.
- c) Mechanical Tests
 - i) Trial run of engine
 - ii) Governor testing
 - iii) Over speed trip test
 - iv) Load pick up and load rejection tests
 - v) Load test and temperature rise measurement of engine and alternator
 - vi) Fuel consumption test

1.6.15 Electro Static Precipitator (ESP) – Electrical				
1.6.15.1 General				
Items/ Components	Tests/ Checks			
	Visual	Make, Type, Rating etc.	Final Inspection as per IS/ IEC /BS	Remarks
TR Set	Y	Y		Refer table for TR
ESP insulator (IEC 168/ 273, IS 2544)	Y	Y	Y	ESP insulators shall be additionally subjected to high temperature test on sample basis as per mutually agreed upon procedure.
Electrostatic Precipitation Management System	Y	Y		Refer table for annunciation, control, PLC Panel
Microprocessor based rapper controller	Y	Y		Refer table for annunciation, control, PLCPanel
Disconnecting switch (IS 13947)	Y	Y	Y	
Heaters (IS 4159/ BS 6351)	Y	Y	Y	

1.6.15.2 Transformer Rectifier (TR) Set								(Table 1/2)				
Items/ Components	Tests/ Checks											
	Visual	Dimensional and physical	Make, Type, Rating	Mechanical, chemical Properties	Electrical	Electronics	Welding	NDT	Pretreatment of tank	Painting	All tests as per IS2026	All tests as per IEC-146
Thyristor	Y	Y	Y			Y						Y
Contactor	Y	Y	Y		Y							
Switch Fuse Unit	Y	Y	Y		Y							
HRC Fuse	Y	Y	Y		Y							
Current Transformer	Y	Y	Y		Y							
Overvoltage Protector	Y	Y	Y		Y							
Measuring Instruments	Y	Y	Y		Y							
Control Transformer	Y	Y	Y		Y							
Bushings	Y	Y	Y		Y							
Dial Thermometer	Y	Y	Y		Y							
Resistor wire wound	Y	Y	Y		Y							
Sudden Pressure Relay	Y	Y	Y		Y							
PVC insulated copper wire (ISI Marked)	Y	Y	Y		Y							
Terminal Block	Y	Y	Y		Y							
Gasket	Y	Y	Y	Y								
Electrolytic copper	Y	Y	Y	Y								
Capacitor, Resistor	Y	Y	Y	Y	Y	Y						
PCB	Y	Y	Y		Y							
Insulated conductor	Y	Y	Y	Y	Y							
Laminations	Y	Y	Y	Y	Y							
Press board, paper	Y	Y	Y	Y	Y							
Insulating oil (Silicon)	Y		Y	Y	Y							

1.6.15.2 Transformer Rectifier (TR) Set					(Table 2/2)							
Items/ Components	Tests/ Checks											
	Visual	Dimensional and physical	Make, Type, Rating	Mechanical, Chemical Properties	Electrical	Electronics	Welding	NDT	Pretreatment of tank	Painting	All tests as per IS2026	All tests as per IEC-146
	Radiator	Y	Y	Y	Y				Y			
	Transformer Tank	Y	Y	Y	Y			Y	Y		Y	
	Panel Fabrication	Y	Y	Y	Y			Y		Y	Y	
	Electronic Cards	Y	Y	Y		Y	Y					
	Linear Reactor, Choke	Y	Y	Y	Y	Y						
	Transformer Assembly	Y	Y		Y	Y						
	Control Panel	Y	Y	Y	Y	Y	Y				Y	
	HV, TR Set	Y	Y	Y		Y	Y				Y	Y
ESP Controller (Separate QP)	Y	Y	Y		Y	Y	Y		Y		Y	

1.6.15.3 Annunciation, Control, PLC Panel													
Items/ Components	Tests/ Checks												
	Visual	GA, BOM ,Lay out of components	Dimensions, General physical inspection	Paint shade, thickness, adhesion	Component Rating, Make, Type	Wiring	IR and HV	Review of TC for instruments	Accessibility of devices	Illumination	Functional check for control element ,	Test as per IEC:1131 *	Routine and acceptance tests as per IS 8623
Annunciation, Control, PLC Panel	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Notes													
1)	Detailed procedure of Burn-in and Elevated temperature test shall be as per Quality Assurance Programme												
2)	(*) Applicable for PLC												
3)	MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.												

1.6.15.4 VFD Module				
Attributes/ Characteristics Items/ Components/ Sub-system assembly	Visual & Dimensional checks	Make / Type/ Rating etc.	Final Inspection as per IS/ IEC	Remarks
HT Breaker (IEC-56)	Y	Y	Y	
DC Reactor	Y	Y		For details refer table for DC Reactor
Transformer	Y	Y	Y	For details refer table for Transformer
Motor	Y	Y	Y	For details refer separate table for Motor
VFD Panel	Y	Y	Y	For details refer table for VFD Panel

Note:

- 1) This is an indicative list of tests/checks. The manufacture is to furnish a detailed Quality Plan indicating the practices & Procedure followed along with relevant supporting documents during QP finalization
- 2) Make of all major Bought-out Items will be subject to purchaser's approval.
- 3) Note: MQP shall be submitted by the manufacturer comprising the tests/checks as stipulated in the matrix/table attached for the specific item/component by the respective manufacturer.

ANNEXURE D

QUALITY ASSURANCE PLAN OF MAJOR EQUIPMENTS USED IN DISTRIBUTION SYSTEM

Quality Assurance Plan

The Utility shall be solely responsible & accountable for assuring quality in the project works. Utility shall formulate a detailed Comprehensive Quality Assurance (QA) plan for the works to be carried out with an objective to create quality infrastructure works. The QA Plan shall be an integral part of the contract agreement with the contractor/ equipment supplier and erection agency as the case may be, in case of turnkey/ partial turnkey or departmental execution of works. Utility has to ensure that the quality of materials/ equipment supplied at site and execution of works carried out at field is in accordance to Quality Assurance Plan/ Guaranteed Technical Particulars GTP) and Technical specifications/ Approved Drawings/ Data Sheets etc.

The Utility, through Contractor/ Project Monitoring Agency (PMA)/ Third Party appointed by Utility, shall strictly ensure Quality Assurance checks during the day to day course of project execution, and ensure the quality of material and equipment as per Quality Plan/ Approved Drawings/ Technical Specifications/ Data sheet/ GTP/ applicable national & international standards etc.

The Utility should ensure that Quality Assurance Plan should be prepared keeping in view the following:

- All equipment/ materials shall comply with the relevant Indian Standard/ International Standards, if Indian Standards are not available.
- All Type Tests, Routine tests & Acceptance tests shall be as per relevant IS.
- Pre-Dispatch Inspection at factory site shall be limited to Power Transformers only. The Power transformers, of all capacities of primary voltage level of 33kV and below, shall be subjected to Pre-Dispatch Inspection at factory site. The details are at Annexure-I.
- Random sampling of material/ equipment from field site/ stores shall be taken up for testing at NABL accredited labs (3rd Party or Utility owned). The minimum sample size for the material/ equipment shall be as specified in Annexure – II of these guidelines.
- The Utility / Contractor (*to be specified in tender document*) shall be responsible for safe handling, packing and transportation of the sealed sample material up to the identified testing laboratory and back to field site/ store. However, in the event the Utility requests to test the material/ equipment after it has been commissioned at site, all applicable costs shall be borne by the Utility.
- The choice of lot and random sampling shall be done by the Utility in the presence of the manufacturer/ supplier, the Contractor and any third party, if applicable.

- Utility shall endeavor to pick the random samples from a lot within ten (10) days of receipt of such lot at the field site/ stores and send it for testing, failing which such lot shall be treated as accepted by the Utility. Further, it should be ensured that testing reports are made available within 2 weeks of submission of samples. However, during the period of random sampling and testing, the installation and commissioning of material/ equipment at the field site may continue.
- In case, the NABL lab test report for random sample from field site/ stores are found significantly mismatched with Routine Test report submitted by manufacturer/ supplier, the entire lot of materials/ equipment supplied by the defaulting manufacturer/ supplier shall be rejected whether supplied materials/ equipment is lying in site stores, in transit or under erection, testing & commissioning or has already been commissioned. All costs related to removal of such rejected materials/ equipment and replacement with new materials/ equipment shall be borne by Contractor/ Manufacturer/ Supplier without any cost implication to the Utility.
- Applicable norms for failure rate of equipment/ material supplied by vendors should be prescribed in the tender documents in case of turnkey/ partial turnkey/ departmental execution of works. Non adherence to such norms should cause black listing of such vendors.
- Further, a system of reporting should be put in place by the Utility to monitor failure rates of key equipment. Suppliers/ vendors/ contractors whose equipment have failed (repeatedly in last 2-3 years with a failure rate of more than 10%) and their promoters shall be blacklisted for a period of 5 years from participating in future tenders for similar works
- All coordination activities including reporting etc. should be through an IT based solution rather than deploying manpower for this purpose.
- The cost of quality monitoring may be optimized including the manpower cost for coordination activities.

TYPE TEST & ROUTINE TESTS:

All the Type Tests & Routine Tests shall be carried out as per Indian Standards.

SITE ACCEPTANCE TEST:

Random sampling of following material/ equipment shall be picked from field site/stores for testing at third party NABL accredited laboratory. Minimum sample size shall be as specified in Annexure-II:

1. Distribution Transformer
2. Ring Main Unit
3. CT/PT Unit (Outdoor) & Metering Cubicle (Indoor)
4. Circuit Breaker
5. Insulators
6. Cables
7. Control & Relay Panel
8. Overhead Conductor
9. Energy Meter
10. Poles
11. Capacitor
12. Distribution Box

The Utility may add any other important material in the above list. The site acceptance tests shall be as per approved Drawings/ Technical Specifications/ Data sheet/ GTP/QA Plan and applicable national & international standard.

In case of power transformer, after installation and commissioning at site, the power transformers shall be further tested to demonstrate satisfactory operation at full load/ as per field requirements, for minimum duration of 5 days without any breakdown or fault. If the power transformer fails within such duration, the same shall be replaced by the manufacturer/ bidder/ supplier without any additional cost to the Utility.

PRE-COMMISSIONING TESTS

On completion of erection of the equipment and before charging, each item of the equipment shall be thoroughly cleaned and then inspected jointly by the Utility and the contractor for correctness and completeness of installation and acceptability for charging, leading to initial pre- commissioning tests at Site. The list of pre-commissioning tests to be performed should be included in the Contractor's quality assurance programme.

COMMISSIONING TESTS

The contractor will use all required instrumentation and control equipment during commissioning tests and such measuring equipment and devices should be duly calibrated as far as practicable. The tests will be conducted at the specified load points and as near the specified cycle condition as practicable. The contractor will apply proper corrections in calculation, to take into account conditions, which do not correspond to the specified conditions. Any special equipment, tools and tackles required for the successful completion of the Commissioning tests shall also be provided by the contractor.

The specific tests to be conducted on equipment should be included in the technical specifications. However, where the pre-commissioning tests have not been specified specifically, they shall be as per relevant IS code of practice or as mutually agreed. The Contractor shall also be responsible for ensuring compliance to all statutory requirements for commissioning and successful operation of the system.

FIELD QUALITY PLAN

The Utility should prepare a separate Field Quality Plan (FQP) for civil, electrical, mechanical etc. works supported with drawings which shall be approved by their competent authority and may be uploaded at web portal. The contractor should adhere to FQP while carrying out physical works. The Utility should also prepare a comprehensive FQP for testing & commissioning of Grid Substations, Distribution transformer Substation, 33 KV, 11 KV line, LT line etc. The system should be energized only after performing all tests as described in the FQP and after clearance from Electrical Inspector. Proper records in this regard, including tests on earth resistance, insulation resistance of 11 kV line & Distribution Transformer etc. shall be maintained, jointly signed by Utility and contractor. All the quality assurance checks conducted in the field should be documented properly and signed by the quality engineer of the contractor & Utility and shall be kept for future reference. These documents shall be maintained by the Utility in proper order and shall be made available at site for verification during inspection.

QUALITY ASSURANCE PROGRAMME OF THE BIDDER/ CONTRACTOR

To ensure that the equipment and services under the scope of the tender, whether manufactured or performed within the Contractor's Works or at his Sub-contractor's premises or at the Utility site or at any other place of work, are in accordance with the specifications/ tender conditions, the Contractor shall adopt a suitable 'Quality Assurance Programme'. Such programme shall be broadly outlined by the contractor and finalized after discussions with the Utility before the award of the contract. The detailed programme shall be submitted by the contractor after the award of contract and acceptance by the Utility.

A Quality Assurance Programme of the contractor shall generally cover the following:

- Organization structure of the Contractor for management and implementation of the proposed quality assurance program
- Documentation control system
- Qualification data for bidder's key personnel
- The procedure for purchases of materials, parts components and selection of sub- Contractor's services including vendor analysis.
- System for shop manufacturing and site erection controls including process controls and fabrication and assembly control
- Control of non-conforming items and system for corrective actions
- Test procedure for field activities
- Control of calibration and testing of measuring instruments and field activities
- System for indication and appraisal of field inspection status
- System for quality audits
- System for authorizing release of manufactured product
- System for maintenance of records
- System for handling storage and delivery
- A manufacturing quality plan detailing out the specific quality control measures and procedures adopted for controlling the quality characteristics relevant to each item of equipment furnished and/ or services rendered.
- A Field quality Plan covering field activities

The Utility or his duly authorized representative like PMA reserves the right to carry out quality audit and quality surveillance of the system and procedure of the Contractor/ his vendor's quality management and control activities.

The Contractor would be required to submit all the Quality Assurance documents as stipulated in the Quality Plan. The contractor shall supply the materials/ equipment of type & design which has already been Type Tested. Contractor/ Manufacturer shall provide copy of such tests at the time of bidding and also at site in support of type-tested materials supplied under the contract.

The Utility should share their Quality Assurance Plan/ Field Quality Plan (FQP) with the Contractor. The schedule for submission/ approval of document as per QAP/ FQP shall be finalized before placement of the contract, keeping in view the overall project schedule. The Contractor shall be responsible for any time delay, misinterpretation, error and conflict during design, manufacturing, testing and erection of the Works resulting from non- compliance with the requirements of the approved Specification.

QUALITY ASSURANCE PLAN FOR STAGE INSPECTION-I OF POWER TRANSFORMER
(33 kV & 22 kV level)

S No.	Particulars	As offered	As observed	Deviation and Remarks
(A)	<u>Inspection of Core :</u>			
	(I) Core Material			
	1) Manufacturer's characteristic certificate in respect of grade of lamination used. (Please furnish test certificate)			
	2) Thickness of core lamination			
	3) Remarks regarding Rusting and smoothness of core.			
	(II) Core Construction :			
	(1) Core Diameter (mm)/Maximum step width (mm)			
	(2) Total cross sectional area of core			
	(3) Effective cross sectional area of core			
	(4) Whether top yoke is cut for LV connection.			
	(5) If yes, at 4 above, whether Reinforcement is done.			
	(6) Core length (leg center to leg center)			
	(7) Window height.			
	(8) Core height			
	(9) Core weight only including channel weight. (Channel weight can be taken separately) Tolerance as agreed in GTP			
(B)	INSPECTION OF WINDING			
	(I) Winding material			
	(1) Material used for			
	a) HV winding			
	b) LV winding			
	(2) Grade of material for			
	a) HV winding			

	b) LV winding			
	(3) Test certificate of manufacturer (enclosed copy) for winding material of:			
	a) HV			
	b) LV			
	(II) Construction Details			
	1) Size of Cross sectional area of conductor for :			
	a) HV winding			
	b) LV winding			
	2) Type of insulation for conductor of :			
	a) HV winding			
	b) LV winding			
	3) Diameter of coils in:			
	a) LV winding			
	a) Internal Diameter (mm) / Inner dimensions			
	ii) Outer diameter (mm) / Outer dimensions			
	b) HV winding			
	a) Internal diameter (mm)/Inner dimensions			
	ii) Outer diameter (mm)/Outer dimensions			
	4) Current density of winding material used for:			
	a) HV			
	b) LV			
	5) Total No. of turns			
	a) HV coils			
	b) LV coils			
	6) Total weight of coils of			
	a) LV winding (Kg)			
	b) HV winding (Kg)			
(C)	INSULATION MATERIALS			

	(I) DPC Paper Insulation			
	a) Type of Paper (Dotted Kraft or Diamond Dotted Kraft)			
	b) Make			
	c) Thickness (mm)			
	d) DPC laying direction			
	e) Percentage Overlapping			
	II) Interlayer Insulation			
	a) Type of Paper			
	b) Make			
	c) Thickness (mm)			
	III) Between HV and LV winding			
	a) Type of Paper			
	i. Make			
	ii. Thickness (mm) (all size)			
	b) Type of Pressboards			
	i. Make			
	ii. Thickness (mm) (all size)			
	IV) Between core and LV			
	Type of Paper			
	i. Make			
	ii. Thickness (mm) (all size)			
	Type of Pressboards			
	i. Make			
	ii. Thickness (mm) (all size)			
	V) Material used for top and bottom yoke and insulation			
	a) Type of Material			
	i. Make			
	ii. Thickness (mm) (all size)			

	VI) Material used for Spanner, wedge and Axial for insulation			
	a) Type of Material			
	i. Make			
	ii. Thickness (mm) (all size)			
	iii. Visual condition(i.e free from dust, burr, damage and sharp edges)			
	VII) Test certificate of manufacturer (enclose copy for all type of papers and pressboard used)			
(D)	TANK :			
	(I) Construction Details:			
	1) Shape			
	2) Thickness of side wall (mm)			
	3) Thickness of top and bottom plate (mm)			
	4) Provision of sloping top cover			
	5) Tank internal dimensions (mm)			
	a) Length X Width			
	b) Height			
	(II) General Details :			
	1) Inside painted by oil corrosion resistant paint (please specify which type of coating done)			
	2) Provision of lifting lugs.			
	a) Numbers			
	b) Reinforcement done by welding all side of Lug			
	3) Provision of air release plug			
	4) Provision of hot dip galvanized GI Nuts Bolts with 1no. plain and 1no. spring washer.			
	5) Deformation of side wall of tank when subject to:			
	a) Vacuum of (-) 0.7 Kg/sq.cm for 30 minutes.			
	b) Pressure of 0.8 Kg/sq.cm. for 30 minutes.			
(E)	TERMINALS:			
	1) Material whether of Brass Rods			
	a) HV			

	b) LV			
	2) Size (dia. In mm)			
	a) HV			
	b) LV			
(F)	BUSHINGS – Two part			
	1) Whether HV & LV bushings mounted as per drawing.			
	a) HV- Top Inclined			
	b) LV – Side			
	2) Bushing Clearance: (mm)			
	a) LV to Earth			
	b) HV to Earth			
	3) Bushing are two part and inner part shall be sealed and external part is replaceable without affecting sealing and need of opening of top cover.			
(G)	TANK BASE:			
	1) Whether tank base is welded folded upwards, as Specified in specification.			
(H)	OIL:			
	1) Name of supplier			
	2) Breakdown voltage of oil: (kV)			
	a) Filled in tanked transformer			
	b) In storage tank (to be tested by Inspecting officer)			
	3) Supplier's test certificate (enclose copy)			
(I)	ENGRAVING:			
	1) Engraving of Sl. No. and name of firm and YoM.			
	a) On bottom of clamping channel of core-coil assembly.			
	b) On Body of tank (on Yellow base with Black paint)			
(J)	NAME PLATE DETAILS:			
	Whether Name Plate is as per approved drawing			
(K)	COLOUR OF TRANSFORMER			
	1) Tank body (Inner side)			
	2) Tank body (Outer side)			

QUALITY ASSURANCE PLAN FOR STAGE INSPECTION- II OF POWER TRANSFORMER**(33 kV & 22 kV level)**

S. No.	Name of test	Specified value(Range)	Test Result	Pass/Fail
1	Visual inspection for material used, it's finish and workmanship	Free from cracks, nicks, protrusion and other visible defects.		
2	Physical Verification of complete CCA with all fittings including insulation used, packing used, Bus bars, Flats, Channels etc.	GTP Values		
3	Clearances and Dimension measurements.	GTP Values		
4	Physical Verification Leads positions	GTP Values		
5	Ratio Measurement at all Taps	GTP Values		
6	Magnetic Balance Test	GTP Values		
7	2KV/Isolation Test for one minute	Should Withstand		

QUALITY ASSURANCE PLAN FOR FINAL INSPECTION OF POWER TRANSFORMERS**(33 kV & 22 kV level)**

S. No.	Name of test	Specified value(Range)	Test Result	Pass/Fail
1	Visual inspection for material, finish and workmanship	Free from cracks, nicks, protrusion and other visible defects.		
2	Physical Verification of complete Transformer with all assembly including test rollers, radiators, cable boxes etc. and Checking of weights, Dimensions.	Approved GTP Values		
3	Measurement of Winding Resistance	Approved GTP Values		
4	Measurement of voltage ratio and phase displacement	Approved GTP Values		
5	Verification of vector group relationship	DYn11		
6	Measurement of short-circuit impedance and Load Loss.	Approved GTP Values		

7	Measurement of No-Load Loss and Current (Losses at 90, 100 and 110% of rated voltage).	Approved GTP Values		
8	Measurement of insulation resistance.	Approved GTP Values		
9	Dielectric Test	GTP Values		
10	Test on ON-Load Tap Changer	GTP Values		
11	Zero-Phase sequence Measurement	Approved GTP Values		
12	Oil Pressure/leakage test on completely assembled transformer at 0.35kg/sq.cm for 8 hrs.	Should withstand		
13	Bushing shall be tested for Capacitance and Power factor and shall meet the manufacture's requirement.	GTP		
14	All CTs and resistance of image coil for winding temperature indicator shall be checked for ratio test, polarity and knee point voltage test	GTP		
15	Determination of Capacitances and dissipation factor winding-to-earth and between windings.	GTP		
16	Magnetic balance test	GTP		
17	Measurement of Magnetizing current at low voltage			
18	Voltage Regulation at rated load and at unit, 0.9, 0.8 lagging power factor	GTP		
19	Measurement of Acoustic Noise Level	GTP		
20	Measurement of the power taken by the fans	GTP		
21	Functional tests on auxiliary equipment: i. Test on OTI and WTI ii. High Voltage test on insulation test for Auxiliary Wiring.	GTP		

22	Test on Oil filled in Transformer i. Dielectric Strength of Oil ii. Water Content. iii. Dielectric Dissipation factor (tan delta at 90° C. iv. Resistivity	GTP		
23	Temperature rise test	GTP		
24	Short Circuit withstand test	Should withstand		
25	Lightning Impulse voltage test with chopped wave.	GTP		

SAMPLE SIZE FOR RANDOM SAMPLING OF MATERIAL/ EQUIPMENT

Sample size for random sampling of material/ equipment to be picked from field site/ stores for testing at third party NABL accredited laboratory:

1. Distribution Transformer: 5% of each Lot
2. Ring Main Unit: 5% of each Lot
3. CT/PT Unit (Outdoor) & Metering Cubicle (Indoor): 5% of each Lot
4. Circuit Breaker: 5% of each Lot
5. Insulators: 1% of each Lot
6. Cables/ABC: 1 sample (as per applicable standard) from each Lot (Lot limited to 25 Drums)
7. Control & Relay Panel: 5% of each Lot
8. Overhead Conductor: 1 sample (as per applicable standard) from each Lot (Lot limited to 25 Drums)
9. Energy Meter: 1% or 32 meters per Lot whichever is lower
10. Poles: 1% of each Lot
11. Capacitor: 5% of each Lot
12. Distribution Box: 1% or 32 nos. per Lot whichever is lower.

ANNEX-B.2.18

Captive Power Plant Generation (to be furnished by State Entity)							
S.No.	Name of State/Uts	Installed Capacity of Captive Power Plants	Gross Generation	Net Generation	Electricity Utilization		
					Power Drawl from Grid	Injection of power to Grid	Captive Consumption
			(kwh)	(kwh)	(kwh)	(kwh)	(kwh)
1							
2							

Open Access Details			
S.No.	Name of State/UTs	Open Access Consumption	
		Included in Energy Requirement of State	Not included in Energy Requirement of State
		(kWh)	(kWh)
1			
2			

Rooftop Solar Generation Details				
S.No.	Name of State/UTs	Installed Capacity of State/UTs	Rooftop Solar Generation	Exchange with Grid
			(kWh)	(kWh)
1				